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SECOND EDITION

FOOD MICROBIOLOGY

An Introduction

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Preface

Food microbiology is an exciting field that reaches into every home and supports a multibillion-dollar food industry. This book provides a taste of its complexity and challenge. The safety of food requires more than memorization of microbiological minutiae. It calls for critical thinking, innovative approaches, and healthy skepticism. We have tried to foster these skills so that today's students will be able to solve tomorrow's problems.

We would have never attempted to write a textbook on such a wide and complex topic as food microbiology "from scratch." Fortunately, ASM Press had published an advanced text for researchers, graduate students, and professors who needed the most up-to-date and in-depth treatment of food microbiology. *Food Microbiology: Fundamentals and Frontiers* was written by an army of subject area experts who presumed that the reader had a working knowledge of microbiology, biochemistry, and genetics. The success of the first two editions of that book gave us the courage (and the resource) to write a food microbiology textbook for undergraduates. *Food Microbiology: an Introduction* is the child of the "big book." For the first edition of this book, we rewrote the experts' chapters to make them accessible to an undergraduate with a semester of microbiology and no biochemistry. Now we have rewritten them again in response to student input about the first edition. In some cases, this meant adding foundational material; in others, it entailed deleting details that only an expert needs to know. The chapters in this book are, therefore, quite different from those originally written for the "big book." In all cases, we have tried to write in a style, at a level, and in language appropriate for undergraduates. To enhance its utility as a textbook, we have added case studies, word puzzles, chapter summaries, questions for critical thinking, a glossary, and even a few cartoons.

The book is divided into five sections. Students should be aware that there is a substantial amount of material in the second edition that is not covered in the first edition. The first section covers the foundational material, describing how bacteria grow in food, how the food affects their growth, control of microbial growth, spores, detection, and microbiological criteria. Instructors may choose to use the other four sections in virtually any order. The gram-negative and gram-positive foodborne pathogens are covered in sections II and III, respectively. Section IV contains chapters on beneficial microbes and spoilage organisms. This edition has taken the single chapter on fermentations and split it into a chapter about lactic acid

bacteria and one about yeast fermentations. Molds are covered both as spoilage organisms and as potential toxin producers. Since viruses may cause more than half of all foodborne illnesses, treatment of viruses has been expanded to include explanations of lytic and temperate phages, the importance of bacteriophage infection prevention to the dairy industry, and the recent adoption of phages for pathogen control. Prions are not bacteria, molds, or viruses; in fact, they are not "microbes" at all. However, they are a major biological concern to the public and food safety experts. Section V covers the chemical, biological, and physical methods of controlling foodborne microbes and closes by examining industrial and regulatory strategies for ensuring food safety.

Although only our names appear on the cover of this book, many people have made important contributions to it. First and foremost, we acknowledge the subject experts whose chapters in *Food Microbiology: Fundamentals and Frontiers* were important sources of information for our writing. They are Gary R. Acuff, John W. Austin, J. Stan Bailey, Dane Bernard, Larry R. Beuchat, Gregory A. Bohach, Robert E. Brackett, Robert L. Buchanan, Herbert J. Buckenhüskes, Lloyd B. Bullerman, Iain Campbell, Michael L. Chikindas, Dean O. Cliver, Jean-Yves D'Aoust, P. Michael Davidson, James S. Dickson, Michael P. Doyle, Józef Farkas, Peter Feng, Graham H. Fleet, Joseph F. Frank, H. Ray Gamble, Per Einar Granum, Paul A. Hartman, Eugene G. Hayunga, Craig W. Hedberg, Ailsa D. Hocking, Lynn M. Jablonski, Timothy C. Jackson, Eric A. Johnson, Mark E. Johnson, James B. Kaper, Jimmy T. Keeton, Charles W. Kim, Sylvia M. Kirov, Todd R. Klaenhammer, Keith A. Lampel, Alex S. Lopez, Douglas L. Marshall, Anthony T. Maurelli, John Maurer, Bruce A. McClane, Jianghong Meng, Kenneth B. Miller, Irving Nachamkin, James D. Oliver, Ynes R. Ortega, Merle D. Pierson, John I. Pitt, Steven C. Ricke, Roy M. Robins-Browne, Peter Setlow, L. Michele Smoot, James L. Steele, Bala Swaminathan, Sterling S. Thompson, Richard C. Whiting, Karen Winkowski, Irene Zabala Díaz, Tong Zhao, and Shaohua Zhao.

The reader should thank the students who reviewed each chapter for level and depth of coverage, writing style, and "what an undergraduate could be expected to know," in addition to grammar and usage. We thank Marcelo Bonnet, Jon Cruz, Rebecca Dengrove, Sylvia Dominguez, Siobain Duffy, Megha Gandhi, Callie Gunnet, Glynis Kolling, Wendy M. Iwanyshyn, Jennifer McEntire, Karla Mendoza, Rebecca I. Montville, Mohamed Badaoui Najjar, June Oshiro, Hoan-Jen Pang, Ethan Solomon, Sarah Smith-Simpson, and Ruth Wiranan for their reviews. Any errors, omissions, or oversimplifications fall on our shoulders.

All of these people, as well as Eleanor Riemer and Ken April (our editor and production editor, respectively, at ASM), helped make this text "student friendly." We hope you find it so and encourage you to explore careers in food microbiology. Remember, there will always be people who have to eat and there will always be microbes. Food microbiologists have great long-term job security.

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KARL R. MATTHEWS

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SECTION I Basics of Food Microbiology

1 The Trajectory of Food Microbiology
2 Factors That Influence Microbes in Foods
3 Spores and Their Significance
4 Detection and Enumeration of Microbes in Food
5 Rapid and Automated Microbial Methods
6 Indicator Microorganisms and Microbiological Criteria

1

The Trajectory of Food Microbiology

Introduction
Who's on First?
Food Microbiology, Past and Present
To the Future and Beyond
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- increase awareness of the antiquity of microbial life and the newness of food microbiology as a scientific field
- appreciate how fundamental discoveries in microbiology still influence the practice of food microbiology
- understand the origins of food microbiology and thus, anticipate its forward path

INTRODUCTION

A former president of the American Society for Microbiology (Box 1.1) defined "microbiology" as an artificial subdiscipline of biology based on size. This suggests that basic biological principles hold true, and are often discovered, in the field of microbiology. "Food microbiology" is a further subdivision of microbiology. It studies microbes that grow in food and how food environments influence microbes. In some ways, food microbiology has changed radically in the last 20 years. The number of recognized foodborne pathogens has doubled. "Safety through end product testing" has given way to the "safety by design" provided by Hazard Analysis Critical Control Points (HACCPs). Genetic and immunological probes have replaced biochemical tests and reduced testing time from days to minutes. In other ways, food microbiology is still near the beginning. Louis Pasteur would find his pipettes in a modern laboratory. Julius Petri would find his plates (albeit plastic rather than glass). Hans Christian Gram would find all the reagents required for his stain. Food microbiologists still study only the microbes that we can see under the microscope and grow on agar media in petri dishes. Experts suggest that only 1% of all the bacteria in the biosphere can be detected by cultural methods.

This chapter's discussion of microbes per se sets the stage for a historical review of food microbiology. The bulk of it concerns bacteria. Most of this book deals with bacteria. Less is known about viruses and prions; so little that they are covered later in a single chapter. This chapter ends with some thoughts about future developments in the field.

WHO'S ON FIRST?

Let there be no doubt about it, the microbes were here first. It is a microbial world (Fig. 1.1). If the earth came into being at 12:01 a.m. of a 24-h day, microbes would arrive at dawn and remain the only living things until well

BOX 1.1

Preparing for the future

Membership in professional societies is a great way to advance professionally, even as a student (who benefits from reduced membership fees and is eligible for a variety of scholarships). Professional societies have many functions: they provide continuing education and employment services to their members, provide expertise to those making laws and public policies, have annual meetings for presentation of the latest science, and publish a variety of books and journals. The three main societies for food microbiologists are listed below. Applications for membership can be obtained from their websites.

The American Society for Microbiology (ASM) (www.asm.org) is the largest life science society in the world. It has 27 divisions covering all facets of microbiology from microbial pathogenesis to immunology to antimicrobial agents to food microbiology, and it publishes 11 scholarly journals. ASM Press is the publisher of this book.

The Institute of Food Technologists (IFT) (www.ift.org) devotes itself to all areas of food science by discipline (food microbiology, food chemistry, and food engineering) as well as by commodity (cereals, fruits and vegetables, and seafood) and by processing method (refrigerated foods). IFT is a nonprofit scientific society with 22,000 members working in food science, food technology, and related professions in industry, academia, and government. IFT publishes two journals, sponsors a variety of short courses, and contributes to public policy and opinion at national, state, and local levels.

The International Association for Food Protection (www.foodprotection.org) is the only professional society devoted exclusively to food safety microbiology. IAFP is dedicated to the education and service of its members, as well as industry personnel. Members keep informed of the latest scientific, technical, and practical developments in food safety and sanitation. IAFP publishes two scientific journals, *Food Protection Trends* and the *Journal of Food Protection*.

after dusk. Around 9 p.m., larger animals emerge, and a few seconds before midnight, humans (Fig. 1.2) appear. The microbes were here first, they cohabit the planet with us, and they will be here after humans are gone. Life is not sterile. We live in a microbial world. Microbes can never be (nor should they be) conquered, once and for all. The food microbiologist can only create foods that microbes do not "like," manipulate the growth of microbes that are in food, kill them, or exclude them by physical barriers.

Bacteria live in airless bogs, in thermal vents, in boiling geysers, in us, and in foods. We are lucky that they are here, for microbes form the foundation of the biosphere. We could not exist without microbes, but they would do just fine without us. Photosynthetic bacteria fix carbon into usable forms and make much of our oxygen. *Rhizobium* bacteria fix air's elemental nitrogen into ammonia that can be used for a variety of life processes. Degradative enzymes allow ruminants to digest cellulose. Microbes recycle the dead into basic components that can be used again and again. Microbes in our intestine aid in digestion, produce vitamins, and prevent colonization by pathogens. For the most part, microbes are our friends.

Figure 1.1 It's a microbial world.

FOOD MICROBIOLOGY, PAST AND PRESENT

From the dawn of civilization until about 10,000 years ago, humans were hunter-gatherers. Humans were lucky to have "enough." There was neither surplus nor a settled place to store it. "Preservation" was not an

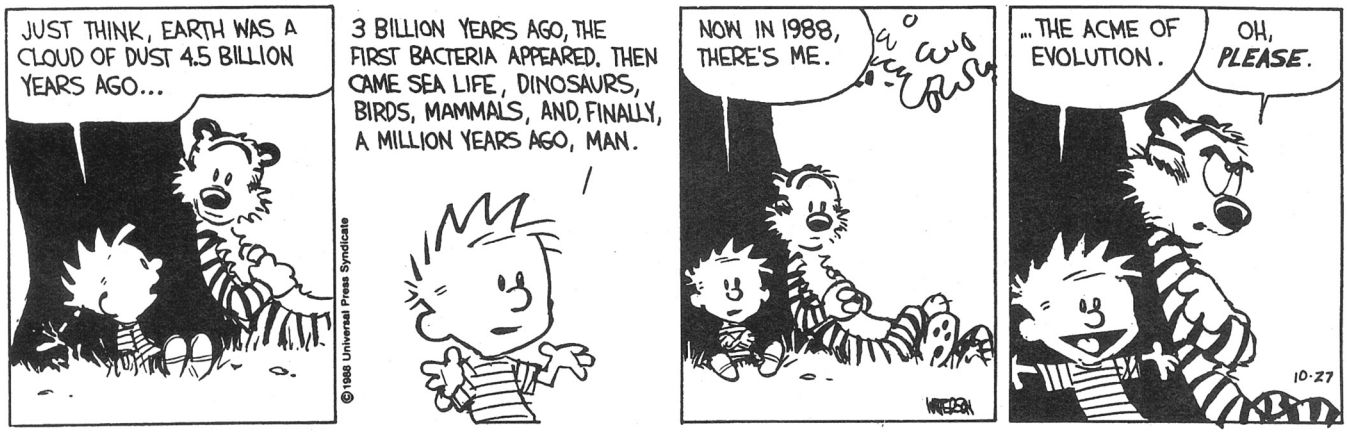

Figure 1.2 Evolution as seen through the eyes of Calvin and Hobbes. *Calvin and Hobbes* © 1988 Bill Watterson. Distributed by Universal Press Syndicate. Reprinted with permission. All rights reserved.

issue. With the shift to agricultural societies, storage, spoilage, and preservation became important challenges. The first preservation methods were undoubtedly accidental. Sun-dried, salted, or frozen foods did not spoil. In the classic "turning lemons into lemonade" style, early humans learned that "spoiled" milk could be acceptable, or even desirable, if viewed as "fermented." Fermenting food became an organized activity around 4000 B.C.E. (Table 1.1). Breweries and bakeries had sprung up long before the idea of yeast was conceived.

Humans remained ignorant of microbes for thousands of years. In 1665, Robert Hooke published *Micrographia*, the first illustrated book on microscopy that detailed the structure of *Mucor*, a microscopic fungus. In 1676, Antony van Leeuwenhoek (Fig. 1.3) used a crude microscope (Fig. 1.4) of Hooke's design to see small living things in pond water. *Micro*biology (Fig. 1.5) was born!

Nonetheless, it took another 200 years to prove that microbes exist and cause fermentative processes. In the mid-1700s, Lazzaro Spallanzani showed that boiled meat placed in a sealed container did not "spoil." Advocates of spontaneous generation, however, argued that air is needed for life and that the air was sealed out. It took another 100 years for Louis Pasteur's elegant "swan-neck flask" experiment (see Box 25.1) to replicate Spallanzani's experiment in a way that allowed access to air, but not microbes. Napoleon, needing to feed his troops as they traveled across Europe, offered a prize to anyone who could preserve food. Nicolas Appert (Fig. 1.6) won this prize when he discovered that foods would not spoil if they were heated in sealed containers, i.e., were canned. Thus, canning was invented without any knowledge of microbiology. Indeed, it was not until the 1900s that mathematical bases for canning processes were developed.

Robert Koch was a giant of microbiology. In the late 1800s, he established criteria for proving that a bacterium caused a disease. To satisfy "Koch's postulates," one had to (i) isolate the suspect bacterium in pure culture from the diseased animal, (ii) expose a healthy animal to the bacterium and make it sick, and (iii) reisolate the bacterium from the newly infected animal. This approach was so brilliant that it is still used today to prove that a disease has a microbial origin. Microbiology remains based on the study of single bacterial species in pure culture. Unfortunately, most

Authors' note

I never appreciated van Leeuwenhoek's skill as a microscopist until I tried to use this replica microscope.

Table 1.1 Significant events in the history of food microbiology[a]

Decade	Event
4000 B.C.E.	Fermentation of food becomes an organized activity.
1670 C.E.	Hooke and van Leeuwenhoek observe microscopic fungi and bacteria; *micro*biology is born.
1760	Spallanzani's experiments with boiled beef strike a blow against spontaneous generation.
1800	Nicolas Appert invents the canning process. Amazingly, this is still a mainstay of food processing 200 years later.
1810	P. Dumond patents the tin can, making Appert's life much easier.
1850	Appert and R. Chevallier are issued a patent for steam sterilization (retort). The use of steam under pressure increases process temperatures, decreases process times, and radically improves the quality of canned food.
	Louis Pasteur demonstrates that living organisms cause lactic and alcoholic fermentations.
1860	Pasteur disproves "spontaneous generation." Life can come only from other life.
	Joseph Lister develops concept of antiseptic practice. Persuading surgeons to wash their hands saves thousands of lives.
1880	Robert Koch postulates bacteria as causative agent of disease. To this day, "Koch's postulates" remain the "gold standard" for proving that bacteria cause a disease.
	Hans Christian Gram invents the Gram stain.
	J. R. Petri invents the petri dish. Petri worked in Koch's lab, where the usefulness of agar was also discovered.
	A. A. Gartner isolates *Salmonella enterica* serovar Enteritidis from a food-poisoning outbreak. A century later, salmonellae are still the leading cause of death among foodborne microbes.
1890	Pasteurization of milk begins in the United States.
1900	The Food and Drug Act is passed in response to Upton Sinclair's exposé of the meat industry in *The Jungle*.
1920	U.S. Public Health Commission publishes methods to prevent botulism.
	Alexander Fleming discovers antibiotics. Less than 100 years later, multiply antibiotic resistant pathogens threaten possible epidemics.
	Clarence Birdseye introduces frozen foods to the retail marketplace.
1930	G. M. Dack confirms that *Staphylococcus aureus* makes toxin.
	Widespread introduction of home refrigerators.
	The Food, Drug, and Cosmetic Act strengthens regulation over foods.
	Viruses are discovered.
1940	The first freeze-dried food is developed.
	The supermarket replaces assorted shops as the place to buy food.
1950	Watson and Crick discover the structure of DNA. Rosalind Franklin played a large, but uncredited role.
	Research on food irradiation begins.
1960	C. Duncan and D. Strong demonstrate that perfringens food poisoning is caused by a toxin.
	The role of fungal toxins is discovered when "turkey X" disease breaks out in poultry.
1970	S. Cohen discovers genetic recombination in bacteria.
	Larry McKay reports plasmids in gram-positive bacteria.
	Introduction of good manufacturing practices for low-acid foods.
	Invention of monoclonal antibodies lays the foundation for mass-produced antibody-based tests.
1980	The first recognized outbreak of listeriosis occurs.
	Escherichia coli O157:H7 is first recognized as a pathogen.
	The first genetic probe for detection of *Salmonella* is developed.
	Polymerase chain reaction is invented.
	Prions are discovered.
1990	Irradiation is approved for pathogen control in meat and poultry.
	The "mad cow" crisis hits the United Kingdom.
	Hazard Analysis Critical Control Points is required by U.S. Department of Agriculture.
2000	Irradiation is approved for shell eggs.
	Regulatory concerns about bioterroristic contamination of food lead to new laws on facility registration, product traceability, and prenotification of imports.
	First report of "mad cow" disease in the United States.
	You are here.

[a]Compiled from Brock (1999), Hartman (2001), and other sources.

microbes cannot be cultured, and they exist in nature as community members, not in pure culture. Two other innovations from Koch's laboratory are still with us. Julius Petri, an assistant in Koch's laboratory, invented the petri dish as a minor modification of Koch's plating method. Workers in the Koch laboratory were often frustrated by the gelatin used in the plating method. Gelatin would not solidify when the temperature was too warm and was dissolved by certain organisms (that produced enzymes that degrade gelatin). Walter Hesse, a physician who had joined the Koch lab to study infectious bacteria in the air, shared this frustration with his wife, Fanny. Frau Hesse had been using agar to thicken jams and jellies for years and suggested that they try agar in the laboratory. You know the rest of the story.

The first half of the 20th century was marked by the discovery of the "traditional" foodborne pathogens. *Salmonella* species came from warm-blooded animals. *Clostridium botulinum* became a problem with improperly canned foods. *Staphylococcus aureus* became associated with poor hygiene, *Bacillus cereus* contaminated starchy foods, and other food/organism combinations became associated with certain illnesses. Viruses were first crystallized and associated with disease in the 1930s and are a major cause of foodborne illness. However, relatively little is known about them and they are "understudied" relative to bacteria. In the middle of the 20th century, biology made a quantum leap forward; James Watson and Francis Crick discovered the structure of deoxyribonucleic acid (DNA) (Fig. 1.7). This gave birth to the era of molecular genetics. This has led to better understanding of how bacteria cause illness, revolutionary genetic and antibody-based detection methods for bacteria, genetic "fingerprints" as epidemiological tools, and the ability to genetically alter fermentation organisms to improve their industrial characteristics. The tools of molecular biology are becoming increasingly important to food microbiology.

The movement away from massive end product testing to safety by design also started in this era. "Good manufacturing practices" provide manufacturers with procedures that should yield safe products. HACCP codified this into a safety assurance system. Food irradiation is approved as a "kill" step in the processing of raw poultry and meat. Risk assessment

Figure 1.3 Antony van Leeuwenhoek.

Figure 1.4 Replica of van Leeuwenhoek microscope.

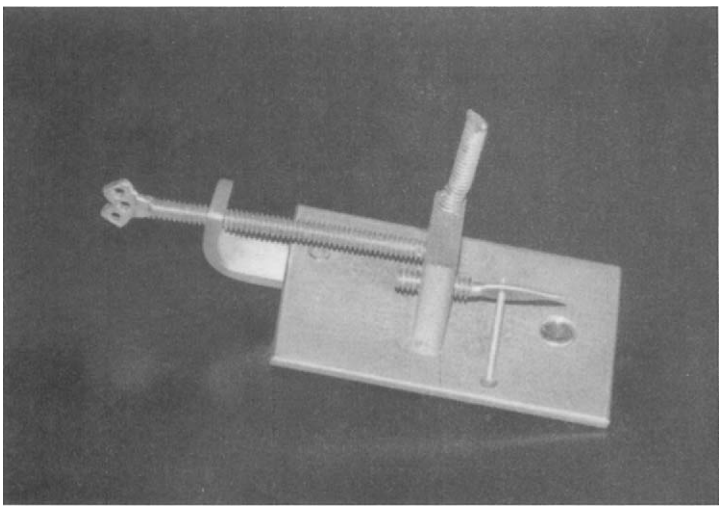

Figure 1.5 The invention of the microscope made laboratories much more interesting. *In the Bleachers* © 2005 Steve Moore. Reprinted with permission of Universal Press Syndicate. All rights reserved.

provides the basis for more sophisticated regulatory approaches based on the end result (fewer sick people) rather than dictating processes that give fewer microbes.

TO THE FUTURE AND BEYOND

My mentor once told me, "If you stare into the crystal ball too long, you end up with shards of glass in your face." It is hard to predict the future. However, it is safe to say that there will be greater demand for more, safer

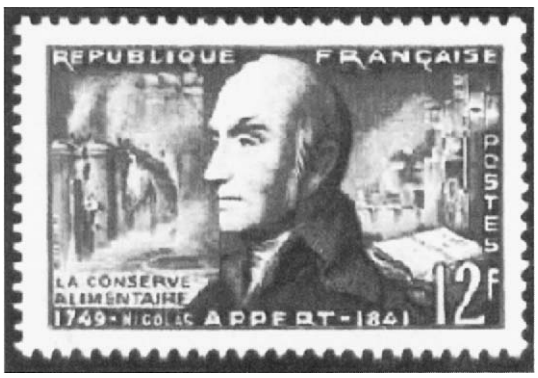

Figure 1.6 Stamp honoring Nicolas Appert.

foods, that innovations in medical microbiology will continue to migrate into food microbiology, and that entirely new concepts will alter the course of food microbiology.

About one-third of the world's food supply is lost to spoilage. An earth's population that will double in your lifetime cannot afford this loss. If everyone is to eat, that level of spoilage must be reduced. Hunger hovers over humanity and may come to dwarf safety as an issue. Fermentation as a low-energy "appropriate technology" may become more important as a preservation method. Microbes themselves may become a food source. (Yeast was touted as "single-cell protein" in the 1970s.) Probiotic bacteria may help people maintain health through their diet. Perhaps, just as pesticides have been cloned into plants to kill insects, antimicrobials will be cloned into food to make them kill bacteria.

While 30% of the world's population is worried about getting *enough* food, the third of the world's population that worries about "overnutrition" demands *safer* food. Zero risk is impossible, and the cost of safety increases exponentially as one approaches zero. How safe is "safe enough"? When should a society shift resources from increased safety to increased availability? The requirements of international trade mandate the harmonization of safety standards. "Globalization" of the food supply creates new challenges to food safety. Ideally, first- and third-world countries should have similar sanitary standards. But how can this be achieved in countries where indoor plumbing and rudimentary sanitation are scarce and refrigeration is almost nonexistent?

Scientific innovations usually appear first in medical technology and impact the food system relatively later. This can be seen through the adaptation of rapid and automated methods, food-processing facilities increasingly adapting the methods of pharmaceutical houses, and increased emphasis on molecular understanding. Look at what's "hot" in the biomedical area today, and you will see the food microbiology of tomorrow.

This chapter highlights the scientific limits of microbiology. They are starting to fall. The causes of one-half of the cases of foodborne illness are unknown. Could they be caused by viruses? Could they be due to the microbes that are there but unculturable? Will new molecular techniques for studying these bacteria allow us to understand and control them? The first 200 years of microbiology has been devoted to the study of bacteria as if they were noninteracting ball bearings. We now know that they "talk" to each other using chemical signals and sensors. This "quorum sensing" helps microbial communities decide when they will take action (for example, to mount an infectious attack). If bacteria talk to each other, what are they saying? If we knew, could we prevent spoilage and disease by scrambling the signal, drowning the signal out with other biochemical noise, or inactivating the receivers? What about the 99% of bacteria that remain undiscovered? What is in their gene pool? How might we use those genes and their products to make food better and safer? What are the roles of prions and viruses in foodborne disease? How do we participate in the ethical debate that surrounds genetic engineering, food irradiation, and other new technologies?

The most exciting era of food microbiology may lie in its future. It is the period that you will live in. You may create the next epoch in food microbiology. To prepare for this, gain a foundational understanding of the field today, learn how to think critically and creatively, and learn to love these little creatures that are so small but so smart.

Figure 1.7 The discovery of DNA's double helix marked a quantum leap in the history of microbiology.

Summary

- Microbes were the first living things on earth and remain the foundation of the biosphere.
- The fundamental biological discoveries of the 1800s, such as the disproof of spontaneous generation, Koch's postulates, and pure culture of microbes, continue to influence the field.
- Molecular biology and genetics have begun to impact food microbiology and will shape its future.
- Food microbiology is a relatively young field of study.

Suggested reading

Brock, T. D. 1999. *Milestones in Microbiology: 1546 to 1940*. ASM Press, Washington, DC.

Hartman, P. 2001. The evolution of food microbiology, p. 3–12. *In* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. ASM Press, Washington, DC.

Needham, C., M. Hoagland, K. McPherson, and B. Dodson. 2000. *Intimate Strangers: Unseen Life on Earth*. ASM Press, Washington, DC.

Questions for critical thought

1. What would have happened if there had been spores in Pasteur's swan-necked flask?
2. Can you think of another experiment that would disprove spontaneous generation? How would food safety be impacted if there were spontaneous generation?
3. Extend Table 1.1 50 years into the future. Enter what you think might happen.
4. Discuss three historical discoveries in microbiology that still influence food microbiology.
5. It is said that if Pasteur walked into a food microbiology laboratory today, he could be comfortably at work in about 15 min. Is this true? Is it good?
6. Why did the invention of the plastic petri dish set food microbiology back 20 years?
7. If 99% of bacteria are unculturable, how do we know that they are there?
8. Looking back at Table 1.1, what historical development do you think has had the greatest impact? Why?

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2

Factors That Influence Microbes in Foods

Introduction

Food Ecosystems, Homeostasis, and Hurdle Technology
Foods as Ecosystems

Classical Microbiology and Its Limitations
Limitations of Detection and Enumeration Methods
Homeostasis and Hurdle Technology
Growth Kinetics

Microbial Physiology and Metabolism
Carbon Flow and Substrate Level Phosphorylation
The Tricarboxylic Acid Cycle Links Glycolysis to Aerobic Respiration

Conclusion

Summary

Suggested reading

Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- distinguish among different methods of culturing foodborne microbes and choose the correct technique for a given application
- recognize how intrinsic and extrinsic factors are used in controlling microbial growth
- use the water activity concept to distinguish between "bulk water" and water that is unavailable for microbial growth
- understand the influence of water activity on microbial growth and microbial ecology
- quantitatively predict the size of microbial populations and their rate of growth using equations that describe the kinetics of microbial growth
- qualitatively describe how altering the variables in the equations for the kinetics of microbial growth alters growth patterns
- relate biochemical pathways to energy generation and metabolic products of foodborne bacteria

INTRODUCTION

Never say that you are *just* a food microbiologist. Food microbiology is a specialty area in microbiology that is regulated by many government agencies (Box 2.1). Just as a heart surgeon is more than a mere surgeon, a food microbiologist is a microbiologist with a specialty. Food microbiologists must understand microbes. They must understand complex food systems. Finally, they must integrate both to solve microbiological problems in complex food ecosystems. Even the question "How many bacteria are in that food?" has no simple answer. The first part of this chapter examines the question of how we detect and quantify bacteria in food. We present the concept of culturability and ask if it really reflects *viability*. Even though injured or otherwise nonculturable cells cannot form colonies on petri dishes, they do exist and can cause illness.

The second part of the chapter presents the importance of factors intrinsic (inherent) and extrinsic (external) to food for bacterial growth. Ecology teaches that the *interaction* of factors determines which organisms grow in an environment. The use of multiple environmental factors (i.e., pH, salt concentration, temperature, etc.) to inhibit microbial growth is called *hurdle technology*. Hurdle technology is an increasingly important preservation strategy.

The kinetics of microbial growth are covered in the third part of this chapter. All four phases of the microbial growth curve are important to

BOX 2.1

Agencies involved in food safety

The U.S. Department of Agriculture (USDA) regulates meat and poultry through its Food Safety and Inspection Service (FSIS). All meat and poultry must be processed under continuous USDA inspection, but the USDA does not have the legal authority to issue a recall. The USDA's Economic Research Service is responsible for the regulation of organic food, while the Agricultural Research Service provides the science that the regulations are founded on and that is required to promote U.S. agriculture.

The Food and Drug Administration (FDA) regulates all foods except meat, poultry, and alcohol (which is regulated by the Bureau of Alcohol, Firearms, Tobacco and Explosives). The FDA periodically inspects food-manufacturing facilities, must approve all thermal processes for low-acid canned foods, and has the authority to issue mandatory recalls. The FDA also regulates drugs and medical devices. The food side of the agency resides in the Center for Food Safety and Applied Nutrition (CFSAN).

The Centers for Disease Control and Prevention (CDC) is the part of the Department of Health and Human Services that provides epidemiological expertise to investigate outbreaks of foodborne illness and tracks the incidence of foodborne illness over time. It oversees PulseNet, a "DNA fingerprinting clearinghouse" that can link geographically dispersed cases to a common outbreak. The CDC has no regulatory authority.

The Environmental Protection Agency (EPA) has regulatory authority over insecticides, fungicides, and herbicides that are applied to crops.

Consider this: a "bargain" pot pie with chicken containing <3% chicken would be regulated by the FDA, but if the company started an "upscale" pot pie with 30% chicken, it would be regulated by the USDA. It the upscale pot pie were so popular that the company added an upscale vegetarian pot pie to its product lineup, that would be regulated by the FDA. Similarly, when benzoic acid is added to yogurt or cottage cheese to inhibit microbial growth, it is regulated by the FDA. When the same compound is used in the field to prevent fungi from growing on grains, the EPA has regulatory authority.

food microbiologists. If the lag phase can be extended beyond the normal shelf life, the food is microbially safe. If the growth rate can be slowed so that the population size is too small to cause illness, the product is safe. If the rate of the death phase can be controlled, it can be used to ensure safety, as in the case of cheddaring cheese made from unpasteurized milk. The ability to quantify microbial growth rates and predict microbial population sizes is an important skill for students of foodborne microbes.

The fourth part of this chapter introduces the physiology and metabolism of foodborne microbes. This book does not assume a background in biochemistry, but the biochemistry associated with foodborne microbes can be very important. The biochemical pathways used by the lactic acid bacteria determine the characteristics of fermented foods. Spoilage is also caused by biochemicals. Lactic acid makes milk sour; carbon dioxide causes cans to swell. The ability of bacteria to grow under adverse conditions is governed by their ability to generate different amounts of energy (in the form of adenosine triphosphate [ATP]) from different biochemical pathways. Bacteria grown in the presence of oxygen (aerobically) produce more energy (ATP) than those grown in the absence of oxygen (anaerobically) (Box 2.2). Thus, organisms such as *Staphylococcus aureus* can withstand hostile environments, such as relatively high salt concentration, better under aerobic conditions than under anaerobic conditions.

FOOD ECOSYSTEMS, HOMEOSTASIS, AND HURDLE TECHNOLOGY

Foods as Ecosystems

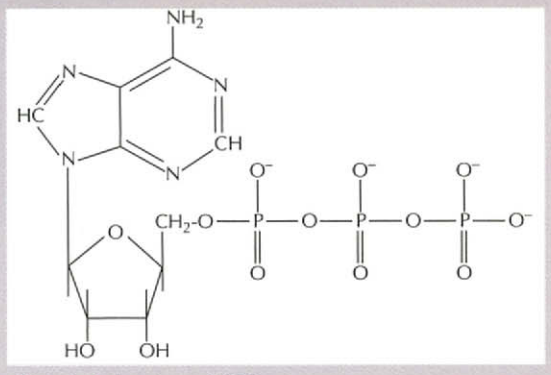
Foods are ecosystems composed of the environment and the organisms that live in it. The food environment is composed of intrinsic factors (inherent to the food) (i.e., pH, water activity [a_w], and nutrients) and extrinsic factors (external to it) (i.e., temperature, gaseous environment, and the

BOX 2.2

Mechanisms of energy generation

Anaerobes cannot liberate all of the energy in a substrate. They typically make one or two ATP molecules by burning 1 mol of glucose. This process is called *substrate level phosphorylation*. Strict anaerobes are killed by air.

Aerobes can burn glucose completely in the presence of oxygen. They generate 34 ATP molecules from 1 mol of glucose through a process called *respiration*. Strict aerobes cannot grow in the absence of oxygen.



Chemical structure of ATP.

Facultative anaerobes can generate energy by both substrate level phosphorylation and respiration. They can grow in a wider variety of environments than strict aerobes or strict anaerobes.

presence of other bacteria). Intrinsic and extrinsic factors can be manipulated to preserve food. When applied to microbiology, ecology can be defined as "the study of the *interactions* between the chemical, physical, and structural aspects of a niche and the composition of its specific microbial population" (International Commission on Microbiological Specifications for Foods, 1980). The term "interactions" emphasizes the dynamic complexity of food ecosystems.

Foods can be heterogeneous on a microscopic scale. Heterogeneity and gradients of pH, oxygen, nutrients, etc., are key ecological factors in foods. Foods may contain multiple microenvironments. This is well illustrated by the food poisoning outbreaks in "aerobic" foods caused by the "strict anaerobe" *Clostridium botulinum*. *C. botulinum* growth in potatoes, sautéed onions, and coleslaw has caused botulism. The food's oxygen is driven out during cooking and diffuses back in so slowly that most of the product remains anaerobic.

CLASSICAL MICROBIOLOGY AND ITS LIMITATIONS

Limitations of Detection and Enumeration Methods

All methods based on the plate count have the same limitations. The plate count assumes that every cell forms one colony and that every colony originates from only one cell. (A colony-forming unit [CFU] ideally would be a single cell but in reality could be a chain of 10 or a clump of 100.) The ability of a given cell to grow to the size of a macroscopic colony depends on many things. These include the cell's physiological state, the growth medium used (and the presence or absence of selective agents), the incubation temperature, etc.

Figure 2.1 illustrates the concepts of plate counting, qualitative detection, and most-probable-number (MPN) methodologies. Consider a 50-ml sample of milk that contains 10^6 CFU of anaerobic bacteria/ml, 10^5 aerobic bacteria, a significant number of *Staphylococcus aureus* organisms (a pathogen), some *Escherichia coli* organisms, and a few salmonellae. Since the bacteria are either too numerous to count directly or too few to find easily, samples have to be diluted or enriched.

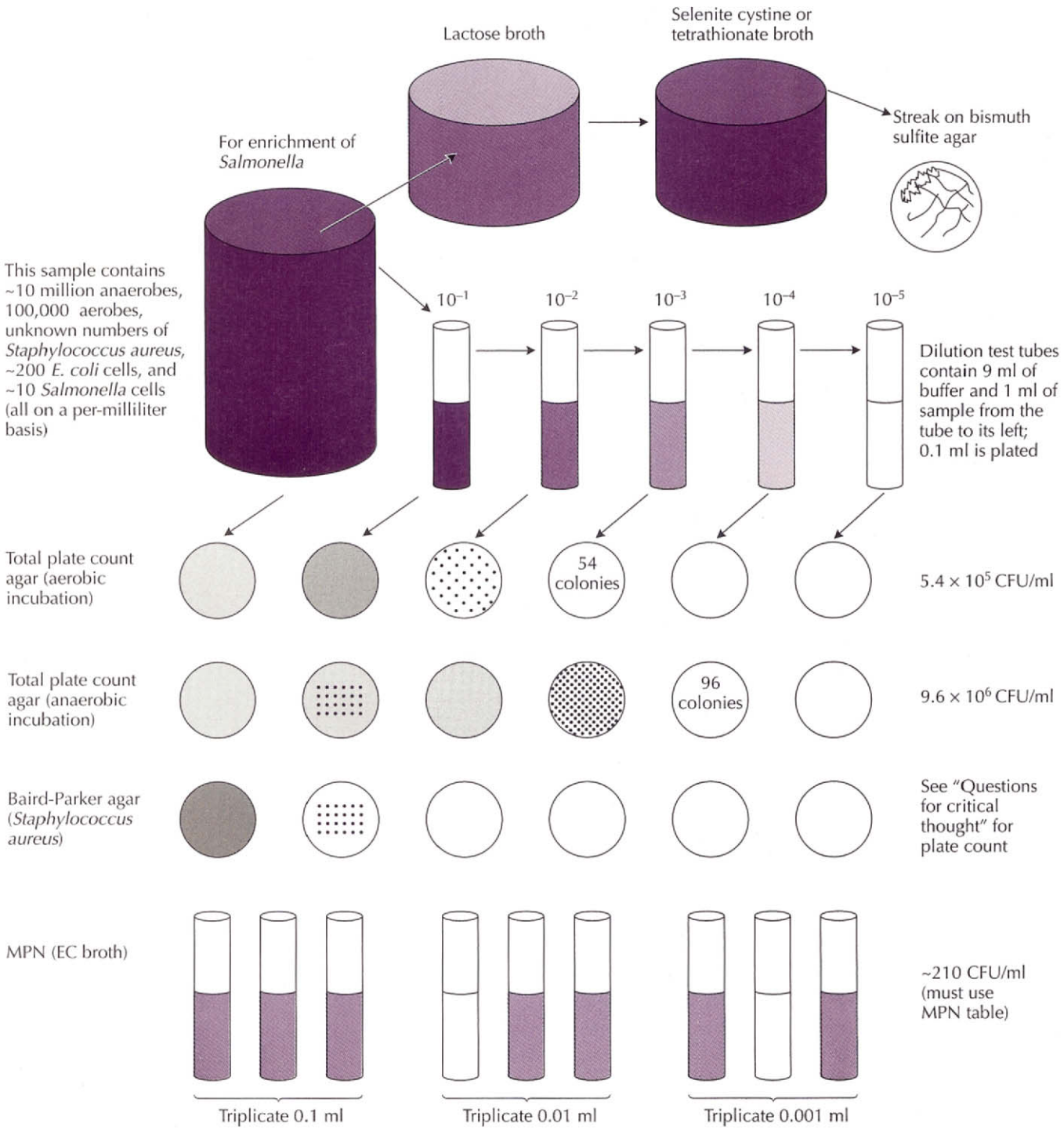


Figure 2.1 Methods of enumerating bacteria. The sample (which represents a 10^0 dilution) is diluted through a series of 10-fold dilutions before enumeration. To enumerate the bacteria, 0.1 ml from each dilution tube is plated on an agar plate of appropriate media. After 24- to 48-h incubations, the colonies are counted and the number is multiplied by the dilution factor to give the microbial load of the original sample. Gray scale indicates culture density. If the number of cells present is expected to be low, the MPN technique is used. In this case, the sample is diluted into triplicate series of test tubes containing the media, and the pattern of positive tubes is used to determine the MPN from an MPN table (Table 2.3).

Plate Counts

Plate counts are used to quantify bacterial populations of >250 CFU/ml (for liquids) or >2,500 CFU/ml (for solids, which must be diluted 1:10 into a liquid to be pipettable). The food sample is first homogenized 1:10 (weight:volume) in a buffer to give a 10-fold dilution. This is then diluted through a series of 10-fold dilutions to give 10-, 100-, 1,000-, 10,000-fold, etc., dilutions and spread onto an agar medium. After incubation at 35°C, the average number of colonies per plate (on plates having between 25 and 250 colonies) is determined and multiplied by the dilution factor to give the number of bacteria in the original sample. This sounds simple, right? Wrong. At one time, the procedure above was referred to as the *total plate count*. However, it counts only those bacteria that can grow on that agar medium at 35°C. The larger population of anaerobes is not detected because they cannot grow in the presence of air. Bacteria that grow only below 15°C might be a good predictor of refrigerated milk's shelf life. They are not detected either. *S. aureus*, which can make you sick, would be lost in the crowd of harmless bacteria. Variations of the standard aerobic plate count consider these issues. Plate counts can be determined at different temperatures or under different atmospheres to increase their specificity for certain types of bacteria.

Selective, or Differential, Media

The concept of selective media is simple. Some ingredients of the medium (in the case of an unusual sugar) favor the growth of the targeted bacteria. Other medium components (such as antibiotics, salt, surfactants, etc.) can inhibit the growth of competing bacteria. Some examples of selective media are given in Table 2.1. In the case of Baird-Parker agar (Fig. 2.2), lithium chloride, glycine, and tellurite allow *S. aureus* to grow while suppressing a larger number of other organisms. However, if injured, target organisms may be killed by the selective agent. In addition, the selective media are not entirely selective. *Listeria monocytogenes* can grow on *Lactobacillus* deMann-Rogosa-Sharp medium. Thus, the identity of suspect pathogens isolated on selective media must be confirmed by biochemical or genetic methods.

Differential media are those designed to make a colony of the target organism look different and stand out from the background of other organisms. For example, *S. aureus* colonies are black with a precipitation zone on Baird-Parker agar. *L. monocytogenes* colonies have a blue sheen on McBride's agar and turn Fraser broth black.

The CFU-per-gram values derived from plate counts are also influenced by intrinsic factors. Table 2.2 illustrates these points by providing *D* values (time to kill 90% of the population) at 55°C for *L. monocytogenes* with different thermal histories (heat shocked for 10 min at 80°C or not heat

Table 2.1 Some selective media used in food microbiology

Bacterium	Medium	Selective agents
Staphylococcus aureus	Baird-Parker agar[a]	Lithium chloride, glycine, and tellurite
Listeria species	LPM	Lithium chloride, phenyl-ethanol, antibiotic
Lactic acid bacteria	deMann-Rogosa-Sharp (MRS)	Low pH, surfactant
Campylobacter species	Abeyta-Hunt-Bark agar	Microaerobic incubation, antibiotic

[a]The biggest honor for a microbiologist is to have an organism named after oneself. Not far behind, and much easier to obtain, is to have a medium named after oneself.

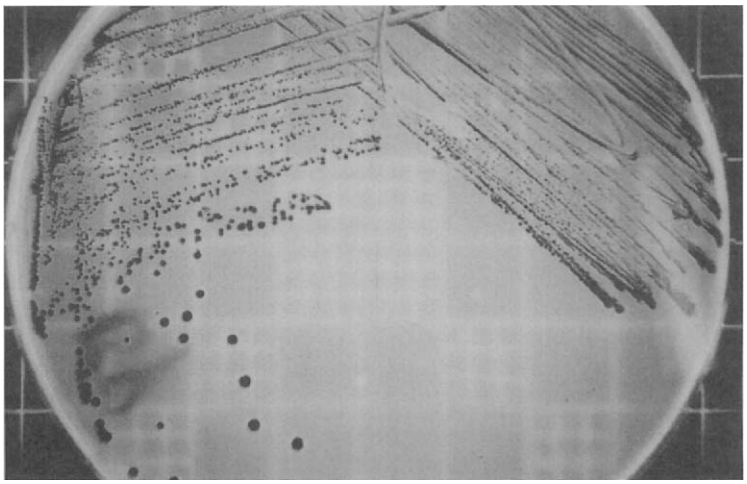


Figure 2.2 *S. aureus* forms black colonies when grown on Baird-Parker agar.

shocked), on selective (McBride's) or nonselective tryptic soy agar with yeast extract (TSAY) media under aerobic or anaerobic atmospheres. Cells that have been heat shocked and recovered on TSAY are about four times more heat resistant (as evidenced by the higher *D* values) than cells which have not been heat shocked and recovered on McBride's medium under aerobic conditions. Similarly, the sporulation temperature influences spore heat resistance.

MPN Methods

MPN methods are used to estimate low numbers of organisms (<30 CFU/ml) in a sample (Fig. 2.1). For example, *E. coli* and coliform bacteria might be present in milk at very low levels. Selective agents are added to broth medium. One milliliter of sample is transferred to triplicate test tubes of media. At least three 10-fold serial dilutions are also inoculated. The samples are incubated, and the pattern of turbid tubes with microbial growth is recorded. This pattern is used to derive the "most probable" number of bacteria from a statistical chart (Table 2.3). The lower sensitivity limit (for example) for a three-tube MPN is <3.6 cells per g. Put differently, 3.6 cells/g is the lowest concentration that can accurately be quantified by a three-tube MPN. Using 10 tubes at each dilution instead of 3 can lower the detection level to <0.96 cells per g, but the 10-tube MPN is rarely cost-effective and seldom used.

Enrichment Techniques

There is a "zero tolerance" (actually 0 cells/25 g) for *Salmonella*, *L. monocytogenes*, and *E. coli* O157:H7 in ready-to-eat foods. The good news is that this

Table 2.2 The influence of thermal history and enumeration protocols on experimentally determined *D* values at 55°C for *L. monocytogenes*

Atmosphere	D_{55} value (min)			
	TSAY medium		McBride's medium	
	With heat shock	Without heat shock	With heat shock	Without heat shock
Aerobic	18.7	8.8	9.5	6.6
Anaerobic	26.4	12.0	No growth	No growth

Table 2.3 MPN table[a]

No. of positive tubes			MPN/g	95% Confidence interval limits		No. of positive tubes			MPN/g	95% Confidence interval limits	
0.10	0.01	0.001		Low	High	0.10	0.01	0.001		Low	High
0	0	0	<3.0		9.5	2	2	0	21	4.5	42
0	0	1	3.0	0.15	9.6	2	2	1	28	8.7	94
0	1	0	3.0	0.15	11	2	2	2	35	8.7	94
0	1	1	6.1	1.2	18	2	3	0	29	8.7	94
0	2	0	6.2	1.2	18	2	3	1	36	8.7	94
0	3	0	9.4	3.6	38	3	0	0	23	4.6	94
1	0	0	3.6	0.17	18	3	0	1	38	8.7	110
1	0	1	7.2	1.3	18	3	0	2	64	17	180
1	0	2	11	3.6	38	3	1	0	43	9	180
1	1	0	7.4	1.3	20	3	1	1	75	17	200
1	1	1	11	3.6	38	3	1	2	120	37	420
1	2	0	11	3.6	42	3	1	3	160	40	420
1	2	1	15	4.5	42	3	2	0	93	18	420
1	3	0	16	4.5	42	3	2	1	150	37	420
2	0	0	9.2	1.4	38	3	2	2	210	40	430
2	0	1	14	3.6	42	3	2	3	290	90	1,000
2	0	2	20	4.5	42	3	3	0	240	42	1,000
2	1	0	15	3.7	42	3	3	1	460	90	2,000
2	1	1	20	4.5	42	3	3	2	1,100	180	4,100
2	1	2	27	8.7	94	3	3	3	>1,100	420	

[a]For 3 tubes with 0.1-, 0.01-, and 0.001-g inocula. Source: www.cfsan.fda.gov/~ebam/bam-a2.html#tables.

makes it unnecessary to count these pathogens. The bad news is that one must be able to find one salmonella hiding among a million other bacteria. This is done by coupling preenrichment media with selective enrichment, followed by plating on selective media and biochemical confirmation or genetic authentication (Fig. 2.1). The preenrichment in a nonselective medium allows injured pathogens to repair, recover, and regain resistance to selective agents. The enrichment step uses selective agents to promote pathogen growth to high levels while inhibiting growth of the competing organisms. The pathogen can then be isolated on a selective agar, and its identity can be confirmed by other methods.

Injury

Injured cells and cells that are "viable but nonculturable" (VNC) pose additional problems to food microbiologists. Although they cannot form colonies and be detected, they can make people sick and be recovered by standard methods once they recover from the injury. *Sublethal* levels of heat, radiation, acid, or sanitizers may injure rather than kill cells. Injured cells are less resistant to selective agents or have increased nutritional requirements. The degree of injury is influenced by time, temperature, selective agent concentration, experimental methodology, and other factors. For example, a sanitizer test may appear to kill listeria cells. However, these cells might be recovered using listeria repair broth. Media for the recovery of injured cells are often specific for a given bacterium or stress condition.

Data that illustrate injury are given in Fig. 2.3. Cells subjected to a mild stress are plated on a rich nonselective medium and on a selective medium containing 6% salt. The difference between the population of cells able to

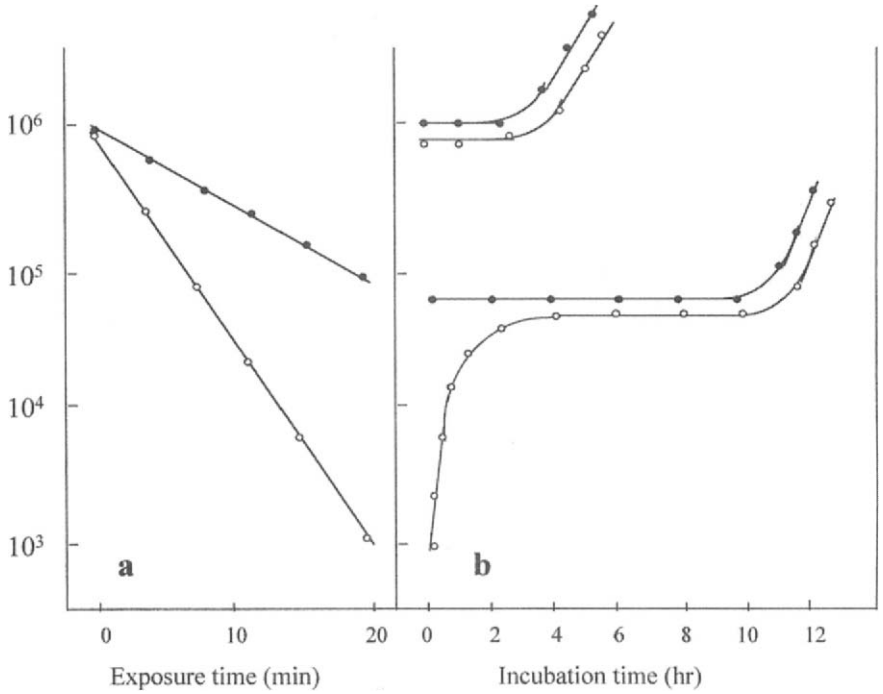


Figure 2.3 Data indicative of injury and repair. When bacteria are plated on selective (○) or nonselective (●) media during exposure to some stressor (e.g., heat [a]), the decrease in CFU on a nonselective medium represents the true lethality, whereas the difference between the values obtained on each medium is defined as injury. During repair (b), resistance to selective agents is regained and the value obtained on the selective medium approaches that of the nonselective medium. Unstressed controls are shown on the top of panel b. (Source: M. P. Doyle, L. R. Beuchat, and T. J. Montville [ed.], *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. [ASM Press, Washington, DC, 2001].)

form colonies on each medium represents the number of injured cells. (If 10^7 CFU of a population/ml are found on the selective medium and 10^4 CFU/ml can grow on the nonselective medium, then 10^3 CFU/ml are injured.)

Cell injury is important to food safety for several reasons. (i) If injured cells are classified as dead during heat resistance determinations, the effect of heating will be overestimated and the resulting heat process will be ineffective. (ii) Injured cells that escape detection at the time of postprocessing sampling may repair before the food is eaten and cause illness. (iii) The selective agent may be common food ingredients such as salt or organic acids or even suboptimal temperature. These would prevent injured cells from growing and result in an underestimation of the microbial levels.

During repair, cells recover from injury. Repair requires ribonucleic acid (RNA) and protein synthesis and often lengthens the lag phase. Environmental factors influence the extent and rate of repair. The microbial stability of cured luncheon meat is due to the extended lag period required for the repair of spoilage organisms.

Viable but Nonculturable

Salmonella, Campylobacter, Escherichia, Shigella, and *Vibrio* are among the genera that can be viable but not be culturable by normal methods. The differentiation of vegetative cells into a dormant VNC state is a survival strategy for many nonsporulating bacteria. During the transition to the VNC

state, rod-shaped cells shrink and become small spherical bodies. These are totally different from bacterial spores. It takes from 2 days to several weeks for an entire population of vegetative cells to become VNC.

Although VNC cells cannot be cultured, their viability can be determined through microscopic methods. Fluorescent probes are used to determine the integrity of the bacterial membrane. Bacteria with intact cell membranes stain fluorescent green, but bacteria with damaged membranes stain fluorescent red. Dyes can also be used to identify VNC cells. Respiring cells reduce the colorless soluble dye (tetrazolium red) to a colored insoluble compound that can be seen with a microscope.

Because the VNC state is most often induced by nutrient limitation in aquatic environments, it might appear irrelevant to nutrient-rich food environments. However, changes in salt concentration, exposure to sanitizers, and shifts in temperature can also cause the VNC state. When *Vibrio vulnificus* (10^5 CFU/ml) is shifted to refrigeration temperatures, it becomes nonculturable (<0.04 CFU/ml) but is still lethal to mice. The bacteria resuscitate in the mice and can be isolated postmortem by culture methods. Other foodborne pathogens can become VNC when shifted to cold temperature in nutritionally rich media, and presumably food. This has chilling implications for the safety of refrigerated foods.

Thus, our awareness of the VNC state should lead to the reexamination of our concept of viability, our dependence on enrichment culture to isolate pathogens, and our reliance on cultural methods to monitor microbes in the environment. We do not understand the mechanisms of VNC formation, the mechanisms that make VNC cells resistant to stressors, and what event signals the cell to resuscitate. The relationship between viability and culturability needs to be better understood.

Intrinsic Factors That Influence Microbial Growth

Characteristics of the food itself are called intrinsic factors. These include pH, a_w, oxidation-reduction potential, naturally occurring compounds that influence microbial growth, and compounds added as preservatives. Most of these factors are covered separately in the chapters on physical and chemical methods of food preservation. Because of their importance, pH and a_w deserve special attention.

pH. The pH value of a food is a log scale measurement of its acidity. pH is defined by the relationship $pH = -\log [H^+]$. Because each unit on the pH scale represents a 10-fold difference, a food with a pH of 6 is 10 times more acidic than one with a pH of 7; pH 5.0 is 100 times more acidic. That a low pH number represents a high acidity invariably causes "misspeaks" even among professionals. Try to avoid this error. The lower the pH, the more energy the cell has to spend on maintaining its intracellular pH near neutrality and the less energy it has to grow, produce toxins, etc. Table 2.4

Table 2.4 Typical pH values of foods

pH	Food
>7.0	Egg albumen
7.0–6.5	Milk, ham, bacon, poultry, fish
6.5–5.3	Raw beef, vegetables, vacuum-packed meat, melons
5.3–4.5	Cottage cheese, fermented vegetables, fermented meats (e.g., summer sausage), many sauces and soups
<4.5	Tomatoes, fruits and fruit juices, yogurt, pickles, sauerkraut

lists typical pH values for several foods. Note that foods with lower pH values generally require less processing than foods with higher pH values. Most foodborne pathogens cannot grow below pH 4.4. They may persist at lower pH values or die off very slowly. Spoilage organisms such as lactic acid bacteria and fungi can grow in acid products, but there are rarely microbial problems in foods with pH values of <3.0. Environmental extremophiles such as *Thiobacillus thiooxidans* and *Sulfolobus acidocaldarius* grow at pH values of <1.0! Fortunately, these organisms are not associated with foods. The optimum pH for most foodborne microbes is near neutrality. The upper limit for growth is pH 8 to 9, but no foods are this alkaline.

Authors' note

"pH" is a noun, not a verb. Your instructors will cringe if they hear you say, "I pHed the sample."

In addition to the energy expense required to maintain intracellular neutrality, pH influences gene expression. The expression of genes governing proton transport, amino acid degradation, adaptation to acidic or basic conditions, and even virulence can be regulated by the external pH. Cells sense changes in acidity (pH) through several mechanisms:

1. Organic acids enter the cell only in the protonated (undissociated) form. Once inside the cell, they dissociate, releasing H^+ (the proton) in the cell cytoplasm. The cell may sense the accumulation of the resulting anions, which cannot pass back across the cell membrane. Additionally, they may sense the released protons, which cause the cytoplasm to become more acidic.
2. The change in pH results in a change in the transmembrane proton gradient (i.e., the difference in pH between the inside and outside of the cell), which can also serve as a sensor to start or stop energy-dependent reactions.
3. The changes in acidity inside the cell may result in the protonation or deprotonation of amino acids in proteins. This may alter the secondary or tertiary structure of the proteins, changing their function and signaling the cell to the change in pH.

Cells must maintain their intracellular pH (pH_i) above some critical pH_i at which intracellular proteins become denatured. *Salmonella enterica* serovar Typhimurium has three progressively more stringent mechanisms to maintain pH_i that supports life. These three mechanisms are the *homeostatic response*, the *acid tolerance response*, and the synthesis of *acid shock proteins*.

At pH_o (external pH) of >6.0, salmonella cells adjust their pH_i through the homeostatic response. The homeostatic response maintains pH_i by increasing the activity of proton pumps to expel more protons from the cytoplasm. The homeostatic mechanism is always on and functions in the presence of protein synthesis inhibitors.

The acid tolerance response (ATR) is triggered by pH_o of 5.5 to 6.0. This mechanism is sensitive to protein synthesis inhibitors; there are at least 18 ATR-induced proteins. ATR appears to involve the membrane-bound ATPase proton pump and maintains pH_i of >5.0 at pH_o values as low as 4.0. The loss of ATPase activity caused by gene disruption mutations or metabolic inhibitors abolishes the ATR but not the pH homeostatic mechanism.

The synthesis of acid shock proteins is the third way that cells regulate pH_i. The synthesis of these proteins is triggered by pH_o from 3.0 to 5.0. They constitute a set of regulatory proteins distinct from the ATR proteins.

Water activity. Water is a major factor in controlling both microbial growth and chemical reactions in food. The *availability* of water in the food rather than its amount is what causes the inhibition. Water can be chemically bound to food molecules or immobilized by capillary action in food

microstructure. The water is bound by ionic interactions, hydrogen bonding, capillary action, etc., with constituents of the food. You may have experienced the difficulty of removing water from a capillary tube. The water bound by capillary action is not available for chemical reactions or microbial growth. The amount of unbound or available water determines whether microbes can grow.

Water activity (a_w) is the measure of available water or, strictly speaking, the energy of water, in foods. Water activity is defined as the ratio of the vapor pressure of water in a food, P, to the vapor pressure of pure water, P_0, at the same temperature:

$$a_w = \frac{P}{P_0}$$

The movement of water vapor from a food to the air depends on the moisture content of the food, the food composition, the temperature, and the humidity of the air. At constant temperature, water in the food equilibrates with water vapor in the air. This is the food's *equilibrium moisture content*. Think of potato chips at a picnic on a humid day; they gain moisture and become soggy. Similarly, bread left out in dry winter air loses moisture and becomes stale. At the equilibrium moisture content, the food neither gains nor loses water to the air. The relative humidity of the air surrounding the food is thus the equilibrium relative humidity (ERH). So, a_w can also be defined as follows:

$$a_w = \text{ERH}\ (\%)/100$$

The relative humidity relationship is used by instruments that use dew point to measure a_w (Fig. 2.4). The food is put in a small cup, and the water in the food is allowed to equilibrate (i.e., to its ERH) with the headspace water vapor. The dew point (the temperature at which the vapor in the air condenses to visible water) is a primary measurement of vapor pressure. So the vapor pressure can be measured by cooling the chamber

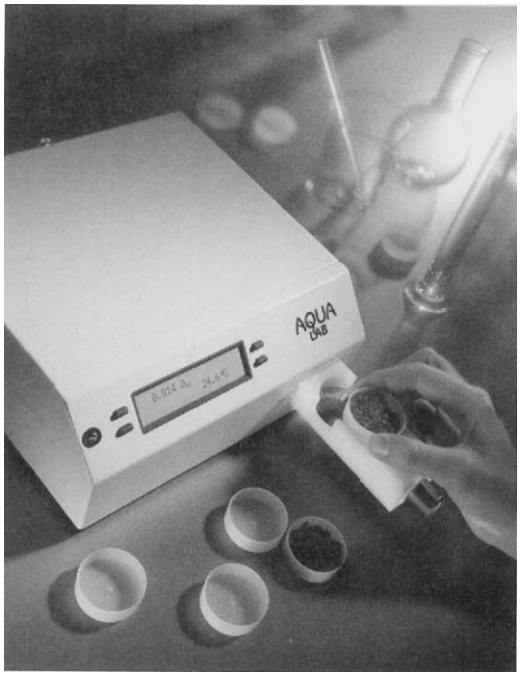


Figure 2.4 The Aqualab Series 3 is a meter that uses the dew point to measure a_w. Courtesy of Decagon Devices.

until condensation is detected optically. The a_w of the sample is then the ratio of the condensation vapor pressure at dew point temperature to the saturation vapor pressure at the sample temperature. These measurements can be made to an a_w of 0.001 in less than 5 min.

The third way to express a_w is on a mole fraction basis as a_w = moles of solvent/(moles of solvent + moles of solute). In symbolic language, this is expressed as:

$$a_w = n_1/(n_1 + n_2)$$

where n_1 is the moles of solvent (usually water) and n_2 is the moles of solute. (Can you see why a_w values must be <1?) Since there are 55.5 moles (mol) of water in 1 liter (1,000 g/18 $gmol^{-1}$), the equation for calculating a_w becomes

$$a_w = 55.5/(55.5 + n_2)$$

The theoretical amounts of sugar and salt required to achieve a given a_w are shown in Table 2.5. While it appears that half as much sodium chloride is required as sucrose (on a molar basis), 1 mol of sodium chloride disassociates into 1 mol of sodium and 1 mol of chloride, to give 2 mol of solute.

These equations can be used to calculate the a_w of a known solute concentration or to determine how much solute should be added to obtain a given a_w as shown below.

Example 1

What is the a_w of a solution containing 10% (weight/weight) glucose?

1. Assume 1 liter of solution.
2. Calculate the molal concentration of glucose.
   a. 100 g/900 g of water,
   b. For molal concentration, this equals 111 g of glucose/1,000 g of water.
   c. 111 g/180 $gmol^{-1}$ (the formula weight for glucose) = 0.61 mol

Table 2.5 Theoretical amounts of salt and sugar required to achieve a given a_w

a_w	Salt		Sugar	
	Molality[a]	%	Molality	%
0.99	0.3	1.7	0.6	17.0
0.98	0.5	2.8	1.1	27.3
0.96	1.1	6.0	2.3	44.0
0.94	1.0	9.5	3.6	55.3
0.92	2.4	12.3	4.9	62.6
0.90	3.1	15.3		
0.88	3.7	17.8		
0.86	4.5	20.8		
0.84	5.3	23.7		
0.80	6.9	28.8		
0.75	9.2	34.9		
0.70	11.9	41.0		
0.65	15.0	46.7		
0.60	18.5	52.0		

[a]Do you remember the difference between molality and molarity?

3. Calculate the a_w.

$$a_w = 55.5/(55.5 + n_2) = 55.5/(55.5 + 0.61) = 55.5/56.1 = 0.989$$

Example 2

What percent glucose would it take to reach an a_w of 0.90?

1. Assume 1 liter of solution.
2. Calculate the molal concentration of glucose required to produce an a_w of 0.90.
   a. $a_w = 55.5/(55.5 + n_2)$
   b. $0.90 = 55.5/(55.5 + n_2)$
   c. $0.90(55.5) + 0.90(n_2) = 55.5$
   d. $49.95 + 0.90(n_2) = 55.5$
   e. $0.90(n_2) = 5.5$
   f. $n_2 = 6.1$ molal
3. Convert molal concentration to a percentage.
   a. 6.1 molal solution = 6.1 mol × 180 g/mol = 1,098 g of glucose
   b. 1,098 g/(1,098 g of glucose + 1,000 g of water) = 52%

Foods at the same a_w may have different moisture contents due to their chemical composition and water binding capacity (Table 2.6).

Dehydration preserves food by removing available water, i.e., reducing a_w. Hot air removes water by evaporation. Freeze-drying removes water by sublimation (the conversion of ice to vapor without passing through the liquid stage) after freezing. These processes reduce the a_w of the food to levels that inhibit growth. Freezing reduces a_w, and when the food thaws, the a_w increases (Fig. 2.5).

a_ws of various foods are compared in Table 2.7. High-moisture foods such as fruits, vegetables, meats, and fish have an a_w of ≥0.98. Intermediate-moisture foods (jams, sausages, etc.) have a_w levels of 0.7 to 0.85. Additional preservative factors (e.g., reduced pH, preservatives, and pasteurization) are required for their microbiological stability.

Figure 2.6 illustrates the interplay among a_w, pathogen growth, and shelf life. Various microbes have different a_w requirements. Decreasing the a_w increases the lag phase of growth, decreases the growth rate, and decreases the number of cells at stationary phase. Foodborne microbes are grouped by their minimal a_w requirements in Table 2.8. Gram-negative

Table 2.6 Moisture content of various dry or dehydrated food products when their a_w is 0.7 at 20°C[a]

Food	% Moisture content
Grains	4–9
Milk powder	7–10
Cocoa powder	7–10
Whole egg powder	10–11
Skim milk powder	10–15
Dried, fat-free meat	10–15
Rice and legume seeds	12–15
Dehydrated vegetables	12–22
Dried soups	13–21
Dried fruits	18–25

[a]Reprinted from J. Farkas, p. 567–591, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

Figure 2.5 When water turns from solid to liquid, it becomes more available for microbial growth. *Lucky Cow* © 2004 Mark Pett. Distributed by Universal Press Syndicate. Reprinted with permission. All rights reserved.

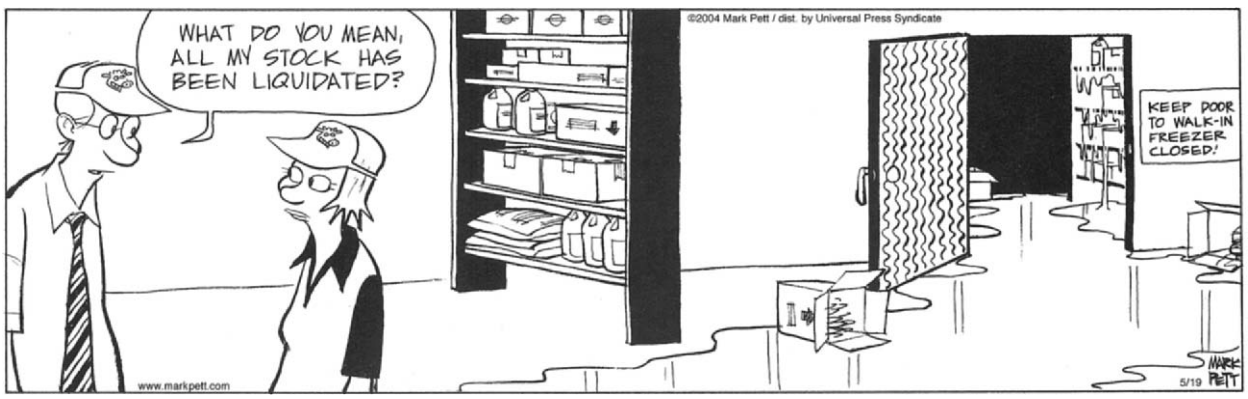


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Table 2.7 Typical a_ws of various foods[a]

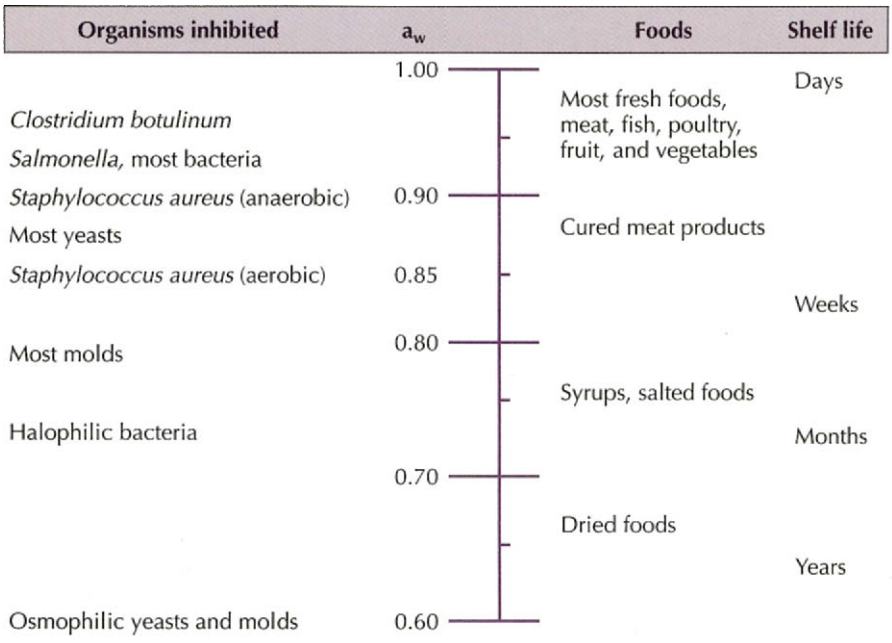
Food	a_w
Fresh, raw fruits, vegetables, meat, and fish	>0.98
Cooked meat, bread	0.95–0.98
Cured meat products, cheeses	0.91–0.95
Sausages, syrups	0.87–0.91
Rice, beans, peas	0.80–0.87
Jams, marmalades	0.75–0.80
Candies	0.65–0.75
Dried fruits	0.60–0.65
Dehydrated vermicelli, spices, milk powder	0.20–0.60

[a]Reprinted from J. Farkas, p. 567–591, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

species usually require the highest a_w. Gram-negative bacteria such as *Pseudomonas* spp. and most *Enterobacteriaceae* usually grow only at a_w values above 0.96 and 0.93, respectively. Gram-positive non-spore-forming bacteria are less sensitive to reduced a_w. Many *Lactobacillaceae* have a minimum a_w near 0.94. Some *Micrococcaceae* grow below a_w of 0.90. Staphylococci are unique among foodborne pathogens because they can grow at a minimum a_w of about 0.86. However, they do not make toxins below a_w of 0.93. Most spore-forming bacteria do not grow below a_w of 0.93. Spore germination and outgrowth of *Bacillus cereus* is prevented at a_w of 0.97 to 0.93. The minimum a_w for *Clostridium perfringens* spore germination and growth is between 0.97 and 0.95.

Several yeast species grow at a_w levels lower than those of bacteria. Salt-tolerant species such as *Debaryomyces hansenii*, *Hansenula anomala*, and *Candida pseudotropicalis* grow well on cured meats and pickles at NaCl con-

Figure 2.6 Illustration of the effect of a_w on shelf life and microbial growth.



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3

Spores and Their Significance

Introduction

Spores in the Food Industry
Low-Acid Canned Foods
Bacteriology of Sporeformers of Public Health Significance
Heat Resistance of *C. botulinum* Spores
Spoilage of Acid and Low-Acid Canned and Vacuum-Packaged Foods by Sporeformers

Spore Biology
Structure
Macromolecules
Small Molecules
Dormancy
Resistance
Freezing and Desiccation Resistance
Pressure Resistance
γ-Radiation Resistance
UV Radiation Resistance
Chemical Resistance
Heat Resistance

The Cycle of Sporulation and Germination
Sporulation
Activation
Germination
Outgrowth

Summary

Suggested reading

Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- discuss how the modern canned food industry has developed to overcome the challenge of spore heat resistance
- define the traits of a low-acid canned food and how it must be processed
- identify, characterize, and differentiate among spore-forming bacteria that cause illness and spoilage
- understand the fundamental difference between a bacterial spore and a vegetative cell
- correlate the unique properties of spores with the challenge they present for food preservation
- draw a diagram of a spore and compare it to a diagram of a bacterial cell
- explain the physical and chemical basis for spore heat resistance
- compare the cycles of sporulation and germination with the gain and loss of spore resistance characteristics

INTRODUCTION

This chapter examines the spore, a unique life form of tremendous significance in food processing. The importance of spores in the food industry and the steps used to control them are discussed first. Then, the structure and unique properties of the spore are described. Finally, the complex life cycle of a spore, which consists of sporulation (the conversion of a vegetative cell to a spore) and germination (the conversion of a spore to a vegetative cell), is presented.

SPORES IN THE FOOD INDUSTRY

Spore-forming bacteria and heat-resistant fungi cause big problems for the food industry. Three species of sporeformers, *Clostridium botulinum*, *Clostridium perfringens*, and *Bacillus cereus*, are infamous for producing toxins. Other sporeformer species cause spoilage. Sporeformers causing foodborne illness and spoilage are particularly important in low-acid foods (equilibrium pH, ≥4.6) packaged in cans, bottles, pouches, or other hermetically (vacuum) sealed containers ("canned" foods), which are processed by heat. Other sporeformers cause spoilage of high-acid foods (equilibrium pH, <4.6). Psychrotrophic sporeformers cause spoilage of refrigerated foods. Fungi that produce heat-resistant ascospores cause spoilage of acidic foods and beverages.

BOX 3.1

Koch's postulates for proof for the bacterial origin of a disease

1. The bacteria must be isolated from a sick animal.
2. The bacteria must be identified in pure culture.
3. When the pure culture of bacteria is reintroduced into a healthy animal, the animal becomes sick.

Louis Pasteur discovered that spore-forming bacteria caused food spoilage during investigations of butyric acid fermentation in wines (see chapter 19). Pasteur isolated an organism he termed *Vibrion butyrique*, which is probably what we now call *Clostridium butyricum*. Spores were discovered independently by Ferdinand Cohn and Robert Koch in 1876. Investigations by Pasteur and Koch linked microbial activity with food quality and safety. During investigation of the anthrax bacillus, *Bacillus anthracis*, the famous Koch's postulates (Box 3.1) were born. These stated that a disease is caused by a specific microorganism and led to the development of pure-culture techniques. Diseases and spoilage caused by spore-formers are associated with thermally processed foods, since heat kills vegetative cells but allows survival and growth of spore-forming organisms.

In the late 1700s, Nicolas Appert invented the process of appertization, which is now known as canning. This process places food in a hermetically (airtight) sealed container and preserves it by heating. He believed that the elimination of air was responsible for the stability of canned foods. The empirical use of thermal processing gradually developed into modern-day thermal processing industries. In the late 1800s and early 1900s, scientists in the United States developed scientific principles to ensure the safety of thermally processed foods. Because of these studies, thermal processing of foods in hermetically sealed containers became an important industry. S. C. Prescott and W. L. Underwood at the Massachusetts Institute of Technology and H. L. Russell at the University of Wisconsin found that spore-forming bacilli caused the spoilage of thermally processed clams, lobsters, and corn. The classic study by J. R. Esty and K. F. Meyer in California provided definitive values of the heat resistance of *C. botulinum* type A and B spores. C. Olin Ball of Rutgers University provided the mathematical foundation for commercial canning of foods. These efforts helped rescue the U.S. canning industry from its near demise in the 1940s due to botulism in commercially canned olives and other foods. Quantitative thermal processes, understanding of spore heat resistance, aseptic processing, and implementation of the Hazard Analysis Critical Control Point concept also grew out of the canned food industry.

Low-Acid Canned Foods

The U.S. Food and Drug Administration (FDA) and the U.S. Department of Agriculture (USDA) Food Safety Inspection Service define a low-acid canned food as one with a final pH of >4.6 and a water activity (a_w) of >0.85. The regulations regarding thermal processing of canned foods are described in the U.S. *Code of Federal Regulations* (21 CFR, parts 108 to 114). The USDA has regulatory control of foods that contain at least 3% raw red meat or 2% cooked poultry. The FDA regulates everything else.

Low-acid foods are packaged in hermetically sealed containers. These are usually cans or glass jars but can also be pouches and other types of containers. These "cans" (as collectively defined in this chapter) containing low-acid food products must be processed by heat to achieve commercial sterility. "Commercial sterility" uses heat to inactivate foodborne pathogens, as well as any spoilage microorganisms that can grow in the food under normal nonrefrigerated conditions. "Commercial sterility" indicates a shelf-stable product with a negligible level of microbial survival. Preservation procedures such as acidification or lowering a_w can also be used to attain commercial sterility. These preservation techniques are often combined to reduce heat treatments. A description of the process for achieving commercial sterility, the facility, equipment, and formulations must be filed

Food microbiologists should be able to answer Calvin's question. *Calvin and Hobbes* © 1988 Bill Watterson. Distributed by Universal Press Syndicate. Reprinted with permission. All rights reserved.

with the FDA or USDA prior to commercial processing. The processor must report process deviations or instances of spoilage.

Inactivation of *C. botulinum* spores is the primary processing goal for canned low-acid foods. It is the most heat-resistant microbial pathogen. The severity of the heat treatment required depends on the class of food, its spore content, pH, storage conditions, and other factors. For example, canned low-acid vegetables and uncured meats usually receive a 12*D* process (see below) or "botulinum cook." Lesser heat treatments are applied to shelf-stable canned cured meats as well as to foods with reduced a_w or other antimicrobial factors. Certain foods and ingredients such as mushrooms, potatoes, spices, sugars, and starches may contain high spore levels and require more than a 12*D* process to prevent spoilage. When spore loads influence the process, they must be monitored.

Researchers in the early and mid-1900s described thermal processes to prevent botulism and spoilage in canned foods. I. J. Pflug refined the semilogarithmic microbial death model and designed a strategy to achieve the required heat process. This is defined as the number of minutes (F) at a given temperature T, and is known as the F_T value, or generically, F. He also introduced the concept of the probability of a nonsterile unit on a one-container basis. This reasoning logically explains the traditional 12*D* term, the time required in a thermal process for a 12-log reduction of *C. botulinum* spores. In thermal processing, two values are used to describe the thermal inactivation. The D value is the time required for a 1-log reduction in viability of a microbial population (a 1-log reduction equates to killing 90% of the population). The z value is the temperature change required to change the D value by a factor of 10. The D value represents the resistance of an organism to a specific temperature. The z value represents the relative resistance of an organism to inactivation at different temperatures. The thermal process for the 10^{-11} to 10^{-13} level of probability of a botulism incident occurring depends on the initial number of *C. botulinum* spores present in a container. This value can be quite high (for example, 10^4 for a container of mushrooms) or very low (such as 10^{-1} or less for a container of meat product). Pflug emphasized that the thermal processing industry should prioritize process design to protect against (i) public health hazard (botulism) from *C. botulinum* spores, (ii) spoilage from mesophilic spore-forming organisms, and (iii) spoilage from thermophilic organisms in containers stored in warm environments. Generally, low-acid foods in hermetically sealed containers are heated to achieve 3 to 6 min at a temperature of 121°C

(250°F) or equivalent at the coldest spot of the food. This ensures the inactivation of the most heat-resistant *C. botulinum* spores with a $D_{121°C}$ of 0.21 min and a z of 10°C (18°F). Economic spoilage is avoided by achieving ~5*D* killing of mesophilic spores that typically have a $D_{121°C}$ of ~1 min. Foods distributed in warm climates require a particularly severe thermal treatment of ~20 min at 121°C to achieve a 5*D* killing of *Clostridium thermosaccharolyticum*, *Geobacillus* (formerly *Bacillus*) *stearothermophilus*, and *Desulfotomaculum nigrificans*. These organisms have a $D_{121°C}$ of ~3 to 4 min. Such severe treatment decreases nutrient content and sensory qualities, but it ensures a shelf-stable food.

The quality of low-acid canned foods can be improved through aseptic processing. In aseptic processing (Fig. 3.1) the following occur:

1. The product is commercially sterilized outside the container. Since the product contacts the heat source, it rapidly achieves a uniform process temperature.
2. The processed product is cooled and filled into presterilized containers.
3. The container receives an aseptic hermetic sealing in a sterile environment.

This technology was initially used for commercial sterilization of milk and creams in the 1950s. It is now used for other foods such as soups, eggnog, cheese spreads, sour cream dips, puddings, and high-acid products such as boxed fruit drinks. Aseptic processing and packaging systems reduce energy, packaging material, and distribution costs.

Bacteriology of Sporeformers of Public Health Significance

Three species of sporeformers, *C. botulinum*, *C. perfringens*, and *B. cereus*, cause foodborne illness and are covered extensively in other chapters. There have also been devastating incidences of intestinal anthrax caused by

Figure 3.1 Comparison of conventional (top) and aseptic (bottom) processing of food. In conventional processing, the food is placed in the container, sealed, and sterilized. In aseptic processing, the food and the container are sterilized separately, filled, and sealed under aseptic conditions.

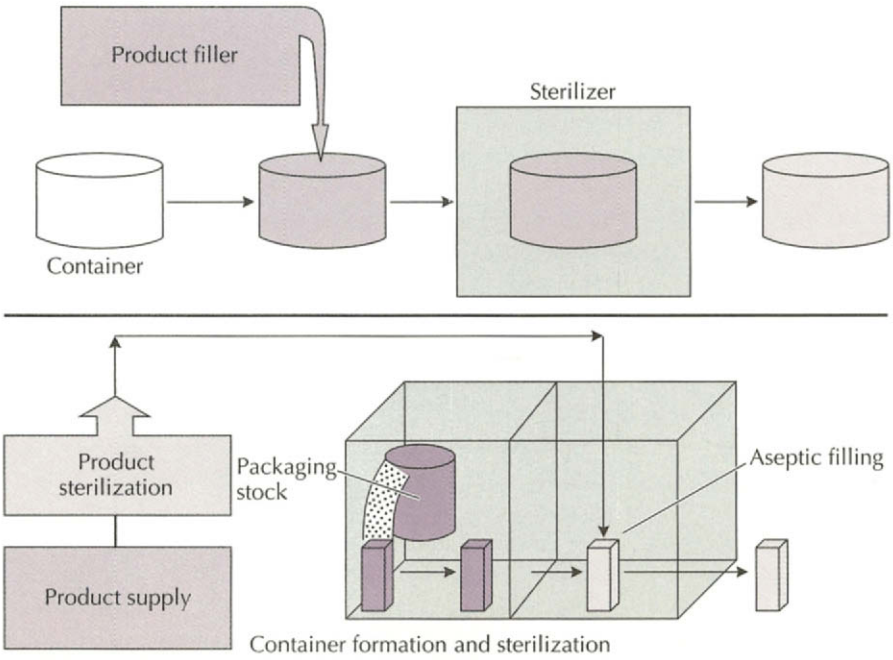


Table 3.1 Genera of gram-positive spore-forming bacteria

Genus	Characteristics
Clostridium	Anaerobic spore-forming rods
Bacillus	Aerobic spore-forming rods
Geobacillus	Thermophilic spore-forming rods
Sporolactobacillus	Spore-forming lactobacilli
Sulfolobus	Sulfur-oxidizing thermophilic aciduric spore-forming bacteria
Desulfotomaculum	Sulfate-reducing spore-forming bacteria
Amphibacillus	Facultatively anaerobic xylan-degrading spore-forming bacteria
Filobacillus	Previously unknown halophilic aerobic spore-forming bacteria

ingestion of contaminated raw or poorly cooked meat. However, *B. anthracis* is not considered a foodborne pathogen. There are other sporeformers associated with food and nonfood environments (Table 3.1). In some cases, advances in molecular biology have led to the division of a genus (e.g., *Geobacillus stearothermophilus* was once *Bacillus stearothermophilus*; nothing has changed, except the name). In other cases, such as *Filobacillus*, the organism does not fit in any known genus, so a new one is initiated. The thermal resistance parameters for many foodborne sporeformers are shown in Table 3.2.

The principal microbial hazard in heat-processed vacuum-packed foods and in minimally processed refrigerated foods is *C. botulinum*. The genus *Clostridium* consists of gram-positive, anaerobic, spore-forming

Table 3.2 Heat resistance of sporeformers of importance in foods[a]

Type of spore	Approx $D_{100°C}$ value (min)
Spores of public health significance	
Group I *Clostridium botulinum* types A and B	7–30
C. botulinum type E	0.01
Bacillus cereus	3–200
Clostridium perfringens	0.3–18
Mesophilic aerobes	
Bacillus subtilis	7–70
Bacillus licheniformis	13.5
Bacillus megaterium	1
Bacillus polymyxa	0.1–0.5
Bacillus thermoacidurans	2–3
Thermophilic aerobes	
Geobacillus stearothermophilus	100–1,600
Bacillus coagulans	20–300
Mesophilic anaerobes	
Clostridium sporogenes	80–100
Thermophilic anaerobes	
Desulfotomaculum nigrificans	<480
Clostridium thermosaccharolyticum	400

[a]Source: P. Setlow and A. E. Johnson, p. 33–70, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

bacilli that obtain energy by fermentation. *C. botulinum* spores swell the mother sporangium, giving a "tennis racket" or club-shaped appearance. The species of *C. botulinum* are heterogeneous and produce antigenically distinct toxins (A through G) that are species specific (see chapter 15). Organisms that produce these toxin types fall into two larger groups. Type I *C. botulinum* strains are mesophiles, produce heat-resistant spores, and make proteases that cause overt spoilage. Type II *C. botulinum* strains are nonproteolytic, make spores that are less heat resistant, but can grow at low temperatures. This has generated regulatory concern that minimally processed refrigerated food could lead to botulism.

C. perfringens is widespread in soils and in the intestinal tracts of humans and certain animals. *C. perfringens* grows extremely rapidly in high-protein foods such as meats that have been cooked to eliminate competitors and are then inadequately cooled. This time in the 40 to 140°F zone allows growth and production of a diarrheal enterotoxin. *C. perfringens* produces other extracellular toxins and enzymes, but these are important mainly for gas gangrene and animal diseases. Its ubiquitous distribution in foods and food environments, formation of spores that survive cooking, and an extremely rapid growth rate in warm foods (doubling times of 6 to 9 min at 43 to 45°C) contribute to the ability of *C. perfringens* to cause foodborne illness.

The heat resistance of *C. perfringens* spores varies among strains. In general, two classes of heat sensitivity are common. Heat-resistant spores have $D_{90°C}$ ($D_{194°F}$) values of 15 to 145 min and z values of 9 to 16°C (16 to 29°F), compared to heat-sensitive spores, which have a $D_{90°C}$ of 3 to 5 min and z of 6 to 8°C (11 to 14°F). The spores of the heat-resistant class generally require a heat shock of 75 to 100°C (167 to 212°F) for 5 to 20 min in order to germinate. The spores of both classes may survive cooking of foods and may be stimulated by heat shock for germination during the heating procedures. Both classes cause diarrheal foodborne illness.

The genus *Bacillus* contains only two pathogenic species, *B. anthracis* and *B. cereus*. *B. cereus* can produce a heat-labile enterotoxin causing diarrhea and a heat-stable toxin causing emesis (vomiting) in humans. Generally, the organism must grow to very high numbers ($>10^6$ cells/g of food) to cause human illness. *B. cereus* is closely related to *B. megaterium*, *B. thuringiensis*, and *B. anthracis*. *B. cereus* can be distinguished from these species by biochemical tests and the absence of toxin crystals. Other bacilli including *B. licheniformis*, *B. subtilis*, and *B. pumilis* have caused foodborne outbreaks, primarily in the United Kingdom.

Spores of *B. cereus* are ellipsoidal and central to subterminal and do not distend the sporangium. Spore germination occurs over a range of 8 to 30°C (46.4 to 86°F). Spores from strains associated with food poisoning have a heat resistance of $D_{95°C}$ (95°C = 203°F) of ~24 min. Other strains have a wider range of heat resistances. The strains involved in food poisoning may have higher heat resistances and therefore be more apt to survive cooking.

Heat Resistance of *C. botulinum* Spores

Heat resistance of spores among species and even among strains can vary greatly (Table 3.2). Group I *C. botulinum* type A and proteolytic B strains produce extremely heat-resistant spores. They are the most important sporeformers threatening the public health safety of canned foods. Esty and Meyer in California carried out the classic investigation of botulinal heat resistance in the wake of commercial outbreaks of botulism in canned

Scrambled word	Answer	Clue
1. TAREPP	_____	Inventor of canning
2. SIETPCA	_____	In the absence of microorganisms
3. UIBCLALS	_____	Aerobic spore-forming genus
4. MUUNITBLO	_____	Type of cook required for low-acid foods
5. MILTIRDCUSO	_____	Anaerobic spore-forming genus
6. LREAICMOMC	_____	Industrial type of sterility that refers to absence of microorganisms of public health significance
7. XRETOC	_____	Center part of spore that is relatively dehydrated and analogous to the cytoplasm of bacteria
8. EPMOOXNUIGSAR	_____	Outermost structure of the spore
9. TGONMIANIRE	_____	The process by which a vegetative cell becomes a spore
10. CENARTESSI	_____	The property of spores that makes them troublesome to food processors
11. ELAGIOSP	_____	In addition to public health concerns, spores can cause _____, which causes considerable economic damage
12. OSRTLOUNAIP	_____	The process by which a vegetative cell becomes a spore

olives and certain other canned vegetables. They examined 109 type A and B strains at five heating temperatures over the range of 100 to 120°C (212 to 248°F). They found that the inactivation rate is logarithmic between 100 and 120°C and depends on the spore concentration, the pH, and the food they are heated in. Esty and Meyer demonstrated that 0.15 M phosphate buffer (Sorensen's buffer), pH 7.0, gave the most consistent heat inactivation results, and their use of a standardized system enables comparisons of heat resistance by researchers today. The data of Esty and Meyer can be extrapolated to give a maximum value for $D_{121.1°C}$ of 0.21 min for *C. botulinum* type A and B spores in phosphate buffer. The canning industry uses $D_{121°C}$ as a standard in calculating process requirements. Proteolytic type F *C. botulinum* spores have heat resistances as follows: $D_{98.9°C}$ = 12.2 to 23.2 min and $D_{110°C}$ = 1.45 to 1.82 min. This is much lower than that of type A spores. Spores of nonproteolytic type B and E *C. botulinum* have much lower heat resistance than proteolytic A and B strains. Type E spores have a $D_{70°C\ (158°F)}$ varying from 29 to 33 min and a $D_{80°C\ (176°F)}$ from 0.3 to 2 min depending on the strains. The z value ranges from 13 to 15°F. The heat resistance of *C. botulinum* spores depends on environmental and recovery conditions. Heat resistance is markedly affected by acidity. Esty and Meyer found that spores have maximum resistance at pH 6.3 and 6.9, and resistance decreases markedly at pH values below 5 or above 9. Increased levels of sodium chloride or sucrose and decreased a_w increase the heat resistance of *C. botulinum* spores. *C. botulinum* spores coated in oil are more resistant to heat. Sporulation of *C. botulinum* at higher temperatures results in spores with greater heat resistance, possibly through the formation of heat shock proteins.

C. botulinum spores of groups I and II are highly resistant to irradiation compared with most vegetative cells. It is probably not practical to inactivate them in foods by irradiation. *C. botulinum* spores have a *D* of 2.0 to 4.5 kilograys. *C. botulinum* spores are also highly resistant to ethylene oxide but

are inactivated by halogen sanitizers and by hydrogen peroxide. Hydrogen peroxide is commonly used for sanitizing surfaces in aseptic packaging. Halogen sanitizers are used in cannery cooling waters.

Spoilage of Acid and Low-Acid Canned and Vacuum-Packaged Foods by Sporeformers

Thermally processed low-acid foods receive a heat treatment adequate to kill spores of *C. botulinum* but not sufficient to kill more-heat-resistant spores. Acid and acidified foods with an equilibrium pH of ≤4.6 are not heated enough to inactivate all spores. Most species of sporeformers do not grow under acid conditions, and inactivation of all spores would decrease food quality and nutrition. Other foods, such as cured meats and hams, that are not heated enough to kill spores must be kept refrigerated for microbial stability. Nonpathogenic sporeformers cause spoilage in these foods.

The spores naturally present in foods and the cannery environment contribute to spoilage problems. Dry ingredients such as sugar, starches, flours, and spices often contain high spore levels. Spore populations can also accumulate in a food plant, such as thermophilic spores on heated equipment and saccharolytic clostridia in plants processing sugar-rich foods such as fruits.

The principal spoilage organisms and spoilage manifestations are presented in Table 3.3. The principal classes of sporeformers causing spoilage are thermophilic flat-sour organisms, thermophilic anaerobes not producing hydrogen sulfide, thermophilic anaerobes forming hydrogen sulfide, putrefactive anaerobes, facultative *Bacillus* mesophiles, butyric clostridia, lactobacilli, and heat-resistant molds and yeasts. *Alicyclobacillus acidoterrestris*

Table 3.3 Spoilage of canned foods by sporeformers[a]

Type of spoilage	pH	Major sporeformers responsible	Spoilage defects
Flat-sour	≥5.3	*B. coagulans, B. stearothermophilus*	No gas, pH lowered. May have abnormal odor and cloudy liquor.
Thermophilic anaerobe	≥4.8	*C. thermosaccharolyticum*	Can swells, may burst. Anaerobic end products give sour, fermented, or butyric odor. Typical foods are spinach, corn.
Sulfide spoilage	≥5.3	*D. nigrificans, C. bifermentans*	Hydrogen sulfide produced, giving rotten egg odor. Iron sulfide precipitate gives blackened appearance. Typical foods are corn, peas.
Putrefactive anaerobe	≥4.8	*C. sporogenes*	Plentiful gas. Disgusting putrid odor. pH often increased. Typical foods are corn, asparagus.
Psychrotrophic clostridia	>4.6		Spoilage of vacuum-packaged chilled meats. Production of gas, off flavors and odors, discoloration.
Aerobic sporeformers	≥4.8	*Bacillus* spp.	Gas usually absent except for cured meats; milk is coagulated. Typical foods are milk, meats, beets.
Butyric spoilage	≥4.0	*C. butyricum, C. tertium*	Gas, acetic and butyric odor. Typical foods are tomatoes, peas, olives, cucumbers.
Acid spoilage	≥4.2	*B. thermoacidurans*	Flat (*Bacillus*) or gas (butyric anaerobes). Off odors depend on organism. Common foods are tomatoes, tomato products, other fruits.
	<4	*Alicyclobacillus acidoterrestris*	Flat spoilage with off flavors. Most common in fruit juices, acid vegetables, and also reported to spoil iced tea.

[a]Source: P. Setlow and E. A. Johnson, p. 33–68, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

and psychrotrophic clostridia can spoil meat and fruit products, respectively. These organisms are controlled by monitoring raw foods entering the cannery to limit the initial spore load in a food product, thermal processing appropriate for storage and distribution conditions, cooling products rapidly, chlorination of cooling water, and good manufacturing practices (GMPs).

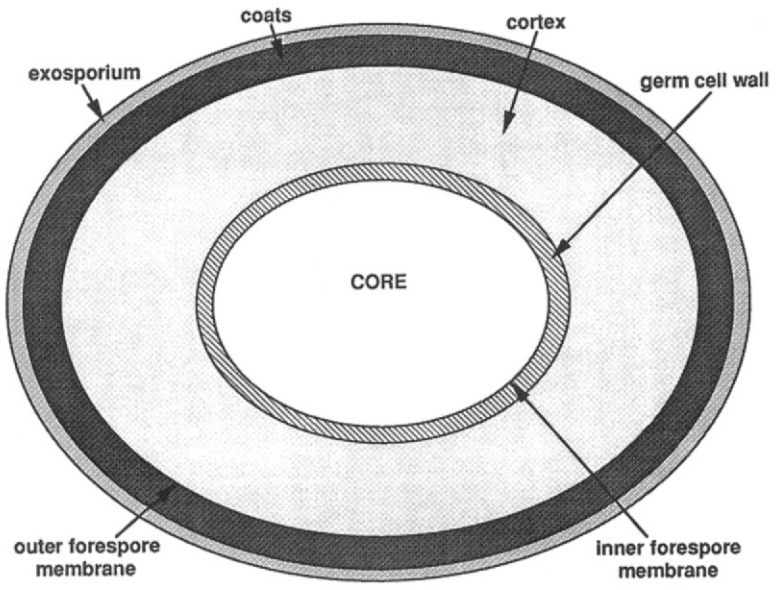
Heat-resistant fungi cause spoilage of acidic foods, particularly fruit products. Although heating for a few minutes at 60 to 75°C kills most filamentous fungi and yeasts, heat-resistant fungi produce thick-walled ascospores that survive heating at ≥85°C for 5 min. The most common heat-resistant spoilage fungi are *Byssochlamys*, *Neosartorya*, *Talaromyces*, and *Eupenicillium*. Certain heat-resistant fungi also produce toxic secondary metabolites called mycotoxins. To prevent spoilage of heat-treated foods, raw materials should be screened for heat-resistant fungi and strict GMPs and sanitation programs should be followed. Manipulation of a_w and oxygen tension and the application of antimycotic agents can also prevent fungal growth.

SPORE BIOLOGY

Structure

The spore is released from the mother cell at the end of sporulation. Spores are biochemically, structurally, and physiologically different from vegetative cells. The spore has seven layers: the exosporangium, coat, outer membrane, cortex, germ cell wall, inner membrane, and core (Fig. 3.2). Many spore structures have no counterparts in the vegetative cell.

Figure 3.2 Structure of a dormant spore. The various structures are not drawn precisely to scale, especially the exosporium, whose size varies tremendously between spores of different species. The relative size of the germ cell wall is also generally smaller than shown. The positions of the inner and outer forespore membranes, between the core and the germ cell wall and between the cortex and coats, respectively, are also noted. Source: P. Setlow and E. A. Johnson, p. 33–70, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).



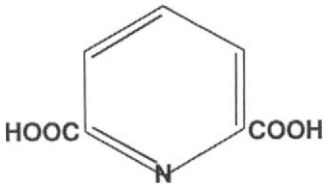


Figure 3.3 Structure of DPA. Note that at physiological pH both carboxyl groups are ionized.

The outermost spore layer, the exosporium, varies in size between species. Underlying the exosporium are the spore coats. The spore coats protect the spore cortex from attack by lytic enzymes. Underlying the spore coats is the outer spore membrane. It helps keep the spore impermeable to small molecules.

The cortex is under the membrane. The peptidoglycan that makes up the cortex is structurally similar to cell wall peptidoglycan, but with several differences. The spore peptidoglycan always contains dipicolinic acid (pyridine-2,6-dicarboxylic acid [DPA]), which is not found in the vegetative cell wall (Fig. 3.3). The spore cortex mechanically dehydrates the spore core and is responsible for much of spore resistance (see below).

Between the cortex and the inner membrane is the germ cell wall. Its structure may be identical to that in vegetative cells. The next structure, the inner spore membrane, is a complete membrane. It is a very strong permeability barrier. This membrane's phospholipid content is similar to that of vegetative cells.

Finally, the spore core contains the spore's DNA, ribosomes, and most enzymes, as well as the deposits of DPA and divalent cations. There are many unique compounds in the dormant spore, including the large pool of small acid-soluble proteins (SASP), which comprises 10 to 20% of spore protein. Much of the SASP is bound to spore DNA. Spore cores have a very low water content. While vegetative cells have ~4 g of water per g (dry weight), the spore core has only 0.4 to 1 g of water per g (dry weight). The core's low water content is responsible for spore dormancy and spore resistance. The spore cortex generates and maintains spore core dehydration, but the precise mechanism(s) involved is unknown.

Macromolecules

Spores are biochemically different from vegetative cells. Some spore proteins are absent in the vegetative cell. The SASP play a major role in spore resistance. The binding of these proteins to DNA provides resistance to chemical and enzymatic cleavage of the DNA backbone and alters the DNA's ultraviolet (UV) photochemistry. Conversely, many proteins present in vegetative cells are absent in spores. These include amino acid and nucleotide biosynthetic enzymes. The catabolic enzymes for using amino acid and carbohydrate are present in both spores and cells. Spore DNA appears to be identical to cell DNA.

Small Molecules

The spore's small molecules, located in the cortex, are different from the cell's small molecules. The low amount of spore core water and the huge deposit of DPA and divalent cations have already been noted (Table 3.4). The ions in the spore core are immobile, since there is no free water. The pH in the spore core is 1 to 1.5 units lower than that in a growing cell. In contrast to growing cells, spores have little, if any, "high-energy" compounds such as deoxynucleoside triphosphates, ribonucleoside triphosphates, reduced pyridine nucleotides, and acyl coenzyme A (acyl-CoA) (Table 3.4).

Dormancy

Spores are metabolically dormant. They have no detectable metabolism. The major cause of the spore's metabolic dormancy is undoubtedly its low water content. This precludes enzyme action. Dormancy is further demonstrated by enzyme-substrate pairs in the core that are stable for months to years but are degraded in the first 15 to 30 min of germination.

Table 3.4 Small molecules in cells and spores of *Bacillus* species

Molecule	Content (mmol/g [dry wt]) in: Cells[a]	Spores[b]
ATP (adenosine 5′-triphosphate)	3.6	≤0.005
ADP (adenosine 5′-diphosphate)	1	0.2
AMP (adenosine 5′-monophosphate)	1	1.2–1.3
Deoxynucleotides	0.59[c]	<0.025[d]
NADH (reduced nicotinamide adenine dinucleotide)	0.35	<0.002[e]
NAD (nicotinamide adenine dinucleotide)	1.95	0.11[e]
NADPH (reduced nicotinamide adenine dinucleotide phosphate)	0.52	<0.001[e]
NADP (nicotinamide adenine dinucleotide phosphate)	0.44	0.018[e]
Acyl-CoA	0.6	<0.01[e]
CoASH[f]	0.7	0.26[e]
CoASSX[g]	<0.1	0.54[e]
3PGA	<0.2	5–18
Glutamic acid	38	24–30
DPA	<0.1	410–470
Ca^{2+}		380–916
Mg^{2+}		86–120
Mn^{2+}		27–56
H^+	7.6–8.1[h]	6.3–6.9[h]

[a]Values for *B. megaterium* in mid-log phase.
[b]Values are ranges from spores of *B. cereus*, *B. subtilis*, and *B. megaterium*.
[c]Value is the total of all four deoxynucleoside triphosphates.
[d]Value is the sum of all four deoxynucleotides.
[e]Values are for *B. megaterium* only.
[f]CoASH, free CoA.
[g]CoASSX, CoA in disulfide linkage to CoA or a protein.
[h]Values are expressed as pH and are ranges for *B. cereus*, *B. megaterium*, and *B. subtilis*.

Resistance

The spore's dormancy helps it survive extremely long periods in the absence of nutrients (Box 3.2). A second factor in long-term spore survival is the spore's extreme resistance to heat, radiation, chemicals, and desiccation. Spores are much more resistant than vegetative cells. Spore resistance is due to a variety of factors such as spore core dehydration, SASP, and impermeability. Since different factors contribute to different types of spore resistance, it is not surprising that different types of resistance are gained at different times in sporulation. The mechanisms of spore heat, UV, and H_2O_2 resistance have been determined. The following discussion of spore resistance concentrates on *B. subtilis* because of the detailed mechanistic data available for this organism. However, factors involved in resistance of *B. subtilis* spores are undoubtedly involved in resistance of other species.

Freezing and Desiccation Resistance

Some bacterial populations are reduced during freezing. More lethality occurs during drying. The precise mechanism(s) of killing is not clear, but one cause may be DNA damage; freeze-drying cells can cause significant mutagenesis. Spores, however, are resistant to multiple cycles of freeze-drying. A complete explanation for spore drying resistance is not yet available. However, the SASP contribute to spore drying resistance by preventing DNA damage.

BOX 3.2

Jurassic spores

Just how long can spores be dormant and still be viable? Think along geological time lines. The earth was formed about 6 billion years ago. Bacteria came on the scene about 3 billion years ago. Dinosaurs went extinct 65 million years ago. "Humanity" is about 6 million years old. Bacteria as a class clearly have staying power. Individual organisms also have staying power. Raúl Cano, a microbiologist at California Polytechnic University, resurrected a spore that had been dormant for 25 million to 40 million years. *Bacillus* species have a symbiotic relationship with bees, which are frequently embedded in amber. Spores had been seen in these bees using electron microscopes, and spore DNA had been isolated, but no one had ever determined if the spores were still alive. Cano rigorously sterilized the amber samples and all of the experimental material before pulverizing it into broth where spores could normally grow. Grow they did! The spores produced bacilli. Analysis of their DNA showed that they were ancient and unrelated to any modern bacilli that could have been potential contaminants.

Cano, R. J., and M. K. Borucki. 1995. Revival and identification of bacterial spores in 24–40 million year old Dominican amber. *Science* **228**:1060–1064.

Pressure Resistance

Spores are much more resistant to high pressures (≥12,000 atmospheres) than are cells. Although spores are more resistant than cells to lower pressures, spores are killed more rapidly at lower pressures than at higher pressures. This apparent anomaly occurs because lower pressures promote spore germination; the germinated spores are then rapidly killed by the pressure treatment. Spore germination is not promoted by high pressures.

γ-Radiation Resistance

Spores are generally more resistant to γ-radiation than are vegetative cells. In the few species studied, γ-radiation resistance is gained during sporulation 1 to 2 h before heat resistance is acquired. The precise factors involved in spore γ-radiation resistance are not known, although SASP are not involved. The low water content in the spore core may provide protection against γ-radiation. However, the amount of water in the spore core has not been correlated with spore γ-radiation resistance. Radiation presumably damages spore DNA through an unknown mechanism.

UV Radiation Resistance

Spores are 7 to 50 times more resistant than are vegetative cells to UV. During sporulation, UV resistance is acquired 2 h before heat resistance, in parallel with synthesis of SASP. These proteins are essential for spore UV resistance. Coats, the cortex, and core dehydration are not necessary for spore UV resistance.

Chemical Resistance

Spores are much more resistant than cells to many chemicals, pH extremes, and lytic enzymes such as lysozyme. Resistance of spores to chemicals is acquired at different times in sporulation. For some compounds, spore coats play a role in chemical resistance, possibly by providing an initial barrier against attack. This is clearly true for lytic enzymes, as spores with coats removed are sensitive to lysozyme. The inability of most molecules to penetrate into the spore core plays an important role in chemical resistance.

Spore Heat Resistance

Heat resistance has huge implications for the food industry. It is probably the best-studied form of resistance. Spore heat resistance is truly remarkable; many spores withstand 100°C for several minutes. Heat resistance is

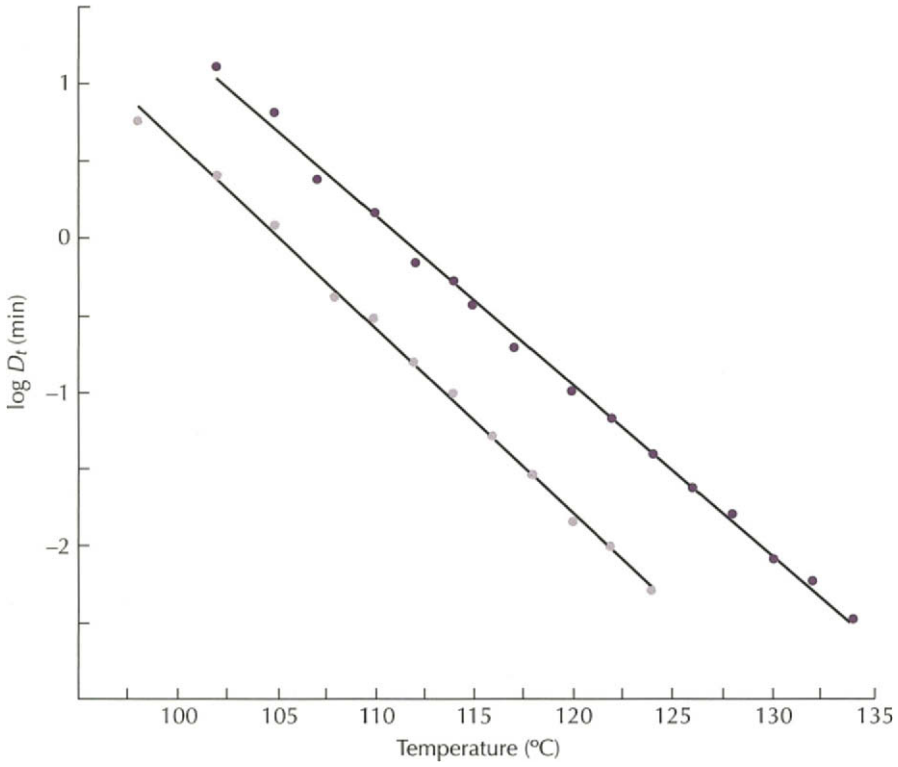
often quantified as a D_t value. This is the time in minutes at the temperature (t) needed to kill 90% of a population. Generally, D values for spores at a temperature of t + 40°C are about equal to those for their vegetative cell counterparts at temperature t. An often-overlooked feature of spore heat resistance is that the extended survival of spores at elevated temperatures is paralleled by even longer survival times at lower temperatures. Spore D values increase 4- to 10-fold for each 10°C decrease in temperature. Consequently, a spore with a D value at 90°C ($D_{90°C}$) of 30 min may have a $D_{20°C}$ value of many years.

We do not know what target(s) is damaged by heat to kill the spores. The target is probably not spore DNA. Spore heat killing is not caused by DNA damage or mutagenesis. Protein(s) may be the target of spore heat killing. Sublethal heat treatment can also injure spores. This damage can be repaired during spore germination and outgrowth. Several factors that modulate spore heat resistance are discussed below.

Sporulation Temperature

Elevated sporulation temperatures increase spore heat resistance (Fig. 3.4). Indeed, spores of thermophiles are generally more heat resistant than spores of mesophiles. Since spore macromolecules are identical to cell macromolecules, spore macromolecules are not intrinsically heat resistant. Presumably, the total macromolecular content of spores from thermophiles is more heat stable than that from mesophiles, accounting for the higher heat resistance. However, spores of the same strain prepared at various temperatures are most heat resistant when prepared at the highest temperature.

Figure 3.4 Influence of sporulation temperature on D values of *Bacillus subtilis* sporulated at 32°C (pale purple dots) or 52°C (dark purple dots). Redrawn from F. Sala et al., *J. Food Prot.* **58**:239–243, 1995, with permission from the *Journal of Food Protection*. Copyright International Association for Food Protection, Des Moines, IA.



In growing bacteria, adaptation to heat stress involves the proteins of the heat shock response. The levels of these proteins increase with increasing sporulation temperature. Although heat shock proteins could play some role in spore heat resistance, this appears not to be the case.

α/β-Type SASP

Surprisingly, neither general mutagenesis nor DNA damage occurs during the killing of spores by wet heat. This is despite the fact that the high temperatures would be expected to cause DNA damage. Therefore, spore DNA must be well protected against heat damage, and the thermal inactivation of spores must be due to some other mechanisms. The major cause of spore DNA protection against heat damage appears to be the saturation of spore DNA by α/β-type SASP. Spores from *B. subtilis* $\alpha^-\beta^-$ mutants are more heat sensitive than those of wild-type strains.

Dry heat and aqueous (wet) heat affect spores differently. First, spores are much more resistant to dry heat than to aqueous heat. *D* values are 100- to 1,000-fold higher in dry than in hydrated spores. Second, spores have a rather high level of mutagenesis (~12% of survivors) when killed by dry heat. This mutagenesis damages spore DNA.

Core Water Content and Heat Resistance

Low core water content is a major factor in spore heat resistance. The spore becomes more heat resistant as it is dehydrated. The spore cortex is essential for creating and maintaining the dehydrated state of the spore core. This is undoubtedly due to the cortex's ability to change its volume upon changes in ionic strength and/or pH. If an expansion in cortex volume were restricted to one direction, i.e., towards the spore core, a mechanical action in the opposite direction would express core water.

There is a good correlation between spore core water content and heat resistance over a 20-fold range of *D* values. However, at the extremes of core water contents, *D* values vary widely, presumably reflecting the importance of other factors such as sporulation temperature, cortex structure, etc. There is little if any free water in the core. Presumably, water-driven chemical reactions are inhibited by the spore core's low water content. This also contributes to heat resistance. Low water content stabilizes macromolecules such as proteins against denaturation by restricting their molecular motion.

THE CYCLE OF SPORULATION AND GERMINATION

Sporulation

Spores are made in response to environmental stress or nutrient depletion. The molecular biologies of sporulation and spore resistance have been extensively studied for the genus *Bacillus*. Spores formed by *Alicyclobacillus*, *Bacillus*, *Clostridium*, *Desulfotomaculum*, and *Sporolactobacillus* also pose problems in foods.

An unequal cell division is the first event in sporulating cells. This creates the smaller spore compartment and the larger mother cell compartment. As sporulation proceeds (Fig. 3.5), the mother cell engulfs the forespore, resulting in a cell (the forespore) within a cell (the mother cell). These both have a complete genome. Since the spore is formed within the mother cell, it is called an endospore.

Genes are expressed at specific times and places during sporulation. Some genes are expressed only in the mother cell. Other genes are expressed

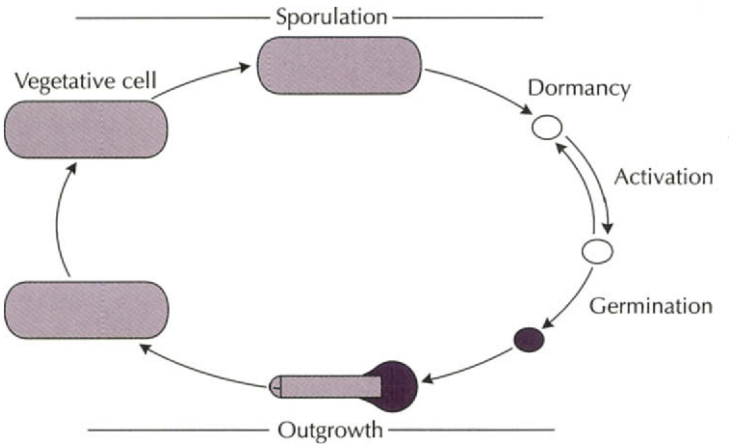


Figure 3.5 Cycle of sporulation, dormancy, activation, and outgrowth.

only in the spore. Gene expression is controlled by the ordered synthesis and activation of new sigma (specificity) factors for RNA polymerase. Many DNA-binding proteins, both repressors and activators, also regulate gene expression during sporulation.

As sporulation proceeds, there are striking morphological and biochemical changes in the developing spore. It becomes encased in two novel layers, the spore cortex and the spore coat. The spore also accumulates a huge store (≥10% of its dry weight) of DPA (Fig. 3.3), found only in spores, and a large amount of divalent cations. The spore becomes both metabolically dormant and extremely resistant to heat, radiation, and chemicals.

Activation

Dormant spores resume metabolism when they germinate (Fig. 3.5). Spores germinate more rapidly and completely if activated prior to exposure to a nutrient which induces germination (i.e., a germinant). There are many ways to activate spores. The most widely used agent is sublethal heat (e.g., 10 min at 80°C). The precise changes induced by spore activation are not clear. In some species, the activation is reversible. In most species, heat activation releases a small amount of the spore's DPA.

Authors' note

If one enumerated a population of 850,000 spores per ml on agar medium, the plate count might be as low as 75,000. If the spores were heated for 10 min at 80°C ("heat shocked"), this might increase to 775,000.

Germination

Spores lose most of their unique attributes within minutes of encountering a germinant. They lose DPA and SASP. The spore's resistance is also lost in the first minutes of germination. During this time, active metabolism begins and synthesis of large molecules starts. Eventually, the germinated spore is converted back to a growing vegetative cell through the process of outgrowth.

Spore germination occurs during the first 20 to 30 min after mixing of spores and germinant. During this period, a resistant dormant spore with a cortex and a large pool of DPA, minerals, and SASP is transformed to a sensitive, actively metabolizing germinated spore in which the cortex and SASP have been degraded. These degradative changes can occur in the absence of nutrients. However, further conversion of the germinated spore into a growing cell requires biosynthetic reactions and added nutrients.

Spore germination can be triggered by a wide variety of compounds, including amino acids, sugars, salts, and DPA. How these compounds trigger

spore germination is not clear. Metabolism of the germinant is not required. Indeed, inorganic salts germinate some spores. The stereospecificity exhibited by germinants (e.g., L-alanine is a germinant, whereas D-alanine often inhibits germination) suggests that some germinants interact directly with specific proteins.

In the first minutes of spore germination, protons and some divalent cations, in particular Zn^{2+}, are released. Release of DPA, loss of spore refractility, and cortex degradation follow. During germination, the spore loses up to 30% of its dry weight and the core increases its water content to that of the vegetative cell. These events require large changes in the permeability of the inner spore membrane. However, neither the nature nor the mechanism of these changes is understood. The changes accompanying the initiation of spore germination may be extremely fast. An individual spore can lose refractility in as little as 30 s to 2 min. However, for a spore population, the time can be much longer. Individual spores initiate germination after widely different lag times.

RNA synthesis starts in the first few minutes of germination. It uses nucleotides stored in the spore or generated by breakdown of preexisting spore RNA. Protein synthesis begins shortly after RNA synthesis. Amino acids produced from SASP breakdown support most protein synthesis early in germination.

Outgrowth

The transition from spore germination to spore outgrowth is not distinct. Outgrowth is the time from ~25 min after the start of germination until the first cell division. Most processes during germination are supported from stored reserves. Outgrowth requires outside resources. During outgrowth, the spore regains the ability to synthesize amino acids, nucleotides, and other small molecules.

DNA replication is not initiated until at least 60 min after the start of germination. DNA repair can occur well before DNA replicative synthesis. Even in the first minute of germination, spores contain deoxynucleoside triphosphates such as ATP. During spore outgrowth, the volume of the outgrowing spore continues to increase, requiring the synthesis of membrane and cell wall component.

Summary

- Spores are unique life forms that are resistant to many stresses.
- The heat resistance of spores in food is measured as their *D* value, the number of minutes that it takes at a given temperature to kill 90% of the spores. The z value is the number of degrees that it takes to change the *D* value by a factor of 10.
- Low-acid canned foods are those with pH values of >4.8 and a_w of >0.85.
- Commercial sterility is achieved through the application of a 12*D* "bot cook."
- Many sporeformers cause economic spoilage.
- Sporulation is the process by which a vegetative cell produces a spore.
- Germination and outgrowth enable the spore to resume life as a vegetative cell.
- The spore cortex is a unique form of peptidoglycan that contributes to spore resistance.

- Dipicolinic acid and SASP contribute to spore resistance.
- The thermal resistance of a spore crop is influenced by many factors, including its sporulation temperature.
- The relatively low water content of the spore core helps make it resistant.
- Sporulation, germination, and outgrowth are parts of the spore's life cycle.

Suggested reading

Driks, A. 2002. Overview: development in bacteria: spore formation in *Bacillus subtilis*. *Cell. Mol. Life Sci.* **59**:389–391.

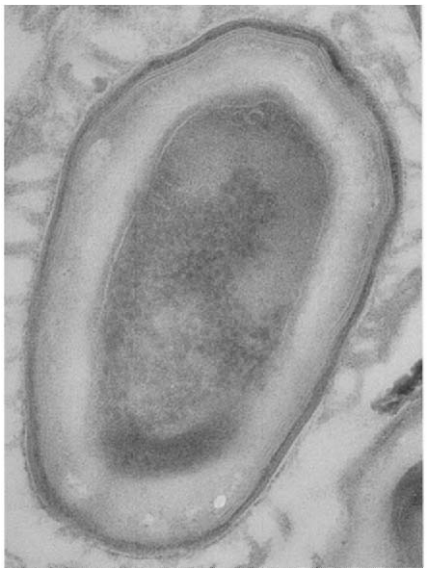
Nicholson, W. L., P. Fajardo-Cavazo, R. Rebeil, T. A. Slieman, P. J. Riesenman, J. F. Law, and Y. Xue. 2002. Bacterial endospores and their significance in stress resistance. *Antonie Leeuwenhoek* **81**:27–32.

Setlow, P., and E. A. Johnson. 2007. Spores and their significance, p. 35–67. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. You may remember that low-acid canned foods must receive a thermal process equivalent to an F_o of 2.4 min. This is based on a 12*D* reduction of *C. botulinum* spores, which have a $D_{250°F}$ of 0.2 min. Imagine two process deviations. In deviation A, the time is 10% less than it should be (i.e., the product receives 2.16 min at 250°F. In deviation B, the temperature is 10% too low (i.e., the product receives 2.40 min at 225°F). Which process deviation represents a greater threat to public safety? Be quantitative in your answer (i.e., calculate the log reduction in *C. botulinum* spores as the result of each deviation, assuming a *z* value of 18°F). Show all of your work.
2. Define *z* value. If an organism has a $D_{80°C}$ of 20 min, a $D_{95°C}$ of 2 min, and a $D_{110°C}$ of 0.2 min, what is its *z* value? (Hint: if you understand the concept, you won't need a calculator.)
3. If you had to choose, would you rather be a spore or a vegetative cell? Why?
4. What are two components found in spores but not vegetative cells? What is their function?
5. Is it okay for cans of tomatoes to contain viable *C. botulinum* spores? Why or why not?
6. Explain what "gene expression is regulated in time and space" means in your own words.
7. Discuss how spore water content influences heat resistance. How does this integrate with what else you know about thermal lethality (e.g., wet versus dry heat, heat resistance at low a_w, etc.)?
8. Figure 3.2 shows an idealized diagram of a spore. How many of the structures can you identify in the real transmission electron micrograph shown here?
9. Many mechanisms in this chapter are described as "not understood," "unknown," or "yet to be determined." Identify one such mechanism and use an online search engine, such as Google Scholar, to find at least one article dealing with that mechanism. Read the introduction and discussion of the paper and explain what is known and what is unknown.

Micrograph of a section through a spore and sporangium of *Bacillus subtilis*. Courtesy of Lee Simon and the Montville laboratory, Rutgers University.



10. Speculate: if spore formation is such a good mechanism for surviving adverse conditions, why don't all bacterial species form spores?
11. Figure 3.4 illustrates the influence of sporulation temperature on D value. What is the effect of sporulation temperature on the z value in this case?
12. *Bonus question:* Researchers can present their data in a variety of formats; tables, line graphs, and bar graphs are just a few. Present the data in Fig. 3.4 as a table. Follow these steps.

    1. "Eyeball" the pairs of log time/temperature data.
    2. Convert the $\log_{10} D$ values to the actual values.
    3. Construct a table with three columns: temperature, D at lower temperature, and D at higher temperature.

    Look at the table and the graph. They both show the same data. What are the advantages and disadvantages of each format?

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4

Detection and Enumeration of Microbes in Food

Introduction
Sample Collection and Processing
Analysis
Metabolism-Based Methods
Surface Testing
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- discuss methods available for microbiological analysis of food
- compare methods of analysis, indicating advantages and disadvantages of each method
- detail procedures for collection and processing of food samples
- calculate the microbial load of a sample
- recognize the difference between conventional and rapid microbiological methods

INTRODUCTION

Microorganisms inhabit our bodies and our environment. Therefore, it is not surprising to find many types of microorganisms (yeasts, molds, viruses, bacteria, and protozoa) in food. Often, food microbiologists refer to microorganisms in food as "the good, the bad, and the ugly." Good microorganisms are those that are used in the production of food (e.g., yogurt, wine, and beer), bad microorganisms cause spoilage, and ugly microorganisms cause human illness. Determining the types and numbers of microorganisms in a food is an important aspect of food microbiology.

The *aerobic plate count* (APC) provides an estimation of the number of microorganisms in a food. As the name implies, strict anaerobes do not grow on the plate and therefore are not included in the APC. The APC changes—increases or decreases—during the processing, handling, and storage of foods. In foods that are held raw, such as refrigerated meats, the APC increases during storage, whereas in dried or frozen foods, the APC remains unchanged or decreases. Depending on the food, the APC can be as low as 10 to as high as 1,000,000 microorganisms per g.

The levels of microorganisms in products from the same origin can differ greatly. For instance, the microbial load of ground meat is much greater than that of whole cuts of meat. During the handling, grinding, and packaging of meat pieces, bacteria may multiply or bacteria on equipment can be transferred to the product. The starting number of microorganisms associated with a food will have a significant impact on the number of microorganisms present in the finished product, even for foods that receive heat treatment. Poor-quality ingredients, poor sanitation, recontamination, and improper handling can result in high levels of microorganisms even in foods that were properly heat treated.

In order to determine if a product meets the microbial levels prescribed in specifications, guidelines, or standards, an estimation of the number of

microorganisms in or on a food is needed. A rule of thumb is that as the microbial count increases, the quality of the food decreases. Of course there are exceptions to this rule; for instance, in fermented foods the level of microorganisms would be expected to increase. This chapter focuses on some of the methods for determining the microbial levels of foods. Note that for the most part microbiological analysis of food samples refers to determining the levels of bacteria.

SAMPLE COLLECTION AND PROCESSING

The methods used for sample collection and processing vary from food to food and for specific microorganisms. The Food and Drug Administration outlines the methods that they use in the *Bacteriological Analytical Manual,* and the Food Safety Inspection Service of the U.S. Department of Agriculture publishes the *Microbiology Laboratory Guidebook*. Regardless of the protocol used, a sample will only yield significant and meaningful information if it represents the mass of material being examined, if the method of collection used protects against microbial contamination, and if the sample is handled in a manner that prevents changes in microbial numbers between collection and analysis.

Care must be taken during sample collection to prevent the introduction of microbes into the food sample. If possible, individual containers of food should be submitted for analysis. However, this is not always practical, and therefore a representative sample must be collected using an aseptic technique. Instruments used for sample collection should be sterilized in the laboratory rather than at the place of sampling. Sterile containers (plastic bags or wide-mouth jars) should be labeled with the name of the food, the date collected, and other information that may be useful in the analysis of results and tracking of the sample.

Authors' note

Bacteria readily stick to fat molecules in food. Therefore, foods high in fat (for example, ground beef [80% lean, 20% fat]) must be properly processed, or else the number of bacteria will be underestimated.

Once the sample is collected, it should be analyzed as quickly as possible to prevent a change in the microbial population. However, this is not always possible, and in that case, the sample should be refrigerated and frozen samples should be kept frozen. Products that are normally refrigerated should not be frozen, since freezing may cause death or damage to some bacterial cells, producing incorrect results.

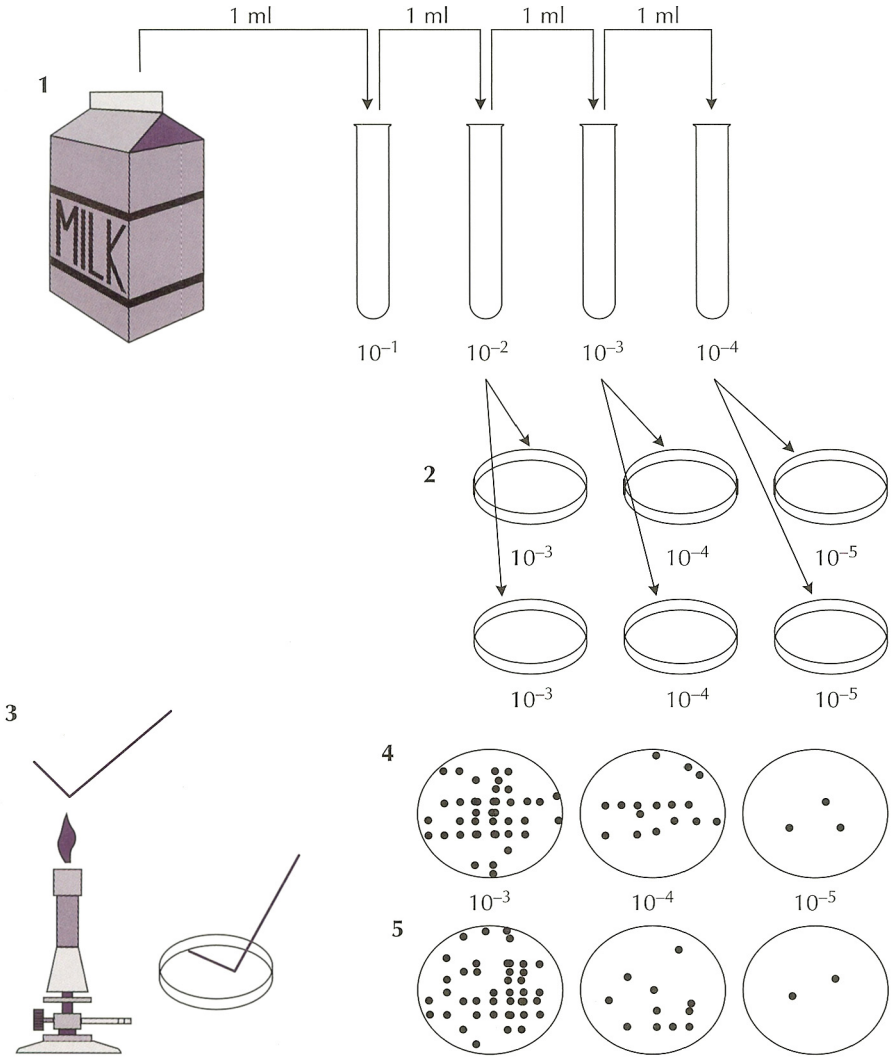
Prior to analysis, some preparation of the sample is generally required (Fig. 4.1). For meaningful results, the sample should be processed to produce a homogeneous suspension of bacteria so they can be pipetted. If the sample is solid, the food is generally mixed with a sterile diluent, such as Butterfield's buffered phosphate or 0.1% peptone water. If an appropriate buffer is not used, the bacteria may multiply in the dilution medium prior to being plated. To make a homogeneous suspension, the sample is added to a sterile plastic stomacher bag, diluent is added, and the sample is processed in a *stomacher* (a device that has two paddles that move rapidly back and forth against the sample). A typical-size food sample of 25 g is added to the sterile bag containing 225 ml of sterile diluent and mixed (placed in the stomacher), and a 1:10 dilution of the food and associated bacteria is obtained.

ANALYSIS

The APC, discussed above, provides a good estimate of the number of microorganisms associated with a sample. Drawbacks of surface plating are the *coalescence* of colonies (two or more colonies growing together) and the

growth of spreaders. In the APC, the hardened agar surfaces must be dried thoroughly to prevent the spread of bacteria by moisture droplets. A 0.1-ml aliquot of the selected dilution of the sample is dispensed onto the surface of the agar and spread with the aid of "hockey sticks," or bent glass rods. For the standard plate count (SPC; also called a pour plate), a measured aliquot (1 ml) of the appropriate dilution is dispensed onto the surface of a sterile petri plate and sterile, melted, and tempered (45°C) agar is added to the plate. The agar and sample are mixed, and the agar is allowed to solidify. The agar plate is then incubated, and isolated colonies known as

Figure 4.1 Preparation of sample for determination of microbial load. Milk, which initially contains millions of microbes, is diluted, and samples are spread plated so that individual colonies can be counted. Appropriate calculations are then made to determine the actual number of microbes in the milk. Step 1: since a liquid sample is being tested, a 1-ml aliquot can be removed and added to 9 ml of buffer for direct preparation of serial 10-fold dilutions. Step 2: 100 µl of each dilution is transferred to replicate plates. Step 3: a glass "hockey stick" is flame sterilized, allowed to cool, and used to spread the sample on the surface of the agar; the plate should be rotated to ensure distribution across the entire surface. Step 4: the plates are incubated under specified conditions. Step 5: dilutions yielding 30 to 300 colonies per plate are counted and expressed as numbers of CFU per milliliter of milk.



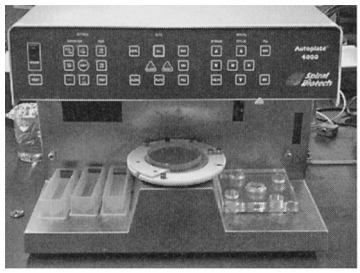
colony-forming units (CFU) are counted. This method has many disadvantages, including the time required for setup, expense, and lack of accuracy.

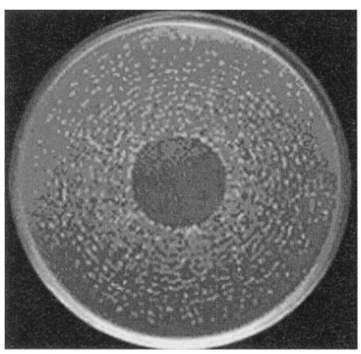
A CFU in theory arises from a single bacterial cell; however, this may not always hold true. In all likelihood, colonies arise from clusters or chains of bacteria, resulting in an underestimation of the number of bacteria in a sample. Therefore, samples may need to be processed to separate the cells into individual reproductive units. This is achieved through homogenization of a sample and by making a series of 1:10 dilutions (Fig. 4.1). A 1:10 dilution of a 1:10 dilution is a 1:100 dilution. For samples with unknown microbial populations, an extensive (10^{-1} to 10^{-7}) series of dilutions may need to be made and plated. In general, standard 100-mm-diameter plates should contain between 30 and 300 colonies. Fewer or more colonies per plate can affect accuracy. The total number of bacteria is determined by multiplying the colony count by the dilution factor to yield the number of bacteria per gram of food.

The *roll tube* method is basically the same as the pour plate method, but screw-cap tubes are used in place of petri plates. Test tubes containing 2 to 4 ml of plate count agar or other appropriate medium are sterilized, and the melted agar is allowed to cool to 45°C (tempered). A 0.1-ml aliquot of the appropriate dilution of the sample is added, and the tube is immersed in cold water in a horizontal position and rolled. The agar then solidifies, forming a thin layer on the inner wall of the tube. The tubes are then incubated upside down to prevent the possibility that any condensation which may form will cause smearing of the colonies. A decided advantage of this method is the savings associated with the use of less plate count agar.

Mechanical devices are available to facilitate the determination of bacterial numbers in food samples. The *spiral plater* uses a robotic arm to dispense a sample in an Archimedes spiral onto the surface of a rotating agar plate (Fig. 4.2). A continuously decreasing volume of the inoculum is deposited from near the center of the plate to the outside of the plate so that a concentration range of up to 10,000:1 is achieved on a single plate. The spiral plater compares favorably with the SPC. A specialized counting grid is used to count colonies on a plate, although the colonies are tightly packed together and may be difficult to count. The method is best suited for liquid samples, such as milk, since food particles may cause blocking of the dispenser. As with any method, there are advantages and disadvantages. The advantages include the use of fewer plates, fewer dilution bottles, and less agar and the ability to process a greater number of samples. The disadvantages include limitation of the types of samples that can be processed and difficulty in counting colonies.

Figure 4.2 (Top) A typical spiral plater. These units are fully automated and have a small footprint. **(Bottom)** Colonies formed on a plate in which the test sample was plated with a spiral plater.





Fluid samples (water or diluent) can be passed through membrane filters to collect bacteria. Membranes with a pore size of 0.45 μm are typically used. The filter can either be placed onto a plate containing a culture medium of choice or be used in the direct microscopic count (DMC) assay. This *membrane filter technique* is useful for examination of water or other dilute liquid samples that can easily pass through the filter. Therefore, even low numbers of microbes can be detected, since large volumes (1 liter or more) can be passed through a single filter. The method can also be used to determine microbial numbers in air. Small amounts of viscous liquid samples, such as milk or juice, can be processed without clogging the membrane.

A number of commercial methods based on the membrane filter technique are available. Some of these methods are discussed in chapter 5. The *direct epifluorescent filter technique* uses fluorescent dyes to stain bacteria that

are then counted using a fluorescence microscope. The number of cells per gram or milliliter is calculated by multiplying the average number per field by the microscope factor. This method has been used for estimating numbers of bacteria in milk, meat, and poultry products and on food contact surfaces. The *hydrophobic grid membrane filter* method is also used to estimate microbial numbers associated with a variety of foods. For this method, specially designed filters containing 1,600 grids that restrict microbial growth and colony size are used. Generally, 1 ml of a 1:10 sample homogenate is passed through the filter. Grids that contain colonies are counted, and a most-probable number is determined.

The DMC is faster than most methods, since no incubation period is required for cells to metabolize and grow (multiply). With this method, a smear of a food sample or a liquid sample (0.01 ml) is uniformly spread over a 1-cm^2 area on a microscope slide. Generally, liquid foods can be analyzed directly, but solid foods must be put into a suspension (1:10 dilution) before analysis. Once the sample is fixed, defatted, and stained, the cells are counted. A calibrated microscope must be used, since the diameter of the field to be examined must be known in order to calculate the number of organisms per gram. A stage micrometer can be used to measure the diameter of the circle (on the slide described above) to the nearest 0.001 mm. The average number of cells or clumps is calculated on a per-field basis and divided by the area of the field to determine the number per square millimeter.

The DMC is perhaps the simplest and most rapid method for estimating bacterial numbers in a sample. Therefore, adjustments or changes in processing can be implemented immediately to correct any problems. The test has other advantages, including little processing of samples, limited equipment required, a slide(s) that can be maintained as a record, and the need for only a small amount of sample. The disadvantages include fatigue (eyes becoming tired from peering through a microscope) and detection of live and dead cells without the ability to distinguish live cells from dead ones, and the method has little or no value for foods with low microbial numbers.

Fluorescent dyes are now available that can be used to differentially stain live and dead bacteria. Using such stains, the total bacterial count and a live-dead count can be obtained. Since the stains are intensely fluorescent, they can be used in samples that may contain low numbers of bacteria. The disadvantages are that the fluorescent signal fades over time and food constituents may also stain or fluoresce, interfering with the ability to count the bacterial cells.

Metabolism-Based Methods

In general microbiology, the metabolism of microorganisms is used to determine starch hydrolysis, sugar fermentation, production of hydrogen sulfide and indole, or nitrate reduction. Measurements of microbial metabolism or production of metabolic products in foods can be used to estimate the bacterial population. Today many of these tests have been miniaturized and are the bases of a number of rapid test methods.

Microorganisms obtain energy through chemical reactions, i.e., oxidation-reduction reactions, where the energy source becomes oxidized while another compound is reduced. Oxidation-reduction reactions consist of electron transfers and can therefore be measured electrically with a potentiometer. The measured oxidation-reduction potential is known as the *redox*

potential. The redox potential can also be determined with indicators and dyes. The addition of such compounds to metabolizing bacteria results in the transfer of electrons to the indicator and subsequent change in color. Several oxidation-reduction indicators or dyes, including methylene blue, resazurin, and tetrazoliums, can be used. The rate at which the indicator shows a change in color is related to the metabolic rate of the microbial culture (sample). Basically, the larger the number of bacteria, the faster the color change occurs. Tests using these compounds are often referred to as *dye reduction tests*, but this is not strictly correct, since resazurin and tetrazoliums are not dyes but indicators.

Reductase tests can be used for a variety of products, and the results compare favorably with those of the SPC. However, for some food samples, including raw meat, the test may not be appropriate, since meat contains inherent high levels of reductive substances. The test is commonly used in the dairy industry for determining the microbial quality of raw milk. The advantage of reductase tests over the SPC is that they can be completed in a shorter time.

Surface Testing

This chapter has concentrated on methods designed to detect and enumerate microorganisms associated with food samples. However, to ensure that food contact surfaces are being maintained in a hygienic state, they should be tested regularly for microorganisms. The inherent problem with surface testing is the consistent removal of microorganisms from the test surface. A number of methods are available; however, each may be appropriate only in specified areas of a food-processing plant.

The *swab test* method is perhaps the most widely used method for microbiological examination of surfaces in food-processing facilities. The basics of the method include swabbing a given area with a moistened cotton or calcium alginate swab. The area to be examined can be defined through the use of templates that have predefined openings (usually 1 cm^2). The template should be sterilized prior to use. After the defined area is swabbed, the swab is returned to a test tube containing a suitable diluent and agitated to dislodge the bacteria. To facilitate the process, calcium alginate swabs can be used and dissolved with the addition of sodium hexametaphosphate to the diluent. The inoculated diluent can be used in both conventional and rapid microbiological assays. The SPC is generally used to determine the number of bacteria in the inoculated diluent. The swab test method is ideal for testing rough, uneven surfaces.

Other methods for examining food contact surfaces include direct contact of agar with a test surface and the use of sticky film or tape. In the *replicate organism direct area contact* method, specially designed petri plates are used so that the medium, when poured, produces a raised agar surface. The agar can then make direct contact with the test surface. Once the plate has been used to test the desired surface area, the lid is replaced and the plate is incubated. Selective media can be used to reduce the growth of bacteria that may spread across the surface of the plate. This method is not suitable for heavily contaminated or rough surfaces.

Sterile sticky tape is commercially available for sampling of surfaces from equipment to beef carcasses. The tape and dispenser are similar in appearance to a standard roll of adhesive tape used for sealing packages. The tape is withdrawn from the roll and held over a premeasured (in square centimeters) area located on the tape dispenser. The sticky surface of the

tape is then pressed against the area to be tested, and then the tape is pressed onto the surface of an agar plate. The method is comparable to the swab test except on wooden surfaces.

A sponge system has been employed for examination of animal carcasses and food contact surfaces. Similar to other contact systems, a moistened sponge is wiped across the test area and then placed into a container (test tube, sterile plastic bag, etc.) containing an appropriate diluent. The container is agitated to dislodge the bacteria from the sponge, and an aliquot of the inoculated diluent is used in the SPC or other microbiological assay.

The methods outlined in this chapter are all commonly referred to as conventional methods. To facilitate the detection and enumeration of microorganisms from food or food-processing facilities, selective medium is commonly used and is commercially available. Selective medium may contain antibiotics or other inhibitors to permit the growth of select microorganisms. Some selective media contain specific substrates that are used by only certain microorganisms. A few of these media are discussed in chapter 5. Often, conventional methods are combined with rapid methods to decrease the time required for identification of microorganisms found in food.

Summary

- Samples must be collected using aseptic techniques and held in such a way as to prevent the growth of microorganisms.
- Food samples should be blended or homogenized prior to examination.
- The SPC is typically used to provide an estimate of the number of microorganisms in a sample.
- Surface test methods are essential in determining whether food contact surfaces are being maintained in a hygienic state.
- Rapid methods (miniaturized tests) based on conventional biochemical methods are now available, reducing the time required to identify a microorganism.

Suggested reading

Corry, J. E. L., B. Jarvis, S. Passmore, and A. Hedges. 2007. A critical review of measurement uncertainty in the enumeration of food micro-organisms. *Food Microbiol.* **24:**230–253.

Food and Drug Administration. 2001. *Bacteriological Analytical Manual Online.* http://vm.cfsan.fda.gov/~ebam/bam-mi.html.

Questions for critical thought

1. The DMC method has little or no value for foods with low microbial numbers. Why?
2. What is the difference between the surface and pour plate SPCs?
3. How can the buffer used for dilution of a sample influence the estimation of bacterial numbers associated with a sample?
4. The swab test and sticky-tape test are both methods for direct sampling of contact surfaces. List advantages and disadvantages for each.
5. Outline a protocol for the collection of a sample of ground beef from a 25-kg block. Ultimately, you want to perform an SPC.

6. The spiral plater is a convenient method for microbial analysis of a sample. Why?
7. Select a food and, using the *Bacteriological Analytical Manual,* available through the Food and Drug Administration website, outline the methodology for determining total bacterial numbers and the tests required for detection of a food pathogen of concern in the selected food.
8. Many infections occur in the home. Do you believe that your home is "clean" and free of pathogens or large numbers of microbes? Evaluate the microbial load in your home, using methods discussed in the chapter. Select several rooms or sites within your home. For example, swab the surface of the kitchen sink and each of the bathroom sinks and either the tub or the shower. Determine which site has the greatest microbial load.

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5

Rapid and Automated Microbial Methods

Introduction
Sample Processing
Requirements and Validation of Rapid Methods
Rapid Methods Based on Traditional Methods
Immunologically Based Methods
Molecular Methods
Potpourri of Rapid Methods
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- identify the various categories of rapid identification methods
- explain the basis of immunological, nucleic acid, and biochemical methods
- recognize when rapid methods are suitable to use
- understand the advantages and disadvantages associated with the use of rapid methods

INTRODUCTION

Microbiological analysis of food is not a "piece of cake." Foods are complex matrices of fats, carbohydrates, proteins, preservatives, and other chemicals. Foods also vary with respect to their physical natures: solid, dry, liquid, or semisolid. Collectively, these attributes can make it difficult to process a sample for microbiological analysis. Even if hurdles associated with processing of a sample are overcome, often foodborne pathogens are present at extremely low levels, further complicating the detection process. To decrease the time required to detect and identify target microbes in a food, an array of rapid microbiological methods have been developed.

Significant advances in the development of rapid methods have been made during the last decade, although rapid methods in some form have been available for more than 20 years. Initially, the medical community led these advances, but as a result of increased consumer awareness of the microbial safety of foods, food microbiologists have closed the gap. Significant advances have been made in the development of real-time sensors for the detection of specific toxins including Shiga toxin and botulinum toxin in food, based on concerns of deliberate contamination by bioterrorists. The task was made more difficult by the increased range of microorganisms now associated with food compared to 20 or 30 years ago. In the "old" days, food microbiologists tested for *Salmonella*, *Clostridium botulinum*, and *Staphylococcus aureus;* now *Listeria monocytogenes*, *Escherichia coli* O157:H7, *Campylobacter jejuni*, and *Vibrio parahaemolyticus* are also of concern. The change in the complexity of microorganisms associated with food is linked to many factors, including changes in processing and consumer preference and a global marketplace.

The basis for ensuring that safe and wholesome foods are available for the consumer is the testing of foods for pathogens and spoilage microorganisms. Conventional test methods (see chapter 4), although not lacking in sensitivity and cost-effectiveness, can be laborious and require several

days before results are known. Products that are minimally processed have an inherently short shelf life, which prevents the use of many conventional methods that may require several days to complete, ultimately limiting product time on the shelf. Rapid assays are based on immunological, biochemical, microbiological, molecular, and serological methods for isolation, detection, enumeration, characterization, and identification.

The times required to complete different assays vary greatly. Indeed, "rapid" may imply seconds, minutes, hours, or even days. When considering whether an assay is rapid, all steps in the assay must be included, not just the time required to complete the rapid test itself. For instance, a rapid assay may require only 15 min to complete, but the food sample may need to be mixed with a general growth medium (*preenrichment*) and incubated for several hours, with an aliquot then transferred to a selective enrichment medium and incubated before an aliquot is finally used in the rapid 15-min assay. Most rapid methods are adaptable to a wide range of foods, since they rely on culture methods to recover injured cells and increase the number of target cells.

The fast pace at which rapid methods are being developed precludes discussion of all available methods. In this chapter, the breadth of rapid methods available and the scientific principles of the methods used for detection of specific microbes in foods are examined. Existing rapid methods and those that show promise for the future but that are not yet commercially available are discussed.

SAMPLE PROCESSING

As mentioned above, food samples generally require some type of processing before a rapid method can be employed. The methods mentioned in this chapter are generally accepted for a broad range of foods, including milk, yogurt, apple juice, ground beef, and tomatoes. The composition of the food—the presence of fats, carbohydrates, and protein—can have a dramatic impact on successfully determining whether a target organism is present. The way that the food is processed, such as cooling, drying, heating, and addition of chemical additives, can also influence the ability to detect the presence of a target microorganism. In general, foodborne pathogens are present in low numbers in foods. Therefore, most rapid methods include enrichment, preenrichment, and/or selective enrichment to facilitate detection of the target pathogen.

Rapid tests have detection limits ranging from 10^2 to 10^5 colony-forming units (CFU)/g or 10^2 to 10^5 CFU/ml; therefore, enrichment increases the probability of detecting low levels of a target organism. The duration of the enrichment incubation varies considerably, based on the manufacturer's directions. Often, the period is relatively short, <12 h, but it can be as long as 24 h. The rapid-test manufacturer may specify the medium to be used for enrichment, or a standard growth medium may be appropriate. Prior to selecting a rapid method, the assay protocol should be read carefully to determine whether an enrichment step is required, since this may dramatically increase the time needed to complete the assay.

REQUIREMENTS AND VALIDATION OF RAPID METHODS

Considerable effort goes into the development of rapid methods. Before rapid methods can be discussed, an understanding of how the accuracy and validity of a test is determined is necessary. To develop a reliable

method, basic information about the target microbe is required. This facilitates the identification of unique features that can then be used for rapid detection. The cornerstone of any method is its *accuracy*. This consists of the *sensitivity*, or the ability of the assay to detect low numbers of the target microorganism, and the *specificity*, or its capability to differentiate the microbe of interest from other microorganisms. A *false-negative* result occurs when an assay fails to detect a target pathogen that is present upon culture. Similarly, a *false-positive* result occurs when the test system gives a positive result for a culture-negative sample. R. R. Beumer et al. give standard equations to calculate sensitivity and specificity rates: sensitivity = $(p \times 100)/(p$ + number of false negatives) and specificity = $(n \times 100)/(n$ + number of false positives), where p is the number of true positives and n is the number of true negatives. Without question, the assay must be as sensitive as possible and the detection limit must be as low as possible. For microorganisms that cause disease, the criterion is <1 cell per 25 g of food.

The intent in developing a rapid assay is to reduce the time required to obtain an accurate result. Conventional testing may require several days, whereas an ideal rapid test should provide results within an 8-h time frame, or a typical workday. In an ideal world, a rapid test would provide results nearly instantaneously, but this has yet to be realized. At the end of this chapter, new technologies that may be capable of delivering nearly instantaneous results are discussed, but for now, we must operate within the constraints of the currently available systems. These have limits of detection of 10^2 to 10^5 CFU/g or ml and require enrichment for 6 to 24 h. Another factor in the selection of a rapid method is the number of samples that can be processed and analyzed at one time. Many tests have adopted the use of *microtiter plates* (small rectangular plates having 96 wells) or similar multiwell formats that can perform many tests for a single sample or that can be used to screen multiple samples simultaneously.

Although the method selected may be rapid and able to be completed within a time frame acceptable for a given operation, other factors, including the speed of sample processing, accuracy, and cost, must be considered. With respect to speed, single diagnostic tests may be acceptable if few tests are to be performed. If large numbers of samples must be processed, a high-throughput system, even though more costly, would be the best choice. The cost of the system selected must also be appropriate for the company using it. Many factors can influence cost, including the training of personnel, the purchase of specialized equipment to conduct the assay, service and maintenance contracts, disposable supplies, and reagents.

A significant factor to consider is whether specialized skills are required for laboratory personnel to perform the assay. Ideally, the assay should be technically easy to perform, the equipment should be easy to operate, and the results should be easy to interpret. If you have not guessed by now, all aspects of the assay should be easy. Of course, the assay should be suitable for the food matrix that is being tested. In many cases, food constituents interfere with performance of the assay. Natural microflora of a sample and other debris may also interfere with the accuracy of the test. Finally, the method should be deemed acceptable by industry or government agencies.

There are several organizations, including the International Standards Organization, the International Dairy Federation, and AOAC International (formerly the Association of Official Analytical Chemists), that validate the effectiveness of testing methods for foods. The Food and Drug Administration (FDA) and U.S. Department of Agriculture, which do not validate

methods, have manuals that outline standard methods used by each organization. In some instances, rapid methods are incorporated into the protocols; however, this does not imply an official endorsement or approval of those tests. The AOAC publishes the *FDA Bacteriological Analytical Manual,* and the Food Safety Inspection Service of the U.S. Department of Agriculture publishes the *Microbiology Laboratory Guidebook.*

AOAC International is the most widely recognized and used service that provides third-party performance testing for manufacturers of test kits. There are two programs, the collaborative study program and the peer-verified program, used by AOAC to validate microbiological and chemical assays designed for the testing of foods. The collaborative study program is more rigorous. Methods that pass examination are reviewed by the AOAC Official Methods Board for approval as a first-action method. After 3 years of use by the scientific community, the methods are eligible for voting for final-action status. Rapid-test kit manufacturers generally list on the product or in product information whether the test has undergone AOAC validation, i.e., AOAC first action or AOAC final action.

RAPID METHODS BASED ON TRADITIONAL METHODS

Traditionally, bacteria were differentiated and identified based on their biochemical profiles. To accomplish this, bacteria needed to be separated from a food matrix. This requires a number of labor-intensive steps to obtain a culture that can then be plated onto or inoculated into various differential media to determine a specific bacterial response. To facilitate the isolation of bacteria from food samples, the samples are mixed with diluent in sterile plastic bags and massaged using a stomacher. Previously, individual sterile blender cups were used, which meant that each cup required cleaning and sterilization after use. Additional dilution of samples can now be done using automated diluters. To further facilitate the process, automated plating systems have been developed that eliminate the need to make dilutions and that use a robotic arm to distribute liquid sample onto a rotating plate. This allows a >1,000-fold dilution range to be counted on a single plate. The process of counting colonies on a plate is time-consuming and open to human error, particularly if a large number of plates must be counted. Colony-counting systems use scanners and specialized software. An added advantage of these systems is that images of plates can be stored for future viewing and analysis.

A variety of products that replace the standard agar plate are available (Table 5.1). One of the most widely used is the Petrifilm system (3M). This system consists of rehydratable nutrients and a gelling agent embedded in disposable cardboard. Petrifilms have been developed for enumerating yeasts, molds, total bacteria, and specific bacteria. To rehydrate the film to support microbial growth, a 1-ml aliquot of sample is dispensed onto the center of the film. The film is incubated under the appropriate conditions, and colonies are counted directly. The Petrifilm system for determining coliforms in foods is comparable to the widely used violet red bile agar and has greater sensitivity.

The *hydrophobic grid membrane filter* (HGMF) (ISO-GRID system; QA Life Science) is more complex than the Petrifilm system and requires specialized equipment to complete. The HGMF requires filtration of the sample through a filter that contains a set of 1,600 grid cells. The HGMF is then

Table 5.1 Representative manual and automated miniaturized biochemical tests

Organism	System	Manufacturer
Enterobacteriaceae	API	BioMerieux
	MICRO-ID	Remel
	BBL crystal	Becton Dickinson
	Vitek	BioMerieux
Coliforms	Bactometer	BioMerieux
Gram-negative bacteria	Vitek	BioMerieux
	Microlog	Biolog
	Omnilog ID	Biolog
Gram-positive bacteria	Vitek	BioMerieux
	Microlog	Biolog
	Omnilog ID	Biolog
Anaerobes	IDS RapID	Remel
	BBL crystal	Becton Dickinson
Listeria	MICRO-ID	Remel
	API	BioMerieux
Streptococci	RapID STR	Remel

placed onto an appropriate medium and incubated, and the colonies are counted. Grids have been designed for *Salmonella*, *E. coli* O157:H7, coliform-*E. coli*, yeast, and mold counts and aerobic plate counts.

Similar to the Petrifilm system and the HGMF system, the SimPlate system is based on the use of specialized dehydrated medium that is reconstituted with sterile water prior to use. The system is a modification of the most-probable-number method. The petri dish device contains numerous wells, and the positive wells are counted and compared to a most-probable-number chart. The system is well suited for testing a variety of products from fresh vegetables to ice cream.

The methods discussed thus far are basically kits and must be used as such. However, in recent years, a staggering number of media have been developed for the detection, enumeration, and identification of specific bacteria (Table 5.2), and media are now available for the detection and enumeration of yeasts and molds. For instance, media marketed by CHROMagar (Paris, France) contain specific chemicals that permit the differentiation and counting of specific bacteria. On CHROMagar ECC, coliform colonies appear pink and *E. coli* colonies appear blue. For the detection of *E. coli* O157:H7, the list of media just keeps growing. These media include MacConkey sorbitol agar, phenol red sorbitol agar containing 4-methylumbelliferyl-β-D-glucuronide, Levine eosin-methylene blue agar, Fluorcult *E. coli* O157:H7 agar (EM Sciences), BCM O157:H7(+) (Biosynth Biochemica and Synthetica), and MacConkey sorbitol agar containing either 5-bromo-4-chloro-3-indoxyl-β-D-glucuronic acid or 4-methylumbelliferyl-β-D-glucuronide for the detection of β-D-glucuronidase activity.

IMMUNOLOGICALLY BASED METHODS

The most widely used rapid methods are based on immunoassay technology. The systems are acceptable for the screening and identification of specific bacteria associated with a variety of foods. The antibodies used in these systems may detect either many cellular targets (polyclonal antibodies) or

Table 5.2 Representative specialty medium rapid kits for detection of bacteria associated with food

Organism	Trade name	Assay format[a]	Manufacturer
E. coli	Petrifilm	Medium film	3M
	Redigel (Colichrome)	Medium	3M
	Coligel	MUG–X-Gal	Charm Science
	E-Colite	MUG–X-Gal	Charm Science
	Pathogel	MUG–X-Gal	Charm Science
Coliforms	Petrifilm	Medium film	3M
	Redigel (Colichrome)	Medium	3M
	Redigel (Violet Red Bile)	Medium	3M
	Coligel	MUG–X-Gal	Charm Science
	E-Colite	MUG–X-Gal	Charm Science
	Pathogel	MUG–X-Gal	Charm Science
Salmonella	SM ID	Medium	BioMerieux
Enterobacteriaceae	Petrifilm	Medium film	3M
	Pathogel	MUG–X-Gal	Charm Science
Staphylococci	Microdase Disk	Medium	Remel
	Bactistaph kit	Medium	Remel
	Novobiocin Disk	Medium	Remel
S. aureus	Petrifilm	Medium film	3M

[a]MUG, 4-methylumbelliferyl-β-D-glucuronide; X-Gal, 5-bromo-4-chloro-3-indolyl-β-D-galactopyranoside.

specific targets (monoclonal antibodies). The use of monoclonal antibodies should eliminate some problems associated with cross-reaction with bacteria other than the target organism. The systems range from simply mixing antibody-coated beads with a test sample to a multistep procedure that includes addition of an enzyme-labeled antibody to a well, followed by incubation, washing, and addition of a chromogen that is acted on and that can be detected either visually or using a spectrophotometer. The typical limits of detection for these systems range from 10^3 to 10^5 CFU/ml. Table 5.3 contains a list of some of the commercially available methods.

Coating of latex particles (small beads) or magnetic beads permits screening of samples for a specific bacterium or separation of a target bacterium from a sample, respectively. For the *latex agglutination test,* latex particles are coated with specific antibodies that upon contact with specific antigens of the target organism produce a visible clumping, or agglutination, reaction. The method is not that sensitive, since $>10^6$ CFU are required for a visible reaction. However, the method is rapid and provides a convenient means to screen for a target microbe. *Immunomagnetic separation* differs from latex agglutination in that the method is designed to separate the target organism from a sample. Basically, an aliquot of magnetic beads coated with a specific antibody is added to a sample (usually broth culture), the sample is incubated to facilitate binding of the target cells to the antibody, and then the complex is isolated from the sample by using a magnet. The beads can then be used to inoculate broth, be plated onto selective agar medium, or be used directly in polymerase chain reaction (PCR) assays or in enzyme-linked immunosorbent assays (ELISAs). Immunomagnetic separation systems are available for separation of *E. coli* O157:H7, *Listeria,* and *Salmonella* from a range of samples, including feces, raw milk, and ice cream.

The *Salmonella* 1-2 Test (BioControl) is a popular and widely accepted method based on *immunodiffusion* (the movement of proteins through agar)

Table 5.3 Representative list of antibody-based assays

Organism	Trade name	Format[a]	Manufacturer
EHEC O157[b]	Reveal	Ab-ppt	Neogen
	Alert	ELISA	Neogen
	VIDAS	ELFA	BioMerieux
	Assurance EHEC EIA	ELISA	BioControl
	VIP	Ab-ppt	BioControl
	TECRA	ELISA	TECRA
E. coli O157:H7	DETEX	ELISA	Molecular Circuitry Inc.
Salmonella spp.	Reveal	Ab-ppt	Neogen
	Alert	ELISA	Neogen
	VIDAS	ELFA	BioMerieux
	1-2 Test	Diffusion	BioControl
	Assurance *Salmonella* EIA	ELISA	BioControl
	VIP	Ab-ppt	BioControl
	Assurance Gold *Salmonella* EIA	ELISA	BioControl
	TECRA	ELISA	TECRA
	UNIQUE	Capture EIA	TECRA
	DETEX	ELISA	Molecular Circuitry Inc.
Listeria	Reveal	ELISA	Neogen
	VIP	Ab-ppt	BioControl
	Assurance	ELISA	BioControl
	TECRA	ELISA	TECRA
	UNIQUE	Capture EIA	TECRA
	DETEX	ELISA	Molecular Circuitry Inc.
L. monocytogenes	DETEX	ELISA	Molecular Circuitry Inc.
Campylobacter	Alert	ELISA	Neogen
	Uni-Lite XCEL	ELISA	Neogen
	VIDAS	ELFA	BioMerieux
	VIP	Ab-ppt	Biocontrol
	Campylobacter	ELISA	Biocontrol
	DETEX	ELISA	Molecular Circuitry Inc.
	TECRA	ELISA	TECRA
Bacillus thuringiensis	Agri-Screen	S-ELISA	Neogen
Pseudomonas	TECRA	ELISA	TECRA
S. aureus	TECRA	ELISA	TECRA

[a]EIA, enzyme immunoassay; S-ELISA, sandwich ELISA; Ab-ppt, antibody precipitation; ELFA, enzyme-linked fluorescent assay.
[b]EHEC, enterohemorrhagic *E. coli*.

and the formation of an antibody-antigen complex that forms a visible line of precipitation in a chamber. The antibody is specific for *Salmonella* flagellar antigen; therefore, nonmotile *Salmonella* organisms are not detected. Many immunoassays are self-contained units that are based on the migration of a sample along a chromatographic strip. Systems using this or similar technology include the VIP tests (BioControl), Reveal tests (Neogen Corp.), SafePath (SafePath Laboratories LLC), and PATH-STIK (Lumac). Tests are available for *Salmonella, E. coli* O157:H7, *Listeria, Campylobacter,* and a variety of toxins. The systems have been used to screen fecal, beef, milk, and environmental samples. The specificity and sensitivity vary with the sample type and assay used.

There are several ELISA-based methods, such as TECRA for detection of *Salmonella, Campylobacter,* and *E. coli* O157:H7; Listeria-TEK (Organon Teknika); and the Assurance EIA system (BioControl). ELISA-based systems

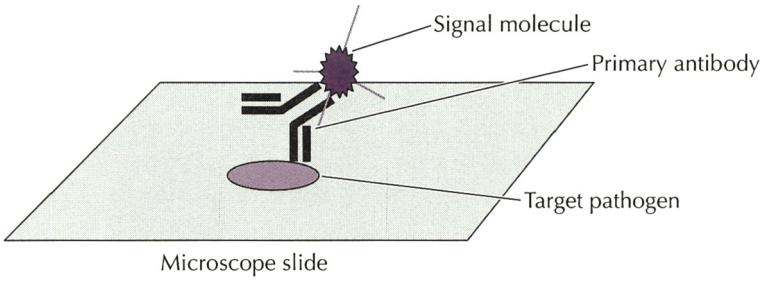

Figure 5.1 Detection of a target foodborne microbe by direct immunoassay.

are also available for the detection and identification of fungi at the genus or species level. Mycotoxins can also be detected by commercially available ELISA systems. The basis for detection of a target microbe or antigen is depicted in Fig. 5.1.

MOLECULAR METHODS

There has been an explosion in the past 10 years in the introduction of nucleic acid (RNA and DNA)-based assays for the differentiation and identification of foodborne pathogens (Table 5.4). DNA methods include, but are not limited to, PCR, pulsed-field gel electrophoresis, ribotyping, plasmid typing, randomly amplified polymorphic DNA, and restriction fragment length polymorphism. Some of these methods have been automated, and kits are available to facilitate the recovery of pure DNA. Perhaps the most widely recognized and used method is PCR.

PCR is a basic three-step process that is based on the amplification of a specific segment of cellular DNA (Fig. 5.2). In the first step, double-stranded DNA, also known as *template DNA,* is denatured into single strands, followed by annealing of *primers* (short segments of DNA complementary to a specific region on the template DNA strand) to the template DNA. The annealing temperature is critical, since it determines the *stringency* of the reaction, or how specific the attachment will be. The final step is extension of the primers, which is accomplished using a thermostable DNA polymerase. Within 2 h, PCR can amplify a single copy of DNA a millionfold. The PCR products can be visualized as a band on an agarose gel stained with ethidium bromide.

Table 5.4 Representative nucleic acid-based rapid methods

Organism	System	Format	Manufacturer
Salmonella	PCR-Light Cycler	PCR-ELISA	Biotecon Diagnostics
	Gene-Trak	Probe	Gene-Trak
	BAX	PCR	Qualicon
E. coli O157:H7	BAX	PCR	Qualicon
E. coli	Gene-Trak	Probe	Gene-Trak
L. monocytogenes	PCR-Light Cycler	PCR-ELISA	Biotecon Diagnostics
	Gene-Trak	Probe	Gene-Trak
Listeria spp.	BAX	PCR	Qualicon
Campylobacter	Gene-Trak	Probe	Gene-Trak
S. aureus	Gene-Trak	Probe	Gene-Trak

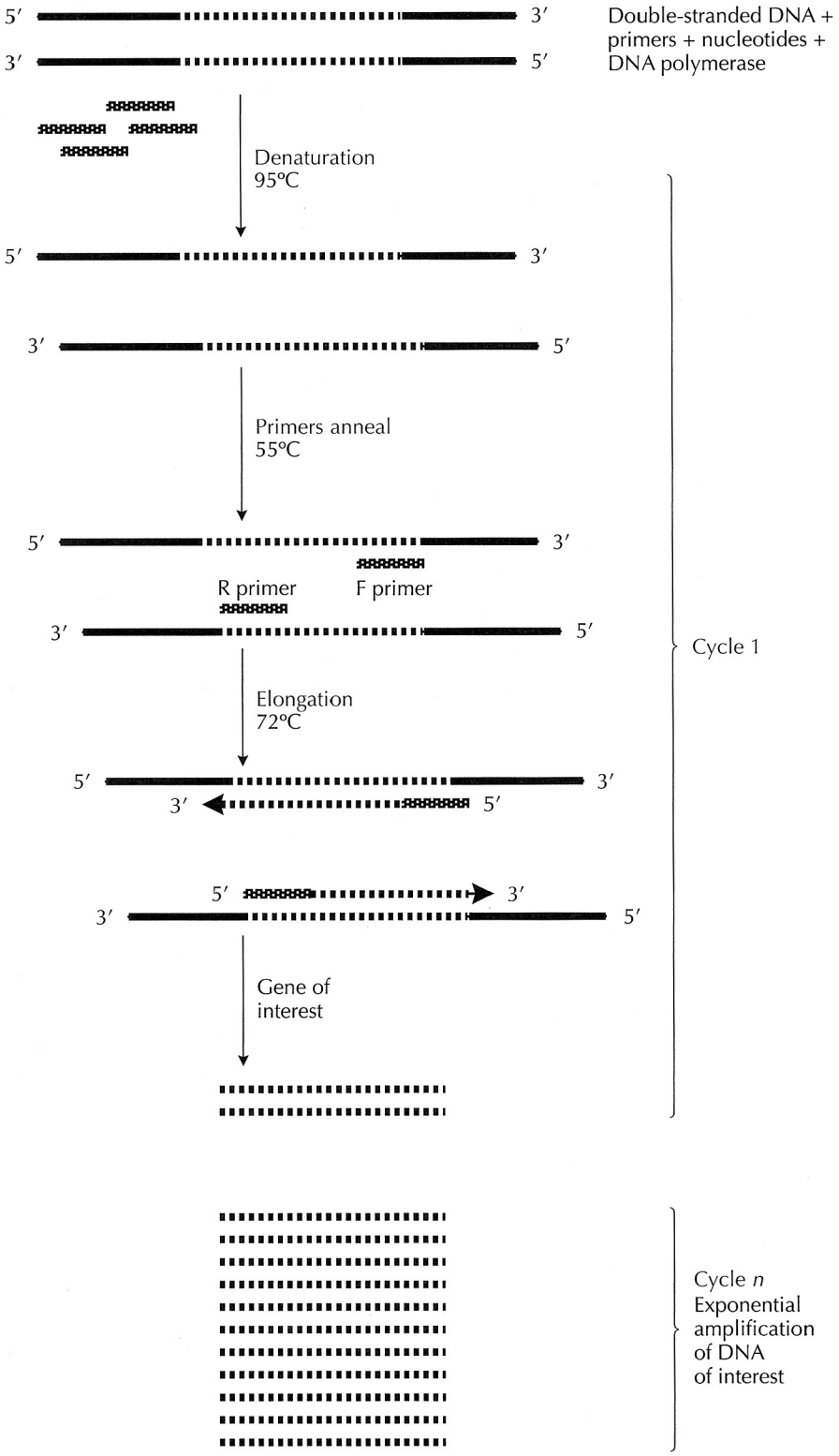

Figure 5.2 Representation of PCR cycle. Initially double-stranded template DNA (from a microbe) is denatured to form two single-stranded pieces of DNA. Primers designed to amplify a specific region on the template DNA are allowed to anneal to the single-stranded DNA. Elongation and extension of the primer make a complementary copy of the DNA template. These three steps make up a single cycle. The cycle is repeated a specified number of times.

Currently, there are only a few commercially available PCR kits for the identification of foodborne pathogens. Although the test is extremely sensitive, equipment required to run the test is expensive, technical expertise is required, and PCR assays are affected by complex components of food. High levels of fats, protein, humic substances, and iron can interfere with the reaction. To overcome these problems, an enrichment step is often included in the assay, effectively diluting inhibitors and increasing cell numbers. The BAX system (Qualicon) is a commercially available PCR assay for the detection of *E. coli* O157:H7, *Salmonella*, and *Listeria*. The kit is unique in that all reagents required for the PCR (primers, enzyme, and deoxyribonucleosides) are *lyophilized* (freeze-dried) and included in a reaction tube. This eliminates potential problems with the introduction of contaminants or the use of incorrect amounts of a reagent.

Authors' note

DNA-based methods are extremely sensitive. Caution must be exercised with use of some of these tests, since the DNA detected may be associated with dead bacteria that would present no health hazard.

There are a number of alternatives to conventional PCR, including reverse transcription-PCR (RT-PCR), real-time PCR, and the nucleic acid sequence-based amplification system. Commercial kits are available for nucleic acid sequence-based amplification (NucliSens Basic Kit; Organon Teknika) and for real-time PCR (Biotecon Diagnostics, Hamilton Square, NJ). For assays for which kits are not available, the methods are well defined. Reagent packages that have been optimized for the isolation of RNA and DNA are available. RT-PCR uses the enzyme reverse transcriptase to convert RNA into DNA, which then serves as the template in PCR. RT-PCR is a useful method, since it is based on RNA, which has a very short half-life. When performed properly, RT-PCR detects only live cells, whereas in PCR, DNA from live and dead cells can potentially be detected.

The commercially available Riboprinter from Qualicon makes it easy to conduct ribotyping for the characterization and fingerprinting of strains and for epidemiological investigation. Results using the automated method can be obtained within 16 h, and riboprint patterns can be stored to create a unique database. Conducting ribotyping without the aid of an automated system is time-consuming and requires a high level of skill. Ribotyping has excellent reproducibility and discriminatory power.

POTPOURRI OF RAPID METHODS

Only a few of the many rapid methods that exist have been discussed. There are a few others that deserve mention, including luminescence-based assays, flow cytometry, and impedance. Of particular importance to the food industry are the bioluminescence-based methods, which can provide a rapid estimate of total microbial numbers. The presence of specific types of bacteria cannot be determined, but the production of light can be correlated to the number of microbes. The method makes use of the ubiquitous presence of adenosine triphosphate (ATP) in all living cells and the ability for it to react with the luciferase enzyme complex found in fireflies. The Uni-lite system (Biotrace, Inc., Plainsboro, NJ), the Lighting bioluminescence system (IDEXX Laboratories, Inc., Westbrook, ME), and the Lumac Hygiene Monitoring kit (Integrated BioSolutions, Inc., Monmouth Junction, NJ) all make use of this technology. For example, the Uni-lite system consists of a transportable luminometer reading unit and swabs prepackaged in tubes containing reagents. The surface to be tested is swabbed, the reaction is activated by placing the swab into an enzyme solution, and the swab and tube are inserted into a chamber in the luminometer to obtain a reading. Results are obtained in <1 min.

The *direct epifluorescence filter technique* is another technique that is designed to illuminate bacteria. The method is simple and involves membrane filtration of a food sample to collect cells associated with the sample. The membrane is then processed by using fluorescent antibodies to stain target bacteria. Bacteria are visualized using an epifluorescence microscope. Problems can occur if food material trapped on the filter also fluoresces (i.e., it is difficult to differentiate bacteria from food).

Other methods that are available or in the developmental stages are based on nanotechnology, impedance, flow cytometry, and bacterial ice nucleation. The Bactometer (BioMerieux) and the Malthus System V (Malthus Diagnostics) are commercially available automated systems based on impedance. As microbes grow, they metabolize substrates with low conductivity into products with high conductivity and therefore decrease impedance. Impedance systems are good for processing a large number of samples and can detect low levels of a given microbe.

The future development of rapid methods will likely exploit the use of nanotechnology. Nanotechnology will permit the development of real-time sensors capable of detecting several different target microbes of concern. Although many different formats are available, silicon chips can be imprinted with specific DNA sequences for the detection and identification of target organisms. Chips could be imprinted with antibodies, RNA, or a host of other agents to target microorganisms. Microbial sensors are particularly applicable in fluid systems with limited organic substances. Presently, problems exist with sensor efficacy in food systems containing fats and proteins that coat the sensor and render it inoperable. These problems will be overcome as technology advances.

Concerns related to bioterrorism and the safety of the food supply have already inspired a massive effort for the development of rapid methods capable of detecting not only typical foodborne pathogens but also microbes of concern for human health that may be intentionally used to contaminate food. Handheld PCR units have been developed for the detection of *Bacillus anthracis* and *Yersinia pestis*. Research is ongoing to modify the units for detection of *Salmonella, E. coli* O157:H7, and *L. monocytogenes*. These units have potential applications in the screening of food at ports of entry and border crossings and on farms.

Summary

- Specialized media that target specific biochemical reactions of the target microbe are available.
- The most widely used format for rapid methods is immunologically based assays.
- An enrichment step is required for most assays to increase the number of bacteria to a detectable level.
- Most assays require 10^2 to 10^5 CFU per g or ml.
- PCR-based assays are extremely sensitive but are subject to interference from the presence of inhibitors in food.
- Future rapid assays will likely be based on nanotechnology.

Suggested reading

Beumer, R. R., E. Brinkman, and F. M. Rombouts. 1991. Enzyme-linked immunoassays for the detection of *Salmonella* spp.: a comparison with other methods. *Int. J. Food Microbiol.* **12:**363–374.

Feng, P. 1995. Rapid methods for detecting foodborne pathogens, p. App.1.01–App.1.16. *In* R. I. Merker (ed.), *FDA Bacteriological Analytical Manual*, ed. 8A. AOAC International, Gaithersburg, MD.

Food and Drug Administration. 2001. *Bacteriological Analytical Manual Online.* http://vm.cfsan.fda.gov/~ebam/bam-mi.html.

Matthews, K. R. 2003. Rapid methods for microbial detection in minimally processed foods, p. 151–163. *In* J. S. Novak, G. M. Saper, and V. K. Juneja (ed.), *Microbial Safety of Minimally Processed Foods.* CRC Press, Boca Raton, FL.

Questions for critical thought

1. The FDA has requested proposals for the development of PCR-based methods for the purpose of providing rapid confirmatory identification of microbial pathogens that are not usually associated with food and foodborne illness. Develop an outline for a proposal and include the pathogens that you would focus on and the type of PCR method you would use. Justify each of your responses.
2. Select a rapid assay listed in one of the tables. Gather literature from the company and find peer-reviewed journal articles related to the assay's efficacy. Based on your review of the literature, write a one-page abstract of the assay. Pretend the abstract is for a rapid-methods newsletter.
3. Using a table format, compare PCR, RT-PCR, and real-time PCR. Provide a brief narrative concerning when each method would be most appropriate to use.
4. Food constituents often interfere with the function of a rapid assay. For example, fat in a sample may coat a biosensor, inhibiting the binding of bacteria present in a sample to the sensor. Using immunologically based assays as an example, indicate from a scientific viewpoint why a given food constituent would interfere with the assay. Now, provide a protocol that would eliminate or prevent such interference.
5. You are working for a small development group that is interested in marketing a new rapid method for detection of foodborne pathogens. The only problem is, they do not have a new rapid method and have asked you to develop one. What type of method would you develop? It must be novel.
6. Many of the rapid bioluminescence assays are based on the detection of ATP. Since ATP is found in all living cells, what is a potential problem(s) associated with this assay in determining whether a surface is contaminated with bacteria?

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6

Indicator Microorganisms and Microbiological Criteria

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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- differentiate among the various microbiological criteria
- identify national and international agencies involved in establishing microbiological criteria
- recognize how indicator organisms are used in microbiological criteria
- understand why some sampling plans are more stringent than others
- identify organisms and types of foods for which zero tolerance has been established
- identify and list steps required to manage microbiological hazards in foods

INTRODUCTION

The Purpose of Microbiological Criteria

Microorganisms cause foodborne illness and spoil food. Trained food microbiologists could probably isolate microbes from most raw and finished food products. Does that mean that the product is not safe to eat or will spoil rapidly? Not necessarily. Indeed, it is okay to have certain bacteria, even pathogenic bacteria, in food, depending on the product. Microbiological criteria ensure that a product has been produced under sanitary conditions and is microbiologically safe to consume.

Microbiological criteria are used to distinguish between an acceptable and an unacceptable product or between acceptable and unacceptable food-processing practices. The numbers and types of microorganisms associated with a food may be used to judge its microbiological safety and quality. Safety is determined by the absence, presence, or level of pathogenic microorganisms or their toxins and their expected control or destruction. The level of spoilage microorganisms reflects the microbiological quality, or wholesomeness, of a food, as well as the effectiveness of measures used to control or destroy such microorganisms. Indicator organisms may be used to assess either the microbiological quality or safety. Specifically, microbiological criteria are used to assess (i) the safety of food, (ii) adherence to good manufacturing practices (GMPs), (iii) the keeping quality (shelf life) of perishable foods, and (iv) the suitability of a food or ingredient for a particular purpose. Appropriately applied microbiological criteria ensure the safety and quality of foods. This in turn increases consumer confidence.

The Need To Establish Microbiological Criteria

A microbiological criterion should be established only in response to a need and when it is both effective and practical. There are many considerations to be taken into account when establishing microbiological criteria. Listed

below are a few important factors for assessing the need for microbiological criteria.

- Evidence of a health hazard based on epidemiological data or a hazard analysis
- The nature of the food's normal microbial makeup and the ability of the food to support microbial growth
- The effect of processing on the microflora of the food
- The potential for microbial contamination and/or growth during processing, handling, storage, and distribution
- Spoilage potential, utility, and GMPs

Definitions

The National Research Council (NRC) of the U.S. National Academy of Sciences addressed the issue of microbiological criteria in their 1985 report entitled *An Evaluation of the Role of Microbiological Criteria for Foods and Food Ingredients*. The report indicates that a microbiological criterion should state what microorganism, group of microorganisms, or toxin produced by a microorganism is covered. The criterion should also indicate whether it can be present or present in only a limited number of samples or a given quantity of a food or food ingredient. In addition, a microbiological criterion should include the following information:

- Statement describing the identity of the food
- Statement identifying the contaminant
- Analytical method to be used for the detection, enumeration, or quantification of each contaminant
- Sampling plan (discussed below)
- Microbiological limits considered appropriate to the food and commensurate with the sampling plan

Criteria may be either mandatory or advisory. A *mandatory criterion* is a criterion that may not be exceeded. Food that does not meet the specified limit is required to be rejected, destroyed, reprocessed, or diverted. An *advisory criterion* permits acceptability judgments to be made. It serves as an alert to deficiencies in processing, distribution, storage, or marketing. For application purposes, there are three categories of criteria: standards, guidelines, and specifications. The following definitions were recommended by the NRC Subcommittee on Microbiological Criteria for Foods and Food Ingredients.

Standard: A microbiological criterion that is part of a law, ordinance, or administrative regulation. A standard is a mandatory criterion. Failure to comply constitutes a violation of the law, ordinance, or regulation and will be subject to the enforcement policy of the regulatory agency having jurisdiction.

Guideline: A microbiological criterion often used by the food industry or regulatory agency to monitor a manufacturing process. Guidelines function as alert mechanisms to signal whether microbiological conditions prevailing at critical control points or in the finished product are within the normal range. Hence, they are used to assess processing efficiency at critical control points and conformity with good manufacturing practices. A microbiological guideline is advisory.

Specifications: Microbiological criteria that are used as a purchase requirement whereby conformance becomes a condition of purchase between buyer and vendor of a food ingredient. A microbiological specification may be advisory or mandatory.

Who Establishes Microbiological Criteria?

Different scientific organizations are involved in developing general principles for the application of microbiological criteria. The scientific organizations which have most influenced the U.S. food industry include the Joint Food and Agricultural Organization and World Health Organization Codex Alimentarius International Food Standards Program, The International Commission on Microbiological Specifications of Foods (ICMSF), the U.S. National Academy of Sciences, and the U.S. National Advisory Committee on Microbiological Criteria for Foods. The Codex Alimentarius Program first formulated *General Principles for the Establishment and Application of Microbiological Criteria* in 1981, and it has since been revised. In 1984, the NRC Subcommittee on Microbiological Criteria for Foods and Food Ingredients formulated general principles for the application of microbiological criteria to food and food ingredients as requested by four U.S. regulatory agencies. The *Codex Alimentarius* contains standards for all principal foods in the form (i.e., processed, semiprocessed, or raw) in which they are delivered to the consumer. Fresh perishable commodities not traded internationally are excluded from these standards. The World Trade Organization provides a framework for ensuring fair trade and harmonizing standards and import requirements for food traded through the Agreements on Sanitary and Phytosanitary Measures and Technical Barriers to Trade. Countries are required to base their standards on science, to base programs on risk analysis methodologies, and to develop ways of achieving equivalence among the different methods of inspection, analysis, and certification used by various trading countries. The World Trade Organization recommends the use of standards, guidelines, and recommendations developed by the Codex Alimentarius Program to make all standards similar.

Authors' note

The Codex Alimentarius *is a set of food standards and guidelines developed to protect the health of consumers and to ensure fair trade practices in the food trade.*

SAMPLING PLANS

A sampling plan includes both the sampling procedure and the decision criteria. To examine a food for the presence of microorganisms, a representative sample is examined by defined methods. A *lot* is a quantity of product produced, handled, and stored within a limited time under uniform conditions. Since it is impractical to examine the entire lot, statistical concepts of population probability and sampling are used to determine the number and size of sample units required from the lot and to provide conclusions drawn from the analytical results. The sampling plan is designed so that inferior lots will be rejected.

In a simplified example of a sampling plan, let us assume that 10 samples were taken and analyzed for the presence of a particular microorganism. Based on the decision criterion, only a certain number of the sample units could be positive for the presence of that microorganism for the lot to be considered acceptable. If in the criterion the maximum allowable number of positive units had been set at 2 ($c = 2$), then a positive result for >2 of the 10 sample units ($n = 10$) would result in rejection of the lot. Ideally, the decision criterion is set to accept lots that are of the desired quality and to reject lots that are not. However, since only part of the lot is examined, there

is always the risk that an acceptable lot will be rejected or that an unacceptable lot will be accepted. The more samples examined, or the larger the size of n, the lower the risk of making an incorrect decision about the lot quality. However, as n increases, sampling becomes more time-consuming and costly. Generally, a compromise is made between the size of n and the level of risk that is acceptable.

Types of Sampling Plans

Sampling plans are divided into two main categories: variables and attributes. A variables plan depends on the frequency distribution of organisms in the food. For correct application of a variables plan, the organisms must have a lognormal distribution (i.e., counts transformed to logarithms are normally distributed). When the food is from a common source and it is produced and/or processed under uniform conditions, a lognormal distribution of the organisms present is assumed. Attributes sampling is the preferred plan when microorganisms are not homogeneously distributed throughout the food or when the target microorganism is present at low levels. This is often the case with pathogenic microorganisms. Attributes plans are also widely used to determine the acceptance or rejection of products at ports or other points of entry, since there is little or no knowledge of how the food was processed and past performance records are not available. Attributes sampling plans may also be used to monitor performance relative to accepted GMPs. Attributes sampling, however, is not appropriate when there is no defined lot or when random sampling is not possible, as might occur when monitoring cleaning practices. Because attributes sampling plans are used more frequently than variables sampling plans, this text does not cover variables sampling in detail.

Attributes Sampling Plans

Two-class plans. A two-class attributes sampling plan assigns the concentration of microorganisms of the sample units tested to a particular attribute class, depending on whether the microbiological counts are above or below some preset concentration, represented by the letter m. The decision criterion is based on (i) the number of sample units tested, n, and (ii) the maximum allowable number of sample units yielding unsatisfactory tests results, c. For example, when n is 5 and c is 2 in a two-class sampling plan designed to make a presence-absence decision about the lot (i.e., $m = 0$), the lot is rejected if more than two of the five sample units tested are positive. As n increases for the set number c, the stringency of the sampling plan also increases. Conversely, for a set sample size n, as c increases, the stringency of the sampling plan decreases, allowing a higher probability of accepting, P_a, food lots of a given quality. Two-class plans are applied most often in qualitative (semiquantitative) pathogen testing in which the results are expressed as the presence or absence of the specific pathogen per sample weight analyzed.

Three-class plans. Three-class sampling plans use the concentrations of microorganisms in the sample units to determine levels of quality and/or safety. Counts above a preset concentration M for any of the n sample units tested are considered unacceptable, and the lot is rejected. The level of the test organism acceptable in the food is represented by m. This concentration in a three-class attribute plan separates acceptable lots (i.e., those with counts less than m) from marginally acceptable lots (i.e., those with counts greater than m but not exceeding M). Counts above m and up to and includ-

ing M are not desirable, but the lot can be accepted provided the number (n) of samples that exceed m is no greater than the preset number, c. Thus, in a three-class sampling plan, the food lot will be rejected if any one of the sample units exceeds M or if the number of sample units with contamination levels above m exceeds c. Similar to the two-class sampling plan, the stringency of the three-class sampling plan is also dependent on the two numbers represented by n and c. The larger the value of n for a given value of c, the better the food quality must be to have the same chance of passing, and vice versa. From n and c, it is then possible to find the probability of acceptance, P_a, for a food lot of a given microbiological quality.

ESTABLISHING LIMITS

Microbiological limits, as defined in a criterion, represent the level above which action is required. Levels should be realistic and should be determined based on knowledge of the raw materials and the effects of processing, product handling, storage, and the end use of the product. Limits should also take into account the likelihood of uneven distribution of microorganisms in the food, the inherent variability of the analytical procedure, the risk associated with the organisms, and the conditions under which the food is handled and consumed. Microbiological limits should include the sample weight to be analyzed, the method reference, and the confidence limits of the referenced method where applicable.

The shelf life of a perishable product is often determined by the number of microorganisms initially present. As a general rule, a food containing a large population of spoilage organisms will have a shorter shelf life than the same food containing fewer of the same spoilage organisms. However, the relationship between total counts and shelf life is not carved in stone. Some types of microorganisms have a greater impact on the sensory characteristics of a food than others.

Foods produced and stored under GMPs may be expected to have a better microbiological profile than those foods produced and stored under poor conditions. The use of poor-quality materials, improper handling, or unsanitary conditions may result in higher bacterial numbers in the finished product. However, low counts in the finished product do not necessarily mean that GMPs were adhered to. Processing steps, such as heat treatment, fermentation, freezing, or frozen storage, can reduce the counts of bacteria that have resulted from noncompliance with GMPs. Other products, such as ground beef, may normally contain high microbial counts even under the best conditions of manufacture due to the growth of psychrotrophic bacteria during refrigeration.

The use of quantitative risk assessment techniques to scientifically determine the probability of occurrence and the severity of known human exposure to foodborne hazards is rapidly gaining acceptance. The process consists of (i) hazard identification, (ii) hazard characterization, (iii) exposure assessment, and (iv) risk characterization. Though quantitative risk assessment techniques are well established for chemical agents, their application to food safety microbiology is relatively new.

INDICATORS OF MICROBIOLOGICAL QUALITY

Examination of a product for indicator organisms can provide simple, reliable, and rapid information about process failure, postprocessing contamination, contamination from the environment, and the general level of hygiene under which the food was processed and stored.

Ideal indicators of product quality or shelf life should meet the following criteria.

- They should be present and detectable in all foods whose quality is to be assessed.
- Their growth and numbers should have a direct negative correlation with product quality.
- They should be easily detected and enumerated and be clearly distinguishable from other organisms.
- They should be enumerable in a short period of time, ideally within a workday.

Indicator Microorganisms

Indicator microorganisms can be used in microbiological criteria. The presence of indicator microbes may suggest a microbial hazard. For example, the presence of generic *Escherichia coli* in a sample indicates possible fecal contamination. These criteria might be used to address existing product quality or to predict the shelf life of the food. Some examples of indicators and the products in which they are used are shown in Table 6.1. Those microorganisms listed in the table are the primary spoilage organisms of the specific products listed. Loss of quality in other products may not be limited to one organism but may involve a variety of organisms. In those types of products, it is often more practical to determine the counts of groups of microorganisms most likely to cause spoilage in each particular food.

The aerobic plate count (APC) is commonly used to determine the total number of microorganisms in a food product. By modifying the environment of incubation or the medium used, the APC can be used to preferentially screen for groups of microorganisms, such as those that are thermodurics, mesophiles, psychrophiles, thermophiles, proteolytic, or lipolytic. The APC may be a component of microbiological criteria assessing product quality when those criteria are used to (i) monitor foods for compliance with standards or guidelines set by various regulatory agencies, (ii) monitor foods for compliance with purchase specifications, and (iii) monitor adherence to GMPs.

Microbiological criteria as specifications are used to determine the usefulness of a food or food ingredient for a particular purpose. For example,

Table 6.1 Organisms highly correlated with product quality[a]

Organism	Product(s)
Acetobacter spp.	Fresh cider
Bacillus spp.	Bread dough
Byssochlamys spp.	Canned fruits
Clostridium spp.	Hard cheeses
Flat-sour spores	Canned vegetables
Lactic acid bacteria	Beers, wines
Lactococcus lactis	Raw milk (refrigerated)
Leuconostoc mesenteroides	Sugar (during refining)
Pectinatus cerevisiiphilus	Beers
"Pseudomonas putrefaciens"	Butter
Yeasts	Fruit juice concentrates
Zygosaccharomyces bailii	Mayonnaise, salad dressing

[a]Source: J. M. Jay, *Modern Food Microbiology*, 4th ed. (Springer, New York, NY, 1991).

specifications are set for thermophilic spores in sugar and spices intended for use in the canning industry. Lots of sugar that fail to meet specifications may not be suitable for use in low-acid canning but could be diverted for other uses. The APC of refrigerated perishable foods, such as milk, meat, poultry, and fish, may be used to indicate the condition of the equipment and utensils used, as well as the time-temperature profile of storage and distribution of the food.

When evaluating the results of an APC for a particular food, it is important to remember that (i) APCs measure only live cells, and therefore, a grossly spoiled product may have a low APC if the spoilage organisms have died; (ii) APCs are of little value in assessing sensory quality, since high microbial counts are generally required for sensory-quality loss; and (iii) since the biochemical activities of different bacteria vary, quality loss may also occur at low total counts. With any food, specific causes of unexpectedly high counts can be identified by examination of samples at control points and by inspection of processing plants. Interpretation of the APC of a food requires knowledge of the expected microbial population at the point where the sample is collected. If counts are higher than expected, this indicates the need to determine why there has been a violation of the criterion.

The direct microscopic count (DMC) is used to give an estimate of the numbers of both viable and nonviable cells in samples containing a large number of microorganisms (i.e., $>10^5$ colony-forming units [CFU]/ml). Considering that the DMC does not differentiate between live and dead cells (unless a fluorescent dye, such as acridine orange, is employed) and that it requires that the total cell count exceed 10^5, the DMC has limited value as a part of the microbiological criteria for quality issues. The use of the DMC as a part of microbiological criteria for foods or ingredients is restricted to a few products, such as raw non-grade A milk, dried milk, liquid and frozen eggs, and dried eggs.

Other methods commonly used to indicate the quality of different food products include the Howard mold count, yeast and mold count, heat-resistant-mold count, and thermophilic-spore count. The Howard mold count (microscopic analysis) is used to detect the inclusion of moldy material in canned fruit and tomato products, as well as to evaluate the sanitary condition of processing machinery in vegetable canneries. Yeasts and molds grow on foods when conditions for bacterial growth are less favorable. Therefore, they can be a problem in fermented dairy products, fruits, fruit beverages, and soft drinks. Yeast and mold counts are used as parts of the microbiological standards for various dairy products, such as cottage cheese and frozen cream and sugar. Heat-resistant molds (e.g., *Byssochlamys fulva* and *Aspergillus fischerianus*) that may survive the thermal processes used for fruit and fruit products may need limits in purchase specifications for ingredients such as fruit concentrates. The canning industry is concerned about thermophilic spores in ingredients because they cause defects in foods held at elevated temperatures (i.e., due to inadequate cooling and/or storage at too high temperatures). Purchase specifications and verification criteria are often used for thermophilic-spore counts in ingredients intended for use in low-acid, heat-processed canned foods.

Metabolic Products

Bacterial levels in a food product can sometimes be estimated by testing for metabolic products made by the microorganisms in the food. When a correlation is established between the presence of a metabolic product and

Table 6.2 Some microbial metabolic products that correlate with food quality[a]

Metabolite(s)	Applicable food product(s)
Cadaverine and putrescine	Vacuum-packaged beef
Diacetyl	Frozen juice concentrate
Ethanol	Apple juice, fishery products
Histamine	Canned tuna
Lactic acid	Canned vegetables
Trimethylamine	Fish
Total volatile bases, total volatile nitrogen	Seafoods
Volatile fatty acids	Butter, cream

[a]Source: J. M. Jay, *Modern Food Microbiology*, 4th ed. (Springer, New York, NY, 1991).

product quality loss, tests for the metabolite may be part of a microbiological criterion.

An example of the use of a test for metabolic products as part of a microbiological criterion is the sensory evaluation of imported shrimp. Trained personnel are able to classify the degree of decomposition (i.e., quality loss) into one of three classes through sensory examination. The shrimp are placed in one of the following quality classes: class 1, passable; class 2, decomposed (slight but definite); class 3, decomposed (advanced). The limits of acceptability of a lot are based on the number of shrimp in a sample that are placed in each of the three classes. Other commodities in which organoleptic examination (examination for changes in appearance, color, texture, and odor) is used to determine quality deterioration include raw milk, meat, poultry, and fish and other seafoods. The food industry also uses these examinations to classify certain foods into quality grades. Other examples of metabolic products used to assess product quality are listed in Table 6.2.

INDICATORS OF FOODBORNE PATHOGENS AND TOXINS

Microbiological criteria for product safety should be developed only when the application of a criterion can reduce or eliminate a foodborne hazard. Microbiological criteria verify that the process is adequate to eliminate the hazard. Each food type should be carefully evaluated through risk assessment to determine the potential hazards and their significance to consumers. When a food is repeatedly implicated as a vehicle in foodborne disease outbreaks, the application of microbiological criteria may be useful. Public health officials and the dairy industry responded to widespread outbreaks of milk-borne disease that occurred around the turn of the 20th century in the United States. By imposing controls on milk production, developing safe and effective pasteurization procedures, and setting microbiological criteria, the safety of milk supplies was greatly improved.

Microbiological criteria may also be applied to food products, such as shellfish, which are frequently subject to contamination by harmful microorganisms. The National Shellfish Sanitation Program utilizes microbiological criteria in this manner to prevent the use of shellfish from polluted waters, which may contain various intestinal pathogens. Depending on the type and level of contamination anticipated, the imposition of microbiological criteria may or may not be justified. Contamination of food with patho-

gens that cannot grow to harmful levels does not warrant microbiological criteria. However, microbiological criteria would be warranted for a pathogen that has a low infectious dose. Though fresh vegetables are often contaminated with small numbers of *Clostridium botulinum, Clostridium perfringens,* and *Bacillus cereus* organisms, epidemiological evidence indicates that this contamination presents no health hazard. Thus, imposing microbiological criteria for these microorganisms would not be beneficial. However, criteria may be appropriate for enteric pathogens on produce, since there have been several outbreaks of foodborne illness resulting from fresh produce contaminated with enteric pathogens.

Often food processors alter the intrinsic or extrinsic parameters of a food (nutrients, pH, water activity, inhibitory chemicals, gaseous atmosphere, temperature of storage, and presence of competing organisms) to prevent the growth of undesirable microorganisms. If control over one or more of these parameters is lost, then there may be risk of a health hazard. For example, in the manufacture of cheese or fermented sausage, a lactic acid starter culture is used to produce acid quickly enough to inhibit the growth of *Staphylococcus aureus* to harmful levels. Process critical control points (CCPs), such as the rate of acid formation, are implemented to prevent the growth of harmful microorganisms or their contamination of the food and to ensure that process control is maintained.

Depending on the pathogen, low levels of the microorganism in the food product may or may not be of concern. Some microorganisms have such a low infectious dose that their mere presence in a food presents a significant public health risk. For such organisms, the concern is not the ability of the pathogen to grow in the food but its ability to survive in the food. Foods having intrinsic or extrinsic factors that prevent the survival of pathogens or toxigenic microorganisms may not be candidates for microbiological criteria related to safety. For example, the acidity of certain foods, such as fermented meat products, might be assumed to be sufficient for pathogen control. However, while the growth and toxin production of *S. aureus* might be prevented, enteric pathogens, such as *E. coli* O157:H7, could survive and produce a product that is unsafe for consumption.

An important and often overlooked variable in microbiological risk assessment is the consumer. More rigid microbiological requirements may be needed if a food is intended for use by infants, the elderly, or immunocompromised people, since they are more susceptible to infectious agents than are healthy adults. The ICMSF proposed a system for classification of foods according to risk into 15 hazard categories called cases, with suggested appropriate sampling plans, shown in Table 6.3.

The stringency of sampling plans is based on either the hazard to the consumer from pathogenic microorganisms and their toxins or toxic metabolites or the potential for quality deterioration. It should take into account the types of microorganisms present and their numbers. Foodborne pathogens are grouped into one of three categories based on the severity of the potential hazard (i.e., severe hazards, moderate hazards with potentially extensive spread, and moderate hazards with limited spread [Table 6.4]). Pathogens with the potential for extensive spread are often initially associated with specific foods. An example is fresh beef contaminated with *E. coli* O157:H7. One or a few contaminated pieces of meat can lead to widespread contamination of the product during processing, such as grinding to produce ground beef. There can also be cross-contamination if the fresh beef is improperly stored with ready-to-eat foods. Microbial pathogens in the

Table 6.3 Plan stringency (case) in relation to degree of health hazard and conditions of use[a]

Type of hazard	Conditions in which food is expected to be handled and consumed after sampling, in the usual course of events[b]		
No direct health hazard	**Increased shelf life**	**No change**	**Reduced shelf life**
Utility (e.g., general contamination, reduced shelf life, and spoilage)	Case 1 3-class $n = 5, c = 3$	Case 2 3-class $n = 5, c = 2$	Case 3 3-class $n = 5, c = 1$
Health hazard	**Reduced hazard**	**No change**	**Increased hazard**
Low, indirect (indicator)	Case 4 3-class $n = 5, c = 3$	Case 5 3-class $n = 5, c = 2$	Case 6 3-class $n = 5, c = 1$
Moderate, direct, limited spread[c]	Case 7 3-class $n = 5, c = 2$	Case 8 3-class $n = 5, c = 1$	Case 9 3-class $n = 5, c = 0$
Moderate, direct, potentially extensive spread	Case 10 2-class $n = 5, c = 0$	Case 11 2-class $n = 10, c = 0$	Case 12 2-class $n = 20, c = 0$
Severe, direct	Case 13 2-class $n = 15, c = 0$	Case 14 2-class $n = 30, c = 0$	Case 15 2-class $n = 60, c = 0$

[a]Reprinted with permission from International Commission on Microbiological Specifications of Foods, *Microorganisms in Foods 2. Sampling for Microbiological Analysis: Principles and Applications*, 2nd ed. (University of Toronto Press, Toronto, Canada, 1986).

[b]More stringent plans would generally be used for sensitive foods destined for susceptible populations.

[c]n, number of sample units drawn from lot; c, maximum allowable number of positive results.

lowest-risk group (moderate hazards with limited spread) are found in many foods, usually in small numbers. Generally, illness is caused only when ingested foods contain large numbers of a pathogen, e.g., *C. perfringens*, or have at some point contained large enough numbers of a pathogen, e.g., *S. aureus*, to produce sufficient toxin to cause illness. Outbreaks are usually restricted to consumers of a particular meal or a particular kind of food.

The effects of conditions to which the food will likely be exposed in relation to the growth or death of a relevant organism(s) must be considered before the appropriate case (Table 6.3) can be determined. In general, if a food will be subjected to treatment (e.g., *C. perfringens* in cooked meats held at inappropriate temperature) that would permit growth of the microorganism and thereby increase the hazard, then case 3, 6, 9, 12, or 15 would apply. Case 2, 5, 8, or 11 would apply if no change in the number of organisms present were expected, for example, in freezing of the product. Finally, case 1, 4, 7, or 10 would apply if the product was intended to be fully cooked, reducing the microbial hazard. Consider the case of foods derived from dried whole eggs which are cooked and eaten immediately (scrambled) or made into products (pasta). This would diminish the microbiological hazard. If the dried eggs were consumed dry or reconstituted and consumed immediately, the microbiological hazard would not change (no opportunity was provided for microorganisms present to multiply). A third scenario involves reconstituting the dried eggs and waiting for an extended period before consuming them, in which case any microorganisms present would have an opportunity to multiply and potentially cause illness.

Indicator Organisms

Microbiological criteria for food safety may use tests for indicator organisms that suggest the possibility of a microbial hazard. *E. coli* in drinking water, for example, indicates possible fecal contamination and therefore the potential presence of other enteric pathogens.

Table 6.4 Hazardous microorganisms and parasites grouped on the basis of risk severity[a]

Category I (severe hazards)
Clostridium botulinum types A, B, E, and F
Shigella dysenteriae
Salmonella enterica serovars Paratyphi A and B
Enterohemorrhagic *E. coli* (EHEC)
Hepatitis A and E viruses
Brucella abortus, Brucella suis
Vibrio cholerae O1
Vibrio vulnificus
Taenia solium

Category II (moderate hazards, potentially extensive spread[b])
Listeria monocytogenes
Salmonella spp.
Shigella spp.
Other enterovirulent *E. coli* (EEC)
Streptococcus pyogenes
Rotavirus
Norwalk virus group
Entamoeba histolytica
Diphyllobothrium latum
Ascaris lumbricoides
Cryptosporidium parvum

Category III (moderate hazards, limited spread)
Bacillus cereus
Campylobacter jejuni
Clostridium perfringens
Staphylococcus aureus
V. cholerae, non-O1
Vibrio parahaemolyticus
Yersinia enterocolitica
Giardia lamblia
Taenia saginata

[a]Source: A. W. Randell and A. J. Whitehead, *Rev. Sci. Tech.* **16**:313–321, 1997.
[b]Although these are classified as moderate hazards, complications and sequelae may be severe in certain susceptible populations.

Microbial indicators should have the following characteristics:

- Easily and rapidly detectable
- Readily distinguishable from microbes commonly associated with the food
- A history of constant association with the pathogen whose presence it is to indicate
- Present when the pathogen of concern is present
- Numbers that ideally correlate with those of the pathogen of concern
- Growth requirements and a growth rate equaling those of the pathogen
- A die-off rate that at least parallels that of the pathogen and ideally has a slightly longer persistence than the pathogen of concern
- Absence from foods that are free of the pathogen except perhaps at certain minimum numbers

Additional criteria for fecal indicators used in food safety have been suggested and include the following. Ideally, the bacteria selected should demonstrate specificity, occurring only in intestinal environments. They should occur in very high numbers in feces so as to be encountered in high dilutions. They should possess a high resistance to the external environment, the contamination of which is to be assessed. They should permit relatively easy and fully reliable detection even when present in low numbers.

With the exception of those for *Listeria monocytogenes*, *E. coli* O157:H7, *Salmonella*, and *S. aureus*, most tests for ensuring safety use indicator organisms rather than direct tests for the specific hazard. An overview of some of the more common indicator microorganisms used for ensuring food safety is given below.

Fecal Coliforms and *E. coli*

Fecal coliforms, including *E. coli*, are easily destroyed by heat and may die during freezing and storage of frozen foods. Microbiological criteria involving *E. coli* are useful in those cases where it is desirable to determine if fecal contamination may have occurred. Contamination of a food with *E. coli* implies a risk that other enteric pathogens may also be present. Fecal coliform bacteria are used as a component of microbiological standards to monitor the wholesomeness of shellfish and the quality of shellfish-growing waters. The purpose is to reduce the risk of harvesting shellfish from waters polluted with fecal material. The fecal coliforms have a higher probability of containing organisms of fecal origin than do coliforms which are comprised of organisms of both fecal and nonfecal origins. Fecal coliforms can become established on equipment and utensils in the food-processing environment and contaminate processed foods. At present, *E. coli* is the most widely used indicator of fecal contamination.

The value of testing for fecal coliforms has recently come under scrutiny. Fecal coliforms include *Klebsiella*, *Enterobacter*, and *Citrobacter* species, but these organisms may be considered false-positive indicators of fecal contamination since they can grow in nonfecal niches including water, food, and waste. The current interpretation is that *E. coli* is the only valid indicator microbe for monitoring food containing fresh vegetables for fecal contamination. Using only the *E. coli* assay should serve to eliminate misinterpretation of results of the fecal coliform assay by some physicians and public health officials.

The presence of *E. coli* in a heat-processed food means either process failure or, more commonly, postprocessing contamination. In the case of refrigerated ready-to-eat (RTE) products, such as shrimp and crabmeat, coliforms are indicators of process integrity with regard to the reintroduction of pathogens from environmental sources and maintenance of adequate refrigeration. The source of coliforms after thermal processing appears to be the processing environment, inadequate sanitation procedures, and/or poor temperature control. Coliforms are recommended over *E. coli* and APC, because coliforms are often present in higher numbers than *E. coli* and the levels of coliforms do not increase over time when the product is stored properly.

Metabolic Products

Certain microbiological criteria related to safety rely on tests for metabolites that indicate a potential hazard. Examples of metabolites as a component of microbiological criteria include (i) tests for thermonuclease or thermostable

deoxyribonuclease in foods containing or suspected of containing $\geq 10^5$ *S. aureus* organisms per ml or g; (ii) illuminating grains under ultraviolet (UV) light to detect the presence of aflatoxin produced by *Aspergillus* spp.; and (iii) assaying for the enzyme alkaline phosphatase, a natural constituent of milk that is inactivated during pasteurization, to determine whether milk has been pasteurized or whether pasteurized milk has been contaminated with raw milk.

APPLICATION AND SPECIFIC PROPOSALS FOR MICROBIOLOGICAL CRITERIA FOR FOOD AND FOOD INGREDIENTS

In this section, examples of the application of microbiological criteria to various foods are presented. There are no general criteria suitable for all foods. The relevant background literature that relates to the quality and safety of a specific food product should be consulted before implementation of microbiological criteria.

The utilization of microbiological criteria, as stated previously, may be either mandatory (standards) or advisory (guidelines or specifications). One obvious example of a mandatory criterion (standard) is the zero tolerance set for *Salmonella* in all RTE foods. The U.S. Department of Agriculture (USDA) Food Safety and Inspection Service has mandated zero tolerance for *E. coli* O157:H7 in fresh ground beef. Since the emergence of *L. monocytogenes* as a foodborne pathogen, the Food and Drug Administration and USDA have also applied zero tolerance for the organism in all RTE foods. However, there is considerable debate as to whether zero tolerance is warranted for *L. monocytogenes*. The ICMSF has recommended that the international community allow tolerance of *L. monocytogenes* as outlined in Table 6.5. These recommendations are examples of advisory criteria. Other advisory criteria that have been established to address the safety of foods are shown in Table 6.6.

Mandatory criteria in the form of standards have also been employed for quality issues. Presented in Table 6.7 are examples of foods and food ingredients for which federal, state, and city, as well as international, microbiological standards have been developed.

Table 6.5 Summary of cases and sampling plans for *L. monocytogenes*[a]

Intended consumer	Health hazard	Conditions in which food is expected to be handled and consumed after sampling, in the usual course of events		
		Reduce degree of hazard	Cause no change in hazard	May increase hazard
Healthy individuals	Moderate, direct, potentially extensive spread	Case 10 $n = 5$ $\leq 100/g$	Case 11 $n = 10$ $\leq 100/g$	Case 12 $n = 20$ $\leq 100/g$
Highly susceptible individuals	Severe, direct	Case 13 $n = 15$ $<1/375$ g	Case 14 $n = 30$ $<1/750$ g	Case 15 $n = 60$ $<1/1{,}500$ g

[a]Reprinted with permission from International Commission on Microbiological Specifications of Foods, *Int. J. Food Microbiol.* **22**:89–96, 1994.

Table 6.6 Examples of various food products for which advisory microbiological criteria have been established

Product category	Test parameter(s)	Case	Plan class	Limit (per g)			
				n	*c*	*m*	*M*
Roast beef[a]	*Salmonella*	12	2	20	0	0	
Pâté	*Salmonella*	12	2	20	0	0	
Raw chicken	APC	1	3	5	3	5×10^5	10^7
Cooked poultry, frozen, RTE	*S. aureus*	8	3	5	1	10^3	10^4
Cooked poultry, frozen, to be reheated	*S. aureus*	8	3	5	1	10^3	10^4
	Salmonella	10	2	5	0	0	
Chocolate or confectionery	*Salmonella*	11	2	10^b	0	0	
Dried milk	APC	2	3	5	2	3×10^4	3×10^5
	Coliforms	5	3	5	1	10	10^2
	Salmonella[c]	10	2	5	0	0	
	(normal routine)	11	2	10	0	0	
		12	2	20	0	0	
	Salmonella[c]	10	2	15	0	0	
	(high-risk populations)	11	2	30	0	0	
		12	2	60	0	0	
Fresh cheese[d]	*S. aureus*			5	2	10^2	10^3
	Coliforms			5	2	10^2	10^3
Soft cheese[d]	*S. aureus*			5	2	10^2	10^3
	Coliforms			5	2	10^2	10^3
Pasteurized liquid; frozen and dried egg products	APC	2	3	5	2	5×10^4	10^6
	Coliforms	5	3	5	2	10^3	10^3
	Salmonella[b] (normal routine)	10	2	5	0	0	
		11	2	10	0	0	
		12	2	15	0	0	
	Salmonella[b]	10	2	15	0	0	
	(high-risk populations)	11	2	30	0	0	
		12	2	60	0	0	
Fresh and frozen fish, to be cooked before being eaten	APC	1	3	5	3	5×10^3	10^7
	E. coli		4		3	11	500
	Salmonella[e]	10	2	5	0	0	
	Vibrio parahaemolyticus[e]	7	3	5	2	10^2	10^3
	S. aureus[e]	7	3	5	2	10^3	10^4
Coconut	*Salmonella*	1	3	5	3	5×10^5	10^7
	Growth not expected	11	2	10	0	0	
	Growth expected	12	2	20	0	0	

[a]Product is cooked, not a roast of beef.
[b]The 25-g analytical unit may be composited.
[c]The case is to be chosen based on whether the hazard is expected to be reduced, unchanged, or increased.
[d]Requirements only for fresh and soft cheese made from pasteurized milk.
[e]For fish known to derive from inshore or inland waters of doubtful bacteriological quality or where fish are to be eaten raw, additional tests may be desirable.

CURRENT STATUS

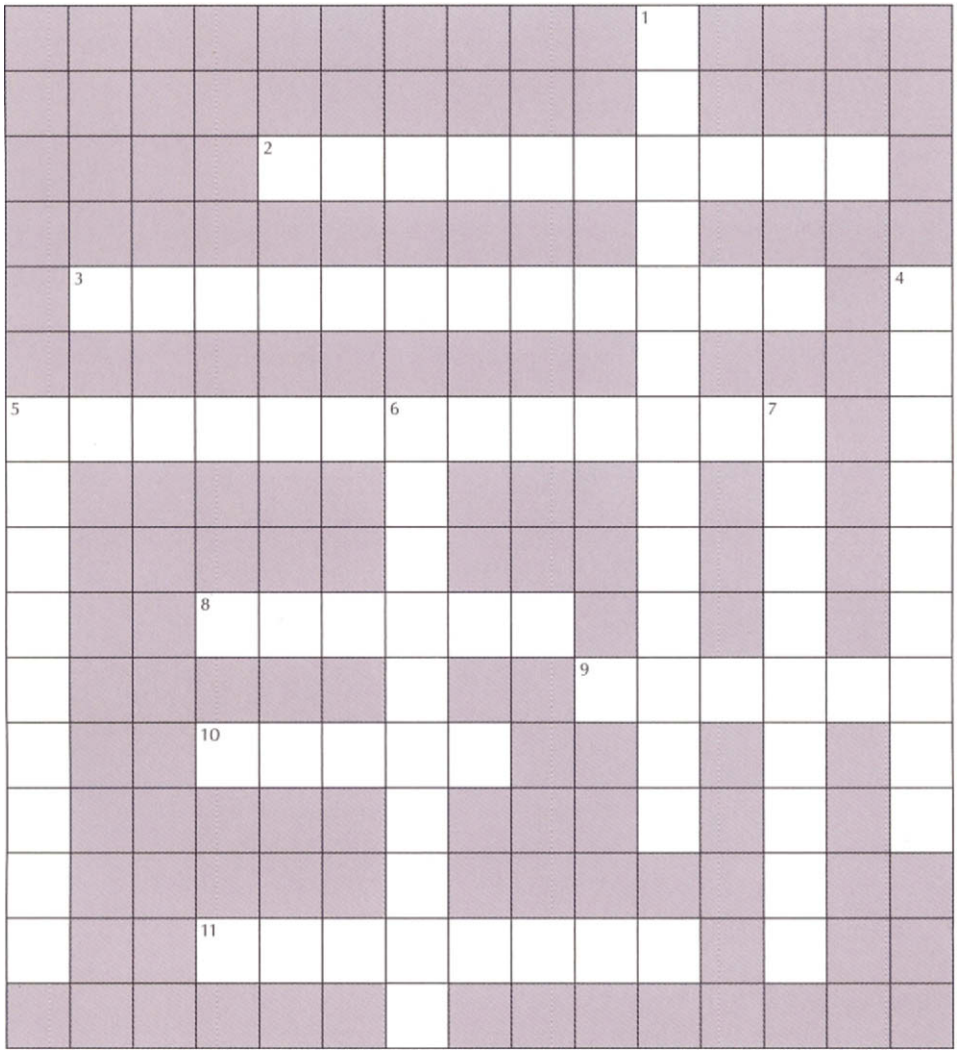
The ICMSF recommends that a series of steps be taken to manage microbiological hazards for foods intended for international trade. These steps include conducting a risk assessment and an assessment of risk management options, establishing a food safety objective (FSO), and confirming that the FSO is achievable by application of GMPs and Hazard Analysis CCPs (HACCPs).

An *FSO* is a statement of the frequency or maximum concentration of a microbiological hazard in a food that is considered acceptable for consumer protection. Examples of FSOs include staphylococcal enterotoxin levels in

Table 6.7 Examples of various food products for which mandatory microbiological criteria have been established

Product category	Test parameter	Comments
United States		
Dairy products		
Raw milk	Aerobic bacteria	Recommendations of U.S. Public Health Service
Grade A pasteurized milk	Aerobic bacteria Coliforms	Recommendations of U.S. Public Health Service
Grade A pasteurized (cultured) milk	Aerobic bacteria Coliforms	Recommendations of U.S. Public Health Service
Dry milk (whole)	SPC Coliforms	Recommendations of U.S. Public Health Service
Dry milk (nonfat)	SPC Coliforms	Standards of Agricultural Marketing Service (USDA)
Frozen desserts	SPC Coliforms	Recommendations of U.S. Public Health Service
Starch and sugars	Total thermophilic-spore count Flat-sour spores Thermophilic anaerobic spores Sulfide spoilage spores	National Canners Association (National Food Processors Association)
Breaded shrimp	APC *E. coli* *S. aureus*	Food and Drug Administration compliance policy guide
International		
Caseins and caseinates	Total bacterial count Thermophilic organisms Coliforms	Europe
Natural mineral waters	Aerobic mesophilic count Coliforms *E. coli* Fecal streptococci Sporulating sulfite-reducing anaerobes *Pseudomonas aeruginosa*, parasites, and pathogenic organisms	*Codex Alimentarius*
Hot meals served by airlines	*E. coli* *S. aureus* *B. cereus* *C. perfringens* *Salmonella*	Europe
Tomato juice	Mold count	Canada
Fish protein	Total plate count *E. coli*	Canada
Gelatin	Total plate count Coliforms *Salmonella*	Canada

cheese that must not exceed 1 μg/100 g or an aflatoxin concentration in peanuts that should not exceed 15 μg/kg. FSOs are broader in scope than microbiological criteria and are intended to communicate the acceptable level of hazard. *Control measures* are actions and activities that can be used to prevent or eliminate a food safety hazard or to reduce it to an acceptable level. The food industry is responsible for applying GMPs and establishing appropriate control measures, including CCPs, in their processes and HACCP plans. For foods in international commerce, quantitative FSOs provide an objective basis to identify what an importing country is willing to accept in relation to the microbiological safety of its food supply.



Across

2. Determines the number of samples tested
3. A type of risk assessment that helps establish microbiological criteria
5. Good _____ practices are prerequisite to HACCP plans
8. A man who must have loved to count molds
9. The _____ microscopic count is limited by low sensitivity
10. _____ Alimentarius
11. A criterion that is part of a law or regulation

Down

1. A criterion that is used as a purchase requirement
4. An organism that substitutes for the real organism of interest
5. A type of microbiological criterion that is not advisory
6. A microbial metabolite that smells like dead bodies
7. A criterion used by a regulatory agency for monitoring

Summary

- Microbiological criteria are most effectively applied as part of quality assurance programs in which HACCP and other prerequisite programs are in place.
- Microbiological criteria for foods continue to evolve as new information becomes available.
- Quantitative risk assessment facilitates the establishment of microbiological criteria for foods in international trade and of guidelines for national standards and policies.
- Microbiological criteria differ based on the food and microorganism of concern.
- Microbiological criteria may be based on the presence or absence of specific metabolites.

Suggested reading

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Pierson, M. D., and D. A. Corlett, Jr. (ed.). 1992. *HACCP: Principles and Applications.* Van Nostrand Reinhold, New York, NY.

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Questions for critical thought

1. Violation of which of the following could result in government action: a microbiological standard, guideline, or specification?
2. Fecal coliform counts are more informative under many circumstances than coliform counts. Briefly, describe the difference between the two indicator methods and provide a scenario for the use of each.
3. The fecal coliform assay may not accurately reflect whether a sample contains fecal material. Why? Aside from the information provided in the chapter, perform a literature search to provide support for the above statement.
4. The microbiological criteria for *L. monocytogenes* in shrimp are as follows: $n = 5$, $c = 0$, and $m = M = 0$. Indicate what each term represents. Based on the information provided, if one sample had 10 CFU, would the product be accepted or rejected?
5. Explain why it is acceptable for some products to be positive for *S. aureus* but not for *Salmonella*.
6. How can the APC be used to screen specifically for a group of microorganisms? Explain your answer.

7. Explain why foods produced and stored under GMPs may be expected to have microbiological profiles different from those of foods produced and stored without the benefit of GMPs.
8. *C. botulinum* is a dangerous pathogen. Why is it acceptable for fresh vegetables to be contaminated with low numbers of this pathogen?
9. Why are microbiological criteria needed for products that will ultimately be cooked?
10. Zero tolerance is set for several pathogens. To what pathogens, and under what conditions, does zero tolerance apply? Speculate why zero tolerance was established for each of the pathogens.
11. How do microbiological criteria affect a food processor? Provide an example of a food and describe the microbiological criteria associated with the product and how those criteria could be met.

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SECTION II Gram-Negative Foodborne Pathogenic Bacteria

7 *Salmonella* Species
8 *Campylobacter* Species
9 Enterohemorrhagic *Escherichia coli*
10 *Yersinia enterocolitica*
11 *Shigella* Species
12 *Vibrio* Species

7

Salmonella Species*

Outbreak

Introduction

Characteristics of the Organism
Biochemical Identification
Taxonomy and Nomenclature
Serological Identification
Physiology

Reservoirs

Characteristics of Disease
Symptoms and Treatment
Preventative Measures
Antibiotic Resistance

Infectious Dose

Pathogenicity and Virulence Factors
Specific and Nonspecific Human Responses
Attachment and Invasion
Growth and Survival within Host Cells
Virulence Plasmids
Other Virulence Factors

Summary

Suggested reading

Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Salmonella*
- understand what conditions in foods favor *Salmonella* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *Salmonella*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *Salmonella*
- identify environmental sources of the organism
- understand the roles of *Salmonella* toxins and virulence factors in causing foodborne illness

Outbreak

On a cool fall evening, nearly 1,400 people gathered in a church meeting hall in Chaptico, MD, for the annual fall dinner. The meal included stuffed ham, turkey, fried oysters, mashed potatoes, and gravy. Many of the attendees commented that they had stuffed themselves and could not eat another bite. Others enjoyed the meal so much that they bought carryout platters. The gastric pleasure of the meal soon faded to gastric (stomach) upset for nearly half of the attendees. People who dined or bought carryout became sick, experiencing nausea, cramps, dehydration, and fever, classic symptoms of foodborne illness. Two elderly people died, 100 people sought care at the hospital emergency room, and 700 people reported becoming ill. Microbiological testing of each food demonstrated that the ham was positive for the strain of *Salmonella* associated with the outbreak. Improper cooking and holding of the stuffed ham likely resulted in growth of *Salmonella* to levels capable of causing illness.

INTRODUCTION

National epidemiologic registries continue to underscore the importance of *Salmonella* spp. as leading causes of foodborne bacterial illnesses in humans. Incidents of foodborne salmonellosis tend to dwarf those associated with most other foodborne pathogens (Table 7.1). It is noteworthy that the problem of human salmonellosis from the consumption of contaminated foods generally remains on the increase worldwide (Table 7.2). Notwithstanding recent improvements in procedures for the epidemiologic investigation of foodborne outbreaks in many countries, the global increases in foodborne salmonellosis are considered real. They are not a result of enhanced surveillance programs and/or greater resourcefulness of medical and public

*The chapter on *Salmonella* in the first edition of this book contained a footnote acknowledging J. Stan Bailey and John J. Maurer as the authors of a chapter in *Food Microbiology: Fundamentals and Frontiers*, 2nd ed., that was adapted for use in the textbook. It should have acknowledged Jean-Yves D'Aoust, John Maurer, and J. Stan Bailey as the authors. The publisher regrets the error.

Table 7.1 Examples of major foodborne outbreaks of salmonellosis worldwide since 1981

Yr	Location(s)	Vehicle	Serovar(s)	No. of cases[a]	No. of deaths
1981	The Netherlands	Salad base	Indiana	600[b]	0
1981	Scotland	Raw milk	Typhimurium PT 204	654	2
1984	Canada	Cheddar cheese	Typhimurium PT 10	2,700	0
1984	France, England	Liver pâté	Goldcoast	756	0
1984	International	Aspic glaze	Enteritidis PT 4	766	2
1984	United States	Salad bars	Typhimurium	751	0
1985	United States	Pasteurized milk	Typhimurium	16,284	7
1987	People's Republic of China	Egg drink	Typhimurium	1,113	NS[c]
1987	Norway	Chocolate	Typhimurium	361	0
1988	Japan	Cuttlefish	Champaign	330	0
1988	Japan	Cooked eggs	*Salmonella* spp.	10,476	NS
1991	United States, Canada	Cantaloupes	Poona	>400	NS
1991	Germany	Fruit soup	Enteritidis	600	NS
1993	France	Mayonnaise	Enteritidis	751	0
1993	United States	Tomatoes	Montevideo	100	0
1993	Germany	Paprika chips	Saint-Paul, Javiana, Rubislaw	>670	0
1994	Finland, Sweden	Alfalfa sprouts	Bovismorbificans	492	0
1994	United States	Ice cream	Enteritidis PT 8	740	0
1995	United States	Orange juice	Hartford, Gaminara	62	0
1995	United States	Restaurant foods	Newport	850	0
1996	United States	Alfalfa sprouts	Montevideo, Meleagrides	481	1
1997	United States	Stuffed ham	Heidelberg	746	1
1998	United States	Toasted oat cereal	Agona	209	0
1998	Canada	Cheddar cheese	Enteritidis PT 8	700	0
1999	Japan	Dried squid	*Salmonella* spp.	>453	0
1999	Australia	Orange juice	Typhimurium	427	NS
1999	Japan	Cuttlefish chips	Oranienburg	≥1,500	NS
1999	Canada	Alfalfa sprouts	Paratyphi B (Java)	>53	NS
1999	Japan	Peanut sauce	Enteritidis PT 1	644	NS
1999	United States, Canada	Orange juice	Muenchen	>220	0
2000	United States	Orange juice	Enteritidis	>74	0
2000	United States	Mung bean sprouts	Enteritidis	>45	0
2000	Canada	Bakery goods	Enteritidis	15	0
2004	Great Britain	Lettuce	Newport	>350	0
2004	United States	Roma tomatoes	Braenderup	125	0
			Javiana	390	0
			Typhimurium	27	0
			Anatum	5	0
			Thompson	4	0
			Muenchen	4	0
2005	Great Britain	Imported shell eggs	Enteritidis	68	0
2005	Canada	Roast beef	*Salmonella* spp.	155	0
2006	Great Britain	Chocolate	Montevideo	>46	0
2006	United States	Peanut butter	Tennessee	>288	0

[a]Confirmed cases unless stated otherwise.
[b]Estimated number of cases.
[c]NS, not specified.

Table 7.2 Trends in foodborne salmonellosis

Country	Annual no. of reported incidents								No. of cases[a]
	1985	1987	1989	1991	1993	1995	1998	2000	
Austria	124	151	440	963	922	873	880	NA[b]	80.8–92.4
Bulgaria	15	10	12	13	12	14	NA	7	2.1–31.9
Canada	59	53	51	28	43	27	NA	NA	0.5–3.8
Denmark	12	5	5	11	7	17	27	NA	27.0–94.90
England and Wales	372	421	935	936	136	90	57	35	22.0–45.0
Finland	11	3	2	4	3	8	1	5	44.9–153.2
France	7	178	462	477	200	185	267	163	21.9–31.9
Germany	ND[c]	ND	23	62	150	117	108	69	103.5–242.7
Hungary	116	122	131	110	158	266	269	155	18.0–88.9
Japan	—[d]	90	146	159	NA	NA	757	518	2.9–8.2
Poland	380	690	709	625[e]	398	365	325	295	44.1–93.9
Scotland	133	180	151	115	NA	NA	11	6	4.0–46.0
United States	79	52	117	122	68	90	NA	NA	1.1–4.3

[a]Range of cases per 100,000 population within the review period (1985 to 2000).
[b]NA, not available.
[c]ND, no data; German reunification did not occur until 1989.
[d]—, data not published.
[e]Estimated from graphic representation of data.

health officials. Events such as the ongoing pandemic of egg-borne *Salmonella enterica* serovar Enteritidis infection clearly have an impact on current disease statistics. Major outbreaks of foodborne salmonellosis in the last few decades are of interest because they underline the multiplicity of foods and *Salmonella* serovars that have been implicated in human illness. In 1974, temperature abuse of egg-containing potato salad served at an outdoor barbecue led to an estimated 3,400 human cases of *S. enterica* serovar Newport infection in which cross-contamination of the salad by an infected food handler was suspected. The large Swedish outbreak of serovar Enteritidis phage type (PT) 4 infection in 1977 was attributed to the consumption of a mayonnaise dressing in a school cafeteria. In 1984, Canada experienced its largest outbreak of foodborne salmonellosis. It was attributed to the consumption of Cheddar cheese manufactured from heat-treated and pasteurized milk, and it resulted in >2,700 confirmed cases of serovar Typhimurium PT 10 infection. Manual override of a flow diversion valve reportedly led to the entry of raw milk into vats of thermized and pasteurized cheese milk. The following year, the largest outbreak of foodborne salmonellosis in the United States occurred, involving 16,284 confirmed cases of illness. Although the cause of this outbreak was never determined, a cross-connection between raw and pasteurized milk lines was likely the cause. A large outbreak of salmonellosis affecting >10,000 Japanese consumers was attributed to a cooked egg dish. One of the largest (>650 cases in Ontario alone) sprout-related salmonellosis outbreaks in the world occurred in 2005 in Canada and was associated with consumption of contaminated mung bean sprouts. In 1993, paprika imported from South America was incriminated as the contaminated ingredient in the manufacture of potato chips distributed in Germany. A major outbreak of foodborne salmonellosis that occurred in the United States involved ice cream contaminated with serovar Enteritidis. The transportation of ice cream mix in an

unsanitized truck that had previously carried raw eggs was the source of contamination. Despite the general perception about the involvement of meat and egg products as the primary sources of human *Salmonella* infections, many outbreaks in recent years have been associated with tomatoes, orange juice, and vegetable sprouts.

CHARACTERISTICS OF THE ORGANISM

In the early 19th century, clinical pathologists in France first documented the association of human intestinal ulceration with a contagious agent. The disease was later identified as typhoid fever. During the first quarter of the 20th century, great advances occurred in the serological detection of somatic (O) and flagellar (H) antigens within the *Salmonella* group. An antigenic scheme for the classification of salmonellae was later developed and now includes >2,500 serovars.

Biochemical Identification

Salmonella spp. are facultatively anaerobic gram-negative rod-shaped bacteria belonging to the family *Enterobacteriaceae*. Although members of the genus are motile via peritrichous flagella (flagella distributed uniformly over the cell surface), nonflagellated variants, such as *S. enterica* serovars Pullorum and Gallinarum, and nonmotile strains resulting from dysfunctional flagella do occur. Salmonellae are chemoorganotrophic (able to utilize a wide range of organic substrates), with an ability to metabolize nutrients by both respiratory and fermentative pathways. The bacteria grow optimally at 37°C and catabolize D-glucose and other carbohydrates with the production of acid and gas. Salmonellae are oxidase negative and catalase negative and grow on citrate as a sole carbon source. They generally produce hydrogen sulfide, decarboxylate lysine, and ornithine and do not hydrolyze urea. Many of these traits have formed the basis for the presumptive biochemical identification of *Salmonella* isolates. Accordingly, a typical *Salmonella* isolate would produce acid and gas from glucose in triple sugar iron (TSI) agar medium and would not utilize lactose or sucrose in TSI or in differential plating media, such as brilliant green, xylose lysine deoxycholate, and Hektoen enteric agars. Additionally, typical salmonellae readily produce an alkaline reaction from the decarboxylation of lysine to cadaverine in lysine ion agar, generate hydrogen sulfide gas in TSI and lysine ion media, and fail to hydrolyze urea. The dynamics of genetic variability (the result of bacterial mutations and conjugative intra- and intergeneric exchange of plasmids encoding determinant biochemical traits) continue to reduce the proportion of "typical" *Salmonella* biotypes. *Salmonella* utilization of lactose and sucrose is plasmid mediated, and the occurrence of *lac*$^+$ and/or *suc*$^+$ biotypes in clinical isolates and food materials is of public health concern. These isolates could escape detection on the disaccharide-dependent plating media that are commonly used in hospital and food industry laboratories. Bismuth sulfite agar remains a medium of choice for isolating salmonellae because, in addition to its high level of selectivity, it responds effectively to the production of extremely low levels of hydrogen sulfide gas. The diagnostic hurdles created by the changing patterns of disaccharide utilization by *Salmonella* are being further complicated by the increasing occurrence of biotypes that cannot decarboxylate lysine, that possess urease activity, that produce indole, and that readily grow in the presence of KCN (potassium cyanide). The variability in bio-

chemical traits is leading to the replacement of traditional testing protocols with molecular technologies targeted at the identification of stable genes and/or their products that are unique to the genus *Salmonella.*

Taxonomy and Nomenclature

According to the World Health Organization (WHO) Collaborating Centre for Reference and Research on *Salmonella* (Institut Pasteur, Paris, France), *S. enterica* and *Salmonella bongori* currently include 2,443 and 20 serovars, respectively (Table 7.3). The Centers for Disease Control and Prevention have adopted as their official nomenclature the following scheme. The genus *Salmonella* contains two species, each of which contains multiple serovars (Table 7.3). The two species are *S. enterica,* the type species, and *S. bongori,* which was formerly subspecies V. *S. enterica* is divided into six subspecies, which are referred to by a roman numeral and a name (I, *S. enterica* subsp. *enterica;* II, *S. enterica* subsp. *salamae;* IIIa, *S. enterica* subsp. *arizonae;* IIIb, *S. enterica* subsp. *diarizonae;* IV, *S. enterica* subsp. *houtenae;* and VI, *S. enterica* subsp. *indica*). *S. enterica* subspecies are differentiated biochemically and by genomic relatedness. *Salmonella* serovars that have been linked to recent cases of foodborne illness include *S. enterica* serovars Enteritidis, Typhimurium, Newport, and Stanley.

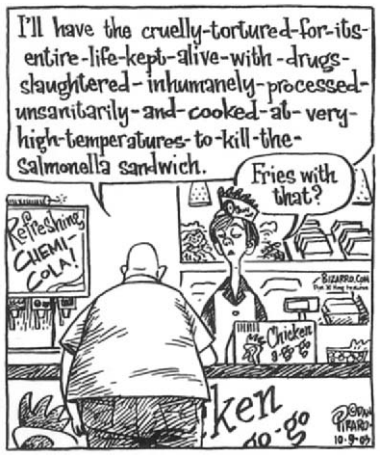
Serological Identification

The aim of serological testing procedures is to determine the complete antigenic formulas of individual *Salmonella* isolates. Antigenic determinants include somatic (O) lipopolysaccharides (LPS) on the external surface of the bacterial outer membrane, flagellar (H) antigens associated with the peritrichous flagella, and the capsular (Vi [virulence]) antigen, which occurs only in serovars Typhi, Paratyphi C, and Dublin. The heat-stable somatic antigens are classified as major or minor antigens. The former category consists of antigens, such as the somatic factors O:4 and O:3, which are specific determinants for the somatic groups B and E, respectively. In contrast, minor somatic antigenic components, such as O:12, are nondiscriminatory, as evidenced by their presence in different somatic groups. Smooth (S) variants are strains with well-developed serotypic LPS that readily agglutinate with specific antibodies, whereas rough (R) variants exhibit incomplete LPS antigens, resulting in weak or no agglutination with *Salmonella* somatic antibodies. Flagellar (H) antigens are heat-labile proteins, and individual *Salmonella* strains may produce one (monophasic) or two (diphasic) sets of flagellar

Table 7.3 The genus *Salmonella*[a]

***Salmonella* species and subspecies (no.)**	**No. of serovars**
S. enterica subsp. *enterica* (I)	1,454
S. enterica subsp. *salamae* (II)	489
S. enterica subsp. *arizonae* (IIIa)	94
S. enterica subsp. *diarizonae* (IIIb)	324
S. enterica subsp. *houtenae* (IV)	70
S. enterica subsp. *indica* (VI)	12
S. bongori (V)	20
Total	2,463

[a]Source: J.-Y. D'Aoust, p. 1233–1299, *in* B. M. Lund, A. C. Baird-Parker, and G. W. Gould (ed.), *The Microbiological Safety and Quality of Food* (Aspen Publishers Inc., Gaithersburg, MD, 2000).



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antigens. Although serovars such as Dublin produce a single set of flagellar antigens, most serovars can elaborate either of two sets of antigens, i.e., phase 1 and phase 2 antigens. Capsular (K) antigens, commonly encountered in members of the family *Enterobacteriaceae*, are limited to the Vi antigen in the genus *Salmonella*.

We will use serovar Infantis (6,7:r:1,5 [see below]) as an example. Commercially available polyvalent somatic antisera each consist of a mixture of antibodies specific for a limited number of major antigens; e.g., polyvalent B (poly-B) antiserum (Difco Laboratories, Detroit, MI) recognizes somatic groups C_1, C_2, F, G, and H. Following a positive agglutination with poly-B antiserum, single-group antisera representing the five somatic groups included in the poly-B reagent would be used to define the serogroup of the isolate. The test isolate would react with the C_1 group antiserum, indicating that antigens 6 and 7 are present. Flagellar antigens would then be determined by broth agglutination reactions using poly-H antisera or the Spicer-Edwards series of antisera. In the former assay, a positive agglutination reaction with one of the five polyvalent antisera (poly-A to -E; Difco) would lead to testing with single-factor antisera to specifically identify the phase 1 and/or phase 2 flagellar antigens present. Agglutination in poly-C flagellar antiserum and subsequent reaction of the isolate with single-group H antisera would confirm the presence of the r antigen (phase 1). The empirical antigenic formula of the isolate would then be (6,7:r). Phase reversal in semisolid agar supplemented with r antiserum would immobilize phase 1 salmonellae at or near the point of inoculation, thereby facilitating the recovery of phase 2 cells from the edge of the zone of migration. Serological testing of phase 2 cells with poly-E and 1-complex antisera would confirm the presence of the flagellar 1 factor. Confirmation of the flagellar 5 antigen with single-factor antiserum would yield the final antigenic formula 6,7:r:1,5, which corresponds to serovar Infantis. A similar analytical approach would be used with the Spicer-Edwards poly-H antisera, in which the identification of flagellar antigens would arise from the pattern of agglutination reactions among the four Spicer-Edwards antisera and with three additional polyvalent antisera, including the L, 1, and e,n complexes.

Physiology

Growth and Survival

Temperature. *Salmonella* spp. are resilient and can adapt to extreme environmental conditions. Some *Salmonella* strains can grow at elevated temperatures (54°C), and others exhibit psychrotrophic properties (they are able to grow in foods stored at 2 to 4°C) (Table 7.4). Moreover, preconditioning of cells to low temperatures can greatly increase the growth and survival of salmonellae in refrigerated food products. Studies of the maximum temperatures for growth of *Salmonella* spp. in foods are generally lacking. Notwithstanding an early report of growth of salmonellae in inoculated custard and chicken à la king at 45.6°C, more recent evidence indicates that prolonged exposure of mesophilic strains to thermal stress conditions results in mutants of serovar Typhimurium capable of growth at 54°C.

Salmonella can survive for extended periods in foods stored at freezing and room temperatures. Several factors can influence the survival of salmonellae during frozen storage of foods: (i) the composition of the freezing matrix, (ii) the kinetics of the freezing process, (iii) the physiological state of

Table 7.4 Physiological limits for the growth of *Salmonella* spp. in foods and bacteriological media

Parameter	Limits (time to double in no.)		Product	Serovar(s)
	Minimum	Maximum		
Temp (°C)	2 (24 h)		Minced beef[a]	Typhimurium
	2 (2 days)		Minced chicken[b]	Typhimurium
	4 (≤10 days)		Shell eggs[b]	Enteritidis
		54.0[c]	Agar medium	Typhimurium
pH	3.99[d]		Tomatoes	Infantis
	4.05[e]		Liquid medium	Anatum, Tennessee, Senftenberg
		9.5	Egg washwater[b]	Typhimurium
a_w	0.93[f]		Rehydrated dried soup[b]	Oranienburg

[a]Naturally contaminated.
[b]Artificially contaminated.
[c]Mutants selected to grow at elevated temperature.
[d]Growth within 24 h at 22°C.
[e]Acidified with HCl or citric acid; growth within 24 h at 30°C.
[f]Growth within 3 days at 30°C.

the foodborne salmonellae, and (iv) serovar-specific responses. The viability of salmonellae in dry foods stored at 25°C decreases with increasing storage temperature and with increasing moisture content.

Many factors may enhance the heat resistance of *Salmonella* and other foodborne bacterial pathogens in food ingredients and finished products. The heat resistance of *Salmonella* spp. increases as the water activity (a_w) of the heating matrix (food) decreases. The type of solutes used to alter the a_w of the heating menstruum can influence the level of acquired heat resistance. Other important features associated with this adaptive response include the greater heat resistance of salmonellae grown in nutritionally rich versus minimal media, of stationary-phase versus logarithmic-phase cells, and of salmonellae previously stored in a dry environment. The ability of *Salmonella* to acquire greater heat resistance following exposure to sublethal temperatures is equally important. This adaptive response has potentially serious implications for the safety of thermal processes that expose or maintain food products at marginally lethal temperatures. Changes in the fatty acid composition of cell membranes in heat-stressed *Salmonella* provide a greater proportion of saturated membrane phospholipids. This results in reduced fluidity of the bacterial cell membrane and an associated increase in membrane resistance to heat damage. The likelihood that other protective cellular functions are triggered by heat shock stimuli cannot be ruled out.

The following scenario involving *Salmonella* contamination of a chocolate confectionery product demonstrates the complex interactions of food, temperature, and organism. The survival of salmonellae in dry-roasted cocoa beans can lead to contamination of in-line and finished products. Thermal inactivation of salmonellae in molten chocolate is difficult. This is because the time-temperature conditions that are required to eliminate the pathogen in this sucrose-containing product with low a_w would likely result in an organoleptically (off taste and off odor) unacceptable product. The problem is further compounded by the ability of salmonellae to survive for many years in the finished product when it is stored at room temperature. Clearly, effective decontamination of raw cocoa beans and stringent in-plant

control measures to prevent cross-contamination of in-line products are of great importance in this food industry.

pH. The physiological adaptability of *Salmonella* spp. is further demonstrated by their ability to grow at pH values ranging from 4.5 to 9.5, with an optimum pH for growth of 6.5 to 7.5 (Table 7.4). Of the various organic and inorganic acids used in product acidification, propionic and acetic acids are more bactericidal than the common food-associated lactic and citric acids. Interestingly, the antibacterial action of organic acids decreases with increasing length of the fatty acid chain. Early research on the ability of *Salmonella* to grow in acidic environments revealed that wild-type strains preconditioned on pH gradient plates could grow in liquid and solid media at considerably lower pH values than the starting wild-type strains. This raises concerns regarding the safety of fermented foods, such as cured sausages and fermented raw milk products. The starter culture-dependent acidification of fermented foods could provide a favorable environment for the growth of salmonellae in the product to a state of increased acid tolerance. The growth and/or enhanced survival of salmonellae during the fermentative process would result in a contaminated ready-to-eat product. Ultimately, this may promote survival of *Salmonella* in the host.

Brief exposure of serovar Typhimurium to mild acid environments of pH 5.5 to 6.0 (preshock) followed by exposure of the adapted cells to a pH of $\leq$4.5 (acid shock) triggers a complex acid tolerance response (ATR). This permits the survival of the microorganism under extreme acid environments (pH 3.0 to 4.0). The response translates into an induced synthesis of 43 acid shock and outer membrane proteins, a reduced growth rate, and pH homeostasis. It is demonstrated by the bacterial maintenance of internal pH values of 7.0 to 7.1 and 5.0 to 5.5 upon sequential exposure of cells to an external pH of 5.0 and 3.3, respectively. Aside from the log-phase ATR, *Salmonella* also possesses a stationary-phase ATR that provides greater acid resistance than the log-phase ATR. This is induced at pHs of $<$5.5 and functions maximally at pH 4.3. This stationary-phase ATR induces the synthesis of only 15 shock proteins. The induction of the remaining acid-protective mechanism associated with stationary-phase *Salmonella* is independent of the external pH and dependent on the alternative sigma factor (σ^S) encoded by the *rpoS* locus. The mechanism enhances the ability of stationary-phase cells to survive in hostile environmental conditions. Thus, three possibly overlapping cellular systems confer acid tolerance on *Salmonella* spp. These are (i) the pH-dependent, *rpoS*-independent log-phase ATR; (ii) the pH-dependent and *rpoS*-independent stationary-phase ATR; and (iii) the pH-independent, *rpoS*-dependent stationary-phase acid resistance. These systems likely operate in the acidic environments that prevail in fermented and in acidified foods and in phagocytic cells of the infected host.

Acid stress can also enhance bacterial resistance to other adverse environmental conditions. The growth of serovar Typhimurium at pH 5.8 increases thermal resistance at 50°C, enhances tolerance to high osmotic stress (2.5 M NaCl), and produces greater surface hydrophobicity and an increased resistance to the antibacterial lactoperoxidase system and surface-active agents, such as crystal violet and polymyxin B.

Osmolarity. High salt concentrations have long been recognized for their ability to extend the shelf life of foods by inhibiting microbial growth. Foods with a_w values of $<$0.93 do not support the growth of salmonellae.

Although *Salmonella* is generally inhibited in the presence of 3 to 4% NaCl, bacterial salt tolerance increases with increasing temperature in the range of 10 to 30°C.

The pH, salt concentration, and temperature of the microenvironment can exert profound effects on the growth kinetics of *Salmonella* spp. It is now accepted that *Salmonella* spp. have the ability to grow under acidic (pH <5.0) conditions or in environments of high salinity (>2% NaCl) with increasing temperature. Interestingly, the presence of salt in acidified foods can reduce the antibacterial action of organic acids in which low concentrations of NaCl or KCl stimulate the growth of serovar Enteritidis in broth medium acidified to pH 5.19 with acetic acid.

Modified atmosphere. Concerns about the ability of *Salmonella* to survive under extremes of pH, temperature, and salinity are further heightened by the widespread refrigerated storage of foods packaged under vacuum or modified atmosphere to prolong shelf life. Gaseous mixtures consisting of 60 to 80% (vol/vol) CO_2 with varying proportions of N_2 and/or O_2 can inhibit the growth of aerobic spoilage microorganisms, such as *Pseudomonas* spp., without promoting the growth of *Salmonella* spp. The safety of modified-atmosphere- and vacuum-packaged foods that contain high levels of salt is coming under question, since anaerobic conditions may enhance *Salmonella* salt tolerance.

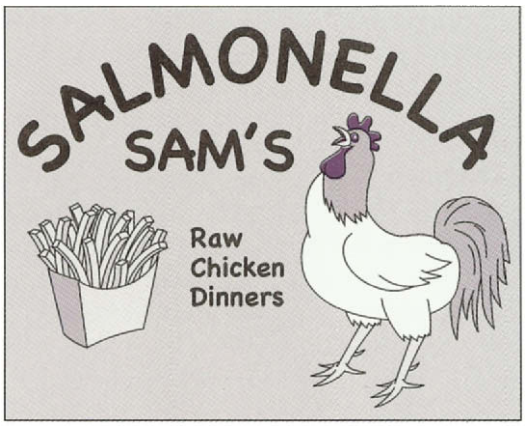
RESERVOIRS

Salmonella spp. will continue to be significant human pathogens in the global food supply for several reasons, including their presence in the environment; intensive farming practices used in the meat, fish, and shellfish industries (which promotes spread among animals); and the recycling of slaughterhouse by-products into animal feeds. Poultry meat and eggs are predominant reservoirs of *Salmonella* spp. in many countries (Fig. 7.1). This overshadows the importance of other animal meats, such as pork, beef, and mutton, as potential vehicles of infection. To address the problem of *Salmonella* in meat products, the U.S. Department of Agriculture Food Safety

Authors' note

Several government agencies are responsible for controlling salmonellosis. The Food and Drug Administration (FDA) inspects imported foods and milk pasteurization plants, promotes better food preparation techniques in restaurants and food processing plants, and regulates the sale of turtles. The U.S. Department of Agriculture (USDA) monitors the health of food animals, inspects egg pasteurization plants, and is responsible for the quality of slaughtered and processed meat. The U.S. Environmental Protection Agency (EPA) regulates and monitors the safety of our drinking water supplies.

Figure 7.1 Consumption of improperly cooked chicken can result in salmonellosis. Consumers must pay attention to the food whether eating at home or at a restaurant to be certain it is cooked properly; in the case of chicken, consumers should make sure it is not pink in color or raw in the center. Salmonellosis is the second leading cause of bacterial foodborne illness in the United States.



Inspection Service requires the meat and poultry industries to implement Hazard Analysis Critical Control Point (HACCP) plans in all of their plants. The Food Safety Inspection Service conducts *Salmonella* testing to verify that the implemented HACCP is helping to control *Salmonella* on finished products. The rate of *Salmonella* contamination has been reduced for all meat animals since the implementation of HACCP.

The continuing pandemic of human serovar Enteritidis infections associated with the eating of raw or lightly cooked shell eggs and egg-containing products further emphasizes the importance of poultry as vehicles of human salmonellosis. The need for sustained and stringent bacteriological control of poultry husbandry practices is urgent. This egg-related pandemic is of particular concern, because the problem arises from transovarian transmission of serovar Enteritidis into the interior of the egg prior to shell deposition. The viability of these internalized serovar Enteritidis organisms remains unaffected by egg surface sanitizing practices. Increased efforts to intervene and reduce the prevalence of serovar Enteritidis have led to significant reductions in egg-borne outbreaks in the United Kingdom and the United States.

Rapid depletion of wild stocks of fish and shellfish in recent years have greatly increased the importance of the international aquaculture industry. The feeding of raw meat scraps and offal, feces potentially contaminated with typhoid and paratyphoid salmonellae, and animal feeds that may harbor *Salmonella* to reared species is common in developing countries. Accordingly, aquaculture farmers are relying heavily on antibiotics applied at subtherapeutic levels to safeguard the vigor of farmed fish and shellfish. The use of antibiotics creates a serious public health concern, since these antimicrobial agents are the mainstay treatment for systemic salmonellosis in humans.

Fruits and vegetables have gained notoriety in recent years as vehicles of human salmonellosis. The situation has developed from the increased global export of fresh and dehydrated fruits and vegetables from countries that enjoy tropical and subtropical climates. The prevailing hygienic conditions during the production, harvesting, and distribution of products in these countries do not always meet minimum standards and may facilitate product contamination. Operational changes in favor of field irrigation with treated effluents, washing of fruits and vegetables with disinfected water, education of local workers on the hygienic handling of fresh produce, and greater protection of products from environmental contamination during all phases of handling would greatly enhance the safety of fresh fruits and vegetables. These recommendations must be more widely adopted since recent large outbreaks of salmonellosis have been linked to the consumption of contaminated lettuce and bean sprouts.

CHARACTERISTICS OF DISEASE

Symptoms and Treatment

Human *Salmonella* infections can lead to several clinical conditions, including enteric (typhoid) fever, uncomplicated enterocolitis, and systemic infections by nontyphoid microorganisms (Fig. 7.1). Enteric fever is a serious human disease associated with the typhoid and paratyphoid strains. Symptoms of enteric fever appear after a period of incubation ranging from 7 to 28 days and may include diarrhea, prolonged and spiking fever,

abdominal pain, headache, and prostration. Diagnosis of the disease relies on the isolation of the infective agent from blood or urine samples in the early stages of the disease or from stools after the onset of clinical symptoms. An asymptomatic chronic carrier state commonly follows the acute phase of enteric fever. The treatment of enteric fever is based on supportive therapy and/or the use of chloramphenicol, ampicillin, or trimethoprim-sulfamethoxazole to eliminate the systemic infection. Marked global increases in the resistance of typhoid and paratyphoid organisms to these antibacterial drugs in the last decade have limited their effectiveness in human therapy.

Human infections with nontyphoid *Salmonella* strains commonly result in symptoms 8 to 72 h after ingestion of the invasive pathogen. The illness is usually self-limiting, and remission of the characteristic nonbloody diarrheal stools and abdominal pain usually occurs within 5 days of the onset of symptoms. The successful treatment of uncomplicated cases may require only supportive therapy, such as fluid and electrolyte replacement. Antibiotics are not used in such episodes, because they prolong the carrier state. Asymptomatic persistence of salmonellae in the gut is probably due to antibiotic-dependent repression of native gut microflora. Human infections with nontyphoid strains can also degenerate into systemic infections and precipitate various chronic conditions.

Salmonella can induce chronic conditions, such as aseptic reactive arthritis, Reiter's syndrome, and ankylosing spondylitis. Bacterial prerequisites for the onset of these chronic diseases include the ability of the bacterial strain to infect mucosal surfaces, the presence of outer membrane LPS, and the ability to invade host cells.

Preventative Measures

The global impact of typhoid and paratyphoid salmonellae on human health led to the early development of parenteral vaccines consisting of heat-, alcohol-, or acetone-killed cells. The phenol- and heat-killed typhoid vaccine is widely used. However, this vaccine can cause adverse reactions, including fever, headache, pain, and swelling at the site of injection. The pandemic of poultry- and egg-borne serovar Enteritidis infections that continues to afflict consumers underscores the potential benefit of vaccines for human and veterinary applications. Live attenuated vaccines continue to generate great research interest because such preparations induce strong and durable immune responses in vaccinees.

Antibiotic Resistance

Current global increases in the number of antibiotic-resistant *Salmonella* serovars in humans and farm animals are alarming. The major public health concern is that *Salmonella* will become resistant to antibiotics used in human medicine, thereby greatly reducing therapeutic options and threatening the lives of infected individuals. The liberal administration of antimicrobial agents in hospitals and other treatment centers contributes significantly to the emergence of many resistant strains. The emergence of antimicrobial resistance in bacterial pathogens, many of which can be traced to food-producing animals, has generated much controversy and public media attention.

Because evolutionary processes offer genetic and phenotypic variability for resistance genes, there will always be some level of background

resistance in bacterial populations. However, without selective pressures, resistance levels are low, and so are the chances for therapy failures. Resistance in *Salmonella* to a single antibiotic was first noted in the early 1960s. The appearance of serovar Typhimurium definitive type 104 (DT104) in the late 1980s raised major concerns because of its multiple resistances against ampicillin, chloramphenicol, streptomycin, sulfonamides, and tetracycline. Some strains are also resistant to gentamicin, trimethoprim, fluoroquinolones, and other antibiotics. Fluoroquinolones belong to a class of newer antimicrobial agents with a broad spectrum of activity, high efficiency, and widespread application in human and veterinary medicine. An increase in the resistance of DT104 strains to fluoroquinolones has led many European countries to ban the nonhuman use of fluoroquinolones.

The genetic determinants for *Salmonella* virulence and antibiotic resistance can occur on the same plasmid. This increases the disease potential of *Salmonella* DT104. In the next few decades (or perhaps sooner), we may well witness the international scientific community feverishly trying to develop new drugs to replace those currently in use, since they will no longer be effective against the multiple-antibiotic-resistant *Salmonella.*

No studies have conclusively linked the subtherapeutic use of antibiotics in agriculture with the development of resistance. The obvious need for data led to the joint development in 1996 of the National Antimicrobial Resistance Monitoring System by the Food and Drug Administration's Center for Veterinary Medicine, the Centers for Disease Control and Prevention, and the U.S. Department of Agriculture. The purpose of this program is to provide baseline data so that changes in the antimicrobial resistance and susceptibility of bacteria can be identified. Additionally, in June 2000 the World Health Organization released a set of global principles aimed at reducing the risks related to the use of antimicrobials in food animals. An Institute of Food Technologists report (2006) provides new recommendations designed to fill knowledge gaps associated with the implications of antimicrobial resistance for the food system.

INFECTIOUS DOSE

Newborns, infants, the elderly, and immunocompromised individuals are more susceptible to *Salmonella* infections than healthy adults. The incompletely developed immune system in newborns and infants, the frequently weak and/or delayed immunological responses in the elderly and debilitated persons, and the generally low gastric acid production in infants and seniors facilitate the intestinal colonization and systemic spread of salmonellae in these populations.

The ingestion of only a few *Salmonella* cells can be infectious (Table 7.5). Determinant factors in salmonellosis are not limited to the immunological heterogeneity within human populations and the virulence of infecting strains but may also include the chemical composition of contaminated food. A common denominator of foods associated with low infectious doses is a high fat content (e.g., cocoa butter in chocolate, milk fat in cheese, and animal fat in meat). *Salmonella* organisms may become entrapped within hydrophobic lipid micelles and be protected against the bactericidal action of gastric acidity. Food producers, processors, and distributors need to be reminded that low levels of salmonellae in a finished food product could lead to serious public health consequences. This could undermine the reputation and economic viability of the incriminated food manufacturer.

Table 7.5 Human infectious doses of *Salmonella*[a]

Food	Serovar(s)	Infectious dose (CFU)[b]
Eggnog	Meleagridis	10^6–10^7
	Anatum	10^5–10^7
Goat cheese	Zanzibar	10^5–10^{11}
Carmine dye	Cubana	10^4
Imitation ice cream	Typhimurium	10^4
Chocolate	Eastbourne	10^2
Hamburger	Newport	10^1–10^2
Cheddar cheese	Heidelberg	10^2
Chocolate	Napoli	10^1–10^2
Cheddar cheese	Typhimurium	10^0–10^1
Chocolate	Typhimurium	$\leq 10^1$
Paprika potato chips	Saint-Paul, Javiana, Rubislaw	$\leq 4.5 \times 10^1$
Alfalfa sprouts	Newport	$\leq 4.6 \times 10^2$
Ice cream	Enteritidis	$\leq 2.8 \times 10^1$

[a]Adapted from J.-Y. D'Aoust, *Int. J. Food Microbiol.* **24**:11–31, 1994.
[b]CFU, colony-forming units.

PATHOGENICITY AND VIRULENCE FACTORS

Specific and Nonspecific Human Responses

The presence of viable salmonellae in the human intestinal tract confirms the successful evasion of ingested organisms from nonspecific host defenses. Antibacterial lactoperoxidase in saliva, gastric acidity, mucoid secretions from intestinal cells, intestinal peristalsis, and sloughing of luminal epithelial cells synergistically inhibit bacterial colonization of the intestine. In addition to these hurdles to bacterial infection, the antibacterial action of phagocytic cells coupled with the immune responses mounts a formidable defense against the systemic spread of *Salmonella*. The human diarrheagenic response to foodborne salmonellosis results from the migration of the pathogen in the oral cavity to intestinal tissues and mesenteric lymph follicles (enterocolitis). The failure of host defense systems to hold the invasive *Salmonella* in check can degenerate into septicemia and other chronic clinical conditions.

Attachment and Invasion

The establishment of a human *Salmonella* infection depends on the ability of the bacterium to attach (colonization) and enter (invasion) intestinal cells. Salmonellae must successfully compete with indigenous gut microflora for suitable attachment sites on the intestinal wall. Upon contact with epithelial cells, *Salmonella* produces proteinaceous appendages on its surface. After bacterial attachment, signaling between the pathogen and the host cell results in *Salmonella* invasion of intestinal cells. Histologically, *Salmonella* invasion is characterized by membrane ruffling of epithelial cells and programmed cell death (apoptosis) of epithelial and phagocytic cells. Diarrhea associated with salmonellae now appears to be in response to bacterial invasion of intestinal cells rather than the action of a putative enterotoxin.

Several sets of genes are involved in the invasion process. The *inv* pathogenicity island is a multigenic locus consisting of 30 genes. Many of these genes encode the synthesis of enzymes and transcriptional activators responsible for the regulation, expression, and translocation of important

effectors to the surfaces of host cells. Genes in the pathogenicity island are useful targets for polymerase chain reaction (PCR)-based detection of *Salmonella* in foods.

Growth and Survival within Host Cells

In contrast to several bacterial pathogens, such as *Yersinia, Shigella,* and enteroinvasive *Escherichia coli,* which replicate within the cytoplasm of host cells, salmonellae are confined to endocytotic vacuoles in which bacterial replication begins within hours following internalization. The infected vacuoles move from the apical to the basal pole of the host cell, where *Salmonella* organisms are released into the tissue. A bacterial surface mechanism that facilitates the migration of salmonellae deeper into layers of tissue upon their release has been tentatively identified.

The systemic migration of *Salmonella* exposes it to phagocytosis and to the antibacterial conditions in the cytoplasm of the host defense cells. Survival of the bacterial cell within the hostile confines of the phagocytic cells determines the host's fate. Whether the host develops enteric fever from *Salmonella* infection is determined partly by the genetics of the host and the salmonellae. Intracellular pathogens have developed different strategies to survive within phagocytic cells. These include (i) escaping the phagosomes, (ii) inhibiting acidification of the phagosomes, (iii) preventing phagosome-lysosome fusion, and (iv) withstanding the toxic environment of the phagolysosome. Although there is evidence to support the ability of *Salmonella* to prevent acidification of the phagosomes, fusion with the lysosome, or maturation of the phagolysosome, *Salmonella* is a hardy bacterium that can survive and replicate within the bactericidal milieu of the acidified phagolysosome.

Virulence Plasmids

Virulence plasmids are large DNA structures that replicate in synchrony with the bacterial chromosome. These plasmids contain many virulence loci ranging from 30 to 60 megadaltons in size and occur with a frequency of one or two copies per chromosome. The presence of virulence plasmids within the genus *Salmonella* is limited and has been confirmed in serovars Typhimurium, Dublin, Gallinarum-Pullorum, Enteritidis, Choleraesuis, and Abortusovis. The highly infectious serovar Typhi does not carry a virulence plasmid. Although limited in its distribution in nature, the plasmid is self-transmissible. Gene products from the transcription of this plasmid aid the systemic spread and infection of tissues other than the intestine but not *Salmonella* adhesion to and invasion of epithelial cells. More specifically, virulence plasmids may enable carrier strains to rapidly multiply within host cells and overwhelm host defense mechanisms. These plasmids also confer on the salmonellae the ability to induce lysis of macrophages and elicit inflammatory response and enteritis in their animal hosts. The *Salmonella* virulence plasmids contain highly conserved nucleotide sequences and exhibit functional homology in which plasmid transfer from a wild-type serovar to another, plasmid-cured serovar restores the virulence of the recipient strain. The *Salmonella* virulence plasmid also contains a fimbrial operon that encodes an adhesin involved in colonization of the small intestine.

Other Virulence Factors

Siderophores are yet another part of the *Salmonella* virulence weaponry. These elements retrieve essential iron from host tissues to drive key cellular functions, such as the electron transport chain and enzymes associated

with iron cofactors. To this end, *Salmonella* must compete with host transferrin, lactoferrin, and ferritin ligands for available iron. For example, transferrin scavenges tissue fluids for Fe^{3+} ions to form Fe^{3+}-transferrin complexes that bind to surface host cell receptors. Upon internalization, the complexes dissociate and the released Fe^{3+} is complexed with ferritin for intracellular storage. Siderophores do not appear to be the only mechanism by which *Salmonella* can acquire iron from its host.

Diarrheagenic enterotoxin may be a *Salmonella* virulence factor. The release of toxin into the cytoplasm of infected host cells precipitates an activation of adenyl cyclase localized in the epithelial cell membrane and a marked increase in the cytoplasmic concentration of cyclic adenosine monophosphate (AMP) in host cells. The simultaneous fluid exsorption into the intestine results from a net secretion of Cl^{2+} ions and depressed Na^{+} absorption at the level of the intestinal villi. Enterotoxigenicity is a virulence phenotype of *Salmonella,* including serovar Typhi, which is expressed within hours following bacterial contact with the targeted host cells.

In addition to enterotoxin, *Salmonella* strains generally elaborate a thermolabile cytotoxic protein, which is localized in the bacterial outer membrane. Hostile environments, such as acidic pH and elevated (42°C) temperature, cause release of the toxin, possibly as a result of induced bacterial lysis. The virulence attribute of cytotoxin stems from its inhibition of protein synthesis and its lysis of host cells, thereby promoting the dissemination of viable salmonellae into host tissues.

Three additional virulence determinants located within or on the external surface of the *Salmonella* outer membrane are worth mentioning. The capsular polysaccharide Vi antigen occurs in most strains of serovar Typhi, in a few strains of serovar Paratyphi C, and rarely in serovar Dublin. The length of LPS that protrudes from the bacterial outer membrane not only defines the rough (short-LPS) and smooth (LPS) phenotypes, but also plays an important role in preventing attack by the host immune system. Porins are outer membrane proteins that function as transmembrane (outer membrane) channels in regulating the influx of nutrients, antibiotics, and other small molecular species. Low osmolarity, low nutrient availability, and low temperature can regulate the expression of these genes.

Summary

- Intensive animal husbandry practices in meat and poultry production and processing industries continue to make raw poultry and meats principal vehicles of human foodborne salmonellosis.
- Unless changes in agricultural and aquacultural practices are implemented, the prevalence of human foodborne salmonellosis will continue.
- Salmonellae are classified as serovars of two species.
- The emergence of multiple-antibiotic-resistant strains may be linked to the use of antibiotics in agriculture.
- Salmonellae quickly adapt to environments of high salinity and low pH, creating a problem with respect to control in food matrices.
- Virulence genes are located on the chromosome in pathogenicity islands and in large virulence plasmids.
- Epidemiologic evidence suggests a low infective dose.
- *Salmonella* invasion proteins may activate apoptosis in epithelial and phagocytic cell types.

Suggested reading

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Questions for critical thought

1. The typing scheme for *Salmonella* is based on two antigens. What is meant by the nomenclature (6,7:r:1,5)?
2. Serovar Enteritidis has been difficult to control in poultry. What unique characteristic(s) of this pathogen contributes to the control problem?
3. *Salmonella* DT104 is resistant to multiple antibiotics. Within the bacterial cell, where are the genes located and why is that important?
4. How might the ATR contribute to bacterial survival in food and subsequently in the host?
5. How could motility and antibodies against the flagella be used for detection of *Salmonella?*
6. To prevent foodborne illness associated with large meals, such as holiday dinners, what food-handling practices should be employed?
7. Individuals on antibiotic therapy are more prone to infection with *Salmonella.* Explain why.
8. Based on the impacts of pH, temperature, salinity, and a_w on the growth and survival of *Salmonella,* develop a hurdle technology to prevent the growth of *Salmonella* in the food of your choice.

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8

Campylobacter Species

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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Campylobacter* spp.
- understand what conditions in foods favor *Campylobacter* sp. growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by a *Campylobacter* sp.
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *Campylobacter* spp.
- identify environmental sources of the organism
- understand the roles of *Campylobacter* sp. toxins and virulence factors in causing foodborne illness

Outbreak

In many states, the sale of unpasteurized milk is illegal, since it is an important vehicle for the transmission of a number of foodborne pathogens, including *Campylobacter jejuni*. In Wisconsin, to circumvent the law against the sale of unpasteurized milk, an organic dairy operation initiated a cow-leasing program. Customers paid a fee to lease "part" of a cow, which farm operators milked; the milk from all leased cows was stored in a bulk tank (a large stainless steel tank). Customers could either pick up the milk at the farm or have the milk delivered to their homes. The farm operation was well run, meeting grade A standards for a farm shipping milk to a pasteurization plant. Customers participating in the cow lease program found out the hard way about campylobacteriosis. A total of 75 persons developed diarrhea, abdominal cramps, fever, and nausea after drinking unpasteurized milk from the organic dairy farm. The genetic fingerprints of *C. jejuni* isolates obtained from stool samples of patients and from milk samples collected on the farm were identical. *C. jejuni* can survive in unpasteurized milk held at 4°C. Finding *C. jejuni* in a farm environment is not unusual, and there are many routes by which milk can be contaminated. Pasteurization effectively kills *C. jejuni*.

INTRODUCTION

Campylobacter jejuni subsp. *jejuni* (hereafter referred to as *C. jejuni*) is one of many species and subspecies within the genus *Campylobacter*, family *Campylobacteraceae*. During the 1980s, an explosion of information on these bacteria was published, as is evident from the 1984 edition of *Bergey's Manual of Systematic Bacteriology*, which was published at a time when there were only eight species and subspecies within the genus *Campylobacter*. Since then, many investigators have become interested in studying the taxonomy

and clinical importance of campylobacters. This has greatly expanded the number of genera and species associated with this group of bacteria.

CHARACTERISTICS OF THE ORGANISM

Authors' note

Campylobacter *is a fragile organism. It cannot tolerate drying and can be killed by oxygen. It grows only if there is less than the atmospheric amount of oxygen present. Freezing reduces the number of* Campylobacter *bacteria present on raw meat.*

The genera *Campylobacter* and *Arcobacter* are included in the family *Campylobacteraceae.* The family *Campylobacteraceae* includes at present 18 species and subspecies within the genus *Campylobacter* and 4 species in the genus *Arcobacter. Campylobacter* is the type genus within the family *Campylobacteraceae.* The organisms are curved, S-shaped, or spiral rods that are 0.2 to 0.9 μm wide and 0.5 to 5 μm long. They are gram-negative, non-spore-forming rods that may form spherical or coccoid bodies in old cultures or cultures exposed to air for prolonged periods. The organisms are motile by means of a single polar unsheathed flagellum at one or both ends. The various species are microaerobic (5% oxygen and 10% carbon dioxide), with a respiratory type of metabolism. Some strains grow aerobically or anaerobically. An atmosphere containing increased hydrogen may be required by some species for microaerobic growth. As determined by pulsed-field gel electrophoresis (PFGE), *C. jejuni* and *Campylobacter coli* have genomes ~1.7 megabases in size, which is about one-third the size of the *Escherichia coli* genome.

ENVIRONMENTAL SUSCEPTIBILITY

C. jejuni is susceptible to a variety of environmental conditions that make it unlikely to survive for long periods of time outside the host. The organism does not grow at temperatures below 30°C, is microaerobic, and is sensitive to drying, high-oxygen conditions, and low pH. The decimal reduction time for campylobacters varies and depends on the food source and temperature. Thus, the organism should not survive in food products brought to adequate cooking temperatures. The organisms are susceptible to gamma irradiation (1 kilogray), but the rate of killing depends on the type of product being processed. Irradiation is less effective for frozen materials than for refrigerated or room temperature meats. Early-log-phase cells are more susceptible than cells grown to log or stationary phase. *Campylobacter* spp. are more radiation sensitive than other foodborne pathogens, such as salmonellae and *Listeria monocytogenes.* Irradiation treatment that is effective against salmonellae and *L. monocytogenes* should be sufficient to kill *Campylobacter* spp.

Campylobacter spp. are susceptible to low pH and are killed at pH 2.3. Campylobacters remain viable and grow in bile at 37°C and survive better in feces, milk, water, and urine held at 4°C than in material held at 25°C. The maximum periods of viability of *Campylobacter* spp. at 4°C were 3 weeks in feces, 4 weeks in water, and 5 weeks in urine. Freezing reduces the number of *Campylobacter* organisms in contaminated poultry, but even after being frozen to −20°C, small numbers of *Campylobacter* organisms can be recovered. Given how "fragile" the microbe is, surprisingly in 2005 *Campylobacter* was second only to *Salmonella* in number of laboratory-confirmed cases of foodborne illness.

RESERVOIRS AND FOODBORNE OUTBREAKS

Many animals serve as reservoirs of *C. jejuni* for human disease. Reservoirs for infection include rabbits, rodents, wild birds, sheep, horses, cows, pigs, poultry, and domestic pets (Table 8.1). Contaminated vegetables and shell-

Table 8.1 Reservoirs and disease-associated species in the family *Campylobacteraceae*[a]

Organism	Reservoir(s)	Disease, sequelae, or comments	
		Humans	Animals
C. jejuni subsp. *jejuni*	Humans, other mammals, birds	Diarrhea, systemic illness, GBS	Diarrhea in primates
C. jejuni subsp. *doylei*	Unknown	Diarrhea	
C. fetus subsp. *fetus*	Cattle, sheep	Systemic illness, diarrhea	Abortion
C. fetus subsp. *venerealis*	Cattle		Infertility
C. coli	Pigs, birds	Diarrhea	
C. lari	Birds, dogs	Diarrhea	
C. upsaliensis	Domestic pets	Diarrhea	Diarrhea
C. hyointestinalis subsp. *hyointestinalis*	Cattle, pigs, hamsters, deer	Rare; proctitis, diarrhea	Proliferative enteritis
C. hyointestinalis subsp. *lawsonii*	Pigs		
C. mucosalis	Pigs	Rare; diarrhea	Proliferative enteritis
C. hyoilei	Pigs		
C. sputorum biovar sputorum	Humans	Oral cavity abscesses	Genital tract of bulls; abortion in sheep
C. sputorum biovar paraureolyticus	Cattle	Diarrhea	
C. sputorum biovar faecalis	Cattle, sheep		Enteritis
C. concisus	Humans	Periodontal disease	
C. curvus	Humans	Periodontal disease	
C. rectus	Humans	Periodontal disease, pulmonary infections	
C. showae	Humans	Periodontal disease	
C. helveticus	Domestic pets		Diarrhea
C. gracilis	Humans	Infections of head, neck, other sites	
Arcobacter butzleri	Cattle, pigs	Diarrhea	Diarrhea, abortion
Arcobacter cryaerophilus	Cattle, sheep, pigs	Diarrhea, bacteremia	Abortion
Arcobacter skirrowii	Cattle, sheep, pigs		Abortion, diarrhea; isolated from genital tract of bulls
Arcobacter nitrofigilis	Plants		

[a]Some data are from M. B. Skirrow, *J. Comp. Pathol.* 111:113–149, 1994.

fish may also be vehicles of infection. *Campylobacter* is frequently isolated from water and water supplies, and water has been the source of infection in some outbreaks. *C. jejuni* can remain dormant in water in a state that has been called *viable but nonculturable* (VBNC). Under unfavorable conditions, the organism essentially remains dormant and cannot be easily recovered on growth media. The role of these forms as a source of infection for humans is not clear.

From 1978 to 1996, there were more than 100 outbreaks of *Campylobacter* enteritis reported to the Centers for Disease Control and Prevention, affecting more than 10,000 individuals (Table 8.2). Campylobacteriosis is not confined to the United States; over 30,000 cases were associated with 900 outbreaks worldwide from 1978 to 2003. The vehicles of *Campylobacter* outbreaks have changed over the past 2 decades. Water and unpasteurized milk were responsible for over one-half of the outbreaks between 1978 and 1987, whereas other foods accounted for >80% of outbreaks between 1988 and 1996. In a 10-year review of outbreaks from 1981 to 1990, 20 outbreaks

Table 8.2 Foodborne and waterborne outbreaks of *Campylobacter* infections reported in the United States, by vehicle, 1978 to 1996[a]

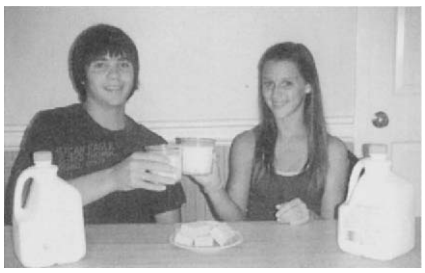
Origin	No. of outbreaks	No. of outbreak-associated cases
Foodborne		
Milk	30	1,212
Chicken	2	16
Turkey	1	11
Beef	1	24
Other meat	2	30
Eggs	1	26
Fruits	4	227
Other foods	4	251
Multiple foods	10	411
Unknown food	42	2,775
Waterborne		
Community water supply	8	5,068
Other water supply	4	104
Total	109	10,155

[a]Reprinted from C. R. Friedman, J. Neimann, H. C. Wegener, and R. V. Tauxe, p. 121–138, *in* I. Nachamkin and M. J. Blaser (ed.), *Campylobacter*, 2nd ed. (ASM Press, Washington, DC, 2000), with permission.

occurred, affecting 1,013 individuals who drank raw milk (Fig. 8.1). The attack rate was 45%. At least one outbreak occurred each year, and most of the outbreaks involved children who had gone on field trips to dairy farms.

The seasonal distribution of outbreaks is somewhat different from that of sporadic cases. Milk- and waterborne outbreaks tend to occur in the spring and fall but do not occur frequently in the summer. Sporadic cases are usually most frequent during the summer. In contrast to the relatively low occurrence of outbreaks caused by campylobacters, the bacteria have been identified in many studies as being among the most common causes of foodborne illness in the United States. In 1995, the Centers for Disease Control and Prevention, the U.S. Department of Agriculture, and the Food and Drug Administration developed an active surveillance system for foodborne diseases, including *Campylobacter* infections, called the Foodborne Diseases Active Surveillance Network, also known as FoodNet. On the basis of a variety of data, including FoodNet incidence data, U.S. Census Bureau data, and data from other sources, it is estimated that there are 2.4 million *Campylobacter* infections in the United States each year. Since campylobacteriosis is self-limiting, the true number of cases that occur each year can only be estimated. This is similar to the incidence of infection in the United Kingdom and other developed nations. Cases occur more often during the summer months and usually follow the ingestion of improperly handled or cooked food, primarily poultry products. Other exposures include drinking raw milk, contaminated surface water, overseas travel, and contact with domestic pets (Table 8.3). FoodNet data show that the incidence of *Campylobacter* infections decreased by 30% from 1996 to 2005 in the United States. The decrease is likely related to several factors including implementation of government programs (Hazard Analysis Critical Control Points [HACCP] in meat and poultry industry), enhanced monitoring of finished product by industry, and consumer knowledge.

Figure 8.1 Drinking a large glass of cold milk can be refreshing. However, if the milk has not been pasteurized, it may contain a number of microbes, including *Campylobacter jejuni*, that when ingested result in illness (diarrhea, abdominal cramps, and fever). The risks of drinking unpasteurized milk far outweigh any benefits that may be attributed to consumption of such milk.



CHARACTERISTICS OF DISEASE

C. jejuni and *C. coli*

Most *Campylobacter* species are associated with intestinal tract infection. Infections in other areas of the body are common with some species, such as *Campylobacter fetus* subsp. *fetus* (hereafter referred to as *C. fetus*). *C. jejuni* and *C. coli* have been recognized since the 1970s as agents of gastrointestinal tract infection.

C. jejuni and *C. coli* are the most common *Campylobacter* species associated with diarrheal illness, and they are indistinguishable. The ratio of *C. jejuni* to *C. coli* is not known, because most laboratories do not routinely distinguish between the organisms. In the United States, an estimated ~5 to 10% of cases attributed to *C. jejuni* are actually due to *C. coli;* this figure may be higher in other parts of the world.

A spectrum of illness can occur during *C. jejuni* or *C. coli* infection, and patients may be *asymptomatic* (show no sign of illness) to severely ill. Symptoms and signs usually include fever, abdominal cramping, and diarrhea (with or without blood) that lasts several days to >1 week. Symptomatic infections are usually self-limited, but relapses may occur in 5 to 10% of untreated patients. Recurrence of abdominal pain is common. Other infections and diseases include bacteremia, bursitis, urinary tract infection, meningitis, endocarditis, peritonitis, erythema nodosum, pancreatitis, abortion and neonatal sepsis, reactive arthritis, and Guillain-Barré syndrome (GBS). Deaths directly attributable to *C. jejuni* infection are rarely reported.

C. jejuni and *C. coli* are susceptible to many antibiotics, including macrolides, fluoroquinolones, aminoglycosides, chloramphenicol, and tetracycline. Erythromycin is the drug of choice for treating *C. jejuni* gastrointestinal tract infections, but ciprofloxacin is a good alternative drug. Early therapy of *Campylobacter* infection with erythromycin or ciprofloxacin is effective in eliminating the campylobacters from stool and may also reduce the duration of symptoms associated with infection.

C. jejuni is generally susceptible to erythromycin, with resistance rates of <5%. Rates of erythromycin resistance in *C. coli* vary, with up to 80% of strains having resistance in some studies. Although ciprofloxacin is effective in treating *Campylobacter* infections, resistant strains have emerged. The resistance may be linked to the use of similar antibiotics (fluoroquinolones) in poultry. In 2005, the Food and Drug Administration Center for Veterinary Medicine banned the use of fluoroquinolones, such as enrofloxacin, in poultry.

Other *Campylobacter* Species

In contrast to *C. jejuni, C. fetus* is primarily associated with bacteremia and non-intestinal-tract infections. *C. fetus* is also associated with abortions, septic arthritis, abscesses, meningitis, endocarditis, mycotic aneurysm, thrombophlebitis, peritonitis, and salpingitis. Although intestinal tract infections occur with this species, the number of cases is low, because the organism does not grow well at 42°C. *C. fetus* is susceptible to the antibiotic cephalothin. *C. fetus* subsp. *venerealis* is not associated with human infection.

EPIDEMIOLOGIC SUBTYPING SYSTEMS USEFUL FOR INVESTIGATING FOODBORNE ILLNESSES

Many typing systems have been devised to study the epidemiology of *Campylobacter* infections. These methods include biotyping, serotyping, bacteriocin sensitivity, detection of preformed enzymes, lectin binding,

Table 8.3 Isolation of *Campylobacter* from different food sources[a]

Product	% Positive samples
Chicken	14–98
Turkey	3–25
Duck	48
Goose	38
Cow's milk	0–12.3
Goat's milk	0
Ewe's milk	0
Beef	0–23.6
Pork	1–23.5
Lamb	0–15.5
Sheep	3
Offal	47
Mussels	47–69
Oysters	6–27
Vegetables	<5

[a]Source: W. Jacobs-Reitsma, p. 467–481, *in* I. Nachamkin and M. J. Blaser (ed.), *Campylobacter,* 2nd ed. (ASM Press, Washington, DC, 2000).

phage typing, multilocus enzyme electrophoresis, and molecularly based methods, such as PFGE, ribotyping, and restriction fragment length polymorphism combined with polymerase chain reaction (PCR) methods (see chapter 5).

The most frequently used phenotypic systems are biotyping and serotyping. The two major serotyping schemes used worldwide detect heat-labile and O antigens. Serotyping systems are simple to perform and have good ability to discriminate between strains.

During the past 5 years, molecularly based methods have come to the forefront for studying the epidemiology of *Campylobacter* infections. PFGE is one of the most powerful methods for molecular analysis of *Campylobacter* species. It has excellent discriminatory power. A variety of PCR-based methods, including ribotyping, random amplified polymorphic DNA, amplified fragment length polymorphism, multiplex PCR-restriction fragment length polymorphism, and real-time PCR, have all been used in various applications to study *Campylobacter* spp. Since the sequencing of the genome the use of microarray technology has been developed as a typing system (see chapter 5 for explanations of these methods).

INFECTIVE DOSE AND SUSCEPTIBLE POPULATIONS

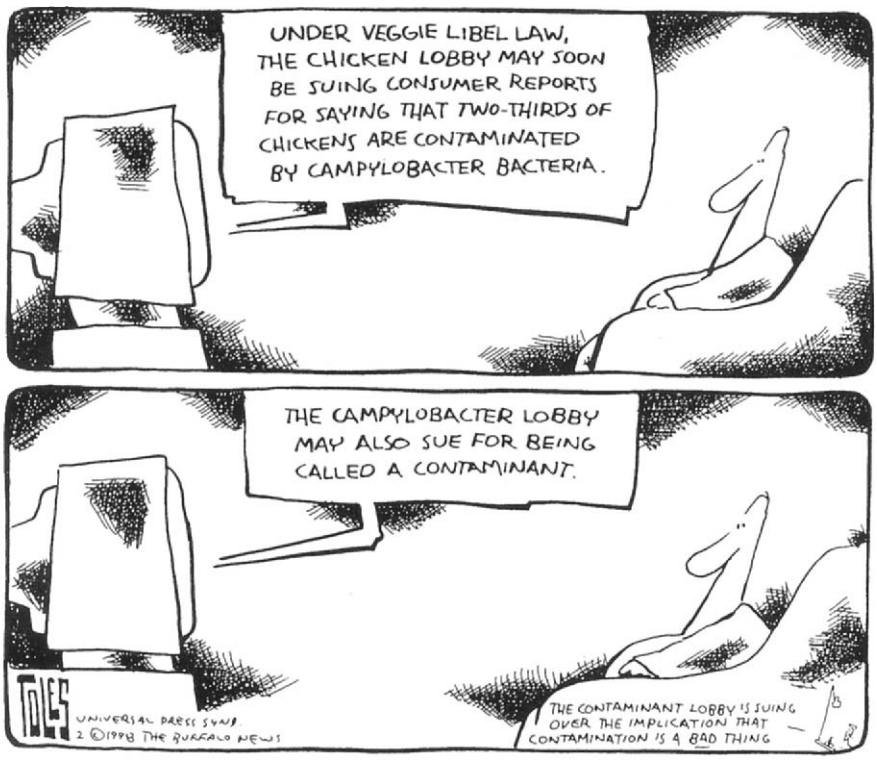
C. jejuni is susceptible to low pH, and hence, stomach acidity kills most campylobacters. The infective dose of *C. jejuni* is not high, with <1,000 organisms causing illness. The ability to cause infection may vary greatly between strains. Human volunteers (18%) became ill when infected with 10^8 colony-forming units (CFU) of a strain designated A3249, but 46% of the volunteers became ill when infected with another strain, 81-176. In an outbreak of *Campylobacter* infection after the ingestion of raw milk, the number of individuals who became ill and the severity of illness seemed to be linked to the amount of milk ingested (the more milk consumed, the greater the number of *Campylobacter* organisms consumed) (Fig. 8.1).

The greatest number of infections in the United States occur in young children and young adults 20 to 40 years of age. The incidence of *Campylobacter* infection in developing countries, such as Mexico and Thailand, may be 10 to 100 times higher than in the United States. In developing countries, in contrast to developed countries, campylobacters are frequently isolated from individuals who may or may not have diarrheal disease. Most infections occur in infancy and early childhood, and the incidence decreases with age. As a child matures, his or her immune system can fight off *Campylobacter* infections, thereby decreasing the number of illnesses. Travelers to developing countries may become infected with *Campylobacter.*

The elderly are more likely to get *bacteremia* (presence of bacteria in the blood) than any other group (infants, young adults, etc.). Continual diarrheal illness and bacteremia may occur in immunocompromised individuals, such as in patients with human immunodeficiency virus infection.

VIRULENCE FACTORS AND MECHANISMS OF PATHOGENICITY

Little is known about the mechanism by which *C. jejuni* causes human disease. *C. jejuni* can cause an enterotoxigenic-like illness with loose or watery diarrhea or an inflammatory intestinal disease with fever. The mechanism of disease may be *invasive* (entering into intestinal cells), since some indi-



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viduals have blood in their feces and may develop bacteremia. A major problem in understanding the pathogenesis of *Campylobacter* infection is the lack of suitable animal models.

Cell Association and Invasion

The uptake of *C. jejuni* by host cells occurs by intimate binding of the bacterium with the host cell surface by using many types of adhesins. Following contact, *C. jejuni* produces at least 14 new proteins. Production corresponds to a rapid increase in uptake and changes in the host cell membrane. At a later point, additional proteins are made and secreted into the cytoplasm of host target cells. Following attachment and internalization, *Campylobacter*-infected cells release molecules that promote the recruitment of white blood cells to the site of the infection. Induction of host cell death by *C. jejuni* may enhance the survival and spread of the pathogen.

Flagella and Motility

Campylobacter species are motile and have a single polar, unsheathed flagellum at one or both ends. Motility and flagella are important determinants for the entry process. *Campylobacter* colonization and/or infection in a variety of animal models is dependent on intact motility and full-length flagella; however, other factors may be involved. Two genes are involved in the expression of the flagellar filament, and motility is essential for colonization. Components of intestinal mucin, particularly L-fucose, attract *C. jejuni*. Movement of *C. jejuni* toward such compounds may be important in the pathogenesis of infection.

Toxins

C. jejuni may produce an enterotoxin similar to cholera toxin. No genetic evidence is available to support the presence of this toxin. The role of cytolethal distending toxin in pathogenesis is unknown. Most strains of *C. jejuni* have *cdt* genes, but the amounts of toxin produced by the strains vary. The toxin may have several roles in pathogenesis. Greater information on toxins of the pathogen should be forthcoming, since the genome has been sequenced.

Other Factors

The regulation of genes in response to environmental changes is becoming an important area of *Campylobacter* research. Campylobacters are microaerobic, and *C. jejuni* has a higher temperature for optimal growth, 42°C, than other intestinal bacterial pathogens. Understanding the effects of environmental signals on the growth, metabolism, and pathogenicity of *Campylobacter* strains will have a major impact on the ability to control campylobacters in the environment and food chain. Such pathways include response to iron; oxidative stress; temperature regulation, including cold and heat shock responses; and starvation.

Autoimmune Diseases

Campylobacter infection is a major trigger of GBS, an acute immune-mediated paralytic disorder affecting the peripheral nervous system. The pathogenesis of GBS induced by *C. jejuni* is not clear. The bacterial lipopolysaccharide causes an immune response. The antibodies produced recognize not only the lipopolysaccharide but also peripheral nerve tissue. This is likely a major mechanism of *Campylobacter*-induced GBS.

IMMUNITY

Infection with *Campylobacter* species results in protective immunity and is likely antibody mediated. Rechallenge with the same strains 28 days after the initial challenge resulted in protection against illness but not necessarily against colonization by *C. jejuni.* People who are regularly exposed to the pathogen through drinking raw milk are less likely to become ill. In developing countries, where *Campylobacter* infections are endemic, immunity to *Campylobacter* infection appears to be age dependent. As a child matures, he or she is less likely to become ill.

Flagellin is an important immunogen during *Campylobacter* infection. Antibodies against this protein correlate to some degree with protective immunity. Breast-feeding provides protection against *Campylobacter* infection in developing countries. Antibodies against flagellin present in breast milk appear to be associated with protection of infants against infection.

Other strategies to prevent the spread of infection to humans include improved hygiene practices during broiler chicken production, such as decontamination of water supplies; use of competitive exclusion flora, which may prevent *C. jejuni* colonization of young chicks; and immunological approaches through the use of animal vaccines.

Summary

- Campylobacteriosis linked to consumption of contaminated food is the most prevalent form of foodborne illness in the United States.

- *C. jejuni* is commonly isolated from poultry.
- Thermoprocessing easily inactivates campylobacters.
- *Campylobacter* infection is a major trigger of GBS.
- Motility is essential for colonization.
- Five to 10% of cases attributed to *C. jejuni* are actually due to *C. coli*.
- The infective dose of *C. jejuni* may be <1,000 organisms.

Suggested reading

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Questions for critical thought

1. How does oxygen concentration influence the growth of *C. jejuni?*
2. How is *C. jejuni* infection linked to GBS?
3. What is the impact of antibiotic-resistant *C. jejuni* on human health? How might the use of antibiotics in poultry production practices influence resistance?
4. Campylobacters can exist in a VBNC state. In what environments do they enter this state? Can the organism recover from this state? Why is entry into the VBNC state important for survival of the organism?
5. What practices can be implemented to control campylobacters in food?
6. Describe how you would differentiate *C. jejuni* from *E. coli* O157:H7 microbiologically and based on signs and symptoms of illness.
7. The incidence of *Campylobacter* infection in developing countries, such as Mexico and Thailand, is severalfold higher than in the United States. What factors may account for such a great difference in the infection rate?
8. *C. jejuni* is zoonotic. What does this statement mean?

9. You are a new employee at a start-up biotechnology firm and have been asked to develop a vaccine against *C. jejuni*. Outline your plan of attack. Consider what you know about the pathogen and the illness it causes and what others have done in the past with respect to vaccine development.
10. Explain how the banning of fluoroquinolone antibiotics in poultry may impact resistance of *Campylobacter* species. What, if any, impact might this have on poultry health?

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9

Enterohemorrhagic *Escherichia coli*

Outbreak
Introduction
Categories of *E. coli*
Characteristics of *E. coli* O157:H7 and Non-O157 EHEC
Acid Tolerance
Antibiotic Resistance
Inactivation by Heat and Irradiation
Reservoirs of *E. coli* O157:H7
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Escherichia coli* O157:H7
- understand what conditions in foods favor *E. coli* O157:H7 growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *E. coli* O157:H7
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *E. coli* O157:H7
- identify environmental sources of *E. coli* O157:H7
- understand the role of *E. coli* O157:H7 toxins and virulence factors in causing foodborne illness

Outbreak

In June 1998, physicians in a small Wyoming town noted an increase in bloody diarrhea among town residents. Further investigation revealed that visitors attending a family reunion and tourists from surrounding states had also become ill. A total of 157 people became ill, but no deaths occurred. Epidemiological evidence demonstrated that illness was significantly associated with nonchlorinated water from the municipal water system. This outbreak underscores the ability of *E. coli* O157:H7 to survive in water and the need to use only protected waters that have been sufficiently processed.

Attending the local county fair is often the highlight of the summer for people living in rural Lane County, Oregon. Walking through the barns to observe the sheep, goats, and other farm animals is entertaining and fascinating. However, for 82 people who attended the fair during the summer of 2002, the visit left them sick from *E. coli* O157:H7 infection. The culprit was not food, water, or even direct contact with animals but rather *E. coli* O157:H7 organisms that were in the air, based on microbiological analysis of dust and grime samples taken inside the goat and sheep barn. There were 74 confirmed cases (based on positive stool culture) and 8 presumed cases; approximately two-thirds of the cases involved children <6 years old. Other outbreaks linked to airborne *E. coli* O157:H7 in barns have occurred.

Consuming a fresh, cool, crisp spinach salad on a hot summer day pleases the palate and cools the body. However, during the summer of 2006, consuming fresh spinach resulted in 206 cases of illness and three deaths in 26 states. Investigators determined that the spinach was contaminated with *E. coli* O157:H7. How, when, and where the spinach became contaminated could not be conclusively determined. Wild pigs captured in the area where the spinach was grown tested positive for the same strain of *E. coli* O157:H7 that was associated with the bagged spinach and isolated from patients. The U.S. Food and Drug Administration emphasized the need to implement good agricultural practices.

INTRODUCTION

Large outbreaks of *E. coli* O157:H7 infection involving hundreds of cases have occurred in the United States, Canada, Japan, and the United Kingdom. The largest outbreak worldwide occurred from May to December 1996 in Japan, involving >11,000 reported cases. In the same year, 21 elderly people died in a large outbreak involving 501 cases in central Scotland. Although *E. coli* O157:H7 is still the predominant serotype of enterohemorrhagic *E. coli* (EHEC) in the United States, Canada, the United Kingdom, and Japan, an increasing number of outbreaks and sporadic cases related to other EHEC serotypes have occurred. A large epidemic involving several thousand cases of *E. coli* O157:NM infection occurred in Swaziland and South Africa following consumption of contaminated surface water. In continental Europe, Australia, and Latin America, non-O157 EHEC infections are more common than *E. coli* O157:H7 infections. Details of many reported foodborne and waterborne outbreaks of EHEC infection are provided in Table 9.1. There are no distinguishing biochemical phenotypes for non-O157 EHEC serotypes, making screening for these bacteria difficult and labor-intensive. The prevalence of non-O157 EHEC infections may be greatly underestimated.

Categories of *E. coli*

Diarrheagenic *E. coli* is categorized into specific groups based on virulence properties, mechanisms of pathogenicity, clinical syndromes, and distinct O:H serotypes. *E. coli* isolates are serologically differentiated based on three major surface antigens that allow serotyping: the O (somatic), H (flagellar), and K (capsule) antigens. The categories of diarrheagenic *E. coli* are enteropathogenic *E. coli* (EPEC), enterotoxigenic *E. coli* (ETEC), enteroinvasive *E. coli* (EIEC), diffusely adhering *E. coli* (DAEC), enteroaggregative *E. coli* (EAEC), and EHEC. All categories of *E. coli* described may be shed in the feces of infected humans, creating the potential for spread to other humans, animals, and the environment (Box 9.1). This chapter focuses primarily on EHEC, which causes the most severe illness.

EPEC

EPEC can cause severe diarrhea in infants, especially in developing countries. EPEC was previously associated with outbreaks of diarrhea in nurseries in developed countries. The major O serogroups associated with illness include O55, O86, O111ab, O119, O125ac, O126, O127, O128ab, and O142. Humans are an important reservoir. EPEC organisms have been shown to induce lesions in cells to which they adhere, and they can invade epithelial cells.

ETEC

ETEC is a major cause of infantile diarrhea in developing countries. It is also the bacterium most frequently responsible for traveler's diarrhea. ETEC colonizes the small intestine via fimbrial colonization factors and produces a heat-labile or heat-stable enterotoxin that causes fluid accumulation and diarrhea. The most frequent ETEC serogroups include O6, O8, O15, O20, O25, O27, O63, O78, O85, O115, O128ac, O148, O159, and O167. Humans are the principal reservoir of ETEC strains that cause human illness.

EIEC

EIEC causes nonbloody diarrhea and dysentery similar to that caused by *Shigella* spp. by invading and multiplying within intestinal epithelial cells. Like that of *Shigella*, the invasive capacity of EIEC is associated with the

Table 9.1 Representative foodborne and waterborne outbreaks of *E. coli* O157:H7 and other EHEC infections[a]

Yr	Mo	Location	No. of cases (deaths)	Setting	Vehicle
1982	2	Oregon	26	Community	Ground beef
1982	5	Michigan	21	Community	Ground beef
1985		Canada	73 (17)	Nursing home	Sandwiches
1987	6	Utah	51	Custodial institution	Ground beef; person to person
1988	10	Minnesota	54	School	Precooked ground beef
1989	12	Missouri	243	Community	Water
1990	7	North Dakota	65	Community	Roast beef
1991	11	Massachusetts	23	Community	Apple cider
1991	7	Oregon	21	Community	Swimming water
1992	12	Oregon	9	Community	Raw milk
1993	1	California, Idaho, Nevada, Washington	732 (4)	Restaurant	Ground beef
1993	3	Oregon	47	Restaurant	Mayonnaise?
1993	8	Oregon	27	Restaurant	Cantaloupe
1994	11	Washington, California	19	Home	Salami
1995	10	Kansas	21	Wedding	Punch, fruit salad
1995	7	Montana	74	Community	Leaf lettuce
1995	9	Maine	37	Camp	Lettuce
1996	5, 6	Connecticut, Illinois	47	Community	Mesclun lettuce
1996	7	Osaka, Japan	7,966 (3)	Community	White radish, sprouts
1996	10	California, Washington, Colorado	71 (1)	Community	Apple juice
1996	11	Central Scotland	501 (21)	Community	Cooked meat
1997	7	Michigan	60	Community	Alfalfa sprouts
1997	11	Wisconsin	13	Church banquet	Meatballs, coleslaw
1998	6	Wyoming	114	Community	Water
1998	7	North Carolina	142	Restaurant	Coleslaw
1998	9	California	20	Church	Cake
1999	8	New York	900 (2)	Fair	Well water
2000	11	Iowa, Minnesota, Wisconsin	52	Community	Lettuce
2002	7	10 States	38	Community	Ground beef
2002	8	Washington	32	Camp	Ground beef
2005	9, 10	Minnesota	23	Community	Prepackaged lettuce
2006	8–10	26 States, Canada	206 (3)	Community	Prepackaged fresh spinach

[a] *E. coli* O157:H7 unless otherwise noted.

presence of a large plasmid (ca. 140 megadaltons [MDa]) that encodes several outer membrane proteins involved in invasiveness. The antigenicities of these outer membrane proteins and the O antigens of EIEC are closely related. The principal site of bacterial localization is the colon, where EIEC organisms invade and grow within epithelial cells, causing cell death. Humans are a major reservoir, and the serogroups most frequently associated with illness include O28ac, O29, O112, O124, O136, O143, O144, O152, O164, and O167. Among these serogroups, O124 is the serogroup most commonly encountered.

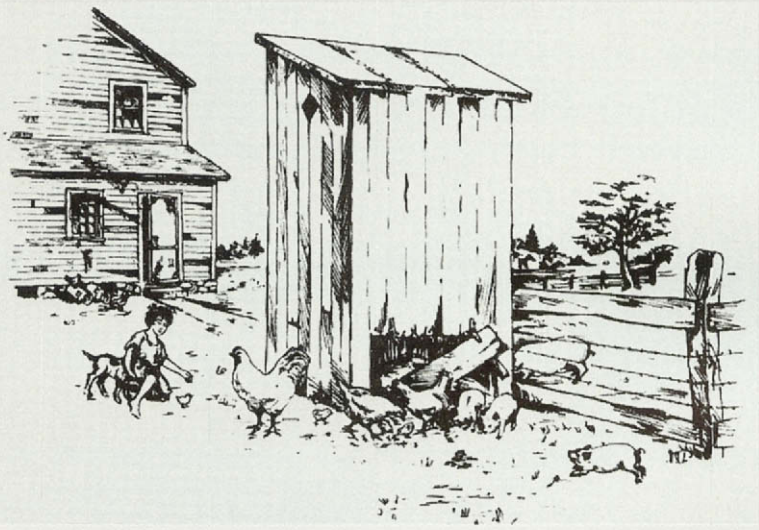
DAEC

DAEC has been associated with diarrhea primarily in young children who are older than infants. The relative risk of DAEC-associated diarrhea increases with age from 1 to 5 years. The reason for this age-related infection

BOX 9.1

Conditions that can lead to the spread of pathogenic *E. coli*

This historic illustration of an old farm outhouse (toilet) underscores what science has demonstrated: pathogens present in human feces such as *E. coli* (EPEC, ETEC, and EHEC) can readily be spread to farm animals, humans, and water through improper controls. In this case, the placement of a dilapidated outhouse near the main residence allows easy access to livestock that may eventually become dinner.



is unknown. DAEC is most commonly of serogroup O1, O2, O21, or O75. Typical symptoms of DAEC infection are mild diarrhea without blood. DAEC strains adhere in a random fashion to HEp-2 or HeLa cell lines. DAEC does not usually produce heat-labile or heat-stable enterotoxin or elevated levels of Shiga toxin (Stx).

EAEC

EAEC is associated with persistent diarrhea in infants and children in several countries worldwide. These *E. coli* strains are different from the other types of pathogenic *E. coli* because of their ability to produce a characteristic pattern of aggregative adherence on HEp-2 cells. EAEC adheres with the appearance of stacked bricks to the surfaces of HEp-2 cells. Serogroups associated with EAEC include O3, O15, O44, O77, O86, O92, O111, and O127. A gene probe derived from a plasmid associated with EAEC strains has been developed to identify *E. coli* strains of this type. More epidemiological information is needed to determine the significance of EAEC as a cause of diarrheal disease.

EHEC

EHEC was first recognized as a human pathogen in 1982, when *E. coli* O157:H7 was identified as the cause of two outbreaks of bloody diarrhea. Since then, other serotypes of *E. coli*, such as O26, O111, and sorbitol-fermenting O157:NM, also have been associated with cases of bloody diarrhea and are classified as EHEC. Serotype O157:H7 is the main cause of EHEC-associated disease in the United States and many other countries.

C O L O N I Z A T I O N

G Y A D I M S A L P D R

R T C L X U R E M I C S

O H I P O A H C A F P D

U J M Q T F A R B D W I

N H E M O R R H A G I C

D E H N R H M X R F I A

B L C B E B L O O D Y T

E P O A T C E F E X S T

E O I T N R S R P Y T L

F E B H E T S C S O O E

E P Y T O R E S O C T Z

Clues

Condition to which enterohemorrhagic *E. coli* is unusually tolerant. _____

Enterohemorrhagic *E. coli* has many unusual _____ characteristics in comparison with "normal" *E. coli.*

An especially nasty type of diarrhea caused by enterohemorrhagic *E. coli.* _____

The primary reservoir of enterohemorrhagic *E. coli.* _____

One of the first nonmeat foods associated with enterohemorrhagic *E. coli,* whose presence led the federal government to require that it be pasteurized. _____

The _____ of the colon by enterohemorrhagic *E. coli* is one of the first steps by which it causes illness.

If a foodborne pathogen does not cause vomiting, it probably causes _____.

The first food implicated in disease caused by *E. coli* O157:H7. _____

If one cooks contaminated hamburgers (or for that matter, salads) *really* well, they become _____.

Something that causes bleeding is called _____.

The second leading source of illness caused by enterohemorrhagic *E. coli.* _____

Element that carries the genes for enterohemorrhagic *E. coli's* pathogenicity. _____

O157:H7 is the _____ of one of the most common kinds of enterohemorrhagic *E. coli.*

Part of HUS indicative of kidney failure. _____

Type of toxin produced by ETEC. _____

All EHEC strains produce factors *cytotoxic* (deadly) to African green monkey kidney (Vero) cells. Thus, they are named verotoxins or Stxs because of their similarity to the Stx produced by *Shigella dysenteriae.* Stx-producing *E. coli* infections are linked to a severe and sometimes fatal condition, hemolytic-uremic syndrome (HUS). *E. coli* organisms of many different serotypes produce Stxs; hence, they have been named Stx-producing *E. coli* (or EHEC). More than 200 serotypes of EHEC have been isolated from humans, but only those strains that cause bloody diarrhea are considered to

be EHEC. Major non-O157:H7 EHEC serotypes include O26:H11, O111:H8, and O157:NM. Several outbreaks of infection with the EHEC *E. coli* O111 have occurred worldwide. In June 1999, an outbreak of *E. coli* O111:H8 infection involving 58 cases occurred at a teenage cheerleading camp in Texas. Contaminated ice was the implicated vehicle. In Australia, from January to February 1995 a cluster of 23 cases of HUS were associated with consumption of *E. coli* O111:NM-contaminated semidry fermented sausage. Since *E. coli* O157:H7 is the most common serotype of EHEC and because more is known about this serotype than about other serotypes of EHEC, this chapter focuses on *E. coli* O157:H7 (Box 9.2).

CHARACTERISTICS OF *E. COLI* O157:H7 AND NON-O157 EHEC

E. coli is a common part of the normal microbial population in the intestinal tracts of humans and other warm-blooded animals. Most *E. coli* strains are harmless; however, some strains are pathogenic and cause diarrheal disease. At present, a total of 167 O antigens, 53 H antigens, and 74 K antigens have been identified. It is considered necessary to determine only the O and H antigens in order to serotype strains of *E. coli* associated with diarrheal disease. The O antigen identifies the serogroup of a strain, and the H antigen identifies its serotype. The application of serotyping to isolates associated with diarrheal disease has shown that particular serogroups often fall into one category of diarrheagenic *E. coli*. However, some serogroups, such as O55, O111, O126, and O128, appear in more than one category.

E. coli O157:H7 was first identified as a foodborne pathogen in 1982, although there had been prior isolation of the organism, which was identified later among isolates at the Centers for Disease Control and Prevention. The earlier isolate was from a California woman who had bloody diarrhea in 1975. In addition to production of Stx(s), most strains of *E. coli* O157:H7 also have several characteristics uncommon in most other *E. coli* strains, i.e., the inability to grow well, if at all, at temperatures of ≥44.5°C in *E. coli* broth; inability to ferment sorbitol within 24 h; inability to produce β-glucuronidase (i.e., inability to hydrolyze 4-methylumbelliferyl-D-glucuronide); possession of an attaching-and-effacing (AE) gene (*eae*); and carriage of a 60-MDa plasmid. Non-O157 EHEC strains do not share these growth and metabolic characteristics, although they all produce Stx(s) and most carry the *eae* gene and large plasmid.

Acid Tolerance

Unlike most foodborne pathogens, many strains of *E. coli* O157:H7 are unusually tolerant of acidic environments. The minimum pH for *E. coli* O157:H7 growth is 4.0 to 4.5, but this is dependent upon the interaction of the pH with other factors. For instance, organic acid sprays containing acetic, citric, or lactic acid do not affect the level of *E. coli* O157:H7 on beef. *E. coli* O157:H7, when inoculated at high levels, has survived fermentation, drying, and storage of fermented sausage (pH 4.5) for up to 2 months at 4°C and has survived in mayonnaise (pH 3.6 to 3.9) for 5 to 7 weeks at 5°C and for 1 to 3 weeks at 20°C and in apple cider (pH 3.6 to 4.0) for 10 to 31 days or for 2 to 3 days at 8 or 25°C, respectively.

Three systems in *E. coli* O157:H7 are involved in acid tolerance: an acid-induced oxidative system, an acid-induced arginine-dependent sys-

BOX 9.2

E. coli O157:H7

E. coli O157:H7 emerged as a pathogen in 1982. The initial cases of illness were predominantly linked to the consumption of beef products, but recently, illnesses have been linked to contaminated fruits and vegetables, recreational water (lakes, swimming pools, and water parks), and person-to-person spread. The organism has acquired many tools (e.g., production of Stx and the ability to produce AE lesions) that facilitate its survival in a host but that can cause debilitating illness. As with *Shigella*, illness may be caused by <100 cells. In the United States, *E. coli* O157:H7 is the predominant EHEC strain. In other parts of the world, this is not necessarily the case.

tem, and a glutamate-dependent system. The oxidative system is less effective in protecting the organism from acid stress than the arginine-dependent and glutamate-dependent systems. The alternate sigma factor RpoS is required for oxidative acid tolerance but is only partially involved with the other two systems. Once induced, the acid-tolerant state can persist for a prolonged period (≥28 days) at refrigeration temperature. More importantly, induction of acid tolerance in *E. coli* O157:H7 can also increase tolerance of other environmental stresses, such as heating, radiation, and antimicrobials.

Antibiotic Resistance

When *E. coli* O157:H7 was first associated with human illness, the pathogen was susceptible to most antibiotics affecting gram-negative bacteria. However, it is becoming increasingly resistant to antibiotics. *E. coli* O157:H7 strains isolated from humans, animals, and food have developed resistance to multiple antibiotics, with streptomycin-sulfisoxazole-tetracycline being the most common resistance profile. Non-O157 EHEC strains isolated from humans and animals also have acquired antibiotic resistance, and some are resistant to multiple antimicrobials commonly used in human and animal medicine. Antibiotic-resistant EHEC strains possess a selective advantage over other bacteria colonizing the intestines of animals that are treated with antibiotics (therapeutically or subtherapeutically). Therefore, antibiotic-resistant EHEC strains may become the primary *E. coli* strains present under antibiotic selective pressure and thus more prevalent in feces. Perhaps fortunately, antibiotic therapy is not recommended in cases of human illness, since the disease may become worse through the release of endotoxin and Stxs from the dead bacteria.

Inactivation by Heat and Irradiation

E. coli O157:H7 is not more heat resistant than other pathogens. The presence of certain compounds in food can protect the organism. The presence of fat protects *E. coli* O157:H7 in ground beef, with *D* values (see chapter 2) for lean (2.0% fat) and fatty (30.5% fat) ground beef of 4.1 and 5.3 min at 57.2°C and 0.3 and 0.5 min at 62.8°C, respectively. Pasteurization of milk (72°C; 16.2 s) is an effective treatment that kills $>10^4$ *E. coli* O157:H7 organisms per ml. Proper heating of foods of animal origin, e.g., heating foods to an internal temperature of at least 68.3°C for several seconds, is an important critical control point to ensure inactivation of *E. coli* O157:H7.

Many countries have approved the use of irradiation to eliminate foodborne pathogens in food. Unlike other processing technologies, irradiation eliminates foodborne pathogens yet maintains the raw character of foods. In the United States, 4.5 kilograys (kGy) is approved for refrigerated raw ground beef and 7.5 kGy is approved for frozen raw ground beef. Foodborne bacterial pathogens are relatively sensitive to irradiation. D_{10} values for *E. coli* O157:H7 in raw ground beef patties range from 0.241 to 0.307 kGy, depending on the temperature, with D_{10} values significantly higher for patties irradiated at −16 than at 4°C. Hence, an irradiation dose of 1.5 kGy should be sufficient to eliminate *E. coli* O157:H7 at the levels that may occur in ground beef. Following the large produce-related outbreaks in the United States in 2006, irradiation of produce may became a viable means to ensure the safety of fresh produce.

RESERVOIRS OF *E. COLI* O157:H7

Detection of *E. coli* O157:H7 and EHEC on Farms

Undercooked ground beef and, less frequently, unpasteurized milk have caused many outbreaks of *E. coli* O157:H7 infection, and hence, cattle have been the focus of many studies of their role as a reservoir of *E. coli* O157:H7. The first link between *E. coli* O157:H7 and cattle was observed in Argentina in 1977 and was associated with a <3-week-old calf. Rates of EHEC carriage as high as 60% have been found in bovine herds in many countries, but in most cases, the rates range from 10 to 25%. The isolation rates of *E. coli* O157:H7 are much lower than those of non-O157 EHEC. Rates of EHEC carriage by dairy cattle on farms in Canada range from 36% of cows and 57% of calves in 80 herds tested. In the United States, 31 (3.2%) of 965 dairy calves and 191 (1.6%) of 11,881 feedlot cattle tested were positive for *E. coli* O157:H7. Young, weaned animals more frequently carry *E. coli* O157:H7 than adult cattle. Cattle are more likely to test positive for *E. coli* O157:H7 during warmer months of the year. This agrees with the seasonal variation in human disease.

Factors Associated with Bovine Carriage of *E. coli* O157:H7

There may be an association between fecal shedding of *E. coli* O157:H7 and feed or environmental factors. For example, some calf starter feed regimens or environmental factors and feed components, such as whole cottonseed, are linked with a reduction of *E. coli* O157:H7. In contrast, keeping calves in groups before weaning, sharing of calf-feeding utensils without sanitation, and early feeding of grain are associated with increased carriage of *E. coli* O157:H7. Grain and hay feeding programs may influence *E. coli* O157:H7 carriage by cattle.

Environmental factors, such as water and feed sources, or farm management practices, such as manure handling, may play important roles in influencing the prevalence of *E. coli* O157:H7 on dairy farms. The pathogen is frequently found in water troughs on farms. *E. coli* O157:H7 can survive for weeks or months in bovine feces and water.

The susceptibility of cattle to intestinal colonization with *E. coli* O157:H7 is largely a function of age. Young animals are more likely to be positive than older animals in the same herd. For many cattle, *E. coli* O157:H7 is transiently carried in the gastrointestinal tract and is intermittently excreted for a few weeks to months by young calves and heifers. Cows can carry more than one strain of *E. coli* O157:H7.

Cattle Model for Infection by *E. coli* O157:H7

E. coli O157:H7 is not a pathogen of weaned calves and adult cattle; hence, animals that carry the pathogen are not ill. However, there is evidence that *E. coli* O157:H7 can cause diarrhea and lesions in newborn calves. The initial sites of localization of *E. coli* O157:H7 in cattle are the forestomachs (rumen, omasum, and reticulum).

Domestic Animals and Wildlife

Although cattle are thought to be the main source of EHEC in the food chain, EHEC strains have also been isolated from other domestic animals and wildlife, such as sheep, goat, deer, dogs, horses, swine, and cats. *E. coli* O157:H7 was also isolated from seagulls and rats. The prevalence of *E. coli*

O157:H7 and EHEC in sheep is generally higher than in other animals. In a survey of seven animal species in Germany, EHEC strains were isolated most frequently from sheep (66.6%), goats (56.1%), and cattle (21.1%), with lower prevalence rates in chickens (0.1%), pigs (7.5%), cats (13.8%), and dogs (4.8%). Carriage of the pathogen by feral animals represents a significant risk to the contamination of food crops (lettuce, spinach, and leafy greens) in the field.

Humans

Fecal shedding of *E. coli* O157:H7 by patients with hemorrhagic colitis or HUS usually lasts 13 to 21 days following the onset of symptoms. However, in some instances, the pathogen can be excreted in feces for >3 weeks. A child infected during a day care center outbreak excreted the pathogen for 62 days. People living on dairy farms have elevated antibody titers against *E. coli* O157:H7; however, the pathogen was not isolated from feces. An asymptomatic long-term carrier state has not been identified. Fecal carriage of *E. coli* O157:H7 by humans is significant because of the potential for person-to-person spread of the pathogen. A factor contributing to person-to-person spread is the bacterium's extraordinarily low infectious dose. Fewer than 100 cells, and possibly as few as 10 cells, can cause illness. Inadequate attention to personal hygiene, especially after using the bathroom, can transfer the pathogen through contaminated hands, resulting in secondary transmission.

DISEASE OUTBREAKS

Geographic Distribution

E. coli O157:H7 is a cause of many major outbreaks worldwide. At least 30 countries on six continents have reported *E. coli* O157:H7 infection in humans. In the United States, 350 outbreaks of *E. coli* O157:H7 infection were documented in 49 states from 1982 to 2002, accounting for 8,598 cases. Although the number of outbreaks has increased, the median outbreak size has declined from 1982 to 2002. This increase may be due in part to improved recognition of *E. coli* O157:H7 infection following the publicity of a large multistate outbreak in the western United States in 1993. Since 1 January 1994, individual cases of *E. coli* O157:H7 infection have been reportable to the National Notifiable Diseases Surveillance System. The numbers of reported outbreaks, clusters, and cases of infection in the United States between 1982 and 1998 are presented in Table 9.2. The exact number of *E. coli* O157:H7 illnesses in the United States is not known because infected persons with mild or no symptoms and persons with nonbloody diarrhea often do not seek medical attention, and these cases would not be reported. The Foodborne Diseases Active Surveillance Network (FoodNet [http://www.cdc.gov/ncidod/dbmd/foodnet/]) reports that the annual rate of *E. coli* O157:H7 infection at several surveillance sites in the United States ranged from 2.1 to 2.8 cases per 100,000 population for 1996 to 1999. However, the number of cases declined to 0.9 cases per 100,000 population from 2000 to 2004. The Centers for Disease Control and Prevention estimates that *E. coli* O157:H7 causes 73,480 illnesses and 61 deaths annually in the United States and that non-O157 EHEC strains account for an additional 37,740 cases with 30 deaths. Eighty-five percent of these cases are due to foodborne transmission.

Table 9.2 Number of reported outbreaks, clusters, and cases of *E. coli* O157:H7 infection in the United States, 1982 to 1998[a]

Yr	No. of outbreaks	No. of cases
1982	2	47
1984	2	70
1986	2	52
1987	1	52
1988	3	153
1989	2	246
1990	2	75
1991	4	54
1992	4	75
1993	17	1,000
1994	32	543
1995	32	455
1996	29	488
1997	22	298
1998	42	477
Total	196	4,085

[a]Source: P. M. Griffin, personal communication, 2000.

Seasonality of *E. coli* O157:H7 Infection

Outbreaks and clusters of *E. coli* O157:H7 infection peak during the warmest months of the year. Approximately 89% of outbreaks and clusters reported in the United States occur from May to November. The reasons for this seasonal pattern are unknown but may include (i) an increased prevalence of the pathogen in cattle during the summer, (ii) greater human exposure to ground beef or other *E. coli* O157:H7-contaminated foods during the cookout months, and/or (iii) more improper handling (temperature abuse and cross-contamination) or incomplete cooking of products such as ground beef during warm months than during other months.

Age of Patients

All age groups can be infected by *E. coli* O157:H7, but the very young and the elderly most frequently experience severe illness with complications. HUS usually occurs in children, whereas thrombotic thrombocytopenic purpura (TTP) occurs in adults. Children 2 to 10 years of age are more likely to become infected with *E. coli* O157:H7. The high rate of infection in this age group is likely the result of increased exposure to contaminated foods, contaminated environments, and infected animals, as well as more opportunities for person-to-person spread between infected children with relatively undeveloped hygiene skills and undeveloped immune systems.

Transmission of *E. coli* O157:H7

Many foods are identified as vehicles of *E. coli* O157:H7 infection. Examples include ground beef, roast beef, cooked meats, venison jerky, salami, raw milk, pasteurized milk, yogurt, cheese, ice cream bars and cake, lettuce, spinach, unpasteurized apple cider and juice, cantaloupe, potatoes, radish sprouts, alfalfa sprouts, fruit or vegetable salad, and coleslaw. Among the 183 outbreaks reported in the United States for which a vehicle

has been identified, ground beef was the most frequent food vehicle (41%), while produce accounted for 21% and 23% were of unknown origin (Fig. 9.1). Contact of foods with meat or feces (human or bovine) contaminated with *E. coli* O157:H7 is a likely source of cross-contamination. Outbreaks attributed to person-to-person (including secondary) transmission (25.5%) and waterborne (particularly recreational water) transmission (12.4%) have occurred.

Examples of Foodborne and Waterborne Outbreaks

The Original Outbreaks

The first documented outbreak of *E. coli* O157:H7 infection occurred in Oregon in 1982, with 26 cases and 19 persons hospitalized. All of the patients had bloody diarrhea and severe abdominal pain. This outbreak was associated with eating undercooked hamburgers from fast-food restaurants of a specific chain. *E. coli* O157:H7 was recovered from the stools of patients. A second outbreak occurred 3 months later and was associated with the same fast-food restaurant chain in Michigan, with 21 cases and 14 persons hospitalized, with an age range of 4 to 58 years. Contaminated hamburgers again were the cause. *E. coli* O157:H7 was isolated both from patients and from a frozen ground beef patty. *E. coli* O157:H7 was identified as the cause by its association with the food and by recovery of bacteria with identical microbiological characteristics from both the patients and the meat from the implicated supplier.

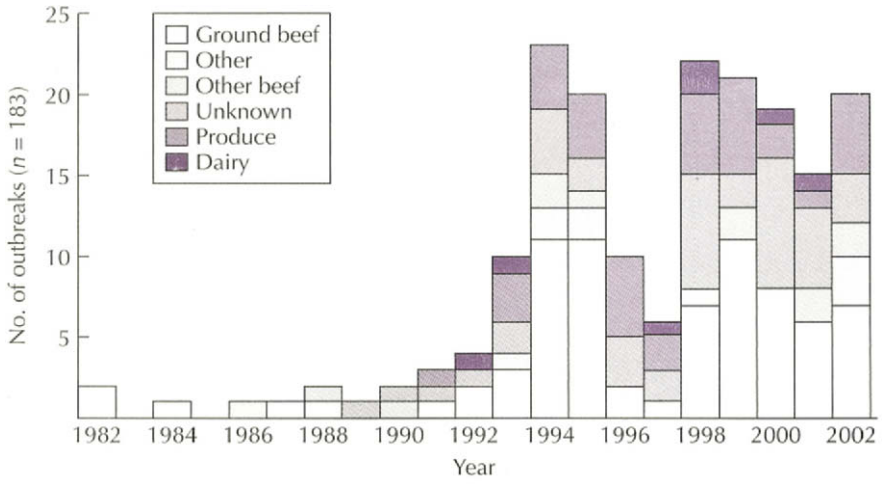
Waterborne Outbreaks

Reported waterborne outbreaks of *E. coli* O157:H7 infection have increased alarmingly in recent years. Swimming water, drinking water, well water, and ice have all been implicated. Investigations of lake-associated outbreaks revealed that in some instances the water was likely contaminated with *E. coli* O157:H7 by toddlers defecating while swimming. Swallowing lake water was subsequently identified as the cause. A large waterborne outbreak of *E. coli* O157:H7 among attendees of a county fair in New York

Authors' note

E. coli O157:H7 infections are associated with more than food and water. Each year children develop E. coli *O157:H7 infection following a visit to the local farm. Children under the age of 5 years should be extra cautious around cattle (including those in petting zoos). Washing hands thoroughly with running water and soap after contact with cattle is recommended.*

Figure 9.1 Epidemiology of *E. coli* O157:H7 outbreak, United States, 1982 to 2002. Adapted from J. M. Rangel, P. H. Sparling, C. Crowe, P. M. Griffin, and D. L. Swerdlow, *Emerg. Infect. Dis.* **11**:603–609, 2005.



occurred in August 1999. More than 900 persons were infected, 65 of whom were hospitalized. Two persons, including a 3-year-old girl, died from HUS (kidney failure), and a 79-year-old man died from HUS-TTP. Unchlorinated well water used to make beverages and ice was identified as the vehicle, and *E. coli* O157:H7 was isolated from samples of well water. Waterborne outbreaks of *E. coli* O157 infections have also been reported in Scotland, southern Africa, and Japan.

Outbreaks from Apple Cider and Juice

The first confirmed outbreak of *E. coli* O157:H7 infection associated with apple cider occurred in Massachusetts in 1991, involving 23 cases. In 1996, three outbreaks of *E. coli* O157:H7 infection associated with unpasteurized apple juice or cider were reported in the United States. The largest of the three occurred in three western states (California, Colorado, and Washington) and British Columbia, Canada, with 71 confirmed cases and one death. *E. coli* O157:H7 was isolated from the apple juice involved. An outbreak also occurred in Connecticut, with 14 cases. Manure contamination of apples was the suspected source of *E. coli* O157:H7 in several of the outbreaks. Using apple drops (i.e., apples picked up from the ground) for making apple cider is a common practice. Apples can become contaminated by resting on soil contaminated with manure. Apples also can become contaminated if transported or stored in areas that contain manure or if they are treated with contaminated water. Investigation of the 1991 outbreak in Massachusetts revealed that the cider press processor also raised cattle that grazed in a field adjacent to the cider mill. These outbreaks led to the U.S. Food and Drug Administration Hazard Analysis Critical Control Point regulations for juice. These require processors to achieve a 5-log-unit reduction in the numbers of the most resistant pathogen in their finished products.

A Large Multistate Outbreak

A large multistate outbreak of *E. coli* O157:H7 infection in the United States occurred in Washington, Idaho, California, and Nevada in early 1993. Approximately 90% of the primary cases were associated with eating at a single fast-food restaurant chain (chain A), at which *E. coli* O157:H7 was isolated from hamburger patties. The number of people who became ill increased because of secondary spread (48 patients in Washington alone) from person to person. One hundred seventy-eight people were hospitalized, 56 developed HUS, and four children died. The outbreak resulted from insufficient cooking of hamburgers by chain A restaurants. Hamburgers cooked according to chain A's cooking procedures in Washington State had internal temperatures below 60°C, which is substantially less than the minimum internal temperature of 68.3°C required by the state of Washington. Cooking the patties to an internal temperature of 68.3°C would have killed the low numbers of *E. coli* O157:H7 organisms in the ground beef.

Outbreaks Associated with Vegetables

Raw vegetables, particularly lettuce and alfalfa sprouts, have been implicated in outbreaks of *E. coli* O157:H7 infection in North America, Europe, and Japan. In the United States, outbreaks associated with lettuce have continued to increase during the past decade. In May 1996, mesclun lettuce was associated with a multistate outbreak of 47 cases in Illinois and Connecticut. In 2006, a large outbreak in multiple states and Canada was associated with consumption of contaminated prebagged spinach.

Between May and December 1996, multiple outbreaks of *E. coli* O157:H7 infection occurred in Japan, involving 11,826 cases and 12 deaths. The largest outbreak affected 7,892 schoolchildren and 74 teachers and staff in Osaka in July 1996, among whom 606 individuals were hospitalized, 106 had HUS, and 3 died. Contaminated white radish sprouts were the source of *E. coli* O157:H7.

CHARACTERISTICS OF DISEASE

The spectrum of human illness caused by *E. coli* O157:H7 infection includes nonbloody diarrhea, bloody diarrhea (hemorrhagic colitis), kidney disease (HUS), and TTP (the adult form of HUS). Some persons may be infected but exhibit no signs or symptoms of illness (this is known as *asymptomatic infection*). Ingestion of the organism is followed by a 3- to 4-day incubation period (range, 2 to 12 days), during which colonization of the large intestine occurs. Illness begins with nonbloody diarrhea and severe abdominal cramps for 1 to 2 days. This progresses in the second or third day of illness to bloody diarrhea that lasts for 4 to 10 days. Many outbreak investigations revealed that >90% of microbiologically documented cases of diarrhea caused by *E. coli* O157:H7 were bloody, but in some outbreaks 30% of cases have involved nonbloody diarrhea. Symptoms usually end after 1 week, but ~6% of patients progress to HUS. The case-fatality rate from *E. coli* O157:H7 infection is ~1%.

HUS largely affects children, for whom it is the leading cause of acute kidney failure. The syndrome is characterized by three features: acute renal insufficiency, microangiopathic hemolytic anemia, and thrombocytopenia. TTP affects mainly adults and resembles HUS histologically. It is accompanied by distinct neurological abnormalities resulting from blood clots in the brain.

INFECTIOUS DOSE

The infectious dose of *E. coli* O157:H7 is thought to be extremely low. For example, between 0.3 and 15 colony-forming units (CFU) of *E. coli* O157:H7 per g were enumerated in lots of frozen ground beef patties associated with a 1993 multistate outbreak in the western United States. Similarly, 0.3 to 0.4 CFU of *E. coli* O157:H7 per g were detected in several intact packages of salami that were associated with a foodborne outbreak. These data suggest that the infectious dose of *E. coli* O157:H7 may be <100 cells. Additional evidence for a low infectious dose is the capability for person-to-person and waterborne transmission of EHEC infection. Age, immune status, and preexisting debilitating health conditions can individually or collectively influence whether an individual becomes ill.

MECHANISMS OF PATHOGENICITY

The exact mechanism of pathogenicity of EHEC is not known. General knowledge about the pathogenicity of EHEC indicates that the bacteria cause disease by their ability to adhere to the host cell membrane and colonize the large intestine. They then produce one or more Stxs. However, the mechanisms of intestinal colonization are not well understood. The roles of various other potential virulence factors and host factors remain to be determined. The virulence factors involved in the pathogenesis of EHEC are summarized in Table 9.3.

Table 9.3 Proteins and genes involved in pathogenesis of EHEC[a]

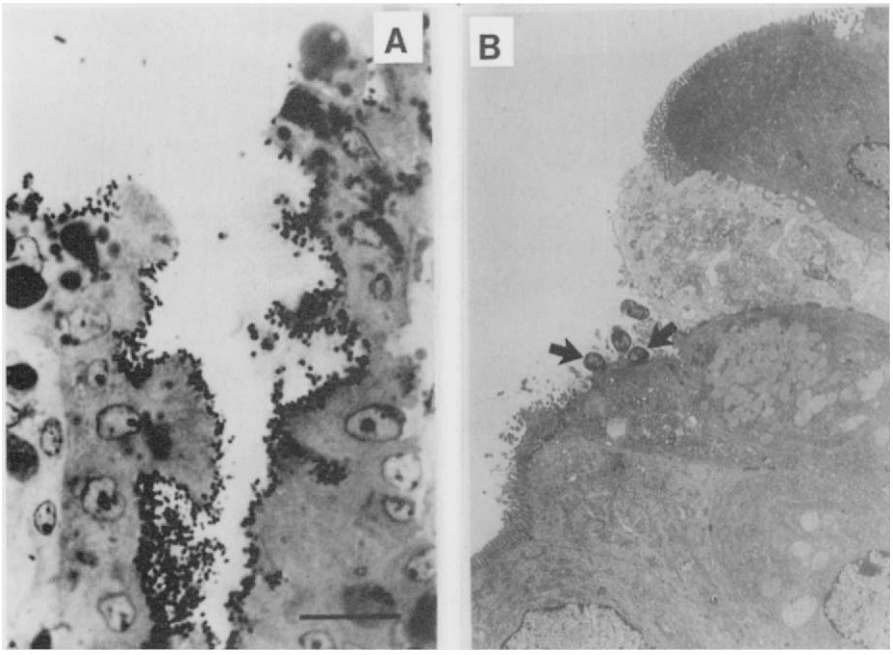
Genetic locus	Protein description	Gene(s)	Function
Chromosome (locus of enterocyte effacement)	Intimin	*eae*	Adherence
	Tir	*tir*	Intimin receptor
	Secretion proteins	*espA, espB, espD*	Induces signal transduction
	Type III secretion system	*escC, escD, escF, escJ, escN, escR, escS, escT, escU, escV, sepQ, sepZ*	Apparatus for extracellular protein secretion
Phage	Stx	*stx1, stx2, stx2c, stx2d*	Inhibits protein synthesis
Plasmid	EHEC hemolysin	EHEC-*hylA*	Disrupts cell membrane permeability
	Catalase-peroxidase	*katP*	

[a]Sources: A. D. O'Brien et al., *Curr. Top. Microbiol. Immunol.* **180:**65–94, 1992; N. T. Perna et al., *Infect. Immun.* **66:**3810–3817, 1998.

Attaching and Effacing

Adherence factors and their genes have been characterized, but the mechanisms of adherence and colonization by *E. coli* O157:H7 and other EHEC strains have not been characterized completely. Studies with animal models revealed that the pathogen can colonize the colons of orally infected animals by an AE mechanism (Fig. 9.2). The AE lesion is characterized by intimate attachment of the bacteria to intestinal cells. This results in loss of microvilli on the epithelial cells and accumulation of actin (F actin) in the cytoplasm. It is believed that similar mechanisms regulate the processes of AE formation by *E. coli* O157:H7.

Figure 9.2 AE lesion caused by *E. coli* O157:H7. **(A)** Irregularly shaped cuboidal to low-columnar crypt neck cells with bacteria adherent to the luminal plasma membrane of a gnotobiotic pig (toluidine blue stain). Reproduced with permission from D. H. Francis, J. E. Collins, and J. R. Duimstra, *Infect. Immun.* **51:**953–956, 1986. **(B)** Transmission electron photomicrograph of rabbit ileum with adherent bacteria (arrow). Reproduced with permission from P. Sherman, R. Soni, and M. Karmali, *Infect. Immun.* **56:**756–761, 1988.



The Locus of Enterocyte Effacement

Genes on the locus of enterocyte effacement of the EPEC chromosome are involved in producing the AE lesion. The locus of enterocyte effacement is not present in *E. coli* strains that are part of the normal intestinal flora, *E. coli* K-12, or ETEC strains. The locus of enterocyte effacement is found in those EPEC and EHEC strains that produce the AE lesion. The locus of enterocyte effacement consists of three segments with genes of known function. Genes associated with the locus of enterocyte effacement encode proteins involved in attachment and recognition that are known to interact directly with the host.

Intimin is an outer membrane protein encoded by the *eae* (from *E. coli* AE) gene. The protein is produced by EPEC, EHEC, *Hafnia alvei*, and *Citrobacter rodentium*. Intimin is the only described adherence factor of *E. coli* O157:H7 and is important for intestinal colonization in animal models.

The Tir protein is produced by the bacterial cell and moves to the host cell, where it serves as a receptor for intimin, via the type III secretion system. Most bacterial pathogens use existing host cell receptors. The ability of EPEC and EHEC to supply their own receptors may provide additional advantages for initiating the process of adhesion, motility, and signal transduction. The type III secretion system, present in many gram-negative bacterial pathogens, is induced upon contact with the host cells and only exports virulence factors to eukaryotic cells.

The 60-MDa Plasmid (pO157)

E. coli O157:H7 isolates carry a plasmid (pO157) of ~60 MDa (unrelated to the 60-MDa plasmid present in EPEC). Because of the association of the 60-MDa plasmid with EHEC, the plasmid is linked to the pathogenesis of EHEC infections. However, its exact role in virulence has not been determined. The plasmid carries potential virulence genes, including those encoding EHEC hemolysin and catalase-peroxidase. The EHEC catalase-peroxidase is an enzyme that protects the bacterium against oxidative stress, a defense mechanism of mammalian cells during bacterial infection.

Stxs

EHEC produces one or two Stxs. The nomenclature of the Stx family and their important characteristics are listed in Table 9.4. Molecular studies of Stx1 proteins from different *E. coli* strains revealed that Stx1 is either

Table 9.4 Nomenclature and biological characteristics of Stx[a]

Nomenclature	Genetic locus	% Amino acid homology to Stx2 subunit:		Receptor	Activated by intestinal mucus	Disease
		A	B			
Stx	Chromosome	55	57	Gb_3	No	Human diarrhea, HC[b], HUS
Stx1	Phage	55	57	Gb_3	No	Human diarrhea, HC, HUS
Stx2	Phage	100	100	Gb_3	No	Human diarrhea, HC, HUS
Stx2c	Phage	100	97	Gb_3	No	Human diarrhea, HC, HUS
Stx2d	Phage	99	97	Gb_3	Yes	Human diarrhea, HC, HUS
Stx2e	Chromosome	93	84	Gb_4[c]	No	Pig edema disease

[a]Source: A. R. Melton-Celsa and A. D. O'Brien, p. 121–128, *in* J. B. Kaper and A. D. O'Brien (ed.), Escherichia coli *O157:H7 and Other Shiga Toxin-Producing* E. coli *Strains* (ASM Press, Washington, DC, 1998).

[b]HC, hemorrhagic colitis.

[c]Gb_4, glycolipid globotetraosylceramide.

completely identical to the Stx of *S. dysenteriae* type 1 or differs by only one amino acid. Unlike Stx1, toxins of the Stx2 group are not neutralized by serum raised against Stx and do not cross-react with Stx1-specific DNA probes. There is sequence and antigenic variation within toxins of the Stx2 family produced by *E. coli* O157:H7 and other EHEC strains. The subgroups of Stx2 are Stx2, Stx2c, Stx2d, and Stx2e.

Structure of the Stx Family

Stxs are composed of a single enzymatic 32-kDa A subunit in association with a pentamer of 7.7-kDa receptor-binding B subunits. The Stx A subunit can be split by trypsin into an enzymatic A1 fragment (~27 kDa) and a carboxyl-terminal A2 fragment (~4 kDa) that links A1 to the B subunits.

Genetics of Stxs

While all *stx1* operons appear to be identical and are located on the genomes of lysogenic lambdoid bacteriophages, there is considerable heterogeneity in the *stx2* family. Unlike the genes for other Stx2 proteins, which are located on bacteriophages that integrate into the chromosome, Stx of *S. dysenteriae* type 1 and Stx2e are encoded by chromosomal genes. The genes of additional variants of Stx2 have been isolated from EHEC. Finally, Stx and Stx1 production are negatively regulated at the transcriptional level by an iron-Fur protein complex that binds at the *stx1* promoter. This is not affected by temperature, whereas Stx2 production is neither iron nor temperature regulated.

Receptors

All members of the Stx family bind to glycolipids on the eukaryotic cell surface. The alteration of binding specificity between Stx2e and the rest of the Stx family is related to the carbohydrate specificities of receptors. Stx1 binds preferentially to Gb_3 (glycolipid globotriaosylceramide) containing C20:1 fatty acid, whereas Stx2c prefers Gb_3 containing C18:1 fatty acid.

Mode of Action of the Stxs

Stxs act by inhibiting protein synthesis. While it appears that transfer of the toxin to the Golgi apparatus is essential for intoxication, the mechanism of entry of the A subunit and particularly the role of the B subunit remain unclear. Although the entire toxin is necessary for its toxic effect on whole cells, the A1 subunit can cleave the *N*-glycoside bond in one adenosine position of the 28S ribosomal RNA (rRNA) that comprises 60S ribosomal subunits. This elimination of a single adenine nucleotide inhibits the elongation factor-dependent binding to ribosomes of aminoacyl-bound transfer RNA (tRNA) molecules. Peptide chain elongation is stopped, and overall protein synthesis is suppressed, resulting in cell death.

The Roles of Stxs in Disease

The precise roles of Stxs in mediating colonic disease, HUS, and neurological disorders are not understood. There is no satisfactory animal model for bloody diarrhea (hemorrhagic colitis) or HUS. The severity of the diseases prevents the study of experimental infections in humans. Therefore, our understanding of the role of Stxs in causing disease is obtained from histopathology of diseased human tissues, animal models, and endothelial tissue culture cells. It appears that Stxs contribute to pathogenesis by di-

rectly damaging vascular endothelial cells in certain organs, thereby disrupting the homeostatic properties of the cells.

There is a correlation between infection with *E. coli* O157:H7 and development of HUS in humans. Histopathologic examination of kidney tissue from HUS patients revealed extensive structural damage in the glomeruli, the basic filtration unit of the kidney. Endotoxin in the presence of Stxs also can activate phagocytic cells to synthesize and release cytokines, superoxide radicals, or proteinases and can amplify endothelial cell damage.

E. coli O157:H7 strains isolated from patients with bloody diarrhea usually produce both Stx1 and Stx2 or Stx2 only; isolates producing only Stx1 are uncommon. Patients infected with *E. coli* O157:H7 producing only Stx2 or Stx2 in combination with Stx1 are more likely to develop serious kidney or circulatory complications than patients infected with EHEC strains producing Stx1 only.

CONCLUSION

The serious nature of hemorrhagic colitis (bloody diarrhea) and HUS caused by *E. coli* O157:H7 places this pathogen in a category apart from other foodborne pathogens that typically cause only mild symptoms. The severity of the illness, combined with its low infectious dose (<100 cells), qualifies *E. coli* O157:H7 to be among the most dangerous of foodborne pathogens. Illness caused by this pathogen has been linked to the consumption of a variety of foods. Recreational and drinking water are also vehicles of transmission of *E. coli* O157:H7 infection. The pathogenic mechanisms of *E. coli* O157:H7 are not fully understood; however, production of one or more Stxs and AE adherence are important virulence factors. Although other EHEC strains cause disease, *E. coli* O157:H7 is still by far the most important serotype of EHEC in North America. Isolation of non-O157:H7 EHEC strains requires techniques not generally used in clinical laboratories; hence, these bacteria are rarely detected in routine practice. Recognition of non-O157 EHEC strains in foodborne illness requires the identification of serotypes of EHEC other than O157:H7 in persons with bloody diarrhea and/or HUS and preferably in the implicated food. The increased availability in clinical laboratories of techniques such as testing for Stxs or their genes and identification of other virulence markers unique to EHEC may enhance the detection of disease attributable to non-O157 EHEC.

Summary

- Cattle are a major reservoir of *E. coli* O157:H7, with undercooked ground beef being the single most frequently implicated vehicle of transmission.
- Acid tolerance is an important feature of this pathogen.
- EHEC strains other than O157:H7 have been increasingly associated with cases of HUS.
- Production of Stx is an important virulence feature of *E. coli* O157:H7.
- *E. coli* O157:H7 can be readily distinguished from other *E. coli* strains based on biochemical characteristics, such as inability to ferment sorbitol and lack of production of β-glucuronidase.

Suggested reading

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Questions for critical thought

1. The infectious dose of *E. coli* O157:H7 may be as low as 10 cells. What characteristics of the pathogen contribute to the low infectious dose?
2. *E. coli* O157:H7 produces Stx. Explain how Stxs affect host cells and the role they play in disease.
3. How might acid tolerance contribute to foodborne illness associated with *E. coli* O157:H7?
4. On-farm feeding practices can influence shedding of the pathogen by cattle. How could this information be used to reduce contamination of beef products?
5. Based on the outbreaks described, what measures could be implemented to reduce cases of foodborne illness linked to *E. coli* O157:H7?
6. Provide a scenario that may result in cross-contamination of lettuce (or other types of produce) with *E. coli* O157:H7.
7. Why are young children (10 years of age and younger) more likely than adults to develop complications (HUS) when infected with *E. coli* O157:H7?
8. Based on characteristics of *E. coli* O157:H7, what tests could be performed to distinguish the pathogen from other *E. coli* strains?
9. Speculate as to why produce-linked *E. coli* O157:H7-related outbreaks have increased during the past decade.

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10 Yersinia enterocolitica

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Pathological Changes
Virulence Determinants
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Yersinia*
- understand what conditions in foods favor *Yersinia* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *Yersinia*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *Yersinia*
- identify environmental sources of the organism
- understand the role of *Yersinia* toxins and virulence factors in causing foodborne illness

Outbreak

One of the largest outbreaks of foodborne disease caused by *Yersinia enterocolitica* occurred between June and August of 1982. At least 172 *Y. enterocolitica* culture-positive cases were identified, and 41% occurred among children <5 years of age. Consumption of pasteurized milk was epidemiologically implicated as the vehicle of transmission of *Y. enterocolitica*. Although *Y. enterocolitica* was not isolated from the milk, the same serotype found in the outbreak was isolated from a milk crate on a hog farm where the outdated milk from the implicated processing plant was fed to hogs. The milk crates, if reused without proper sanitizing, might have resulted in cross-contamination of the outside of milk cartons destined for consumers. The symptoms of *Yersinia* enteritis, severe abdominal pain and fever, mimic those of appendicitis. Indeed, 17 patients underwent unnecessary *appendectomies* (removal of the appendix).

INTRODUCTION

Y. enterocolitica first emerged as a human pathogen during the 1930s. *Y. enterocolitica* shows between 10 and 30% DNA homology with other genera in the family *Enterobacteriaceae* and is ~50% related to *Yersinia pseudotuberculosis* and *Yersinia pestis* (the cause of plague). The last two species show >90% DNA homology, and genetic analysis has revealed that *Y. pestis* is a clone of *Y. pseudotuberculosis* that evolved some 1,500 to 20,000 years ago, shortly before the first known pandemics of human plague.

Characteristics of the Organism

The genus *Yersinia* has 11 species and is classified within the family *Enterobacteriaceae*. As with other members of this family, yersiniae are gram-negative, oxidase-negative, rod-shaped, facultative anaerobes that ferment

glucose. The genus includes three well-characterized pathogens of mammals, one pathogen of fish, and several other species whose etiologic role in disease is uncertain. The four known pathogenic species are *Y. pestis*, the causative agent of bubonic and pneumonic plague (the black death); *Y. pseudotuberculosis*, a rodent pathogen that occasionally causes disease in humans; *Yersinia ruckeri*, a cause of enteric disease in salmonids and other freshwater fish; and *Y. enterocolitica*, a versatile intestinal pathogen, which is the most prevalent *Yersinia* species among humans.

Y. pestis is transmitted to its host by the bites of fleas or respiratory aerosols, whereas *Y. pseudotuberculosis* and *Y. enterocolitica* are foodborne pathogens. Nevertheless, these three species share a number of essential virulence determinants that enable them to overcome host defenses. Analogs of these virulence determinants occur in several other enterobacteria, such as enteropathogenic and enterohemorrhagic *Escherichia coli* and *Salmonella* and *Shigella* species, as well as in various pathogens of animals (e.g., *Pseudomonas aeruginosa* and *Bordetella* species) and plants (e.g., *Erwinia amylovora*, *Xanthomonas campestris*, and *Pseudomonas syringae*). This provides evidence for the horizontal transfer of virulence genes among diverse bacterial pathogens. In addition, yersiniae may have acquired a number of human genes that enable them to undermine key aspects of the physiological response to infection.

Classification

Authors' note

Identification of Y. enterocolitica *is time-consuming, since commercial rapid test kits are not available.*

Y. enterocolitica is a heterogeneous species that is divisible into a large number of subgroups, largely according to biochemical activity and lipopolysaccharide (LPS) O antigens (Table 10.1). Biotyping is based on the ability of *Y. enterocolitica* to metabolize selected organic substrates and provides a convenient means to subdivide the species into subtypes of clinical and epidemiologic significance (Table 10.1). Most pathogenic strains of humans and domestic animals occur within biovars 1B, 2, 3, 4, and 5. By contrast, *Y. enterocolitica* strains of biovar 1A are commonly obtained from terrestrial and freshwater ecosystems. For this reason, they are often referred to as environmental strains, although some of them may be responsible for intestinal infections. Not all isolates of *Y. enterocolitica* obtained from soil, water, or unprocessed foods can be assigned to a biovar. These strains lack the characteristic virulence determinants of biovars 1B though 5 (see below)

Table 10.1 Biotyping scheme of *Y. enterocolitica*

	Reaction of biovar[a]:					
Test	**1A**	**1B**	**2**	**3**	**4**	**5**
Lipase (Tween hydrolysis)	+	+	−	−	−	−
Esculin hydrolysis	D	−	−	−	−	−
Indole production	+	+	(+)	−	−	−
D-Xylose fermentation	+	+	+	+	−	D
Voges-Proskauer reaction	+	+	+	+	+	(+)
Trehalose fermentation	+	+	+	+	+	−
Nitrate reduction	+	+	+	+	+	−
Pyrazinamidase	+	−	−	−	−	−
β-D-Glucosidase	+	−	−	−	−	−
Proline peptidase	D	−	−	−	−	−

[a] +, positive; (+), delayed positive; −, negative; D, different reactions.

and may represent novel nonpathogenic subtypes or even new *Yersinia* species.

Serotyping of *Y. enterocolitica*, based on LPS surface O antigens, coincides to some extent with biovar typing and provides a useful additional tool to subdivide this species in a way that relates to its pathologic significance. Serogroup O:3 is the variety most frequently isolated from humans. Almost all of these isolates belong to biovar 4. Other serogroups commonly obtained from humans include O:9 (biovar 2) and O:5,27 (biovar 2 or 3), particularly in northern Europe. The most frequent *Y. enterocolitica* biovar obtained from human clinical samples worldwide is biovar 4. Biovar 1B bacteria are usually isolated from patients in the United States and are referred to as American strains. They have caused several foodborne outbreaks of yersiniosis in the United States.

At least 18 flagellar (H) antigens of *Y. enterocolitica*, designated by lowercase letters (a,b; b,c; b,c,e,f,k; m; etc.), have also been identified. There is some overlap between the H antigens of *Y. enterocolitica* and those of related species, but complete O and H serotyping is seldom done.

Other schemes for subtyping *Yersinia* species include bacteriophage typing, multienzyme electrophoresis, and the demonstration of restriction fragment length polymorphisms of chromosomal and plasmid DNA. These techniques can be used to facilitate epidemiologic investigations of outbreaks or to trace the sources of sporadic infections.

Susceptibility and Tolerance

Y. enterocolitica is unusual because it can grow at temperatures below 4°C. The doubling time at the optimum growth temperature (ca. 28 to 30°C) is ~34 min, which increases to 1 h at 22°C, 5 h at 7°C, and ~40 h at 1°C. *Y. enterocolitica* readily withstands freezing and can survive in frozen foods for extended periods even after repeated freezing and thawing. *Y. enterocolitica* generally survives better at room temperature and refrigeration temperature than at intermediate temperatures. *Y. enterocolitica* persists longer in cooked foods than in raw foods, probably due to increased availability of nutrients in cooked foods. Also, the presence of other psychrotrophic (growing between 5 and 35°C) bacteria, including nonpathogenic strains of *Y. enterocolitica*, in unprocessed food may restrict bacterial growth. The number of viable *Y. enterocolitica* organisms may increase more than a millionfold on cooked beef or pork within 24 h at 25°C or within 10 days at 7°C. Growth rates are slower on raw beef and pork. *Y. enterocolitica* can grow at refrigeration temperature in vacuum-packed meat, boiled eggs, boiled fish, pasteurized liquid eggs, pasteurized whole milk, cottage cheese, and tofu (soybean curd). Growth also occurs in refrigerated seafoods, such as oysters, raw shrimp, and cooked crabmeat, but at a lower rate than in pork or beef. Yersiniae can also persist for extended periods in refrigerated vegetables and cottage cheese.

Y. enterocolitica and *Y. pseudotuberculosis* can grow over a pH range of approximately pH 4 to 10, with an optimum pH of ca. 7.6. They tolerate alkaline conditions extremely well, but their acid tolerance is less apparent and depends on the acidulent used, the environmental temperature, the composition of the medium, and the growth phase of the bacteria. The acid tolerance of *Y. enterocolitica* is enhanced by the production of urease, which hydrolyzes urea to release ammonia and elevates the cytoplasmic pH.

Y. enterocolitica and *Y. pseudotuberculosis* are susceptible to heat and are easily killed by pasteurization at 71.8°C for 18 s or 62.8°C for 30 min. Expo-

sure of surface-contaminated meat to hot water (80°C) for 10 to 20 s reduced bacterial viability by at least 99.9%. *Y. enterocolitica* is also easily killed by ionizing and ultraviolet (UV) irradiation and by sodium nitrate and nitrite added to food, although it is relatively resistant to these salts in solution. The pathogen can also tolerate NaCl at concentrations of up to 5%. *Y. enterocolitica* is susceptible to organic acids, such as lactic and acetic acids, and to chlorine. However, some resistance to chlorine occurs among yersiniae grown under conditions that are similar to natural aquatic environments.

CHARACTERISTICS OF INFECTION

Infections with *Y. enterocolitica* typically manifest as nonspecific, self-limiting diarrhea, but they may lead to a variety of autoimmune diseases (Table 10.2). The risk of these diseases is determined partly by host factors, in particular, age and immune status. *Y. enterocolitica* enters the gastrointestinal tract after ingestion of contaminated food or water. The median infectious dose for humans is not known, but it likely exceeds 10^4 colony-forming units (CFU). Stomach acid is a significant barrier to infection with *Y. enterocolitica*, and in individuals with decreased levels of stomach acid, the infectious dose may be lower.

Most symptomatic infections with *Y. enterocolitica* occur in children, especially in those <5 years of age. In these patients, yersiniosis causes diarrhea, often accompanied by low fever and abdominal pain. The diarrhea varies from watery to mucoid. The illness typically lasts from a few days to 3 weeks, although some patients develop diarrhea that may persist for several months.

Table 10.2 Clinical symptoms and diseases associated with *Y. enterocolitica* infections

Common symptoms and diseases
Diarrhea (gastroenteritis), especially in young children
Enterocolitis
Pseudoappendicular syndrome due to terminal ileitis; acute mesenteric lymphadenitis
Pharyngitis
Postinfection autoimmune sequelae
Arthritis, especially associated with HLA-B27
Erythema nodosum
Uveitis, associated with HLA-B27
Glomerulonephritis (uncommon)
Myocarditis (uncommon)
Thyroiditis (uncommon)
Less common symptoms and diseases
Septicemia
Visceral abscesses, e.g., in liver, spleen, or lung
Skin infection (pustules, wound infection, pyomyositis, etc.)
Pneumonia
Endocarditis
Osteomyelitis
Peritonitis
Meningitis
Intussusception
Eye infection (conjunctivitis, panophthalmitis)

In children older than 5 years and adolescents, acute yersiniosis often causes pain in the abdomen that is mistaken for appendicitis. This is referred to as pseudoappendicular syndrome. Some children who experience this also have fever but little or no diarrhea. The pseudoappendicular syndrome happens more frequently with the more virulent strains of *Y. enterocolitica,* notably strains of biovar 1B. *Y. enterocolitica* is rarely found in patients with true appendicitis.

Although *Y. enterocolitica* is seldom isolated outside the intestine, there appears to be no tissue in which it cannot grow. Factors that may make an individual more susceptible to *Yersinia* bacteremia include decreased immune function, malnutrition, chronic kidney disease, liver disease, alcoholism, diabetes, and acute and chronic iron overload states. Spread of the organism in the body can lead to various diseases, including abscesses, catheter-associated infections, heart disease, and meningitis. *Yersinia* bacteremia has a fatality rate between 30 and 60%.

Although most individuals with yersiniosis recover without long-term complications, infections with *Y. enterocolitica* are noteworthy for the large variety of immunological complications, such as reactive arthritis, carditis, and thyroiditis, which follow acute infection. Of these, reactive arthritis is the most widely recognized.

RESERVOIRS

Infections with *Yersinia* species are *zoonoses* (diseases transmitted from animals to humans). The subgroups of *Y. enterocolitica* that commonly occur in humans also occur in domestic animals, whereas those which are infrequent in humans generally reside in wild rodents. *Y. enterocolitica* can survive in many environments and has been isolated from the intestinal tracts of many different mammalian species, as well as from birds, frogs, fish, flies, fleas, crabs, and oysters.

Foods that are frequently positive for *Y. enterocolitica* include pork, beef, lamb, poultry, and dairy products (notably milk, cream, and ice cream). *Y. enterocolitica* is also commonly found in terrestrial and freshwater systems, including soil, vegetation, lakes, rivers, wells, and streams. It can survive for extended periods in soil, vegetation, streams, lakes, wells, and spring water, particularly at low environmental temperatures. Many environmental isolates of *Y. enterocolitica* lack markers of bacterial virulence and may not be a risk to human or animal health.

Although *Y. enterocolitica* has been recovered from a variety of wild and domesticated animals, pigs are the only animal species from which *Y. enterocolitica* of biovar 4, serogroup O:3 (the variety most commonly associated with human disease), has been isolated with any degree of frequency. Pigs may also carry *Y. enterocolitica* of serogroups O:9 and O:5,27, particularly in regions where human infections with these varieties are common. In countries with a high incidence of human yersiniosis, *Y. enterocolitica* is commonly isolated from pigs at slaughterhouses. The tissue most frequently culture positive at slaughter is the tonsils. This appears to be the preferred site of *Y. enterocolitica* infection in pigs. Yersiniae have also been isolated from tongue, cecum, rectum, feces, and gut tissue. *Y. enterocolitica* is seldom isolated from meat offered for retail sale. However, standard methods of bacterial isolation and detection may underestimate the true incidence of contamination. Further evidence that pigs are a significant reservoir of human infections is based on epidemiologic studies linking consumption of

raw or undercooked pork with yersiniosis. Infection also occurs after handling contaminated pig intestines while preparing chitterlings.

Food animals are seldom infected with biovar 1B strains of *Y. enterocolitica*, the reservoir of which remains unknown. The relatively low incidence of human yersiniosis caused by these strains, despite their high virulence, suggests limited contact between their reservoir and humans. Yersiniae of this biovar are pathogens of rodents; therefore, rats or mice are likely the natural reservoir of these strains.

FOODBORNE OUTBREAKS

Considering the widespread occurrence of *Y. enterocolitica* in nature and its ability to colonize food animals, to persist within animals and the environment, and to grow at refrigeration temperature, outbreaks of yersiniosis are surprisingly uncommon. Most foodborne outbreaks in which a source was identified have been traced to milk (Table 10.3). *Y. enterocolitica* is easily destroyed by pasteurization. Therefore, infection results from the consumption of raw milk or milk that is contaminated after pasteurization. During the mid 1970s, two outbreaks of yersiniosis caused by *Y. enterocolitica* O:5,27 occurred among 138 Canadian schoolchildren who had consumed raw milk, but the organism was not recovered from the suspected source. In 1976, serogroup O:8 *Y. enterocolitica* was responsible for an outbreak in New York State which affected 217 people, 38 of whom were culture positive. The source of infection was chocolate milk, which evidently became contaminated after pasteurization.

In 1981, an outbreak of infection with *Y. enterocolitica* O:8 affected 35% of 455 individuals at a diet camp in New York State. Seven patients were hospitalized as a result of infection, five of whom underwent appendectomies. The source of the infection was reconstituted powdered milk and/or chow mein. An infected food handler probably contaminated the food during preparation.

More recently, an outbreak of infection with *Y. enterocolitica* O:3 affected 15 infants and children in metropolitan Atlanta. In this instance, bacteria were transmitted from raw chitterlings to the affected children on the hands of food handlers. Other foods that have been responsible for outbreaks of yersiniosis include pork cheese (a type of sausage prepared from chitterlings), bean sprouts, and tofu. In the outbreaks associated with bean

Table 10.3 Selected foodborne outbreaks of infection with *Y. enterocolitica*

Location	Yr	Mo	No. of cases	Serogroup	Source
Canada	1976	April	138	O:5,27	Raw milk?[a]
New York	1976	September	38	O:8	Chocolate-flavored milk
Japan	1980	April	1,051	O:3	Milk
New York	1981	July	159	O:8	Powdered milk, chow mein
Washington	1981	December	50	O:8	Tofu and spring water
Pennsylvania	1982	February	16	O:8	Bean sprouts and well water
Southern United States	1982	June	172	O:13a,13b	Milk?
Hungary	1983	December	8	O:3	Pork cheese (sausage)
Georgia (United States)	1989	November	15	O:3	Pork chitterlings
Northeastern United States	1995	October	10	O:8	Pasteurized milk?

[a]?, bacteria were not isolated from the suspected source.

sprouts and tofu, contaminated well or spring water was the probable source of yersiniae. Water was also the likely source of infection in a case of *Y. enterocolitica* bacteremia in a 75-year-old man in New York State and a small family outbreak in Ontario, Canada. Several outbreaks of presumed foodborne infection with *Y. enterocolitica* O:3 have occurred in the United Kingdom and Japan, but in most cases, the source of these outbreaks was not identified.

MECHANISMS OF PATHOGENICITY

Y. enterocolitica is an invasive *enteric* (intestinal) pathogen whose virulence determinants have been the subject of intensive investigation. Not all strains of *Y. enterocolitica* are equally virulent. *Y. enterocolitica* strains of biovars 1B, 2, 3, 4, and 5 possess many interactive virulence determinants.

Pathological Changes

Examination of surgical specimens from patients with yersiniosis shows that *Y. enterocolitica* is an invasive pathogen. The distal ileum (part of the intestine) is the main infection site. Human volunteers cannot be used to study yersiniosis, since they may develop other diseases. Thus, most information regarding the pathogenesis of yersiniosis has been obtained from animal models, in particular mice and rabbits. Studies of experimentally infected rabbits and pigs show that after penetrating the epithelium of the intestine, the pathogens can spread throughout the body. If the bacteria reach the lymph nodes, they can enter the bloodstream and can disseminate to any organ. However, they preferentially localize in the liver and spleen.

VIRULENCE DETERMINANTS

Chromosomal Determinants of Virulence

The ability of *Y. enterocolitica* to enter mammalian cells is associated with invasin, an outer membrane protein encoded by the chromosomal *inv* gene. Invasins are related to intimin, an essential virulence determinant of enteropathogenic and enterohemorrhagic strains of *E. coli*, which require the protein in order to produce the distinctive attaching-effacing lesions that characterize infection with these bacteria.

The amino terminus of invasin is inserted in the bacterial outer membrane, while the carboxyl terminus is exposed on the surface. The carboxyl terminus binds the host cell integrins. The internalization process is controlled entirely by the host cell, because dead bacteria and even latex particles coated with invasin are internalized. Although DNA sequences the same as that of *inv* occur in all *Yersinia* species (except *Y. ruckeri*), the gene is functional only in *Y. pseudotuberculosis* and the classical pathogenic biovars (1B through 5) of *Y. enterocolitica*. This suggests that invasin plays a key role in virulence.

Most strains of *Y. enterocolitica* secrete a heat-stable enterotoxin, known as Yst (or Yst-a), which is active in infant mice. The contribution of Yst to diarrhea associated with yersiniosis is uncertain. Some strains of *Y. enterocolitica* can produce Yst or the other enterotoxins over a wide range of temperatures, from 4 to 37°C. Since these toxins are relatively acid stable, they could resist inactivation by stomach acid. If they were ingested preformed in food, they could cause foodborne illness. In artificially inoculated foods, however, these toxins are produced mainly at 25°C during the stationary

phase of bacterial growth. The storage conditions required for their production in food generally result in severe spoilage, making the ingestion of preformed Yst unlikely.

Under laboratory conditions, *Y. enterocolitica* is motile when grown at 25°C but not when grown at 37°C. The expression of the genes encoding both flagellin and invasin appears to be regulated at the level of transcription. Mutants of *Y. enterocolitica* that are defective in the expression of invasin are extremely motile. Flagellar proteins are made in individuals with yersiniosis, based on assays for the detection of flagellar proteins. Motility does not appear to contribute to the virulence of *Y. enterocolitica* in mice.

Other Virulence Determinants

The observation that patients suffering from iron overload have increased susceptibility to severe infections with *Y. enterocolitica* suggests that the availability of iron in tissues may determine the outcome of yersiniosis. Some isolates of *Y. enterocolitica* are *hemolytic* (they lyse red blood cells) due to the production of phospholipase A. A strain of *Y. enterocolitica* in which the *yplA* gene encoding this enzyme was deleted had decreased virulence for inoculated mice. YplA is secreted by *Y. enterocolitica* via the same export system used for flagellar proteins.

In *Y. enterocolitica,* acid tolerance is associated with the production of urease. This enzyme catalyzes the release of ammonia from urea and enables the bacteria to resist pHs as low as 2.5. Urease also contributes to the survival of *Y. enterocolitica* in host tissues, but the mechanism by which this occurs is not known. The urease of *Y. enterocolitica* is also unusual in that it displays optimal activity at pH 3.5 to 4.5, suggesting a physiological role in protecting the bacteria from acid.

All fully virulent, highly invasive strains of *Y. enterocolitica* carry a ca. 70-kilobase plasmid, called pYV (for plasmid for *Yersinia* virulence), which is found in *Y. enterocolitica, Y. pestis,* and *Y. pseudotuberculosis.* pYV permits the bacteria to resist phagocytosis (ingestion by specific host cells) and lysis. Thus, the pathogen can grow within tissues. When pYV-negative strains of *Y. enterocolitica* are incubated with host epithelial cells or phagocytes, they penetrate the cells in large numbers without causing damage. By contrast, pYV-positive bacteria generally remain outside host cells (i.e., resist phagocytosis).

Enteropathogenic yersiniae use a special secretory pathway (the type III secretory pathway) to inject proteins into the cytosol of eukaryotic cells. This facilitates pathogenesis. The transport of *Yersinia* outer membrane proteins from the bacterial cytoplasm into the host cell cytosol may occur in one step from bacteria that are closely bound to the host cell.

Pathogenesis of *Yersinia*-Induced Autoimmunity

Following an infection with pYV-positive *Y. enterocolitica,* a small number of patients develop *autoimmune* (the person's own immune system starts to attack them) arthritis. A similar condition can also occur after infections with *Campylobacter, Salmonella, Shigella,* or *Chlamydia* species. The pathogenesis of reactive arthritis is poorly understood. Men and women are affected equally. Arthritis typically follows the onset of diarrhea or the pseudoappendicular syndrome by 1 to 2 weeks, with a range of 1 to 38 days. The joints most commonly involved are the knees, ankles, toes, tarsal joints, fingers, wrists, and elbows. The duration of arthritis is typically <3 months.

Many other autoimmune complications have been linked to *Y. enterocolitica* infection including various thyroid disorders, autoimmune thyroiditis, and Graves' disease (hyperthyroidism). The latter is an immune system disorder caused by a person producing antibodies to the thyrotropin (TSH) receptor. The main link between *Y. enterocolitica* and thyroid diseases is that patients with these disorders frequently have elevated levels of antibodies to *Y. enterocolitica* O:3.

Other autoimmune complications of yersiniosis, including Reiter's syndrome, uveitis, acute proliferative glomerulonephritis, collagenous colitis, and rheumatic-like carditis, have been reported, mostly from Scandinavian countries. Yersiniosis has also been linked to various thyroid disorders, including nontoxic goiter and Hashimoto's thyroiditis, although the causative role of yersiniae in these conditions is uncertain. In Japan, *Y. pseudotuberculosis* has been linked to Kawasaki's disease.

Summary

- *Y. enterocolitica* expresses factors, such as urease, flagella, and smooth LPS, which facilitate its passage through the stomach and the mucus layer of the small intestine.
- Once *Y. enterocolitica* begins to replicate in the intestine at 37°C, LPS becomes rough, exposing proteins on the bacterial surface that are involved in attachment and invasion.
- The higher infectivity of *Y. enterocolitica* when grown at ambient temperature compared with its infectivity when grown at 37°C may account for the small number of reports of human-to-human transmission of yersiniosis.
- The well-defined life cycle of *Y. enterocolitica* with its distinctive temperature-induced phases is reminiscent of the flea-rat-flea cycle of *Y. pestis.*
- Outbreaks of yersiniosis are surprisingly uncommon, considering the ubiquity of *Y. enterocolitica* in nature and its abilities to colonize food animals, to persist within animals and the environment, and to proliferate at refrigeration temperature.

Suggested reading

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Questions for critical thought

1. Why is *Y. enterocolitica* considered unusual among pathogenic enterobacteria? Explain.

2. What domestic animal is considered a reservoir for *Y. enterocolitica?* How was this conclusion drawn?
3. Which group is most likely to develop pseudoappendicular syndrome? What is pseudoappendicular syndrome?
4. What four key factors are associated with the virulence of *Y. enterocolitica?*
5. Infection with *Y. enterocolitica* can result in other diseases. What diseases can develop, and how does infection with *Y. enterocolitica* play a role in those diseases?
6. Identify unique characteristics of *Y. enterocolitica* that would facilitate identification of the pathogen.
7. In reference to the large *Y. enterocolitica* outbreak linked to consumption of pasteurized milk, what measure(s) should have been implemented to prevent such an outbreak?
8. *Y. enterocolitica* is motile when grown at 25°C but not when grown at 37°C. This seems counter to properties required for an enteric pathogen. Provide several reasons why the organism would prefer motility at 25°C to motility at 37°C.
9. Select two of the autoimmune diseases from Table 10.2 and use your investigative skills to determine the characteristics of each disease.
10. Virulent strains of *Yersinia* carry the plasmid for *Yersinia* virulence (pYV). What properties that enhance virulence does the plasmid impart to strains? Outline a set of experiments that could be done to prove the role of pYV in virulence.
11. You work for the Centers for Disease Control and Prevention (CDC), and there is a large budget cut. The director wants you to eliminate one foodborne pathogen from the top five list for which data are collected in an effort to save money. You decide to remove *Y. enterocolitica* from the list. Write a one-page justification for your decision based on your knowledge of other pathogens discussed in this book.

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11

Shigella Species

Outbreak
Introduction
Classification and Biochemical Characteristics
Shigella in Foods
Survival and Growth in Foods
Characteristics of Disease
Foodborne Outbreaks
Virulence Factors
Genetic Regulation
Conclusions
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Shigella*
- understand what conditions in foods favor *Shigella* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *Shigella*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *Shigella*
- identify environmental sources of the organism
- understand the role of *Shigella* toxins and virulence factors in causing foodborne illness

Outbreak

In a 2-month period during the summer of 1998, seven outbreaks of *Shigella sonnei* infection associated with eating fresh parsley occurred. Fresh parsley is often used as a garnish or sprinkled on top of a meal prior to serving to add taste and color. Hundreds of people became ill, with most experiencing severe short-term diarrhea. Most of the patients ate at restaurants that served chopped, uncooked parsley. The causative agent was identified as *S. sonnei*, and based on pulsed-field gel electrophoresis, epidemiologic traceback, and data from other investigations, one farm in Mexico was implicated as the source of the contaminated parsley. As a result of these outbreaks, changes in food-handling practices were proposed, eliminating practices such as chopping and holding large quantities of parsley and chopping of parsley at room temperature. *Shigella* is spread through contaminated water used for irrigation and postharvest processing of fresh produce. Transmission occurs through the fecal-oral route, with as few as 10 to 100 organisms capable of causing infection. Humans and other primates are the only reservoirs for *S. sonnei*.

INTRODUCTION

Bacillary dysentery, or shigellosis, is caused by *Shigella* species. Dysentery was the term used by Hippocrates to describe an illness characterized by frequent passage of stools containing blood and mucus accompanied by painful abdominal cramps. Perhaps one of the greatest historical impacts of this disease has been its powerful influence on military operations. Protracted military campaigns and sieges have almost always spawned epidemics of dysentery, causing large numbers of military and civilian casualties. With a low infectious dose required to cause disease coupled with oral

transmission by fecally contaminated food and water, it is not surprising that dysentery caused by *Shigella* spp. follows in the wake of many natural (earthquakes, floods, and famine) and man-made (war) disasters. Apart from these special circumstances, shigellosis remains an important disease in developed and developing countries.

During the past 2 decades, several large outbreaks of shigellosis (listed below) have been linked to the consumption of contaminated food. Disease is caused by ingestion of these contaminated foods, and in some instances it subsequently leads to rapid dissemination through contaminated feces from infected individuals.

1989 and 1994: Shigellosis aboard cruise ships. In October 1989, 14% of passengers and 3% of crew members aboard a cruise ship reported having gastrointestinal symptoms. A multiple-antibiotic-resistant strain of *Shigella flexneri* 2a was isolated from several ill passengers and crew. The vehicle of the outbreak was German potato salad. Contamination was introduced by infected food handlers, initially in the country where the food was originally prepared and subsequently by a member of the galley crew on the cruise ship. Another outbreak of shigellosis occurred in August 1994 on the cruise ship S.S. *Viking Serenade*. Thirty-seven percent (586) of the passengers and 4% (24) of the crew reported having diarrhea, and one death occurred. *S. flexneri* 2a was isolated from patients, and the suspected vehicle was spring onions.

1990: Operation Desert Shield. Diarrheal diseases during a military operation can be a major factor in reducing troop readiness. Enteric pathogens were isolated from 214 U.S. soldiers in Operation Desert Shield, and of those, 113 cases were diagnosed as shigellosis; *S. sonnei* was the most prevalent species isolated. Shigellosis accounted for more time lost from military duties and was responsible for more severe morbidity than enterotoxigenic *Escherichia coli*, the most common enteric pathogen isolated from U.S. troops in Saudi Arabia. The suspected vehicle was contaminated fresh vegetables, specifically, lettuce. Twelve heads of lettuce were tested, and enteric pathogens were isolated from all of them.

1991: Moose soup in Alaska. In September 1991 in Galena, AK, 25 people who participated in a gathering of local residents contracted shigellosis associated with eating homemade moose soup. One of five women who made the soup reported having gastroenteritis while preparing it. *S. sonnei* was isolated from a hospitalized patient.

1994: Contaminated produce. Lettuce and green onions have been implicated in illness. An outbreak in Norway of 110 culture-confirmed cases of shigellosis caused by *S. sonnei* was reported in 1994. Iceberg lettuce from Spain, served in a salad bar, was suspected as the source of the outbreak in Norway and was likely responsible for increases in shigellosis in other European countries, including the United Kingdom and Sweden. *S. sonnei* was isolated from patients from several northwest European countries but was not isolated from any foods. Epidemiologic evidence indicated that imported lettuce was the vehicle of these outbreaks. An outbreak of *S. flexneri* serotype 6 (mannitol-negative) infection occurred in the Midwest in 1994. Although not confirmed, the suspected vehicle was Mexican

green onions (scallions, or spring onions). Seventeen cases of shigellosis occurred at a church potluck meal in Indiana, 29 cases occurred at an anniversary reception in Indiana, and 26 culture-confirmed mannitol-negative *S. flexneri* or *Shigella* sp. cases were reported to the Illinois State Department of Health. *S. flexneri* serotype 6 was also isolated from patients in Missouri, Minnesota, Wisconsin, Michigan, and Kentucky. Ingestion of green onions was implicated as the vehicle of infection. An infected worker most likely contaminated the onions at the time of harvest or packing.

2000: Five-layer bean dip. An outbreak of shigellosis associated with contaminated five-layer (bean, salsa, guacamole, nacho cheese, and sour cream) party dip occurred in three West Coast states. The causative agent, *S. sonnei*, was isolated from at least 30 patients. The pathogen was isolated from only one layer (cheese) of the dip and was initially detected by a polymerase chain reaction (PCR) assay targeting shigellae. *Shigella* was isolated subsequently by enrichment, followed by plating on selective agar.

2001: Tomatoes. An outbreak involving more than 880 people was linked, based on epidemiological information, to the use by several restaurants of overripe and bruised tomatoes purchased from a single distributor. The strain was isolated from ill and symptomatic workers, but they reported not being ill prior to the tomatoes' arrival.

One of the striking features regarding foodborne outbreaks of shigellosis is that contamination of foods usually does not occur at the processing plant but rather through an infected food handler. As is evident from the examples above and in Table 11.1, these incidents can be anything from contamination of foods by infected food handlers at small-town gatherings and picnics to large-scale outbreaks, such as those on cruise ships and at institutions.

Table 11.1 Examples of foodborne outbreaks caused by *Shigella* spp.

Yr	Location	Source of contamination[a]	Isolate
1986	Texas	Shredded lettuce	*S. sonnei*
1987	Rainbow Family gathering	Food handlers	*S. sonnei*
1988–1989	Monroe, NY	Multiple sources	*S. sonnei*
1988	Outdoor music festival, Michigan	Food handlers	*S. sonnei*
1988	Commercial airline	Cold sandwiches	*S. sonnei*
1989	Cruise ship	Potato salad	*S. flexneri*
1990	Operation Desert Shield (U.S. troops)	Fresh produce	*Shigella* spp.
1991	Alaska	Moose soup	*S. sonnei*
1992–1993	Operation Restore Hope, Somalia (U.S. troops)		*Shigella* spp.
1994	Europe	Shredded lettuce from Spain	*S. sonnei*
1994	Midwest	Green onions	*S. flexneri*
1994	Cruise ship		*S. flexneri*
1998	Various U.S. locations	Fresh parsley	*S. sonnei*
2000	West Coast	Bean dip	*S. sonnei*
2001	New York	Tomato	*S. flexneri*

[a]The source of contamination is listed when known.

Classification and Biochemical Characteristics

There are four species in the genus *Shigella,* serologically grouped (41 serotypes) based on their somatic (O) antigens: *Shigella dysenteriae* (group A), *S. flexneri* (group B), *Shigella boydii* (group C), and *S. sonnei* (group D). As members of the family *Enterobacteriaceae,* they are genetically almost identical to the escherichiae and closely related to the salmonellae. *Shigella* spp. are nonmotile, oxidase-negative, gram-negative rods. An important biochemical characteristic that distinguishes these bacteria from other enterics is their inability to ferment lactose; however, some strains of *S. sonnei* may ferment lactose slowly or utilize citric acid as a sole carbon source. They do not produce H_2S, except for *S. flexneri* 6 and *S. boydii* serotypes 13 and 14, and do not produce gas from glucose. *Shigella* spp. are inhibited by potassium cyanide and do not synthesize lysine decarboxylase. Enteroinvasive *E. coli* (EIEC) has pathogenic and biochemical properties similar to those of *Shigella* spp. This similarity poses a problem in distinguishing these pathogens. For example, EIEC is nonmotile and is unable to ferment lactose. Some serotypes of EIEC also have O antigens identical to those of *Shigella.*

Shigella spp. are not particular in their growth requirements and are routinely cultivated in the laboratory on artificial medium. Cultures of *Shigella* are easily isolated and grown from analytical samples, including water and clinical specimens. In the latter case, *Shigella* spp. are present in fecal specimens in large numbers (10^3 to 10^9 per g of stool) during the acute phase of infection, and therefore, identification is readily accomplished using culture media, biochemical analysis, and serological typing. Shigellae are shed by, and continue to be detected from, convalescent patients (10^2 to 10^3 per g of stool) for weeks or longer after the initial infection. Isolation of *Shigella* at this stage of infection is more difficult because a selective enrichment broth for shigellae is not available, and therefore, shigellae can be outgrown by resident bacterial fecal flora.

Isolation of *Shigella* spp. from foods is not as easy as from other sources. Foods have many different physical attributes that may affect the recovery of shigellae. These factors include composition, such as the fat content of the food; physical parameters, such as pH and salt; and the natural microbial flora of the food. In the last case, other microbes in a sample may overgrow shigellae during culture in broth media. The amount of time from the clinical report of a suspected outbreak to the analysis of the food samples can be considerable, thus lessening the chances of identifying the causative agent. The physiological state of shigellae present in the food is a contributing factor in the successful recovery of the pathogen. *Shigella* spp. may be present in low numbers or in a poor physiological state in suspected food samples. Under these conditions, special enrichment procedures are required for successful isolation and detection of shigellae.

Shigella in Foods

Shigella spp. are not associated with any specific foods. Common foods that have been implicated in outbreaks caused by shigellae include potato salad, chicken, tossed salad, and shellfish. Establishments where contaminated foods have been served include the home, restaurants, camps, picnics, schools, airlines, sorority houses, and military mess halls. In many cases, the source (food) was not identified. From 1983 to 1987, 2,397 foodborne outbreaks representing 54,453 cases were reported to the Centers for Disease Control and Prevention (CDC). In only 38% of the cases was the

source of the etiological agent identified. Whereas epidemiologic methods may strongly imply a common food source, *Shigella* spp. are not often recovered from foods and identified by using standard bacteriological methods. Also, since shigellae are not commonly associated with any particular food, routine testing of foods to identify these pathogens is not usually performed.

The traditional approach to address the problem of microbially contaminated foods in the processing plant is to inspect the final product. There are several drawbacks to this approach. Current bacteriological methods are often time-consuming and laborious. An alternative to end product testing is the Hazard Analysis Critical Control Point (HACCP) system. The HACCP system identifies certain points of the processing system that may be most vulnerable to microbial contamination and chemical and physical hazards.

In contrast, establishing specific critical control points for preventing *Shigella* contamination of foods is not always suitable for the HACCP concept. The pathogen is usually introduced into the food supply by an infected person, such as a food handler with poor personal hygiene. In some cases, this may occur at the manufacturing site, but more likely it happens at a point between the processing plant and the consumer. Another factor is the fact that foods, such as vegetables (lettuce is a good example), can be contaminated at the site of collection and shipped directly to market. Although the HACCP system is a method for controlling food safety and preventing foodborne outbreaks, pathogens such as *Shigella* that are not indigenous to, but rather introduced into, foods are most likely to be undetected.

Survival and Growth in Foods

Depending upon growth conditions, *Shigella* spp. can survive in media with a pH range of 2 to 3 for several hours. However, shigellae do not usually survive well in low-pH foods or in stool samples. Studies using citrus juices (orange and lemon), carbonated beverages, and wine revealed that shigellae were recovered after 1 to 6 days. In neutral-pH foods, such as butter or margarine, shigellae were recovered after 100 days when stored frozen or at 6°C.

Shigella can survive a temperature range of −20°C to room temperature. However, shigellae survive longer in foods stored frozen or at refrigeration temperature than in those stored at room temperature. In foods such as salads containing mayonnaise and some cheese products, *Shigella* has survived for 13 to 92 days. *Shigella* can survive for an extended period on dry surfaces and in foods such as frozen shrimp, ice cream, and minced pork. Growth of *Shigella* is impeded in the presence of 3.8 to 5.2% NaCl at pH 4.8 to 5.0, in 300 to 700 mg of $NaNO_2$/liter, and in 0.5 to 1.5 mg of sodium hypochlorite (NaClO)/liter of water at 4°C. *Shigella* is sensitive to ionizing radiation, with a reduction of 10^7 colony-forming units (CFU)/g at 3 kilograys.

CHARACTERISTICS OF DISEASE

Shigellosis is differentiated from diseases caused by most other foodborne pathogens described in this book by at least two important characteristics: (i) the production of bloody diarrhea or dysentery and (ii) the low infectious dose. Dysentery involves bloody diarrhea, but the passage of bloody mucoid stools is accompanied by severe abdominal and rectal pain,

BOX 11.1

Shigella is easily transmitted

Shigella is easily passed from person to person by food. Outbreaks are often associated with poor hygiene, especially improper hand washing after use of the bathroom. The pathogen multiplies rapidly in food at room temperature. *Shigella* produces a powerful toxin, Shiga toxin, once in the host. Shigellosis is an infection, not an intoxication. That is, vegetative cells must be ingested, after which the cells multiply and cause illness. As few as 100 cells may cause illness.

cramps, and fever. While abdominal pain and diarrhea are experienced by nearly all patients with shigellosis, fever occurs in about one-third and gross blood in the stools occurs in ~40% of cases. The clinical features of shigellosis range from a mild watery diarrhea to severe dysentery. The dysentery stage caused by *Shigella* spp. may or may not be preceded by watery diarrhea. During the dysentery stage, there is extensive bacterial colonization of the colon and invasion of the cells of the colon. As the infection progresses, dead cells of the mucosal surface slough off. This leads to the presence of blood, pus, and mucus in the stools.

Shigella is a serious pathogen that causes disease in otherwise healthy individuals (Box 11.1). The greatest frequency of illness is among children <6 years of age. The incubation period for shigellosis is 1 to 7 days, but the symptoms usually begin within 3 days. The severity of illness differs depending on the strain involved; however, regardless of the severity of the illness, shigellosis is self-limiting. If left untreated, clinical illness usually persists for 1 to 2 weeks (although it may last as long as 1 month), and the patient recovers. Since the infection is self-limited in normally healthy patients and full recovery occurs without the use of antibiotics, drug therapy is usually not indicated. However, the antibiotic of choice for treatment of shigellosis is trimethoprim-sulfamethoxazole. Complications arising from the disease include severe dehydration, intestinal perforation, septicemia, seizures, hemolytic-uremic syndrome, and Reiter's syndrome.

FOODBORNE OUTBREAKS

Although the number of reported foodborne outbreaks of shigellosis in the United States has declined recently, shigellosis continues to be a major public health concern. Data from 1982 to 1997 estimated that there were approximately 448,000 cases of shigellosis in the United States during that period. This made it the third leading cause of foodborne outbreaks by bacterial pathogens. Worldwide, the World Health Organization (WHO) estimates that *Shigella* spp. are responsible for 164.7 million cases of shigellosis annually in developing countries and 1.5 million cases in developed countries. The CDC, WHO, and other international agencies have instituted surveillance systems to monitor foodborne outbreaks caused by *Shigella.*

Humans are the natural reservoir of *Shigella.* Human-to-human transmission of *Shigella* is through the fecal-oral route. Most cases of shigellosis result from the ingestion of fecally contaminated food or water. With foods, the major cause of contamination is poor personal hygiene of food handlers. From infected carriers, shigellae are spread by several routes, including food, fingers, feces, and flies. The highest incidence of shigellosis occurs during the warmer months of the year. Improper storage of contaminated foods is the second most common factor contributing to foodborne outbreaks of shigellosis. Other contributing factors are inadequate cooking, contaminated equipment, and food obtained from unsafe sources. To reduce the spread of shigellosis, infected patients should be monitored until stool samples are negative for *Shigella.* The low infectious dose of *Shigella* underlies the high rate of transmission. As few as 100 cells of *Shigella* can cause illness, facilitating person-to-person spread, as well as foodborne and waterborne outbreaks of diarrhea.

Shigellosis can be widespread in institutional settings, such as prisons, mental hospitals, and nursing homes, where crowding and/or insuffi-

ciently hygienic conditions create an environment for direct fecal-oral contamination. The occurrence of disasters that destroy the sanitary waste treatment and water purification infrastructure is often associated with large outbreaks of shigellosis.

VIRULENCE FACTORS

The clinical symptoms of shigellosis can be directly attributed to the hallmarks of *Shigella* virulence: the ability to induce diarrhea, invade epithelial cells of the intestine, multiply intracellularly, and spread from cell to cell. The production of enterotoxins by the bacteria while they are in the small bowel probably causes the diarrhea that precedes dysentery. The ability of *Shigella* to invade epithelial cells and move from cell to cell is regulated by an array of genes. A bacterium's mechanism for regulating expression of the genes involved in virulence is important for pathogenicity. *S. dysenteriae* produces a thermolabile toxin, designated Shiga toxin, that is involved in the pathogenesis of *Shigella* diarrhea. Strains that are invasive and produce the toxin cause the most severe infection.

The growth temperature is an important factor in controlling virulence. Virulent strains of *Shigella* spp. are invasive when grown at 37°C but noninvasive when grown at 30°C. This strategy ensures that the organism conserves energy by synthesizing virulence products only when the bacterium is in the host.

Genetic Regulation

Given the complexity of the interactions between host and pathogen, it is not surprising that *Shigella* virulence requires several genes. These include both chromosomal and plasmid-carried genes. The large plasmid has an indispensable role in invasion by *S. sonnei* and *S. flexneri.* Other *Shigella* spp., as well as strains of EIEC, contain similar plasmids, which are functionally interchangeable and show significant degrees of DNA relatedness. The plasmids of *Shigella* and EIEC are probably derived from a common ancestor.

The *ipa* group of genes encode invasion plasmid antigens, the main antigens detected with sera from convalescent patients and experimentally challenged monkeys. These genes are required for the invasion of mammalian cells. Ultimately, the proteins form a complex on the bacterial cell surface and are responsible for transducing the signal leading to entry of *Shigella* into the host cells via bacterium-directed phagocytosis. The products of the *ipa* genes have also been postulated to be the contact hemolysin responsible for lysis of the phagocytic vacuole minutes after entry of the bacterium into the host cell. The ability of *S. flexneri* to induce programmed cell death in infected macrophages is an additional property assigned to IpaB.

In contrast to the genes of the virulence plasmid that are responsible for the invasion of mammalian tissues, most of the chromosomal loci associated with *Shigella* virulence are involved in regulation or survival within the host. Although *Shigella* and *E. coli* are very closely related at the genetic level, there are significant differences beyond the presence of the virulence plasmid in *Shigella.* In addition to extra genes in the *Shigella* chromosome, there are genes present in the closely related *E. coli* that are missing from the chromosome of *Shigella;* genetic divergence accounts for differences in the virulence and pathogenicity of *Shigella* and *E. coli.*

Authors' note

Genes that encode virulence are highly regulated by temperature. At typical ambient temperatures (below 30°C, or 86°F) at which foods would be expected to be held, virulence gene expression is suppressed. Once the organism is ingested and exposed to body temperature (37°C, or 98.6°F), virulence genes are activated and the organism is now capable of causing severe illness.

CONCLUSIONS

While foodborne infections due to *Shigella* spp. may not be as frequent as those caused by other foodborne pathogens, they have the potential for explosive spread due to the extremely low infectious dose that can cause overt clinical disease. In addition, cases of bacillary dysentery frequently require medical attention (even hospitalization), resulting in time lost from work, as the severity and duration of symptoms can be incapacitating. There is no effective vaccine against dysentery caused by *Shigella*. These features, coupled with the wide geographical distribution of the strains and the sensitivity of the human population to *Shigella* infection, make *Shigella* a formidable public health threat.

Summary

- The infective dose may be as low as 100 cells.
- Shigellosis is self-limiting.
- Humans are the natural reservoir of *Shigella*.
- *Shigella* is spread through the fecal-oral route; therefore, hand washing is one of the most effective control measures.
- Expression of virulence genes is regulated by temperature.
- *Shigella* shares many virulence genes with *E. coli*.
- Genes responsible for virulence are located on a virulence plasmid.

Suggested reading

Lampel, K. A., and A. T. Maurelli. 2007. *Shigella* species, p. 323–342. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Niyogi, S. K. 2005. Shigellosis. *J. Clin. Microbiol.* **43:**133–143.

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Questions for critical thought

1. Shigellosis is distinguished from diseases caused by other foodborne pathogens by at least two characteristics. What are the two key characteristics?
2. Why is the use of antibiotics to treat shigellosis controversial?
3. What is meant by the term infectious dose? Compared to other foodborne pathogens, does *Shigella* have a low infectious dose?
4. How does temperature play a role in *Shigella* pathogenesis?
5. Once *Shigella* has invaded epithelial cells of the host, what unique function can the pathogen perform and how does that influence pathogenesis?
6. What is the most likely route by which food becomes contaminated with *Shigella*?
7. What characteristics of the pathogen with respect to reservoir and shedding pattern may exacerbate its spread and the subsequent outbreak of shigellosis?
8. Many cases of shigellosis are linked to consumption of fresh produce. What measures could be implemented to reduce the risk of shigellosis from consumption of fresh vegetables?

9. Extra-hard question. The Food and Drug Administration (FDA) approved the use of *Listeria*-specific bacteriophage preparation on ready-to-eat meat and poultry products as a means to control the foodborne pathogen *Listeria monocytogenes*. Provide a brief narrative to address each point listed below with respect to the use of phages to control *Shigella* associated with a food of your choice. The points to be addressed are phage source, specificity of phages, phage titer, allergenicity, immunogenicity, method of treating food, determining efficacy of phage treatment, benefit compared to existing methods, and potential pitfalls associated with utilization of phages as a natural decontamination system.

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12 *Vibrio* Species

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Summary
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Vibrio*
- understand what conditions in foods favor *Vibrio* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *Vibrio*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent the growth of *Vibrio*
- identify environmental sources of the organism
- understand the role of *Vibrio* toxins and virulence factors in causing foodborne illness

Outbreak

Raw oysters are consumed for their taste and because of the old wives' tale that they act as an aphrodisiac. People rarely consider the hazards associated with the consumption of raw shellfish. During a 2-month period (July and August) in 1997, one of the largest outbreaks of culture-confirmed *Vibrio parahaemolyticus* infection occurred in North America. At least 209 persons became ill with gastroenteritis from eating raw oysters. Symptoms included diarrhea, abdominal cramps, nausea, and vomiting. Although symptoms of gastroenteritis may attract the attention of others, they would certainly not be considered attractive. Oysters sampled from the beds implicated in the outbreak yielded <200 *V. parahaemolyticus* colony-forming units (CFU)/g of oyster meat, suggesting that human illness may occur at levels of bacteria lower than the current action level. The United States and Canada permit the sale of oysters if there are $<10^4$ CFU of *V. parahaemolyticus* per g of oyster meat.

INTRODUCTION

Over 20 *Vibrio* species have now been described. At least 12 are capable of causing infection in humans, although with the exception of *Vibrio cholerae* and *V. parahaemolyticus*, little is known about the virulence mechanisms they employ. Of the 12 pathogens, 8 are directly food associated, and these are the subject of this chapter. (*Vibrio carchariae* and *Vibrio damsela* infections appear to result solely from wound infections, and the routes of infection for *Vibrio metschnikovii* and *Vibrio cincinnatiensis* are unclear.)

One of the most consistent aspects of vibrio infections is a recent history of seafood consumption. Vibrios, the predominant bacterial genus in estuarine waters, are associated with a great variety of seafoods. Approximately 40

to 60% of finfish and shellfish at supermarkets may contain *Vibrio* spp., with *V. parahaemolyticus* and *Vibrio alginolyticus* being the most commonly isolated. Vibrios are most frequently isolated from molluscan shellfish during the summer months. The estimated annual incidence of *Vibrio* infections increased 78% from baseline (1996–1998 period) to 2006. Aside from additional measures to reduce contamination of seafood, consumers should be informed that they are at risk for *Vibrio* infections when they consume raw seafood.

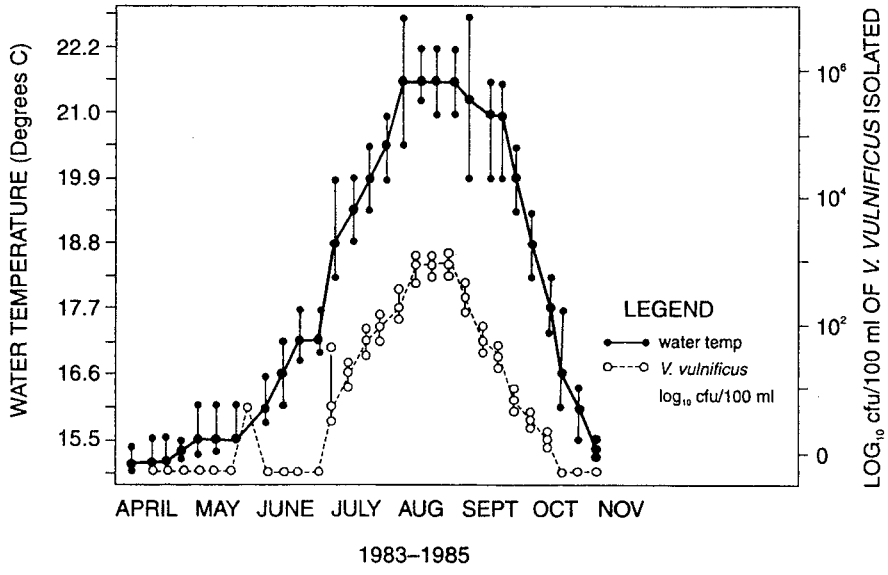
CHARACTERISTICS OF THE ORGANISM

Many different enrichment broths for the isolation of vibrios have been described, although alkaline peptone water remains the most commonly used. These are frequently coupled with thiosulfate-citrate-bile salts-sucrose (TCBS) agar or other plating media. In order to distinguish vibrios from the *Enterobacteriaceae*, sucrose-positive colonies are subjected to the oxidase test (used to detect the presence of cytochrome *c*); however, erroneous results may arise when colonies are obtained directly from TCBS agar. Therefore, sucrose-positive colonies on TCBS should be subcultured by heavy inoculation onto a nonselective medium, such as blood agar, and allowed to grow for 5 to 8 h before being tested for oxidase activity.

Epidemiology

The numbers of vibrios in both surface waters and shellfish correlate with seasonality, generally being greater during the warm-weather months between April and October (Fig. 12.1). Similarly, vibrios are more commonly isolated from the warmer waters of the Gulf and East Coasts than from those of the West and Pacific Northwest. Seasonality is most notable for *Vibrio vulnificus* and *V. parahaemolyticus* infections, whereas some vibrios, such as *Vibrio fluvialis*, occur throughout the year.

Figure 12.1 Numbers of CFU of *V. vulnificus* (± standard deviations) isolated from northeastern United States coastal waters as a function of time of year and water temperature. Reprinted from R. C. Tilton and R. W. Ryan, *Diagn. Microbiol. Infect. Dis.* 6:109–117, 1987, with permission of the publisher.



CHARACTERISTICS OF DISEASE

There is considerable variation in the severities of the various *Vibrio*-associated diseases. The outcomes of infections depend on an individual's underlying health (e.g., patients with chronic liver disease are more likely to succumb to severe illness). An exception to this generalization is *V. cholerae* O1/O139, which can readily make noncompromised individuals sick. Most cases are associated with a recent history of seafood consumption. Because vibrios are part of the normal estuarine microflora caused by fecal contamination, *Vibrio* infections cannot be controlled through shellfish sanitation programs. It is therefore essential that raw seafood be kept cold to prevent significant bacterial growth and that it be properly cooked before consumption.

Symptoms of foodborne illness for the two most common *Vibrio* species differ and are worth noting. Symptoms of *V. vulnificus* foodborne illness generally include fever, chills, and nausea. Diarrhea is generally not experienced. Mortality occurs in 40 to 60% of cases. The most common symptoms of *V. parahaemolyticus* infection are diarrhea, abdominal cramps, nausea, and vomiting. Mortality is extremely low.

SUSCEPTIBILITY TO PHYSICAL AND CHEMICAL TREATMENTS

With the exception of *V. cholerae* and *V. parahaemolyticus*, relatively little is known about the effects of preservation methods on inactivation of vibrios. Generally, vibrios are sensitive to cold, and seafoods can be protective for vibrios at refrigeration temperatures. Vibrios have survived for extended periods in products stored at refrigeration temperatures and after prolonged frozen storage. Thermal processing is a very effective means of reducing populations of *Vibrio* in foods. The U.S. Food and Drug Administration recommends steaming shellstock oysters, clams, and mussels for 4 to 9 min; frying shucked oysters for 10 min at 375°F; or baking oysters for 10 min at 450°F. Thorough heating of shellfish to an internal temperature of at least 60°C for several minutes should kill pathogenic vibrios. Irradiation and high hydrostatic pressure also reduce the numbers of *Vibrio* organisms. High hydrostatic pressure has an additional advantage in that the process perfectly shucks the oyster. A large variety of dried spices, the oils of several herbs, tomato sauce, and several organic acids exhibit bactericidal activity. The process of depuration, in which filter-feeding bivalves are purified of certain bacteria by pumping bacterium-free water through the bivalve tissues, removes *Salmonella* and *Escherichia coli*, but not *Vibrio*.

V. cholerae

V. cholerae O1 causes cholera, one of the few foodborne illnesses with *epidemic* (many people in one area become sick at the same time) and *pandemic* (similar to an epidemic but over a larger area) potential. Important distinctions within the species are made on the basis of production of cholera enterotoxin (cholera toxin [CT]), serogroup, and potential for epidemic spread. Not all *V. cholerae* strains of the O1 serogroup produce CT; however, *V. cholerae* O1 CT-producing strains have long been associated with epidemic and pandemic cholera. Nearly 200 serogroups of *V. cholerae* have been described. Serotyping of *V. cholerae* strains is based on the lipopolysaccharide (LPS), also known as the somatic (O) antigen; H antigens are not useful in serotyping.

Isolation and Identification

Suspected *V. cholerae* isolates can be cultured on a standard series of biochemical media used for identification of *Enterobacteriaceae* and *Vibrionaceae*. Both conventional tube tests and commercially available enteric identification systems work well.

Nucleic acid probes are not routinely used for the identification of *V. cholerae* due to the ease of identifying this species by conventional methods. DNA probes are useful in distinguishing CT-producing strains of *V. cholerae* through the detection of CT (*ctx*) genes. Restriction fragment length polymorphism analysis has been useful in epidemiological studies of *V. cholerae*.

Reservoirs

V. cholerae is part of the normal, free-living bacterial flora in estuarine areas. Non-O1/non-O139 strains are much more commonly isolated from the environment than are O1 strains, even in epidemic settings in which fecal contamination of the environment might be expected. Periodic introduction of such environmental isolates into the human population through the ingestion of uncooked or undercooked shellfish appears to be responsible for localized outbreaks along the U.S. Gulf Coast and in Australia. The persistence of *V. cholerae* within the environment may be facilitated by its ability to assume survival forms, including a viable but nonculturable (VBNC) state. In this dormant state, the cells are reduced in size and become ovoid. Although VBNC vibrios are not culturable with nonselective enrichment broth or plates, studies have demonstrated that VBNC *V. cholerae* O1 organisms injected into ligated rabbit ileal loops or ingested by volunteers have yielded culturable *V. cholerae* O1 in the intestinal contents or stool specimens, respectively.

Long-term carriage of *V. cholerae* in humans is extremely rare and is not important in the transmission of disease. However, even after the cessation of symptoms, patients who have not been treated with antibiotics may continue to excrete vibrios for 1 to 2 weeks. Asymptomatic carriers are most commonly identified among household members of persons with acute illness: in various studies, the rate of asymptomatic carriage in this group has ranged from 4 to almost 22%.

Foodborne Outbreaks

The critical role of water in the transmission of cholera has been recognized for more than a century. In 1854, the London physician and epidemiologist John Snow determined that illness was associated with consumption of water derived from the Thames River at a point below major sewage inflows. The range of food items implicated in the transmission of cholera includes crabs, shrimp, raw fish, mussels, cockles, squid, oysters, clams, rice, raw pork, millet gruel, cooked rice, food bought from street vendors, frozen coconut milk, and raw vegetables and fruit. One shared characteristic of the foods is their neutral or nearly neutral pH. When foodborne outbreaks occur and a number of different foods are suspected, foods with an acid pH can be eliminated. Food acts to buffer *V. cholerae* O1 against killing by gastric acid. In the United States, crabs, shrimp, and oysters have been the most frequently implicated vehicles of *V. cholerae* illness.

Characteristics of Disease

The explosive, dehydrating diarrhea characteristic of cholera actually occurs in only a minority of persons infected with CT-producing *V. cholerae* O1/O139. Most infections with *V. cholerae* O1 are mild or even asympto-

matic. The incubation period of cholera can range from several hours to 5 days and is dependent in part on the *inoculum* size (the number of cells ingested). The onset of illness may be sudden, with watery diarrhea, or there can be loss of appetite, abdominal pain, and simple diarrhea. Initially, the stool is brown with fecal matter, but once diarrhea starts, it becomes a pale gray color with a slightly fishy odor. Mucus in the stool gives the characteristic "rice water" appearance. Vomiting can occur a few hours after the onset of diarrhea. In healthy North American volunteers, doses of 10^{11} CFU of *V. cholerae* were required to cause diarrhea when the inoculum was given in buffered saline (pH 7.2). However, a lower number of bacteria caused illness in volunteers when 10^6 vibrios were given, with food (fish and rice) acting as a buffer.

Virulence Mechanisms

Infection due to *V. cholerae* O1/O139 begins with the ingestion of food or water contaminated with the pathogen. After passage through the acid barrier of the stomach, vibrios colonize the small intestine using one or more adherence factors. Production of CT (and possibly other toxins) disrupts ion transport by intestinal epithelial cells. This leads to the severe diarrhea characteristic of cholera with excessive loss of water and electrolytes. Cholera halotoxin is composed of five identical B subunits and a single A subunit; the individual subunits are not sufficient to cause secretogenic activity (an increase in secretion of cellular components) in animals or intact cell culture systems.

V. cholerae produces a variety of extracellular products that are harmful to eukaryotic cells. In addition to the well-characterized toxins (soluble hemagglutinin and hemolysins), *V. cholerae* can produce a number of other toxic factors, but the responsible proteins and genes have not yet been purified or cloned. A number of other virulence factors are involved in illness, including colonization factors, flagella, LPS, and polysaccharide capsule.

V. mimicus

Before 1981, *Vibrio mimicus* was known as sucrose-negative *V. cholerae* non-O1. This species was determined to be a distinctly different species on the basis of biochemical reactions and DNA hybridization studies, and the name *mimicus* was given because of its similarity to *V. cholerae.* This organism is isolated mainly from cases of gastroenteritis but can also cause ear infections. The reservoir of *V. mimicus* is the aquatic environment. Besides being present free in the water, *V. mimicus* was also isolated from the roots of aquatic plants, from sediments, and from plankton at levels of up to 6×10^4 CFU per 100 g of plankton.

Foodborne Outbreaks

Gastroenteritis due to *V. mimicus* has been associated only with consumption of seafood. In the United States, consumption of raw oysters is the main cause of illness due to this species. In Japan, consumption of raw fish has resulted in at least two outbreaks involving *V. mimicus* of serogroup O41.

Characteristics of Disease

Disease due to *V. mimicus* is characterized by diarrhea, nausea, vomiting, and abdominal cramps in most patients. In some individuals, fever, headache, and bloody diarrhea also occur. There are no volunteer or epidemiologic data to enable estimation of an infectious dose for *V. mimicus.*

There is no particularly susceptible population, other than people who eat raw oysters.

Virulence Factors

V. mimicus does not produce unique enterotoxins, but many strains produce toxins that were first described for other *Vibrio* species, including CT, thermostable direct hemolysin (TDH), Zot, and a heat-stable enterotoxin apparently identical to the NAG-ST produced by *V. cholerae* non-O1/non-O139 strains. There is little information about potential intestinal colonization factors of *V. mimicus.*

V. parahaemolyticus

Along with *V. cholerae, V. parahaemolyticus* is the best described of the pathogenic vibrios, with numerous studies reported since the first description of its involvement in a major outbreak of food poisoning in 1950. Between 1973 and 1998, a total of 40 outbreaks of *V. parahaemolyticus* infection in the United States were reported to the Centers for Disease Control and Prevention, with >1,000 persons involved. In the United States the percentage of *Vibrio* infections attributed to *V. parahaemolyticus* continues to increase, from 54% in 2005 to 64% in 2006.

Classification

V. parahaemolyticus is serotyped according to both its somatic (O) and capsular polysaccharide (K) antigens. Currently 12 O (LPS) antigens and 59 K (capsular polysaccharide) antigens are recognized. Although many environmental and some clinical isolates are untypeable by the K antigen, most clinical strains can be classified based on their O types. A special consideration in the taxonomy of *V. parahaemolyticus* is the ability of certain strains to produce a hemolysin, called the TDH, or Kanagawa, hemolysin, which is linked to virulence in the species.

Reservoirs

V. parahaemolyticus occurs naturally in estuarine waters throughout the world and is easily isolated from coastal waters of the United States, as well as from sediment, suspended particles, plankton, and a variety of fish and shellfish. The last source includes at least 30 different species, among them clams, oysters, lobsters, scallops, shrimp, and crabs. A high percentage of seafood samples tested positive for the species. *V. parahaemolyticus* counts are season dependent; samples analyzed in January and February are often free of *V. parahaemolyticus.*

Foodborne Outbreaks

Gastroenteritis from *V. parahaemolyticus* is almost exclusively associated with seafood that is consumed raw, inadequately cooked, or cooked but recontaminated. In Japan, *V. parahaemolyticus* is a major cause of foodborne illness. Approximately 70% of all bacterial foodborne illnesses in the 1960s were the result of this pathogen. Seafood, including fish, crabs, shrimps, lobsters, and oysters, is primarily associated with U.S. outbreaks. The first major outbreak (with 320 persons ill) in the United States occurred in Maryland in 1971, a result of eating improperly steamed crabs. Subsequent outbreaks occurred throughout U.S. coastal regions and Hawaii. The largest U.S. outbreak of *V. parahaemolyticus* infection occurred during the summer of 1978 and affected 1,133 of 1,700 persons attending a dinner in Port Allen, LA.

Characteristics of Disease

V. parahaemolyticus has generation times of 8 to 9 min at 37°C and 12 to 18 min in seafood. Hence, *V. parahaemolyticus* has the ability to grow rapidly, both in vitro and in vivo, contributing to the infectious dose required for illness. Symptoms may begin 4 to >30 h after the ingestion of contaminated food, with a mean onset time of 23.6 h. Primary symptoms include diarrhea and abdominal cramps, along with nausea, vomiting, and fever. The symptoms subside in 3 to 5 days in most individuals. Approximately 10^5 to 10^7 CFU is required for illness. The numbers of *V. parahaemolyticus* organisms present in fish and shellfish are usually no greater than 10^4 Kanagawa phenotype-positive cells, suggesting that temperature abuse of contaminated food occurs prior to consumption. The temperature abuse permits the growth of the pathogen to levels that result in illness.

Authors' note

Determining whether strains produce Kanagawa hemolysin or TDH requires the use of a special blood agar called Wagatsuma agar. With the explosion of polymerase chain reaction (PCR)-based methods, the presence of the gene (tdh) *that encodes the toxin can now be easily determined.*

Virulence Mechanisms

Although the epidemiologic linkage between virulence for humans and the ability of *V. parahaemolyticus* isolates to produce the Kanagawa hemolysin has long been established, the molecular mechanisms by which this factor can cause diarrhea have only recently been elucidated. *V. parahaemolyticus* possesses at least three hemolytic components: a thermolabile hemolysin gene (*tlh*), a TDH gene (*tdh*), and a TDH-related gene (*trh*), linked to disease.

While much is understood regarding the toxins of *V. parahaemolyticus*, little is known of the adherence process. This is an essential step in the pathogenesis of most enteropathogens (pathogens that infect the intestines). Several adhesive factors have been proposed, including the outer membrane, lateral flagella, pili, and a mannose-resistant, cell-associated hemagglutinin; however, the importance of any of these factors in human disease is unknown.

V. vulnificus

V. vulnificus is the most serious of the pathogenic vibrios in the United States. In Florida, *V. vulnificus* is the leading cause of reported deaths due to foodborne illness. Approximately 86% of people experiencing *V. vulnificus* foodborne illness require hospitalization. Among the susceptible population at risk for infection by this bacterium, primary septicemia cases resulting from raw oyster consumption typically have fatality rates of 60%. This is the highest death rate for any foodborne disease agent in the United States. The Centers for Disease Control and Prevention estimates that there are ~50 cases of foodborne *V. vulnificus* infection annually. The bacterium can produce wound infections in addition to gastroenteritis and primary septicemias. Wound infections carry a 20 to 25% fatality rate and are also seawater and/or shellfish associated. Surgery is usually required to clean the infected tissue. In some cases, amputation of the infected limb is required.

Authors' note

Raw oysters have been enjoyed for centuries. However, food microbiologists know that nearly all cases of Vibrio parahaemolyticus *and* V. vulnificus *infection result from consumption of raw oysters. A fresh oyster is not necessarily uncontaminated.*

Classification

The phenotypic traits of this species have been fully described in several studies. The isolation of *V. vulnificus* from blood samples is straightforward, as the bacterium grows readily on TCBS, MacConkey, and blood agars. Isolation from the environment is much more difficult. Vibrios comprise 50% or more of estuarine bacterial populations, and most have not been characterized. Considerable variation exists in the phenotypic traits of *V. vulnificus*, including lactose and sucrose fermentation, considered among the most important characteristics in identifying the species.

Susceptibility to Control Methods

V. vulnificus is susceptible to freezing, low-temperature pasteurization, high hydrostatic pressure, and ionizing radiation. The pathogen can be killed through exposure to horseradish-based sauces. Application of these sauces to raw oysters will not make the oysters safe to eat, since they do not kill bacteria within the oyster.

Reservoirs

V. vulnificus is widespread in estuarine environments, from the Gulf, Atlantic, and Pacific coasts of the United States to locations around the world. The presence of *V. vulnificus* in water is not associated with the presence of fecal coliforms. Water temperature does have an impact on the presence and levels of the pathogen in water. *V. vulnificus* is seldom isolated from water or oysters when water temperatures are low. This was thought to result from cold-induced death during cold-weather months. However, it may be related to a cold-induced VBNC state, in which the cells remain viable but are no longer culturable on the routine media normally employed for their isolation.

Foodborne Outbreaks

There is no report of more than one person developing *V. vulnificus* infection following consumption of the same lot of oysters. Raw oysters from the same lot, or even the same serving, may have very different levels of the pathogen. In fact, two oysters taken from the same estuarine location may have vastly different populations of *V. vulnificus*. In most cases *V. vulnificus* illness occurs between April and October, when the water is warmer. Nearly all cases of *V. vulnificus* infection result from consumption of raw oysters, and most of these infections result in primary septicemias.

Characteristics of Disease

Discussion here is limited to the primary foodborne form of infection caused by *V. vulnificus*. Symptoms vary considerably among cases, with onset times ranging from 7 h to several days, with a median of 26 h. Symptoms include fever, chills, nausea, and hypotension. Surprisingly, symptoms typical of gastroenteritis—abdominal pain, vomiting, and diarrhea—are not common. In severe cases, survival depends to a great extent directly on prompt antibiotic administration. The infectious dose of *V. vulnificus* is not known. *V. vulnificus* is susceptible to most antibiotics.

Virulence Mechanisms

The polysaccharide capsule, LPS, and a large number of extracellular compounds contribute to virulence. The polysaccharide capsule, which is produced by nearly all strains of *V. vulnificus*, is essential to the bacterium's ability to initiate infection. Elevated levels of serum iron appear to be essential for *V. vulnificus* to multiply in the human host. Avirulent strains produce "translucent" acapsular colonies (Fig. 12.2). Symptoms which occur during *V. vulnificus* septicemia, including fever, tissue edema, hemorrhage, and especially the significant hypotension, are those classically associated with endotoxic shock caused by gram-negative bacteria.

V. fluvialis, V. furnissii, V. hollisae, and *V. alginolyticus*

V. fluvialis has been isolated frequently from brackish and marine waters and sediments in the United States, as well as other countries. It also has been isolated from fish and shellfish from the Pacific Northwest and Gulf

Figure 12.2 Opaque (encapsulated) and translucent (acapsular) colonies of *V. vulnificus*. Virulent strains are generally encapsulated, whereas nonencapsulated cells are usually avirulent. Reprinted from L. M. Simpson, V. K. White, S. F. Zane, and J. D. Oliver, *Infect. Immun.* **55:**269–272, 1987, with permission.

Coast. *Vibrio furnissii* has been isolated from river and estuarine waters, marine mollusks, and crustacea throughout the world. The distribution of *Vibrio hollisae* is not well documented, although it is likely a marine species. It appears that the bacterium prefers warm waters. *V. alginolyticus* is often found in high numbers in seawater and seafood obtained throughout the world. It is easily isolated from fish, clams, crabs, oysters, mussels, and shrimp, as well as water. Human infections caused by vibrios such as *V. fluvialis, V. furnissii, V. hollisae,* and *V. alginolyticus* are less common and usually less severe than those caused by the pathogens discussed above, although deaths have been reported. Consumption of contaminated seafood is usually the source of infection.

Summary

- Illness associated with *V. cholerae* is not common in the United States.
- *V. vulnificus* and *V. parahaemolyticus* are most commonly associated with foodborne illness.
- *V. vulnificus* is solely responsible for 95% of all deaths due to seafood-borne infections in the United States.
- Numbers of vibrios in water are usually largest during the warm-weather months between April and October.
- Proper thermal processing of seafood kills *Vibrio* species.

Suggested reading

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Questions for critical thought

1. *Vibrio* may enter a VBNC state in water. What are key characteristics of this state, and how is this state associated with survival of the organism?
2. Why would treatment of oysters with Tabasco sauce or other similar agents be ineffective in eliminating all *Vibrio* spp. from an oyster?
3. Sanitation programs are often of limited value in the control of vibrios associated with seafood. Why?
4. Which *Vibrio* species is associated with cutaneous infection? What group of people is most at risk? Are cutaneous infections a significant health risk?
5. What do the common virulence characteristics of *V. parahaemolyticus* and *V. vulnificus* include? How do these contribute to pathogenesis?
6. Upon review of a patient's symptoms, how could you presumptively state whether the individual was ill from *V. vulnificus* or *V. parahaemolyticus?*
7. There are many preservation methods that could be used to ensure the safety of seafood. What methods could you use to ensure the safety of fresh seafood? Suggest advantages and disadvantages of those methods.

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SECTION III Gram-Positive Foodborne Pathogenic Bacteria

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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Listeria monocytogenes*
- understand what conditions in foods favor *L. monocytogenes* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *L. monocytogenes*
- choose appropriate interventions (heat, preservatives, or formulation) to prevent *L. monocytogenes* growth
- identify environmental sources of *L. monocytogenes*
- understand the role of *L. monocytogenes* toxin(s) and virulence factor(s) in causing foodborne illness

Outbreak

The time line of foodborne outbreaks, from the moment of contamination until the last case is reported, can be very long. This case study examines important questions related to these outbreaks. How long does it take to detect an outbreak of a sporadic disease? How much evidence does it take to launch a recall? What do you do with 35 million pounds of recalled hot dogs?

These are just a few of the questions asked in a listeriosis outbreak that spanned 1998 and 1999, killed 12 people, and made 79 people in 17 states ill. The sequence of events is outlined below:

Spring and summer, 1998: A hot-dog-processing plant had an unusually high isolation rate of listeriae from the environment.

July 4 weekend, 1998: An overhead air-conditioning unit thought to be the source of listeriae was cut apart and removed from the plant.

October 1998: The Centers for Disease Control and Prevention (CDC) began investigating four listeriosis cases from Tennessee. A 74-year-old mother of four who loved hot dogs was dead. Her flu-like illness made her drowsy and headachy and then killed her. Soon, more cases were reported from Ohio, New York, and Connecticut.

November 1998: A 31-year-old camp counselor died in Ohio. The genetic fingerprints of bacteria isolated from the listeriosis victims in the different states were identical. They must have eaten a common food.

Early December 1998: The food was statistically identified as hot dogs.

Mid-December 1998: Hot dog brands packed at Sara Lee's Bil Mar plant were implicated. With four people dead, the plant stopped shipping product.

22 December 1998: Sara Lee announced the recall of all meat made at the Bil Mar plant.

23 December 1998: The outbreak strain of *Listeria monocytogenes* was isolated from an unopened package of Bil Mar hot dogs. This meets the U.S. Department of Agriculture (USDA) criterion for a recall.

25 December 1998: Another victim died in upstate New York.

January 1999: The USDA issued its first press release on the recall. It was of no help to a pregnant 27-year-old video store manager. She survived a severe "case of the flu" only to see her premature twin babies delivered stillborn. *Listeria monocytogenes* was isolated from her placenta.

May 1999: The USDA advised immunocompromised consumers not to eat hot dogs or deli meats unless they were thoroughly heated.

This is the story of but one listeria recall. The USDA typically issues about 30 recalls for *L. monocytogenes* contamination each year.

INTRODUCTION

Authors' note

An early step in the investigation of foodborne disease is to determine what food(s) was eaten by people who got sick, but not by those who did not get sick. These numbers can statistically implicate a food.

Over the last 25 years, listeriosis has become a major foodborne disease. The first outbreak was traced to coleslaw in 1981. Outbreaks were soon linked to cheeses, lunch meats, milk, chicken nuggets, and fish. There are now about 2,000 cases of listeriosis every year, with 500 fatalities. Many factors contribute to the emergence of listeriosis and other "new" pathogens. These include the following.

- Eliminating the spoilage bacteria that prevent listeriae from growing. *L. monocytogenes* is a relatively poor competitor with other bacteria but thrives under refrigerated conditions when competing bacteria have been eliminated.
- Changing demographics, with more people at risk due to old age, immunosuppressants, organ transplants, etc. Some pathogens, such as *Clostridium botulinum*, can make anyone sick. Others, like *L. monocytogenes*, are "opportunistic" pathogens that attack only people whose defenses are down.
- Changing food production practices, particularly centralization and consolidation. Small institutional kitchens are increasingly consolidated into larger centralized facilities which make good hygienic practices more challenging.
- Increasing use of refrigeration to preserve foods. *L. monocytogenes* grows better than other organisms in the cold.
- Changing eating habits with increased consumer demand for "fresh," "minimally processed," and "natural" foods that require little cooking or preparation and do not contain preservatives that would prevent listerial growth.
- Changing awareness (have you ever noticed that once you buy a silver Nissan Altima, the road is full of silver Nissan Altimas?) and ability to detect sporadic outbreaks. Computerized databases, DNA fingerprinting, and the Internet have played key roles in enabling us to detect previously unrecognized sporadic outbreaks.

Listeriosis, the disease, is not a typical foodborne illness. The disease is sporadic and rare, but severe. It can cause meningitis, septicemia, and abortion (Table 13.1). Twenty to 30% of people who get listeriosis die from it (i.e., the fatality:case ratio is high), and there is a long time lag from when the food is eaten to the onset of illness. *L. monocytogenes*, the organism, differs from most other foodborne pathogens. The organism is widely distrib-

Table 13.1 Symptoms caused by *L. monocytogenes*

Low-grade flu-like infection—not serious, except in pregnant woman (who abort)
Listeric meningitis—headache, drowsiness, coma
Perinatal infection
Encephalitis
Psychosis
Infectious mononucleosis
Septicemia

uted in nature, resistant to adverse environmental conditions, and psychrotrophic and grows in human phagocytes. In addition, *Listeria* survives for long periods in or on food soil, plants, and hard surfaces. These traits make listeria control difficult for the food industry. It is unclear how *L. monocytogenes* should be viewed. The disease's severity and case-fatality rate require that it be controlled. But given the organism's characteristics, it is unrealistic to make all food *Listeria*-free.

Authors' note

Once the cell is phagocytized, it can continue growing and move from cell to cell without reentering the bloodstream.

CHARACTERISTICS OF THE ORGANISM

Classification

The Genus *Listeria*

Joseph Lister (Fig. 13.1), the inventor of medical disinfection, had a mouthwash and a genus of bacteria named after him. This genus, *Listeria*, is related to *Clostridium*, *Staphylococcus*, *Streptococcus*, *Lactobacillus*, and *Brochothrix*. *L. monocytogenes* is one of six species in the genus *Listeria*. The other species are *L. ivanovii*, *L. innocua*, *L. seeligeri*, *L. welshimeri*, and *L. grayi*. Within the genus *Listeria*, only *L. monocytogenes* and *L. ivanovii* are pathogens. *L. monocytogenes* is a human pathogen. *L. ivanovii* is primarily an animal pathogen.

Listeria species are differentiated by a few biochemical traits. The biochemical tests that characterize the species are acid production from D-xylose, L-rhamnose, alpha-methyl-D-mannoside, and D-mannitol. The ability to lyse red blood cells differentiates *L. monocytogenes* from nonpathogenic *Listeria* species. Tumbling motility due to peritrichous flagella is another characteristic trait of *L. monocytogenes*, which is a facultative anaerobe. The nonpathogenic species are frequently isolated from the same foods and environments as *L. monocytogenes*. "Rapid" tests based on genetic or immunological traits identify listeria isolates to the genus or species level. Special media, such as Oxoid's Chromogenic Listeria Agar (Fig. 13.2), also allow its rapid identification.

Public health officials investigating outbreaks can characterize *L. monocytogenes* isolates below the species level to help identify common sources of the organism. This can be done by genetic fingerprinting or serotypes. There are 13 serotypes of pathogenic *L. monocytogenes*. Ninety-five percent of the organisms isolated from listeriosis victims belong to 3 serotypes: 1/2a, 1/2b, and 4b. In the United States, the CDC has established a network (PulseNet) of public health laboratories to subtype foodborne pathogenic bacteria by using genetic fingerprinting. PulseNet laboratories use standardized methods for subtyping bacterial DNA by pulsed-field gel electrophoresis (PFGE), which uses alternating currents to increase the resolution of the DNA fingerprints. By comparing PFGE patterns of foodborne pathogens

Figure 13.1 Joseph Lister.

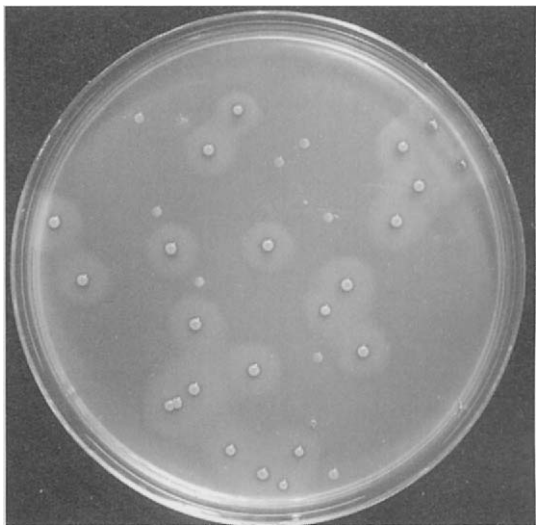


Figure 13.2 *L. monocytogenes* colonies on Oxoid Chromogenic Listeria Agar. (Courtesy of Oxoid/Remel.)

from different locations via the Internet, they can identify outbreaks having a common source.

Susceptibility to Physical and Chemical Agents

L. monocytogenes grows from 0 to 45°C but grows slowly at colder temperatures (Table 13.2). Freezing does not significantly reduce the size of the bacterial population. Survival and injury during frozen storage depend on the food and the freezing rate. *L. monocytogenes* is killed at temperatures of >50°C.

L. monocytogenes grows in laboratory media at pH values as low as 4.4. At pH values below 4.3, cells may survive but do not grow. Organic acids such as acetic, citric, and lactic acids at 0.1% can inhibit *L. monocytogenes* growth. The antilisterial activity of these acids is related to their degree of dissociation (see chapter 23). Not all acids are created equal; citric and lactic acids are less harmful at an equivalent pH than acetic acid. HCl is least effective of all.

L. monocytogenes grows best at water activity (a_w) values of ≥0.97. For most strains, the minimum a_w for growth is 0.93, but some strains may grow at a_w values as low as 0.90. The bacterium may survive for long periods at a_w values as low as 0.83. *L. monocytogenes* heat resistance increases as the a_w of the food in which it is heated decreases. This is problematic for food manufacturers who combine low a_w and heat treatments to maintain the safety of foods such as cured meats.

L. monocytogenes grows to high levels in moderate salt concentrations (6.5%). It can even grow in the presence of 10 to 12% sodium chloride. The bacterium survives for long periods in higher salt concentrations. Lowering the temperature increases the survival in high salt environments. Thus, cured meats (like hot dogs, bologna, and ham) are very hospitable environments for listeria growth.

Table 13.2 Influence of temperature on doubling time of *L. monocytogenes*

Temp (°C)	t_d (hours)[a]
4	43
10	6.6
37	1.1

[a] t_d, doubling time.

LISTERIOSIS AND SPECIFIC FOODS

Ready-to-Eat Foods

Ready-to-eat foods (such as soft cheeses, frankfurters, delicatessen meats, and poultry products) can pose a high listeriosis risk for susceptible populations. Some foods that have caused listeriosis are shown in Table 13.3. Because 20% of refrigerators have temperatures of >50°F, refrigeration

cannot ensure the safety of ready-to-eat foods. In addition, refrigeration offers an environment where *L. monocytogenes* can outcompete mesophilic pathogens.

Milk Products

Raw milk is a source of *L. monocytogenes*. Mexican-style cheese made with unpasteurized milk caused one of the first listeriosis outbreaks in California. Pasteurization is considered an adequate safety intervention. The World Health Organization states, "Pasteurization is a safe process which reduces the number of *L. monocytogenes* organisms in raw milk to levels that do not pose an appreciable risk to human health." U.S. regulatory authorities share this view. However, *L. monocytogenes* grows well in pasteurized milk. This makes postprocess contamination (the contamination of food by environmental bacteria after the food has received its antimicrobial processing) a major concern. Listeriae grow more rapidly in pasteurized milk than in raw milk at 7°C because there are fewer competitors in pasteurized milk. Milk contaminated after pasteurization and refrigerated can reach very high populations after 1 week. Temperature abuse can make things even worse. In a recent outbreak, chocolate milk was contaminated in the plant and then held at room temperature (in the summer) for several days. This allowed *L. monocytogenes* populations to reach the very high level of 10^9 colony-forming units (CFU)/ml. The victims told the epidemiologists to check the chocolate milk because "it tasted bad." Nonetheless, in spite of the off flavor, they still drank it.

Table 13.3 Some foods that have been associated with listeriosis

Cheese
Alfalfa tablets
Fish
Chicken nuggets
Raw milk
Cod roe
Turkey frankfurters
Cook-and-chill chicken
Smoked fish
Deli salads
Deli meat
Hot dogs
Vegetable rennet
Human breast milk
Ice cream
Pork sausage
Homemade sausage
Salted mushrooms

Cheeses

L. monocytogenes survives cheese manufacturing and ripening because of its temperature hardiness, ability to grow in the cold, and salt tolerance. *L. monocytogenes* is concentrated in the cheese curd during manufacturing. The behavior of listeriae in the curd is influenced by the type of cheese, ranging from growth in feta cheese to significant death in cottage cheese. For example, during cheese ripening, *L. monocytogenes* can grow in Camembert cheese, die gradually in Cheddar or Colby cheeses, or decrease rapidly during early ripening and then stabilize, as in blue cheese. The reasons for this are unknown. Consumption of soft cheeses by susceptible persons is a risk factor for listeriosis. The CDC recommends that pregnant women and other immunocompromised people avoid eating soft cheeses.

Authors' note

10^9 CFU/ml is the level of bacteria that can be reached in a laboratory on a good day.

Meat and Poultry Products

L. monocytogenes growth in meat and poultry depends on the type of meat, pH, and the presence of other bacteria. *L. monocytogenes* grows better in poultry than in other meats. Roast beef and summer sausage support the least growth. Contamination of animal muscle tissue can be caused by the presence of *L. monocytogenes* in or on the animal before slaughter or by contamination of the carcass after slaughter. Because *L. monocytogenes* concentrates and multiplies in the kidney, lymph nodes, liver, and spleen, eating organ meat may be more hazardous than eating muscle tissue. *L. monocytogenes* attaches to the surface of raw meats, where it is difficult to remove or kill. It also grows readily in processed meat products, including vacuum-packaged beef, at pH values near 6.0. There is little or no growth around pH 5.0. Ready-to-eat meats that are heated and then cooled in brine before packaging may favor *L. monocytogenes* growth because competitive bacteria are reduced and listeriae have a high salt tolerance.

A 1993–1996 USDA monitoring program for *L. monocytogenes* in cooked ready-to-eat meats found it in 0.2 to 5.0% of beef jerky, cooked sausages, salads, and spreads. In sliced ham and sliced luncheon meats the incidence was 5.1 to 81%. This has decreased dramatically over the last 10 years. In Canada, the incidence of *L. monocytogenes* in domestic ready-to-eat, cooked meat products was 24% in 1989 and 1990 but declined to 3% or less during 1991 and 1992. However, in environmental specimens obtained from these establishments the contamination level remained constant from 1989 to 1992. This suggests that once established in a food plant environment, *L. monocytogenes* may persist for years.

Seafoods

L. monocytogenes has been isolated from fresh, frozen, and processed seafood products, including crustaceans, molluscan shellfish, and finfish. Shrimp, smoked mussels, and imitation crabmeat have all been vehicles for illness. Rainbow trout were implicated in two outbreaks. A U.S. Food and Drug Administration (FDA) survey of refrigerated or frozen cooked crabmeat showed contamination levels of 4.1% for domestic products and 8.3% for imported products. Crab and smoked fish were contaminated with *L. monocytogenes* at 7.5 and 13.6%, respectively.

Among the seafoods that may be high-risk foods for listeriosis are molluscs, including mussels, clams, and oysters; raw fish; lightly preserved fish products including salted, marinated, fermented, and cold-smoked fish; and mildly heat-processed fish products and crustaceans.

Other Methods of Food Preservation

A survey of vacuum-packaged processed meat revealed that 53% of the products were contaminated with *L. monocytogenes* and that 4% contained >1,000 CFU/g (a high level for a food that is ready to eat). This corroborates experimental evidence that the growth of *L. monocytogenes* (a facultative anaerobe) is not significantly affected by vacuum packaging. Studies with meat juice, raw chicken, and precooked chicken nuggets revealed that modified atmospheres do not protect against *L. monocytogenes* growth.

Many surviving cells are injured by heating, freezing, or various other treatments. Heat-stressed *L. monocytogenes* may be less pathogenic than nonstressed cells because energy normally devoted to pathogenesis is diverted to dealing with stressors. On nonselective agar (or food), injured cells can repair the damage induced by stress and grow. When injured cells are subjected to additional stress in selective agar (or food), they may not be recovered.

Sources of *L. monocytogenes* in the Environment

L. monocytogenes is ubiquitous in the environment. *Listeria* species survive and grow in many water environments: surface water of canals and lakes, ditches or polders in The Netherlands, freshwater tributaries draining into a California bay, and sewage. *Listeria* organisms have been isolated from alfalfa plants and crops grown on soil treated with sewage sludge. One-half of the radish samples grown in soil inoculated with *L. monocytogenes* were still contaminated 3 months later. *L. monocytogenes* is also present in pasture grasses and grass silages. Decaying plant and fecal material probably contribute to *L. monocytogenes* in soil. The soil provides a cool, moist environment, and the decaying material provides the nutrients.

L. monocytogenes has been isolated from the feces of healthy animals. Many animal species can get listeriosis. Humans having listeriosis and

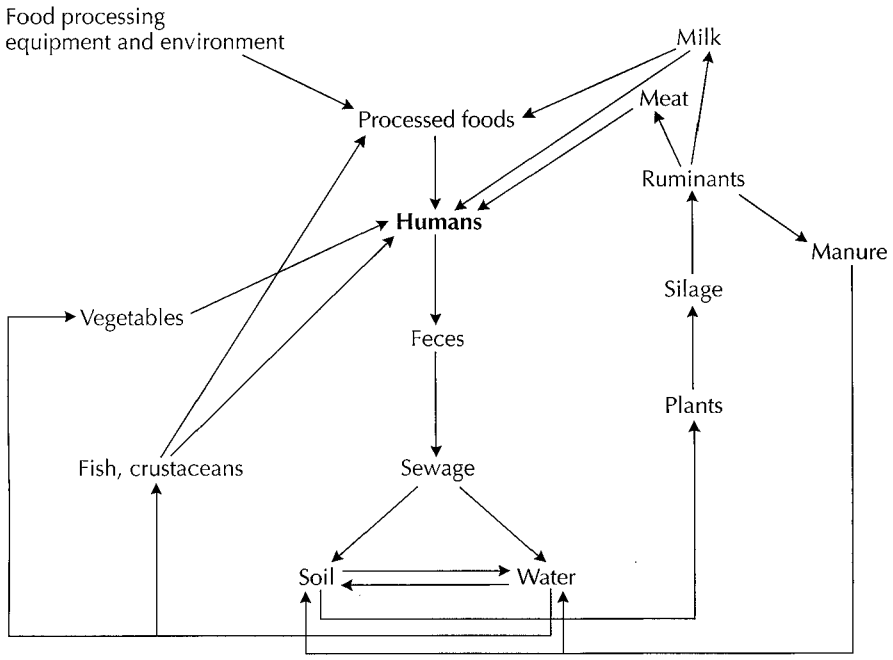


Figure 13.3 Potential routes of transmission of *L. monocytogenes*. (Based on B. Swaminathan, p. 383–409, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville [ed.], *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. [ASM Press, Washington, DC, 2001].)

humans who carry the organism without becoming sick both shed the organism in their feces. Figure 13.3 illustrates how *L. monocytogenes* is spread from the environment to animals and humans and back to the environment.

Food-Processing Plants

L. monocytogenes enters food-processing plants through soil on workers' shoes and clothing and on vehicles. It can also enter the facility with contaminated raw vegetable and animal tissue and with human carriers. The high humidity and nutrient levels of food-processing environments promote listerial growth. *L. monocytogenes* is often detected in moist areas such as floor drains, condensed and stagnant water, floors, residues, and processing equipment. *L. monocytogenes* attaches to surfaces, including stainless steel, glass, and rubber.

L. monocytogenes contaminates carcasses from feces during slaughter. A high percentage (11 to 52%) of healthy animals are fecal carriers. *L. monocytogenes* is present in both unclean and clean zones in slaughterhouses, especially on workers' hands. The most heavily contaminated working areas are cow dehiding, pig stunning, and hoisting. Studies in turkey and poultry slaughterhouses found *L. monocytogenes* in feather plucker drip water, chill water overflow, and recycled cleaning water.

L. monocytogenes does not survive heat processing. It gets into processed foods primarily by postprocessing contamination. *L. monocytogenes* is particularly hard to eliminate from food plants because it adheres to food contact surfaces and forms biofilms in hard-to-reach areas such as drains. This makes proper sanitation difficult. *L. monocytogenes* is often in raw materials used in food plants, so there are ample opportunities for reintroduction of listeriae into processing facilities that have previously eradicated it.

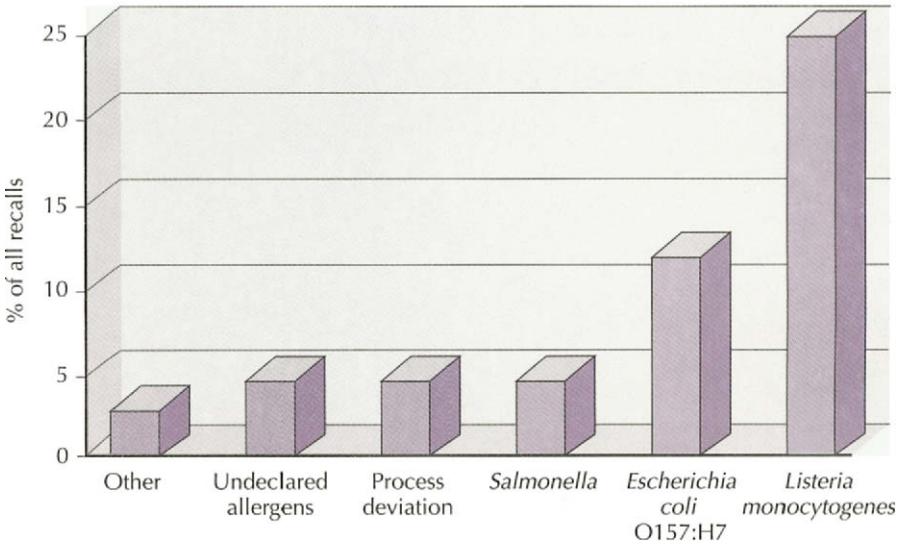


Figure 13.4 *L. monocytogenes* is a leading cause of recalls. The graph shows the leading causes of USDA recalls in 1997.

Prevalence and the Regulatory Status of *L. monocytogenes*

L. monocytogenes contamination of food is widespread. It is a major cause of USDA recalls (Fig. 13.4). Contamination levels range from 0% in bakery goods to 16% in some ready-to-eat foods. The contamination of raw foods is much higher, up to 60% in raw chicken. Preserved, but not heat-treated fish and meat products are more frequently contaminated than heat-treated meat foods. *L. monocytogenes* can be isolated from many vegetables including bean sprouts, cabbage, cucumbers, leafy vegetables, potatoes, prepackaged salads, radishes, salad vegetables, and tomatoes.

There is an international debate about how *L. monocytogenes* should be regulated (Box 13.1). Because *L. monocytogenes* is so common, regulatory agencies in many countries assert that it is impossible to produce *L. monocytogenes*-free foods. They have established "tolerance levels" for *L. monocytogenes*. Products that have caused human listeriosis are placed in a special category and regulated more strictly than foods that have never caused listeriosis. Most European Union countries feel that foods should be free of *L. monocytogenes*, if possible, or have the lowest level possible. Thus, foods intended for susceptible populations must be listeria free. Other foods may contain *L. monocytogenes* at <100 CFU/g. In contrast, the United Kingdom and the United States have a "zero tolerance" (in a 25-g sample) for *L. monocytogenes*. Both countries argue that the infective dose must be known before an "acceptable" level for *L. monocytogenes* can be set. The infective dose is unknown, and it may differ for different types of people.

Authors' note

There is a zero tolerance for L. monocytogenes *in ready-to-eat foods, but no detection method that is 100% accurate.*

A "citizens' petition" (by which anyone can ask the U.S. government to change its regulations) has proposed a tolerance for *L. monocytogenes* in low-risk foods (Table 13.4). The argument is that if *L. monocytogenes* cannot grow in the food, is not held at temperatures that would support growth (yes, this is redundant with the first condition), and has a low probability of making someone ill, there is no justification for zero tolerance. There are two additional arguments against the U.S. zero-tolerance policy. The first is that the FDA only has legal authority to issue regulations that are based on sound science to protect the health of the American people. However, dif-

BOX 13.1

Group exercise: industrial and regulatory responses to *Listeria monocytogenes*

Objective

To understand some of the real-world factors involved in setting industrial policies through use of a simulation.

Scenario

You work for MSF Foods, a large cookie company that added a premium pie line 8 months ago. This line has been a huge success, causing a 4% increase in corporate profits. Marketing has determined that adding cheesecake and cream pie products to the line could fill a new market niche and increase next year's profits by a similar extent.

The product line manager questioned if these products might pose a *Listeria* risk, and what kind of end product testing would be required to ensure product safety. (In the cookie line, the company does "test and hold" for *Salmonella*. Should a similar program be instituted for *Listeria* in the pies?) The production side of the company says that a test-and-hold policy would be the kiss of death for this short-shelf-life, refrigerated project. The microbiology department insists that end product testing is the only way to ensure the safety of the product.

A meeting has been called to resolve this issue. Your job is, at the end of the meeting, to write a two-page recommendation to the Vice President of Cookies, Creams, and Cakes.

Your assignment

1. THE PRE-MEETING: Split into groups and discuss the issues. Decide what additional information is required for you to make a decision. Decide which side of the testing controversy you stand on and will fight for during the meeting.
2. THE MEETING: The group tries to reach consensus. (Self-explanatory.)
3. THE WORK PRODUCT: Write a two-page memo to the Vice President of Cookies, Creams, and Cakes (who knows a lot about cookies, but nothing about microbiology) outlining your recommendation as to what the company should do, and why.

ferent parts of the population require different degrees of protection (Table 13.5). The incidence of listeriosis (~0.7 per 100,000 people) is the same in European countries with a <100-CFU/g tolerance as it is in the United States, where there is a zero tolerance. Thus, the zero-tolerance regulation appears to offer no additional benefit. Secondly, international trade requires harmonization of microbial specifications. When the United States bars the import of foods meeting the European Union's <100-CFU/g tolerance, it can be charged with inhibition of free trade through "nontariff trade barriers." However, politically, no one wants to be seen as "in favor of allowing germs in food." The debate over zero tolerance continues. The current dilemma is discussed in Box 13.2.

Human Carriers

L. monocytogenes can be carried asymptomatically in the feces of many groups. These include healthy people, pregnant women, patients with gastroenteritis, slaughterhouse workers, laboratory workers handling *Listeria*, food handlers, and patients undergoing hemodialysis. *L. monocytogenes* has been isolated from 2 to 6% of fecal samples from healthy people. Patients with listeriosis often excrete high numbers of *L. monocytogenes*. Specimens

Table 13.4 Citizens' proposal for low-risk versus high-risk foods

Low-risk food	High-risk food
Does not support growth of *L. monocytogenes* to high numbers	Food supports the growth of *L. monocytogenes* to high numbers
Is not held for extended periods at temperatures that permit growth	Held for long periods at refrigeration temperatures
Probability of illness is 1 in 100,000 for at-risk populations	Probability of illness is 1 in 6,000 for at-risk populations
Tolerance of <100 CFU/g	Remains zero tolerance

Table 13.5 Relative risk of listeriosis for different populations

Population	Relative risk
Total	1[a]
Over 70 years old	3
Pregnant	17
HIV[b] positive	200

[a]Absolute risk, seven cases per million people.
[b]HIV, human immunodeficiency virus.

from 21% of patients had $>10^4$ *L. monocytogenes* organisms/g of feces. The same strain of *L. monocytogenes* was fecally shed by 18% of their housemates. Fecal carriers amplify outbreaks through secondary transmission. (The "primary" transmission is through the food to the first victim. The "secondary" transmission is from the feces of the first victim to another person.) This secondary fecal-oral transmission of bacteria that originate in foodborne outbreaks also occurs with other pathogens.

FOODBORNE OUTBREAKS

Foodborne transmission of listeriosis was first documented in 1981 during a Canadian outbreak in Nova Scotia. A case-control study and strain typing proved that food was the source. There were 34 cases among pregnant women and 7 cases among nonpregnant adults over a 6-month period. A case-control study implicated coleslaw. The epidemic strain was later isolated from an unopened package of coleslaw. Cabbage fertilized with manure from sheep suspected to have had *Listeria* meningitis was the probable source. Harvested cabbage was stored over the winter and spring in an unheated shed. The cold provided a growth advantage for the psychrotrophic *L. monocytogenes*.

Contaminated Mexican-style cheese caused a 1985 listeriosis outbreak in California. There were 142 cases over 8 months. Pregnant women accounted for 93 cases. The remaining 49 were nonpregnant adults. Most of the victims had a predisposing condition for listeriosis. Almost one-third of the people infected died. However, about 10,000 people consumed the cheese, so the overall attack rate was very low. Inadequate pasteurization of milk and mixing of raw milk with pasteurized milk caused the outbreak.

L. monocytogenes in soft cheese was responsible for a 4-year outbreak of 122 cases in Switzerland. A contaminated pâté caused a 300-case outbreak

BOX 13.2

The current dilemma

The United States is currently spending hundreds of millions of dollars to decrease the incidence of listeriosis. But is the money being spent effectively? Is the obsession with listeriosis cost-effective?

It is hard to say. The organism is widespread in the environment and difficult to eradicate, yet relatively few foods have been implicated in listeriosis outbreaks. Should all ready-to-eat foods or only those with a historical problem be tested for *Listeria*? There are relatively few cases of listeriosis, and when outbreaks occur, only a tiny percentage of people who eat the product get sick. Although the attack rate is low, the fatality rate is high. Might it be more cost-effective to educate at-risk populations to avoid certain foods rather than test for listeriae? The methods for isolating and identifying *Listeria* leave much to be desired. They have a 10 to 15% false-negative rate; i.e., if 100 samples containing *Listeria* were tested, it would not be detected in 10 to 15 of the samples. Is it fair to demand zero tolerance when there is no 100% reliable test? Does it make sense to test products in which the presence of listeriae is unlikely or that have received listericidal treatments? The International Commission for Microbial Specifications for Foods has suggested that these foods need not be tested. The current regulatory emphasis *has* dramatically decreased the isolation of *Listeria* from food, but the number of cases of listeriosis has not had a corresponding decrease. Regulatory agencies, food industry groups, and consumer advocates all agree that the current regulations need to be reexamined, but they disagree on how or why. For the moment, there is emphasis on developing better surveillance for listeriosis so that progress (or lack of progress) can be monitored. Foods that have been associated with listeriosis are under increased scrutiny, but this has not prevented additional recalls. There are efforts to educate at-risk groups, especially pregnant women who can be reached through their obstetricians, but these need to be more effective. The debate is sure to continue for years.

S Y O W K C A D F A A D L

W M A C R O P H A G E Z I

X P P R E G N A N T G P S

X O E U F W B Z A N U W T

S S M Y N O Z R I B D C E

S T W O R Q E P I C K A R

I P F T R G Y Q I C V G I

J R I S I T U D Y J O E O

H O X R O I A X O T J E L

N C F R T R A L R L G A Y

D E E O O V S U I T W W S

R S U P A H J W L T G X I

I S S Z R U Z T N J Y B N

Clues

Enzyme that helps *Listeria monocytogenes* break through membranes. _____

Symptom of listeriosis in some pregnant women. _____

Human cell that *Listeria* can grow inside of. _____

Listeriosis is a disease characterized by low morbidity but high _____.

Most contamination with *Listeria monocytogenes* occurs _____.

_____ women are a high-risk group for listeriosis.

The increased popularity of _____ foods over the last 20 years has led to an increased prevalence of listeriosis.

_____ is a method for identifying bacteria below the species level on the basis of their antigens.

Unlike most foodborne diseases which occur in large outbreaks, cases of listeriosis are _____.

Listeria monocytogenes is difficult to control because it is _____ in the environment.

in the United Kingdom. Recalling the food, advising people not to eat contaminated product, and taking action to prevent *L. monocytogenes* contamination at processing facilities controlled these large outbreaks.

There was a 10-state outbreak of listeriosis between May and November 2000. The *L. monocytogenes* isolates were all serotype 1/2a and had identical PFGE patterns. Eight perinatal and 21 nonperinatal cases were reported. (The perinatal period includes the time before birth to about 1 month after.) Among the 21 nonperinatal-case patients, the median age was 65 years (range, 29 to 92 years); 62% were female. This outbreak resulted in four deaths and three miscarriages or stillbirths. A case-control study implicated deli turkey meat.

CHARACTERISTICS OF DISEASE

L. monocytogenes causes listeriosis in well-defined high-risk groups. These include pregnant women, newborns, and immunocompromised adults. Listeriosis occasionally occurs in otherwise healthy people. In nonpregnant adults, *L. monocytogenes* causes septicemia, meningitis, and meningoencephalitis. The mortality rate is 20 to 25%. Cancer, organ transplants, immunosuppressive therapy, infection with human immunodeficiency virus, and advanced age predispose people to listeriosis. *L. monocytogenes* infection may cause only mild flu-like symptoms in pregnant women but cause stillbirth or abortion of the fetus. While listeriosis *outbreaks* attract the most attention, most cases of human listeriosis are *sporadic*. Some sporadic cases may be unrecognized common-source outbreaks. Although foods may be implicated in some sporadic cases, the mode of infection is usually unknown. Because they are so hard to investigate, many sporadic cases cannot be associated with food. Incubation times of up to 5 weeks make it hard to get accurate food histories and examine suspect foods. Understanding the epidemiology of sporadic cases would help develop more effective control strategies.

Sporadic listeriosis is a rare disease. In Europe and Canada, passive surveillance suggests that there are 2 to 7 cases per million people. An active surveillance study in the United States gave a more precise estimate of 7.4 cases per million people. Since 1990, there has been an almost 50% decrease in sporadic listeriosis in the United States. This suggests that current preventive measures are making a difference.

Human listeriosis is poorly understood. Although exposure to *L. monocytogenes* is common, listeriosis is rare. This may be due to high human resistance or low pathogenicity of most strains. Some healthy humans are asymptomatic fecal carriers of *L. monocytogenes*. The risk of clinical disease in carriers is unknown. Although *Listeria* seems to target pregnant women, some pregnant women also have asymptomatic fecal carriage of listeriae and have normal pregnancies. Women who give birth to infected infants may not be infected in later pregnancies.

L. monocytogenes can also cause a second illness, feverish gastroenteritis. The gastroenteritis outbreaks differ from the invasive outbreaks. They affect people with no predisposing risk factors. The infectious dose is higher (1.9×10^5 to 1×10^9 CFU/g or ml) than that for invasive listeriosis. Finally, the symptoms appear within several hours (18 to 27 h) of exposure in contrast to the weeks observed for invasive listeriosis. The first case of listeric gastroenteritis was reported in 1994. It was caused by the postprocess contamination of chocolate milk held in an unrefrigerated processing vat. The contaminated milk was then stored unrefrigerated (i.e., incubated) by the buyers who served it at a large summer gathering; ~75% of those who drank the milk got sick.

INFECTIOUS DOSE

The infectious dose of *L. monocytogenes* depends on the immunological status of the human, the virulence of the microbe, and the food. Studies with human volunteers are impossible (i.e., they would be unethical), but in studies with animals, reducing exposure levels reduces clinical disease. There are usually >100 CFU of *L. monocytogenes*/g in foods responsible for outbreaks. However, the frankfurters implicated in the 1998 listeriosis outbreak had <0.3 CFU/g. More epidemiologic data are needed to accurately assess the infectious dose.

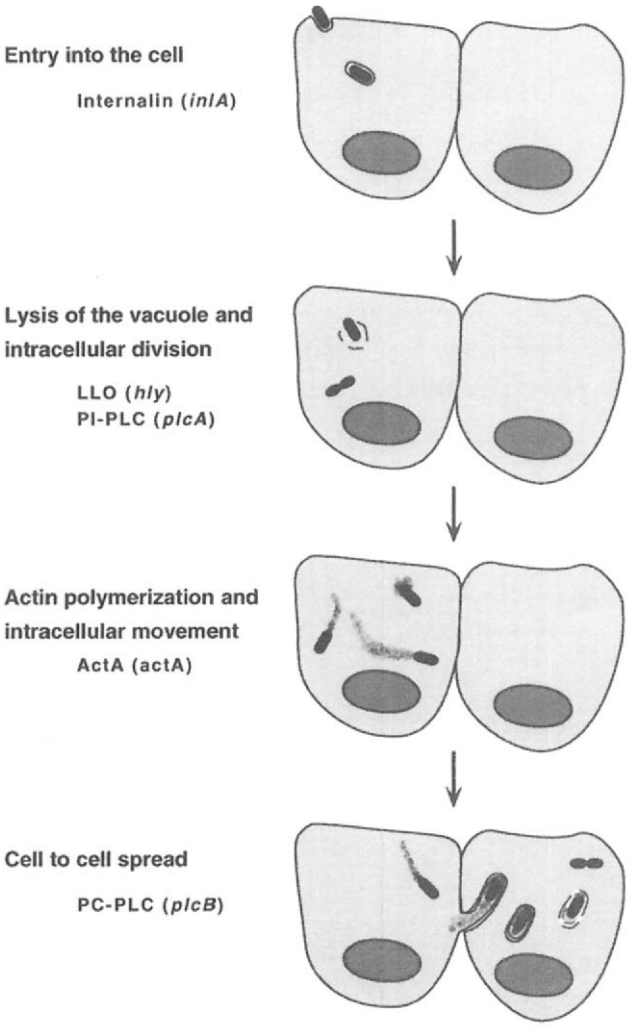
VIRULENCE FACTORS AND MECHANISMS OF PATHOGENICITY

L. monocytogenes is unique among foodborne pathogens. Other pathogens excrete toxins or multiply in the blood. In contrast, *L. monocytogenes* enters the host's cells, grows inside the cell, and passes directly into nearby cells (Fig. 13.5). Cell-to-cell transmission reduces the bacterium's exposure to antibiotics and circulating antibodies. This membrane-penetrating ability allows *L. monocytogenes* to cross into the brain and placenta.

Pathogenicity of *L. monocytogenes*

There are many tests for studying *L. monocytogenes* pathogenicity, including a fertilized hen egg test, tissue culture assays, and tests using laboratory animals, particularly mice. In these studies, mice are infected by injection into the stomach cavity, intravenously, or by mouth. Virulence is then evaluated by comparing the 50% lethal dose or by counting bacteria in the spleen or liver. Fifty percent lethal dose values range from 10^3 to 10^7 CFU.

Figure 13.5 Schematic representation of *L. monocytogenes* cell-to-cell spread. (Reprinted from B. Swaminathan, p. 383–409, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville [ed.], *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. [ASM Press, Washington, DC, 2001].)



When mice are exposed to *L. monocytogenes*, the infection follows a well-defined course, lasting for about 1 week. Within 10 min after intravenous injection, 90% of the bacteria are taken up by the liver and 5 to 10% by the spleen. During the first 6 h, the number of viable listeriae in the liver decreases 10-fold, indicating their rapid destruction. Surviving listeriae then infect susceptible macrophages and multiply. They grow exponentially in the spleen and liver for the next 48 h. If inactivation ensues during the next 3 to 4 days, the host recovers.

L. monocytogenes enters humans through ingestion, crosses the intestinal barrier, and is internalized by macrophages, where it replicates. The blood then transports the listeriae to lymph nodes. When they reach the liver and the spleen, most listeriae are rapidly killed. The survivors travel by the blood to the brain or the placenta.

Specific Genes Mediate Pathogenicity

L. monocytogenes cells can fight their way into nonphagocytic human cells. Our understanding of how gene regulation influences pathogenicity is in its infancy. A series of *L. monocytogenes* genes gives the instructions for each step of this infectious process. The protein internalin is needed to start the process. Internalin is coded for by the gene *inlA*. This gene is thermoregulated and expressed more readily at 37°C than at 20°C. Mutants lacking this gene cannot form the phagosome that introduces the bacteria to the inside of the mammalian cell. To do real damage, the listeria must be released from the phagosome and liberated into the cytoplasm of the host cell. The *hly* gene is responsible for this. It codes for listeriolysin, an enzyme that breaks open the phagosome. Listeriolysin breaks down red blood cells; its activity is measured as hemolytic activity. *L. monocytogenes* mutants lacking the *hly* gene are not pathogenic. These cells can be entrapped by phagosomes but are not freed into the cytoplasm.

Once the listeria cells are free in the host cytoplasm, the genes that regulate growth and multiplication work in the normal fashion. The listeria cells have the unusual ability to use the host's actin molecules. Actin, together with myosin, is the major component of human muscles. It is in nonmuscle cells at lower concentrations. The *actA* gene in listeria codes for a protein, ActA, which is made at one end of the bacterium. ActA makes the actin propel the cell forward. The force generated by the actin pushes the cell across the infected host's membrane and into the membrane of the adjacent cell. The final gene needed for pathogenicity is *plcB*. This gene codes for a membrane-hydrolyzing enzyme that helps listeriolysin O liberate the bacterium into the cytoplasm of the adjacent human cell.

Summary

- *L. monocytogenes* is a ubiquitous organism that can grow at refrigeration temperature.
- There are relatively few (5,000/year) cases of listeriosis, but there is a high (~25%) fatality rate among people who get it.
- *L. monocytogenes* is a hearty organism that is relatively resistant to dehydration, low pH, and low a_w.
- Ready-to-eat foods that are preserved by refrigeration pose a special challenge with regard to *L. monocytogenes*.
- Symptomatic and asymptomatic people can shed *L. monocytogenes* in their feces.

- The United States has a "zero tolerance" for *L. monocytogenes* in ready-to-eat foods, but in Europe there is some tolerance for it in certain foods.
- Outbreaks of listeriosis generally extend over a long period, are caused by refrigerated foods, have high case-fatality rates, and involve a disproportionate number of pregnant women.
- *L. monocytogenes* enters the host's cells, grows inside the cell, and passes directly into nearby cells.
- Internalin, listeriolysin, ActA, and a membrane-hydrolyzing enzyme are the four major proteins that govern the pathogenic process.

Suggested reading

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Swaminathan, B., D. Cabanes, W. Zhang, and P. Cossart. 2007. *Listeria monocytogenes,* p. 457–491. *In* M. P. Doyle, and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers,* 3rd ed. ASM Press, Washington, DC.

Tompkin, R. B. 2002. Control of *Listeria monocytogenes* in the food-processing environment. *J. Food Prot.* **65:**709–725.

Questions for critical thought

1. What reasons might explain our unawareness of *L. monocytogenes* prior to 1980?
2. What are the pros and cons of a tolerance versus "zero tolerance" for *L. monocytogenes* in ready-to-eat food?
3. Make up an outbreak of listeriosis. Include food, setting, population, symptoms, the chain of events that caused the outbreak, and how it might have been prevented.
4. Diagram the life of *L. monocytogenes* as an intracellular pathogen, noting important gene products.
5. What characteristics make *L. monocytogenes* a "successful" foodborne pathogen?
6. Why do listeriae grow more rapidly in pasteurized milk than in raw milk?
7. Write down one thing you do not understand about *L. monocytogenes.* Discuss it with a classmate.
8. Choose one of the "more questions than answers" issues, and write a page on it.
9. Use the CDC, USDA, or FDA website to find a recent listeria outbreak or recall. How do its characteristics fit with those described in the text?
10. Write a really good question about *Listeria* that might be used on your exam.

More questions than answers

1. Are all *L. monocytogenes* strains equally pathogenic?
2. Are all of them capable of causing outbreaks?
3. What is the infectious dose of *L. monocytogenes* for nonpregnant adults, persons with immune system deficiencies, and pregnant women?

4. Do the infectious doses vary significantly between strains?
5. Is the expression of virulence genes influenced by the food the listeriae are in?
6. Do all strains cause feverish gastroenteritis in humans?
7. Are some *L. monocytogenes* strains better able to form biofilms and persist in the environment?
8. Is it realistic to expect all ready-to-eat foods to be free of *Listeria*?
9. How can we ensure that ready-to-eat salads are not contaminated with *L. monocytogenes*?
10. How can we prevent *Listeria* contamination of other ready-to-eat foods such as frankfurters, soft cheeses, and smoked fish?

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14 *Staphylococcus aureus*

Outbreak
Characteristics of the Organism
Historical Aspects and General Considerations
Sources of Staphylococcal Food Contamination
Resistance to Adverse Environmental Conditions
Foodborne Outbreaks
Incidence of Staphylococcal Food Poisoning
A Typical Large Staphylococcal Food Poisoning Outbreak
Characteristics of Disease
Infective Dose and Susceptible Populations
Numbers of Staphylococci Required
Toxin Dose Required
Microbiology, Toxins, and Pathogenicity
Nomenclature, Characteristics, and Distribution of Staphylococcal Enterotoxin-Producing Staphylococci
Introduction to, and Nomenclature of, the Staphylococcal Enterotoxins
Staphylococcal Regulation of Staphylococcal Enterotoxin Expression
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Staphylococcus aureus*
- understand what conditions in foods favor of *S. aureus* growth
- recognize, from symptoms and time of onset, a case of *S. aureus* food poisoning
- choose appropriate interventions (heat, preservatives, or formulation) to prevent the growth of *S. aureus*
- identify sources of *S. aureus*
- understand the role of staphylococcal enterotoxins in causing foodborne illness

Outbreak

My fear of flying is related to bad food, not hijackings, crashes, or boring in-flight movies. As a college student, I boarded a flight in Tokyo. It would stop in Anchorage, AK, for a fresh crew and (as it turned out, not so fresh) food. The first leg of the trip was uneventful. An hour out of Alaska, we ate peanuts and soda. Breakfast would break the boredom of the flight—in a way I could only imagine. Breakfast smelled good. It was an omelet with ham and cheese, served nice and hot. But before the service cart reached the back of the plane, there were rumblings in the front. About 30 min after eating, a passenger rushed to the rear of the plane, ignored the attendant's admonition that the seat belt sign was on, and vomited on a blue-haired woman before finally reaching the toilet. Although the air was calm, most of the passengers started to turn green. The trickle of passengers trying to reach the toilet turned into a stream. The flow crested 2.5 h after breakfast. The first man ill was lucky. He got a toilet. There were only four bathrooms for the remaining 250 people with vomiting and diarrhea! A burly first-class passenger blocked the way to the front of the cabin. In first class, the passenger-to-toilet ratio was 20 to 2, and first-class food had not even made them sick! About 80% of the passengers had diarrhea, 70% had vomiting, and 85% had nausea. I will spare you the details of what happened when they ran out of "discomfort bags." Fortunately, the crew was served a "microbiologically insensitive" meal. Only one of them got sick. She will never pass up steak in favor of an omelet again.

After we landed, two men in lab coats told me that I would be exempt from the customs line if I would "do them a favor." My rectal swabs tested positive for *Staphylococcus aureus* toxin and the organism. They found staphylococcal toxin and the organism in the cheese omelets and the ham, too. Other investigators visited the kitchen. They discovered that the meal had been prepared 14 h before the flight and had been held at room temperature. It was then held at inadequate refrigeration during the flight and microwaved to heat just before serving. When the public

health investigators went to shake hands with the head chef, they saw a pus-filled boil on his hand. They cultured it, and guess what they found—*S. aureus*.

CHARACTERISTICS OF THE ORGANISM

Historical Aspects and General Considerations

Authors' note

The pathogenicity of many foodborne pathogens was first proven by drinking the cultures. This was done in the days before the creation of institutional review boards for the use of human subjects. Do not try this at home.

The link between staphylococci and foodborne illness was made in 1914 when researchers found that drinking contaminated milk caused vomiting and diarrhea. In 1930, G. M. Dack and coworkers voluntarily consumed fluids from cultures of "a yellow hemolytic *Staphylococcus*" isolated from contaminated cake. They became ill with "vomiting, abdominal cramps and diarrhea."

S. aureus is well characterized. It excretes a variety of compounds. Many of these, including the staphylococcal enterotoxins, are virulence factors. The staphylococcal toxins cause at least two human diseases, toxic shock syndrome and staphylococcal food poisoning.

S. aureus is now recognized as the main agent of staphylococcal food poisoning, a common cause of gastroenteritis. Unlike many other forms of gastroenteritis, one does not get staphylococcal food poisoning by eating live bacteria. One gets it by eating staphylococcal toxins that have already been made in the contaminated food. This form of food poisoning is known as an "intoxication" or "poisoning" because it does not require that the bacteria infect and grow in the victim. Indeed, outbreaks have been caused by foods in which the organism has been killed but the toxin remains. Staphylococcal toxin is unique because it is not destroyed by heating, even by canning. This is illustrated by a large outbreak caused by consumption of canned mushrooms. *S. aureus* grew and produced toxin in the mushrooms before they were canned (Box 14.1). The canned mushrooms were processed at 121°C for 2.4 min (see chapter 25) to produce a product that was "commercially sterile." However, people who ate the mushrooms became ill with typical symptoms of staphylococcal food poisoning. Staphylococcal enterotoxin was subsequently isolated from the food.

Sources of Staphylococcal Food Contamination

People are the main reservoir of *S. aureus*. Humans are "natural" carriers and spread staphylococci to other people and to food. In humans, the nose interior is the main colonization site. *S. aureus* also occurs on the skin. *S. aureus* spreads by direct contact, by skin fragments, or through respiratory droplets produced when people cough or sneeze. Most staphylococcal food poisoning is traced to food contaminated by humans during preparation. In addition to contamination by food handlers, meat grinders, knives, storage utensils, cutting blocks, and saw blades may also introduce *S. aureus* into food. Conditions often associated with outbreaks of staphylococcal illness are inadequate refrigeration, preparing foods too far in advance, poor personal hygiene, inadequate cooking or heating of food, and prolonged use of warming plates when serving foods.

Animals are also *S. aureus* sources. For example, mastitis (the infection of cow teats) is a serious problem for the dairy industry. It is often caused by *S. aureus*. Bovine mastitis is the single most costly agricultural disease in the United States. Mastitis is also a public health concern because the bacteria can contaminate milk and dairy products. This can be controlled through strict hygiene of automated milking machines, milk handlers, and facilities.

BOX 14.1

Everything affects everything else

By now, the reader should realize that microbes grow in food ecosystems and that the foods' intrinsic and extrinsic factors determine their safety. Sometimes, these factors can reach far beyond the realm of food microbiology, even into politics and economic systems. Consider the case of the Chinese canned mushrooms.

Late in 1989, the Centers for Disease Control and Prevention received four reports of staphylococcal food-poisoning outbreaks from Mississippi, New York, and Pennsylvania. Almost 100 people were ill with nausea, vomiting, diarrhea, and cramps. All of the victims had eaten omelets, salads, pizza, or tomato sauce containing mushrooms. All of the mushrooms were traced to large institutional cans of mushrooms that had been imported from the People's Republic of China. The U.S. FDA then banned the importation of institutional-size cans of mushrooms from China, creating an international incident by disrupting this 50 million-pound-per-year mushroom business. A team of U.S. investigators was quickly sent to China to determine the cause of the tainted mushrooms and to help correct the situation.

Processing records indicated that the mushrooms had been properly canned and not contaminated after processing. This meant that the heat-resistant staphylococcal enterotoxin had been formed in the mushrooms before they were canned. The Chinese had a long history of safe canning for export. What had changed?

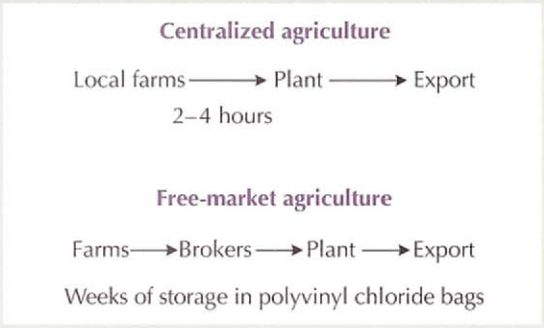


Illustration of mushroom canning practices.

The chain "from farm to fork" was short and fast in the era when the communist state centralized growing and canning (see figure). The mushrooms were grown in fields around the cannery, harvested, and canned within a few hours. There was total vertical integration and control. When free-market agriculture was introduced to the region, brokers who bought and sold mushrooms entered the scene. The brokers would pack mushrooms in polyvinyl chloride bags and buy and sell them like any other commodity. The cannery usually ended up getting mushrooms that had been in storage rather than premium fresh mushrooms. The *S. aureus* organisms, which did not compete with the aerobic spoilage organisms in fresh mushrooms, thrived in the anaerobic conditions of the plastic bags. They made toxin without causing obvious spoilage to the mushrooms. The mystery was solved. The *S. aureus* cells grew and made toxin in the mushrooms before they were canned. When the mushrooms were heated during the canning process, the cells were killed but the toxin survived.

Although the case of the Chinese canned mushrooms may seem unique, food safety always depends on every link of the food chain that reaches from farm to fork. Whenever any part of the chain is changed, the effect on the rest of the chain should be considered.

The source of staphylococcal food contamination for a specific outbreak is not always known. Regardless of its source, *S. aureus* is present in many foods (Table 14.1). The staphylococcal levels are usually low initially, <100 colony-forming units (CFU)/g. However, the organism can grow to high levels (>10^6 CFU/g) and cause staphylococcal food poisoning under favorable conditions.

Resistance to Adverse Environmental Conditions

S. aureus's unique resistance to the inhibitors normally found in food helps it grow under conditions where other pathogens cannot. It is one of the most resistant non-spore-forming pathogens. Thus, foods with high salt content or low water activity (a_w) are often implicated in outbreaks. In fact, under aerobic conditions, *S. aureus* can grow at a_w values as low as 0.86. These foods include cured meats, puddings, and sauces.

Staphylococci have an efficient osmoprotectant system. Several compounds accumulate in the cell or enhance its growth under osmotic stress.

Table 14.1 Prevalence of *S. aureus* in several common food products[a]

Product	No. of samples tested	% Positive for *S. aureus*	*S. aureus* content (CFU/g)[b]
Ground beef	74	57	≥100
	1,830	8	≥1,000
	1,090	9	>100
Big game	112	46	<10
Pork sausage	67	25	≥100
Ground turkey	50	6	≤10
	75	80	≤3.4
Salmon steaks	86	2	≤3.6
Oysters	59	10	≤3.6
Blue crab meat	896	52	≤3
Peeled shrimp	1,468	27	≤3
Lobster tail	1,315	24	≤3
Assorted cream pies	465	1	≤25
Tuna pot pies	1,290	2	≤10
Delicatessen salads	517	12	≤3

[a]Adapted from L. M. Jablonski and G. A. Bohach, p. 411–434, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

[b]Determined by either direct plate count or most-probable-number technique.

Glycine betaine is the most important osmoprotectant. To varying degrees, other compounds including L-proline, proline betaine, choline, and taurine also act as compatible solutes. Proline and glycine betaine accumulate to very high levels in *S. aureus* in response to low a_w. This lowers the intracellular a_w to match the external a_w. However, the transport of these compounds requires energy, diverting it from other cellular purposes. Under extreme conditions of growth, the organism may not produce toxin.

Because there are so many human and animal reservoirs, controlling staphylococcal food poisoning is especially challenging. *S. aureus* persists in sites such as the mucosal surface because it can bind to, and be internalized by, many different cell types. Once bound, *S. aureus* triggers a series of host-specific changes that are similar to other facultative intracellular pathogens. These include phagosome formation, protein tyrosine kinase activation, and changes in cell morphology, which are explained more fully for the case of *Listeria monocytogenes* in chapter 13.

FOODBORNE OUTBREAKS

Incidence of Staphylococcal Food Poisoning

There is little reason to report staphylococcal food poisoning, because people usually recover in 24 to 48 h and have no reason to go to the doctor. Although there is national surveillance for staphylococcal food poisoning, it is not an officially reportable disease. Only 1 to 5% of all staphylococcal food poisoning cases in the United States are reported. Most of these are highly publicized outbreaks. Sporadic cases in the home are usually unreported. Staphylococcal food poisoning accounts for about 14% of the total outbreaks of foodborne illness within the United States. It is the third most common confirmed bacterial foodborne illness. There are about 25 major outbreaks of staphylococcal food poisoning annually in the United States. The occurrence of staphylococcal food poisoning is seasonal; most cases are in the late summer when temperatures are warm and food is stored im-

properly. A second peak occurs in November and December, presumably due to the mishandling of holiday leftovers. Although international reporting is also poor, staphylococcal food poisoning is a leading cause of foodborne illness worldwide. In one study, 40% of foodborne gastroenteritis outbreaks in Hungary were due to staphylococcal food poisoning. The percentage is slightly lower in Japan (20 to 25%). In Great Britain, meat or poultry products caused 75% of the 359 staphylococcal food poisoning cases reported between 1969 and 1990.

A Typical Large Staphylococcal Food Poisoning Outbreak

The U.S. Food and Drug Administration (FDA) reported an outbreak having many typical elements of staphylococcal food poisoning. The type of food involved, means of contamination, inadequate food handling measures, and symptoms all pointed to *S. aureus*. The outbreak was traced to one meal served to 5,824 schoolchildren at 16 sites in Texas. A total of 1,364 children developed typical staphylococcal food poisoning. Investigations revealed that 95% of the ill children had eaten chicken salad which contained *S. aureus* at high levels.

The meal was prepared in a central kitchen the day before. Frozen chickens were boiled for 3 h. After being cooked, the chickens were deboned, cooled to room temperature using a fan, ground into small pieces, placed into 30.5-cm-deep pans, and stored overnight in a walk-in refrigerator at 5.5 to 7°C. The following morning, the other salad ingredients were added and the mixture was blended. The food was placed in containers and trucked to the schools between 9:30 and 10:30 a.m. It was kept at room temperature until served between 11:30 a.m. and noon.

The chicken was probably contaminated after cooking when it was deboned. The storage of the warm chicken in the deep pans prevented rapid cooling. It provided a good environment for staphylococcal growth and toxin production. Holding the food in warm classrooms gave an additional opportunity for growth. The screening of food handlers to identify *S. aureus* carriers, cooling the chicken more rapidly, and refrigerating the salad after preparation could have prevented the incident.

CHARACTERISTICS OF DISEASE

Staphylococcal food poisoning is a self-limiting illness causing emesis (vomiting) after an unusually short time of onset—as little as 30 min after eating (Box 14.2). However, vomiting is not the only symptom. Likewise, many patients with staphylococcal food poisoning do not vomit. Nausea, cramps, diarrhea, headaches, and/or prostration are other common symptoms. In a summary of clinical symptoms involving 2,992 patients diagnosed with staphylococcal food poisoning, 82% complained of vomiting, 74% felt nauseated, 68% had diarrhea, and 64% exhibited abdominal pain. In all cases of diarrhea, vomiting was always present. The lack of fever is consistent with the illness being caused by a toxin, not an infection.

Symptoms usually develop within 6 h after eating. In one report, 75% of the victims had symptoms of staphylococcal food poisoning within 6 to 10 h after eating. The mean time of onset is 4.4 h, although illness can appear in ~30 min. Death due to staphylococcal food poisoning is rare. The fatality rate ranges from 0.03% for the general public to 4.4% for more susceptible populations such as children and the elderly. Approximately 10% of patients with confirmed staphylococcal food poisoning seek medical

Authors' note

The period between the consumption of tainted food and onset of illness can be an important clue in foodborne illness.

BOX 14.2

Things are not always what they seem

You would immediately suspect *S. aureus* intoxication if you were presented with a scenario where more than 25 people become ill with vomiting and nausea within 30 min of eating at a buffet. Such was the case in an outbreak that originated in a restaurant in Ohio in 2000. However, toxin was not detected in the food or the victims' feces. Statistical analysis implicated the salad. Testing of the salad and a victim's vomitus revealed the presence of methomyl, an organophosphate choline esterase inhibitor. Methomyl was used as "fly bait" distributed around the restaurant by hand or by spoon. The spoon was missing. In a separate case of food-related poisoning, three people who ate roti (an Indian dish) died. The restaurant stored methomyl in an unlabeled tin can. These outbreaks illustrate two important lessons: food utensils should never be used to distribute chemicals, and chemicals and insecticides should be clearly labeled and stored in a locked cabinet.

help. Treatment is usually minimal, although fluids are given when diarrhea and vomiting are severe.

INFECTIVE DOSE AND SUSCEPTIBLE POPULATIONS

Numbers of Staphylococci Required

The FDA suggests that *S. aureus* levels of $>10^5$ cells/g of food may produce enough enterotoxin to cause illness. Other studies suggest that 10^5 to 10^8 cells/g is the typical range, although lower cell populations are sometimes implicated.

Toxin Dose Required

The best data on minimum toxic dose come from analysis of food recovered from outbreaks. Staphylococcal toxins are quite potent, and 1 ng (10^{-9} g) of staphylococcal enterotoxin per g of food can cause illness. However, the staphylococcal enterotoxin level in outbreak foods is relatively large, ranging from 1 to 5 μg of ingested toxin.

Many factors contribute to the severity and likelihood of getting staphylococcal food poisoning. These include individual susceptibility to the toxin and how much food was eaten. The toxin type may also be important. Although enterotoxin A causes more outbreaks, staphylococcal enterotoxin B produces more-severe symptoms. Forty-six percent of 2,291 individuals exposed to staphylococcal enterotoxin B had symptoms severe enough for the patients to be hospitalized. Only 5% percent of 1,813 individuals exposed to enterotoxin A required hospitalization. This may reflect different levels of toxin expression, as enterotoxin B is generally produced at higher levels than enterotoxin A.

The staphylococcal enterotoxin acts at the viscera (gut). The emetic response results from a stimulation of its neural receptors. These transmit impulses through the nerves, ultimately stimulating the brain's vomiting center. Different levels of ingested enterotoxins are required to make human volunteers or monkeys vomit. Monkeys are generally less susceptible than humans. Human volunteers require 20 to 25 μg of staphylococcal enterotoxin B (0.4 μg/kg of body weight) to induce vomiting. In the rhesus monkey, the 50% emetic dose is about 1 μg/kg.

MICROBIOLOGY, TOXINS, AND PATHOGENICITY

Nomenclature, Characteristics, and Distribution of Staphylococcal Enterotoxin-Producing Staphylococci

The term "staphylococci" informally describes a group of small, spherical, gram-positive bacteria. Their cells have a diameter ranging from approximately 0.5 to 1.5 μm (Fig. 14.1). They are catalase positive (i.e., they have enzymes that break down hydrogen peroxide). Staphylococci have typical gram-positive cell walls containing peptidoglycan and teichoic acids. Some characteristics that differentiate *S. aureus* from other staphylococcal species are summarized in Table 14.2.

Bergey's Manual of Determinative Bacteriology (the "dictionary" for the classification of bacteria) puts staphylococci in the family *Micrococcaceae*. This family includes the genera *Micrococcus*, *Staphylococcus*, and *Planococcus*. The genus *Staphylococcus* is further subdivided into more than 23 species and subspecies. Many of these contaminate food. Staphylococci that form black colonies on Baird-Parker agar (Fig. 2.2) are suspected of being *S. aureus*. This is usually confirmed by the "coagulase" test, which is easily conducted in the laboratory. (Coagulase is an enzyme highly associated with toxin production.) However, several species of *Staphylococcus*, including both coagulase-negative and coagulase-positive isolates, can produce staphylococcal enterotoxins. Nonetheless, although several species can cause gastroenteritis, nearly all staphylococcal food poisoning is attributed to *S. aureus*.

Introduction to, and Nomenclature of, the Staphylococcal Enterotoxins

Current Classification Scheme Based on Antigenicity

S. aureus makes multiple toxins, which can be differentiated by immunological tests. Staphylococcal enterotoxins are named by letter in the order of their discovery. Staphylococcal enterotoxin A, enterotoxin B, enterotoxin C, enterotoxin D, and enterotoxin E are the major types. Protein sequencing and recombinant DNA methods have yielded the primary sequences of the

Figure 14.1 Electron micrograph of *S. aureus* cells.

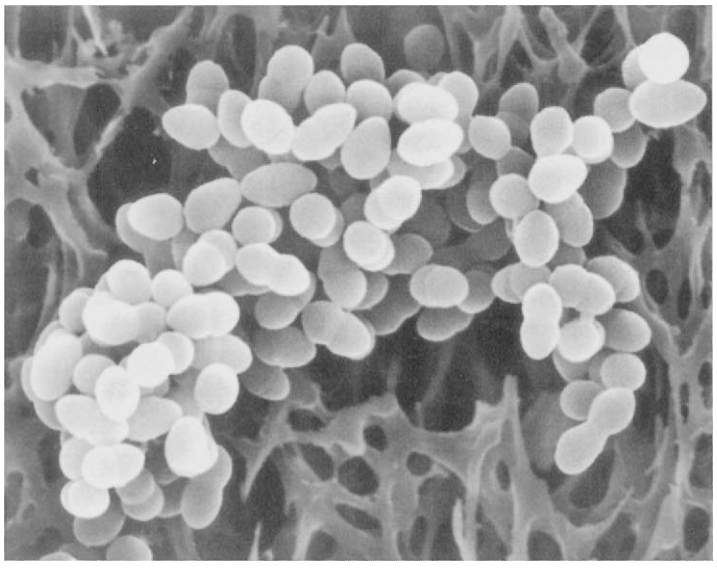


Table 14.2 General characteristics of selected species of *Staphylococcus*[a]

Characteristic	Presence of phenotype[b]					
	S. aureus	*S. chromogenes*	*S. hyicus*	*S. intermedius*	*S. epidermidis*	*S. saprophyticus*
Coagulase	+	−	+	+	−	−
Thermostable nuclease	+	−	+	+	±	−
Clumping factor	+	−	−	+	−	−
Yellow pigment	+	+	−	−	−	±
Hemolytic activity	+	−	−	+	±	−
Phosphatase	+	+	+	+	±	−
Lysostaphin	Sensitive	Sensitive	Sensitive	Sensitive	Slightly sensitive	ND
Hyaluronidase	+	−	+	−	±	ND
Mannitol fermentation	+	±	−	±	−	±
Novobiocin resistance	−	−	−	−	−	+

[a]Reprinted from L. M. Jablonski and G. A. Bohach, p. 411–434, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

[b]ND, not determined; +, present; −, absent; ±, variable.

enterotoxins (Fig. 14.2). They have a high degree of homology (similarity of amino acid sequence). Staphylococcal enterotoxins G, H, J, and I are more recent discoveries. An exotoxin produced by the *S. aureus* associated with toxic shock syndrome was initially called staphylococcal enterotoxin F. When it was discovered that this enterotoxin did not cause vomiting in animals (as the other staphylococcal enterotoxins do), its name was changed to toxic shock syndrome toxin 1 (Box 14.3).

Staphylococcal Regulation of Staphylococcal Enterotoxin Expression

General Considerations

Staphylococcal enterotoxins are produced in low quantities throughout most of the exponential growth phase. The amount made is strain dependent. Staphylococcal enterotoxins B and C are produced in the largest quantities, up to 350 μg/ml. Staphylococcal enterotoxins A, D, and E are easily detectable by gel diffusion assays, which detect as little as 100 ng of enterotoxin per ml of culture. Strains that produce very low levels of toxins require more sensitive detection methods. Of these, staphylococcal enterotoxins D and J are most likely to be undetected.

Molecular Regulation of Staphylococcal Enterotoxin Production

Three genes that regulate expression of *S. aureus* virulence factors are *agr* (accessory gene regulator), *sar* (staphylococcal accessory regulator), and *sae* (*S. aureus* exoprotein expression). The best characterized is *agr*. Mutations in *agr* decrease toxin expression. Gene regulation by *agr* can be transcriptional or translational. Not all staphylococcal toxins are regulated by *agr*. Staphylococcal enterotoxin A expression is not affected by *agr* mutations. Since at least 15 genes are under *agr* control, it is considered a "global regulator."

Regulation of Staphylococcal Enterotoxin Gene Expression

The properties of *agr* explain several aspects of enterotoxin production and, in many cases, explain what was once known as "glucose effects." For example, *agr* expression coincides with expression of enterotoxins B and C during growth. All have highest expression during late-exponential and

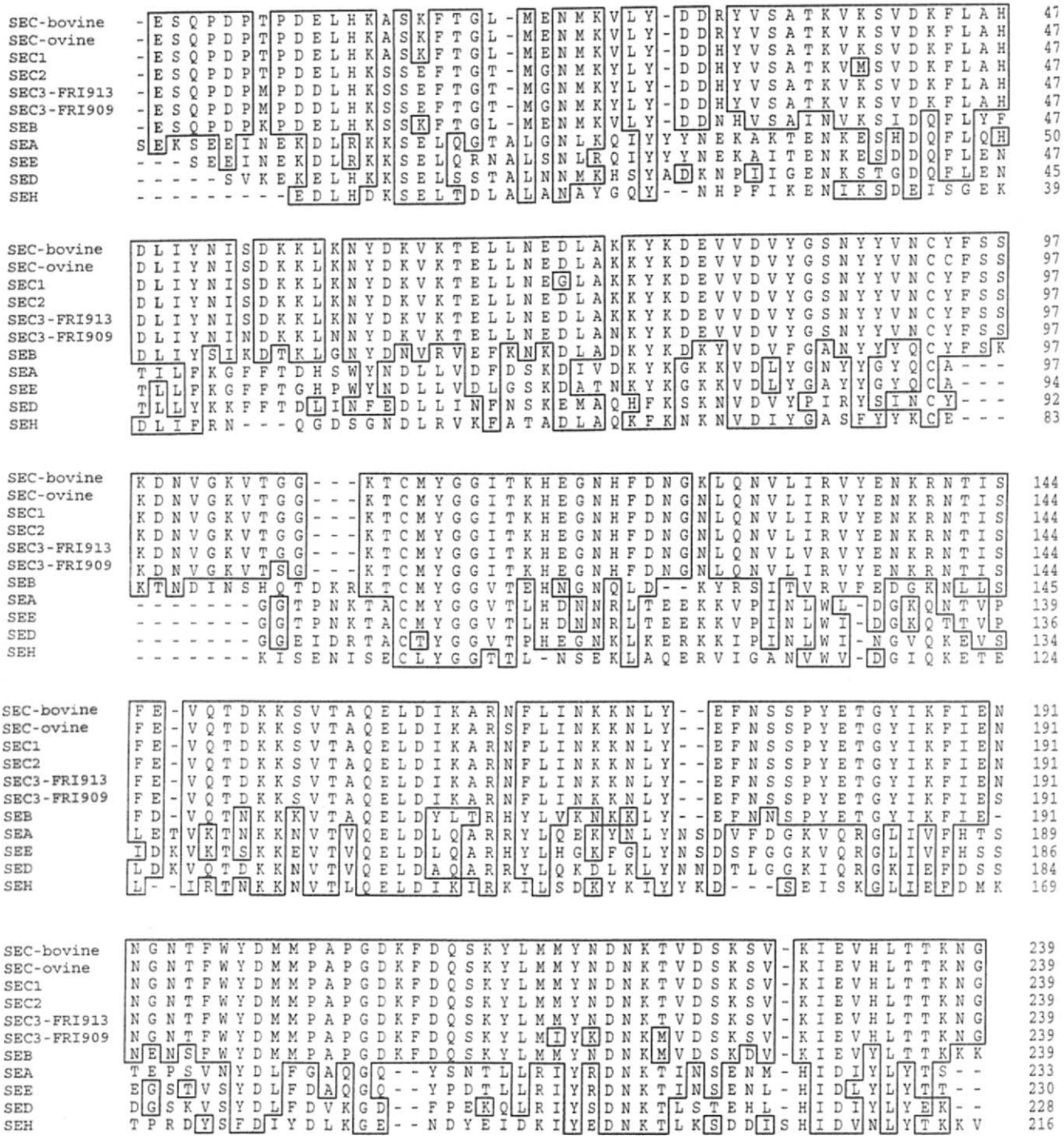


Figure 14.2 Amino acid sequence alignment of staphylococcal enterotoxin (SE) types A, E, J, D, C, B, G, H, and I. Identical residues are boxed. Reprinted from L. M. Jablonski and G. A. Bohach, p. 411–434, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

BOX 14.3

Toxic shock syndrome: changes in microbial ecology give rise to a new threat from an old pathogen

Toxic shock syndrome (TSS) is not caused by food but is caused by *S. aureus*. In women, TSS causes vomiting, diarrhea, renal problems, central nervous system symptoms, and the peeling of skin from the hands and feet. It can be fatal. TSS emerged in the mid-1970s when superabsorbent tampons were introduced into the highly competitive feminine hygiene market. The superabsorbency led to extended periods of usage. This led to extended time for growth of—and toxin production by—*S. aureus* in a moist, warm, nutrient-rich environment. TSS toxin 1 causes 75% of the cases, but staphylococcal enterotoxin B (the same toxin that causes food poisoning) is responsible for 25% of the cases. The federal government now regulates tampon absorbency and advises menstruating women to change their tampons frequently.

postexponential growth. Staphylococcal enterotoxin A, which is not regulated by *agr*, is produced earlier. Furthermore, the production of several staphylococcal enterotoxins is inhibited by growth in media containing glucose; this affects *agr*. Enterotoxin C expression is affected by glucose in two different ways. First, glucose metabolism indirectly influences enterotoxin C production by reducing pH: *agr* is expressed the most at neutral pH, but growth in media containing glucose lowers pH. This directly reduces *agr* expression. Expression of *sec* and other *agr* target genes is also affected. Glucose also reduces *sec* expression in strains lacking *agr*. This suggests a second glucose-dependent mechanism for stopping toxin production, independent of *agr* and not influenced by pH.

In many organisms, a process known as autoinduction coordinates the expression of proteins. In autoinduction, the production of one compound induces the production of others in the same organism. The *agr* locus is at the center of the *S. aureus* autoinduction response. Components of the *agr* operon comprise a quorum-sensing apparatus analogous to the signal transduction pathway of bacterial two-component (receiver and response) systems.

Authors' note

Quorum sensing is the way in which bacteria determine the size of their population and whether it is big enough to act in a certain fashion. This is analogous to legislative bodies that require a quorum (a certain number of members) to do business. Signal transduction is the mechanism by which bacteria "sense" some external factor and transduce it to an internal response.

Other Molecular Aspects of Staphylococcal Enterotoxin Expression

Many factors selectively inhibit enterotoxin expression. Their effects on regulation and signal transduction are only beginning to be defined. *agr* is not the only signal transduction mechanism for *S. aureus*. Glucose's negative effect is not entirely due to higher acid levels. Cultures containing glucose produce less enterotoxin even when the acid is neutralized.

S. aureus is osmotolerant. When it grows in foods with low a_w, less enterotoxin is produced. In experiments with enterotoxin C-producing strains, levels of *sec* messenger ribonucleic acid (mRNA) and enterotoxin C protein are both reduced in response to high NaCl concentrations.

Basic Structural and Biophysical Features

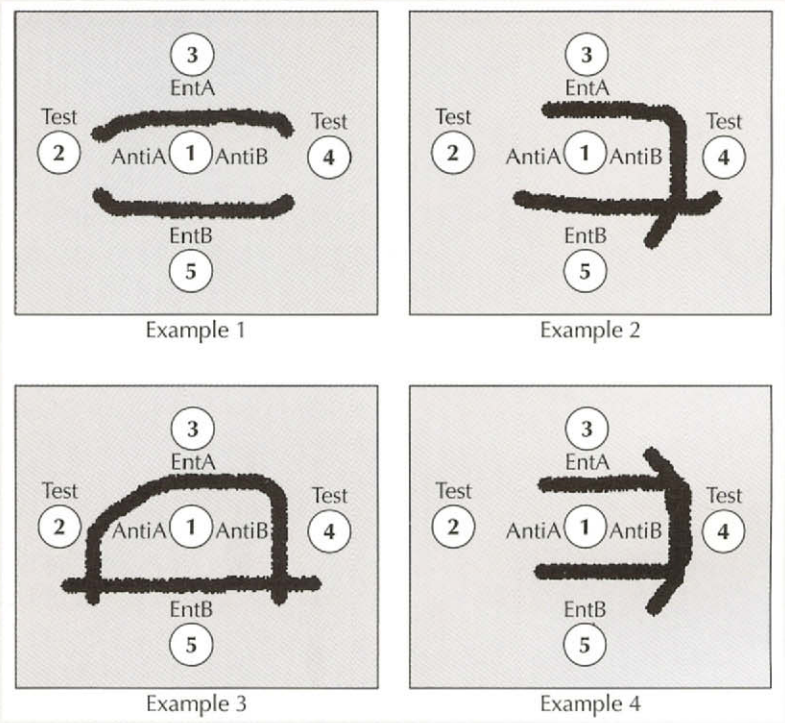
Staphylococcal enterotoxins are single polypeptides of 25 to 28 kilodaltons. Most are neutral or basic proteins that have no net charge at pH values ranging from 7 to 8.6 (i.e., their isoelectric points fall in this range). All staphylococcal enterotoxins are monomeric (single-unit) proteins. They are made as larger precursors with a signal peptide that is cut off as they are excreted from the cell.

BOX 14.4

S. aureus enterotoxin

The characterization of a staphylococcal isolate as an enterotoxin producer is an important task. There are many *Staphylococcus* species in the environment that pose no threat to the safety of foods. These need to be differentiated from the ones that pose a real hazard. The first step in the isolation of *S. aureus* is plating on Baird-Parker medium. Presumptive *S. aureus* colonies are very distinctive. They are black with a halo of precipitation against a clearing zone. Direct tests for enterotoxin are difficult. So suspect colonies are subcultured in broth, and the supernatants are tested for coagulase and/or thermostable nuclease activity. The coagulase test determines the culture's ability to coagulate a special type of rabbit serum. In a strong positive reaction, the tube can be inverted without the clot falling out. Weaker reactions need to be confirmed by another method, usually a thermostable nuclease. This test uses a color change to show the ability to break down DNA. 3M Petrifilm *S. aureus* Count Plates cleverly combine the reactions of Baird-Parker medium with the thermostable nuclease reaction on a single piece of Petrifilm. However, both coagulase and thermostable nuclease activities correlate with, but do not prove, enterotoxin production. The official method for detecting enterotoxin is the microslide diffusion assay (see figure). This method is tedious and complex. The microslide apparatus is constructed by placing a drilled Plexiglas template on a slab of diffusion agar supported by a microscope slide. Antitoxin(s) (usually A and B [AntiA and AntiB in the figure] since these are the most common) is placed in the center well. Positive controls of purified enterotoxin are placed in wells 3 (EntA) and 5 (EntB). Test samples are placed in wells 2 and 4. Thus, the reaction of example 1 has immunoprecipitation lines between the antitoxin and the control enterotoxins, but not the sample. Toxin is absent. When the sample contains toxin, its immunodiffusion line crosses the immunodiffusion line caused by the toxin standard. Thus, in example 2, enterotoxin B (but not A) is present in test sample 4. There are a variety of other toxin tests which come in kit form and are based on enzyme-linked immunosorbent assays. They are simpler to run and easier to interpret. Hopefully, the acceptance of enzyme-linked immunoassays for staphylococcal enterotoxins will make detection easier.

The figure and explanations for the immunodiffusion assay are derived from the FDA's *Bacteriological Analytical Manual*.



The three-dimensional shape of all staphylococcal enterotoxins is similar. Staphylococcal enterotoxins A, B, C, and E have a low α-helix (<10%) content compared to β-pleated sheets/β-turn structures (approximately 60 to 85%). It is the three-dimensional structure that gives staphylococcal enterotoxins their remarkable heat stability.

Staphylococcal enterotoxins are very stable. Toxicity and antigenicity are not completely destroyed by boiling or even canning. The temperatures needed to inactivate staphylococcal toxins are much higher than those needed to kill *S. aureus* cells. In many cases of staphylococcal food poisoning, no live bacteria are found in the food.

Staphylococcal Enterotoxin Antigenic Properties

Each enterotoxin type has enough antigenic distinctness to be differentiated from other toxins using antibodies. However, some cross-reactivity can occur. The level of cross-reactivity generally correlates with similar amino acid sequences. The type C enterotoxin subtypes and their molecular variants are cross-reactive, as are enterotoxins A and E, the two major serological types with the greatest sequence similarity. The two most distantly related enterotoxins recognized by a common antibody are staphylococcal enterotoxins A and D. Box 14.4 illustrates how antigenicity can be used as a toxin detection system.

Summary

- *S. aureus* is commonly associated with humans.
- Poor sanitation causes many outbreaks.
- Symptoms of vomiting and diarrhea develop 0.5 to 6 h after eating.
- *S. aureus* makes several antigenically distinct enterotoxins.
- The cells are heat sensitive, but the toxins are heat resistant.
- *S. aureus* is the most osmotolerant foodborne pathogen.
- There is considerable knowledge of enterotoxin genetics and mechanism of action.

Suggested reading

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Dinges, M. M., P. M. Orwin, and P. M. Schievery. 2000. Exotoxins of *Staphylococcus aureus*. *Clin. Microbiol. Rev.* **13**:16–34.

Seok, K. S., and G. A. Bohach. 2007. *Staphylococcus aureus*, p. 493–518. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. What issues contributed to the outbreak described at the beginning of the chapter? How could it have been prevented?
2. Why did the passengers get sick even though the food was properly heated?
3. What would *you* do if you were served undercooked chicken on a transatlantic flight?
4. What is the difference between enterotoxins and exotoxins? Name an exotoxin (from another chapter) that is not an enterotoxin.
5. Why do food microbiologists sometimes refer to staphylococcal food poisoning as a "double-bucket" syndrome?
6. What is signal transduction? Why do bacteria need it?
7. Briefly discuss genetic aspects of staphylococcal enterotoxin production.
8. Both botulinal toxin and staphylococcal toxin are proteins, but the former is very heat sensitive and the latter is very heat resistant. Why?
9. You work for Chubby Chuck's Chickenery (CCC), which produces (among other chicken specialties) about 1,000 pounds of chicken salad

per day. It is packed in 10-lb tubs and distributed under refrigeration to restaurants and delis. Chubby Chuck heard about the terrible Texas outbreak and tells you it is your job to make sure this never happens at CCC. What do you do?

10. Being the diligent plant manager of CCC, you test product for staphylococcal enterotoxins. On Friday, you get the results of Thursday's testing. Two batches of chicken salad test positive for *Staphylococcus* enterotoxin A. Batch 200498-6 was packaged and shipped out. Batch 200491-6 was batched into 10-gallon tubs which are still in the cold storage room of the plant. What do you do? What (and when) do you tell Chubby Chuck?
11. Interpret the results of the microslide immunodiffusion assays shown in the figure in Box 14.4, examples 3 and 4. (Remember that the test preparations in wells 2 and 4 can contain EntA, EntB, neither, or both.)

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15 *Clostridium botulinum*

Introduction
Four Faces of Botulism
Characteristics of the Disease
Toxic and Infectious Doses and Susceptible Populations
Characteristics of *C. botulinum*
Classification
Tolerance of Preservation Methods
Sources of *C. botulinum*
Occurrence of *C. botulinum* in the Environment
Occurrence of *C. botulinum* in Foods
Virulence Factors and Mechanisms of Pathogenicity
Structure of the Neurotoxins
Genetic Regulation of the Neurotoxins
Mode of Action of the Neurotoxins
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Clostridium botulinum*
- understand what conditions in foods favor the growth of *C. botulinum*
- recognize, from symptoms and time of onset, a case of botulism
- choose appropriate interventions (heat, preservatives, or formulation) to prevent the growth of *C. botulinum*
- identify environmental sources of *C. botulinum*
- understand the role of spores, anaerobic conditions, and heat sensitivity of the toxin in causing or preventing botulism

INTRODUCTION

Botulism is a rare but sometimes deadly disease. There are an average of 24 cases of foodborne botulism, 3 cases of wound botulism, and 71 cases of intestinal (or "infant") botulism reported annually to the Centers for Disease Control and Prevention (CDC). Although botulism from commercial foods is rare, many countries report relatively frequent outbreaks from other sources (Table 15.1).

Four Faces of Botulism

Botulism is traditionally associated with canned foods, usually home canned. A commercial case occurred in New York State in the early 1970s. An elderly couple ate some vichyssoise soup on a hot summer night. By the next morning, they had started to see double, had difficulty swallowing, and experienced paralysis in their arms and legs. If untreated, toxin would have paralyzed their diaphragms and suffocated them. Fortunately, they were quickly diagnosed with botulism and put on ventilators, and they lived. They never fully recovered. The investigation revealed that the workers responsible for ensuring the commercial sterilization of the cans had misinterpreted a new process schedule. They underprocessed the soup, allowing *Clostridium botulinum* to grow and make toxin. In response to this incident, the Food and Drug Administration (FDA) instituted good manufacturing practices (GMPs) for low-acid foods (i.e., those with pH values of >4.6 and water activity [a_w] of >0.85). The GMPs require that all retort operators attend a better-processing-control school and be certified. Processing schedules and changes to them can be made only by recognized processing authorities.

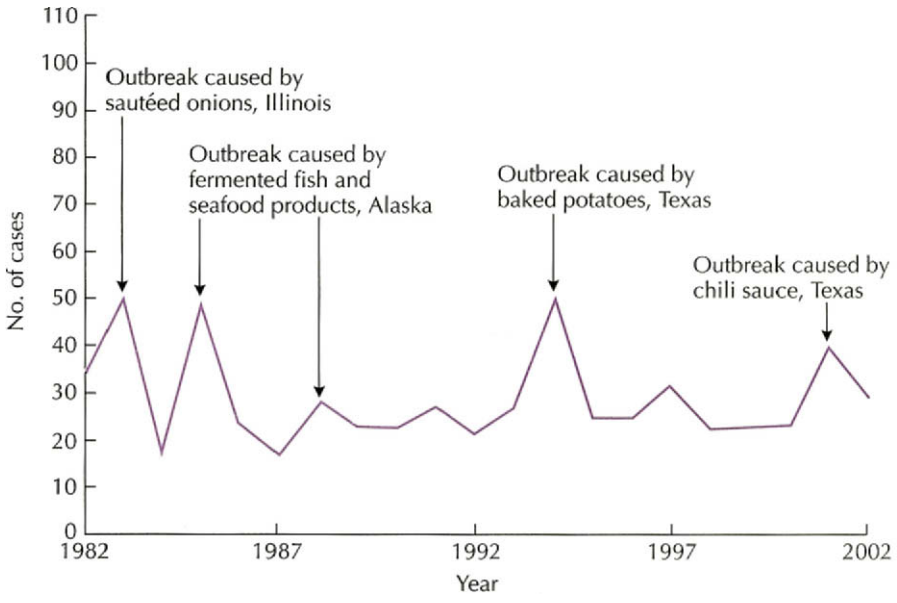
Foodborne botulism is often associated with temperature abuse. Figure 15.1 shows spikes in the incidence of botulism which are caused by

Table 15.1 Reported foodborne botulism cases

Country	Period	No. of cases	Usual type	Usual food
Argentina	1979–1997	277	A	Preserved vegetables
Belgium	1988–1998	10	B	Meats
Canada	1985–1999	183	E	Traditional Inuit fermented marine mammals
China	1958–1989	2,861	A, B	Fermented bean products
France	1988–1998	72	B	Home-cured ham
Germany	1988–1998	177	B	Meats
Iran	1972–1974	314	E	Fish
Italy	1988–1998	412	B	Vegetables preserved in oil or water
Japan	1951–1987	479	E	Fish or fish products
Norway	1975–1997	26	E	*Rakfisk* (traditional fermented fish)
Poland	1988–1998	1,995	B	Home-preserved meats
Russia	1988–1992	2,300	B	Home-preserved mushrooms, fish
Spain	1988–1998	92	B	Vegetables
United States	1950–1996	1,087	A	Vegetables

temperature abuse. One of the biggest such outbreaks occurred in Clovis, NM. Over 40 people who ate at a local salad bar started seeing double, became short of breath and weak, had slurred speech, and showed various signs of paralysis. The local doctor, to his credit, quickly diagnosed botulism. Remembering its association with canned foods, he (prematurely) proclaimed the commercially canned three-bean salad to be the culprit. Public health authorities from the CDC and the FDA did an exhaustive investigation that cleared the beans and implicated the potato salad. They discovered that it is a common practice in restaurants to put leftover potatoes in a box and store them at room temperature. When enough potatoes accumulate, they are used for potato salad with no further heating. This

Figure 15.1 Reported cases of foodborne botulism, United States, 1982 to 2002. The peaks in the graph demonstrate the role of temperature abuse in outbreaks. Courtesy of the CDC.



turns out to be a perfect scenario for botulism poisoning. Botulinal spores are normally found on potatoes. Baking kills any other competing bacteria and drives off oxygen, creating the anaerobic conditions under which *C. botulinum* thrives. The surviving spores turn into cells that grow in the potatoes and make botulinal toxin. Since the potato salad is not cooked, the toxin is not destroyed. Victims of this outbreak required years to recover.

While most commercial products have excellent safety records, commercially bottled garlic in oil has caused many outbreaks. As a result, garlic in oil can be sold in North America only if a second barrier, such as acidification, is present and the product is refrigerated. Failure to refrigerate was probably responsible for a 2006 botulism outbreak associated with pasteurized carrot juice. Three people who drank the carrot juice at a common meal developed facial and then respiratory paralysis. Treatment with botulinum A antitoxin stopped further progression of the disease. Botulinum A toxin was isolated from both the patients and the product.

In the Canadian north, Alaska, Scandinavia, and northern Japan, most botulism outbreaks involve fish. *C. botulinum* type E is implicated in most outbreaks involving northern native foods. Many of these foods are fermented products. However, the level of fermentable carbohydrates is too low to ensure sufficiently rapid acidification to inhibit botulinal growth.

Scientists in the California Department of Public Health in the 1970s discovered the second form of botulism, infant botulism. They noticed infants who were not thriving, could not lift themselves up, and had poor muscle tone (Fig. 15.2). Their deaths resembled "sudden infant death." Early investigation of *hospitalized* infants revealed an association with breast-feeding. It was soon realized, however, that the protective effect of breast-feeding allowed the diagnosis and hospitalization of these infants. Infants who were not breast-fed were much sicker and usually did not make it to the hospital. Eventually, an association with eating honey and other raw agricultural products led to a new scenario for botulism. We all consume botulinal spores but are protected by the bacteria colonizing our gut. When infants, who have no protective intestinal microbiota, eat spores,

Figure 15.2 An infant with flaccid paralysis that is characteristic of human botulism. Courtesy of the CDC.

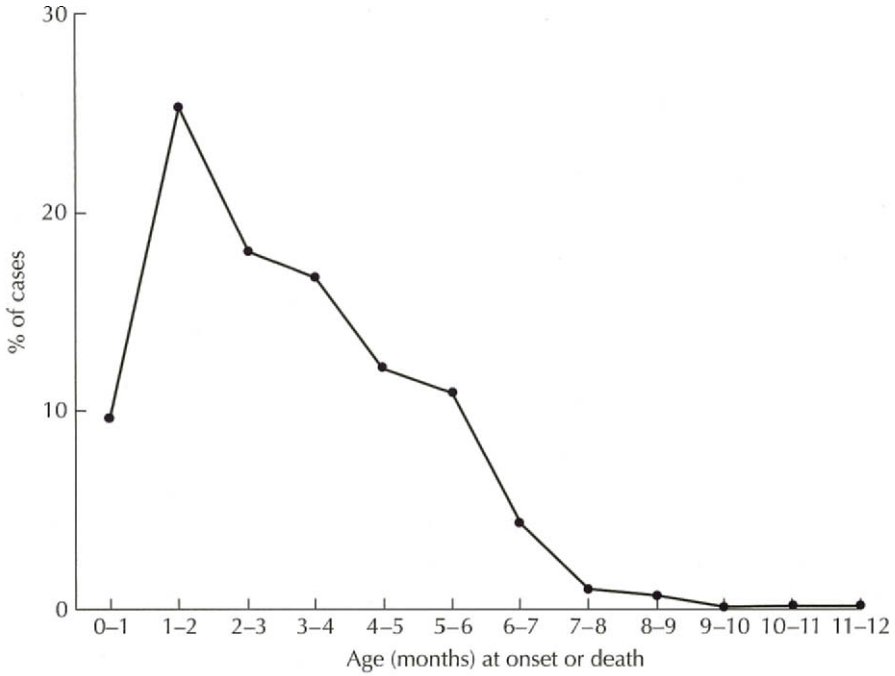


Figure 15.3 Age distribution of infant botulism. The data are for 1,428 cases in the United States from 1976 to 1995. Courtesy of the CDC.

these spores can germinate, become cells, colonize the gut, and produce toxin in place. Adults who are immunocompromised or who have undergone antibiotic therapy that kills their intestinal bacteria can also get infant botulism, so it has been renamed "intestinal botulism." To prevent infant botulism, the American Academy of Pediatrics recommends that children under the age of 2 years should not be fed any raw agricultural products. As shown in Fig. 15.3, botulism in infants older than 6 months has virtually disappeared.

The third form of botulism, wound botulism, is not foodborne but can still be fatal. It is caused when spores are introduced to body tissue below the skin (Fig. 15.4). In one recent year, the number of botulism cases caused by drug users injecting contaminated heroin was greater than the number of foodborne cases. A fourth face of botulism is the intentional use of *C. botulinum* as an agent of bioterrorism (Box 15.1).

Figure 15.4 Wound botulism. Courtesy of the CDC.

BOX 15.1

Biosecurity and biosafety in the age of terrorism

A microbiologist wearing a personal air-purifying apparatus and working in a class IIa biosafety cabinet.

Clostridium botulinum produces the most potent toxin known. Yet there was a time when food microbiologists traded botulinum cultures like baseball cards. If they had nothing to trade, they could buy cultures at minimal cost with minimal red tape. That time ended in October 2001, when a terrorist sent spore-laden letters to politicians and celebrities. *Bacillus anthracis* became a household word, a word associated with terror. Microbiology was suddenly not such a noble field. Microbiologists who worked with spores were "visited" by FBI agents.

By the end of January 2002, Congress passed, and President George W. Bush signed, the Providing Appropriate Tools Required to Intercept and Obstruct Terrorism Act (i.e., the "Patriot Act") and the Public Health Security and Bioterrorism Preparedness and Response Act. Together, these laws give the federal government strong oversight of microbiologists who work with microbes that could be used for bioterrorism, now known as "select agents." Select agents that food microbiologists might normally work with include *C. botulinum*, botulinum toxin, *Clostridium perfringens* epsilon toxin, staphylococcal enterotoxin, and T-2 toxin. It is now a federal offense to possess any of these unless the laboratory has been registered, inspected, and approved by the CDC. The Office of the Attorney General of the United States must run background checks and approve scientists who work with select agents. "Restricted persons," i.e., anyone convicted of a felony or another crime that could carry a penalty of more than 1 year in jail, citizens of countries hostile to the United States, and people who are "mentally defective" or were committed to a mental institution, are prohibited from having access to select agents.

The select agent act requires that the agents be inventoried, kept locked in a room accessible only to people cleared by the Department of Justice, and handled under a biosafety cabinet using BSL-3 work practices. In addition to adhering to the normal BSL guidelines, the laboratory must conduct a risk assessment, have a security plan in place, and maintain detailed records of laboratory entry, culture maintenance, and experimental procedures.

Biosafety levels are designated by how dangerous the organism is and how easy it is to treat the illness it may cause. Microbes that do not consistently cause sickness in healthy adults can be used in a BSL-1 laboratory. BSL-1 labs must use standard microbiological practices such as no smoking, eating, or drinking in the lab and no mouth pipetting. No other special equipment is required. Most foodborne pathogens need a BSL-2 laboratory. These organisms can make people sick if ingested or introduced through cuts or through membranes. These sicknesses are relatively easy to cure. In addition to the BSL-1 requirements, access to BSL-2 labs is restricted to lab personnel, there is a biohazard warning sign, procedures that produce aerosols are done under a biosafety cabinet, and lab personnel must wear lab coats and gloves. In some cases, personal breathing apparatus may provide an additional level of protection (see figure). BSL-3 labs are much more specialized and are required for microbes that can be transmitted through aerosols. Organisms causing serious lethal illnesses require BSL-3. In addition to BSL-2, BSL-3 requires controlled access, decontamination of lab waste and lab clothing, and engineering safeguards. The lab cannot be located off a common hallway and must have negative airflow that is exhausted without recirculation to other parts of the building. Biosafety cabinets are used for all culture manipulations. There are also BSL-4 laboratories for organisms like Ebola virus that are highly infectious and have no cure. There are only a few BSL-4 laboratories in the United States. Fortunately, no foodborne organisms require BSL-4 precautions.

These security and safety precautions will undoubtedly tighten over the next decade. They will make it more difficult, costly, and restrictive to conduct research on select agents that are also contaminants of food. The lengthy time required for Department of Justice clearance and immunizations will probably prevent graduate students and postdoctoral associates from working with these organisms. This will reduce the number of scientists trained to respond to bioterroristic attacks or to the much more probable outbreaks of foodborne botulism. Hopefully, future legislation will make it easier to conduct legitimate research while still protecting us from the misuse of microbes.

CHARACTERISTICS OF THE DISEASE

Foodborne botulism can range from a mild illness, which may be disregarded or misdiagnosed, to a serious disease that can kill within 1 day. Symptoms typically appear 12 to 36 h after ingesting the neurotoxin but may appear within a few hours or not until up to 14 days later. The earlier symptoms appear, the more serious the illness. The first symptoms are generally nausea and vomiting. These are followed by neurological signs and symptoms, including visual impairments (blurred or double vision, drooping eyelids, and fixed and dilated pupils), loss of normal mouth and throat functions (difficulty in speaking and swallowing; dry mouth, throat, and tongue; and sore throat), general fatigue and lack of muscle coordination, and respiratory impairment. Other symptoms may include abdominal pain, diarrhea, or constipation, which is usually the initial symptom of infant botulism. Respiratory failure and airway obstruction are the main causes of death. Fatality rates in the early 1900s were ≥50%. The availability of antisera and modern respiratory support systems have decreased fatality to about 10%. However, it may take 3 or 4 months on a ventilator and then years of therapy to fully recover. If botulinal toxin were used as an agent of biowarfare, there would not be enough ventilators or antitoxin available to mount a credible response.

Botulism is often incorrectly diagnosed as other illnesses, including other forms of food poisoning, stroke, poliomyelitis, organophosphate poisoning, tick paralysis, myasthenia gravis, and carbon monoxide poisoning, but most commonly as Guillain-Barré syndrome.

Initial treatment of botulism involves removing or inactivating the neurotoxin by (i) neutralizing the circulating neurotoxin with antiserum, (ii) using enemas to remove residual neurotoxin from the bowel, and (iii) gastric lavage or treating with emetics if the food might still be in the stomach. Antiserum is most effective in the early stages of illness. Subsequent treatment is mainly mechanical ventilation to counteract paralysis of the respiratory muscles. Optimal treatment for infant botulism consists primarily of high-quality supportive care.

TOXIC AND INFECTIOUS DOSES AND SUSCEPTIBLE POPULATIONS

Since toxicity experiments with human subjects are frowned on, little is known about the minimum toxic dose of botulinal neurotoxins. From a food safety perspective, the presence of neurotoxin or conditions permitting *C. botulinum* growth cannot be tolerated. Botulinal toxin is the most toxic natural substance known. The 50% lethal dose (LD_{50}) (amount required to kill 50% of the subjects) for botulinal toxin injected under the skin in monkeys is ~0.4 ng/kg of body weight. This suggests that the LD_{50} for a 150-lb human would be 0.000000001 (1×10^{-9}) ounce! This toxicity and the relative ease with which *C. botulinum* can be grown make it a potential agent for biological warfare. Indeed, in 2003, the United Nations Special Commission reported that Iraq produced about 50 gallons of concentrated botulinal toxin. This was allegedly weaponized in 16 missiles and 100 bombs, but as of this date no such weapons have been confirmed by forces on the ground.

The only way to prevent foodborne botulism is by preventing neurotoxin production in foods. Immunization of high-risk populations has been

considered, but it is not cost-effective. Currently, only researchers who work with the organism and members of the U.S. military are immunized. The preservation of high-moisture foods is usually designed to prevent *C. botulinum* growth. For example, shelf-stable canned foods are thermally processed to kill *C. botulinum* spores. Control of *C. botulinum* in minimally processed foods is achieved by growth inhibition, typically using a combination of factors such as microbial inhibitors and refrigeration. Preventing *C. botulinum* growth usually controls other foodborne pathogens and spoilage microorganisms.

CHARACTERISTICS OF *C. BOTULINUM*

Classification

C. botulinum is a gram-positive, rod-shaped, obligately anaerobic (oxygen kills it) bacterium. Anaerobic chambers (Fig. 15.5) or other methods of oxygen exclusion must be used when *C. botulinum* is studied in the laboratory. The organism forms oval endospores during the stationary-phase cultures (Fig. 15.6). There are seven types (A through G) of antigenically distinct *C. botulinum* toxins. Human botulism, including foodborne, wound, and intestinal botulism, is associated with types A, B, E, and, very rarely, F. Types C and D cause botulism in animals. To date, there is no direct evidence linking type G to disease.

The species is also divided into four groups based on physiological differences (Table 15.2). Group I contains all type A strains and proteolytic strains (i.e., those that produce enzymes that break down proteins) of types B and F. Group II encompasses all type E strains and nonproteolytic strains of types B and F. Group III contains type C and D strains. Group IV is comprised of *C. botulinum* type G. These groupings agree with results of DNA

Figure 15.5 Anaerobic chambers contain atmospheres devoid of oxygen, which is lethal to *C. botulinum*.

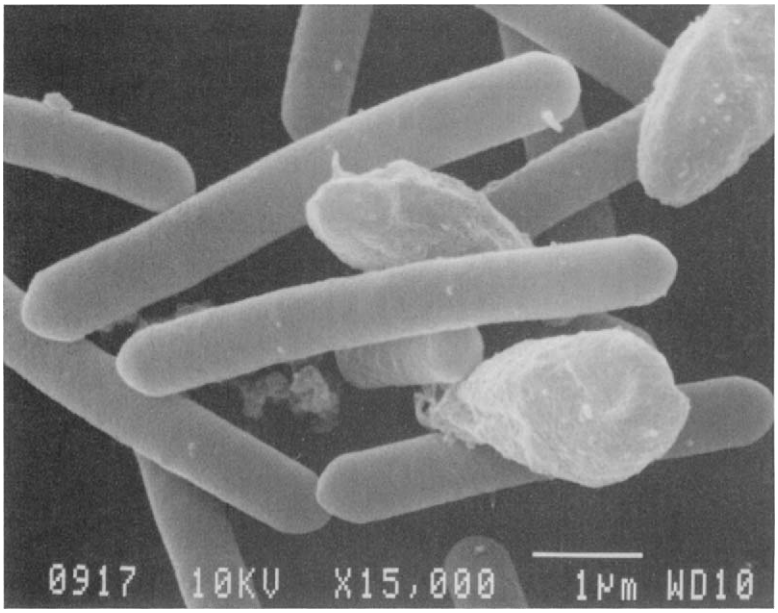


Figure 15.6 Scanning electron micrograph of *C. botulinum* vegetative cells (rods) and spores (irregular ovals). Originally taken by B. Maleeff and kindly provided by P. Cooke, both of the U.S. Department of Agriculture (USDA) Eastern Regional Research Center.

homology studies. Sequence studies using 16S and 23S ribosomal RNA show a high degree of relatedness among strains within each group, but little relatedness between groups.

Group I strains are proteolytic and are typified by strains that produce neurotoxin type A. The optimal growth temperature for members of this group is 37°C, with growth occurring between 10 and 48°C. These cultures typically produce high levels of neurotoxin (10^6 mouse LD_{50}/ml [1 mouse LD_{50} is the amount of neurotoxin required to kill 50% of injected mice within 4 days]). Spores in this group have a high heat resistance, with $D_{100°C}$ values of approximately 25 min (the D value is the time required to inactivate 90% of the population at a given temperature). To inhibit growth, the pH must be below 4.6, or the salt concentration must be above 10%, or the a_w must be below 0.94.

Table 15.2 Grouping and characteristics of strains of *C. botulinum*[a]

Characteristic	Toxin group			
	I	II	III	IV
Neurotoxin type(s)	A, B, F	B, E, F	C, D	G
Minimum temp (°C) for growth	10	3	15	ND
Optimum temp (°C) for growth	35–40	18–25	40	37
Minimum pH for growth	4.6	ca. 5	ND	ND
Inhibitory NaCl concentration (%)	10	5	ND	ND
Minimum a_w for growth	0.94	0.97	ND	ND
$D_{100°C}$ of spores (min)	25	<0.1	0.1–0.9	0.8–1.12
$D_{121°C}$ of spores (min)	0.1–0.2	<0.001	ND	ND

[a]ND, not determined. Reprinted from J. W. Austin, p. 329–349, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers* (ASM Press, Washington, DC, 2001).

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Summary

Suggested reading

Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Clostridium perfringens*
- understand what conditions in foods favor *C. perfringens* growth
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *C. perfringens*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent *C. perfringens* growth
- identify environmental sources of the organism
- understand the role of spores and the sporulation cycle in *C. perfringens* foodborne illness

THE FOODBORNE ILLNESS

A Spore's-Eye View of *C. perfringens* Toxicoinfections

We spores are part of "the great circle of life." Most of the time, we just wait. Dormant. Patient. Then, there is a change. Heat kills our competitors, the vegetative cells. The temperature drops to that of a warm incubator, and we morph into vegetative cells and multiply at an incredibly rapid rate. There are millions of us. A poor human eats us, and we are in perfringens heaven. We have warm temperatures, lots of nutrients, and anaerobic conditions. Our growth produces so much carbon dioxide that the poor human thinks she is having a "gas attack." Oh, the cramps. Then, as nutrients run out, we sporulate back into our dormant form. During sporulation, as we cells break apart, we release a protein toxin that causes diarrhea in our victims. Expelled from our host, we return to the great circle of life. We wait. Dormant. Patient.

A Human View of *C. perfringens* Type A Foodborne Illness

Two outbreaks of *C. perfringens* type A foodborne illness occurred around St. Patrick's Day. One hundred fifty-six people in the Cleveland, OH, outbreak ate corned beef prepared at a local delicatessen. The corned beef was prepared by boiling for 3 h and cooling slowly at room temperature before being put in the refrigerator. Four days later, the corned beef was warmed to 48.8°C and served. The corned beef sandwiches were held at room temperature from late morning until later that afternoon and then consumed along with copious quantities of green brew (which looked better going down than it did coming up).

The St. Patrick's Day outbreak in Virginia involved 86 people at a traditional corned beef dinner. The corned beef was a frozen commercially prepared, brined product. It was thawed, cooked in large (4.54-kg) pieces, stored in a refrigerator, and reheated under a heat lamp for 90 min before serving. Samples of the food were later found to contain large populations of *C. perfringens*.

These two outbreaks illustrate the typical involvement of meat and temperature abuse in *C. perfringens* type A foodborne illness. In the Ohio outbreak, the corned beef was cooled too slowly after cooking and was then reheated to a temperature insufficient to kill *C. perfringens* cells. In the Virginia outbreak, the beef was cooked in overly large portions and not reheated adequately before serving.

Incidence

C. perfringens type A toxicoinfection is the third most common cause of foodborne disease in the United States. From 1993 to 1997, there were 40 outbreaks (representing 4.1% of total foodborne disease outbreaks) of *C. perfringens* type A toxicoinfection in the United States. These accounted for 2,772 cases (6.3% of total cases of bacterial foodborne diseases). For the period from 1998 to 2002, there were 6,742 cases with four fatalities. Other countries report a higher proportion; in Norway, 30% of foodborne illness cases are attributed to *C. perfringens*. However, like all foodborne illnesses, most cases of *C. perfringens* type A toxicoinfection are not recognized or reported. Conservative estimates suggest that 250,000 cases of *C. perfringens* type A foodborne illness occur in the United States each year, causing an average of seven deaths. The economic costs associated with *C. perfringens* type A food toxicoinfections probably exceed $240 million.

Outbreaks of *C. perfringens* type A toxicoinfections are usually large (~50 to 100 cases) and often occur in institutionalized settings. The large size of outbreaks is due to two factors. First, large institutions often prepare food in advance and hold it for later serving. This lets *C. perfringens* grow if the stored food is temperature abused. Second, given the relatively mild and routine symptoms of most *C. perfringens* cases, public health officials usually become involved only when lots of people get sick. *C. perfringens* foodborne illness occurs throughout the year but, like most foodborne illness, is slightly more common during summer months. The warmer weather can facilitate temperature abuse of foods during cooling and holding. Summer also provides more occasions (like picnics, fairs, and carnivals) where food is prepared in casual settings by people who are having too much fun to worry about hygienic practices.

Food Vehicles for *C. perfringens* Foodborne Illness

C. perfringens is unable to make 13 of the 20 amino acids required for growth and survival. This makes high-protein foods the most common vehicles. *C. perfringens* foodborne illness in the United States is most commonly associated with meats and poultry. Other products that are high in protein and subject to temperature abuse (e.g., gravies, stews, and Mexican foods) are also significant vehicles.

Factors Contributing to *C. perfringens* Type A Foodborne Illness

C. perfringens foodborne illness usually results from temperature abuse during the cooking, cooling, or holding of foods. Its growth optimum of 43°C gives it an edge over other mesophilic pathogens in temperature-

abused foods. The Centers for Disease Control and Prevention (CDC) reports that improper storage or holding temperature was a contributor to 100% of recent *C. perfringens* outbreaks. Improper cooking was a factor in ~30% of these outbreaks. Contaminated equipment contributed to ~15% of the outbreaks.

The importance of temperature abuse in *C. perfringens* foodborne illness is not surprising. *C. perfringens* vegetative cells are relatively heat tolerant. However, *C. perfringens* spores are much more heat resistant than vegetative cells. Incomplete cooking can induce the germination of *C. perfringens* spores that survived the heating. If the food is improperly cooled or stored, the vegetative cells that can multiply rapidly are produced from the germinated spores.

Preventing *C. perfringens* Type A Foodborne Illness

Thoroughly cooking food is the best intervention against *C. perfringens* foodborne illness. This is particularly important for large roasts and turkeys. Because of their size, it is hard to reach the high internal temperatures needed to kill *C. perfringens* spores. The difficulty of cooking large pieces of meat to such high internal temperatures helps explains why those foods are such common vehicles for *C. perfringens* outbreaks. A second, and perhaps even more important intervention is to rapidly cool and store cooked foods to temperatures at which *C. perfringens* vegetative cells cannot grow (i.e., below 40°F or above 140°F).

Authors' note

The U.S. Food Code suggests that food be held below 40 or above 140°F to prevent microbial growth. U.S. Department of Agriculture regulations dictate that meats should pass through this range within 2 h.

Identification of *C. perfringens* Type A Foodborne Illness Outbreaks

Public health agencies traditionally use criteria such as incubation time and illness symptoms, or type and history of food vehicles (e.g., is temperature-abused meat or poultry involved?) for identifying *C. perfringens* outbreaks. However, the similarities between the onset times and symptomology of *C. perfringens* foodborne illness and other illnesses such as *Bacillus cereus* toxicoinfection make it unwise to completely rely on clinical and epidemiologic features for identifying outbreaks. The presence of large numbers of spores in the feces or the detection of specific toxins is more reliable in identifying outbreaks of *C. perfringens* illness. Bacteriologic criteria used by the CDC to identify an outbreak include the presence of either 10^5 *C. perfringens* organisms/g of stool from two or more ill persons or 10^5 *C. perfringens* organisms/g of the epidemiologically implicated food. Because *C. perfringens* is widely distributed in the environment and <5% of environmental samples make toxin, simply finding it in suspect food or feces is not proof of an outbreak. Most strains of *C. perfringens* in food or feces do not carry the gene for toxin production and are unable to cause illness.

The CDC and Food and Drug Administration (FDA) now use the detection of *C. perfringens* enterotoxin in feces of multiple people who are sick to identify *C. perfringens* food poisoning outbreaks.

Since *C. perfringens* enterotoxin can be present in feces of people suffering from nonfoodborne gastrointestinal (GI) diseases (such as antibiotic-associated diarrhea), the presence of *C. perfringens* enterotoxin in the feces of a single individual is not sufficient to identify *C. perfringens* toxicoinfection. However, detection of *C. perfringens* enterotoxin in feces from several individuals provides strong evidence for an outbreak. This is particularly true when they consumed a common food, developed illness within typical incubation times, and presented with the characteristic symptoms of *C. perfringens* toxicoinfection. The use of fecal *C. perfringens* toxin detection for

identifying *C. perfringens* outbreaks is limited because fecal samples must be collected soon after the onset of symptoms. It is often difficult to recruit donors under these conditions. However, several commercially available serologic kits are available to detect *C. perfringens* enterotoxin in feces.

CHARACTERISTICS OF *C. PERFRINGENS* TYPE A FOODBORNE ILLNESS

Symptoms of *C. perfringens* type A toxicoinfection develop 8 to 16 h after consumption of contaminated food. They last about 12 to 24 h. Victims of *C. perfringens* toxicoinfection usually suffer only diarrhea and severe abdominal cramps. Vomiting and fever are not typical. While death rates from *C. perfringens* type A toxicoinfection are low, fatalities do occur in debilitated or elderly populations.

The typical pathogenesis of *C. perfringens* type A toxicoinfection is illustrated in Fig. 16.1. Initially, temperature abuse stimulates *C. perfringens* cells to rapidly multiply in food. Large numbers of the bacteria are consumed. Some vegetative cells survive passage through the stomach and remain viable when they enter the small intestine. In the small intestine they multiply, sporulate, and release enterotoxin. *C. perfringens* enterotoxin is made during the sporulation of *C. perfringens* cells in the small intestine. After being released in the intestine, *C. perfringens* enterotoxin quickly binds to, and damages, intestinal epithelial cells. This intestinal damage initiates fluid loss, i.e., diarrhea.

Two factors help explain why *C. perfringens* toxicoinfection is usually relatively mild and self-limited. The diarrhea probably flushes (this word is not chosen lightly) unbound *C. perfringens* toxin cells from the small intestine. Secondly, *C. perfringens* toxin preferentially affects villus tip cells. These are the oldest intestinal cells and are rapidly replaced in young, healthy individuals by normal cell turnover.

INFECTIOUS DOSE FOR *C. PERFRINGENS* TYPE A FOODBORNE ILLNESS

C. perfringens cells are killed by stomach acidity. Hence, cases of *C. perfringens* toxicoinfection usually develop only when a heavily contaminated food (i.e., a food containing $>10^6$ to 10^7 *C. perfringens* vegetative cells/g of food) is eaten. The *C. perfringens* toxin is produced in the victim, and then *C. perfringens* organisms sporulate in their intestines. Therefore, this illness is considered a toxicoinfection.

Everyone is susceptible to *C. perfringens* toxicoinfection; but the illness is more serious in elderly or debilitated individuals. Many people develop a transient serum antibody response to *C. perfringens* toxin following illness. There is, however, no evidence that previous exposure provides future protection.

THE ORGANISM

Overview

C. perfringens became recognized as a foodborne pathogen in the 1940s and 1950s, following the pioneering work of K. Knox, K. Mat-Donald, L. S. McClung, and E. Hobbs. *C. perfringens* is a gram-positive, rod-shaped, encapsulated, nonmotile bacterium that causes a broad spectrum of human

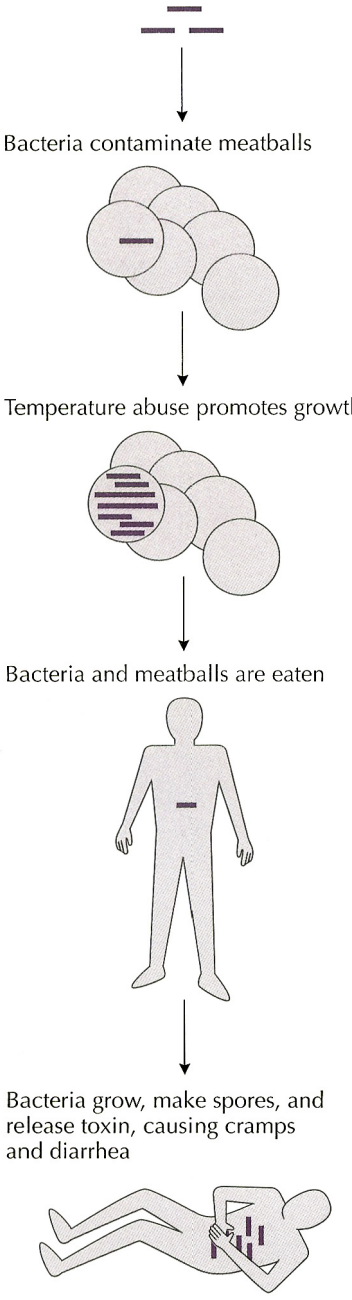

Figure 16.1 Schematic representation of *C. perfringens* food poisoning. Vegetative cells or spores contaminate a meat product and multiply rapidly when food is "incubated." A person consumes the vegetative cells, which then multiply rapidly in the small intestine (producing lots of gas) and sporulate, releasing the toxin at the same time as the spores. The victim is very sick with gas, cramps, and diarrhea but recovers in 24 to 48 h.

and veterinary diseases. *C. perfringens* is considered an *anaerobe* because it does not produce colonies on agar plates exposed to air. However, it tolerates moderate exposure to air. *C. perfringens* virulence results from its prolific toxin-producing ability, including at least two toxins, i.e., *C. perfringens* enterotoxin and β-toxin, that are active on the human GI tract. These cause *C. perfringens* toxicoinfection and necrotic enteritis. Foodborne necrotic enteritis has a 15 to 25% fatality rate but is extremely rare in industrialized societies. We do not discuss it further.

In addition to producing GI-active toxins, *C. perfringens* has other traits that help it cause foodborne disease. *C. perfringens* grows rapidly, doubling

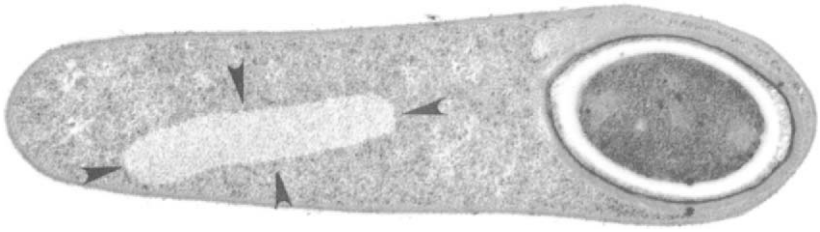

Figure 16.2 Transmission electron micrograph of a sporulating *C. perfringens* cell. Note the terminal fully developed spore and the paracrystalline inclusion body (arrowheads). Courtesy of Ron Labbe, University of Massachusetts.

in number in <10 min. This allows *C. perfringens* to quickly multiply in food. *C. perfringens* forms spores (Fig. 16.2) that are resistant to environmental stresses such as radiation, desiccation, and heat. More specifically, they survive in incompletely cooked or inadequately reheated food.

Classification: Toxin Typing of *C. perfringens*

There are at least 14 different *C. perfringens* toxins. Fortunately, no individual *C. perfringens* cell carries the gene(s) encoding all of these toxins. This limitation provides the basis for a toxin-typing system. This system classifies *C. perfringens* isolates into five types (A through E), depending on an isolate's ability to make 4 (alpha, beta, epsilon, and iota) of the 14 toxins (Table 16.1). This table also identifies the toxins' biological activities and their genes. Toxin typing of *C. perfringens* traditionally involved laborious toxin-antiserum neutralization tests in mice. The development of polymerase chain reaction (PCR)-based schemes makes toxin genotyping of *C. perfringens* isolates much easier.

The two foodborne diseases caused by *C. perfringens* are associated with different *C. perfringens* types. Necrotic enteritis is caused by type C isolates. The β-toxin produced by those isolates is the primary virulence factor for that disease. As implied by its name, *C. perfringens* type A toxicoinfection is associated with type A isolates of *C. perfringens*, even though other *C. perfringens* types sometimes make a toxin with similar properties. *C. perfringens* enterotoxin-producing type A isolates are more widely distributed than other types of *C. perfringens* and are thus responsible for most of the illness.

Table 16.1 Toxin typing and characteristics of *C. perfringens* toxins[a]

***C. perfringens* type classification**	**Biological activity of toxins produced by each type[b]**				
	Alpha phospholipase C	**Beta necrotizing**	**Epsilon permease**	**Iota ADP-ribosylating**	**Enterotoxin**
A	+	−	−	−	+
B	+	+	+	−	+
C	+	+	−	−	+
D	+	−	+	−	+
E	+	−	−	+	+
Gene	*plc*	*cpb1, cpb2*	*ext*	*iap, iba*	*cpe*
Gene location	Chromosome	Plasmid	Plasmid	Plasmid	Chromosome, plasmid

[a]Compiled from S. Brynestad and P. E. Granum, *Int. J. Food Microbiol.* 74:195–200, 2002; C. L. Hatheway, *Clin. Microbiol. Rev.* 3:66–98, 1990; and B. A. McClane, p. 423–444, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

[b]+, positive for this phenotype; −, negative for this phenotype.

Susceptibility of *C. perfringens* to Preservation Methods

The growth of pathogens in food is affected by factors such as temperature, redox potential (E_h), pH, and water activity (a_w) levels. The effect of those factors on *C. perfringens* growth is briefly discussed below.

Temperature

Due to their heat resistance, *C. perfringens* spores survive in undercooked foods. Spore heat resistance is influenced by environmental and genetic factors. The environment in which a *C. perfringens* spore is heated affects heat resistance properties. Spores of some *C. perfringens* strains survive boiling for an hour or longer. There are also genetic variations in heat resistance properties. Perhaps due to selective pressure, spores of toxicoinfectious isolates generally have much greater heat resistance than spores of other *C. perfringens* isolates. Finally, incomplete cooking of foods may not only fail to kill *C. perfringens* spores but actually favor development of *C. perfringens* type A toxicoinfection by inducing spore germination.

Vegetative cells of *C. perfringens* are somewhat heat tolerant. Although not truly thermophilic, they have a relatively high optimal growth temperature (43 to 45°C) and can often grow at 50°C. Vegetative cells of toxicoinfectious isolates are about twofold more heat resistant than vegetative cells of other *C. perfringens* isolates.

C. perfringens cells are not tolerant of refrigeration or freezing. *C. perfringens* growth rates rapidly decrease at temperatures below ~15°C. No growth occurs at 6°C. In contrast, spores of *C. perfringens* are cold resistant. Toxicoinfection can result if viable spores present in refrigerated or frozen foods germinate when that food is warmed for serving.

Other Factors

Growth of *C. perfringens* in food is also affected by a_w, E_h, pH, and (probably) the presence of curing agents, such as nitrates. *C. perfringens* is less tolerant of low-a_w environments than is *Staphylococcus aureus*. The lowest a_w supporting vegetative growth of *C. perfringens* is 0.93 to 0.97, depending on the solute used to control the a_w.

C. perfringens does not require an extremely reduced environment for growth. If the environmental E_h is suitably low for initiating growth, *C. perfringens* can then modify the E_h of its environment (by producing reducing molecules such as ferredoxin) to produce more favorable growth conditions. The E_h of many common foods (e.g., raw meats and gravies) is low enough to support *C. perfringens* growth.

C. perfringens growth is also pH sensitive. Optimal growth occurs at pH 6 to 7, whereas *C. perfringens* grows poorly, if at all, at pH values of ≤ 5 and ≥ 8.3.

Preservation factors such as pH, a_w, and, perhaps, curing agents control *C. perfringens* by inhibiting the outgrowth of *C. perfringens* spores. However, ungerminated spores may remain viable in foods in the presence of preservation factors that prevent cell growth. Those spores may germinate later if the growth-limiting factor(s) is removed during food preparation.

RESERVOIRS FOR *C. PERFRINGENS* TYPE A

C. perfringens is present in natural environments, including soil (at levels of 10^3 to 10^4 colony-forming units [CFU]/g), foods (e.g., approximately 50% of raw or frozen meat contains *C. perfringens*), dust, and the intestinal tract of humans and domestic animals (e.g., human feces usually contain

10^4 to 10^6 *C. perfringens* organisms/g). The widespread distribution of *C. perfringens* is linked to its frequent occurrence in foodborne illness. However, recent studies revealed that the ubiquitous distribution of *C. perfringens* in nature is not especially relevant for understanding the reservoir(s) for this foodborne pathogen. This is because <5% of all *C. perfringens* isolates harbor the *cpe* (*C. perfringens* enterotoxin) gene required for toxin production. Only *C. perfringens* toxin-positive isolates (i.e., with the *cpe* gene) cause foodborne illness. Because most *C. perfringens* environmental isolates are *cpe* negative, it is important to find where the enterotoxigenic *C. perfringens* exists in nature.

The *cpe* gene can be located on either the chromosome or a plasmid. However, the *cpe* gene of most toxicoinfectious isolates is located on the chromosome. This strong association between foodborne illness and isolates that possess chromosomal *cpe* provides another criterion for identifying the location of toxicoinfectious isolates in nature. By learning the location of the *cpe* gene in environmental isolates, microbiologists will be able to answer critical questions about the ecology of *C. perfringens* toxicoinfectious isolates. Questions to be answered include the following. Are those isolates present in low numbers in some healthy human carriers? Are they present in some food animals? Do they enter foods during processing? Do they enter foods during final handling, cooking, or holding?

VIRULENCE FACTORS CONTRIBUTING TO *C. PERFRINGENS* TYPE A FOODBORNE ILLNESS

Heat Resistance

As mentioned earlier, most cases of *C. perfringens* toxicoinfections are caused by isolates carrying a chromosomal *cpe* gene. This association between chromosomal *cpe* isolates and toxicoinfections is odd because *C. perfringens* isolates carrying chromosomal or plasmid *cpe* genes make similar levels of the same toxin. However, cells with chromosomal *cpe* have greater heat resistance than cells with plasmid *cpe*. Since cooked meats cause most outbreaks, the greater heat resistance of isolates with chromosomal *cpe* favors their survival during incomplete cooking.

C. perfringens Enterotoxin

Evidence that *C. perfringens* Enterotoxin Is Involved in Foodborne Illness

C. perfringens toxin is classified as an enterotoxin because it induces fluid and electrolyte losses from the intestinal tract of mammals. There is strong epidemiological evidence that *C. perfringens* enterotoxin causes *C. perfringens* type A foodborne illness. This evidence includes the following.

1. The presence of *C. perfringens* toxin in a victim's feces is strongly correlated to illness.
2. The toxin causes serious intestinal effects in experimental animals.
3. Human volunteers fed purified *C. perfringens* toxin (yum . . .) get sick with the symptoms of *C. perfringens* foodborne illness.
4. The intestinal inflammation caused by *C. perfringens* toxin-positive isolates in experimental animals can be neutralized with *C. perfringens* toxin-specific antisera.

More recently, experiments that fulfilled Koch's postulates on a molecular basis confirmed the importance of enterotoxin for the *C. perfringens* pathogenesis. Those experiments revealed that sporulating (but not vegeta-

tive) culture lysates of a *C. perfringens cpe*$^+$ strain induced fluid accumulation in animals. This is consistent with *C. perfringens* toxin (whose expression is sporulation associated) being necessary for the toxicity of that isolate. Furthermore, neither vegetative nor sporulating cultures of a *cpe*-negative mutant made animals sick. When the *cpe* gene was reintroduced in the mutant, full virulence was restored.

C. perfringens type A toxicoinfection is not the only disease involving *C. perfringens* toxin. *C. perfringens* toxin-producing isolates also cause nonfoodborne human GI illnesses. These include antibiotic-associated diarrhea, sporadic diarrhea, and some veterinary diarrheas. The *cpe*-positive *C. perfringens* isolates causing nonfoodborne GI diseases are genetically distinct from those causing toxicoinfections. That is, *cpe* is located on the chromosome of toxicoinfectious isolates but on a plasmid in nonfoodborne disease isolates.

Expression and Release of *C. perfringens* Enterotoxin

C. perfringens enterotoxin expression and release has three interesting features. The toxin is made only by sporulating vegetative cells. The toxin is not *secreted* by the sporulating cells but is released when the mother cell lyses. Many *C. perfringens* toxin-positive isolates make extremely large amounts of toxin.

Synthesis of *C. perfringens* enterotoxin. Synthesis of *C. perfringens* toxin begins when cells sporulate and increases for the next 6 to 8 h. After 6 to 8 h of sporulation, *C. perfringens* toxin can represent up to 30% of the total cell protein. Why do some *C. perfringens* strains produce so much toxin during sporulation? The amount of *C. perfringens* toxin made by an isolate is not influenced by the location of the *cpe* gene. Nor is the amount of toxin made related to a gene dosage effect; all *cpe*-positive isolates carry only a single copy of the *cpe* gene.

There is a general relationship between an isolate's sporulation ability and its toxin production; i.e., the better a *C. perfringens* isolate sporulates, the more *C. perfringens* toxin is produced. However, this correlation is not absolute.

Authors' note

Cells can have multiple copies of a given gene. One would expect that a higher gene copy number would result in a larger amount of the gene product (i.e., toxin). This is not always true.

Release of *C. perfringens* enterotoxin from *C. perfringens*. *C. perfringens* toxin accumulates in cells during its synthesis, often reaching concentrations high enough to form paracrystalline inclusion bodies (Fig. 16.2). Intracellular *C. perfringens* toxin is released into the intestines at the end of sporulation, when the mother cell lyses to free its mature spore. The need for the mother cell to lyse and release the *C. perfringens* toxin explains, at least in part, why (despite *C. perfringens* toxin's quick intestinal action) symptoms develop only 8 to 24 h after ingestion. *C. perfringens* cells must grow in the intestine and complete sporulation before toxin is released. This takes 8 to 12 h.

C. perfringens Enterotoxin Biochemistry

C. perfringens toxin is heat sensitive. Its biologic activity can be inactivated by heating for 5 min at 60°C. The toxin is also quite sensitive to pH extremes but is resistant to some proteolytic treatments. In fact, limited treatment with the proteolytic enzyme trypsin or chymotrypsin increases *C. perfringens* toxin activity two- to threefold. This suggests that human intestinal proteases may activate *C. perfringens* toxin.

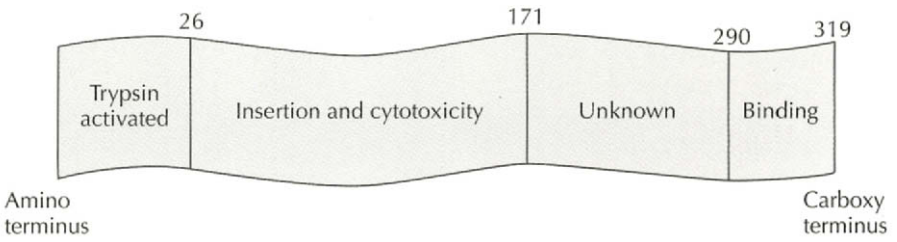

Figure 16.3 Stylized representation of *C. perfringens* enterotoxin showing the region that is cleaved by trypsin, the region responsible for insertion and cytotoxicity, and the binding site. The amino acid sequence numbers are given at the top. Modified from S. Brynestad and P. E. Granum, *Int. J. Food Microbiol.* **74:**195–200, 2002.

The enterotoxin is a single 3.5-kilodalton protein with specific regions conferring trypsin activation, binding, insertion, and cytotoxicity (Fig. 16.3). On a biochemical level, the enterotoxin acts through a multistep process. First it binds to a specific receptor on the intestinal epithelial cells. Then, the entire enterotoxin molecule is inserted into the epithelial cell's membrane. This releases ions and disrupts physiological processes.

C. perfringens Enterotoxin Effects on the Gastrointestinal Tract

The principal target for *C. perfringens* enterotoxin is the small intestine. Several features differentiate the biological activity of *C. perfringens* toxin from those of cholera and *Escherichia coli* heat-labile toxins. *C. perfringens* toxin induces direct cellular damage to the small intestine. The villus tips are particularly sensitive. Although other bacterial toxins (e.g., Shiga toxin and *Clostridium difficile* toxins) are also cytotoxic and cause intestinal tissue damage, *C. perfringens* toxin is unique with respect to how quickly it damages intestinal tissue. *C. perfringens* toxin-induced intestinal damage can develop in 15 to 30 min.

Summary

- *C. perfringens* is an anaerobic spore-forming bacterium commonly found on meat and meat products.
- The toxicoinfection toxin is released as a paracrystalline protein when vegetative cells release their spores in the gut.
- Improper heating or cooling is the major cause of *C. perfringens* type A toxicoinfection.
- The *cpe* gene codes for the enterotoxin; it is chromosomally located on toxicoinfectious isolates but on the plasmids of isolates not associated with foods.
- Victims of *C. perfringens* type A toxicoinfections usually suffer only diarrhea and severe abdominal cramps.

Suggested reading

Food and Drug Administration. 2002. Foodborne pathogenic microorganisms and natural toxins. http://vm.cfsan.fda.gov/~mow/intro.html.

Labbe, R. G. 1989. *Clostridium perfringens*, p. 192–234. *In* M. P. Doyle (ed.), *Microbial Foodborne Pathogens*. Marcel Dekker, Inc., New York, NY.

McClane, B. A. 2007. *Clostridium perfringens*, p. 423–444. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. Create an imaginary *C. perfringens* outbreak. Describe how the food was prepared and handled and the steps that led to the outbreak. Also include the victims' symptoms.
2. Would you rather be a spore or a vegetative cell? Why?
3. You are an eminent professor of food microbiology, awaiting delivery of food for your son's graduation party. The caterer arrives with 10 trays of Swedish meatballs, lasagna, and chicken parmesan. She assures you that "they are good and cold, just heat them up for a couple of hours on the steam tables." Why is this a problem? What do you do?
4. What is the difference between a case and an outbreak? Calculate the number of cases per outbreak for *C. perfringens* and some other foodborne pathogens. What can you infer from these numbers?
5. What is the biggest single causative factor in the occurrence of *C. perfringens* type A toxicoinfections?
6. What characteristics of *C. perfringens* contribute to its ability as a foodborne pathogen?
7. Both *C. perfringens* and *C. botulinum* foodborne diseases are linked to spores. How are these diseases different? How are they similar? Why is their frequency so different?
8. When foods are heated from 40 to 140°F or cooled from 140 to 40°F, it is important to get through this interval as rapidly as possible to prevent microbial growth. Rapid heating is easily achieved by obvious means. However, improper cooling causes as many outbreaks as improper heating and is much harder to control. Think of three ways to cool foods through the danger zone rapidly.
9. What is a "paracrystalline inclusion body"?
10. Here are several facts. Multiple copies of the *cpe* gene do not result in increased pathogenicity. Isolates from foodborne disease usually have *cpe* on the chromosome rather than the plasmid. Spores from isolates with a chromosomal *cpe* gene are more heat resistant than those from isolates that have *cpe* on the plasmid. How do you relate these facts to the role of *cpe* in *C. perfringens* foodborne illness? What experiments could you do to refine this understanding?

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Outbreak
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Environmental Sources
Foodborne Outbreaks
Characteristics of Disease
Dose
Virulence Factors and Mechanisms of Pathogenicity
The Emetic Toxin
Enterotoxins
The Spore
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- use basic biochemical characteristics to identify *Bacillus cereus*
- understand what conditions in foods favor the growth of *B. cereus*
- recognize, from symptoms and time of onset, a case of foodborne illness caused by *B. cereus*
- choose appropriate interventions (heat, preservatives, and formulation) to prevent *B. cereus* from causing foodborne illness
- identify environmental sources of *B. cereus*
- understand the role of spores and toxins in causing this foodborne illness

Outbreak

Even though fried rice is the classic example used to illustrate *B. cereus* food poisoning, it is still causing problems. The *Morbidity and Mortality Weekly Report* (the Centers for Disease Control and Prevention's weekly listing of how many people are getting sick or dying of various diseases) summarized a *B. cereus* food poisoning outbreak at two day care centers in Virginia. Eighty-two children ate a "special" catered international lunch. Sixty-seven children who ate the lunch were identified out of 80 people interviewed. None of the 13 children who did not eat the lunch got sick. Fourteen of the people who ate the lunch became ill with nausea, abdominal cramps, and diarrhea. The median time of onset for the illness was 2 h after eating. The median time for recovery was 4 h after onset.

Chicken fried rice was associated with the outbreak. One-third of the people who ate the rice became ill. None of the people who did not eat the rice became ill. *B. cereus* at $>10^5$ organisms per g was isolated from one child's vomit. Levels in the leftover chicken were $>10^6$ organisms per g.

The investigation uncovered a classic pattern of food mishandling. The rice was cooked the night before the special lunch and cooled at room temperature before being refrigerated overnight. The next morning, the rice and chicken were fried with oil, delivered to the day care center, held without refrigeration, and served without heating. Neither the restaurant staff nor the day care center staff knew that these were dangerous food handling practices.

INTRODUCTION

With only ~27,000 cases per year in the United States, *B. cereus* is considered a minor foodborne pathogen, but one that is increasingly important. The seemingly low incidence may reflect that *B. cereus* foodborne disease is not notifiable.

Authors' note

Notifiable diseases are those (usually infectious) diseases that must be reported by local health officials to the Centers for Disease Control and Prevention.

While *B. cereus* causes <2% of foodborne cases of identifiable origin in the United States, the prevalence is as high as 30% in Europe. *B. cereus* produces two toxins. An emetic (vomiting) toxin is produced by cells growing in the food and acts rapidly (0.5 to 6 h). In contrast, the diarrheal illness is caused by enterotoxin(s) produced during vegetative growth of *B. cereus* in the small intestine. The diarrheal toxin that causes diarrhea, cramps, and rectal tenesmus acts more slowly (6 to 14 h). Both types of illness are mild, brief, and self-limiting. This contributes to their being underreported. *B. cereus* does not compete well with other vegetative cells. But when the other cells are killed by cooking, *B. cereus* spores survive, grow, and cause illness. Outbreaks are commonly associated with meats, gravies, fried rice, pasta, sauces, puddings, and dairy products.

CHARACTERISTICS OF THE ORGANISM

Genus and species designations were originally based on the basic physiology of the organism (as determined by biochemical tests) and what the organism looked like (under the microscope). Thus, *any* spore-forming, aerobic rod-shaped bacterium was classified as a *Bacillus* species. Molecular biology and genetic characterization have now revealed that many of these species identifications by phenotypic characteristics are inaccurate. The genus *Bacillus* as described in *Bergey's Manual of Systematic Bacteriology* (the dictionary of microbial classification) is too diverse for a single genus. According to 16S ribosomal RNA (rRNA) sequences, there are at least five genera or rRNA groups within the genus *Bacillus*. With the subsequent identification of many new species, the number of genera has increased to approximately 16.

Bacillus anthracis, B. cereus, B. mycoides, B. thuringiensis, B. pseudomycoides, and *B. weihenstephanensis* are members of the *B. cereus* group. These bacteria have very similar 16S and 23S rRNA sequences, indicating they have diverged from a common evolutionary line. *B. anthracis* is the most distinctive member of this group, both in its highly virulent pathogenicity and in its notoriety as an agent for biological warfare. In addition, extensive studies of DNA from *B. cereus* and *B. thuringiensis* suggest that there is no scientific reason for them to be in separate species. Indeed, *B. thuringiensis* strains used as a natural bioinsecticide have caused cases of *"B. cereus"* food poisoning, and some *"B. cereus"* strains produce anthrax toxins.

Most *B. cereus* strains are unable to grow below 10°C or in milk products stored between 4 and 8°C. However, psychrotolerant strains grow at temperatures as low as 4 to 6°C. Some strains are motile via peritrichous flagella, while others are not motile at all. The nutritional requirements of these gram-positive spore-forming rods are simple. Hence, they are commonly associated with starchy foods like rice.

ENVIRONMENTAL SOURCES

B. cereus is widespread in nature and frequently isolated from soil and growing plants. It is easily spread from the environment to foods by cross-contamination. *B. cereus* is especially problematic in dairy foods. The spores, which can be present in soil at 10^5/g, spread to the cows' udders and into raw milk. Because *B. cereus* spores are especially hydrophobic, they attach to pipelines and other solid surfaces in the dairy. The spores survive milk pasteurization. After germination, free from competition from

L T Q S U V M S A D Q V J B G

B Q E P X M L A E H R R A I D

V D J N A S E R O P S C M R Z

K E N T E R O T O X I N X N X

Q L C T N S K I E L C G Z N V

N H C I T E M E L J A Q O O P

S R U C R H S U O T R S M X V

U P I H T I S H S K H I G N O

O M R T I C D A B J T V P S F

S Y V J E C P X D O N Y Q F R

Z S K R C S H G J F A T J B Y

Q F E H T I B V C P R H N V N

X U B F E J L R V V U X I W K

S H W U F Z S E K T T C A N B

Clues

Species name for another, not normally foodborne, pathogen in the genus *Bacillus*. _____

Foodborne pathogen that makes both an emetic toxin and a diarrheal toxin. _____

Toxin that causes rapid and uncontrollable release of watery fecal matter from the bowels. _____

Toxin that causes vomiting. _____

Italian food that can be a source of spores. _____

Asian food often implicated in outbreaks of *B. cereus* gastroenteritis. _____

Medium for growth of plants, reservoir for spores, and what you might do to your pants when you consume a diarrheal toxin. _____

Resistant but dormant life form produced by species in the genera *Bacillus* and *Clostridium*. _____

Rectal analog of "dry heaves." _____

What you do after consuming an emetic toxin. _____

Food microbiologists call pathogens that produce emetic and _____ toxins "double-bucket" bacteria.

other vegetative cells, they grow, causing problems in many milk products. *B. cereus*'s psychrotrophic nature makes the situation worse.

FOODBORNE OUTBREAKS

B. cereus outbreaks of foodborne illness are highly underestimated. This can be due to the relatively short duration of illness (usually <24 h), the similarity of symptoms to other foodborne pathogens, and the likelihood that illness involving *B. cereus*-contaminated milk is limited to one or two cases

and not identified as an outbreak. Near the end of milk's shelf life, there frequently are enough *B. cereus* cells to cause illness. The infectious dose is 10^3 to 10^4 organisms. Fortunately, *B. cereus* proteases cause off flavors. One would hope that this spoilage would keep consumers from drinking highly contaminated milk.

Authors' note

The argument that "the food would spoil before it could make someone sick" is invalidated by actual experience. Investigators are often led to the cause of the outbreak by victims who tell them to test a particular food because "it tasted really bad."

It is not possible to compare the incidence of outbreaks in different countries because their methods of tracking foodborne illnesses differ greatly. The percentage of outbreaks and cases attributed to *B. cereus* in Japan, North America, and Europe varies from 1 to 47% for outbreaks and from ~0.7 to 33% for cases. The greatest number of reported *B. cereus* outbreaks and cases are from Iceland, The Netherlands, and Norway. In The Netherlands in 1991 *B. cereus* caused 27% of outbreaks in which a causative agent was identified. However, the actual incidence of *B. cereus* was only 2.8% of the total number of cases because most cases of foodborne illness were of unknown etiology.

CHARACTERISTICS OF DISEASE

There are two types of *B. cereus* foodborne illness. The first type, caused by an emetic toxin, results in vomiting. The second type, caused by enterotoxin(s), results in diarrhea. A few *B. cereus* strains make both toxins and cause both symptoms. The incubation time (>6 h; average, 12 h) is too long for diarrheal illness to be caused by toxin premade in the food. (Preformed toxins act rapidly when ingested.) However, enterotoxin(s) can be formed in food, when the *B. cereus* population is at least 100-fold higher than that necessary for causing food poisoning. Products with such large populations of *B. cereus* are no longer acceptable to the consumer, although food containing $>10^7$ *B. cereus* organisms/ml may not always appear spoiled. The two types of *B. cereus* foodborne illness are characterized in Table 17.1.

DOSE

After the first diarrheal outbreak of *B. cereus* foodborne illness in Oslo, Norway (from vanilla sauce), S. Hauge isolated the causative agent, grew it to 4×10^6 cells/ml, and drank 200 ml of the culture. (Although the use of oneself as a human subject was accepted and even considered courageous in the early days of microbiology, this practice would be unethical by current standards. Do not try this at home.) About 13 h later, he developed abdom-

Table 17.1 Characteristics of the two types of illness caused by *B. cereus*[a]

Characteristic	Diarrheal syndrome	Emetic syndrome
Dose causing illness	10^5–10^7 cells (total)	10^5–10^8 cells per g
Toxin produced	In the small intestine of the host	Preformed in foods
Type of toxin	Protein; enterotoxin(s)	Cyclic peptide; emetic toxin
Incubation period	8–16 h (occasionally >24 h)	0.5–5 h
Duration of illness	12–24 h (occasionally several days)	6–24 h
Symptoms	Abdominal pain, watery diarrhea, and occasionally nausea	Nausea, vomiting, and malaise (sometimes followed by diarrhea, due to production of enterotoxin)
Foods most frequently implicated	Meat products, soups, vegetables, puddings/sauces, and milk/milk products	Fried and cooked rice, pasta, pastry, and noodles

[a]Reprinted from P. E. Granum, p. 373–381, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

inal pain and watery diarrhea that lasted for approximately 8 h. Counts of *B. cereus* ranging from 200 to 10^9 organisms/g (or ml) have been reported in foods incriminated in outbreaks, indicating that the total dose (cells/milliliter times milliliters consumed) ranges from approximately 5×10^4 to 1×10^{11}. The total number of *B. cereus* cells that must be present to produce enough toxin to produce illness is probably 10^5 to 10^8 cells. Food containing $>10^3$ *B. cereus* organisms/g is not safe to eat.

VIRULENCE FACTORS AND MECHANISMS OF PATHOGENICITY

Very different types of toxins cause the two types of *B. cereus* foodborne illness. The emetic toxin, causing vomiting, has been isolated and characterized, whereas the diarrheal disease is caused by one or more enterotoxins.

The Emetic Toxin

The emetic toxin (Table 17.2) causes emesis (vomiting) only. Its structure has been a mystery because the only detection system involved living primates. However, the recent discovery that HEp-2 cells could detect the toxin has led to its purification and structure determination. The emetic toxin is now named "cereulide." Cereulide has a ring structure of three repeats of four amino and/or oxyacids: [D-*O*-Leu-D-Ala-L-*O*-Val-L-Val]$_3$. The structure suggests that cereulide is an enzymatically synthesized peptide and not a gene product. The emetic toxin's biosynthetic pathway and mechanism of action are unknown. Emetic toxin is resistant to autoclaving, a wide range of pH values (2 to 11), and digestion by proteases. It is not antigenic.

Enterotoxins

Cloning and sequencing studies show that *B. cereus* produces at least five different proteins (or protein complexes) known as enterotoxins. The extent to which the different proteins are involved in foodborne illness might be determined through microslide diffusion tests (Fig. 17.1).

Three enterotoxins are probably involved in *B. cereus* foodborne illness. Two of the enterotoxins are multicomponent and related. A third enterotoxin is a single 34-kilodalton protein. A three-component hemolysin (Hbl) with enterotoxin activity has dermonecrotic and vascular permeability

Table 17.2 Properties of the emetic toxin cereulide[a]

Trait	Property, activity, or value
Molecular mass	1.2 kilodaltons
Structure	Ring-shaped peptide
Isoelectric point	Uncharged
Antigenic	No (?)
Biological activity in living primates	Vomiting
Cytotoxic	No
Heat stability	90 min at 121°C
pH stability	Stable at pH 2–11
Effect of proteolysis (trypsin, pepsin)	None
Conditions under which toxin is produced	In food (rice and milk) at 25–32°C

[a]Adapted from P. E. Granum, p. 373–381, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

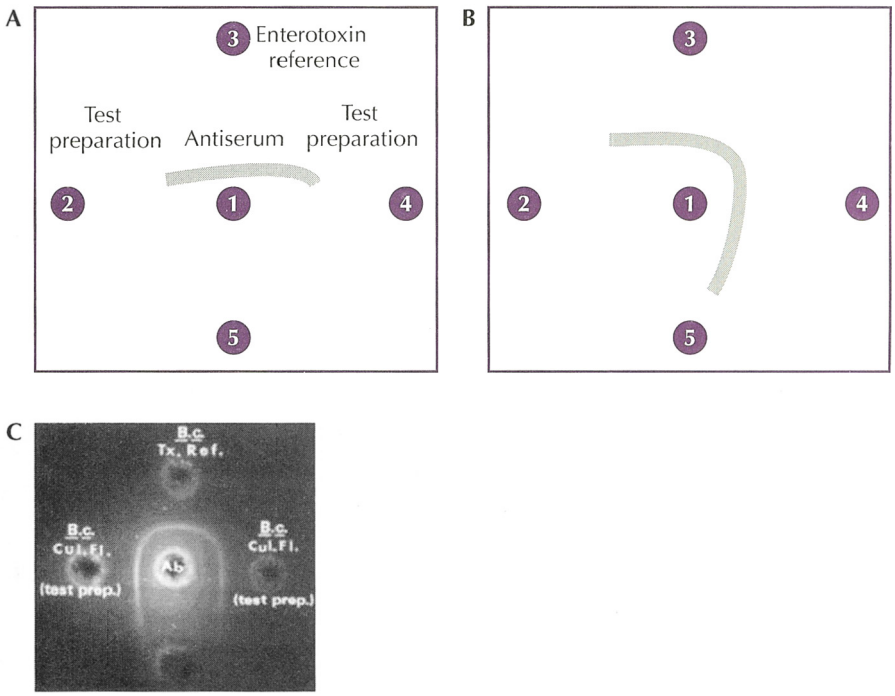

Figure 17.1 Microslide diffusion tests are used to detect *B. cereus* diarrheagenic and staphylococcal enterotoxins. Antiserum to the toxin is put into well 1, and reference toxin (a positive control) is put into well 3. Test samples are put into wells 2, 4, and 5. When the antiserum and toxin diffuse from their respective wells and meet, a precipitation line is formed. **(A)** Test where the food samples are negative but a line is formed with the known enterotoxin. **(B)** Test where the sample in well 4 contains enterotoxin. **(C)** Photo of actual results, where samples in wells 2 and 4 (but not 5) are positive for toxin. Photo courtesy of R. Bennett, U.S. Food and Drug Administration; diagrams based on those supplied by R. Bennett.

Authors' note

In the rabbit ileal loop assay, a rabbit is anesthetized, its surgically exposed intestine is tied off (i.e., ligated) into loops, and the sample is injected. If the sample causes fluid accumulation in the ligated loop, it is positive for toxin.

activities and causes fluid accumulation in ligated rabbit ileal loops. It may be the virulence factor in *B. cereus* diarrhea. All three proteins of the Hbl are transcribed from one operon (*hbl*).

The Spore

The *B. cereus* spore is important in foodborne illness. The *B. cereus* spore is more hydrophobic than any other *Bacillus* species' spores. This enables it to adhere to several types of surfaces. It is difficult to remove during cleaning and is a difficult target for disinfectants. *B. cereus* spores also contain appendages and/or pili that are involved in adhesion. These spore properties enable them to withstand sanitation and remain on surfaces to contaminate foods. They also aid in adherence to epithelial cells.

Summary

- *B. cereus* is a sporeformer that is especially problematic in dairy foods.
- It is a normal inhabitant of soil and is isolated from a variety of foods.
- Two different toxins cause an emetic or a diarrheal type of foodborne illness.
- Desserts, meat dishes, and dairy products are frequently associated with diarrheal illness.

- Rice and pasta are the most common vehicles of emetic illness.
- Some strains of the *B. cereus* group grow at refrigeration temperature.
- *B. cereus* foodborne illness is probably highly underreported.

Suggested reading

Granum, P. E. 2007. *Bacillus cereus*, p. 445–455. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. Speculate as to why *B. cereus* is considered a larger problem in Europe than in the United States.
2. Some people have suggested that *B. cereus* be used as a surrogate to research the deadly bioagent *B. anthracis*. What are the pros and cons of this proposal?
3. Why are *B. cereus* foodborne illnesses more likely to be underreported than other foodborne illnesses?
4. Create a case study of a *B. cereus* foodborne outbreak.
5. What is "rectal tenesmus"?
6. Describe the genetics of enterotoxin production by *B. cereus*.
7. Why is the use of knockout mice a good strategy to study the diarrheal toxins but not the emetic ones?

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SECTION IV Other Microbes Important in Food

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19 Yeast-Based and Other Fermentations
20 Spoilage Organisms
21 Molds
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18

Lactic Acid Bacteria and Food Fermentations

Introduction

The Biochemical Foundation of Food Fermentation
Catabolic Pathways

Dairy Fermentations
Starter Cultures
Production of Aroma Compounds
Genetics of Lactic Acid Bacteria

Fermented Vegetables
Ingredients and Additives Used during Fermentations
Sauerkraut Fermentation
Pickle Fermentation

Meat Fermentations

Summary

Suggested reading

Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- understand the biochemical basis of food fermentations
- appreciate the similarities and differences among fermentations of dairy, vegetable, and meat products
- relate specific bacteria to specific fermentations
- outline the process for making fermented foods
- understand the benefits of using fermentation as a food processing method

INTRODUCTION

Fermentation is a word with many meanings. Louis Pasteur used it to describe life in the absence of oxygen. A more formal biochemical definition of fermentation is the process that bacteria use to make energy from carbohydrates in the absence of oxygen. In common usage, "fermentation" also describes any biological process (for example, fermentations that make vinegar, antibiotics, monosodium glutamate, amino acids, citric acid, etc.) whether oxygen is present or not. With a few exceptions, *food fermentations* are bioprocesses that change food properties while the bacteria generate energy in the absence of oxygen. These changes go far beyond acid production. Fermentations add value to foods by producing flavor compounds and carbonation, altering texture, and increasing nutrient bioavailability (Fig. 18.1; Table 18.1).

This chapter simplifies fermentation biochemistry for students who have not had a biochemistry course. These biochemical principles are applied to the fermentation of dairy, vegetable, and meat products.

THE BIOCHEMICAL FOUNDATION OF FOOD FERMENTATION

Energy is released when a compound is oxidized, i.e., when an electron is "lost" (think back to Chemistry 101). Fermentation products such as alcohol are more oxidized than the starting product (usually a sugar), but they are not completely oxidized. For example, sugar can be partially oxidized to alcohol. The alcohol can be further oxidized when burned, releasing additional energy.

When one compound is oxidized and loses an electron, that electron must go somewhere. It goes to an electron acceptor, which becomes reduced. This can be expressed as follows:

Reduced electron donor (compound A) → Oxidized compound A
Oxidized electron acceptor (compound B) → Reduced compound B

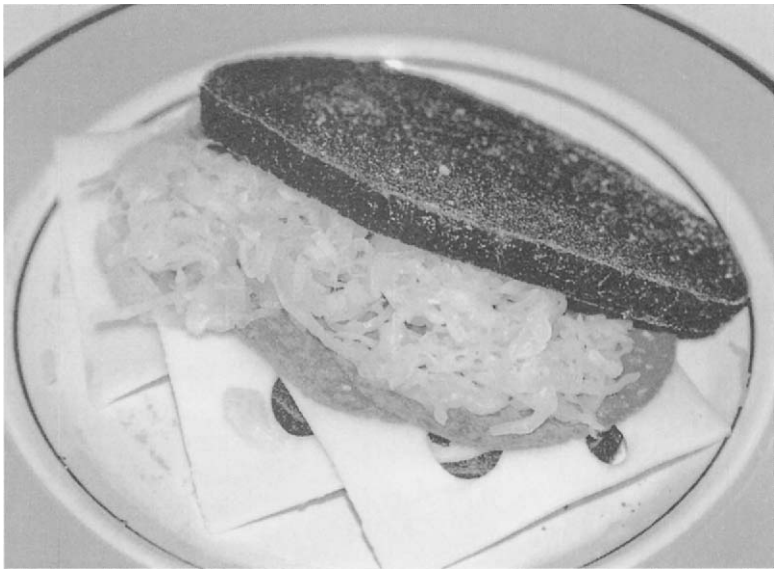


Figure 18.1 The bread, sauerkraut, meat, and Swiss cheese in this sandwich are all fermentation products.

When oxygen is the electron acceptor in the oxidation of sugars, the following reactions occur:

$$\begin{array}{r} \text{Sugar } (C_6H_{12}O_6) \rightarrow CO_2 \\ O_2 \rightarrow H_2O \\ \hline \text{Sugar} + O_2 \rightarrow CO_2 + H_2O \text{ (+ energy)} \end{array}$$

This complete oxidation gives the cell a lot of energy. This energy is stored, by mechanisms more appropriate for a biochemistry textbook, in the form of phosphorylated compounds such as adenosine 5′-triphosphate (ATP). In the example above, the complete oxidation of glucose produces 34 ATP molecules. This energy can be transferred to other compounds by moving the phosphoryl group.

In fermented foods, there is no oxygen to serve as an electron acceptor, so part of the sugar must serve as electron acceptor. This part of the sugar cannot be further oxidized to generate more energy. This incomplete oxidation does not yield very much energy (only one or two molecules of ATP from one molecule of glucose), but in the land of the blind, the one-eyed man is king. The incompletely oxidized fermentation products such as ethanol, acetic acid, and lactic acid are sometimes oxidized by other "scavenging organisms." Ethanol is oxidized to acetic acid with the generation

Table 18.1 Uses of fermentation

To preserve vegetables and fruits
To develop characteristic sensory properties, i.e., flavor, aroma, and texture
To destroy naturally occurring toxins and undesirable components in raw materials
To improve digestibility, especially of some legumes
To enrich products with desired microbial metabolites, e.g., L(+)-lactic acid or amino acids
To create new products for new markets
To increase dietary value

of one ATP molecule by *Acetobacter aceti*. *Propionibacterium freudenreichii* subsp. *shermanii* makes one ATP molecule by oxidizing lactic acid to propionic acid.

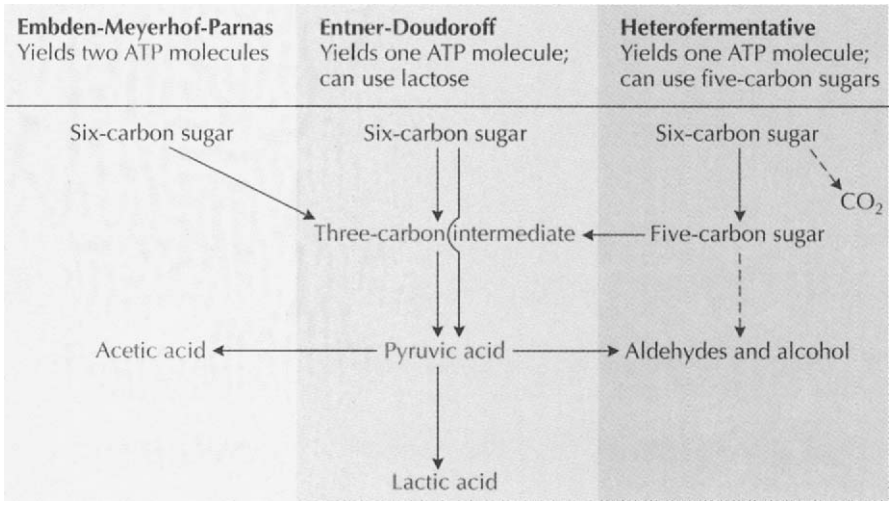
Catabolic Pathways

Figure 2.12 shows the detailed, step-by-step biochemical pathways used by fermentative bacteria. The more generalized Fig. 18.2 allows the pathways to be understood with less biochemical detail. The Embden-Meyerhof-Parnas (EMP) pathway from glucose to pyruvic acid may be the most important catabolic pathway. One molecule of glucose is converted to two molecules of pyruvic acid. At this point, the oxidation-reduction reactions are not balanced. They can be balanced by oxidizing pyruvate to compounds like lactic acid or ethanol. Nonfermentative organisms (like us) completely oxidize the pyruvic acid to carbon dioxide and water, using oxygen as the terminal electron acceptor. In terms of ATP generated per molecule of glucose used (i.e., two molecules), the EMP pathway is the "best" of the fermentative pathways. Since its only product is lactic acid, organisms that use it are called *homolactic*.

The Entner-Doudoroff and heterofermentative pathways yield only one ATP molecule per molecule of glucose. The Entner-Doudoroff pathway is important in dairy fermentations. Lactose, or "milk sugar," is a disaccharide broken down to glucose and galactose during the fermentation. Milk contains 4.5 to 5.0% lactose and no other free carbohydrates. The EMP pathway cannot metabolize the galactose portion of lactose. Organisms that use the EMP excrete galactose, wasting one-half of their energy source. The "wasted" galactose can be taken up and metabolized by organisms using the Entner-Doudoroff pathway, but these organisms produce only one three-carbon intermediate and therefore one ATP molecule.

The heterofermentative pathway releases a carbon dioxide molecule, makes one three-carbon intermediate, and generates one ATP molecule. However, it allows organisms to use five-carbon sugars (pentoses) that other organisms cannot use. Thus, in environments where there are lots of pentoses but no hexoses (six-carbon sugars that can yield two ATP molecules), heterofermentative organisms have a big advantage.

Figure 18.2 Simplified catabolic pathways used in fermented foods.



Many factors determine which pathway an organism uses. These include the genes for specific enzymes, their regulation, and the available sugars. Bacteria in the genera *Lactococcus* and *Pediococcus* are homofermentative, producing only lactic acid. Some *Lactobacillus* species are homofermentative, some are heterofermentative, and some can use both pathways.

DAIRY FERMENTATIONS

Many microbes are used to make fermented milk products (Table 18.2). These are shown in Fig. 18.3. Note that lactococci normally form chains in liquid media rather than the clusters that they form when grown on solid media. Pediococci are often found as tetrads, whereas leuconostocs have a single division plane and form chains. The main bacteria used for acid production are the homofermentative lactic acid bacteria (LAB). Heterofermentative LAB contribute flavor compounds as described below. The homo- and heterofermentative bacteria are often paired to give the flavor and texture characteristic of a given cheese. Paired cultures can often acidify faster than single cultures.

Pasteurization kills most of the natural LAB and other bacteria present in raw milk. This facilitates the addition of LAB "starter cultures" that do not have to compete with large numbers of naturally occurring milk bacteria. Starter cultures allow manufacturers to control the rate and extent of acid development in the fermented food. This results in greater process control and a more consistent product.

Other bacteria, referred to as secondary microbiota, are added to some fermented products to influence flavor and texture. They typically comprise 10 to 20% of the total starter culture. *Leuconostoc* species and strains of *Lactococcus lactis* subsp. *lactis* that metabolize citric acid produce aroma compounds and carbon dioxide in cultured buttermilk and cheeses such as

Table 18.2 Microorganisms involved in the manufacture of cheeses and fermented milks

Product	Principal acid producer	Intentionally introduced secondary microbiota
Cheeses		
Colby, Cheddar, cottage, cream	*Lactococcus lactis* subsp. *cremoris* or *lactis*	None
Gouda, Edam, Havarti	*Lactococcus lactis* subsp. *cremoris* or *lactis*	*Leuconostoc* spp., *Lactococcus lactis* subsp. *lactis*
Brick, Limburger	*Lactococcus lactis* subsp. *cremoris* or *lactis*	*Geotrichum candidum, Brevibacterium linens, Micrococcus* spp.
Camembert	*Lactococcus lactis* subsp. *cremoris* or *lactis*	*Penicillium camemberti*, sometimes *Brevibacterium linens*
Blue	*Lactococcus lactis* subsp. *cremoris* or *lactis*	*Lactococcus lactis* subsp. *lactis*, *Penicillium roqueforti*
Mozzarella, provolone, Romano, Parmesan	*Streptococcus thermophilus, Lactobacillus delbrueckii* subsp. *bulgaricus, Lactobacillus helveticus*	None; animal lipases added to Romano for piquant or rancid flavor
Swiss	*Streptococcus thermophilus, Lactobacillus helveticus, Lactobacillus delbrueckii* subsp. *bulgaricus*	*Propionibacterium freudenreichii* subsp. *shermanii*
Fermented milks		
Yogurt	*Streptococcus thermophilus, Lactobacillus delbrueckii* subsp. *bulgaricus*	None
Buttermilk	*Lactococcus lactis* subsp. *cremoris* or *lactis*	*Leuconostoc* spp., *Lactococcus lactis* subsp. *lactis*
Sour cream	*Lactococcus lactis* subsp. *cremoris* or *lactis*	None

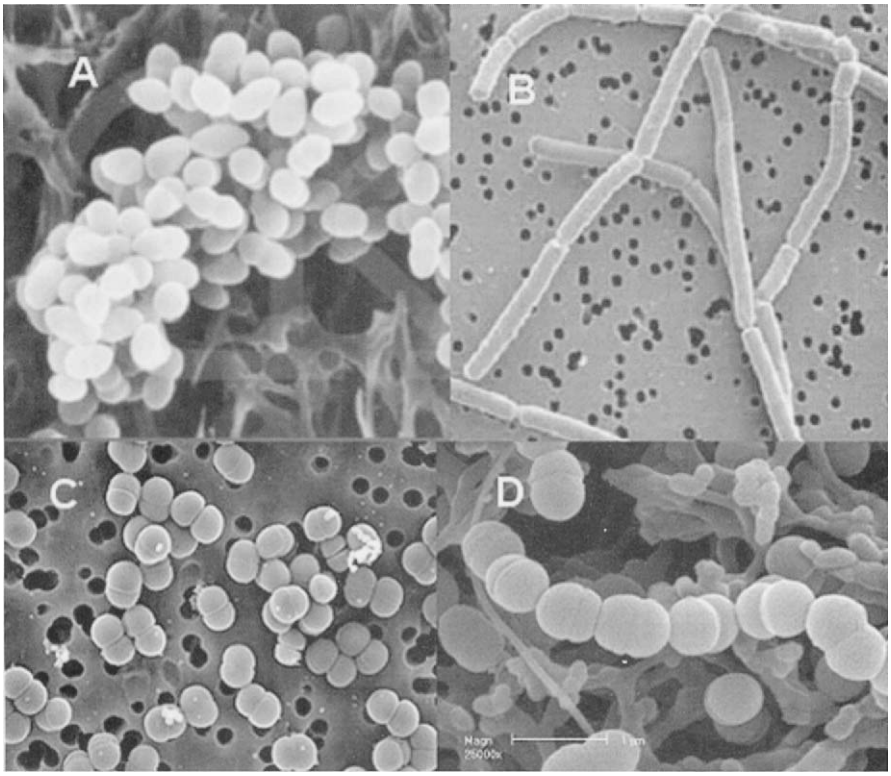


Figure 18.3 LAB associated with food fermentations. **(A)** *Lactococcus;* **(B)** *Lactobacillus;* **(C)** *Pediococcus;* **(D)** *Leuconostoc.* (B and C) Courtesy of the U.S. Department of Energy Joint Genome Institute. (D) Reprinted from G. Kaletunç, J. Lee, H. Alpas, and F. Bozoglu, *Appl. Environ. Microbiol.* **70:**1116–1122, 2004, with permission from the American Society for Microbiology.

Gouda, Edam, blue, and Havarti. Heterofermentative lactobacilli (*Lactobacillus brevis, Lactobacillus fermentum,* and *Lactobacillus kefiri*) are part of the varied microbiota (including several yeast species) that produce ethanol, carbon dioxide, and lactic acid in more-exotic cultured milks such as kefir and koumiss. This ethanol is not very volatile and does not contribute much flavor. However, when esterified to make volatile compounds, it has a major impact on taste. *Propionibacterium freudenreichii* subsp. *shermanii* is added to Swiss-type cheeses. Its scavenging metabolism converts L-lactic acid to propionic acid, acetic acid, and carbon dioxide. The carbon dioxide forms the "eyes" in Swiss cheese. Propionibacteria also ferment citric acid to glutamic acid, a natural flavor enhancer. *Penicillium roqueforti* is a secondary culture for the production of blue cheese, while *P. camemberti* is used to make Camembert cheese.

A generic scheme for making cheese is shown in Fig. 18.4. It is important to note that only high-quality milk should be used in the fermentation. The removal of sediment, by a process called clarification, is important for both esthetic and safety reasons. Pathogens like *Listeria monocytogenes* can be inside the leukocytes shed in the milk. Clarification removes these leukocytes. Other ingredients such as salt, seasoning, or even additional milk solids are added prior to heating, which kills any remaining bacteria. Homogenization stabilizes the product to prevent fat separation during the fermentation and to ensure an even distribution of all the product ingredients. In the actual fermentation step, the starter cultures rapidly (within 4

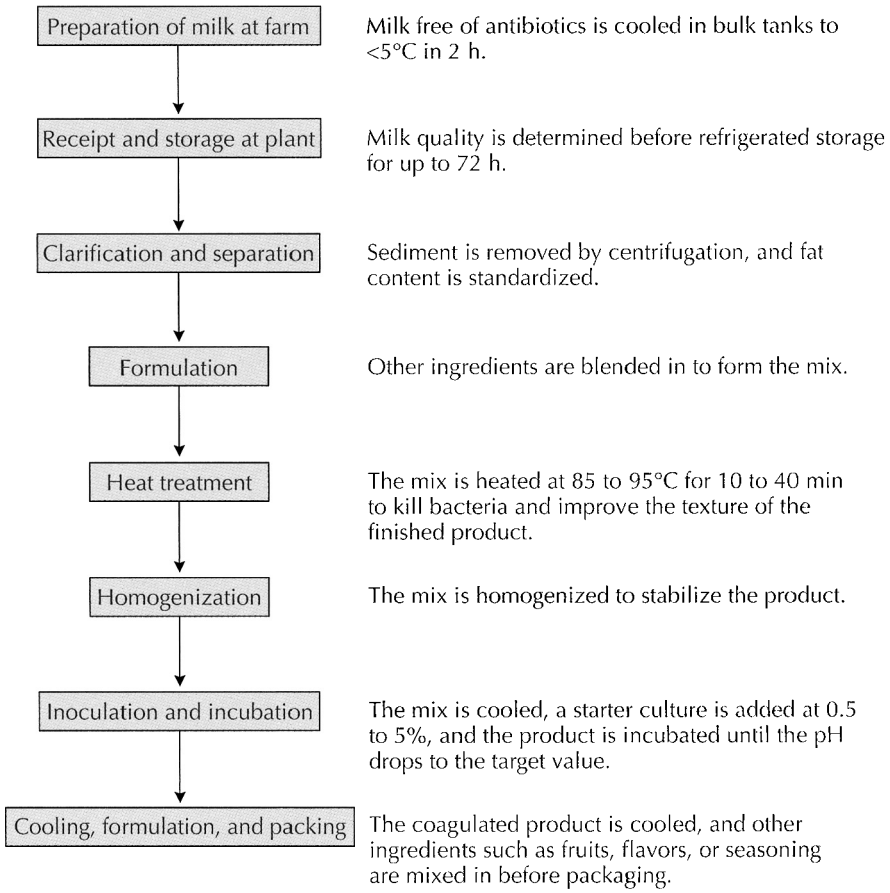


Figure 18.4 Generic scheme for making fermented dairy products.

to 8 h) lower the pH to <5.3 in cheese and <4.6 in fermented milk products. At this pH, only acid-tolerant bacteria can grow. After the fermentation, the product may be packaged, aged, or subjected to secondary fermentations for flavor development.

Yogurt has traditionally been made using *Streptococcus thermophilus* and *Lactobacillus delbrueckii* subsp. *bulgaricus*. *Lactobacillus acidophilus* and *Bifidobacterium* spp. are often added due to their popularity as probiotics (see chapter 24). The first step in yogurt manufacture is to concentrate the milk by 25% using a vacuum dehydrator. Milk solids (5%, wt/wt) are then added, and the mixture is heated to 90°C for 30 to 90 min. After being cooled to 45°C, the starter culture is added at 2% (vol/vol) and incubated for 3 to 5 h. The final product has a titratable acidity of 0.8 to 0.9% and about 10^9 organisms/g. These may die off during cold storage to a population of 10^6 organisms/g, the minimum required to make a "live and active culture" claim for the yogurt.

Starter Cultures

The key to the commercial success of fermented milk products is consistent and predictable rates of acid production (Table 18.3). The addition of defined bacterial cultures, i.e., starter cultures, to initiate the fermentation guarantees predictable acidification. Acidity has profound effects on moisture control, protein retention, mineral loss, protein hydration, and interac-

tions between protein molecules. These, in turn, determine the product's sensory characteristics. This consistency can be undermined if bacteriophages attack the starter cultures. The problem of bacteriophages and methods to counteract them are explained in chapter 22. The adulteration of milk by antibiotics can also wreak havoc with starter cultures. In countries where milk is routinely screened for antibiotics, this is no longer a big issue.

The flavor of cultured milk products is determined by microorganisms used as starter cultures and the secondary microbiota. With cultured milks and some cheeses, such as mozzarella, cream cheese, and cottage cheese, the short time from processing to consumption (1 day to 4 weeks) is not long enough for secondary organisms to generate "ripened" flavors. In fact, cottage cheese can be made by a "direct set" method, where acid is added directly to the milk, rather than fermenting it.

Table 18.3 Benefits of using starter cultures in manufacture of fermented food

Rapid acid production
Decreased rate of contamination
Production of flavor compounds
More consistent product
Greater process predictability

Production of Aroma Compounds

Lactic acid is the main end product in dairy fermentations. It contributes the acid taste, but not the aroma. The main aromas and flavors of fermented milks come from acetic acid, acetaldehyde, and diacetyl. In yogurt, these are formed by *Streptococcus thermophilus* and *Lactobacillus delbrueckii* subsp. *bulgaricus.* In buttermilk and some cheeses, *Leuconostoc* species and citrate-utilizing *Lactococcus lactis* subsp. *lactis* produce the aroma compounds.

Milk contains 0.15 to 0.2% citric acid. Because citric acid is converted to pyruvic acid without the need to regenerate nicotinamide adenine dinucleotide (NAD), virtually all of it can be converted to diacetyl, which has a buttery flavor. Citric acid is metabolized by *Leuconostoc* species, citrate-metabolizing *Lactococcus lactis* subsp. *lactis,* and facultative heterofermentative lactobacilli to make diacetyl, acetic acid, and carbon dioxide. The carbon dioxide causes the holes (eyes) in Gouda and Edam cheeses and the effervescent quality of buttermilk.

Several metabolic pathways form acetaldehyde. The enzyme threonine aldolase cleaves threonine to glycine and acetaldehyde. Acetaldehyde is also formed by citric acid-metabolizing *Lactococcus lactis* subsp. *lactis.* Fermented milks develop a "yogurt" or "green apple" flavor defect when the ratio of diacetyl to acetaldehyde is high. Excessive acetaldehyde in yogurt is caused by prolonged fermentation and is associated with high acid content. The use of *Leuconostoc* that converts acetaldehyde to ethanol prevents these excessive acetaldehyde levels. Rapid cooling and refrigerated storage also reduce off-flavor production.

Proteolysis is required for the growth of many LAB. They require anywhere from 6 to 16 amino acids to grow. Casein constitutes 80% of milk protein, and there are few free amino acids in milk. This makes the ability to break casein down to amino acids a very important trait for LAB. The casein can be hydrolyzed by extracellular proteases, by proteases in the cell membrane, or by intracellular proteases.

Proteolysis also generates flavors in ripened cheeses. The free amino acids and peptides produced by the extracellular proteases can have positive or negative effects in the cheese. A major negative effect of proteolysis is bitterness. This is caused by the breakdown of casein into hydrophobic peptides that are 3 to 27 amino acids long. Other bacterial peptidases can hydrolyze these to nonbitter peptides and amino acids. Therefore, the accumulation of bitter peptides depends on their relative rates of production and destruction.

Genetics of Lactic Acid Bacteria

Research on the genetics of LAB initially focused on plasmids and natural gene transfer in lactococci. The development of a detailed understanding of LAB genetics, natural gene transfer systems, and the development of tools required for genetic engineering have made this an important area of food microbiology.

In LAB, genes can be located on the chromosome, be on insertions to the chromosome (introns), move from one location to another (transposable elements), or be located on extrachromosomal (not on the chromosome) pieces of DNA called plasmids. Lactococci, lactobacilli, and *Streptococcus thermophilus* can gain new genes using transduction, the transfer of bacterial genetic material by a bacteriophage as described in chapter 22. LAB chromosomes are relatively small, in the range of 1.8 to 3.4 megabase pairs. Transposable genetic elements that move from one chromosomal site to another are important in some LAB. Insertion sequences, the simplest of transposable elements, also occur in many LAB. More-complex self-transmissible conjugal transposons code for nisin production, sucrose utilization, and a bacteriophage defense mechanism. These can be moved from one cell to another.

Plasmids are independently replicating extrachromosomal circular DNA molecules. They are of particular importance in lactococci, where they encode characteristics essential for dairy fermentations, including lactose metabolism, proteinase activity, oligopeptide transport, bacteriophage resistance, exopolysaccharide production, and citric acid utilization.

FERMENTED VEGETABLES

The ancients fermented vegetables to retain their nutritive value. Many different microorganisms are used for this purpose. LAB and yeasts are preferentially used in Europe and the Americas. Molds are used to ferment many Eastern foods. LAB are used extensively for biopreservation of vegetables. The market for pickled vegetable products is $2 billion in the United States alone.

Lactic acid fermentation of vegetables was practiced by the Chinese in prehistoric times. The oldest written evidence dates from the first century C.E., when Plinius described the preservation of white cabbage in earthen vessels. This results in the lactic acid fermentation of cabbage into sauerkraut. The development of heat sterilization and refrigeration systems has made fermentations less important as a preservation method in industrialized countries. Nonetheless, in developing countries, fermentation is still a critical means of food preservation.

Vegetable fermentations involve complex microbiological, biochemical, chemical, and physical reactions. Fermentations are also influenced by many external factors. These are classified into four groups: technological factors, ingredients, raw material quality, and native microbiota.

Table 18.4 shows a generic process for manufacturing fermented vegetables. It is important to start with sound vegetables. Bruised or damaged vegetables can have high levels of natural microbiota. Uniform sizing facilitates processing and packaging. Blanching or cooking may be required to inactivate enzymes or degas the vegetables. The placement of the vegetables in a fermentation vessel is straightforward. Starter cultures are not usually added because they cannot compete with the vegetables' natural

Table 18.4 Steps in a typical vegetable fermentation

1. Select vegetables that are sound, undamaged, uniformly sized, and at the proper ripeness.
2. Pretreat, for example, by peeling, blanching, or cooking.
3. Place whole, pierced, shredded, or sliced vegetables in fermentation vessels. These can hold from 100 liters to 100 tons. Starter cultures can also be added.
4. Completely cover the vegetables with brine. Seal the fermentation vessels to exclude oxygen and ensure anaerobic conditions.
5. Let the fermentation take its natural course. The fermentation time depends on the temperature, the type of product, and the bacteria present.
6. Distribute the final fermented product fresh, unpackaged, packaged, or pasteurized.

microbiota. Anaerobic conditions (i.e., the exclusion of oxygen) are required since this is a biochemical fermentation. Anaerobiosis is easily achieved by covering the fermentation vessel. Any residual dissolved oxygen is removed by the respiration of the vegetables and any aerobes present. The fermentation proceeds at whatever speed is dictated by the natural microbiota. The finished product can be distributed in a variety of forms. Because the botanical, physical, and chemical properties of various vegetables differ, the process described above must be tailored to specific products.

Ingredients and Additives Used during Fermentations

Salt is added to fermentations for many reasons. Salty flavors are essential to many fermented foods. The amount of salt used depends on the vegetable and on consumer preference. In sauerkraut production, salt enhances fluid release from shredded cabbage. It also helps create anaerobic conditions in fermentation vessels. In addition, salt has a selective effect on the vegetables' natural microbiota. Increasing amounts of salt favor the growth of LAB and inhibit the undesirable bacteria and fungi. In sauerkraut fermentations, heterofermentative LAB are favored by low salt concentration (~1%) and are greatly inhibited at 3%. Higher salt concentrations favor homofermentative species, accelerating the fermentation. Salt content below 0.8% often results in undesirable fermentation as well as in soft sauerkraut.

Other food additives are used in specific fermented vegetables and are regulated by law in most countries. The addition of ascorbic acid to sauerkraut prevents gray or brown discoloration. Citric acid or sulfur dioxide is also used for this purpose. Sorbic acid prevents the growth of yeasts, molds, and other microbes.

Sauerkraut Fermentation

The overall microbial population, as well as the population of LAB, changes during the course of a vegetable fermentation. The fermentation of cabbage, for example, has four stages:

1. Fermentation starts as soon as chopped cabbage is placed into vessels. In commercial processing, these vessels can contain 100 tons of chopped cabbage. Salt is added to an equilibrium concentration of 2%. The cabbage is tightly packed to form a bowl at the top of the vessel. To create an anaerobic environment, the cabbage is covered with a plastic sheet which is then filled with water. Under the anaerobic conditions, the number of strictly aerobic bacteria decreases,

while facultatively anaerobic enterobacteria grow for the first 2 or 3 days. During this period, dissolved oxygen is consumed by the microorganisms and by plant respiration. The formation of lactic, acetic, formic, and succinic acids lowers the pH. Carbon dioxide may generate foam.

2. During the second stage of fermentation, non-LAB are suppressed by anaerobic conditions and are overgrown by the LAB. Cabbage typically contains 4 to 5% sugar, with about half glucose and half fructose. The fermentation of these sugars to lactic acid is initiated by heterofermentative LAB, namely, *Leuconostoc mesenteroides*, followed by *Lactobacillus brevis*. This succession of microorganisms is complete after 3 to 6 days, and the lactic acid concentration increases to ~1%.
3. The third stage of fermentation is dominated by homofermentative LAB. Their growth is favored by the complete lack of oxygen, low pH, and high salt content. The homofermentative LAB increase the total acid content to 1.5 to 2.0%. Most sauerkraut is pasteurized when it reaches pH 3.8 to 4.1.
4. The fourth stage of fermentation is optional. For economic reasons, sauerkraut may be stored in the fermentation vessel for up to 1 year. During the storage period, *Lactobacillus brevis* and some heterofermentative species metabolize the pentoses released by the breakdown of cell walls. The acid content may increase to 2.5%.

Pickle Fermentation

Pickles are made from cucumbers. Although more than 40 flavor compounds have been isolated from pickles, most of them originate from the cucumber. Indeed, pickles made by direct acidification outsell those made by fermentation. Much dill pickle production provides the vegetable component sold in the food service restaurants that feed the fast food nation (unless they hold the pickles and hold the lettuce).

Commercial cucumber fermentations are conducted outdoors in vessels that contain 8 to 10,000 gallons of product. Starter cultures are rarely used. After the cucumbers are covered with brine, the vessel is sealed with a wooden headboard. Cucumbers contain glucose and fructose at about 1% each; these undergo homofermentation, typically by naturally occurring *Lactobacillus plantarum*. The final pH of ~3.7 is well below the regulatory cutoff for high-acid foods (i.e., pH 4.6). This level of acid gives a 5-log reduction of pathogens and ensures product safety. Like sauerkraut, pickles can be stored in their vessels for up to 1 year.

"Sweet" pickles are fermented in a similar fashion with small cucumbers. After the fermentation, the cucumbers are washed and then soaked in 25 to 30% sugar.

The carbon dioxide produced by cucumber respiration or excessive growth of heterofermentative *Leuconostoc mesenteroides* can accumulate in the pickles and cause "floaters," also known as "bloaters." Bloater formation is a serious problem in cucumber fermentation. The damage is worse with larger cucumbers, at higher fermentation temperatures, and when the dissolved carbon dioxide concentration in the brine is high. The brine's carbon dioxide content can be reduced by limiting growth of heterofermentative LAB and by removing carbon dioxide from the brine by purging with nitrogen or other gases.

MEAT FERMENTATIONS

Fermented meats are not as popular as fermented vegetable or dairy products but still constitute a major class of fermented food that have an economic value in the billions of dollars. The natural microbiota of meat is gram-negative, aerobic putrefying spoilage bacteria. Fermentation by LAB prevents spoilage and turns the raw meat into a totally different product. The most common types of fermented meats are dry and semidry fermented sausages.

The U.S. Department of Agriculture (USDA) Food Safety Inspection Service (FSIS) requires that shelf-stable sausages must contain nitrite and curing agents, be fermented to pH <5.0, and have a moisture:protein ratio of ≤3.1:1.0. The production of such a sausage is straightforward (Table 18.5). Grinding the meat reduces its particle size. It is then blended with nitrites, curing agents, spices, a fermentable carbohydrate, and sometimes glucono-delta-lactone. The nitrite inhibits *Clostridium botulinum,* contributes to the cured meat taste, and converts myoglobin to nitrosomyoglobin (the pink color of cured meat). Fermentable sugars are added because there is not enough glucose in meat to produce much acid. Glucono-delta-lactone is converted to gluconic acid. This is converted by LAB to lactic and acetic acids and accelerates acidification. The starter cultures (10^7 colony-forming units [CFU]/g) are also added at this point. The mixture is stuffed into oxygen-impermeable casing to ensure an anaerobic fermentation. The sausage links are incubated to promote microbial growth. The starter cultures decrease the pH from 5.6 to 4.8 within 8 h. The lactic acid coagulates the protein. This facilitates drying. The finished sausage is sometimes given a heat treatment.

There are several ways to start a meat fermentation. The native bacteria of lactobacilli, coagulase-negative staphylococci, micrococci, and yeast can be used. Unfortunately, because their presence in a given lot of meat is variable, the use of indigenous bacteria makes process control difficult. A *slightly* more sophisticated approach is using some of the last batch to inoculate the next batch. Defined starter cultures are relatively new to the meat industry and represent the third way to start a fermentation. Attempts to use the native meat LAB were unsuccessful because they do not tolerate the freeze-drying process used to make starter cultures. The next step in the starter culture progression was the use of freeze-dried *Pediococcus acidilactici,* an organism not normally associated with meat. However, it is homofermentative, can tolerate 6% salt, and is not proteolytic or lipolytic. It also has the advantage of growing at 43 to 50°C, a temperature that precludes the growth of pathogens. However, freeze-dried cultures were hard

Table 18.5 Steps in the manufacture of fermented sausages

1. Reduce particle size of high-quality meat by grinding.
2. Add curing salts (nitrite at <150 parts per million), 3% glucose as a fermentable sugar, spices, and starter culture at ~10^7 CFU/g.
3. Blend.
4. Vacuum stuff into casing.
5. Incubate (also referred to as "ripen").
6. Heat to inactivate starter culture and pathogens.
7. Age (dry).

to dissolve and had long lag times, so in the 1980s, manufacturers started using frozen cultures. When inoculated into the meat at 10^7 CFU/g, they shorten the lag time and acidify very rapidly.

Summary

- Fermentation is an incomplete oxidation of sugars in the absence of oxygen that uses an internal organic compound as an electron acceptor.
- "Fermentation" can also mean "bioprocess."
- Fermentative metabolism yields only 1 or 2 ATP molecules per molecule of glucose, much less than oxidative metabolism.
- Homofermentative bacteria make only lactic acid.
- Heterofermentative bacteria make lactic acid, ethanol, carbon dioxide, and acetic acid.
- Homofermentative bacteria are used to make acid, whereas heterofermentative bacteria contribute flavor.
- Starter cultures increase the speed and consistency of fermentations.
- LAB have several methods for transferring genes.
- Plasmids carry the genes for many traits important to fermentations.
- Vegetable fermentations rely on indigenous microbiota.

Suggested reading

Breidt, F., Jr., R. F. McFeeters, and I. Díaz-Muñiz. 2007. Fermented vegetables, p. 783–794. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

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Ricke, C. S., I. Z. Diaz, and J. T. Keeton. 2007. Fermented meat, poultry, and fish products, p. 795–816. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Salminen, S., and A. von Wright (ed.). 1998. *Lactic Acid Bacteria.* Marcel Dekker, Inc., New York, NY.

Questions for critical thought

1. How is the saying "In the land of the blind, the one-eyed man is king" relevant to energy production by fermentative metabolism? Can you think of another folk saying that captures the same concept?
2. If you were a *Lactobacillus* species able to use pentoses and hexoses and landed in an environment with both, which would you use? Why?
3. What are three general usages of the word "fermentation"?
4. Why does pH have profound effects on moisture control during cheese manufacture, retention of coagulants, and hydration of proteins?
5. Why are commercial starter cultures rarely used for pickle fermentation? If you wanted to revolutionize the pickle industry by using starter cultures, what would you do?

6. The year is 2017, and you are an associate professor at Big State University. The Peanut Producers and Packers want to develop a fermented product that will add value and increase the demand for peanuts. The Peanut Producers and Packers will allocate $300,000 to this project and have written to several universities requesting one-page research proposals. You would like the money. Write a proposal.
7. Why is the oxidation of ethanol and lactic acid referred to as "scavenging metabolism"?
8. Identify the microorganisms associated with each component of the sandwich shown in Fig. 18.1.

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Yeast-Based and Other Fermentations

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Fermentations That Use Yeast
Bread
Beer
Wine
Vinegar Fermentation
Cocoa and Coffee Fermentations
Cocoa
Coffee
Fermented Foods of Non-Western Societies
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- appreciate the role of yeast in a variety of fermentations
- relate the steps of bread making to its fermentation and final characteristics
- describe the biochemical basis of beer and wine fermentations
- identify the similarities between beer and wine making
- understand the role of different ingredients and process steps in the production of beer
- describe the differences and similarities in the production of red and white wines
- put the evolution of vinegar-processing technologies into a historical context
- describe the role of fermentation in the production of cocoa, coffee, and indigenous fermentations

INTRODUCTION

Enthusiasm is the yeast that makes your hopes shine to the stars.

HENRY FORD

Yeasts have long been associated with raising, fermentation, carbonation, and transformational abilities. This chapter examines these abilities of yeast in the context of bread, beer, and wine. Bread uses the carbon dioxide produced by the yeast to leaven, or raise it. This fermentation takes some time, so those in a hurry (e.g., the exodus of the Israelites from Egypt) eat unleavened bread. The production of beer is more complex and involves several steps. Enzymes must be made, and they must break down starch to fermentable carbohydrates before the carbohydrates can be fermented to alcohol. Since wine grapes contain simple sugars that are easily fermented, wine making is in some ways simpler than beer making, but in other ways, due to the complexity of grapes and their chemistry, more complex. Vinegar production is in theory very easy, since it involves only a single oxidation step. In reality, getting that oxygen into the fermenting liquid can be quite challenging. Finally, the microbiology of some non-Western fermented food is briefly discussed.

FERMENTATIONS THAT USE YEAST

Bread

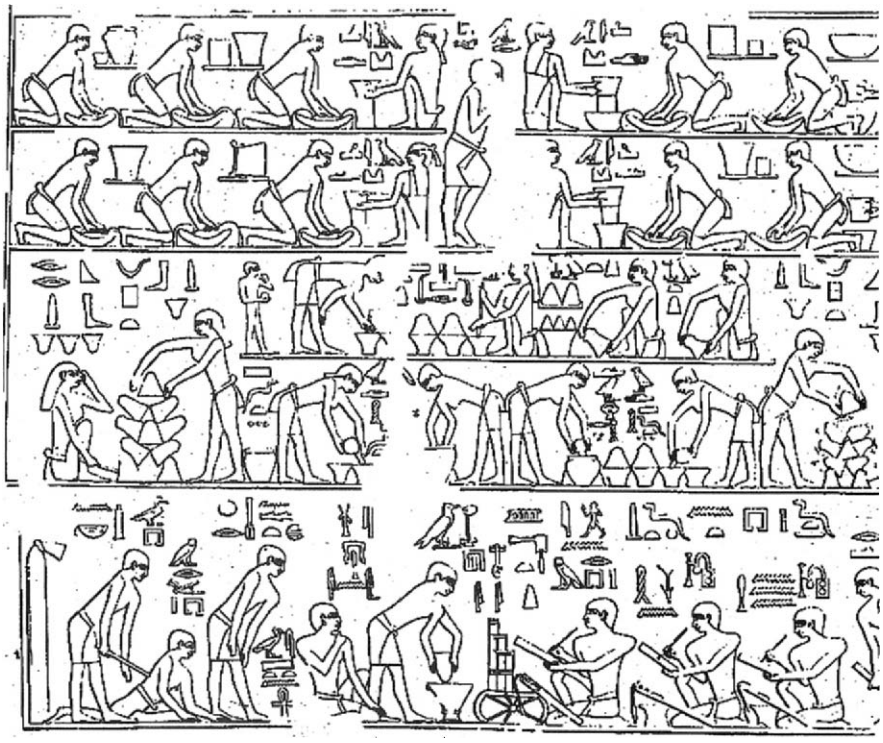
He who has no bread has no authority.

TURKISH PROVERB (ask Marie Antoinette—cake is not a good substitute for bread!)

As ancient societies became more sophisticated, their cereal food progressed from porridges and gruels to unleavened flat breads, and finally to breads leavened with yeast. This occurred as early as 2700 B.C.E., as evidenced by archaeological remains of Egyptian baking ovens (Fig. 19.1). In those early times, bakeries were often attached to breweries so that the yeast by-product of brewing (*Saccharomyces cerevisiae*) could be used in bread making. Today, specialized strains of yeast are used for each purpose.

The first step in bread making is to mix flour, sugar (as a fermentable carbohydrate), fat (for texture), salt, and other ingredients. The yeast *Saccharomyces cerevisiae* (baker's yeast) is added in the form of dried powder, block, or cream at 1 to 6% on a weight basis per weight of flour. The yeast's main role in bread production is to make the carbon dioxide that makes the bread rise. However, the yeast also produces amylases that break down starch to the more fermentable glucose. Water is added, and the dough is kneaded so that gluten protein in the flour stretches and the dough forms a viscoelastic (i.e., both viscous and elastic) mass. The bread is fermented or "proofed" at 28 to 32°C for several hours. The dough is then remixed (or "punched" in home usage) to evenly distribute the gas cells, portioned into

Figure 19.1 Scenes of Egyptian bread making from an excavation by Georg Steindorff. Note the kneading of the bread in the first two rows and the filling of conical bread molds in the next two rows. The finished bread is removed from the molds in the bottom panel. The scribes may be early food microbiologists documenting the process.



loaves, and fermented again so that the loaves double in volume. During the baking at around 200°C for ~30 min, carbon dioxide expansion causes a 40% increase in bread volume, which is captured as the protein "sets" by denaturation.

For sourdough breads, a stable mixture of heterofermentative lactic acid bacteria (LAB) is also added, at levels of 10^7 to 10^9 colony-forming units (CFU)/g. The *Lactobacillus* species, primarily *L. sanfranciscensis*, add a characteristic "bite" to the bread, improve texture, and prevent spoilage. When sourdough breads are made by artisans, the inoculum is usually made by using some of the last batch. The sequential transfer of the cultures can propagate the starter culture for decades.

Beer

Beer is proof that God loves us and wants us to be happy.

BENJAMIN FRANKLIN

Beer production, using methods remarkably similar to those we use today, has been traced to ancient times (Box 19.1). Beer making has reached tremendous proportions in our days. In 2004, $331 billion worth of beer was sold in the United States alone. The essential ingredients of beer are hops (for flavor and antimicrobial activity), yeast (to produce alcohol and carbon dioxide), water (for obvious reasons), and malt (to provide the fermentable carbohydrate). German purity laws forbid the use of any other ingredients. Other countries (including the United States) allow less expensive cereals, grains, or corn to be used as "adjuncts" to augment or replace the malt as the source of the fermentable carbohydrates.

Authors' note

This statement applies to those of us who are at least 21 years old. God's love is manifest to minors as ice cream.

The brewing process consists of malting, mashing, wort boiling, fermentation, and postfermentation treatments (Fig. 19.2). The yeast cannot ferment the starch in the barley, so it is malted and mashed. During malting, the grain is steeped in water for 24 to 48 h to induce germination. This breaks down the cell wall and protein matrix that contains the starch granules and liberates amylases. During mashing, the amylases break the starch into fermentable carbohydrates. At the end of this process, the liquid malt is almost ready to ferment. Herbs, usually hops, are added at this point for flavor and antimicrobial activity. Different hops give different beers their different distinctive flavors. The mixture of hops and liquid malt is boiled and then cooled to ~20°C before adding the yeast.

Different species of yeast are used to produce different types of beer. *S. cerevisiae*, which grows on top of the fermentation mix, is used for ales. *Saccharomyces carlsbergensis*, which settles to the bottom, is used for lagers. In both cases, the inoculum level is quite high (~10^7/ml) and increases eightfold over the course of the fermentation, i.e., growth consists of three doublings. The yeast from one fermentation is used to inoculate the next batch for 10 to 15 fermentations. Then, a new defined inoculum from a standard source is reintroduced. The yeast uses the various sugars in the wort sequentially, with simple sugars being fermented first and maltotriose relatively later. Ethanol production continues after the yeast stops growing and flavor compounds, such as aldehydes and higher alcohols, are produced.

The postfermentation processing of beer includes aging to remove the "green" flavors caused by diacetyl and acetaldehyde. These compounds are present in low concentrations but have a low flavor threshold (i.e., humans can taste them at very low concentrations) and impact taste adversely. Aging in casks, storage in the presence of yeast at 15°C, and avoiding exposure to oxygen help remove these flavor defects. Most beers are

BOX 19.1

The Hymn to Ninkasi shows that the coproduction of bread and beer predates the 19th century B.C.E.

The Hymn to Ninkasi	Contemporary Process
Borne of the flowing water, Tenderly cared for by the Ninhursag, Borne of the flowing water, Tenderly cared for by the Ninhursag,	**Bread**
You are the one who handles the dough [and] with a big shovel, Mixing in a pit, the bappir with sweet aromatics, Ninkasi, you are the one who handles the dough [and] with a big shovel, Mixing in a pit, the bappir with [date]-honey,	Preparing dough for baking
You are the one who bakes the bappir in the big oven, Puts in order the piles of hulled grains, Ninkasi, you are the one who bakes the bappir in the big oven, Puts in order the piles of hulled grains,	Baking the bread
You are the one who waters the malt set on the ground, The noble dogs keep away even the potentates, Ninkasi, you are the one who waters the malt set on the ground, The noble dogs keep away even the potentates,	**Beer**
You are the one who soaks the malt in a jar, The waves rise, the waves fall. Ninkasi, you are the one who soaks the malt in a jar, The waves rise, the waves fall.	Malting
You are the one who spreads the cooked mash on large reed mats, Coolness overcomes, Ninkasi, you are the one who spreads the cooked mash on large reed mats, Coolness overcomes,	Mashing
You are the one who holds with both hands the great sweet wort, Brewing [it] with honey [and] wine (You the sweet wort to the vessel) Ninkasi, (. . .) (You the sweet wort to the vessel)	Making the wort
The filtering vat, which makes a pleasant sound, You place appropriately on a large collector vat. Ninkasi, the filtering vat, which makes a pleasant sound, You place appropriately on a large collector vat.	Brewing
When you pour out the filtered beer of the collector vat, It is [like] the onrush of Tigris and Euphrates. Ninkasi, you are the one who pours out the filtered beer of the collector vat, It is [like] the onrush of Tigris and Euphrates.	Filtering Bottling

Translation by Miguel Civil

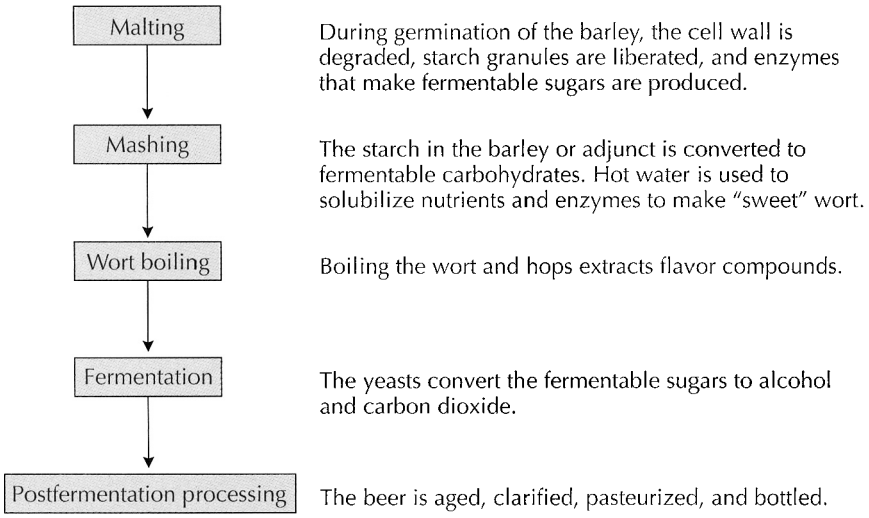


Figure 19.2 Schematic for making beer. Worldwide consumption tops 18 billion gallons per year. Adapted from I. Campbell, p. 735–745, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

filtered until clear, although the cloudiness of boutique beers is part of their charm. Filter aids, such as cellulose fiber or pumice, enhance the removal of the yeast by filtration. Membrane filtration can be used as a means of "cold pasteurization." This retains the beer's "draft" taste and avoids the energy expense of heat pasteurization. For heat pasteurization, the beer is held for 5 to 30 min at 60°C to kill any yeast which remains after filtration. The pasteurized beer is then dispensed into sanitized bottles. Draft beer, which is dispensed into kegs, is not heated and must be refrigerated during shipping and storage.

Wine

> *No nation is drunken where wine is cheap; and none sober, where the dearness of wine substitutes ardent spirits as the common beverage. It is, in truth, the only antidote to the bane of whiskey.*
>
> THOMAS JEFFERSON

Wine making also started in ancient times, probably when the natural yeasts contaminated grape juice and fermented it to an alcoholic beverage. It was not until the late 19th century that the French microbiologist Louis Pasteur discovered the role of microbes in this fermentation. He also discovered the role of bacteria in wine spoilage. Pasteurization was invented to kill these bacteria in wine. Only later was it applied to milk. France remains the world's biggest producer of wine, followed by Italy, Spain, and the United States. Together, these four countries produce more than 15 million tons of wine per year.

Figure 19.3 outlines the wine-making process. Many important aspects of wine making come before the actual fermentation. The most obvious is the grapes. Different varieties of grape are used for white and red wines. Riesling, chardonnay, and sauvignon blanc are among the grape varieties used for white wines. Merlot, pinot noir, and shiraz can be used to make red wines. The sugar-to-acid ratios, climate, soil, vine age, and other factors contribute to the flavor of the grape. The flavor of the grape, of course, is transmitted to the wine. The crushing and pretreatments are different for

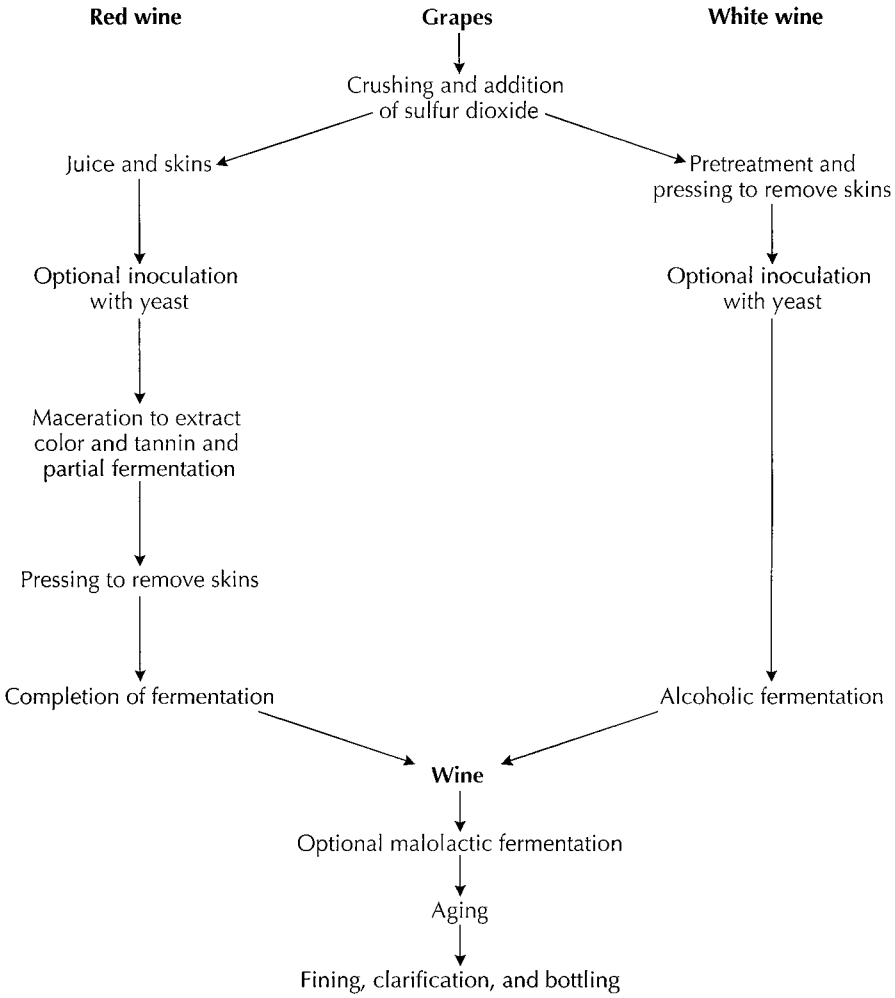


Figure 19.3 Schematic for making wines. Five hundred ninety-five million gallons of wine, worth $21 billion, was consumed in the United States in 2002. Adapted from G. H. Fleet, p. 747–772, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

red and white wines. For white wines, the juice is drained away from the skins immediately after crushing. The juice is then clarified and sent to the fermentation tank. For red wines, the juice and skins go directly to the fermentor. The skins float, and the juice is pumped over them to extract the purple and red anthocyanins and phenolics that give red wines their color, their astringency, and possibly their health benefit. After the fermentation has started, the liquid is pressed through the skins into a fermentation tank.

Grapes can be fermented in barrels, but large stainless steel tanks are more commonly used for mass-produced wines. To inoculate or not to inoculate grape juice with a defined inoculum of yeast is a subject of ongoing debate. As with other fermentations, using a defined inoculum gives a more consistent product and a faster process. *S. cerevisiae* or *Saccharomyces bayanus* is used at levels of 10^6 to 10^7 CFU/g. However, *S. cerevisiae* does not naturally occur at high levels on grapes. If the wine is not inoculated and native yeasts are used, the process is less predictable, but the product can be extraordinarily good as well as very bad. These native yeasts can come from the grapes or the equipment in the winery. The native grape yeasts that cause wine fermentations are mostly *Kloeckera* and *Hansenia* species,

with some *Candida, Pichia,* and *Hansenula* species. Although these yeasts occur in low numbers (~10 to 1,000 CFU/g) on immature fruit, the population increases to 10^4 to 10^6 CFU/g as the grapes ripen.

The fermentation of white wines usually takes 1 to 2 weeks at 10 to 18°C. The relatively low temperature retains important volatile flavors. Red wines are fermented for a shorter period (~7 days) at a higher temperature (20 to 30°C) to extract the red color. As with the growth of other microorganisms, yeast growth is influenced by intrinsic and extrinsic factors of the grape juice. These include the pH, oxygen concentration, sugar concentration, temperature, additional ingredients, and other indigenous microbiota. The fermentation ends when glucose and fructose are completely utilized. In addition to ethanol and carbon dioxide, the fermentation can produce glycerol (which smoothens the taste and imparts viscosity) and higher-alcohol esters and aldehydes as flavor compounds. The production of succinic and acetic acids causes wine defects.

Malic acid occurs naturally in some grapes and can give wine a very sour taste. In fact, it is used to manufacture SweetTarts and Jolly Rancher candies. Malic acid is a by-product of the tricarboxylic acid cycle and generally decreases as the fruit ripens. Wines with too much malic acid are bitter. The malolactic fermentation is used to decrease the acidity of these wines. This secondary fermentation starts 1 to 3 weeks after the alcohol fermentation is finished. It is carried out by *Oenococcus oeni* (formerly classified as a leuconostoc), which decarboxylates malic acid to lactic acid.

$$\underset{\text{Malic acid}}{\text{HOOC-C(OH)-CH}_2\text{-COOH}} \rightarrow \underset{\text{Lactic acid}}{\text{HOOC-C(OH)-CH}_3} + \underset{\text{Carbon dioxide}}{\text{CO}_2}$$

In addition to creating a mellower flavor, the malolactic fermentation increases the pH by 0.3 to 0.5 units.

The postfermentation processes for wines are straightforward. Red wines are stored in oak barrels for 1 to 2 years for flavor development. Some of the flavor comes from the barrels, which may be made with smoked woods, resins, or other flavor-inducing mechanisms. White wines are not usually aged. Potassium sorbate at 100 to 200 μg/ml can be added to red or white wine to control yeast spoilage. The wine is then filtered to clarity and bottled.

VINEGAR FERMENTATION

Men are like wine—some turn to vinegar, but the best improve with age.
POPE JOHN XXIII

The vinegar fermentation is almost as ancient as that of wine. It had to come after the discovery of wine, since vinegar is made from wine (or these days, from other alcoholic substrates). Vinegar was undoubtedly discovered accidentally when a *Gluconobacter* or *Acetobacter* species contaminated wine and turned it to vinegar. This turned out to be not such a bad thing, since the vinegar could then be used to preserve other foods.

Vinegar is made by oxidizing ethanol to acetic acid:

$$\text{CH}_3\text{CH}_2\text{OH} + \text{O}_2 \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$$

This occurs in a two-step bioprocess. The first step in vinegar production is the alcoholic fermentation by yeast to produce alcohol. Ethanol from wine or hard cider is frequently used, but vinegar can be made by the fermentation of almost any fruit or starchy material. In the second step, the

Figure 19.4 A large-scale fermentor used in the modern production of vinegar.

ethanol is oxidized to acetic acid. Since the *Gluconobacter oxidans* used in making vinegar is a strict aerobe and the ethanol must be *oxidized* to acetic acid, oxygen transfer is the rate-limiting step in the process. The oldest processors simply left wine in wooden barrels exposed to the air, inoculated it with *Gluconobacter,* and waited. In several weeks, or perhaps months, the wine became vinegar. The "trickling" fermentor described below radically reduced this time to a day or two by increasing the oxygen transfer rate. The trickling fermentor is a large wooden box with slatted sides to allow oxygenation. The box is filled with wood shavings, on which *Gluconobacter* grows as biofilms. A rotating sprayer at the top of the chamber sprays the alcoholic liquid on the top of the chips and it trickles down over the immobilized *Gluconobacter,* being converted to acetic acid in the process. The large surface area of the bacteria on the wood shavings exposed to oxygen makes this method very efficient. Its demise occurred not due to technical limitations, but because the craftsmen who could build trickling fermentors retired or died. The trickling fermentor is probably the first bioreactor using immobilized cell technology. The most modern method of vinegar production is the submerged culture reactor (Fig. 19.4). By pumping oxygen through these very large fermentors at rates equal to the volume of the fermentor, large quantities of vinegar can be made in a very short time.

COCOA AND COFFEE FERMENTATIONS

Cocoa

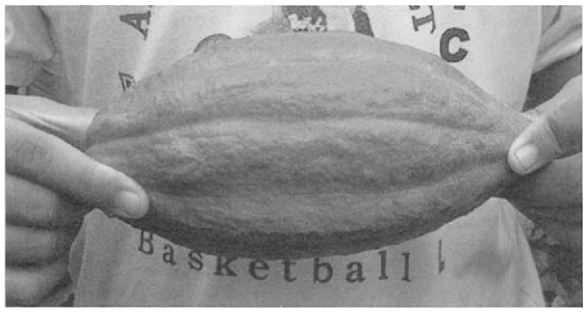
Cocoa? Cocoa! Damn miserable puny stuff, fit for kittens and unwashed boys. Did Shakespeare drink cocoa?

SHIRLEY JACKSON

Good cocoa flavor is the result of a complex fermentation involving microbial succession. Cocoa is made from the plant *Theobroma cacao.* It is native to South America and grows 15° north and south of the equator. The cacao "beans" or seeds are contained in pods about the size of a coconut (Fig. 19.5). The pods are filled with mucilage (a thick gooey substance) and seeds, much like a pumpkin. Without the fermentation, the cocoa seeds are bitter and astringent. To overcome this, the seeds and mucilage are removed from the pods, put into heaps or containers, and fermented for 2 to 8 days. The material is turned periodically to introduce oxygen. The acetic acid and heat generated by the fermentation keep the seeds from germinating.

The mucilage contains sucrose and other sugars at a level of 10 to 12%. The mucilage also contains citric acid, which contributes to a pH of 3.4 to

Figure 19.5 A cocoa pod.

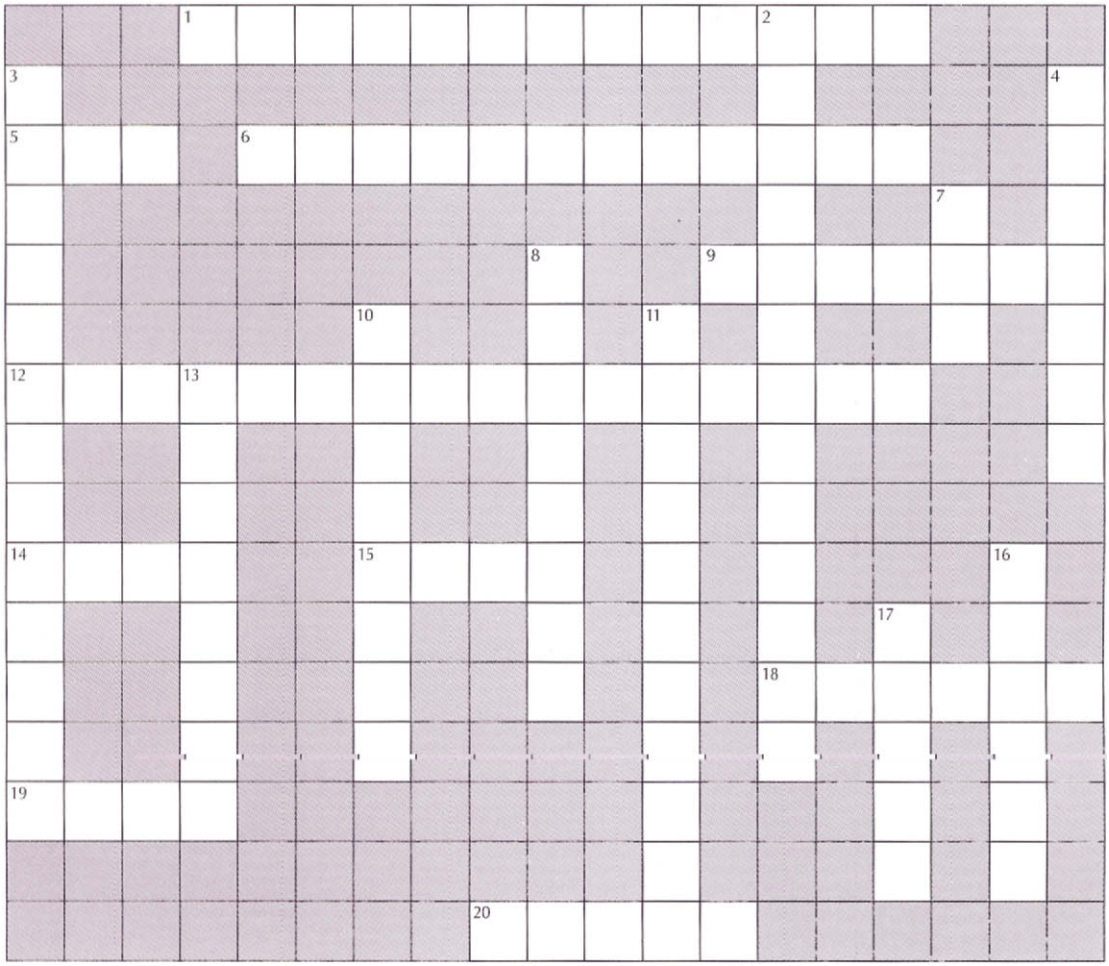


4.0. The fermentative microbes come from the environment, the plants, and the workers. During the first phase of the fermentation, which lasts 2 or 3 days, the yeasts are the most important microbes. They produce alcohols and aldehydes and create anaerobic conditions that favor the growth of LAB to levels as high as 10^7 CFU/ml. The LAB produce large amounts of hydrolytic enzymes, invertases, glycosidases, and proteases. The microbial population is primarily homofermentative but contains some heterofermentative LAB. The turning of the fermenting mass introduces air that favors acetic acid bacteria in the final stage of the fermentation. Levels of *Gluconobacter* species and *Acetobacter* species can reach 10^6 CFU/g. The fermentation provides flavor precursors that are converted to chocolate flavors during drying and roasting.

There are two main methods of fermenting the cocoa. In Ghana and other small-scale cocoa-producing countries, the seeds are put into heaps on the ground, covered with banana leaves, and turned by hand (Fig. 19.6). All of this is very labor-intensive but produces very high quality seed. These seeds yield the very best chocolate. The second fermentation method, favored on large plantations and estates, uses large boxes instead of heaps to contain the fermenting mass. These boxes range from 1 by 1 by 1 m to 7 by 5 by 1 m. These are frequently grouped or arranged in tiers to make turning easier. In both methods, a liquid fermentate called "sweats" is produced and is drained away from the beans. This makes drying easier. The larger scale of the box fermentation makes the seed fermentation quicker and cheaper than that of heap fermentation but yields a lower quality. What chocolate manufacturers would like, of course, is a fermentation that is quick and cheap like the box fermentation but that yields a high-quality seed like the heap fermentation. This will require considerable research on the microbial ecology of both systems.

The development of good cocoa flavor is still very much an art. The complexity of the microbial succession prevents the use of starter cultures. The flavors are also dictated by the genetic potential of the plant. While a good fermentation cannot upgrade an inferior product, a bad fermentation can ruin a superior cocoa.

Figure 19.6 The heap method of fermenting cocoa.



Across

1. Virus that attacks bacteria
5. Homofermentation yields 2, heterofermentation yields 1
6. A bioprocess is also called a _____
9. Used in cured meat for color, taste, and antimicrobial properties
12. Type of fermentation that makes only lactic acid
14. During Prohibition, amateur fermentation microbiologists were often the targets of a _____
15. Propionibacteria produce these in Swiss cheese
18. Fermentation can be described as life in the absence of _____
19. Analogous to the _____ planted to start the growth of a vegetable, starter cultures are used to _____ a new fermentation
20. Used to make beer, wine, and bread

Down

2. Fermented foods are microbially safe due to _____
3. Acids are made from fermentable _____
4. Oxidized alcohol
7. Abbreviation for citric
8. Heterofermentative, but not homofermentative, microbes use these
10. Their breakdown is responsible for bitter flavor in cheese
11. Genus consisting of only homofermentative species
13. Energy is made when compounds are _____
16. The liquid drained away in making chocolate
17. Bacteriophage cycle that breaks open the host

Table 19.1 Other regionally important food fermentations[a]

Product	Organism(s)	Description	Product
Fermentations based on yeasts and molds			
Soy sauce	*Aspergillus oryzae, Aspergillus sojae*, yeast, pediococci	Soy and roasted wheat are fermented at 35 to 40°C for 2–4 mo with 12–19% salt, pH 4.6–4.8, pasteurized at 70–80°C.	Flavoring agent
Miso	*A. oryzae* and secondary fermentation of yeasts and pediococci	Rice and/or soybean paste is fermented for 1 wk–1 yr at 25–30°C with 4–14% salt.	Soup base
Sake	Yeast, mold, and LAB	Rice is fermented by a mold called koji to liberate the sugar from the starch. This is then fermented by yeast to produce alcohol. Some types of sake also undergo a lactic acid fermentation.	Alcoholic beverage
Sufu	*Mucor* species, *Rhizopus chinensis, Actinomucor elegans*	Curd of soybean milk is pressed, cubed, boiled in brine, and inoculated. Then, it is aged for 1–12 mo.	Condiment
Tempeh	Molds and bacteria	Soybeans are soaked and inoculated with *Rhizopus oligosporus*. Incubation is for 24 h at 37°C. Bacteria are sometimes added as a prefermentation to lower pH or produce flavors.	Mold-penetrated and -covered cake yields a high-quality protein that can be deep-fried or used in soup.
Lao-chao	*Rhizopus oryzae, R. chinensis, Endomycopsis* spp.	Inoculated rice is incubated for 1 or 2 days at room temperature to yield a juicy, sweet, slightly alcoholic product.	Dessert
Oncom	*Neurospora intermedia, R. oligosporus*	Deoiled peanut cakes are inoculated and incubated at 25–30°C for 1 or 2 days. Surface becomes covered with colored conidia or spores.	Fermented peanut press-cake
Fermentations based on LAB			
Gari	LAB and yeast	Cassava is peeled, washed, and grated and then fermented for 2 days as water is pressed out.	Staple in stews, or eaten with cold sweet water
Kimchi	LAB	Natural fermentation of cabbage or radish taproot and garlic. A variety of seasonings can also be added.	Korean staple as sour carbonated vegetable
Poi	Initially lactobacilli and lactococci, followed by yeast	Corms of taro plant are ground, hydrated, and fermented by native microbiota for 1–3 days.	Eaten as main or side dish
Fermentations based on other bacteria			
Kefir	Bacteria and yeast	Cow or goat milk is inoculated with kefir grains. The kefir grains contain a variety of yeasts and bacteria, proteins, lipids, and carbohydrates. The lactose is fermented overnight at room temperature to yield a sour, carbonated, and slightly alcoholic beverage.	Beverage
Natto	*Bacillus natto (subtilis)*	Whole soybeans are fermented at 25–40°C for 12 h–2 wk.	Eaten as food

[a]Adapted from L. R. Beuchat, p. 701–720, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

Coffee

Coffee, the finest organic suspension ever devised

Star Trek: Voyager

Several species of *Coffea*, including *C. arabica*, *C. robusta*, and *C. canephora*, are grown in South America, Central America, Hawaii, Ethiopia, and India for use in the commercial production of coffee. It takes about 1 year from the time a tree flowers until it has a ripe coffee cherry. Although much smaller than cocoa pods, coffee cherries that contain the beans are surrounded by fleshy mucilage. The mucilage must be removed to liberate the beans. This can be done mechanically, by enzymes, by dehydration, or by fermentation.

The fermentation plays a much less critical role in coffee production than in cocoa production. Its main role is to liberate the beans. The fermentation contributes little to bean flavor or quality. When the beans are fermented, they are first mechanically depulped. The beans, covered with residual mucilage, are submerged in tanks of water. Native yeast, molds, LAB, and gram-negative bacteria ferment the mucilage to water-soluble products that can then be washed away. Since the main component of mucilage is pectin, the fermentation is dominated by a pectinolytic population. During the fermentation, which takes 12 to 60 h, the pH of the beans drops to ~3.7 from the pH of 5.4 to 6.4 characteristic of the native bean. The beans are then subjected to 10 to 25 days of sun drying characterized by an ill-defined microbial succession. The beans can then be roasted and ground.

Authors' note

It is hard to find the right "politically correct" adjective since many "traditional," "indigenous," and "non-Western" societies are more adept in the use of fermented foods than is mainstream American society.

FERMENTED FOODS OF NON-WESTERN SOCIETIES

There are dozens of fermented foods that play important roles throughout Asia, Africa, and South America. Fermented foods play very important roles in countries that lack refrigeration, energy-intensive methods of food

Figure 19.7 One of the authors (Thomas Montville) enjoying a typical Korean meal composed largely of fermented foods.

preservation, or the rudimentary technology required for an indigenous food industry. The fermentation may be used for preservation (as in the acidification of kimchi or gari), to improve nutritional value or health (as in the case of tempeh or kefir), or simply to create a pleasurable effect (as in the case of sake).

These fermentations have been loosely organized according to their main microbiotas in Table 19.1. The substrates for these fermentations are usually local crops that are starchy or have some naturally occurring fermentable carbohydrate. In almost all cases, native microbiotas serve as the inocula, although some, like tempeh and sake, are inoculated.

The frequent use of molds and fungi, usually associated with spoilage, may strike you as unnatural. But it is no more unnatural than the use of LAB, which spoil meat, to initiate dairy fermentations. LAB are important in many indigenous vegetable fermentations, where their acidity imparts a distinctive taste and serves as a preservative (Fig. 19.7). Many traditional fermentations use mixed, often uncharacterized, microbiotas.

Summary

- Beer and bread both rely on yeast for fermentation to produce carbon dioxide and ethanol.
- The production of beer involves malting to liberate hydrolytic enzymes, mashing to liberate fermentable carbohydrates, the actual fermentation, clarification, pasteurization, and bottling.
- The production of wine is technically simpler than that of beer, since the fermentable carbohydrates are naturally present in the grapes.
- A secondary fermentation by *Oenococcus oeni* can be used to remove malic acid from wines.
- Vinegar production is an oxidative fermentation which has used barrels, trickling wooden boxes, and large fermentors to overcome oxygenation as the rate-limiting step.
- Mucilage is fermented in coffee and cocoa fermentations to liberate the beans for further processing.
- Fermentation is much more important to cocoa quality than coffee quality.
- Indigenous fermentations use the substrates in local crops to preserve them and/or make new products with improved nutritional characteristics.

Suggested reading

Campbell, I. 2007. Beer, p. 851–862. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Fleet, G. H. 2007. Wine, p. 863–890. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Nout, M. J. R., P. K. Sarkar, and L. R. Beuchat. 2007. Indigenous fermented foods, p. 817–835. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Thompson, S. S., K. B. Miller, and A. S. Lopez. 2007. Cocoa and coffee, p. 837–850. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. How do the chemical end products of the yeast fermentation function in bread and beer?

2. Why is mashing required in the production of beer, but not wine?
3. What is an adjunct? Why is it used? Compare this to the use of adjunct professors at colleges and universities.
4. Why are fresh, defined inocula used to start beer fermentations after using the prior-batch inoculation method for the 10 to 15 prior batches?
5. Why is refrigerated storage required for draft beer but not bottled beer?
6. Research the differences between lager beer, ale, and stout. Briefly describe these. (If over 21, which do you prefer? Why?)
7. Trivia question: Benjamin Franklin's illegitimate son, Benjamin, shared his father's love of beer and wine. As governor of New Jersey, he signed the charter of what colonial college?
8. What are the pros and cons of using defined inocula in wine fermentations? If you used all of your savings to buy a winery, would you use natural microbiotas or defined inocula?
9. What is the terminal electron acceptor in the fermentation of wine to vinegar?
10. What common role does fermentation play in cocoa and coffee production?
11. Choose one of the indigenous fermentations shown in Table 19.1 and research it to write a one-page report which describes its history, microbiology, and culinary uses.
12. Use dimensional analysis and simplifying assumptions to estimate how many 1-liter bottles of wine can be made from the 15 million tons produced annually.
13. What is the mechanism by which vinegar preserves food?

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20 Spoilage Organisms

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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- gain knowledge about microorganisms responsible for spoilage of a wide range of food products
- discuss intrinsic mechanisms that inhibit spoilage of food
- understand the impact of food processing on microbial spoilage
- identify the sources of microorganisms responsible for spoilage of specific products
- discuss procedures that can be implemented to minimize contamination of raw materials

INTRODUCTION

The phone rings, and your friend from out of town asks if she can drop by for a visit. You go so far as to suggest that you will make dinner. You open the refrigerator to gather the necessary ingredients to make a delicious meal. Upon opening a jar of spaghetti sauce, you notice a greenish white fuzzy mass on the surface of the sauce. The luncheon meat that you remove to make antipasto has a green sheen and a strange odor, and it feels slimy. For a salad, you take out lettuce, but it has turned brown, and the peppers have large dark spots. In order to gather your thoughts, you decide to sit down and have a glass of milk. When pouring the milk, you notice that it is lumpy and a bit smelly. You start to realize that you should have paid more attention in the food microbiology course that you attended as an undergraduate.

A product (meat, dairy, fruit, or seafood) is considered spoiled if sensory changes make it unacceptable to the consumer. Factors associated with food spoilage include color defects or changes in texture, the development of off flavors or odors, slime, or any other characteristic that makes the food undesirable for consumption. While enzymatic activity within a food contributes to changes during storage, *organoleptically* detectable (i.e., detectable through a change in odor or color) spoilage is generally a result of decomposition and the formation of metabolites resulting from microbial growth. This chapter discusses spoilage of many foods, from meat to dairy to produce. The types of microorganisms involved, conditions conducive to spoilage, and defects associated with spoilage are covered.

MEAT, POULTRY, AND SEAFOOD PRODUCTS

Origin of Microflora in Meat

The levels of bacteria in muscle tissues of healthy live animals are extremely low. High numbers of bacteria are present on the hide, hair, and hooves of red-meat animals, as well as in the gastrointestinal tract. Microorganisms on the hide include bacteria, such as *Staphylococcus, Micrococcus,* and *Pseudomonas* species, and fungi, such as yeasts and molds, which are normally associated with skin microflora, as well as species contributed by fecal material and soil. The numbers and composition of this microflora are influenced by environmental conditions.

The majority of bacteria on a dressed red-meat carcass originate from the hide. During hide removal, bacteria are carried from the hide onto the underlying tissue with the initial incision. Unlike those of cattle and sheep, the skin of hogs is usually not removed but is scalded and left on the carcass. Recontamination can also occur during dehairing due to the presence of debris in dehairing machines. Contamination can occur if the intestinal tract is pierced or if fecal material is introduced from the rectum during the removal of the abdominal contents. Handling can result in cross-contamination of other carcasses. In addition to the hide and viscera, the processing environment, such as floors, walls, contact surfaces, knives, and workers' hands, can be a source of contamination in red meats.

Origin of Microflora in Poultry

The skin, feathers, and feet of poultry harbor microorganisms resident on the skin, as well as from litter and feces. Although present on the skin, psychrotrophic bacteria, consisting primarily of *Acinetobacter* and *Moraxella,* are primarily associated with the feathers. Contamination and cross-contamination with fecal material may occur during transportation of birds from growing houses to slaughter facilities and during processing, when the birds are hung and bled. Following processing, carcasses are chilled rapidly to limit the growth of microbes. Slush ice, continuous-immersion, spray, air, and carbon dioxide chilling systems have been utilized or proposed for this purpose. Potable water should be used in these systems, since bacteria present in untreated water can contribute to spoilage.

Origin of Microflora in Finfish

The numbers and composition of microflora on finfish are influenced by the environment from which the fish are taken, the season, and the conditions of harvesting, handling, and processing. Water temperature has a significant influence on the initial number and types of bacteria on the surface of the fish. Higher numbers of bacteria are generally present on fish from warm subtropical or tropical waters than on fish from colder waters. Fish taken from temperate waters harbor predominantly psychrotrophic bacteria, while mesophilic bacteria predominate on fish taken from tropical areas. Bacteria from the genera *Acinetobacter, Aeromonas, Cytophaga, Flavobacterium, Moraxella, Pseudomonas, Shewanella,* and *Vibrio* dominate on fish and shellfish taken from temperate waters, while *Bacillus,* coryneforms, and *Micrococcus* species frequently predominate on fish taken from subtropical and tropical waters. The initial microflora on fish is influenced by the method of harvesting. Trawled fish generally have higher microbial levels than those that are line caught. In trawling, the dragging of fish and debris along the ocean bottom stirs up mud that contaminates the fish. In addi-

tion, the compaction of fish in trawling nets may cause expression of intestinal contents, with subsequent contamination of the fish surface. A delay in chilling fish also enhances the possibility of rapid microbial growth. In some Scrombridae and Scomberesocidae fish (tuna, mackerel, and skipjack), this can lead to the generation of toxic levels of histamine by *Morganella* (*Proteus*) *morganii* and related gram-negative bacteria that produce histidine decarboxylase. *Photobacterium phosphoreum* may produce histamine during low-temperature storage of these fish.

Origin of Microflora in Shellfish

Unlike other crustacean shellfish (lobsters, crabs, or crayfish) that are kept alive until they are heat processed, shrimp die soon after harvesting. Decomposition begins soon after death and involves bacteria on the shrimp surface that originate from the marine environment or from contamination during handling and washing. Molluscan shellfish (oysters, clams, scallops, and mussels) are stationary filter feeders, and thus, their microfloras depend greatly on the quality of the water in which they reside, the quality of the wash water, and other factors. Bacteria, including *Pseudomonas* spp., *Shewanella putrefaciens, Acinetobacter,* and *Moraxella,* that spoil finfish also cause spoilage of shellfish.

Bacterial Attachment to Food Surfaces

Spoilage of meat, poultry, and seafood generally occurs as a result of the growth of bacteria that have colonized muscle surfaces. The first stage in colonization and growth involves the attachment of microbial cells to the muscle surface. Bacterial attachment to muscle surfaces involves two stages. The first is a loose, reversible sorption that may be related to van der Waals forces or other physicochemical factors. One of the factors that influence attachment at this point is the population of bacteria in the water film. The second stage consists of an irreversible attachment to surfaces involving the production of an extracellular polysaccharide layer known as a *glycocalyx.* Many factors can influence bacterial attachment, including surface characteristics, growth phase, temperature, and the motility of the bacteria. The development of microbial association with muscle foods is illustrated in Fig. 20.1.

Microbial Progression during Storage

The initial microflora of muscle foods is highly variable. It comes from the microorganisms resident in and on the live animal; environmental sources, such as vegetation, water, and soil; ingredients used in meat products; workers' hands; and contact surfaces in processing facilities. Most perishable meat, poultry, and seafood products are stored at refrigeration temperatures to prolong their shelf life. As microbial growth occurs during storage, the composition of the microflora is altered so that it is dominated by a few, or often a single, microbial species, usually of the genera *Pseudomonas, Lactobacillus, Moraxella,* and *Acinetobacter* or the species *Brochothrix thermosphacta* (Fig. 20.1). *Pseudomonas* species are able to compete successfully on aerobically stored refrigerated muscle foods, since they have a competitive growth rate, but *Moraxella* and *Acinetobacter* species are less capable of competing at refrigeration temperatures and lower pH. The growth of the aerobic spoilage microflora is suppressed during storage under vacuum and modified atmospheres. Under these conditions, lactic acid bacteria are favored because of their growth rate, their fermentative metabolism, and their ability

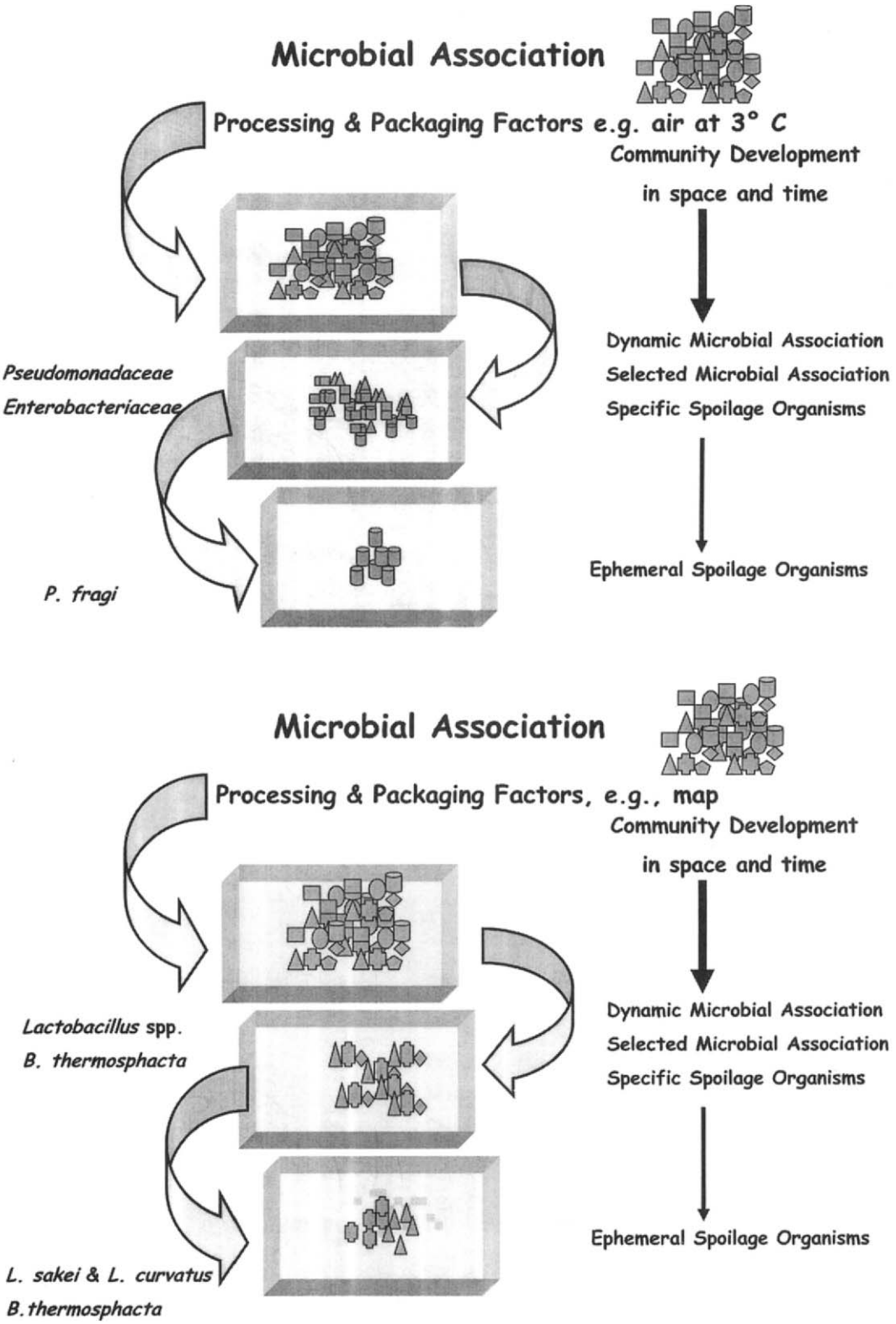


Figure 20.1 Microbial progression in muscle foods under various processing and packaging conditions. Reprinted from G.-J. E. Nychas, D. L. Marshall, and J. N. Sofos, p. 112, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

to grow at the pH range of meat. At higher pH, *B. thermosphacta* and *S. putrefaciens* may grow and contribute to spoilage. The water activity (a_w) of some types of processed meats is lowered by dehydration or the addition of solutes, such as salt or sugar. When low a_w restricts the growth of bacteria, growth of fungi may occur. Microbial growth does not occur on products with a_w values of <0.60.

Muscle Tissue as a Growth Medium

The composition (percent adipose, lean, and carbohydrate) of the meat influences the microbial growth and type of spoilage. Research suggests that spoilage defects in meat become evident when the number of spoilage bacteria on the surface reaches 10^7 colony-forming units (CFU)/cm^2. During aerobic spoilage, off odors are first detected when levels reach 10^7 CFU/cm^2. When numbers reach 10^8 CFU/cm^2, the muscle tissue surface begins to feel tacky, the first stage in slime formation.

Composition and Spoilage of Red Meats

The a_w of red-meat lean muscle tissue is 0.99, with a corresponding water content of 74 to 80%. The protein content may vary from 15 to 22% on a wet-weight basis. The lipid contents of intact red meats vary from 2.5 to 37%, and the carbohydrate compositions range from 0 to 1.2%. Glycolysis leads to the accumulation of lactic acid, and as a result, the pH of the muscle tissue decreases. The spoilage of meats stored at ambient temperature results from the growth of mesophiles, predominantly *Clostridium perfringens* and members of the family *Enterobacteriaceae.* Spoilage deep within muscle tissue, known as "sours" or "bone taint," has been attributed to a slow cooling of carcasses, resulting in the growth of anaerobic mesophiles thought to be already present in the muscle tissues. If the surfaces of whole carcasses or fresh meat cuts become dry, bacterial growth may be restricted and fungal spoilage may occur. *Thamnidium, Mucor,* and *Rhizopus* may produce a whiskery, airy, or cottony gray-to-black growth on beef due to the presence of mycelia. Black spot has been attributed to the growth of *Cladosporidium,* white spot has been attributed to the growth of *Sporotrichum* and *Chrysosporium,* and green patches have been attributed to *Penicillium.* Molds do not grow on beef held at temperatures below −5°C.

Composition and Spoilage of Poultry Muscle

The mechanism of microbial spoilage of poultry muscle is similar to that of red meat. Spoilage is generally restricted to the outer surfaces of the skin and cuts and has been characterized by off odors and sliminess, as well as various types of discoloration. Skin may provide a barrier to the introduction of spoilage microorganisms to the underlying muscle tissue. Although the pH of breast muscle (pH 5.7 to 5.9) differs from that of leg muscle (pH 6.4 to 6.7), the organisms that cause spoilage are similar and include *Pseudomonas, Aeromonas,* and *S. putrefaciens.* For poultry carcasses packaged in oxygen-impermeable films, spoilage may be caused by *Shewanella, B. thermosphacta,* and atypical lactobacilli. The ability to make sulfide compounds, such as hydrogen sulfide, dimethyl sulfide, and methyl mercaptan, makes *S. putrefaciens* an important component of the spoilage microflora.

Composition and Spoilage of Finfish

The internal muscle tissue of a healthy live fish is generally sterile. Bacteria are present on the outer slime layer of the skin and gill surfaces and in the intestines in the case of feeding fish. The composition of fish muscle is

highly variable between species and may fluctuate widely, depending upon size, season, fishing grounds, and diet. The average composition of nonfatty fish, such as cod, has been characterized as 18% protein and <1% lipid, whereas in fatty fish, such as herring, the lipid content may range from 1 to 30%, with the water content varying so that fat and water constitute ~80% of the muscle tissue. As with other muscle foods, the spoilage microflora of fresh ice-stored fish consists largely of *Pseudomonas* spp. *S. putrefaciens* may also contribute to the spoilage of seafood, and *Acinetobacter* and *Moraxella* may constitute a smaller portion of the spoilage microbes. The spoilage characteristics of fresh fish can be divided into four stages. Stage I occurs from 0 to 6 days after death and involves shifting of bacterial populations without odor. Stage II occurs from 7 to 10 days after death, with bacterial growth becoming apparent and the development of a slightly fishy odor. Stage III occurs from 11 to 14 days after death and is characterized by rapid bacterial growth, a sour and fishy odor, and the start of slime formation on the skin. Stage IV occurs >14 days after death, at which point bacterial numbers are stationary, proteolysis begins, the skin is extremely slimy, and the odor is offensive.

Composition and Spoilage of Shellfish

Crustacean and molluscan shellfish generally contain larger amounts of free amino acids than finfish. Trimethylamine oxide is present in crustacean shellfish, with the exception of cephalopods, scallops, and cockles, and is absent in molluscan tissue. Crustaceans possess potent cathepsin-like enzymes, which rapidly break down proteins, leading to tissue softening and the development of volatile off odors. Removal of the head after harvest can extend shelf life by eliminating an organ that releases degradative enzymes. Some crustacean meats (primarily shrimp) suffer from a visual defect known as black spot melanosis, which is due to polyphenol oxidase activity and not to microbial action.

Molluscan shellfish contain a lower total nitrogen concentration in their flesh than do finfish or crustacean shellfish and much more carbohydrate, mostly in the form of glycogen. As a result, the spoilage pattern of molluscan shellfish differs from that of other seafoods and is generally fermentative, with the pH of tissues declining as spoilage progresses, yielding a predominance of lactobacilli and streptococci.

Factors Influencing Spoilage

Proteolytic and Lipolytic Activities

Although *Pseudomonas* and other aerobic spoilage bacteria are able to produce proteolytic enzymes, their production is delayed until the late logarithmic phase of growth. Proteolysis occurs only in populations of $>10^8$ CFU/cm^2, when spoilage is well advanced and the bacteria are approaching their maximum cell density.

Oxidative rancidity of fat occurs when unsaturated fatty acids react with oxygen from the storage environment. Stable compounds, such as aldehydes, ketones, and short-chain fatty acids, are produced, resulting in the eventual development of rancid flavors and odors. Autoxidation, independent of microbial activity, occurs in muscle foods stored in aerobic environments. The rate is influenced by the proportion of unsaturated fatty acids in the fat. Generally, lipase production is restricted while carbohydrate substrates in muscle tissue are being utilized; it is unlikely that microbial lipolytic activity would occur until glucose on the muscle surface is

depleted. At this point, amino acids would also be degraded, and the resulting spoilage characteristics would possibly mask the effects of rancidity.

Spoilage of Adipose Tissue

Adipose tissue consists predominantly of insoluble fat, which cannot be used for microbial growth until it is broken down and emulsified. While the spoilage processes and growth rates of spoilage bacteria are similar for adipose and muscle tissues, the low level of carbohydrates in adipose tissue means that spoilage odors are detected when lower numbers of bacteria are present and most, if not all, available glucose is depleted. This occurs when populations exceed 10^6 CFU/cm^2. The growth rates of some psychrotrophic microorganisms, such as *Hafnia alvei, Serratia liquefaciens,* and *Lactobacillus plantarum,* are higher on fat than on lean beef and pork tissues. Spoilage bacteria, such as *S. putrefaciens,* may also grow. In practice, spoilage of fat before lean tissue is unlikely, given the presence of muscle tissue fluids in vacuum-packaged cuts and the restriction of bacterial growth due to the drying of carcass surfaces.

Spoilage under Anaerobic Conditions

Lactic acid bacteria dominate the spoilage microflora of muscle foods when oxygen is excluded from the storage environment. If the pH of the muscle tissue is high or residual amounts of oxygen are present, other microorganisms, such as *B. thermosphacta* and *S. putrefaciens,* may cause spoilage. The growth rate of bacteria under anaerobic conditions is considerably reduced compared to growth under aerobic conditions. In addition, the maximum cell density achieved under anaerobic conditions ($\sim 10^8$ CFU/cm^2) is considerably less than that achieved under aerobic conditions ($>10^9$ CFU/cm^2). The sour, acid, cheesy odors or cheesy and dairy flavors that develop in muscle tissue under anaerobic conditions can be attributed, at least in part, to the accumulation of short-chain fatty acids and amines. The presence of *B. thermosphacta* in significant numbers causes more rapid spoilage than that of lactic acid bacteria.

Other Meat Characteristics Associated with Spoilage

Animals subjected to excessive stress or exercise before slaughter can have depleted levels of muscle glycogen. This results in a condition known as dark, firm, and dry. Spoilage of dark, firm, and dry meat occurs more quickly than spoilage of meat with normal pH. Rapid spoilage is caused by the absence of glucose, and hence lactic acid, in the tissues. Pale, soft, exudative muscle tissue is a condition that occurs in pork and turkey, and to a lesser extent in beef, in which sugar utilization decreases the muscle pH to its ultimate level while the muscle temperature is still high. There is debate over whether pale, soft, exudative meats spoil more slowly than meat with normal pH.

Comminuted Products

The limited shelf life of comminuted (ground and blended) muscle foods has been attributed to (i) a higher initial microbial load due to use of a poorer-quality product for grinding, (ii) contamination during processing, and (iii) the effects of the comminution of the muscle tissue. On the surfaces of aerobically stored comminuted products, *Pseudomonas, Acinetobacter,* and *Moraxella* species are the main microflora, while in the interior, due to the limited availability of oxygen, lactic acid bacteria are dominant. Occasional contaminants, such as

Aeromonas spp. or members of the family *Enterobacteriaceae,* occur more often on comminuted products than on intact tissue.

Cooked Products

Cooking muscle foods destroys vegetative bacterial cells, although endospores may survive. In perishable, cooked, uncured meats, spoilage is caused by bacteria that survive heat processing or by postprocessing contaminants. Microorganisms responsible for spoilage of these products include psychrotrophic micrococci, streptococci, lactobacilli, and *B. thermosphacta.*

Processed Products

Microbial spoilage of processed products depends on the nature of the product and the ingredients used. Spoilage of processed meats may be characterized as slimy spoilage, souring, or greening. *Slimy spoilage,* which is usually confined to the outside surfaces of product casings, results from the growth of some yeasts; two genera of lactic acid bacteria, *Lactobacillus* and *Enterococcus;* and *B. thermosphacta. Souring* occurs, typically beneath casings, when bacteria, such as lactobacilli, enterococci, and *B. thermosphacta,* utilize lactose and other sugars to produce acids. While *greening* of fresh meats may be the result of hydrogen sulfide production by certain bacteria, greening of cured meat products may also develop in the presence of hydrogen peroxide, which may form on the surfaces of vacuum- or modified-atmosphere-packaged meats when they are exposed to air. *Lactobacillus viridescens* is the most common cause of this type of greening, but species of *Streptococcus* and *Leuconostoc* may also produce the defect. The presence of these bacteria is often a result of poor sanitation.

Cured meats are spoiled by microorganisms that tolerate low a_w, such as lactobacilli or micrococci. If sucrose is added to cured meats, a slimy dextran layer may form due to the activity of *Leuconostoc* spp. or other bacteria, such as *L. viridescens. B. thermosphacta* may also be involved in the spoilage of these products. If dried meats are properly prepared and stored, their low a_w renders them microbiologically stable. To restrict the growth of some species of microbes that can grow at low a_w, the water content of dried meat products must be sufficiently reduced. Microbial spoilage of these products should not occur unless exposure to high relative humidity or other high-moisture conditions results in an uptake of moisture. The addition of antifungal agents may retard the growth of microbes in these products.

Control of Spoilage of Muscle Foods

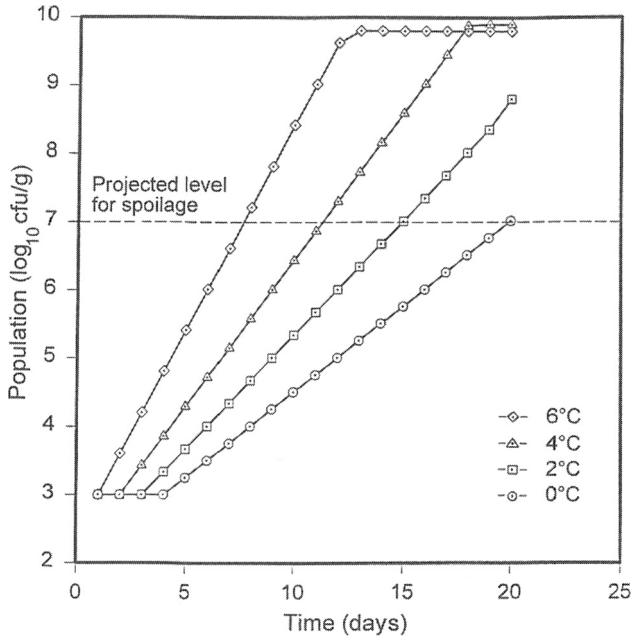
Modifying intrinsic characteristics of products or extrinsic characteristics of the storage environment can control the growth of spoilage microorganisms on muscle foods. The shelf life of processed meats can be extended by processing procedures and ingredients that either prevent the growth of spoilage microorganisms or select for a less offensive spoilage microflora. For fresh meats, the extension of shelf life has become especially important as the food industry moves toward the centralization of processing activities, with products distributed to more distant domestic and international markets. Specific methods to prevent spoilage and prolong the shelf life of muscle foods include effective good manufacturing practices at lower levels for products entering storage and distribution; rinsing by water, immersion, and spray systems to remove physical and microbial contaminants

from carcasses; and incorporation of antimicrobial compounds, such as chlorine and organic acids, in wash water. The use of lactic, acetic, propionic, and citric acids has been investigated, although lactic and acetic acids are most often used to reduce microbial levels on carcass surfaces. The effectiveness of organic acids is influenced by factors such as the type of acid, concentration, temperature, and point of application in processing.

Temperature is the most important environmental parameter influencing the growth of microorganisms in muscle foods. As the temperature is decreased below the optimum for the growth of microorganisms, generation times and lag times are extended, and growth is therefore slowed (Fig. 20.2). Under aerobic storage conditions, the comparatively high growth rate of *Pseudomonas* spp. allows successful competition with mesophiles and other psychrotrophs at temperatures below 20°C. Likewise, under anaerobic conditions at temperatures below 20°C, the rapid growth rate of psychrotrophic lactobacilli allows successful competition with other psychrotrophic spoilage microorganisms. Storage of foods at temperatures at which mesophiles can grow allows these microorganisms to contribute to spoilage. Higher storage temperatures may also allow the growth of psychrotrophic spoilage microorganisms that would otherwise be restricted by the pH or lactic acid concentration of muscle tissue at reduced temperatures. Likewise, the minimum temperature for the growth of psychrotrophic spoilage microorganisms may be increased by other factors that influence growth, e.g., pH, a_w, or oxidation-reduction potential, or by inhibitory agents, such as food additives or increased carbon dioxide.

The shelf life of muscle foods can be extended by storage under vacuum or modified atmospheres. Modified-atmosphere packaging (MAP) involves the storage of products under a high-oxygen barrier film with a

Figure 20.2 Predicted populations of total aerobic bacteria in ground beef as affected by storage temperature. Reprinted from T. C. Jackson, D. L. Marshall, G. R. Acuff, and J. S. Dickson, p. 91–109, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).



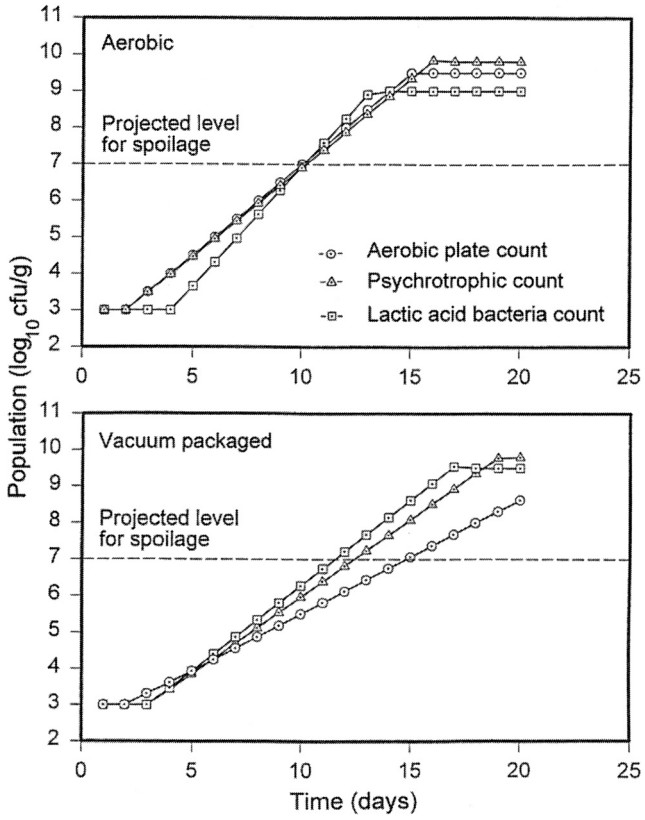


Figure 20.3 Populations of total aerobic, psychrotrophic, and lactic acid bacteria in aerobically packaged and vacuum-packaged ground pork stored at 0°C. The growth curves are based on the derived values for lag phase duration and generation time. Reprinted from T. C. Jackson, D. L. Marshall, G. R. Acuff, and J. S. Dickson, p. 91–109, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers,* 2nd ed. (ASM Press, Washington, DC, 2001).

headspace containing gas with a composition different from air. Typically, this headspace contains elevated amounts of carbon dioxide, and it may also contain nitrogen and oxygen in various proportions. Much of the extended shelf life of such products results from a modification of the spoilage microflora from an aerobic psychrotrophic population, consisting of bacteria such as *Pseudomonas, Moraxella,* and *Acinetobacter,* to one consisting predominantly of lactic acid bacteria and *B. thermosphacta* (Fig. 20.3). Carbon dioxide is commonly used in modified-atmosphere environments due to its bacteriostatic activity. Carbon dioxide increases the lag phases and generation times of many microorganisms. Oxygen is occasionally included in modified atmospheres used to package fresh red meats to maintain the oxymyoglobin responsible for the fresh red, or "bloomed," color desired by the consumer. However, if significant amounts of oxygen are present, aerobic microorganisms and *B. thermosphacta* may grow, resulting in spoilage. Nitrogen may also be included to displace oxygen and delay oxidative rancidity and to inhibit the growth of aerobic microorganisms.

Irradiation of meats enhances microbiological safety and quality by significantly reducing populations of pathogenic and spoilage bacteria. At present, relatively low doses of irradiation, i.e., <5 kilograys (kGy), are receiving the most interest. The doses of radiation required to inactivate 90%

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21 Molds

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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- gain knowledge about common food spoilage molds and those associated with disease
- understand methods used for differentiating and identifying molds
- associate specific mycotoxins with molds that produce toxins of concern
- identify foods that are commonly associated with mold problems
- know conditions that contribute to the growth of molds and toxin production in foods
- discuss methods used for screening of raw and processed materials for the presence of mycotoxins

INTRODUCTION

We have known for centuries that eating certain mushrooms can cause illness or even be fatal. That some common food spoilage molds cause disease and death is a newer realization. Indeed, in 1940, a *Penicillium* species was identified as the source of toxicity in "yellow" rice in Japan. Mycotoxins came to the attention of scientists in the Western world in the early 1960s with the outbreak of turkey "X" disease in England. This disease killed ~100,000 turkeys and other farm animals. The cause of the disease was traced to peanut meal in the feed, which was heavily contaminated with *Aspergillus flavus*. Analysis of the feed revealed that a group of fluorescent compounds, later named aflatoxins, were responsible for the outbreak. Aflatoxins also killed large numbers of ducklings in Kenya and caused hepatoma in hatchery-reared trout in California. This chapter focuses on molds that are commonly associated with the spoilage of foods.

ISOLATION, ENUMERATION, AND IDENTIFICATION

Bacteria grow much faster than molds and are often found in contaminated samples. Therefore, media for the enumeration of fungi contain antibacterial compounds and compounds to inhibit mold colony spreading, such as dichloran rose bengal chloramphenicol agar or dichloran 18% glycerol agar. Czapek agar, a defined medium based on mineral salts, or a derivative such as Czapek yeast extract agar and malt extract agar are used for the identification of *Aspergillus* species from foods. Growth on Czapek yeast extract 20% sucrose agar can also be a useful aid in identifying species of *Aspergillus*. The most effective medium for rapid detection of aflatoxigenic molds is *A. flavus* and *Aspergillus parasiticus* agar. Under incubation at 30°C

for 42 to 48 h, *A. flavus* and *A. parasiticus* produce a bright orange-yellow colony reverse, which is diagnostic and readily recognized. The structures of *A. flavus* and *A. parasiticus* are shown in Fig. 21.1.

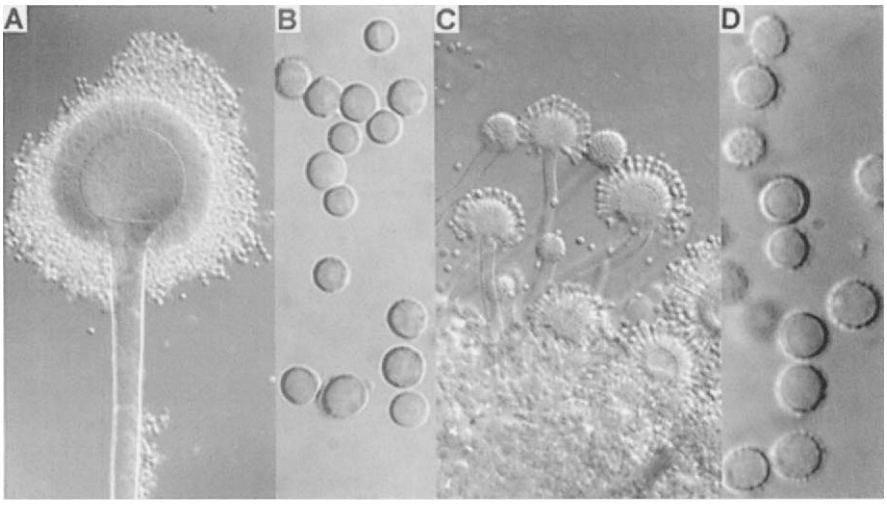
Unlike *Penicillium* species, *Aspergillus* species are conveniently "color coded," and the color of the *conidia* (nonmotile spores) serves as a very useful starting point in identification. The microscopic morphology is also important in identification. *Phialides* (cells producing conidia) may be produced directly from the swollen apex (*vesicle*) of long stalks (*stipes*), or there may be an intermediate row of supporting cells (*metulae*). Correct identification of *Aspergillus* species is an essential prerequisite to assessing the potential for mycotoxin contamination in a commodity, food, or feedstuff.

General enumeration procedures suitable for foodborne molds are effective for enumerating all common *Penicillium* species. Many antibacterial media give satisfactory results. However, some *Penicillium* species grow rather weakly on very dilute media, such as potato dextrose agar. Therefore, dichloran rose bengal chloramphenicol and dichloran 18% glycerol agars are recommended.

Identification of *Penicillium* isolates to the species level is not easy and is preferably carried out under carefully standardized conditions of medium, incubation time, and temperature. In addition to the microscopic morphology, gross physiological features, including colony diameters and the colors of conidia and colony pigments, are used to distinguish species. A selective medium, dichloran rose bengal yeast extract sucrose agar, is used for the enumeration of *Penicillium verrucosum* and *Penicillium viridicatum*. *P. verrucosum* produces a violet-brown reverse coloration on this agar. Figure 21.2 shows penicillia of four *Penicillium* subgenera.

Fusarium species are most often associated with cereal grains; seeds; milled cereal products, such as flour and corn meal; barley malt; animal feeds; and dead plant tissue. To isolate *Fusarium* species from these products, it is necessary to use selective media. The basic techniques for the detection and isolation of *Fusarium* employ plating techniques, either plate

Figure 21.1 Aflatoxigenic fungi. **(A)** *A. flavus* head (magnification, ×300); **(B)** *A. flavus* conidia (magnification, ×1,875); **(C)** young *A. parasiticus* heads (magnification, ×300); **(D)** *A. parasiticus* conidia (magnification, ×1,875). Reprinted from A. D. Hocking, p. 540–550, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).



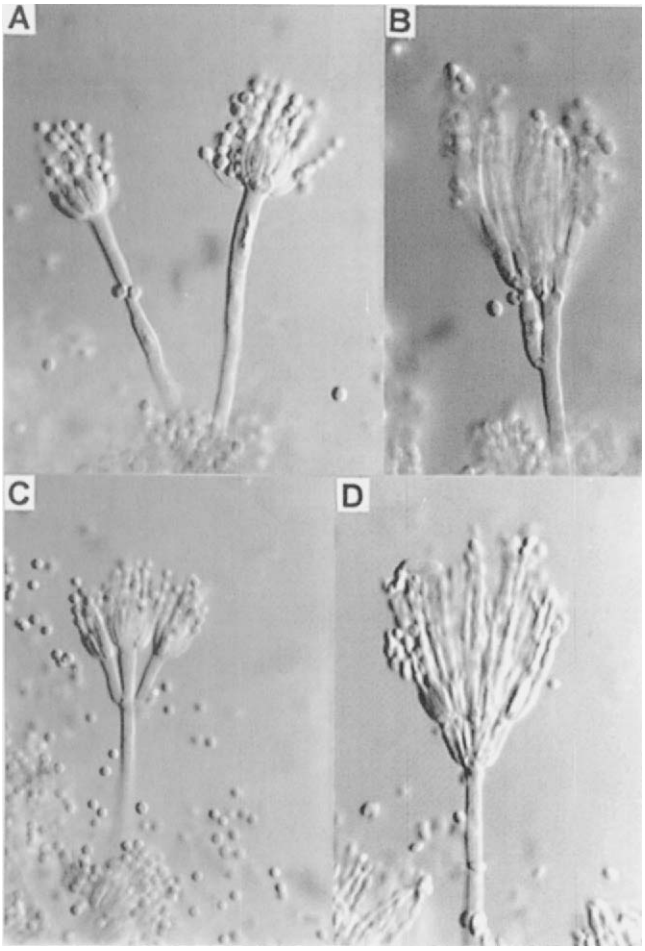


Figure 21.2 Penicillia (magnification, ×750) representative of the four *Penicillium* subgenera. **(A)** *Penicillium* subgenus *Aspergilloides* (*P. glabrum*); **(B)** *Penicillium* subgenus *Penicillium* (*P. expansum*); **(C)** *Penicillium* subgenus *Furcatum* (*P. citrinum*); **(D)** *Penicillium* subgenus *Biverticillium* (*P. variabile*). Reprinted from J. I. Pitt, p. 551–562, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

counts of serial dilutions of products or the placement of seeds or kernels of grain directly on the surface of agar medium in petri dishes, i.e., *direct plating*. Since many *Fusarium* species are plant pathogens and are found in fields where crops are grown, these molds respond to light. Growth, pigmentation, and spore production are most typical when cultures are grown in alternating light and dark cycles of 12 h each. Fluorescent light or diffuse sunlight from a north window is best. Fluctuating temperatures, such as 25°C (day) and 20°C (night), also enhance growth and sporulation.

Several culture media have been used to detect and isolate *Fusarium* species. These include Nash-Snyder medium, modified Czapek Dox agar, Czapek iprodione dichloran (CZID) agar, potato dextrose iprodione dichloran agar, and dichloran chloramphenicol peptone agar. Nash-Snyder medium and modified Czapek Dox agar contain pentachloronitrobenzene, a known carcinogen, and are not favored for routine use in food microbiology laboratories. However, these media can be useful for evaluating samples that are heavily contaminated with bacteria and other fungi. The most

commonly used medium for isolating *Fusarium* from foods is CZID agar, but rapid identification of *Fusarium* isolates to the species level is difficult, if not impossible, on this medium. Isolates must be subcultured on other media, such as carnation leaf agar (CLA), for identification. However, CZID agar is a good selective medium for *Fusarium*. Most molds are inhibited on CZID agar, and *Fusarium* species can be readily distinguished.

Identification of *Fusarium* species is based largely on the production and morphology of macroconidia and microconidia (Fig. 21.3). *Fusarium* species do not readily form conidia on all culture media, and conidia formed on high-carbohydrate media, such as potato dextrose agar, are often more variable and less typical. CLA is a medium that supports abundant and consistent spore production. Carnation leaves from actively growing, disbudded young carnation plants free of pesticide residues are cut into small pieces (5 mm^2), dried in an oven at 45 to 55°C for 2 h, and sterilized by irradiation. CLA is prepared by placing a few pieces of carnation leaf on the surface of 2.0% water agar. *Fusarium* isolates are then inoculated onto the agar and leaf interface, where they form abundant and typical conidia and conidiophores in sporodochia (Fig. 21.4). CLA is low in carbohydrates and rich in other complex, naturally occurring substances that apparently stimulate spore production.

ASPERGILLUS SPECIES

Aspergillus was first described nearly 300 years ago and is an important genus in foods. Although a few species have been used to make food (e.g., *Aspergillus oryzae* in soy sauce manufacture), most *Aspergillus* species occur in foods as spoilage or biodeterioration fungi. They are extremely common in stored commodities, such as grains, nuts, and spices, and occur more frequently in tropical and subtropical than in temperate climates. The genus *Aspergillus* contains several species capable of producing mycotoxins, although the mycotoxin literature of the past 35 years has been dominated by papers on aflatoxins. Since foods contain many toxin-producing *Aspergillus* species, it is important to identify them and determine if they can produce toxin in a particular food.

The genus *Aspergillus* is large, containing >100 recognized species, most of which grow well in laboratory culture. Within the genus *Aspergillus*, species are grouped into six subgenera that are subdivided into sections. There are a number of teleomorphic (sexual-spore-producing) genera that have *Aspergillus* conidial states (*anamorphs*, i.e., asexual states or stages), but the only two of real importance in foods are the xerophilic (growing at low water activity [a_w]) *Eurotium* and *Neosartorya* species. They produce heat-resistant ascospores and cause spoilage in heat-processed foods, mainly fruit products. The most widely used taxonomy for *Aspergillus* is based on traditional morphological taxonomic techniques. However, chemical techniques, such as isoenzyme patterns, secondary metabolites, and ubiquinone systems, and molecular techniques have been used to clarify relationships within the genus.

Almost 50 *Aspergillus* species can produce toxic metabolites, but the *Aspergillus* mycotoxins (Table 21.1) of greatest significance in foods and feeds are aflatoxins (produced by *A. flavus*, *A. parasiticus*, and *Aspergillus nomius*); ochratoxin A from *Aspergillus ochraceus* and related species and from *Aspergillus carbonarius* and occasionally *Aspergillus niger*; sterigmatocystin, produced primarily by *Aspergillus versicolor* but also by *Emericella* species;

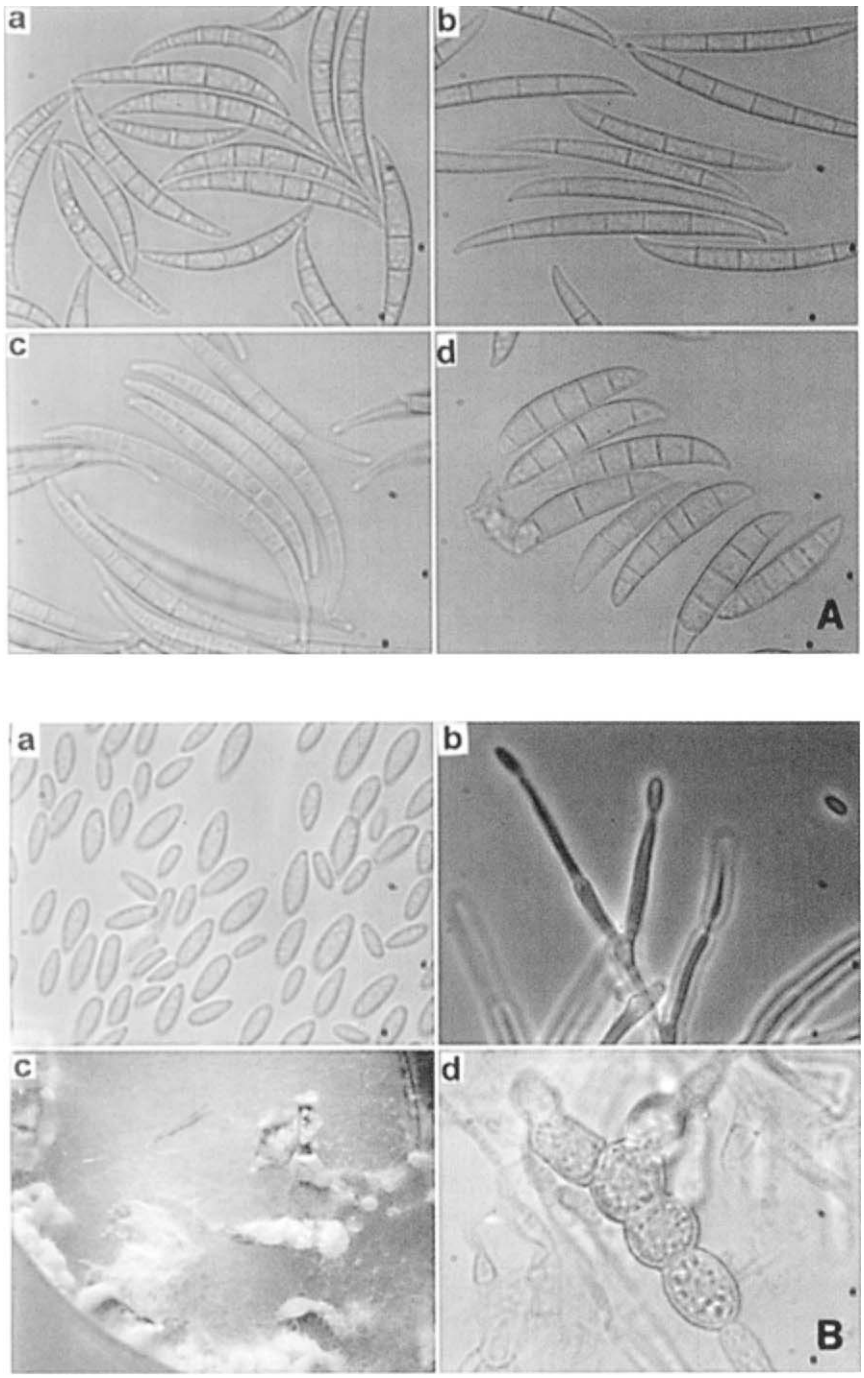
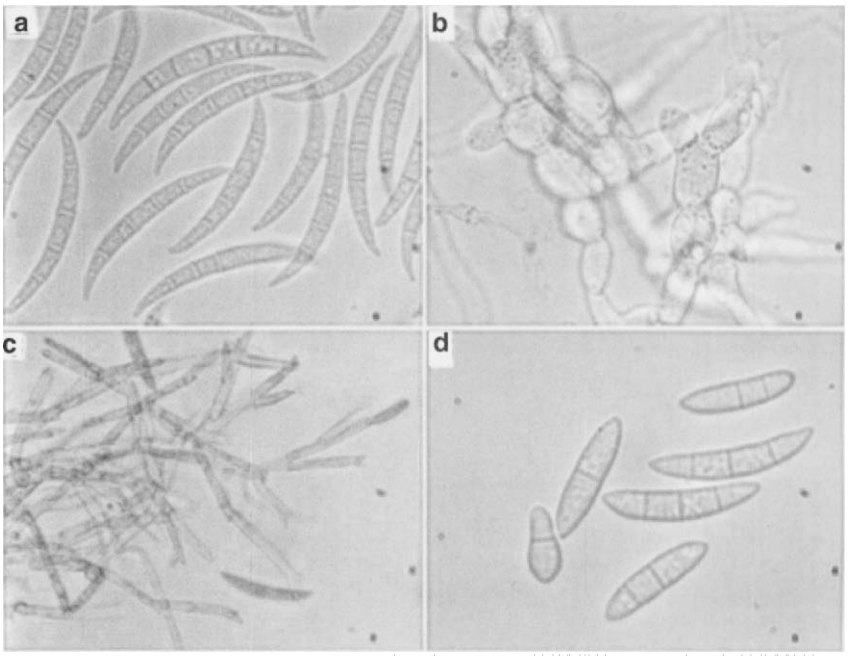
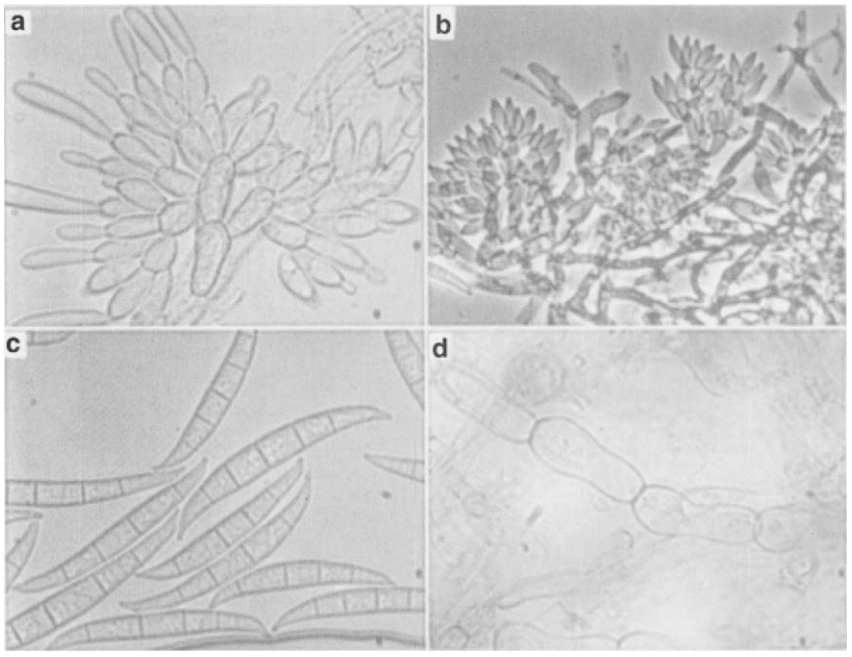


Figure 21.3 **(A)** Examples of macroconidia of *Fusarium* species. **(a)** *F. graminearum*; **(b)** *F. moniliforme*; **(c)** *F. equiseti*; **(d)** *F. culmorum* (magnification, ×1,000). **(B)** Examples of micro- and macroscopic structures of *Fusarium* species. **(a)** Microconidia; **(b)** monophialides; **(c)** sporodochia; **(d)** chlamydospores (magnification, ×1,000 [a, b, and d] and ×10 [c]). Reprinted from L. B. Bullerman, p. 563–578, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

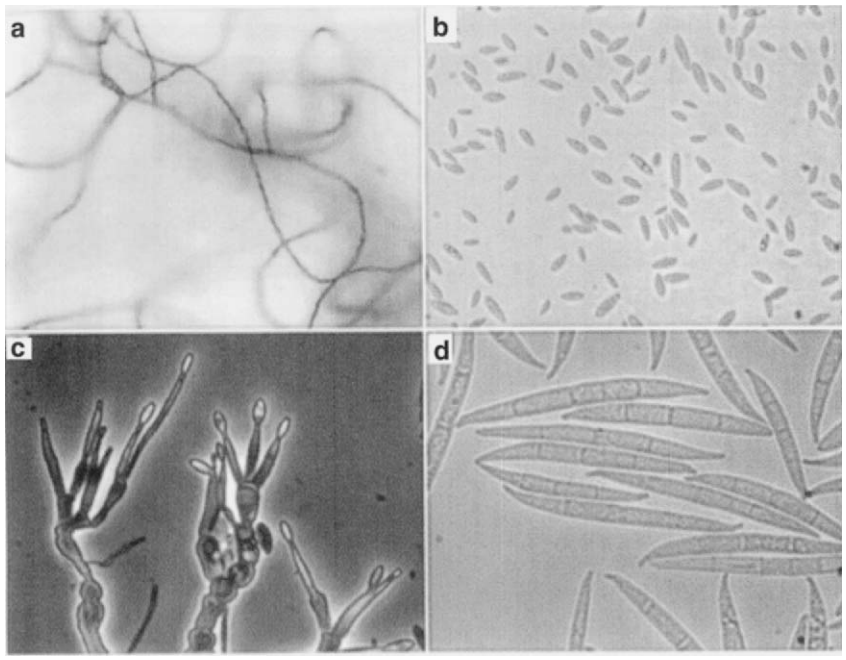


A. Fusarium sporotrichioides.

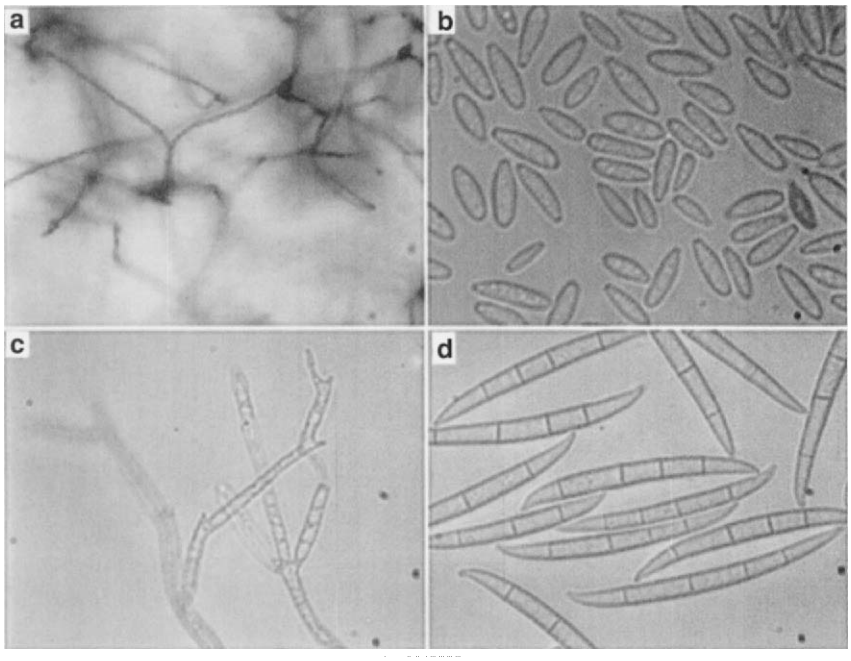


B. Fusarium graminearum.

Figure 21.4 **(A)** Microscopic structures of *F. sporotrichioides.* **(a)** Macroconidia; **(b)** chlamydospores; **(c)** phialides; **(d)** microconidia (magnification, ×1,000 [a, b, and d] and ×550 [c]). **(B)** Microscopic structures of *F. graminearum.* **(a)** Conidiophores (monophialides); **(b)** monophialides in sporodochia; **(c)** macroconidia; **(d)** chlamydospores (magnification, ×1,000 [a, c, and d] and ×550 [b]).



C. Fusarium moniliforme.



D. Fusarium proliferatum.

Figure 21.4 *(continued)* **(C)** Microscopic structures of *F. moniliforme*. **(a)** Microconidia in chains; **(b)** microconidia; **(c)** monophialides producing microconidia; **(d)** macroconidia (magnification, ×165 [a], ×550 [b], and ×1,000 [c and d]). **(D)** Microscopic structures of *F. proliferatum*. **(a)** Microconidia in chains; **(b)** microconidia; **(c)** polyphialides; **(d)** macroconidia (magnification, ×165 [a] and ×1,000 [b to d]). Reprinted from L. B. Bullerman, p. 563–578, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

Table 21.1 Significant mycotoxins produced by *Aspergillus* species and their toxic effects[a]

Mycotoxin(s)	Toxicity	Species
Aflatoxins B_1 and B_2	Acute liver damage, cirrhosis, carcinogenic (liver), teratogenic, immunosuppressive	*A. flavus, A. parasiticus, A. nomius*
Aflatoxins G_1 and G_2	Effects similar to B aflatoxins; G_1 toxicity is less than B_1 but greater than B_2	*A. parasiticus, A. nomius*
Cyclopiazonic acid	Degeneration and necrosis of various organs, tremorgenic, low oral toxicity	*A. flavus, A. tamarii*
Ochratoxin A	Kidney necrosis (especially pigs), teratogenic, immunosuppressive, probably carcinogenic	*A. ochraceus* and related species, *A. carbonarius, A. niger* (occasional)
Sterigmatocystin	Acute liver and kidney damage, carcinogenic (liver)	*A. versicolor, Emericella* spp.
Fumitremorgens	Tremorgenic (rats and mice)	*A. fumigatus*
Territrems	Tremorgenic (rats and mice)	*A. terreus*
Tryptoquivalines	Tremorgenic	*A. clavatus*
Cytochalasins	Cytotoxic	*A. clavatus*
Echinulins	Feed refusal (pigs)	*Eurotium chevalieri, Eurotium amstelodami*

[a]Reprinted from A. D. Hocking, p. 537–550, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

and cyclopiazonic acid (*A. flavus* is the primary source, but it is also reported to be produced by *Aspergillus tamarii* [Fig. 21.5]). *Tremorgenic* (i.e., producing involuntary trembling of the body) toxins are produced by *Aspergillus terreus* (territrems), *Aspergillus fumigatus* (fumitremorgens), and *Aspergillus clavatus* (tryptoquivaline). Citrinin, patulin, and penicillic acid may also be produced by certain *Aspergillus* species.

Aspergillus species produce toxins that exhibit a wide range of toxicities and cause significant long-term effects. Aflatoxin B_1 is perhaps the most potent liver carcinogen known for a wide range of animal species, including humans. Both ochratoxin A and citrinin affect kidney function. Cyclopiazonic acid has a wide range of effects. Tremorgenic toxins, such as territrems, affect the central nervous system. Table 21.1 lists the most significant toxins produced by *Aspergillus* species and their toxic effects.

A. flavus and *A. parasiticus*

Undoubtedly, the most important group of toxigenic aspergilli are the aflatoxigenic molds, *A. flavus, A. parasiticus,* and the recently described but much less common species *A. nomius,* all of which are classified in *Aspergillus* section *Flavi.* Although these three species are closely related and show many similarities, many characteristics may be used in their differentiation (Table 21.2). The toxins produced by these three species are species specific. *A. flavus* can produce aflatoxins B_1 and B_2 and cyclopiazonic acid, but only some isolates are toxigenic. *A. parasiticus* produces aflatoxins B_1, B_2, G_1, and G_2 but not cyclopiazonic acid, and almost all isolates are toxigenic. *A. nomius* is morphologically similar to *A. flavus,* but like *A. parasiticus,* it produces B and G aflatoxins without cyclopiazonic acid. Because this species appears to be uncommon, it has been little studied. The potential toxigenicities of isolates are not known, and the practical importance of the species is hard to assess. *A. oryzae* and *Aspergillus sojae* are related to *A. flavus* and *A. parasiticus* and are used to make fermented foods, but they do not produce toxins. *A. tamarii* produces aflatoxin B and cyclopiazonic acid. Obviously, accurate differentiation of related species within section *Flavi* is important in order to determine the potential for toxin production and the types of toxins likely to be present.

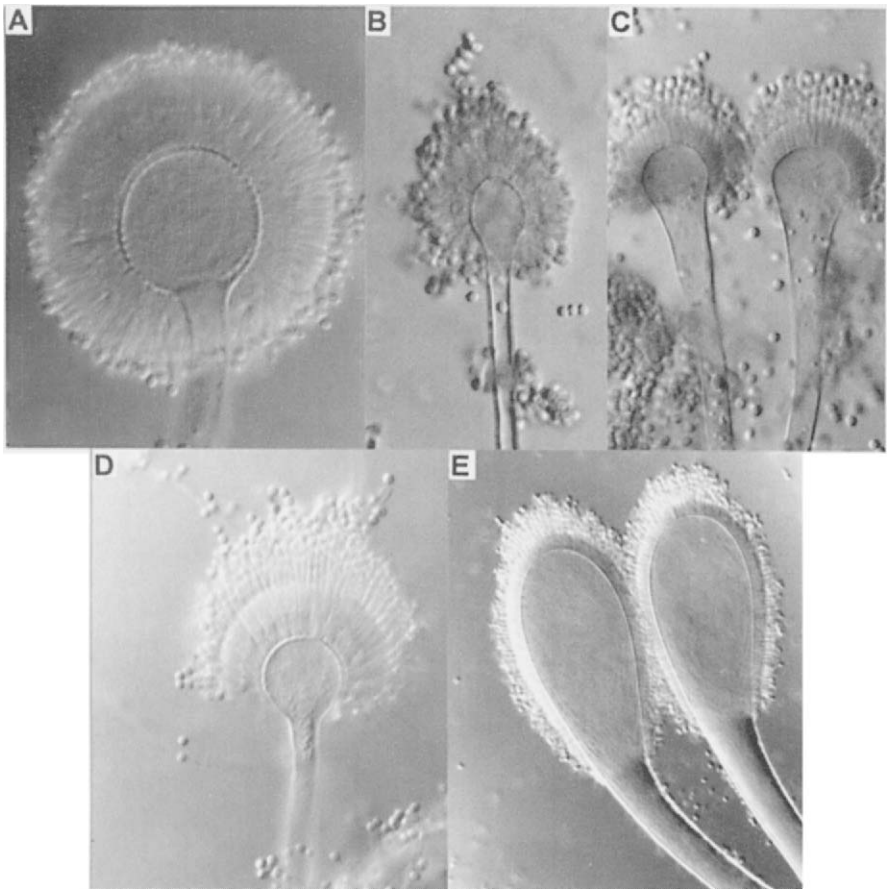


Figure 21.5 Some common mycotoxigenic *Aspergillus* species. **(A)** *A. ochraceus* (magnification, ×750); **(B)** *A. versicolor* (magnification, ×750); **(C)** *A. fumigatus* (magnification, ×750); **(D)** *A. terreus* (magnification, ×750); **(E)** *A. clavatus* (magnification, ×300). Reprinted from A. D. Hocking, p. 537–550, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

Detection and Identification

A. flavus and *A. parasiticus* (Fig. 21.1) are easily distinguished from other *Aspergillus* species by using appropriate media and methods. The texture of the conidial walls is a reliable differentiating feature: conidia of *A. flavus* (Fig. 21.1B) are usually smooth to finely roughened, while those of *A. parasiticus* (Fig. 21.1D) are clearly rough when observed under an oil immersion lens. The combinations of characteristics most useful in differentiating among the three aflatoxigenic species are summarized in Table 21.2.

Table 21.2 Distinguishing features of *A. flavus*, *A. parasiticus*, and *A. nomius*[a]

Species	Conidia	Sclerotia	Toxins
A. flavus	Smooth to moderately roughened, variable in size	Large, globose	Aflatoxins B_1 and B_2, cyclopiazonic acid
A. parasiticus	Conspicuously roughened, little variation in size	Large, globose	Aflatoxins B and G
A. nomius	Similar to *A. flavus* (bullet shaped)	Small, elongated	Aflatoxins B and G

[a]Reprinted from A. D. Hocking, p. 537–550, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

Aflatoxins

Aflatoxins are difuranocoumarin derivatives. Aflatoxins B_1, B_2, G_1, and G_2 are produced in nature by the molds discussed above. The letters B and G refer to the fluorescent colors (blue and green, respectively) observed under long-wave ultraviolet (UV) light, and the subscripts 1 and 2 refer to their separation patterns on thin-layer chromatography plates. Aflatoxins M_1 and M_2 are produced from their respective B aflatoxins by hydroxylation in lactating animals and are excreted in milk at a rate of ~1.5% of ingested B aflatoxins.

Aflatoxins are synthesized through the polyketide pathway, beginning with condensation of an acetyl unit with two malonyl units and the loss of carbon dioxide. Intermediate steps result in the formation of norsolorinic acid. Norsolorinic acid undergoes several metabolic conversions to form aflatoxin B_1. G group aflatoxins are formed from the same substrate as B group aflatoxins, namely, *O*-methylsterigmatocystin, but by an independent pathway. Most of the genes involved in the biosynthesis of aflatoxins are contained within a single gene cluster in the genomes of *A. flavus* and *A. parasiticus*, and their regulation and expression are now relatively well understood.

Toxicity. Aflatoxins are both acutely and chronically toxic in animals and humans, producing acute liver damage, liver cirrhosis, and tumor induction. Even more important to human health are the immunosuppressive effects of aflatoxins, either alone or in combination with other mycotoxins. Immunosuppression can increase susceptibility to infectious diseases, particularly in populations where aflatoxin ingestion is *chronic* (continuous), and can interfere with the production of antibodies in response to immunization in animals, and perhaps also in children.

Acute aflatoxicosis in humans is rare; however, several outbreaks have been reported. In 1967, 26 people in two farming communities in Taiwan became ill with apparent food poisoning. Nineteen were children, three of whom died. Although postmortems were not performed, rice from the affected households contained ~200 μg of aflatoxin B_1/kg, which was probably responsible for the outbreak. An outbreak of hepatitis in India in 1974 that affected 400 people, 100 of whom died, was almost certainly caused by aflatoxins. The outbreak was traced to corn heavily contaminated with *A. flavus* and containing up to 15 mg of aflatoxins per kg. It was calculated that the affected adults may have consumed 2 to 6 mg on a single day, implying that the acute lethal dose for adult humans is on the order of 10 mg. The deaths of 13 Chinese children in the northwestern Malaysian state of Perak were reportedly due to ingestion of contaminated noodles. Aflatoxins were confirmed in postmortem tissue samples. A case of systemic aspergillosis caused by an aflatoxin-producing strain of *A. flavus* has been reported in which aflatoxins B_1, B_2, and M_1 were detected in lung lesions and were considered to have played a role in damaging the immune system of the patient.

The greatest direct impact of aflatoxins on human health is their potential to induce liver cancer. Human liver cancer has a high incidence in central Africa and parts of Southeast Asia. Studies in several African countries and Thailand have shown a correlation between aflatoxin intake and the occurrence of primary liver cancer. No such correlation could be demonstrated for populations in rural areas of the United States, despite the occurrence of aflatoxins in corn. However, evidence supports the hypothesis that high aflatoxin intakes are causally related to high incidences of cancer,

even in the absence of hepatitis B. In animals, aflatoxins cause various syndromes, including cancer of the liver, colon, and kidneys. Regular low-level intake of aflatoxins can lead to poor feed conversion, low weight gain, and poor milk yields in cattle.

Mechanism of action. Aflatoxin B_1 is metabolized in the liver, leading to the formation of highly reactive intermediates, one of which is 2,3-epoxy-aflatoxin B_1. Binding of these reactive intermediates to DNA results in disruption of transcription and abnormal cell proliferation, leading to mutagenesis or carcinogenesis. Aflatoxins also inhibit oxygen uptake in the tissues by acting on the electron transport chain and inhibiting various enzymes, resulting in decreased production of adenosine triphosphate (ATP).

Toxin detection. Mycotoxins can be detected by chemical or biological methods. Chemical determination of aflatoxins has become fairly standardized. Samples are extracted with organic solvents, such as chloroform or methanol, in combination with small amounts of water. The representative sample size(s) depends on the particle size; therefore, a 500-g sample is acceptable for oils or milk powder, but 3 kg of flour or peanut butter is necessary. Extracts can be further cleaned up by passage through a silica gel column, concentrated, and then separated by thin-layer chromatography or high-performance liquid chromatography. Following thin-layer chromatography, aflatoxins are visualized under UV light and quantified by visual comparison with known concentrations of standards. High-performance liquid chromatography with detection by UV light or absorption spectrometry provides a more readily quantifiable (although not necessarily more accurate or sensitive) technique.

Immunoassay techniques, including enzyme-linked immunosorbent assays and dipstick tests, for aflatoxin detection have been developed. These kits are commercially available. Immunoaffinity and fluorimetric detection have been combined in developing biosensors able to detect aflatoxin down to 50 μg/kg in as little as 2 min. The polymerase chain reaction (PCR) has been applied to detect the presence of aflatoxigenic fungi in foods, targeting aflatoxin biosynthetic genes.

Occurrence of aflatoxigenic molds and aflatoxins. While *A. flavus* is widely distributed in nature, *A. parasiticus* is less widespread. The actual extent of *A. parasiticus* occurrence in nature is likely greater than reported, since both species are reported indiscriminately as *A. flavus*. *A. flavus* and *A. parasiticus* have a strong affinity for nuts and oilseeds. Corn, peanuts, and cottonseed are the most important crops invaded by these molds, and in many instances, invasion takes place before harvest, not during storage as was once believed. Peanuts are invaded while still in the ground if the crop suffers drought stress or related factors. In corn, insect damage to developing kernels allows the entry of aflatoxigenic molds, but invasion can also occur through the silks of developing ears. Cottonseeds are invaded through the nectaries.

Significant amounts of aflatoxins can occur in peanuts, corn, and other nuts and oilseeds, particularly in some tropical countries, where crops may be grown under marginal conditions and where drying and storage facilities are limited. *A. flavus* grows well on cereals and spices, but aflatoxin production in these commodities can be prevented with proper drying, handling, and storage.

Control and inactivation. Control of aflatoxins in commodities generally relies on screening techniques that separate the affected nuts, grains, or seeds. In corn, cottonseed, and figs, screening for aflatoxins can be done by examination under UV light: those particles which fluoresce may be contaminated. All peanuts fluoresce when exposed to UV light, so this method is not useful for detecting kernels that are contaminated with aflatoxins. Peanuts containing aflatoxins are removed by electronic color-sorting machines, which detect discolored kernels.

Aflatoxins can be partially destroyed by chemical treatments. Oxidizing agents, such as ozone and hydrogen peroxide, remove aflatoxins from contaminated peanut meals. Although ozone is effective in destroying aflatoxins B_1 and G_1 after exposure at 100°C for 2 h, there is no effect on aflatoxin B_2. The treatment also decreases the lysine content of the meal. Treatment with hydrogen peroxide destroys 97% of aflatoxin in defatted peanut meal. The most practical chemical method of aflatoxin destruction is the use of anhydrous ammonia gas at elevated temperatures and pressures, with a 95 to 98% reduction in total aflatoxin in peanut meal. This technique is used commercially for detoxification of animal feeds in Senegal, France, and the United States.

The ultimate method for the control of aflatoxins in commodities, particularly peanuts, is to prevent the plants from becoming infected with the molds. Progress toward this goal is being made through a biological control strategy, which is the early infection of plants with nontoxigenic strains of *A. flavus* to prevent the subsequent entry of toxigenic strains.

Aflatoxins are among the few mycotoxins covered by legislation. Statutory limits are imposed by some countries on the amounts of aflatoxin that can be present in particular foods. The limit imposed by most Western countries is 5 to 20 μg of aflatoxin B_1/kg (parts per billion) in several human foods, including peanuts and peanut products. The amount allowed in animal feeds varies, but up to 300 μg/kg is allowed in feedstuffs for beef cattle and sheep in the United States. There are no statutory limits for aflatoxins in foods or feeds in Southeast Asia.

Other Toxigenic Aspergilli

Authors' note

Aflatoxin B_1 is considered the most toxic member of the aflatoxin family. The toxin is a potent carcinogen, linked to various types of cancer that are prevalent in China, Saharan Africa, and Southeast Asia.

Although aflatoxins are of greatest concern, there are other toxins produced by other *Aspergillus* species. *A. ochraceus* (Fig. 21.5A) is a widely distributed mold that is particularly common in dried foods, such as nuts (e.g., peanuts and pecans), beans, dried fruit, biltong (a South African dried meat), and dried fish. *A. ochraceus* produces ochratoxin. Ochratoxin A is the major toxin, with minor components of lower toxicity designated ochratoxin B and C. Other reported toxic metabolites are xanthomegnin and viomellein. Other *Aspergillus* species closely related to *A. ochraceus* can also produce ochratoxin A. *Aspergillus sclerotiorum, Aspergillus alliaceus, Aspergillus melleus,* and *Aspergillus sulphureus* have all been reported to produce ochratoxins, but in *A. ochraceus* and related species, only a portion of isolates are toxigenic. Production of ochratoxin A by *Aspergillus niger* var. *niger* was reported in 1994. *A. carbonarius* also produces ochratoxin A.

A. versicolor (Fig. 21.5B) is the most important food spoilage and toxigenic species in *Aspergillus* section *Versicolores*. *A. versicolor* is the major producer of sterigmatocystin, a carcinogenic compound that is a precursor of the aflatoxins. Surprisingly, *A. versicolor* does not produce aflatoxins, but it produces ochratoxin A. Sterigmatocystin has been reported to occur naturally in rice in Japan, wheat and barley in Canada, cereal-based products in

the United Kingdom, and Ras cheese. Sterigmatocystin has low acute oral toxicity because it is relatively insoluble in water and gastric juices. As a liver carcinogen, sterigmatocystin is ~1/150 as potent as aflatoxin B_1 but is still much more potent than most other known liver carcinogens.

A. fumigatus, section *Fumigati* (Fig. 21.5C), is best recognized as a human pathogen that causes aspergillosis of the lung. Its prime habitat is decaying vegetation. *A. fumigatus* is isolated frequently from foods, particularly stored commodities, but is not regarded as a serious spoilage mold.

A. fumigatus is capable of producing several toxins that affect the central nervous system, causing tremors. Fumitremorgens A, B, and C are toxic cyclic dipeptides that are produced by *A. fumigatus* and *Aspergillus caespitosus*. Verruculogen, also produced by these species, has a structure similar to that of the fumitremorgens but with a different side chain. Verruculogen is tremorgenic and appears to cause inhibition of the alpha motor cells of the anterior horn.

A. terreus (Fig. 21.5D) occurs commonly in soil and in foods, particularly stored cereals and cereal products, beans, and nuts, but is not regarded as an important spoilage mold. *A. terreus* can produce a group of tremorgenic toxins known as territrems. *A. clavatus* (Fig. 21.5E) is found in soil and decomposing plant materials. Although *A. clavatus* has been reported to be present in various stored grains, it is especially common in malting barley, an environment particularly suited to its growth and sporulation. *A. clavatus* produces patulin, cytochalasins, and the tremorgenic mycotoxins tryptoquivaline, tryptoquivalone, and related compounds.

PENICILLIUM SPECIES

The discovery of penicillin in 1929 provided impetus to search for other *Penicillium* metabolites with antibiotic properties and ultimately led to the recognition of citrinin, patulin, and griseofulvin as "toxic antibiotics," later called mycotoxins. The literature on toxigenic penicillia is now quite vast. In a comprehensive review of the literature on fungal metabolites, ~120 common mold species were demonstrably toxic to higher animals. Forty-two toxic metabolites are produced by one or more *Penicillium* species. At least 85 *Penicillium* species are listed as toxigenic. Classification within the genus *Penicillium* is based primarily on microscopic morphology (Fig. 21.2). The genus is divided into subgenera based on the number and arrangement of phialides and metulae and *rami* (elements supporting phialides) on the stipes.

Significant *Penicillium* Mycotoxins

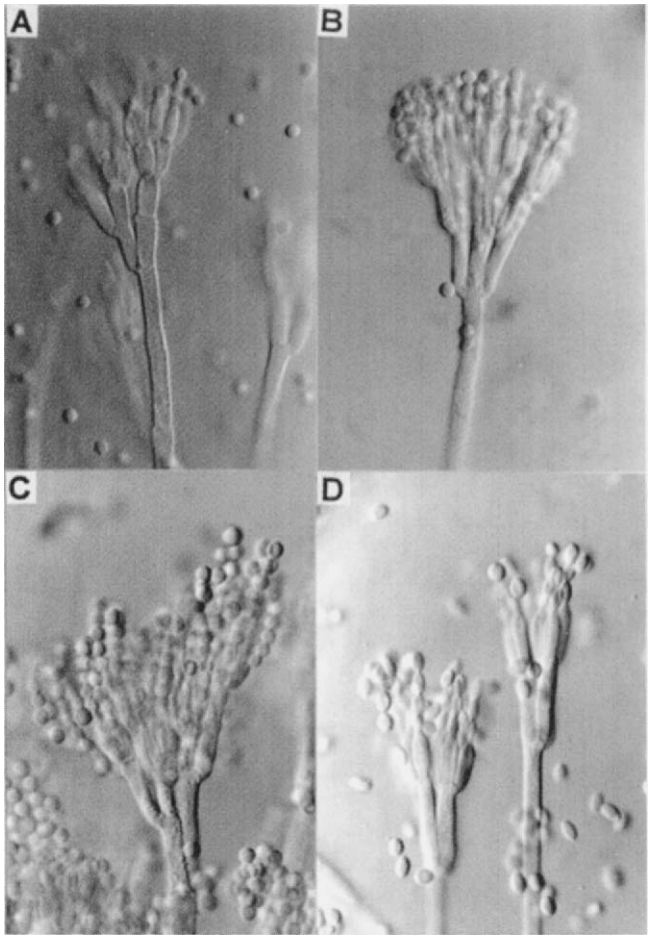
As already noted, >80 *Penicillium* species are reported to be toxin producers. The range of mycotoxin classes produced by *Penicillium* species is broader than for any other fungal genus. Toxicity due to *Penicillium* species is very diverse. However, most toxins can be placed in two broad groups, those that affect liver and kidney function and those that are neurotoxins. Generally, the *Penicillium* toxins that affect liver or kidney function are asymptomatic or can cause generalized debility in humans or animals. In contrast, the toxicity of the neurotoxins in animals is often characterized by sustained trembling. However, individual toxins show wide variations from these generalizations.

In fact, the most important toxin produced by a *Penicillium* species, ochratoxin A, is derived from isocoumarin linked to the amino acid phenylalanine. The target organ of toxicity in all mammalian species is the kidney.

The lesions can be produced by both acute and chronic exposure. Ochratoxin A is produced by *P. verrucosum* (Fig. 21.6A), which grows in barley and wheat crops in cold climates, especially Scandinavia, central Europe, and western Canada. *P. verrucosum* grows most strongly at relatively low temperatures, down to 0°C, with a maximum near 31°C. The major source of ochratoxin A in foods is bread made from barley or wheat in which *P. verrucosum* has grown. Because ochratoxin A is fat soluble and not readily excreted, it also accumulates in the depot fat of animals that eat feeds containing the toxin, and from there it is ingested by humans eating, for example, pig meats. It has been suggested that ochratoxin A is a causal agent of Balkan endemic nephropathy, a kidney disease with a high mortality rate in certain areas of Bulgaria, the former Yugoslavia, and Romania.

Citrinin is a significant renal toxin affecting domestic animals, such as pigs and dogs. It is also an important toxin in domestic birds, in which it produces watery diarrhea, an increase in water consumption, and reduced weight gain due to kidney degeneration. The importance of citrinin in human health is difficult to assess. Primarily recognized as a metabolite of *Penicillium citrinum*, citrinin is produced by more than 20 other fungal

Figure 21.6 Some toxigenic *Penicillium* species (magnification, ×750). **(A)** *P. verrucosum;* **(B)** *P. crustosum;* **(C)** *P. roqueforti;* **(D)** *P. oxalicum.* Reprinted from J. I. Pitt, p. 551–562, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).



species. *P. citrinum* is a *ubiquitous* (i.e., found in many environments) mold and has been isolated from nearly every kind of food surveyed for fungi. The most common sources are cereals, especially rice, wheat, and corn; milled grains; and flour. Citrinin is produced over most of the growth temperature range, but the effect of a_w is unknown.

Patulin is a lactone, and it produces teratogenic effects (that is, it causes fetal malformations [birth defects]) in rodents, as well as neurological and gastrointestinal effects. The most important *Penicillium* species producing patulin is *Penicillium expansum,* best known as a fruit pathogen but also widespread in other fresh and processed foods. *P. expansum* produces patulin as it rots apples and pears. Patulin appears to lack chronic toxic effects in humans, and therefore, low levels in juices are perhaps of little concern. However, because apple juice is widely consumed by children as well as adults, some countries have set an upper limit of 50 μg/liter for patulin in apple juice and other apple products. It is also important as an indicator of the use of poor-quality raw materials in juice manufacture. Patulin is produced by several other *Penicillium* species besides *P. expansum* (e.g., *Penicillium griseofulvum* and *Penicillium roqueforti*), but the potential for production of unacceptable levels in foods appears to be much lower. Patulin is quite stable in apple juice during storage; pasteurization at 90°C for 10 s caused a <20% reduction in levels of patulin.

Cyclopiazonic acid, a highly toxic compound, causes fatty degeneration and hepatic cell necrosis in the liver and kidneys of domestic animals. Cyclopiazonic acid is produced by *A. flavus* and at least six *Penicillium* species (e.g., *Penicillium camemberti* Thom, *P. commune*, and *P. griseofulvum*). Along with *A. flavus, P. commune*, a frequent cause of cheese spoilage, is probably the most common source of cyclopiazonic acid in foods. Other species producing this toxin have a wide range of habitats.

Acute cardiac beriberi was a common disease in Japan in the second half of the 19th century and has been linked to citreoviridin. The symptoms are heart distress, labored breathing, nausea, and vomiting, followed by anguish, pain, restlessness, and sometimes maniacal behavior. In extreme cases, progressive paralysis leading to respiratory failure occurred. The major source of citreoviridin is *Penicillium citreonigrum* (synonyms, *Penicillium citreoviride* and *Penicillium toxicarium*), a species which usually occurs in rice, less commonly in other cereals, and rarely in other foods. Citreoviridin is produced at temperatures from 10 to 37°C, with a maximum near 20°C.

Several tremorgenic mycotoxins are produced by *Penicillium* species. The most important is the highly toxic penitrem A. Verruculogen, which is equally toxic, is not produced by species common in foods. Less toxic compounds include fumitremorgen B, paxilline, verrucosidin, and janthitrems. At doses of ~1 mg/kg of body weight, penitrem A causes brain damage and death in rats, but the mechanism remains unclear. The only common foodborne species producing penitrem A is *P. crustosum* (Fig. 21.6B). Nearly all isolates of *P. crustosum* produce penitrem A at high levels, so the presence of this species in food or feed is a warning signal. Penitrem A appears to be produced only at high a_w. *P. crustosum* is a ubiquitous spoilage mold, occurring almost universally in cereal and animal feed samples. *P. crustosum* causes spoilage of corn, processed meats, nuts, cheese, and fruit juices, as well as being a weak pathogen on pomaceous fruits and cucurbits.

P. roqueforti (Fig. 21.6C), a species which is used in cheese manufacture but which can also be a spoilage mold, produces two toxins of interest.

Roquefortine has a relatively high 50% lethal dose (340 mg/kg intraperitoneally in mice), but it has been reported to be the cause of the deaths of dogs in Canada, with symptoms similar to strychnine poisoning. PR toxin (produced by *P. roqueforti*) is apparently much more toxic (50% lethal dose, 6 mg/kg intraperitoneally in mice) than roquefortine, but it has not been implicated in animal or human disease. Due to their potential occurrence in staple foods, PR toxin and roquefortine are of considerable public health significance.

Secalonic acids are dimeric xanthones produced by a range of taxonomically distantly related molds. Secalonic acid D, the only secalonic acid produced by *Penicillium* species, has significant animal toxicity. Secalonic acid D is produced as a major metabolite of *Penicillium oxalicum* (Fig. 21.6D). A major habitat for *P. oxalicum* is corn at harvest. However, the role of secalonic acid D in human or animal disease remains unclear.

FUSARIA AND TOXIGENIC MOLDS OTHER THAN ASPERGILLI AND PENICILLIA

Authors' note

Saprophytic organisms live on dead plant tissue. They do not affect the health of a plant.

The most important group of mycotoxigenic molds other than *Aspergillus* and *Penicillium* species are species of the genus *Fusarium*. *Fusarium* species are most often found as contaminants of plant-derived foods, especially cereal grains. Many *Fusarium* species are plant pathogens, while others are *saprophytic;* most can be found in the soil. As such, these molds and their toxic metabolites (mycotoxins) find their way into animal feeds and human foods (Table 21.3). In terms of human foods, *Fusarium* species are most often encountered as contaminants of cereal grains, oil seeds, and beans. Another mold genus, other than *Aspergillus* and *Penicillium*, whose members can produce mycotoxins is *Alternaria*. *Alternaria* species are widely distributed in the environment and can be found in soil, decaying plant materials, and dust.

Toxigenic *Fusarium* Species

The most common toxic species, discussed below, belong to the sections *Sporotrichiella* (*Fusarium sporotrichioides* and *Fusarium poae*), *Gibbosum* (*Fusarium equiseti*), *Discolor* (*Fusarium graminearum* and *Fusarium culmo-*

Table 21.3 Major mycotoxins that may be produced by *Fusarium* species of importance in cereal grains and grain-based foods[a]

Section	Species	Potential mycotoxin(s)
Sporotrichiella	*F. poae*	Type A trichothecenes, T-2 toxin, diacetoxyscirpenol
	F. sporotrichioides	T-2 toxin
Gibbosum	*F. equiseti*	Unknown
Discolor	*F. graminearum*	Deoxynivalenol (vomitoxin), 3-acetyldeoxynivalenol, 15-acetyldeoxynivalenol, zearalenone, possibly others
	F. culmorum	Nivalenol, zearalenone
Liseola	*F. verticillioides*	Fumonisins and others
	F. proliferatum	Fumonisins, moniliformin, and others
	F. subglutinans	Moniliformin and others

[a]Reprinted from L. B. Bullerman, p. 563–578, *in* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. (ASM Press, Washington, DC, 2007).

rum), and *Liseola* (*Fusarium verticillioides*, *Fusarium moniliforme*, *Fusarium proliferatum*, and *Fusarium subglutinans*). The major mycotoxins produced by these species are summarized in Table 21.3. The genus *Fusarium* is characterized by production of septate *hyphae* that generally range in color from white to pink, red, purple, or brown due to pigment production. The most common characteristic of the genus is the production of large septate, crescent-shaped, fusiform, or sickle-shaped spores known as *macroconidia*. The macroconidia exhibit a foot-shaped basal cell and a beak-shaped or snout-like apical cell (Fig. 21.3A). Some species also produce smaller one- or two-celled conidia known as *microconidia* (Fig. 21.3B), and some produce swollen, thick-walled chlamydospores in the hyphae or in the macroconidia (Fig. 21.3B).

F. poae and *F. sporotrichioides* are in section *Sporotrichiella*. *F. poae* is widespread in soils of temperate regions and is found on grains, such as wheat, corn, and barley. It exists primarily as a saprophyte but may be weakly parasitic. It is most commonly found in temperate regions of Russia, Europe, Canada, and the northern United States. *F. poae* has also been found in warmer regions, such as Australia, India, Iraq, and South Africa. *F. sporotrichioides* is found in soil and a wide variety of plant materials. The mold is found in the temperate to colder regions of the world, including Russia, northern Europe, Canada, the northern United States, and Japan. It can grow at low to very low temperatures, e.g., at −2°C on grain overwintering in the field.

Of the species in section *Gibbosum*, only *F. equiseti* is discussed here. *F. equiseti* has been found in soils from Alaska to tropical regions and has been isolated from cereal grains and overwintered cereals in Europe, Russia, and North America. However, *F. equiseti* is particularly common in tropical and subtropical areas. For the most part, *F. equiseti* is saphrophytic, but it may be pathogenic to plants such as bananas, avocados, and cucurbits. *F. equiseti* may contribute to leukemia in humans by affecting the immune system.

F. graminearum, in section *Discolor*, is a plant pathogen found worldwide in the soil and is the most widely distributed toxigenic *Fusarium* species. It causes various diseases of cereal grains, including gibberella ear rots in corn and fusarium head blight or scab in wheat and other small grains. These two diseases are important to food microbiology and food safety because the mold and its major toxic metabolites, deoxynivalenol and zearalenone, may contaminate grain and food products made from the grain.

F. culmorum is also widely distributed in the soil and causes diseases of cereal grains, among which ear rot in corn is important in food microbiology and food safety, since the mold and its toxins may contaminate corn-based foods. *F. culmorum* produces the mycotoxins deoxynivalenol, zearalenone, and acetyldeoxynivalenol.

The *Liseola* section contains four species of the genus *Fusarium* that are of interest: *F. verticillioides*, *F. moniliforme*, *F. proliferatum*, and *F. subglutinans*. *F. verticillioides* is a soilborne plant pathogen that is found in corn growing in all regions of the world. It is the most prevalent mold associated with corn. It often produces symptomless infections of corn plants but may infect the grain as well and has been found worldwide on food-grade and feed-grade corn. The presence of *F. verticillioides* in corn is a major concern because of the possible widespread contamination of corn and corn-based foods with its toxic metabolites, especially the fumonisins. The main human disease associated with *F. verticillioides* is esophageal cancer. Mycotoxins produced by *F. verticillioides* include fumonisins, fusaric acid, fusarins, and fusariocins.

Authors' note

Hyphae are branched or unbranched filaments. A mass of hyphae is referred to as a mycelium.

F N V I W N M Z O Q D W A

U S P T A S U V A L F N L

S U D R W F I A L J T L E

A L P O A I L K R I O K I

R L O F T M L A B E G T W

I I L E B P I A T J V O T

U G Y U M Y C O T O X I N

M R K Q A T I V Z W X Q L

W E E O E S N M M U U I B

L P T R A J E O L D R R N

X S I M O R P H O L O G Y

S A D P A T U L I N D R A

L R E T T Y W X F D K R Y

Clues

The first mycotoxin to be a public health concern. _____

Because molds grow so slowly, their media usually contain _____ compounds.

Mold genus that is "color coded" for easy identification. _____

Species of *Aspergillus* that makes aflatoxins. _____

Fungal genus most commonly associated with grains. _____

Molds are identified and their species determined primarily by their _____.

A _____ is a chemical toxin made by molds.

_____ is a relatively benign mycotoxin, but its level in apple juice is regulated because children drink so much apple juice.

_____ is a genus of mold known for its ability to make antibiotics as well as spoiling food.

Aflatoxins are synthesized in the _____ pathway.

Penicillium _____ is used in the production of a cheese that bears its name.

Ingestion of aflatoxin can cause severe damage to the _____.

F. proliferatum is closely related to *F. verticillioides,* yet less is known about this species, possibly because of its frequent misidentification as *F. moniliforme.* It is also frequently isolated from corn, where it probably occurs in much the same way as *F. verticillioides. F. proliferatum* is capable of producing fumonisins but has not been associated with animal or human diseases.

F. subglutinans is very similar to *F. verticillioides* and *F. proliferatum.* It is widely distributed on corn and other grains. Little information is available about this mold, probably because of its misidentification as *F. moniliforme. F. subglutinans* has not been specifically associated with any reported animal

or human diseases, but it has been found in corn from regions with high incidences of human esophageal cancer.

Detection and Quantitation of *Fusarium* Toxins

Fusarium species produce several toxic or biologically active metabolites. The toxins are trichothecenes, a group of closely related compounds. The trichothecenes are divided into three groups: type A (diacetoxyscirpenol, T-2 toxin, HT-2 toxin, and neosolaniol), type B (deoxynivalenol, 3 acetyldeoxynivalenol, 15-acetyldeoxynivalenol, nivalenol, and fusarenon-X), and type C (satratoxins). Of these, the toxin most commonly found in cereal grains and most often associated with human illness is deoxynivalenol. Other *Fusarium* toxins associated with diseases are zearalenone and the fumonisins. The most common chromatographic separation techniques used to identify these toxins are thin-layer chromatography and high-performance liquid chromatography. Gas chromatography also has some applications, particularly when coupled with mass spectrometry. Immunoassays have been developed for *Fusarium* toxins. Qualitative kits for screening, as well as kits for quantitative analysis, are available for deoxynivalenol, zearalenone, T-2 toxin, and fumonisins.

Other Toxic Molds

Other potentially toxic molds that may contaminate foods include species of the genera *Acremonium*, *Alternaria*, *Chaetomium*, *Cladosporium*, *Claviceps*, *Myrothecium*, *Phomopsis*, *Rhizoctonia*, and *Rhizopus*, as well as the species *Diplodia maydis*, *Phoma herbarum*, *Pithomyces chartarum*, *Stachybotrys chartarum*, and *Trichothecium roseum*. However, most of these molds are more likely to be present in animal feeds, and their significance for food safety may be minimal. Some have been shown to produce toxic secondary metabolites in vitro which have yet to be found to occur naturally.

The ergot mold, *Claviceps purpurea*, is the cause of the earliest recognized human mycotoxicosis, known as ergotism. Convulsive ergotism may have been the cause of the behavior that led to the witchcraft trials of 1692 in Salem, MA. *Alternaria* species infect plants in the field and may contaminate wheat, sorghum, and barley. *Alternaria* species also infect various fruits and vegetables, including apples, pears, citrus fruits, peppers, tomatoes, and potatoes.

Summary

- *Aspergillus*, *Penicillium*, and *Fusarium* are the mold genera most commonly associated with foods.
- Toxigenic molds are most often found as contaminants of plant-derived foods.
- The characteristics used to differentiate and identify molds are microscopic morphology and gross physiological features, including colony diameters and the colors of conidia and colony pigments.
- Specific growth media that inhibit the growth of bacteria are required for molds.
- Hydrogen peroxide is used commercially for detoxification of grains and other commodities.
- Molds can grow at low to very low temperatures, e.g., at −2°C on grain overwintering in the field.

Suggested reading

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Questions for critical thought

1. The presence of patulin in apple juice is a human health risk. Why? Detection of patulin may indicate quality control problems. Explain.
2. Which aflatoxins are covered by legislation and why?
3. How would you identify and differentiate *Aspergillus* and *Penicillium*?
4. Food microbiologists are concerned with the presence of *F. verticillioides* in corn because of the possible widespread contamination of corn and corn-based foods with its toxic metabolites. What toxic metabolites does this microorganism produce and what is the main disease of concern for humans?
5. In identifying members of the genus *Fusarium*, what common (distinct) characteristics of the microorganism are used?
6. Compare the characteristics and properties of sterigmatocystin and aflatoxin.
7. List several ways to prevent the growth of mold in grains. List methods that can be used to detoxify grains.
8. Investigate a recent outbreak of foodborne illness related to the growth of toxigenic mold in the product. Describe the characteristics of the product, including processing, handling, and storage practices. Formulate a strategy that could have been implemented to prevent the outbreak.

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22 Viruses and Prions

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Bacteriophages in the Dairy Industry
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Summary
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- differentiate between viruses, bacteria, and prions
- recognize the characteristics of viruses and their life cycles
- understand the positive role of viruses in pathogen control, their detrimental effect in dairy fermentations, and their role in foodborne illness
- compare and contrast control strategies for viruses, bacteria, and prions in foods
- explain what a prion is and how it is transmitted
- appreciate that science is constantly changing

INTRODUCTION

Most of food microbiology, and the bulk of this book, is devoted to *bacteria*. However, viruses cause 50 to 66% of the foodborne illness cases in the United States. In addition to causing human illness, some viruses cause problems in the dairy industry. Still other viruses can be used to control pathogens, thereby preventing human illness.

Prions (pronounced pree-ons), the causative agents of mad cow disease and Creutzfeldt-Jakob disease (CJD), were responsible for 155 cases and 150 deaths in the United Kingdom (as of September 2005). Although they have not caused any illness in the United States, they are covered here because they are biological, can get into the food supply (although the U.S. Department of Agriculture [USDA] is trying to keep them out), and have already alarmed consumers.

Bacteria, viruses, and prions are fundamentally different (Table 22.1). While bacteria are complex, foodborne viruses consist of RNA and a few proteins. A prion is even simpler, consisting of a single protein molecule. There are many foodborne bacteria that grow in food and are easily cultured in the laboratory. There are only two major (and two minor) foodborne viruses. They do not grow in foods, cannot be cultured in the laboratory, and are difficult to detect. They are transferred from feces to food through poor hygiene. The food carries the viruses to the intestine, where they infect the person. Prions are proteins that are part of the food, and they are not inactivated by heat. Bacteria and viruses cause millions of cases of foodborne illness per year. No cases of prion-related CJD in the United States have been reported to date.

Table 22.1 Comparison of bacteria, viruses, and prions

Trait	Disease-causing agent		
	Bacteria	**Viruses**	**Prions**
Composition	Complex structures of lipid, carbohydrates, proteins, and genetic material	Single strand of RNA in a relatively simple structure	A specific type of protein
Types important in food	More than a dozen gram-positive and gram-negative organisms	Two most important are noroviruses and hepatitis A virus	Only one that we know of
Control in food	Control growth by changing intrinsic or extrinsic factors. Kill by applying energy.	Do not grow in food. Inactivate by heat. Prevent entry by good hygiene.	Do not grow in food. Not inactivated by heat. Public health measures prevent consumption of meat from "mad cows."
Disease	Infection or intoxication	Infection	Protein conformational conversion
Transmission	Eating food where bacteria have grown	Fecal-oral	Eating meat that contains prions
Detection methods	Relatively easy to culture in the microbiology lab	Electron microscopy or human cell culture	Examination of tissue at autopsy
No. of cases/yr in United States	75 million, with up to 60% caused by viruses		None yet

VIRUSES

Elementary Virology

There are just a few basics to understanding viruses.

1. Viruses are obligate intracellular parasites. This means that they *can only* live *inside* a host, *to its detriment*. Viruses have no life outside a living cell. There is even some debate as to whether they are alive at all! Thus, viruses are frequently called "particles." For reasons outlined below, viruses are very "picky" about what living cells (or hosts) they infect. Viruses have limited host specificity. A virus whose host is a bacterium cannot infect a human. Nor can a virus like the one that causes foot-and-mouth disease infect a human. However, sometimes a mutation allows a virus to jump the species barrier. Such a jump is the presumed origin of human immunodeficiency virus (HIV). If a similar species jump occurred, avian flu virus could start to infect humans.
2. Viruses are very simple, consisting only of DNA or RNA (but never both), some protein, and occasionally a small amount of glycoprotein.
3. The basic principle of viral infection is that their attachment proteins (also called "ligands") bind to receptor proteins on the host cell (Fig. 22.1). The ligand-receptor protein interaction is specific, much like the enzyme-substrate interaction. Bacterial viruses (bacteriophages) bind only to bacteria. The viruses that cause human intestinal diseases (i.e., the noroviruses that cause diarrhea) can bind only to the epithelial cells of the intestine. It is the ligand-receptor interaction that creates viral specificity.
4. The viral "life" cycle consists of four simple steps. First, the virus attaches to the host cell through the ligand-receptor interaction. Then, the genetic information in the virus (i.e., RNA or DNA) enters the host cell. Once in the host cell, the viral genome commandeers the host's metabolic machinery to replicate viral genes and synthesize viral protein. Finally, the viral parts spontaneously assemble into viruses that rupture the host and are released. Unlike bacteria, viruses do not replicate by doubling, they replicate in a "lytic burst."

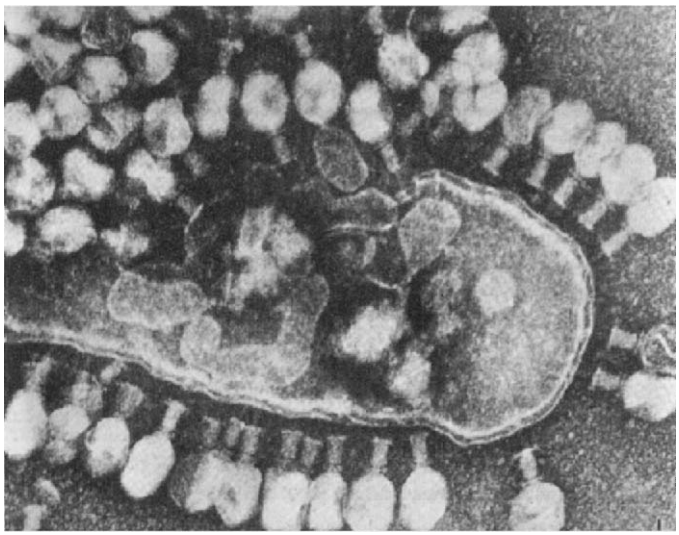


Figure 22.1 A T-even phage caught in the act of infecting a cell. Reprinted from http://www.armageddononline.org with permission.

This simple explanation chronicles the life of a *lytic* phage. The lytic phage enters the host cell, does its dirty work, and bursts out. There is another type of phage, the *temperate* phage. When these tricksters enter a host, their genes enter (or integrate) into the host genes. In this "hiding place," they are replicated as part of the host genome. These integrated phages are in the *lysogenic state*. The lysogenic phages can be dormant for a long time and then reactivate, as in the case of shingles or cold sores. At some point, the lysogenic phages "go lytic." When they leave the host genome, they can take some host DNA with them and transfer these genes into the next host they infect. The process is called *transduction*. Genetic engineering sometimes uses transducing phages as a mechanism of gene transfer.

Viruses that attack bacteria are called *bacteriophages* and have a relatively complex structure. Bacteriophages look like a lunar landing module. The head contains the viral genome, which is pushed through the tail when the receptor protein ligands interact with the host-specific receptors (Fig. 22.2). Human viruses, such as the norovirus (Fig. 22.3), are structurally simpler. Their DNA is surrounded by proteins called capsomes, and sometimes by an envelope. DNA enters the host by a more complicated mechanism, but the cycle still follows the one explained above.

Bacteriophages are counted in a fashion similar to the bacterial plate count. The "plaquing" of viruses exploits their lytic nature. The host bacteria are spread in a confluent layer on an agar plate. (The idea is to create a lawn of bacteria on the plate rather than producing a countable number of bacteria.) The viruses are serially diluted (in the same fashion as bacteria for plate counts) and spread over the surface of bacteria. The plates are incubated. Each virus infects and spreads through all the bacteria around it, lysing them and creating a clearing zone, or plaque (Fig. 22.4). The number of plaques is directly related to the initial number of virus particles and is expressed as plaque-forming units (PFU).

Viruses as Agents of Foodborne Illness

The ultimate source of foodborne viruses is human fecal matter. Humans are the *only* host for human gastrointestinal viruses. Viruses infect humans, grow in intestinal cells, and release large numbers of progeny into the feces.

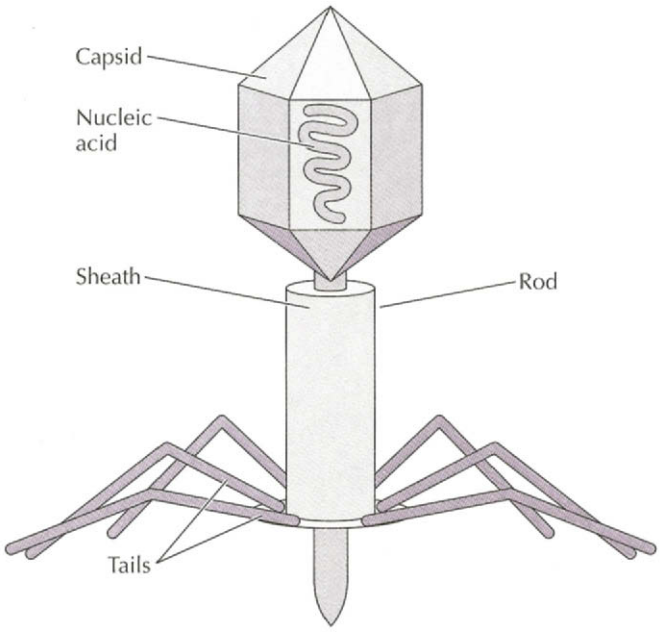


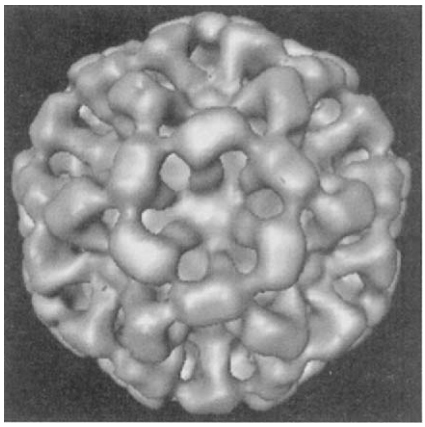
Figure 22.2 The structure of a T-even bacteriophage. Redrawn from http://www.armageddononline.org with permission.

A food handler's feces can easily contain 10^8 viruses per gram. Vomit can also carry viruses, with 10^7 particles in a single droplet. A simple exercise illustrates the power of the fecal-oral route. Imagine a food handler who is negligent in washing his hands after defecating. (Sorry, guys, but studies show that many more men than women fail to wash their hands after toileting.) A 0.01-g piece of feces hiding under his fingernails would contain 10^6 virus particles. If the infectious dose were 1,000 particles, the feces could get mixed into 1,000 servings of salad dressing, doughnut glaze, or whipped cream, causing illness in 1,000 people (1,000 particles/serving × 1,000 servings = the million virus particles originally under the fingernail). This example gives new meaning to the phrase "finger-licking good."

Similar incidents of fecal contamination occurred in the following situations:

- St. Paul, MN, where 3,000 people got sick after eating buttercream frosting prepared by an infected food handler.
- Cruise ships, where hundreds of people per ship became sick from eating contaminated fruits and ice.
- A central school lunch kitchen in Japan, where 3,200 children and 120 teachers were stricken.
- Michigan schools, where frozen strawberries distributed by the school lunch program sickened 230 people.

Figure 22.3 A norovirus. Courtesy of B. V. V. Prasad (Baylor College of Medicine).



Raw shellfish from water polluted with feces are a classic cause of viral food poisoning. Through their filtering action, bivalves concentrate viruses from the water by a factor of 100 to 1,000. The first reported outbreak (630 cases) of hepatitis A from oysters was caused when feces from an outhouse drained into an oyster bed. In 1988, clams harvested from polluted waters in China were eaten raw and caused 300,000 (!) cases of hepatitis A.

There are four types of enteric viruses. The most important are noroviruses. Since there are no official methods for detecting viruses in

foods, outbreak cases are discovered by their symptoms. The clinical criteria for noroviruses are listed in Table 22.2. Rotaviruses are usually transmitted by water, causing 130 million cases of diarrhea and 6 million deaths per year on a worldwide basis. Astroviruses and enteric adenoviruses also cause gastroenteritis.

Enteric viruses have simple structures. A coding strand of RNA is surrounded by a few structural proteins. The RNA makes more RNA inside the host cell by using an RNA-dependent RNA polymerase. The viral genome codes for the polymerase, the capsid proteins, and perhaps other virus-specific enzymes. There is no DNA in the replicative cycle of foodborne viruses.

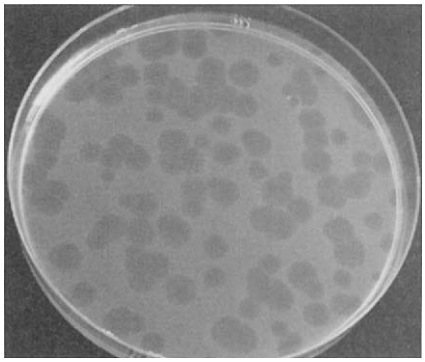


Figure 22.4 Clearing zones on the lawn of sensitive bacteria indicate plaques caused by viruses. From the ASM Microbe Library, courtesy of D. Sue Katz (Rogers State University) and Marie Panec (Moorpark College). Photograph taken by Grisel Quiroz.

Intervention

Viral particles are very persistent in the environment. They survive for hours on hands and for weeks on dry surfaces or vegetables. Nonetheless, it is, in theory, very simple to prevent viral foodborne diseases through a few simple measures. Viruses can be kept out of the food by using good food-handler hygiene and not using shellfish from polluted water. Using potable (i.e., drinking) water in all areas of food processing also prevents viral contamination. Heating food or sanitizing food contact surfaces kills viruses. Nonetheless, 50 million cases of foodborne viral disease per year suggest that these "simple" measures are not being followed.

Other antibacterial food processing methods may, or may not, kill viruses. Viruses are relatively acid resistant and can survive stomach acid. More radiation is needed to kill viruses than bacteria. Traditional heating methods kill viruses as easily as bacteria. The antiviral effectiveness of new, nonthermal methods such as high pressure, pulsed electric light, and electromagnetic waves is largely unknown.

Depuration is a process that attempts to cleanse shellfish of viruses by moving them from the polluted water to clean water. Over time, the shellfish should wash the viruses out with the clean water. This process is monitored using fecal coliforms as indicators. However, the viruses are frequently retained after the indicator coliforms have been washed out. This makes it hard to determine when depuration is complete.

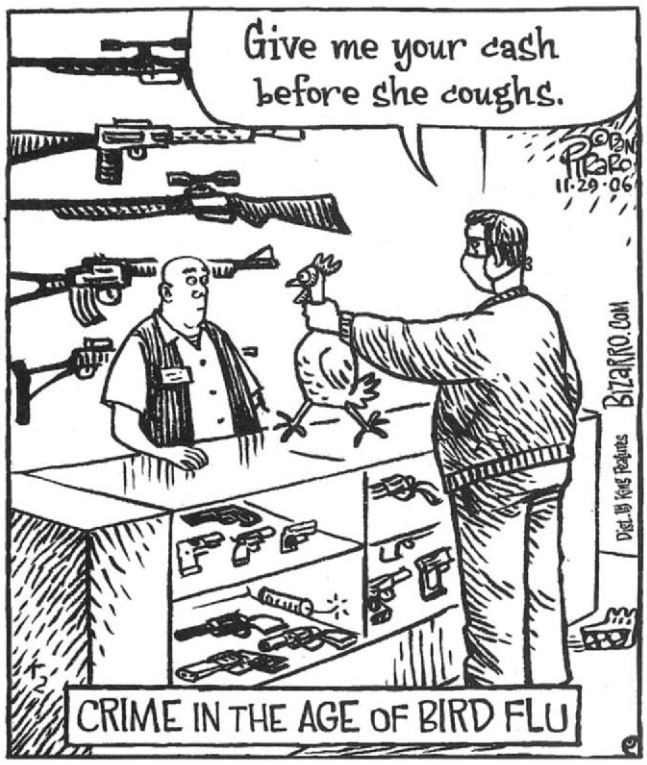
Noroviruses

Noroviruses, previously called Norwalk or Norwalk-like viruses (from their first outbreak at an elementary school in Norwalk, OH), are a major cause of foodborne disease. Outbreaks can have thousands of cases.

Table 22.2 Characteristics of the two major foodborne viruses

Characteristic	Norovirus	Hepatitis A virus
Disease symptoms	Nausea, projectile vomiting, diarrhea due to action on intestinal cells, lasts for 12–60 h	Jaundice, anorexia, vomiting, prolonged malaise, lasts for weeks to months
Incubation period	Usually 24–48 h, but can be as quick as 10 h	15–50 days (median, 29 days)
Shedding of viral particles in feces (or vomit)	Simultaneous with symptoms; continues a few weeks after illness	Shed for 10–14 days before symptoms appear
Immunity	Exposure does not generate immunity	Exposure confers durable immunity[a]
Type of virus	Calicivirus, spherical, 32-nm diam, with depressions on surface	Picornavirus, spherical, 28-nm diam
Estimated no. of foodborne cases/yr	50 million	2,000

[a]However, there can be a relapse years later. It is noteworthy that children get less severe symptoms and can have the disease with no symptoms at all. These children have lifelong immunity. Thus, in developing countries where most people are exposed as children, adult cases are rarer. In industrialized countries, adults are at greater risk.



Obviously this robber does not know about "species specificity."
© Dan Piraro. King Features Syndicate.

Noroviruses occupy a perennial position in the "top 10" agents of foodborne illness, occasionally winning the coveted number 1 spot. They, like all viruses, are difficult to study because they cannot be cultured like bacteria or grown in the laboratory. Noroviruses are detected by electron microscopy, antibodies, and genetic probes.

Noroviruses are small, round caliciviruses that contain a single strand of RNA. They are transmitted by the fecal-oral route. The viral particles are hearty, surviving storage, refrigeration, freezing, and passage through stomach acid. Upon reaching the small intestine, the virus infects mucosal cells, multiplies inside, kills the cell, and contaminates more feces. Viruses continue to be spread by fecal shedding for more than 1 week after the person recovers.

Hepatitis A

Only 5% of all hepatitis cases are food related, making them insignificant in the overall control of hepatitis. From the food safety side, however, this 5% amounts to about 2,000 cases per year. Hepatitis A is transmitted in foods by the feces of infected food handlers.

Hepatitis A has an incubation period of up to 50 days. This is troublesome because an infected person sheds the virus in their feces for 10 to 14 days *before* they get sick. The shedding can also continue for weeks after the person recovers. This makes it easy to unwittingly transmit the virus. Hepatitis is a nasty disease. It causes jaundice (yellowing of the skin), anorexia, vomiting, and a profound melancholy. These symptoms can last several weeks, and since there is no way to culture the virus, the diagnosis can be confirmed only by the detection of hepatitis-specific antibodies in the person's blood. Hepatitis can cause permanent or reoccurring kidney problems. When the virus infects hepatocytes (liver cells), the body's im-

mune system responds with T cells that are toxic to liver cells. This results in liver damage. In countries where fecal contamination is common, most children get a mild case (and acquire immunity!) of hepatitis by age five. In more developed countries, hepatitis A is more common among adults, who get more serious cases.

Hepatitis A virus, like noroviruses, is structurally simple. The spherical coat, consisting of four different proteins, surrounds and protects the coding strand of RNA. The RNA codes for the RNA-dependent RNA polymerase mentioned previously, a protease, and a large protein.

Noroviruses and hepatitis A virus are compared in Table 22.2.

Food Relatedness of Other Viruses

Astroviruses cause ~40,000 illnesses/year in the United States. They are round with star patterns on their coat and carry a single coding strand of RNA. There is an incubation period of 3 to 4 days. Diarrhea is the primary symptom, usually lasting 2 to 3 days; sometimes it can last as long as 2 weeks. Rotaviruses contain double-stranded RNA and are a common cause of infant death in developing countries.

Tick-borne encephalitis virus can be transmitted in raw milk. Poliomyelitis was one of the first foodborne viral diseases. Fortunately, with the advent of vaccination and pasteurization, it is no longer a significant problem in the United States. Parvoviruses, adenoviruses, and coronaviruses also cause gastroenteritis.

It is important to note that many viruses are not transmitted by foods. These include rabies, hantavirus, heptavirus, rhinoviruses (which cause the common cold), and HIV. HIV-positive people pose no risk as food handlers, and even those symptomatic with acquired immunodeficiency syndrome (AIDS) are not a threat if they do not have diarrhea. (Although HIV *is not* transmitted by the fecal-oral route, diarrheal fluid can contain *Salmonella*, *Escherichia coli* O157:H7, and noroviruses that *are* transmitted this way.)

Bacteriophages in the Dairy Industry

We have established that viruses can lyse their hosts. Sometime that host is bacterial. Now imagine a bacteriophage attack on a 5,000-gallon fermentor full of lactic acid bacteria and milk. The bacteriophages attack and infect the lactic acid bacteria. The lactic acid bacteria lyse, and with no bacteria to convert lactose into lactic acid, the fermentation becomes "stuck." In plants that process 1 million pounds per day, bacteriophage attacks become unthinkable.

There are several ways to combat bacteriophage attacks (Table 22.3). The least sophisticated method is asepsis. Fogging the plant with 20 parts per million of chlorine kills environmental bacteriophages. The control of airflow from the processed to the unprocessed side of the plant also helps prevent contamination. Asepsis (keeping organisms out) works well for the production of starter cultures but becomes impractical for large manufacturing facilities.

Culture rotation is the next level of bacteriophage control (Fig. 22.5). The processor starts with a collection of different starter culture strains (A to K). Instead of using a single strain of bacteria for the fermentation, a pool of three strains (A, B, and C) is used. Samples from the fermentor are plaqued against each strain on a regular basis to detect whatever phages are present. If one strain, C, is attacked by bacteriophages (as evidenced by positive plaques), that strain is removed from the starter culture pool and replaced with phage-insensitive strain D. If strain A is subsequently

Table 22.3 Defenses of lactic acid bacteria against bacteriophages

Defense	Mechanism
Biological	
Adsorption interference	Lactic acid bacteria can prevent bacteriophage adsorption if they lack specific proteins or mask the receptors that the bacteriophage recognize.
Restriction/modification	Resistant lactic acid bacteria have enzymes that modify their own DNA to distinguish it from phage DNA, which they destroy using restriction enzymes.
Abortive infection	These systems inhibit infection after the bacteriophage have adsorbed, penetrated, and started the lytic process.
Commercial	
Phage-inhibitory media (PIM)	Starter cultures are grown in media that lack Ca, which is required for phage adsorption.
Frozen concentrated starter cultures	Starters are grown in PIM, concentrated, and added to the fermentation at such high levels that no further growth is required.
Strain rotation	Lactic acid bacteria strains are constantly rotated so that bacteriophage-sensitive strains are identified and replaced with bacteriophage-insensitive strains.
Good manufacturing practices	Minimize contamination with viruses

attacked by phages, it is replaced with strain E, etc. The strains used in the fermentation are "rotated" to keep them ahead of the bacteriophage attack. To keep the pool of bacteriophage-resistant strains from drying up, the phage-sensitive strains are mutagenized to create more phage-insensitive strains that can be added back into the pool.

Keeping the starter culture bacteriophage free is another way to protect the fermentation. Here, phage-inhibitory media are used to grow the starter cultures. Bacteriophages require Ca^{2+} for absorption onto lactic acid bacteria. When phosphate is added to the media to chelate divalent cations, the bacteriophages cannot bind and the starter culture is protected.

The final method of protecting the fermentation is to use starter cultures at such high concentrations that they can carry out the fermentation without needing to multiply. If the bacteria are not growing, the bacteriophages cannot lyse them. This application uses lactic acid bacteria that cannot metabolize proteins and therefore cannot grow. When the bacteria are added at a final concentration of 10^7 to 10^8 CFU/ml, there is enough metabolic activity to conduct the fermentation in the absence of bacterial growth.

Beneficial Uses of Viruses

Bacterial Detection

Bacteria can also be used in positive food safety applications. Because bacteriophages can multiply more rapidly than their bacterial hosts, they can be used in rapid detection systems for specific pathogens. When bacteriophages are treated with a fluorescent dye and are exposed to a phage-sensitive host, they become concentrated (and visible) on the target bacteria. Phages can also carry *reporter genes* such as β-galactosidase. When a temperate phage integrates into the target pathogen's chromosome, the phage reporter genes are expressed and the presence of the pathogen is reported by the production of a color. The luminescence (*lux*) gene also makes a good reporter gene. Such detection systems have been developed for *E. coli* O157:H7, *Listeria monocytogenes*, and *Salmonella* species.

Bacterial Control

Not only can bacteriophages be used to detect pathogens, they can also be used to control them! The Food and Drug Administration (FDA) recently approved the use of a bacteriophage preparation, LMP-102, as a food addi-

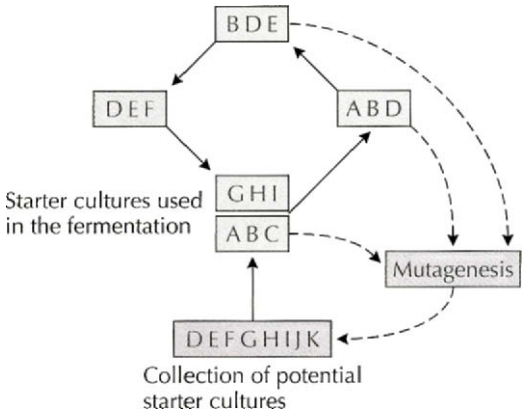


Figure 22.5 A diagrammatic explanation of culture rotation. A pool of bacteriophage-resistant lactic acid bacteria strains is used to construct a three-strain fermentation culture. As one strain becomes attacked by bacteriophages, it is replaced by another member of the pool. The phage-sensitive strain is mutagenized to phage resistance and returned to the pool for later use. (See the text for additional information.)

tive to control *L. monocytogenes* in ready-to-eat meat and poultry. Intralytix, Inc., petitioned for the approval after demonstrating that no toxic material was carried over from the *L. monocytogenes* cells used to produce the phage. The development of phage resistance is minimized by using a mixture of six phages specific to different *L. monocytogenes* strains. FDA had no toxicity or environmental issues with this antimicrobial since the phage is specific to *L. monocytogenes*. Intralytix, Inc., further assured FDA of the additive's safety by providing the complete sequence of this double-stranded DNA virus. Since the bacteriophage preparation contains lytic rather than lysogenic phages, there was no concern that it might transfer undesirable genes into new hosts. The Center for Science in the Public Interest (a consumer watchdog group) commended the safety of the virus and its utility in *Listeria* control. A similar phage product, ECP-100, is under development for control of *E. coli* O157:H7.

The use of bacteriophages as sanitizers is another intriguing application. The application takes an ecological approach. With traditional sanitizers, all the bacteria on a manufacturing surface are killed, leaving a vacant niche. Pathogens can rapidly fill this niche. By using pathogen-specific bacteriophages as a sanitizer, only the pathogen is killed and the remaining bacteria in the niche can prevent the pathogen reintroduction. The Environmental Protection Agency (EPA) has issued an experimental permit so that this application can be tested in a manufacturing setting.

PRIONS

Prions do not replicate like bacteria or commandeer the host's metabolic machinery as do viruses. However, several important lessons about the nature of science can be learned from the prion story:

- Science-based public policy can be based only on what is "known."
- The fact that something has not been discovered does not mean it does not exist.
- Established scientific "fact" is often theory, subject to modification.
- New findings can overthrow the most established dogma.

A Short History of the Prion

The first animal with symptoms of what we now know as transmissible spongiform encephalopathy (TSE) was reported in 1732, but the disease was not recognized as such at the time. Today, most consumers have heard of mad cow disease. This disease, more formally known as bovine spongiform encephalopathy (BSE), brought the British meat industry to its knees when 182,000 suspect cattle were destroyed. Regulators fear its appearance in the United States.

BSE did not appear (or perhaps exist?) in cattle until 1986, but the prion story started much earlier. In 1972 a young doctor named Stanley Prusiner became fascinated with a patient whose brain was being eaten away. However, there was no immune response and the rest of her body was untouched. Prusiner's studies of her disease were considered ridiculous and almost cost him his job. Animals that die from "spongiform encephalopathy" (SE) have decayed, spongy brains. SE was recognized as a symptom of scrapie in sheep, kuru in cannibals, chronic wasting disease in cattle and elk, and CJD in humans. However, until Prusiner, no one had linked the cause of these diseases, which have no treatment. They were thought to be caused by mutations, genetic predispositions, or "slow" viruses (since the incubation period was 3 to 5 years). Prusiner proposed the prion (coined by combining "proteinacous" and "infectious") concept to explain scrapies in 1982.

Authors' note

How or why "proin" became "prion," we do not know, but it did.

According to Prusiner, a prion is an infectious protein particle devoid of nucleic acid. The PrP^{c} is the prion protein normally found in healthy brain and backbone membranes. When it is contacted by the infectious modified prion, PrP^{sc}, the PrP^{c} refolds into the PrP^{sc} configuration.

The prion concept contradicted the dogma that only DNA and RNA transmit biological information. It was met with ridicule and derision. Furthermore, by postulating that prion proteins can exist in two forms, Prusiner broke the holy grail of molecular biology: DNA sequence is transcribed to RNA sequence, which is translated into a protein with a specific amino acid sequence that dictates its (only possible) structure. Protein structure ultimately determines protein function.

This would have remained an obscure piece of scientific history except for one thing: in 1986 BSE broke out in British cattle. Since that time, it has been detected in over 182,000 cattle in the United Kingdom. Although the disease was attributed to "slow viruses" at the time, prions had jumped the species barrier from sheep to cows. The PrP^{sc} is transmitted from animal to animal when they are fed offal or the rendered remains of infected animals. (The United States banned the use of offal in animal feed in 1989.) Over the next 10 years, the "statistical correlation" of beef consumption with a new form of CJD became clear in Britain. There were 155 probable human cases resulting in 150 deaths in the United Kingdom through February 2005. (Note the high fatality rate.) The link between CJD and beef caused the British beef industry to change their feeding regimes and kill all animals in herds afflicted with BSE. Well over 100,000 cattle were sacrificed as a preventive measure. In 1996, the British government admitted that BSE could be transmitted to humans as CJD. All hell broke loose. The dust still has not settled, but in 1997, Stanley Prusiner was awarded the Nobel Prize in Medicine for his discovery of the prion.

The U.S. experience with BSE has been different from that of England due to the United States' early banning of offal in animal feed and the testing of high-risk animals. In 2003, 1 cow of 394,000 tested in the United

States was identified as having BSE. American authorities quickly determined that the cow came from Canada and had never entered the food chain. Canadian authorities destroyed 2,700 head of cattle to prevent the possible spread of BSE-infected cattle. Due to this quick action, countries that had banned Canadian beef in May 2003 eased the ban a few months later. Unfortunately, two more cases of BSE were detected in Canada in 2005.

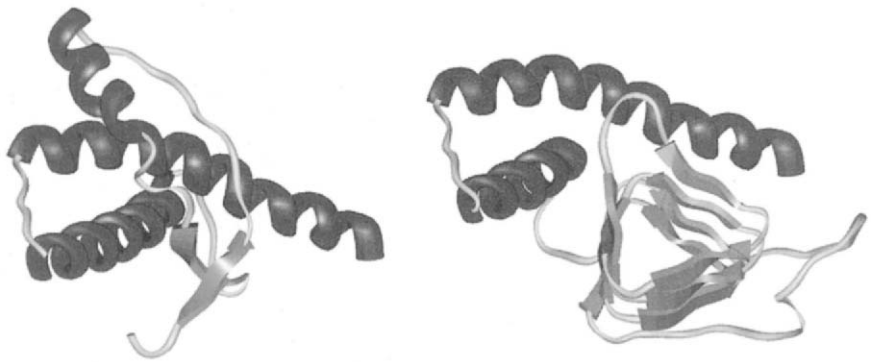
Prion Biology

The prion concept remains controversial, especially the "devoid of nucleic acid" part of the definition. It is difficult to prove that something does not exist. Maybe it exists but cannot be found. That is the argument against prions; there "must" be a small amount of nucleic acid, but it cannot be detected. However, treatments that destroy nucleic acids do not destroy prions, and nucleic acids have not been detected in purified prions. The concept of protein conformational conversion explains all of the *infectiously transmissible* SEs. However, protein conformational conversion is only one of three mechanisms that cause SEs. A hereditary defect in PrP^c, such as the one that causes Libyan Jews to get CJD 30 times more often than other Jews, causes a familial form of CJD. A mutation in PrP^c causes sporadic CJD in the elderly. Finally, there is the conversion of PrP^c into PrP^{sc} by PrP^{sc} consumed in the diet.

One must understand a bit of protein chemistry to understand how prions work. The *primary sequence* of amino acids in a protein determines its *secondary structure*. Common secondary structures are α helices and β-pleated sheets. A protein molecule may contain several helices, pleated sheets, and unordered sections. These arrange themselves to form the protein's overall shape, or conformation. This is called its tertiary structure. PrP^c has a conformation with several α helices. When it contacts PrP^{sc}, it refolds to match that conformation, which has large runs of pleated sheets (Fig. 22.6, arrows). In this conformation, the prion protein destroys brain tissue.

Prions are not inactivated by heat, irradiation, or food preservatives. Once they are in the food, they are there for good. The only way to prevent TSEs is to prevent the converted form from touching the normal form. This is done by making sure that animals do not eat animals. Most industrialized

Figure 22.6 Structures of the normal prion protein, PrP^c **(left)** and the infectious form, PrP^{sc} **(right)**. Note the changes in the secondary structures (coils and arrows) as well as the change in the overall shape (tertiary structure). Both forms have the same amino acid sequence. Reprinted with permission from Sarit Helman.



countries now ban the use of "animal by-products" as animal feed. The United States has banned importation of beef from countries where BSE is present. However, wild animals such as mink and deer also carry TSEs. Theoretically, these prions might jump the species barrier into domestic cattle.

Summary

- Viruses have a simple life cycle of adsorption, entry, replication, assembly, and release in a lytic burst.
- Viruses are host specific due to unique ligand-receptor binding.
- Viruses cannot be grown in culture, but bacteriophages can be plaqued on plates of sensitive bacteria.
- Viruses can cause as much foodborne disease as bacteria (if not more) but cannot grow in foods.
- Fecal-oral transmission is the main route of human infection with enteric viruses.
- Proper heating and good hygiene should keep foods safe from viruses.
- Noroviruses and hepatitis A virus are the major foodborne viruses.
- Viruses have useful applications such as pathogen detection and control.
- There are several methods to prevent bacteriophage attacks on fermenting dairy cultures.
- Prions are infectious proteins devoid of RNA and DNA and were discovered by Stanley Prusiner in the 1980s.
- Prions cause TSEs such as scrapie, kuru, mad cow disease, and CJD disease.
- TSEs are caused when normal PrP^{c} is converted into infectious PrP^{sc} by mutation or by contact with an ingested PrP^{sc} molecule.

Suggested reading

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Questions for critical thought

1. This chapter's chronology of mad cow disease is accurate up to 2005. What has happened with mad cow disease in North America since that time?
2. How are bacteria, viruses, and prions different?
3. What intervention can be used to make food safe from all of them?
4. There are no routine lab tests to detect viruses or prions in foods. How does this make their control different from controlling foodborne pathogens?

5. What is a prion? What are the three ways in which it can be formed?
6. Much of this book deals with how to prevent microbial growth in food. How can viruses be prevented from growing in foods?
7. Think of three terms that could be used to convey to your less sophisticated peers the concept of fecal-oral transmission.
8. Why is dilution *not* the solution for preventing viral contamination by shellfish? Diluting viruses in large volumes of water should bring them below the infectious dose. Consider the case of the outhouse over the oyster bed. Assume 50 people, all infected at 10^8 particles per g of feces, depositing 2 kg of fecal material per day into a body of water that is 1 km by 1 km and 10 m deep. How many viral particles per milliliter of water would there be?
9. How are biological viruses like computer viruses?
10. Compare the ways that computers can be protected from viruses to the ways that humans can be protected.
11. Draw a diagram of the life cycle of a temperate bacteriophage.
12. Write a paragraph (or two) designed to convince consumers that using viruses to kill *L. monocytogenes* is "safe and natural."

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SECTION V

Control of Microorganisms in Food

23 Antimicrobial Preservatives

24 Biologically Based Preservation and Probiotic Bacteria

25 Physical Methods of Food Preservation

26 Industrial Strategies for Ensuring Safe Food

23 Antimicrobial Preservatives

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Sodium Chloride
Disinfectants
Sulfites
Chlorine
Quaternary Ammonium Compounds
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Ozone
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Lysozyme
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Isothiocyanates
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- link chemical preservatives to their ability to prevent food spoilage
- distinguish positive and negative aspects of food preservatives
- characterize traits that distinguish "natural" from "chemical" preservatives
- describe factors that affect antimicrobial preservation systems and how these factors interact
- understand how organic acids inhibit microbes and the influence of pK_a and pH on their antimicrobial action
- identify enzymes and inorganic food preservatives, how they work, and in what foods they are useful

INTRODUCTION

Preservatives are good. Without them, food spoils. Spoilage can be caused by microbiological, enzymological, chemical, and physical changes. Preservatives can prevent or delay these changes. Most antimicrobials inhibit growth without killing the microbe. Chilling, freezing, water activity (a_w) reduction, acidification, fermentation, energy (e.g., thermal, irradiation, and pressure), and antimicrobial compounds all inhibit microbial growth.

Antimicrobials are sometimes called preservatives. However, when used alone, "preservative" is a broader term than "antimicrobial." Preservatives include antibrowning agents, antioxidants, stabilizers, etc. The major antimicrobial targets are foodborne pathogens and spoilage microorganisms. They cause off odors, off flavors, liquefaction, and discoloration and produce slime. Spoilage can be a good thing. It warns people not to eat foods that may contain pathogens. However, many pathogens can grow without obvious signs of spoilage. (If pathogens always spoiled food, no one would eat the food and there would be no foodborne illness.) Preservation systems that allow pathogen growth without spoilage are inherently dangerous.

Most antimicrobials only *inhibit* growth. They are bacterio*static* rather than bacterio*cidal*. Bacteriostatic compounds stop microbial growth; bacteriocidal compounds kill the bacteria. Antimicrobials act at specific metabolic targets. Antimicrobials having different mechanisms can be combined to give a synergistic effect. This is called *hurdle technology*. The cell wall, cell membrane, enzymes, and genes are all targets for antimicrobial action. Because metabolism is an interlocking web, it is often hard to determine an antimicrobial's exact mechanism of action.

Antimicrobial classification is arbitrary. This chapter divides antimicrobials into two classes, traditional "chemical" antimicrobials and naturally occurring ones.

Chemical antimicrobials

- have been used for years
- are approved by many countries
- can be made by synthetic means
- are usually considered "bad" by consumers

Chemical antimicrobials are also found in nature. These include acetic acid from vinegar, benzoic acid from cranberries, and sorbic acid from rowanberries.

Naturally occurring food preservatives refer to compounds that are

- extracted from natural sources
- not usually subject to regulatory approval as food additives
- "label friendly"

FACTORS THAT AFFECT ANTIMICROBIAL ACTIVITY

Foods are complex systems. Many variables influence microbial growth and inhibition. For instance, the food's pH, a_w, storage temperature, and gas composition impact antimicrobial effectiveness. These factors must be considered when developing preservation systems. Antimicrobial activity is affected by microbial, intrinsic, extrinsic, and process factors.

Microbial factors include the following:

- the organism's inherent resistance (e.g., vegetative cells versus spores) to the preservation method
- initial cell number
- growth rate
- growth phase
- interaction with other microorganisms (e.g., antagonism)
- cellular composition (which may be a function of the cells' prior history)
- physiological status (injury, adaptation, and metabolic state)

There is an explosion of knowledge about how cells respond to stress. Tolerance gained by exposure to one stressor, such as cold, may make cells more tolerant of another, such as acid. This is called adaptation, stress response, or acquired tolerance. Special proteins are produced when cells are stressed. They are responsible for the acquired tolerance. Some stress responses are specific to a given stressor. Others give broad resistance to a range of stressors.

Intrinsic factors are part of the food itself. These include composition, pH, buffering capacity, oxidation-reduction potential, and a_w. These can be specified during product development to formulate food safety into the food. When a product is reformulated to reduce salt, acidity, sugar, or other potentially inhibitory substances, the impact on microbial safety should be considered. For example, pathogens might grow in previously stable products when the products are reformulated to replace sucrose with an artificial sweetener. This removes the a_w barrier that prevents pathogen growth.

Extrinsic factors also influence antimicrobial activity. These variables are external to the food. They include temperature, storage time, atmo-

sphere, and relative humidity. The time-temperature combination is perhaps the most important. It is arguably the most difficult to control. (Consider the shopper who forgets that he left his refrigerated entrée in the trunk of his car and then eats it the next day.) Extrinsic factors also need consideration during product development. A manufacturer of frozen foods expanding into refrigerated entrées faces new microbiological challenges.

The microbiology of a *processed food* is often different from that of an unprocessed food. Processing can change food composition. For example, drying changes the a_w of a food. The addition of soluble compounds can also decrease a_w. Heating foods can eliminate vegetative cells and allow spores to thrive. Food processing can also change food microstructure.

ORGANIC ACIDS

The short-chain organic acids (Table 23.1) have similar mechanisms and applications. They are all found in nature but often made chemically for use in foods. They have generally recognized as safe (GRAS) status and are label friendly. Due to their mechanism of action (see below), organic acids are most useful below their pK_a (explained in the next paragraph). They are rarely used in foods with pH values of >5.5. This is a major limitation.

To be effective, organic acids must cross the bacterial cell membrane and get into the cytoplasm (Fig. 23.1). Organic acids exist in charged and uncharged forms. The charged form cannot cross the bacterial membrane and is essentially useless. It is the uncharged (protonated) acid that gets into the bacterium and inhibits growth. The pH at which the concentrations of charged and uncharged acid are equal is the pK_a. Since pH is a log scale, at one pH unit above the pK_a, there is 10-fold less protonated acid. This low concentration would not be inhibitory. This is why organic acids are most useful in acidic foods near or below their pK_a. At low pH, the protonated acid crosses the cell membrane. The neutral cytoplasmic pH then causes the acid to dissociate. The release of the proton acidifies the cytoplasm. The cell uses adenosine triphosphate (ATP) to pump protons out of the cell. This maintains the neutral intracellular pH needed for viability but uses up the ATP energy needed for growth.

Table 23.1 Short-chain organic acids

Organic acid	Formula	pK_a	Found in:	Active against:	Applications	Concn (%)
Acetic	CH_3COOH	4.5	Vinegar	Bacteria and yeasts	Bakery, cheese, olives, gravies, sauces, meat, poultry	<0.1–0.8
Benzoic		4.2	Cranberries	Fungi	Beverages, jams, jellies	<0.1
Lactic	$CH_3OHCOOH$	4.8	Fermented food	Bacteria	Used mainly for pH control and flavor. Also used for sanitation, spraying on meat	<0.4–2.0
Propionic	CH_3CH_2COOH	4.9	Swiss cheese	Yeasts, molds, rope-forming bacteria	Baked goods and cheeses	0.3–no limit
Sorbic	$CH_3CH{=}CHCH{=}CHCOOH$	4.7	Rowanberry	Fungi and certain bacteria	Baked goods, cheese, wine	0.1–0.3

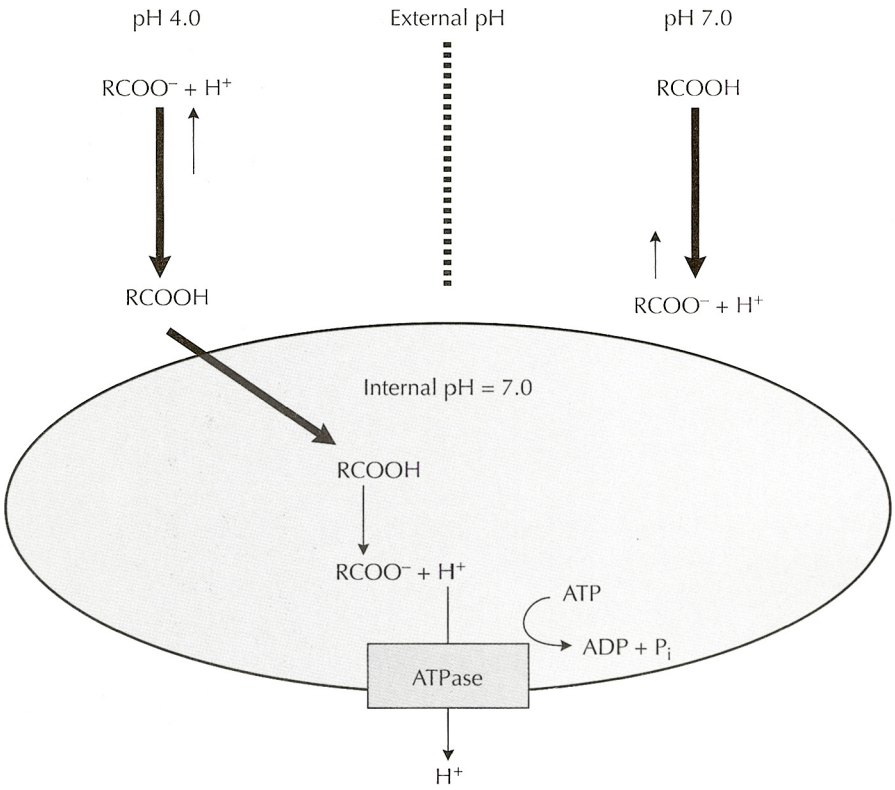


Figure 23.1 Organic acids are effective inhibitors only at low pH, where their protonated form can cross the cell membrane. At the higher pH inside the cell, the acid dissociates and the cell must use energy to pump out the proton to deacidify the cytoplasm. At high external pH, the dissociated acid cannot cross the membrane and get into the cell and therefore is not effective.

PARABENZOIC ACIDS

Benzoic acid has a pK_a similar to those of the organic acids. However, the carbon chains present between the benzene and the acid parts of the molecule increase the pK_a. These parabenzoic acids, commonly called "parabens," remain undissociated up to their pK_a of 8.5. This gives them an effective pH range of 3.0 to 8.0. The methyl, propyl, and heptyl parabens are approved antimicrobials in most countries.

The antimicrobial activity of parabens increases with increasing chain length. Unfortunately, their solubility in water decreases with increasing chain lengths. The oppositional influence of chain length and solubility determines which parabenzoate can be used in a given application. Methyl and propyl parabenzoates are often used in a 2:1 to 3:1 ratio to capitalize on their respective solubility and increased activity. The *n*-heptyl ester is used in beers, noncarbonated soft drinks, and fruit-based beverages. Parabens are generally more active against molds and yeasts than against bacteria (Table 23.1). Parabens are used in baked goods, beverages, fruit products, jams and jellies, fermented foods, syrups, salad dressings, wine, and fillings.

Phenolics and parabens act at microbial membranes, but the specific mechanisms are unclear. The membrane's barrier properties may be degraded, allowing intracellular compounds to leak out. Respiration may also be inhibited, perhaps due to changes in membrane fluidity.

NITRITES

Nitrites are used in cured meats such as bacon, hot dogs, ham, and bologna. Curing solutions contain nitrite, salt, sugar, spices, and ascorbate or erythorbate. Nitrite has many uses in cured meats. Nitrosomyoglobin makes cured meat pink. In addition to being antimicrobials, nitrites contribute to cured meat flavor and texture and serve as antioxidants. Nitrite's effectiveness depends on many factors. Nitrites work better at a lower pH. Nitrite is more inhibitory in the absence of air. The reducing agents ascorbate and isoascorbate enhance nitrite's antibotulinal action. Temperature, salt, and microbial load also influence nitrite's effectiveness.

Sodium nitrite inhibits *Clostridium botulinum* in cured meats. Nitrite also inhibits bacteria such as *Achromobacter, Enterobacter, Escherichia coli, Flavobacterium, Micrococcus, Pseudomonas,* and *Listeria monocytogenes*. Governmental regulations limit its use to 156 parts per million (ppm) for most cured meat products and 100 to 120 ppm in bacon. Sodium erythorbate or isoascorbate is required as a cure accelerator and to inhibit nitrosamine formation.

Nitrosamines are carcinogens formed by reactions of nitrite with amines. The discovery that nitrosamines are produced from nitrite and meat protein at high temperature triggered extensive research on nitrite substitutes. However, no other compound could replace nitrite's antimicrobial, taste, and color functions. Concern about nitrosamines decreased upon learning that vegetables were a larger source of dietary nitrites.

PHOSPHATES

Sodium acid pyrophosphate, tetrasodium pyrophosphate, sodium tripolyphosphate, sodium tetrapolyphosphate, sodium hexametaphosphate, and trisodium phosphate are active against microbes in foods.

Gram-positive bacteria are more susceptible to phosphates than gram-negative bacteria. Tetrasodium pyrophosphate, sodium tripolyphosphate, and sodium hexametaphosphate at 0.5% inhibit *Bacillus subtilis, Clostridium sporogenes,* and *Clostridium bifermentans*. Sodium acid pyrophosphate inhibits *C. botulinum* toxin production but not growth. Outgrowth of *Bacillus* spores is prevented by 0.2 to 1.0% sodium hexametaphosphate.

Polyphosphates' ability to chelate metal ions is responsible for their antimicrobial action. Polyphosphates probably inhibit gram-positive bacteria and fungi by removing essential cations from the cell wall binding sites. Divalent cations can reverse polyphosphate inhibition. Inhibition is also reduced at lower pH due to protonation of the chelating sites. Orthophosphates have no chelating ability and thus no inhibitory activity.

SODIUM CHLORIDE

Sodium chloride (NaCl), or common salt, is the oldest preservative known. A few foods, such as raw meats and fish, are preserved solely by high salt concentrations, i.e., "salting." More often, salt is combined with processes such as canning, pasteurization, or drying. Foodborne pathogens are usually inhibited by an a_w of 0.92 or less (equivalent to 13% [weight/volume] NaCl) under otherwise optimum conditions. The exception is *Staphylococcus aureus*, which can grow at an a_w of 0.83 to 0.86 under aerobic conditions. In anaerobic conditions, the minimum a_w is >0.90. *L. monocytogenes* is also salt tolerant. It survives in saturated salt solutions at low temperatures.

Sodium chloride's primary lethal mechanism is plasmolysis (shrinkage of the cell due to water loss). Salt reduces a_w, creating unfavorable growth conditions. As the a_w decreases, cells undergo osmotic shock and rapidly lose water through plasmolysis. During plasmolysis, a cell dies or remains dormant. To resume growth, the cell must concentrate specific solutes to equilibrate with the external a_w. However, they must use up ATP to concentrate these solutes, and that loss of cellular energy results in growth inhibition or death.

DISINFECTANTS

Disinfectants are not food preservatives in a strict sense, but they do kill or inhibit microorganisms that can spoil food. They can be indirect or incidental food additives used on food contact surfaces. These applications are regulated by the Environmental Protection Agency rather than the Food and Drug Administration (FDA). Disinfectants are becoming increasingly important as the U.S. Department of Agriculture (USDA) begins to require microbial-reduction steps in the processing of raw meat and poultry. Most often, these take the form of an antimicrobial wash.

Sulfites

Sulfur dioxide (SO_2) has been used as a disinfectant since ancient times. Antimicrobial salts of sulfur dioxide include potassium sulfite (K_2SO_3), sodium sulfite (Na_2SO_3), potassium bisulfite ($KHSO_3$), sodium bisulfite ($NaHSO_3$), potassium metabisulfite ($K_2S_2O_5$), and sodium metabisulfite (NaS_2O_5). Sulfites are used primarily in fruit and vegetable products to control spoilage yeasts and molds, and bacteria. Bacteria are generally more sensitive than yeasts or molds. This allows wine to be treated with sulfites to inhibit the bacteria that cause the malolactic fermentation, without interfering with the yeast's alcoholic fermentation. Sulfites also act as antioxidants that inhibit enzymatic browning.

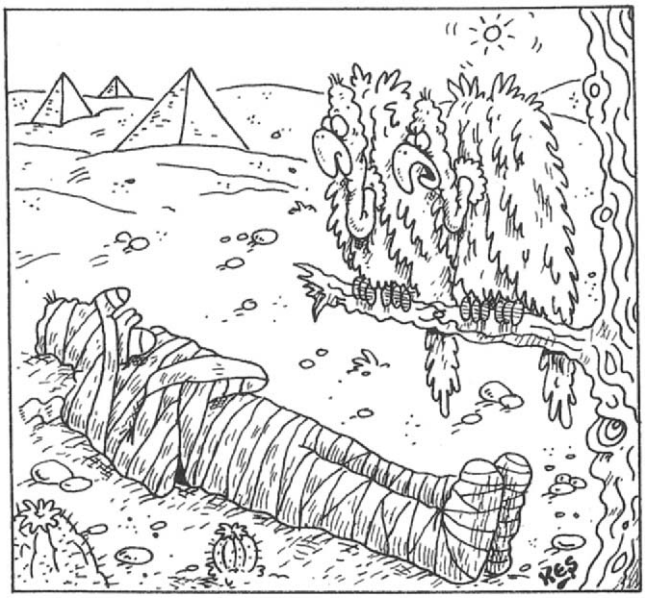
The most important factor for sulfites' antimicrobial activity is pH. In water, sulfur dioxide and its salts exist in a pH-dependent equilibrium:

$$SO_2 \cdot H_2O \leftrightarrow HSO_3^- + H^+ \leftrightarrow SO_3^{2-} + H^+$$

Once again, dissociation of the acid plays a major role. As the pH decreases, the proportion of $SO_2 \cdot H_2O$ increases and the bisulfite (HSO_3^-) ion concentration decreases. The pK_a values for sulfur dioxide are about 1.8 and 7.2. Sulfite is most inhibitory when the acid or $SO_2 \cdot H_2O$ is undissociated. Therefore, their most effective pH range is <4.0. $SO_2 \cdot H_2O$ is 100 to 1,000 times more active than HSO_3^- or SO_3^{2-} against *E. coli*, yeasts, and *Aspergillus niger*. Sulfites, especially the bisulfite ion, are very reactive. They form addition compounds (α-hydroxysulfonates) with aldehydes and ketones.

Low concentrations of sulfur dioxide are fungicidal. The inhibitory concentration of sulfur dioxide is 0.1 to 20.2 ppm against *Saccharomyces*, *Zygosaccharomyces*, *Pichia*, *Hansenula*, and *Candida* species. Sulfur dioxide at 25 to 100 ppm inhibits *Byssochlamys nivea* growth and patulin production in juices. Sulfites may inhibit spoilage bacteria in wines, fruits, and meats.

Sulfur dioxide controls microbial spoilage in fruits, fruit juices, wines, sausages, fresh shrimp, and pickles. It is added at 50 to 100 mg/liter to grape juices used for making wines to inhibit molds, bacteria, and wild yeasts. If the right concentration is used, sulfur dioxide does not interfere with wine yeasts or with wine flavor. During fermentation, sulfur dioxide also serves as an antioxidant, clarifier, and dissolving agent. Sulfur dioxide



"I'm not eating that! It's probably full of preservatives!"

Source: www.CartoonStock.com.

(50 to 75 mg/liter) also prevents postfermentation spoilage by *Acetobacter* spp. that oxidize ethanol into acetic acid (i.e., vinegar).

The targets for sulfite inhibition are the cytoplasmic membrane, DNA replication, protein synthesis, and enzymes. In *Saccharomyces cerevisiae*, *Saccharomycodes ludwigii*, and *Z. bailii*, sulfites diffuse into the cell. Other fungi have active transport systems for sulfites.

Chlorine

Chlorine in the form of hypochlorous acid is probably the most widely used sanitizer. It is added to wash waters and water used to convey fruits and vegetables. It is used to sanitize food contact surfaces and to reduce the microbial load on the surfaces of meat and poultry.

Sodium hypochlorite (bleach) is the most commonly used form. A strong oxidizer, it is active against bacteria, yeasts, molds, and viruses. Unfortunately it is very corrosive, even to metals such as stainless steel.

Chlorine dioxide (ClO_2) has important properties that make it preferable to hypochlorite. While it has similar antimicrobial properties, it is odorless and breaks down undesirable compounds like phenols, sulfites, and mercaptans (which taste and smell bad).

Quaternary Ammonium Compounds

Quaternary ammonium compounds, commonly called "quats," are a group of chemicals in which four compounds surround a nitrogen (Fig. 23.2). They are very versatile because the R groups can contain any number of saturated, unsaturated, cyclic or noncylic, substituted or unsubstituted alkyl groups. The quaternary ammonium compound is positively charged and is accompanied by a negatively charged compound, often chloride. These compounds are colorless, relatively odorless, and noncorrosive.

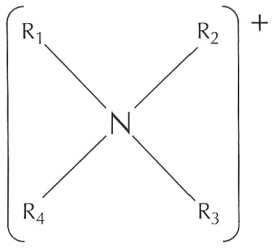


Figure 23.2 Structure of a quaternary ammonium compound, where the R ligands can be the same or different compounds. See the text.

Quats are active against bacteria, yeasts, fungi, and viruses. As you might guess, they are more expensive than hypochloride-based disinfectants.

Peroxides

Peroxides are compounds that have covalently bound oxygens (-O-O-) at their core. When represented as R_1-O-O-R_2, the addition groups (R_1 and R_2) can be inorganic (i.e., hydrogen) or organic (acetic acid) compounds. Hydrogen peroxide and peroxyacetic acid are the two most commonly used peroxides.

The action of both compounds depends on disassociation. In the case of hydrogen peroxide:

$$O_2^- + H_2O_2 \rightarrow OH\cdot + OH^- + O_2$$

The lethal action of both compounds comes from the OH· radical. This very strong oxidizer reacts with membrane lipids, DNA, and proteins.

Hydrogen peroxide is used as a sanitizer that kills a wide variety of bacteria, yeasts, molds, and viruses, with the exception of bacteria that have strong catalase activity. Hydrogen peroxide is used medicinally in 3 to 5% solutions to disinfect wounds. In the foods industry, similar concentrations are used to sterilize equipment and food contact surfaces. Because it whitens meats, it cannot be used in that application. Hydrogen peroxide is not approved as a food additive per se. Thus, the FDA has limited its use to produce whose endogenous catalase activity can remove hydrogen peroxide residues.

Peroxyacids are much stronger than hydrogen peroxides and can be used at much lower concentrations that do not affect poultry or meat quality. Their uses to reduce the microbial load on poultry, meat, and produce are important applications. Peroxyacids are also used to sterilize food contact surfaces and containers used in aseptic processing. Peroxidases and catalase do not affect peroxyacids.

Ozone

Ozone is another disinfectant that is a strong oxidant. Its primary target is nucleic acids. Ozone is GRAS for bottle washes. Ozone has a great deal of potential for use with raw fruits and vegetables, where it can reduce *E. coli* O157:H7 viability over 1,000-fold. However, as of this writing, these applications have not been approved. Ozone is not effective in meats, because meats react to reduce the ozone concentration below the effective level.

NATURALLY OCCURRING ANTIMICROBIALS

Many foods contain natural antimicrobials. They may extend the food's shelf life, but their use as direct food additives presents many challenges. Ideally, they should be effective when added as food ingredients, without purification. The cost of using the natural antimicrobial should be as low as that of chemical preservatives. They should not affect the food's taste, nutrition, or safety. However, most natural antimicrobials are normally present in amounts too low to be effective when mixed into foods. If higher levels are used, taste and smell are affected.

Lysozyme

Lysozyme is an antimicrobial enzyme produced in eggs, milk, and tears. Commercial lysozyme is made from dried egg white. The lysozyme in hens' eggs has 129 amino acids. It is stable to heat (100°C) at low pH (<5.3). It is inactivated at lower temperatures when the pH is increased. The enzyme

breaks the β-1,4 glycosidic bonds between *N*-acetylmuramic acid and *N*-acetylglucosamine of peptidoglycan. This breaks down the bacterial cell wall.

Lysozyme is most active against gram-positive bacteria, probably because their peptidoglycan is more exposed. It inhibits *C. botulinum, Clostridium thermosaccharolyticum, Clostridium tyrobutyricum, Bacillus stearothermophilus, B. cereus,* and *L. monocytogenes.* Lysozyme is the main antimicrobial in egg albumen. Ovotransferrin, ovomucoid, and alkaline pH enhance its activity in eggs. Lysozyme is less effective against gram-negative bacteria because they have less peptidoglycan and a protective outer membrane.

The chelator ethylenediaminetetraacetic acid (EDTA) enhances lysozyme activity. EDTA helps lysozyme penetrate to the peptidoglycan. Gram-negative cell susceptibility can be increased by chelators (e.g., EDTA) that bind Ca^{2+} or Mg^{2+}. These cations are needed to maintain a functional lipopolysaccharide layer. Gram-negative cells are also sensitized to lysozyme by pH shock, heat shock, osmotic shock, drying, and freeze-thaw cycling.

Lysozyme is one of the few natural antimicrobials subject to regulatory approval for food use. In Europe, it is used to prevent gas formation ("blowing") in cheeses. Lysozyme is used in Japan to preserve seafood, vegetables, pasta, and salads.

Lactoferrin and Other Iron-Binding Proteins

Milk and eggs contain antimicrobial iron-binding proteins. Lactoferrin is the main iron-binding protein in milk. Milk also contains low levels of the iron-binding protein transferrin. There are two iron-binding sites per lactoferrin molecule. One bicarbonate (HCO_3^-) is required for each Fe^{3+} bound by lactoferrin. Citrate, another chelator in milk, inhibits lactoferrin's activity, whereas bicarbonate reverses the inhibition. To be effective, lactoferrin must be in a low-iron environment where bicarbonate is present. Milk has these conditions.

Microorganisms with a low iron requirement are not inhibited by lactoferrin. Iron stimulates growth in many genera including *Clostridium, Escherichia, Listeria, Pseudomonas, Salmonella, Staphylococcus, Vibrio,* and *Yersinia.* Lactoferrin's depletion of iron may cause inhibition.

Egg albumen has another iron-binding molecule, ovotransferrin. This is sometimes called conalbumin. It makes up 10 to 13% of the total egg white protein. Ovotransferrin has an amino acid sequence that is very similar to that of lactoferrin. Each ovotransferrin molecule has two iron binding sites. Like lactoferrin, it binds anions such as bicarbonate with each bound ferric iron. To be inhibitory, there must be more ovotransferrin molecules than iron molecules and the pH must be alkaline. Ovotransferrin inhibits gram-positive and gram-negative bacteria. Gram-positive bacteria are generally more sensitive. *Bacillus* and *Micrococcus* species are very sensitive. Some yeasts are also sensitive. Ovotransferrin is bacteriostatic. Like lactoferrin, the primary inhibitory mechanism is probably iron depletion.

Avidin

Avidin is another egg albumen protein. It makes up about 0.05% of the total egg albumen protein. It is stable to heat and a wide pH range. Avidin binds biotin at a ratio of four biotin molecules per avidin molecule. Biotin activates enzymes in some metabolic cycles. Avidin inhibits bacteria and yeasts that require biotin for growth. However, avidin also binds transport proteins in the *E. coli* outer membrane. This suggests that avidin may inhibit bacteria by interfering with transport.

C I T A T S O I R E T C A B N

I P R E S E R V A T I V E I J

L A V O G G G B Y R P B R X J

O N Y D E T A N O T O R P O I

N F H R L Z N I J K E P X N R

E C H N X A I K S F X I K S S

H D O D U H C Y O S Q Z J U N

P H O S P H A T E S J Q C L O

U Z Z D E B C T I G H K K F P

X U A L I A I U Y C H I W I O

Z V A F L R D E J V T D J T B

U T C D T R S E N A R B M E M

E V P I A M Y B C Z N M U S A

Q D N X J O R S R O A F T G Q

Clues

Compounds that prevent the growth of bacteria but do not kill them are called _____.

Some inhibitors act by their ability to _____ metals essential for microbial growth.

An organic acid frequently associated with fermented foods. _____

A proteinaceous microbial inhibitor that acts by chelating iron. _____

Target for action of phenolic compounds. _____

In addition to preventing the growth of *Clostridium botulinum*, these compounds give cured meat its characteristic color and taste. _____

Antimicrobials in this class are among the most common antimicrobials, but they are only effective below their pK_a. _____

The antimicrobial activity of many plant extracts is due to their _____ compounds.

Inorganic antimicrobials that act as chelators. _____

A generic term for compounds that include antimicrobials, antioxidants, texture stabilizers, etc. _____

The form of an organic acid that can cross the microbial membrane. _____

Compounds that prevent browning in addition to their antimicrobial activity but cause allergic reactions in some people. _____

Spices and Their Essential Oils

Spices are usually added to foods as flavoring agents. However, they also have some antimicrobial activity. The ancient Egyptians used spices for food preservation and embalming around 1550 B.C.E. Cloves, cinnamon, oregano, thyme, and to a lesser extent sage and rosemary are the strongest

antimicrobial spices. Table 23.2 lists their active chemical components, concentrations required, and sensitive organisms. Not all spices have antimicrobial activity.

There are not many studies on the inhibitory mechanism(s) of spices. The terpenes in spice oils are the primary antimicrobials. Many of the most active terpenes, e.g., eugenol, thymol, and carvacrol, are phenolic. Therefore, their modes of action are probably related to other phenolic compounds. They interfere with membrane function.

Onions and Garlic

Onions and garlic contain the best-characterized plant antimicrobials. These compounds inhibit *B. subtilis*, *Serratia marcescens*, *Mycobacterium*, *B. cereus*, *C. botulinum* type A, *E. coli*, *Lactobacillus plantarum*, *Leuconostoc mesenteroides*, *Salmonella*, *Shigella*, and *S. aureus*. The fungi *Aspergillus flavus*, *Aspergillus parasiticus*, *Candida albicans*, *Cryptococcus*, *Penicillium*, *Rhodotorula*, *Saccharomyces*, *Torulopsis*, and *Trichosporon* are also inhibited.

The major antimicrobial in garlic is allicin. It is formed by allinase from alliin when garlic cells are disrupted. Allicin probably inhibits sulfhydryl-containing enzymes in bacteria. A similar reaction occurs in onion, forming thiopropanal-S-oxide. Onions also contain the antimicrobial phenolic compounds protocatechuic acid and catechol.

Isothiocyanates

Isothiocyanates (R-N=C=S) are potent antimicrobials. They are made from glucosinolates in plant cells from the *Cruciferae* or mustard family. This family contains cabbage, kohlrabi, Brussels sprouts, cauliflower, broccoli, kale, horseradish, mustard, turnips, and rutabagas. Fungi, yeasts, and bacteria are inhibited by 0.016 to 0.062 μg/ml in the vapor phase or 10 to 600 μg/ml in liquid. Isothiocyanates may inhibit cells by reacting with disulfide bonds or inactivating enzymes that contain sulfhydryl groups.

Phenolic Compounds

Phenolic compounds have an aromatic ring with one or more hydroxyl groups. Phenolic compounds are classified as simple phenols and phenolic acids, hydroxycinnamic acid derivatives, and the flavonoids. Many inhibitors already discussed are phenolics.

Table 23.2 Antimicrobial activity of spices

Spice	Antimicrobial compound(s)	Concn	Target of activity
Cloves Cinnamon	Eugenol Cinnamon aldehyde	Spices are 10 to 20% volatile oil. Clove is active at 1/100 dilution.	Clove inhibits *B. subtilis*, *E. coli*, *Salmonella*, *L. monocytogenes*, *S. aureus*
Oregano Thyme	Carvacrol Thymol	MIC,[a] 0.02–0.05%; MBC,[b] 0.03–0.1% essential oil	*E. coli*, *Salmonella*, *L. monocytogenes*, *S. aureus*, *B. cereus*, some yeasts and molds
Sage, rosemary	Terpenes, borneol, camphor, thujone	MIC, 0.3%; MBC, 0.5%	More effective against gram-positive bacteria; *B. cereus*, *S. aureus*, *L. monocytogenes*
Vanillin	4-Hydroxy-3-methoxy-benzaldehyde	500–2,000 μg/ml	Molds and nonlactic gram-positive bacteria

[a]MIC, minimum inhibitory concentration.
[b]MBC, minimum bacteriocidal concentration.

Simple phenolic compounds include monophenols (e.g., *p*-cresol), diphenols (e.g., hydroquinone), and triphenols (e.g., gallic acid). Gallic acid occurs in plants as quinic acid esters or hydrolyzable tannins (tannic acid). The only preservative use of simple phenols is in wood smoke. Smoking meats, cheeses, fish, and poultry imparts both a desirable flavor and a preservative effect. Phenol and cresol contribute to smoke's flavor and antioxidant and antimicrobial action. Several commercial smoke preparations at 0.25 and 0.5% reduce *L. monocytogenes* viability in buffer. The phenolic glycoside oleuropein, or its aglycone, inhibits *L. plantarum*, *L. mesenteroides*, *Pseudomonas fluorescens*, *B. subtilis*, *Rhizopus* sp., and *Geotrichum candidum*.

Summary

- Microbial spoilage is just one type of spoilage.
- Foods that permit pathogen growth but not spoilage organism growth are especially dangerous.
- Antimicrobials can be "-cidal" or "-static" but are usually the latter.
- Antimicrobial activity is influenced by microbial, intrinsic, and extrinsic factors.
- Organic acids are effective below their pK_a values because they cross the bacterial membrane, dissociate, and acidify the cytoplasm.
- Parabenzoic acids act similarly to organic acids, but have higher pK_a values.
- Lysozyme is an enzyme that kills gram-positive cells by breaking down their cell walls.
- Nitrites prevent *C. botulinum* growth in cured meats and confer color and flavor.
- Sodium chloride lowers a_w and causes cellular plasmolysis.
- Phosphates and lactoferrin chelate metal ions that microbes need to grow.
- Sulfites are used primarily on fruits and vegetables to inhibit yeasts and molds.
- The amount of available water (a_w) determines which organisms can grow in a food and influences lag time, growth rate, and final cell density.
- A water activity of 0.85 is the lowest a_w that allows growth of pathogens. This makes it a cardinal value in food microbiology.

Suggested reading

Ash, M., and I. Ash. 1995. *Handbook of Chemical Preservatives*. Grower, Brookfield, VT.

Branen, A. L., P. M. Davidson, and S. Salminen (ed.). 1990. *Food Additives*. Marcel Dekker, New York, NY.

Davidson, P. M., and T. M. Taylor. 2007. Chemical preservatives and natural antimicrobial compounds, p. 713–746. *In* M. P. Doyle and L. R. Beuchat (ed.), *Food Microbiology: Fundamentals and Frontiers*, 3rd ed. ASM Press, Washington, DC.

Questions for critical thought

1. If an antimicrobial were bacteriostatic and inhibited microbes "forever," would foods last "forever"? Explain.

2. You are Director of Microbiology at Fine Fat Foods. Marketing has directed that all those nasty preservatives be removed from all products. How can you maintain the safety and quality of your products?
3. Diagram the process by which organic acids work in high-acid foods but not in low-acid foods.
4. If you were assigned the task of developing the ideal antimicrobial, what criteria would it have to meet?
5. Calculate the amount of protonated acetic acid at pH 3.5 and pH 5.5 if you started with a 1 molar solution. How much acetic acid would you have to add at pH 5.5 to get as much protonated acetic acid as there is at pH 3.5?
6. You are an extension specialist at Big State University. A local inventor approaches you for advice on marketing a new antimicrobial. Called "Superlator," it chelates all cations, preventing growth of bacteria that require any cations. What advice do you give the inventor?
7. What concepts from Chemistry 101 are important for understanding how preservatives work?
8. Famous Frank's Freshly Frozen Potpies is losing market share and rapidly becoming less famous as the demand for "fresh, never frozen" food increases. Frank knows that he has to change and figures that going fresh is no big deal. (Just skip the freezing step!) He hires you, the company's first microbiologist, to advise him on any special challenges this transition might hold. What do you tell Frank?
9. Take the following ingredient list and make it label friendly. "This product contains dihydrogen oxide, sodium chloride, ascorbic acid, acetic acid, and 4-hydroxy-3-methoxy-benzaldehyde."
10. Why do sugar syrup and salad dressing that are both 50% water present different challenges for microbial stability?
11. According to Raoult's law, how much sucrose would one have to add to 1 liter of water to decrease its a_w to 0.90?

Authors' note

In the world of industry, "challenges" is often a code word for "problems." No one likes to have problems, but we are all excited by challenges (or so the thinking goes).

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Biologically Based Preservation and Probiotic Bacteria

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Biopreservation by Controlled Acidification
Bacteriocins
General Characteristics
Bacteriocin Applications in Foods
Probiotic Bacteria
The Human GI Tract Is a Microbial Ecosystem
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- understand that biological methods can be used to "naturally" enhance food safety without changing the food (as fermentations do)
- appreciate the potential antimicrobial uses of the small proteins called bacteriocins
- relate the organization of bacteriocin genes to the way bacteriocins are made
- describe the characteristics of probiotic bacteria, suggest possible health benefits, and critically evaluate commercial claims for probiotic bacteria
- link the scarcity of validated data about the health benefits of probiotic bacteria to the human gastrointestinal tract's complex ecosystem

INTRODUCTION

It is easy to tell if foods are processed or contain preservatives. It is obvious from changes in the way the food looks or information on the labels. However, this information is often misused. Consumers are wary of preservatives and processed foods, even though they give us a safe and diverse diet. As a result, consumers increasingly rely on refrigeration to ensure the safety of fresh foods. (Note well that the opposite of "fresh" is not "processed" or even "frozen"; it is "spoiled.") There are two reasons that refrigeration should not be used as the sole preservation method. The first is that 20% of refrigerators are at temperatures of >10°C (50°F). (You *do* remember the 40-140 rule that foods should be held below 40 or above 140°F to prevent the growth of most foodborne pathogens.) The second reason is that *Listeria monocytogenes* can grow at refrigeration temperatures of <10°C. Because of this, additional means of keeping refrigerated foods safe are required. Lactic acid bacteria (LAB) can provide this added protection. LAB are accepted by consumers as "natural" and "health promoting." They are, through fermentation, among the oldest forms of food preservation. Biologically based preservation methods that do not ferment the foods are among the newest forms of food preservation.

This chapter provides an overview of biopreservation. *Biopreservation* is the use of LAB, their metabolic products, or both to improve or ensure the safety and quality of foods that are not seen as typical fermented foods.

Some LAB produce antimicrobial proteins called bacteriocins. Bacteriocins may inhibit spoilage and pathogenic bacteria without changing the food (i.e., acidifying it, producing gas, solidification, etc.). The use of bacteriocins is a newer and emerging area of food microbiology.

The food industry is market driven. Advertisers try to turn technical needs that might be seen as marketing negatives (i.e., the need to preserve a food) into something that will help sell the food (i.e., the preservation method improves consumer health). Thus, consumers want foods that are more than nutritious. Foods should be distinctly health promoting. Since LAB are natural, and perceived as health promoting, foods preserved by biological techniques may have an edge if marketed as "probiotic."

BIOPRESERVATION BY CONTROLLED ACIDIFICATION

Organic acids (such as acetic, lactic, and citric acids) can inhibit microbial growth. The acids are usually added to the food. LAB, however, can produce lactic acid *in* the food. Many factors determine the effectiveness of acidification "in place." These include the food's initial pH, its buffering capacity, the pathogen of greatest concern, the nature of the fermentable carbohydrate, and the growth rates of the LAB and of the pathogen at refrigerated and abuse temperatures. Biopreservation can require customization and research for each product application. Bacteriocins, diacetyl, and hydrogen peroxide may enhance acid's inhibition. For example, MicroGard, a generally recognized as safe (GRAS) cultured-milk product, is frequently added to cottage cheese in the United States as a biopreservative. MicroGard is made by fermenting milk using *Propionibacterium freudenreichii* subsp. *shermanii* to produce acetic acid, propionic acid, low-molecular-weight proteins, and a bacteriocin.

The idea of using LAB acid production to prevent the production of botulinal toxin dates back to the 1950s. This technology exploits *Clostridium botulinum*'s inability to grow at pH <4.8 to prevent its growth if bacon is left unrefrigerated. LAB and a fermentable carbohydrate (such as glucose or lactose) are added to the food. The LAB grow and produce acid only when the food is temperature abused. Under proper refrigeration, the LAB cannot grow, no acid is formed, and the preservation system is invisible to the consumer.

Authors' note

The use of controlled acidification may not sound very effective, but in the control experiment using conventional bacon, more than one-half of the samples became toxic. If you like bacon, be sure to refrigerate it!

Nitrites are powerful inhibitors of botulinal spores. When it was found that the nitrites used to cure meats form cancer-causing chemicals called nitrosamines, research was initiated to find nitrite substitutes with antibotulinal activity. Nobi Tanaka of the University of Wisconsin reduced nitrite in bacon by using controlled acidification. When bacon was inoculated with 10^3 botulinal spores/g and incubated at 28°C, toxin was produced in 58% of the bacon samples prepared with the standard 120 parts per million of nitrite. When the nitrite was reduced to 80 or 40 parts per million and supplemented with sucrose and starter cultures, the pH dropped and ≤2% of bacon became toxic. The U.S. Department of Agriculture (USDA) approved the "Wisconsin process" for bacon manufacture in 1986.

BACTERIOCINS

General Characteristics

LAB bacteriocins are a relatively heterogeneous group of small proteins. They act against closely related bacteria but not against the bacteria that make them. "Closely related" can cover a wide range of gram-positive bacteria. Chelating agents, hydrostatic pressure, or injury can make gram-negative bacteria bacteriocin sensitive. Bacteriocins are not enzymes but may have some unknown cellular function.

The term "bacteriocins" was originally used for only those plasmid-mediated proteins produced by bacteria late in the stationary phase of growth that are bactericidal to a narrow range of closely related bacteria having specific binding sites for that bacteriocin.

It is now clear that few LAB bacteriocins meet all of these criteria. Bacteriocins differ in which bacteria they kill, their biochemical traits, and if their genes are located on the chromosome or plasmid. Most bacteriocins are small (3 to 10 kilodaltons), have a high isoelectric point (i.e., they precipitate at a high pH), and contain both hydrophobic (repelled by water) and hydrophilic (water-loving) sections.

The lantibiotics and the "pediocin-like" bacteriocins are the two most important bacteriocin groups. If you know the basic biochemistry of amide linkages in proteins, you can see that the lantibiotic nisin is very strange (Fig. 24.1). Nisin contains unusual amino acids, other than the 20 normal ones. After the protein is made, some amino acids react with cysteine to form thioether (single sulfhydryl) lanthionine rings. Bacteriocins containing lanthionine rings are called *lantibiotics*. There are many structurally similar lantibiotics. Nisin, the first and best-characterized LAB bacteriocin, is produced in two related forms. Nisin A contains a histidine at position 27 and is the most widely used nisin. Nisin Z has an asparagine at position 27 and is also commercially available. Nisins with other amino acid substitutions are not in common usage. Another lantibiotic, subtilin, is produced by *Bacillus subtilis*. It also contains five lanthionine rings and a conformation similar to that of nisin.

Authors' note

A plasmid is an extra piece of DNA, separate from the chromosome.

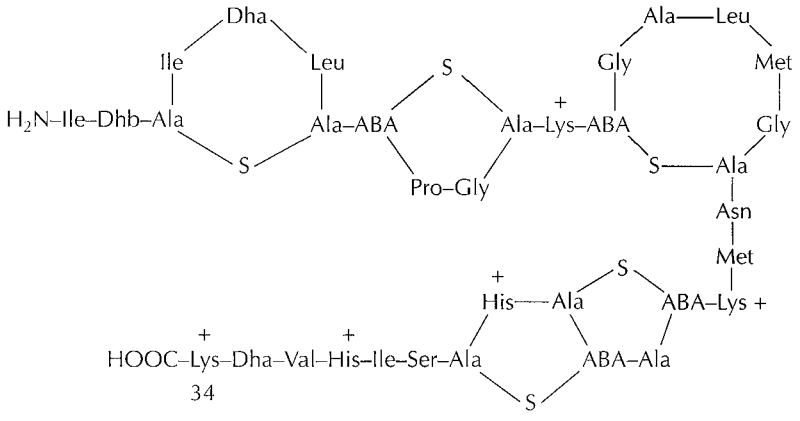
Pediocin-like bacteriocins are small heat-stable proteins made of the usual 20 amino acids. They all have the same leader amino acid sequence containing a [glycine-glycine-↓-any amino acid] cleavage site (where the arrow indicates the cleavage site). The upstream amino acids are cut off during the process of excreting the bacteriocin. Pediocin-like bacteriocins act against *L. monocytogenes*. They have a tyrosine-glycine-asparagine-glycine-valine-(any amino acid)-cysteine amino-terminal sequence (Fig. 24.2). Pediocin PA-1, pediocin AcH, sakacins A and P, leucocin A, bavaricin MN, and curvacin A are members of this group.

Authors' note

Proteins are made by linking amino acids' amide and carboxy groups. There is an amine "left over" at the starting end and an unlinked carboxyl group at the ending end.

The antimicrobial peptide polylysine is in a class by itself. As the name suggests, polylysine is a polymer consisting of 25 to 35 lysine monomers (Fig. 24.3). It has no secondary or tertiary structure. Polylysine is produced

Figure 24.1 Structure of nisin showing positions of unusual amino acids (dehydroalanine [Dha], dehydrobutyrine [DhB], lanthionine [Ala-S-Ala], and methyl lanthionine [ABA-S-Ala]) in addition to regular amino acids.



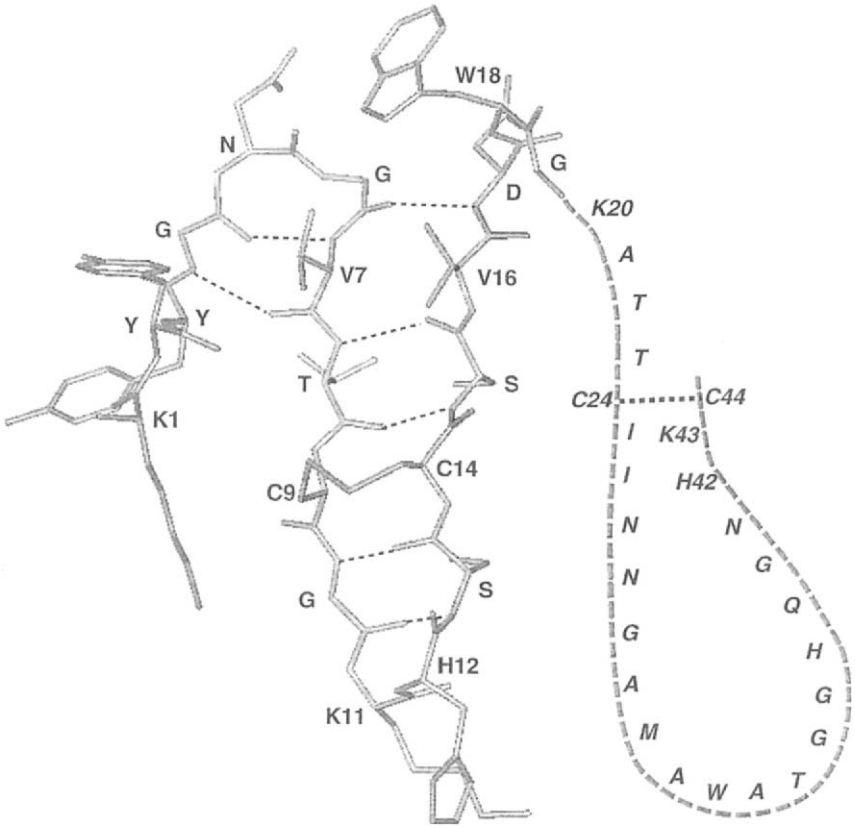


Figure 24.2 Structure of pediocin PA-1 showing the KYYNGV carboxy terminus, disulfide linkages, hairpin loop, and unstructured amino terminus.

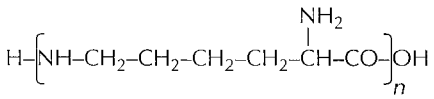
on an industrial scale in Japan by a fermentation process using *Streptomyces albulus*. It is effective at concentrations less than 100 μg/ml against bacteriophages, fungi, yeasts, and gram-positive and gram-negative bacteria.

Bacteriocin Applications in Foods

Bacteriocins or the bacteria that make them can be added to foods to inhibit pathogens. Only nisin is commercially available as an ingredient, but pediocin addition is also effective. Many applications add the LAB bacteria rather than the pure bacteriocin. Bacteriocins can also be used to improve the quality of fermented foods by inhibiting unwanted bacteria and promoting growth of the starter culture. There are many benefits to using defined starter cultures to make fermented foods. However, the native bacteria of the food usually outcompete the starter cultures. Using starter cultures that produce bacteriocins to kill the native bacteria solves this problem.

Nisin is added to milk, cheese and dairy products, many canned foods, mayonnaise, and baby foods throughout the world. It is GRAS for the inhi-

Figure 24.3 Structure of polylysine, where n = 25 to 35 lysine molecules.



bition of botulinal spores in many foods. Nisin also sensitizes spores to heat, so that thermal treatments can be reduced. However, this application is not approved in the United States because it might mask poor process control. Nisin can also prolong the shelf life of foods like sauces, salad dressings, and cheeses.

Nisin is often used with other inhibitors. Nisin may be used with food packaged in a modified atmosphere. Nisin increases the shelf life and delays toxin production by type E botulinal strains in fresh fish packaged in a carbon dioxide atmosphere. The combination of nisin and modified atmosphere to prevent *L. monocytogenes* growth in pork is more effective than either used alone. When nisin (5 mg/liter) is added to liquid whole eggs before pasteurization, their refrigerated shelf life doubles.

Pediocins inhibit *L. monocytogenes* vegetative cells but not clostridial spores. European patents cover the use of pediocin PA-1 as a dried powder or culture liquid to extend the shelf life of salads and salad dressing and as an antilisterial agent in cream, cottage cheese, meats, and salads.

Pediocins are more effective than nisin in meat. Dipping meat in pediocin PA-1 decreases the viability of attached *L. monocytogenes* 100- to 1,000-fold. Pretreating meat with pediocin reduces subsequent *L. monocytogenes* attachment. Pediocin AcH can cause a 10- to 1,000,000-fold reduction of listeriae in ground beef, sausage, and other products. In most cases, pediocin kills listeriae rapidly and delays growth of the survivors. The use of emulsifiers such as Tween 80 or the entrapment in other lipids makes pediocin work better in fatty foods.

Bacteriocin-Producing Starter Cultures Can Improve the Safety of Fermented Foods

Foods that are usually made by LAB are easily improved using bacteriocin-producing starter cultures. For example, including a nisin-producing starter culture to make Cheddar cheese increases the shelf life from 14 days for the control samples to 87 days for samples containing nisin when stored at 22°C.

Pediococci that make pediocins are especially effective in fermented meats. Pediocin production by *Pediococcus acidilactici* PAC 1.0 during the manufacture of fermented dry sausage reduces *L. monocytogenes* viability >10-fold. When *P. acidilactici* H is used to ferment summer sausage, *L. monocytogenes* viability drops more than 1,000-fold.

Organization of Bacteriocin Genes: a Generic Operon

The DNA or amino acid sequences of many bacteriocins are known. There is a general model for the genetic organization of bacteriocin genes, although the genetics of polylysine synthesis are unknown. The organization of a generic *operon* is shown in Fig. 24.4. An operon clusters the genes containing the structural information near the genes involved in immunity, maturation, processing, and export of the bacteriocin molecules. The genes that control bacteriocin biosynthesis are also on the operon. Not all bacteriocin operons contain all of these genes. Nor is the organization of the genes in each operon identical for every bacteriocin. The generic operon shows many of their similarities.

The *structural gene* usually encodes a prepeptide. The prepeptide is made of the excreted bacteriocin plus an amino-terminal "leader sequence" that is cut off during transport. Pediocin-like bacteriocin genes code for a prepeptide containing a glycine-glycine cleavage site. The leader sequence

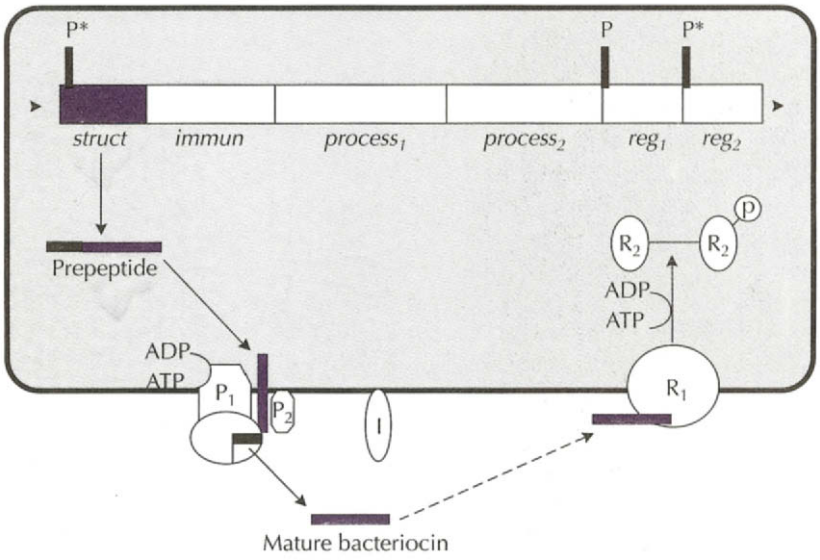


Figure 24.4 A generic bacteriocin operon. The structural gene (*struct*) codes for a prepeptide which is modified and excreted by the products (P1 and P2) of the processing gene (*process*$_1$ and *process*$_2$). The operon also contains the immunity gene (*immun*). The operon may be regulated by a signal transduction pathway coded for by *reg1* and *reg2*, producing a histidine kinase (R_1) and response regulator (R_2) subject to phosphorylation (R_2~P). Regulatory molecules P and P* affect transcription of the operon. ADP, adenosine diphosphate. Adapted from T. J. Montville, K. Winkowski, and M. L. Chikindas, p. 629–647, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

Authors' note

In this case, "immunity" means "protection from" and has nothing to do with antigens, antibodies, or the immune response.

may be needed for recognition by export machinery. The *immunity gene* protects cells that make a given bacteriocin from being killed by it.

Immunity is specific to the bacteriocin being made and is coordinated with its production. The *processing and export genes* code for at least two proteins that mature the bacteriocin and export it from the cell. The *regulatory genes* code for proteins that turn bacteriocin synthesis on and off.

Bacteriocin action. LAB bacteriocins target the cell membrane of sensitive bacteria. They disrupt the membrane by making pores. This increases its permeability to small compounds, causing a rapid efflux of preaccumulated ions, amino acids, and in some cases, adenosine 5′-triphosphate (ATP) molecules. The equilibration of compounds across the membrane destroys the gradients required for many vital cellular functions. The mechanism of polylysine action is quite different. Because it is so cationic, there is electrostatic absorption of polylysine onto the cell surface, where it strips off the cell membrane

Legal status. The legal status of bacteriocins and the bacteria that make them is unclear. LAB are GRAS for the production of fermented foods. GRAS status, which is given by the U.S. Food and Drug Administration (FDA), is especially desirable because it lets a compound be used in a specific application without additional approval. Some people overlook the linkage of GRAS status to a specific application. The GRAS status of LAB for the production of fermented foods does not automatically make LAB, or all of their metabolic products, GRAS for other uses such as the preser-

vation of foods that are not fermented. Nisin is the only bacteriocin that has GRAS status. The 1988 GRAS affirmation for the use of nisin in pasteurized processed cheese was supported by toxicological data. This affirmation is the foundation for many additional GRAS affirmations. Polylysine received GRAS affirmation for use in cooked rice and sushi rice in 2003.

In the nonregulatory sense, bacteriocins are widely considered safe. However, bacteriocins produced by GRAS organisms are not automatically GRAS themselves. Bacteriocins that are not GRAS are regulated as food additives and require premarket approval by the FDA. A food fermented by bacteriocin-producing starters can be used as an ingredient in a second food product. Its use *as an ingredient* might coincidentally extend the shelf life of the product without listing "preservative" on the label. However, if the ingredient were added *for the purpose of extending shelf life,* the FDA would probably consider it to be an additive and require both premarket clearance and label declaration. Purified bacteriocins used as preservatives definitely require premarket approval by the FDA.

Authors' note

GRAS status can be conferred three ways: if the product was in use prior to 1958, if the FDA affirms that a compound is GRAS in response to an industry request, or if a company "self-affirms" that its compound is GRAS. (Welcome to the wild world of food law!)

PROBIOTIC BACTERIA

The idea that eating LAB improves health first appeared in print in 1907. In his book *Prolongation of Life,* the Nobel laureate Elie Metchnikoff attributed the longevity of people in the Balkans to the bacteria in the yogurt they ate. He hypothesized that the bacteria in the yogurt suppress "bad" bacteria in the gut. The term "probiotic" was rediscovered in the 1970s and used to describe the healthful effects of feeding microbial feed supplements to animals. In 1989, R. Fuller proposed a narrower definition whereby viable "good" bacteria are consumed and act in the gastrointestinal (GI) tract to benefit the host organism (i.e., us!). "Probiotics" have been defined in a variety of ways by different authors and organizations. For example, the Food and Agriculture Organization and the World Health Organization (FAO and WHO) redefine probiotics as "Live microorganisms which when administered in adequate amounts confer a health benefit to the host." While we know that fermentations improve digestibility, generate amino acids, and produce vitamins in food, there are not many peer-reviewed data showing that probiotic bacteria promote "intestinal well-being" or are otherwise "health promoting."

Authors' note

There is no agreement as to what constitutes "intestinal well-being."

There are many reports of the probiotic effect of LAB. Some of the species involved are listed in Table 24.1. Applications are given in Table 24.2. However, because there are so few well-controlled studies with humans, the therapeutic use of LAB has not gained acceptance in mainstream medicine (Box 24.1).

Cultures used to make fermented products are usually chosen for their technological traits rather than their health-promoting attributes. For example, *Lactobacillus delbrueckii* subsp. *bulgaricus* and *Streptococcus thermophilus,* traditionally used to make yogurt, are hearty and acidify rapidly. However, their poor resistance to acid and bile salts and inability to survive stomach passage prevent a probiotic effect. To capitalize on the health-promoting potential, many yogurt manufacturers now use *Lactobacillus acidophilus* and *Bifidobacterium* spp. that have been isolated from humans and are resistant to acid and bile salts. Other characteristics important for commercial cultures include the ability to be cultured on a large scale, GRAS status, and demonstrated clinical efficacy.

Table 24.1 Examples of human probiotic species and strains with research documentation[a]

Bifidobacterium breve	*Lactobacillus delbrueckii* subsp. *bulgaricus*	*Lactobacillus reuteri*
Bifidobacterium lactis	*Lactobacillus fermentum*	*Lactobacillus rhamnosus*
Bifidobacterium longum	*Lactobacillus johnsonii*	*Lactobacillus salivarius*
Lactobacillus acidophilus	*Lactobacillus paracasei*	*Saccharomyces boulardii*
Lactobacillus casei	*Lactobacillus plantarum*	*Streptococcus thermophilus*

[a]Adapted from Table 39.2 of T. R. Klaenhammer, p. 797–811, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001). Information in the source table was compiled from M. E. Sanders and J. Huis in't Veld, *Antonie Leeuwenhoek* **76**:293–315, 1999.

The Human GI Tract Is a Microbial Ecosystem

Authors' note

A baby's feces change as its diet and intestinal microbiota change. I was fascinated to watch my first child's poop change from black meconium to mustardy yellow cream to foul brown solid as his diet progressed.

The complexity of the human GI tract makes research on probiotic bacteria difficult. The average human contains over 400 species of bacteria. Thirty or forty of these constitute 99% of the bacteria in healthy humans. It is existentially noteworthy that we contain as many bacterial cells as human cells ($\sim 10^{13}$ to 10^{14}). Because it has so many organisms and environments, the GI tract must be studied as an ecosystem. Humans, like other ecosystems, experience a progression of inhabitants. We are born more or less sterile inside. We are colonized by bifidobacteria when breast-fed or by lactobacilli when fed with cow milk. Strict anaerobes come with the introduction of solid food.

Once established, the composition and distribution of our microbial component are very hard to change. In ecological terms, the microbiota of the human GI tract is a climax community.

The metabolic activity of the climax community is quite variable, and it is strongly influenced by the host physiology and by everything that affects host physiology. (Think about what happens when you eat too many beans.)

Authors' note

A climax community is an ecosystem in which the progression of species has stopped, all niches are filled, and a characteristic group of organisms is maintained.

The human GI tract is about 350 cm long from the oral orifice to the anal orifice. The GI tract is divided into three major sections. Each has its own distinct microbiota. The stomach is highly acidic (pH 1 to 3) and is populated by $<10^3$ colony-forming units (CFU) of aerobic gram-positive organisms per g. The intestinal pH (6.8 to 8.6) favors microbial growth. The

Table 24.2 Possible health benefits and mechanisms of probiotic bacteria

Health effect	Possible mechanisms
Promotes ability to digest lactose	Probiotic cultures make lactose-using enzymes
Fights foodborne disease	Occupies colonization sites, promotes immunity, decreases severity of diarrhea
Anticancer	Binds cancer-causing compounds, decreases levels of natural enzymes that promote cancer, stimulates immune function
Enhances immune system	Strengthens defenses against infections and tumors, increases antigen-specific immune responses, lowers inflammatory responses
Fights heart disease	Increases bile activity, may reduce cholesterol
Reduces blood pressure	Bacterial action on milk proteins produces a compound that lowers blood pressure in animals
Reduces ulcers	Production of inhibitor against *Helicobacter pylori*[a]
Limits urogenital infections	Adhesion to urinary and vaginal cells, competitive exclusion, production of inhibitors (surfactants, hydrogen peroxide)

[a]It is now widely accepted that ulcers are caused by *H. pylori*, not stress.

BOX 24.1

Probiotics in the marketplace

"You can eliminate painful and embarrassing Candida *yeast infections with probiotic bacteria!"*

"Probiotic bacteria inhibit harmful organisms, boost immune function, and increase resistance to infection."

"Promoting a proper balance of bacteria is critical to good health."

These are just a few claims from dozens of Internet sites that promote the consumption of probiotic bacteria. Despite the scientific uncertainty that surrounds dietary supplements, probiotics are big business. Neither foods nor drugs, dietary supplements are legally defined as dietary ingredients taken by mouth to supplement the diet, which are not labeled as a food, are not the sole item of a meal (for then they would be a *food*), and are not to be taken to prevent, cure, or treat a disease (for then they would be a *drug*). (The first claim above is illegal.) Supplements are not regulated as foods, nor are they regulated as drugs. They are hardly regulated at all! There are few government regulations governing the purity, potency, or effectiveness of supplements. Only the manufacturer's reputation assures consumers of the identity and viability of probiotic cultures. Nonetheless, in 2002 probiotic U.S. sales reached $160 million, and they are expected to grow at 10% per year for the next 10 years. There are even probiotics for dogs on the U.S. market! The more mature probiotic market in Japan exceeds $6 billion.

The area of probiotic bacteria is a difficult one in which to demonstrate cause and effect. Its secrets may lie beyond classical pure-culture microbiology, for probiotics are supposed to affect the microbial members of the human intestinal ecosystem. The largest producer of probiotics in the world is Yakult. This company was founded by Minoru Shirota, who developed a strain of *Lactobacillus casei* that was exceptionally hearty and resistant to stomach acid. With its modest claims of this strain being "friendly bacteria" that "promote intestinal well-being," the company grossed $1.5 billion in 23 countries. There is increasing clinical support for the use of probiotics in the management of many gastric disorders. Nestlé, Dannon, Danisco, Stonyfield Farm, and other mainstream food companies have recently introduced probiotic products. It is hard to be "anti*pro*biotics."

small intestine is a transitional zone inhabited by 10^3 to 10^4 CFU of *Lactobacillus, Bifidobacterium, Bacteroides,* and *Streptococcus* per g. Microbial growth in the large intestine is luxuriant. There are 10^{11} to 10^{12} CFU/g, with anaerobes outnumbering aerobes 100- to 10,000-fold. A diverse microbial population populates the large intestine. Bacteria from the genera *Bacteroides, Fusobacterium, Lactobacillus, Bifidobacterium,* and *Eubacterium* are present in large numbers. *Lactobacillus acidophilus* is especially important. The *Enterobacteriaceae* are present at relatively lower levels.

The normal human biota contains two different bacterial populations. There are indigenous bacteria that have colonized the host by adhering to the intestine. There are also foodborne bacteria that are just passing through. Dietary factors such as carnivorous or vegetarian diet, or even starvation, make surprisingly little difference in the distribution of bacteria in the GI tract.

The complexity of the GI ecosystem makes colonization hard to study. Colonization is influenced by gastric acidity, bile salt concentration, peristalsis, digestive enzymes, and the immune response. Feces are 33 to 50% bacteria on a dry weight basis and are their own ecosystem. The fecal ecosystem is subject to dehydration and has high levels of enzyme activities.

The future use of probiotic bacteria is more difficult to predict. *Something* happens in the GI tract when LAB are consumed in large numbers. Exactly *what* happens and *why* are hard to determine. It is very important that these benefits be studied in the context of diet, well-characterized probiotic bacteria, individual human intestinal ecosystems, and the established principles of microbial ecology. Much of this lies outside the scope of pure-culture microbiology and may be the work of future food microbiologists.

Summary

- Refrigeration alone is insufficient to ensure microbial food safety.
- LAB may be used to produce acid in place when a food is held at temperatures between 40 and 140°F. The acidification prevents pathogens from growing and making toxin.
- Bacteriocins are small proteins that kill sensitive bacteria by making pores in them.
- Lantibiotic bacteriocins have unusual amino acids and single sulfur rings.
- Pediocin-like bacteriocins have a common leader sequence and cleavage site and are active against *L. monocytogenes*.
- Bacteriocins can be produced by GRAS LAB in the food or added to the food as "natural preservatives" if they are approved as food additives or have GRAS affirmation.
- The genes for bacteriocins, their regulation, and the enzymes that process them are grouped together in an operon.
- Polylysine is another peptide antimicrobial but is different from lantibiotics of pediocin-like bacteriocins.
- Probiotic bacteria are thought to give health benefits by improving the balance of bacteria in the intestines.
- Possible health benefits for probiotics may include decreased cholesterol, increased ability to use lactose, decreased symptoms of diarrheal illness, and anticancer action.
- The human GI tract is a complex ecosystem. This makes the study of probiotics difficult.

Suggested reading

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Sanders, M. E. 1999. Probiotics—scientific status summary. *Food Technol.* **52:**67–77.

Yoshida, T., and T. Nagasawa. 2003. ε-Poly-L-lysine: microbial production, biodegradation, and application potential. *Appl. Microbiol. Biotechnol.* **62:**21-26.

Questions for critical thought

1. What is the temperature of your refrigerator? (If you are living in a dorm room without a refrigerator, check the temperature of the refrigerator at the store where you buy food.)
2. Why does the pH drop in Wisconsin process bacon?
3. What were the original six characteristics used to determine if an inhibitor was a bacteriocin?

4. Why is it a good thing that a bacteriocin does not act against the organism that makes it?
5. Why does it benefit the bacterium to have all the genes involved in bacteriocin synthesis and regulation clustered in an operon?
6. Make a narrow column and a wide column on a piece of paper. List the genes involved in bacteriocin production in the narrow column. In the wide column, write what would happen if this gene were deleted. Choose one of these genes and describe how it might be altered for commercial benefit.
7. Choose one health benefit of probiotic bacteria. Find, read, and evaluate three research articles that relate to it.
8. If humans contain 10^{13} to 10^{14} human cells and 10^{13} to 10^{14} bacterial cells, why don't we look more like slimy bacterial colonies?
9. List and discuss factors that influence the conclusions of studies on the health benefits of probiotics.
10. What are the differences between Fuller's definition and the FAO/WHO definition of "probiotic"? Why does it matter how "probiotic" is defined?

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Physical Methods of Food Preservation

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Thermobacteriology
Calculating Heat Processes for Foods
Microwave Heat Treatment
Preservation by Irradiation
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LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- appreciate the role of temperature, controlled atmosphere, and freezing in microbial growth
- differentiate among different types of heat processing, their objectives, and their industrial uses
- quantitatively determine, using *D* and *z* values, the lethality of a given heat process on an organism
- use his or her understanding of irradiation preservation to make a technical and social judgment about its use in foods
- gain familiarity with new "nonthermal" methods of food preservation

INTRODUCTION

Conditions too stressful for microbial growth can damage cells. Minor stresses inhibit cell growth. Major stresses kill cells. Thus, food environments can be manipulated to inhibit or kill microbes. Physical manipulation of foods can inhibit microbial growth, kill cells, or mechanically remove them from the food. Dehydration, refrigeration, and freezing inhibit microbial growth. Microbes can be killed by heating or ultraviolet (UV) or ionizing radiation. Nonthermal energy is also lethal. Membrane filtration removes microbes from liquid foods. This chapter discusses all these physical preservation methods except filtration.

PHYSICAL DEHYDRATION PROCESSES

Drying

When foods are being dried, hot air evaporates the water and carries it away. Both the drying temperature and the decreased water activity (a_w) affect the microbes. Factors such as the size and composition of food pieces and the time-temperature combinations used in drying influence the microbiological effect of drying.

During the initial period of drying, the temperature is still low and the relative humidity is high. The size and composition of the food particle determine the length of this phase. If this particle size is large and the material is dense, this phase can be long, allowing microbes to grow. During later phases of drying, the temperature is high but the relative humidity is low. This provides no opportunity for growth but is not very lethal either. The higher temperatures result in lower moisture content, and dry heat is less lethal than wet heat. If the drying lasts long enough (at least 30 min), there

are time-temperature combinations that kill microbial cells. With certain drying methods, the surface temperature may reach 100°C, while internal temperature remains lower. This kills surface microbes, but not those in the food's interior. Lethal combinations of high temperature and high humidity are rare during drying. In fact, the major cause of microbial death during drying is the high-temperature and low-humidity conditions which last long after the actual drying. Cell populations decrease during storage because injured cells that cannot recover at low a_w gradually die.

The a_w values of dried foods are usually much lower than the minimum a_w required for microbial growth. Therefore, dried products are microbiologically stable. If the relative humidity increases or the temperature drops, condensation on the product surface may permit mold growth.

Freeze-Drying

Authors' note

While beyond the scope of this book, the preservation of bacteria in "culture collections" is a neglected aspect of food microbiology. Many microbiologists are amateur naturalists, maintaining their own private culture collections and trading cultures with their colleagues like baseball cards. However, bacterial traits can "drift" over years of repeated transfer. It is better to get cultures in the freeze-dried form from a recognized culture collection such as the American Type Culture Collection. Secure forms of culture preservation are essential for process security and research reproducibility.

Freeze-drying (lyophilization) combines two preservation methods. It freezes the food and then dehydrates the frozen food through vacuum sublimation of the ice. That is, water moves from the ice state to the vapor state without becoming liquid. This is a gentle way to remove water. In other drying methods, the solution moves to the food surface, where the water leaves and the solutes concentrate. During freeze-drying, the sublimation front moves into the food and the ice sublimates where it is formed. Thus, solutes remain inside the food at their original location, and the food retains its original structure. As a result, freeze-dried foods rehydrate rapidly to 90 to 95% of their original moisture content. Unfortunately, the large surface area of freeze-dried foods makes them vulnerable to oxidation during storage. Packaging freeze-dried foods in inert atmospheres delays oxidation. Vapor-impermeable packaging prevents their rehydration.

The freeze-drying of foods inhibits microbes by decreasing a_w during freezing and by the sublimation of the ice. The extents of cell damage in these two phases may be different. Microbial survival also depends on the composition of freeze-dried food. Carbohydrates, proteins, and collodial substances are usually protective. Cell viability slowly decreases during storage of freeze-dried foods.

Freeze-drying is optimized for cell survival during the preservation of stock cultures. Low dehydration temperature, protective additives (such as glycerol or nonfat dried milk solids) in the microbial suspension, and storage of lyophilized cultures under vacuum increase viability. Gram-positive bacteria survive freeze-drying better than gram-negative bacteria.

COOL STORAGE

Authors' note

While the importance of refrigeration is well recognized, 20% of refrigerators have temperatures of >50°F! Do not count on them to ensure food safety.

Cool (or chill) storage refers to storage at temperatures from about 16 to −2°C. While pure water freezes at 0°C, most foods remain unfrozen until they reach a temperature of ≤−2°C. However, many fruits and vegetables suffer from a chilling injury when kept at <4 to 12°C. The length of cool storage may vary from a few days to several weeks (Table 25.1).

Chemical reaction rates decrease as temperatures decrease. This decreases microbial growth rates. Refrigeration temperature is below the minimal growth temperature of most foodborne microbes (see chapter 2). Even psychrotrophic microbes such as *Yersinia enterocolitica*, *Vibrio parahaemolyticus*, *Listeria monocytogenes*, and *Aeromonas hydrophila* grow very slowly at low temperature. A temperature increase of only a few degrees can dramatically increase microbial growth rates. Refrigerated foods can spoil rapidly

Table 25.1 Shelf life extension of raw foods by cool storage[a]

Food	Average useful storage life (days) at: 0°C (32°F)	22°C (72°F)
Meat	6–10	1
Fish	2–7	1
Poultry	5–18	1
Fruits	2–180	1–20
Leafy vegetables	3–20	1–7
Root crops	90–300	7–50

[a]Adapted from J. Farkas, p. 567–592, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

if their initial microbial populations are high. Since the temperature requirements for various microbes differ, refrigeration may change the ecology of the microbial population.

The temperature range over which an organism can grow depends on its ability to regulate membrane fluidity. Psychrotrophs have more unsaturated fatty acid residues and more branched-chain fatty acids in their lipids at low temperatures. This leads to a decrease in lipid melting point (i.e., increased fluidity). The fluid state allows membrane proteins to function. The idea that cells alter their fatty acid compositions to maintain membrane fluidity is termed *homeoviscous adaptation*.

Controlled-Atmosphere Storage

The minimal growth temperature is lowest when other factors are optimal. When other factors are unfavorable, the minimal growth temperature increases. The most important factors are pH, a_w, and oxygen concentration. Adding refrigeration to this suboptimal combination of factors extends food stability. Controlled-atmosphere storage is widely used for certain fruits and vegetables. The oxygen content is reduced (2 to 5%), and the carbon dioxide content is increased (8 to 10%). These conditions are maintained in airtight chilled storage rooms. Modified-atmosphere storage slows respiration of the fruit or vegetable. This limits quality deterioration and microbial growth.

Carbon dioxide inhibits growth for several reasons. When dissolved in water, carbon dioxide reduces the pH. However, the primary mechanism is direct inhibition of microbial respiration (see chapter 2). The reduced oxygen levels contribute to carbon dioxide's inhibitory effect. Psychrotrophic spoilage bacteria such as *Pseudomonas* and *Acinetobacter* are particularly sensitive to carbon dioxide, while lactic acid bacteria and yeasts are not.

Modified-Atmosphere Packaging

Modified-atmosphere packaging (MAP) is similar to controlled-atmosphere storage. In the case of vacuum-packaged products, the air pressure rather than composition is changed. The residual air pressure of only 0.3 to 0.4 bar (1 bar = 10^5 pascals) reduces the available oxygen. If packaging films with very low gas permeability are used for fresh foods, the carbon dioxide concentration may change. This is because oxygen is used and carbon dioxide is produced during respiration. This modified atmosphere is very inhibitory to certain microbes and increases the keeping quality of foods. However, *Clostridium botulinum* and other pathogens grow well in the

absence of O_2 and may be hazardous. Because of this, some vacuum-packed foods, especially pasteurized foods, should be kept below 3.3°C and may require secondary barriers to microbial growth.

MAP for respiring products like fresh fruit and vegetables is complicated. The gas permeability of the packaging film must equilibrate the atmosphere in the package. The package atmosphere is affected by the rate at which respiration and gas permeability change with temperature. MAP could extend the shelf life of many perishable products, but its adaptation is hindered by concern about growth of pathogens at refrigeration temperatures. Fish and fish products pose a risk for *C. botulinum* growth in MAP products. The U.S. National Academy of Sciences recommends that fish not be packed under modified atmospheres unless its safety has been proven experimentally.

FREEZING AND FROZEN STORAGE

Many convenience foods are frozen to maintain their high quality. Freezing is, however, a highly energy-intensive process. Commercially, foods are frozen to <-18°C in cold air, by contact with a cooled surface, by submersion into a cold refrigerant liquid such as liquid nitrogen, or by spraying the refrigerant onto the food.

Foods freeze over a broad temperature range rather than at a single well-defined temperature. Depending on the foods' composition, water starts freezing at -1 to -3°C. This initial freezing increases the solute concentration in the water not yet frozen. This further decreases the freezing point. This continues until the temperature at which the solutes reach their solubility limit is attained, and the residual water freezes. This is called the *eutectic* temperature. The totally frozen state is a complex system of ice crystals and crystallized soluble substances. It occurs at -15 to -20°C for fruits and vegetables and at <-40°C for meats.

Freezing produces an osmotic shock in the microbes. Intercellular ice crystal formation causes mechanical injury. The concentration of cellular liquids changes the pH and ionic strength. This inactivates enzymes, denatures proteins, and inhibits metabolic processes. Cell membranes suffer the major damage. In addition, thawing reexposes surviving cells to these effects. The injury of microbial cells may be reversible or irreversible. The extent of injury, repair, death, and survival varies according to the freezing, frozen-storage, and thawing conditions.

The freezing rate has a major impact on microbial viability. During slow freezing, crystallization occurs extracellularly and the cell's cytoplasm becomes supercooled at -5 to -10°C. Cells lose water that then freezes extracellularly. Water crystallization increases the concentration of the external solution. This also removes water from the cells. Overall, microbes are exposed to osmotic effects for a relatively long time. This increases injury. Increased freezing rates decrease the duration of the osmotic effects. This increases microbial survival. When freezing rates are too high, crystal formation also occurs intracellularly. This injures the cells drastically and decreases survival rates. Freezing rates also impact food quality. Faster freezing results in higher quality. However, while food manufacturers can optimize freezing, they have little control over thawing by the consumer, which has an equally important effect.

The most important influence on the microbial effect of freezing is the food environment. Certain compounds enhance, while others diminish,

freezing's lethal effects. Sodium chloride reduces the freezing point of solutions, thereby extending the time during which cells are exposed to high solute concentrations before freezing occurs. Compounds such as glycerol, saccharose, gelatine, and proteins generally protect cells from freeze damage.

Microbes do not grow below about −8°C. The fate of microbes surviving freezing may change during storage. Often the death of survivors is fast initially and slows gradually, and finally the survival level stabilizes. Death during frozen storage is probably due to the unfrozen, very concentrated residual solution. The residual solution may change during storage. At fluctuating temperatures, such as those in a home freezer with automatic defrost, the size of the ice crystals may increase. Fluctuating temperatures are more lethal than stable ones. Gram-positive microbes survive frozen storage better than gram-negative bacteria.

Although freezing and frozen storage reduce microbial viability, freezing is not considered a lethal process. Under some conditions, survivors can grow during thawing. Their levels may then equal or exceed the level before freezing. Thawing releases a nutrient-rich solution from the food cells. During thawing, microbes can penetrate damaged food tissue more easily, and liquid condenses on the food surface. These conditions favor microbial growth. Thawed products are especially vulnerable to rapid microbial spoilage. Refreezing thawed products can be dangerous.

PRESERVATION BY HEAT TREATMENTS

Heat is the most widely used method for killing microbes. For a food microbiologist, a microbe's heat resistance is one of its most important traits.

Pasteurization (named after Louis Pasteur [Box 25.1]) is a relatively mild heat treatment. Pasteurization's purpose is to kill non-spore-forming *pathogenic* bacteria. It also inactivates enzymes and kills many (99 to 99.9%) spoilage organisms. Because of this, pasteurized foods take longer to spoil especially when refrigerated to delay the growth of surviving organisms.

Sterilization kills *all* microbes. The goal of commercial sterilization is freedom from pathogens and shelf stability rather than absolute sterility. A product may contain a viable spore that cannot grow (due to low pH, low a_w, etc.) and still be "commercially sterile." *C. botulinum* spores, for example, may be present in high-acid foods. Because they cannot germinate and grow, they do not present a hazard.

Technological Fundamentals

Foods can be heat processed after or before packaging. The most common method is to heat after packaging by canning. *Canning* is the commercial sterilization of food in hermetically sealed (airtight) containers. Nicolas Appert invented the canning process in the early 1800s. His research was in response to Napoleon's offer of a prize for the development of a food preservation method that would extend the range of his troops. Canned foods, in the form of "C" rations or today's "MRE" (meal, ready-to-eat), remain the staple of soldiers around the world.

When foods are packaged after heating in the process of *aseptic packaging*, the food and the packaging material are sterilized separately. The food is then placed aseptically (in a sterile environment) into the package. Aseptic technology is widely used for fruit juices, dairy products, creams, sauces, and soups that would suffer quality loss if heated for long periods. These products are most often packaged in boxes (like the juice boxes

Authors' note

Note that foods can be "canned" in glass jars, in foil pouches, and in other containers besides cans.

BOX 25.1

Louis Pasteur, the first food microbiologist

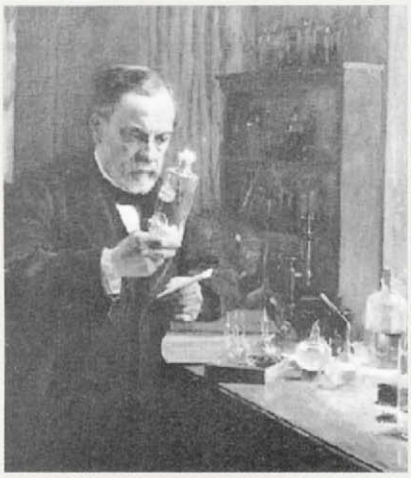


Figure 1. Louis Pasteur in the laboratory.

Louis Pasteur (Fig. 1) is one of the mythic giants of science but was also a man of great humility and integrity. He was not a particularly good student as a child. He would rather go fishing or paint. But his life bent toward science when, at 26, he "discovered" stereochemistry, i.e., that some acids could exhibit "handedness" by bending light to the left or to the right. He concluded that life was asymmetric and abandoned chemistry to study biology.

Pasteur the biologist fathered several other disciplines. Until his time, many people thought that life "popped up" spontaneously and that fermentation was a chemical process. Pasteur disproved this using a simple but elegant experiment with a curved-neck flask (Fig. 2). It was full of culture media that would support bacterial growth but prevented external bacteria from entering. These flasks, some of which are still on exhibit at the Institut Pasteur, remained sterile. By showing that "life comes from life," he killed the idea of spontaneous generation. From this, he postulated that fermentation was a biological rather than chemical process. Milk, wine, and beer spoilage caused by "ferments" could be stopped by heating for a few minutes at relatively low temperatures, a process we now know as "pasteurization." Pasteur was the first food microbiologist!

Pasteur is also the father of immunology, having developed animal vaccines for cholera and anthrax. He did not want to test the vaccination concept on humans, fearing guilt and ignominy if failure caused a human death. His hand was forced, however, when Madame Meister appeared at his laboratory with her nine-year-old son, Joseph. Joseph had been mauled by a rabid dog 2 days earlier and would certainly die without the vaccine. With fear in his heart, Pasteur treated the boy, who recovered and worked at the Institut Pasteur for the rest of his life. Joseph Meister's loyalty to Pasteur is legendary. There is a story that, when the Germans overran France in 1940, they sought to disinter Pasteur in retribution for his refusal to accept an honorary degree from the University of Bonn. Fifty years after his cure, Meister is purported to have taken his own life rather than open the crypt of his savior.

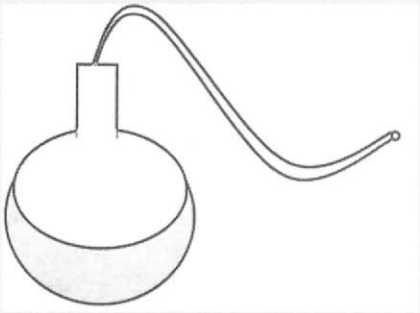


Figure 2. Curved-neck flask of the type used by Pasteur.

We live in an age where there is debate about the worth of "basic" versus "applied" science. Some feel that basic science is "better" science. Others see no need to pay for research unless it addresses some defined problem. It would do us well to remember the words of Louis Pasteur: "There does not exist...a science one can give the name applied science. There are science and the application of science, bound together as the fruit and the tree which bears it."

children take to lunch). Postprocess contamination can occur if bacteria are not rigorously excluded from the packaging area.

High-temperature, short-time (HTST) heating consists of rapid heating to temperatures above 140°C, holding for several seconds, and then rapid cooling. This produces shelf-stable food. HTST processes improve product quality, process efficiency, and the shelf life of a food. HTST preserves nutrients and sensory attributes better than conventional processing, which subjects foods to lower temperatures for long periods. Juice boxes, drink pouches, and a variety of condiments are made using HTST technology.

The mild heating of packaged foods combined with well-controlled refrigeration produces a very-high-quality product. These products, also known as "REPFEDs" (i.e., "refrigerated processed foods of extended durability"), cook-chill products, or *sous vide* meals (foods mildly heated within vacuum packs), are becoming very popular.

However, some *C. botulinum* strains can grow at temperatures as low as 3.0°C and are a hazard in these foods. The Food and Drug Administration (FDA) requires a second hurdle (i.e., a preservative) in addition to refrigeration to ensure the safety of these foods.

Authors' note

Should food be thought of as "durable"?

Thermobacteriology

"Wet" heat (e.g., steam) kills microbes by denaturing nucleic acids, proteins, and enzymes. DNA damage may be the key event in killing vegetative cells. In spores, the germination system is the most heat-sensitive component. During mild heating, cytoplasmic membranes are a major site of injury. Dry heat is less lethal than wet heat and kills more slowly by dehydration and oxidation. Because of these different mechanisms, dry heat needs higher temperatures and longer times to cause the same lethality as wet heat.

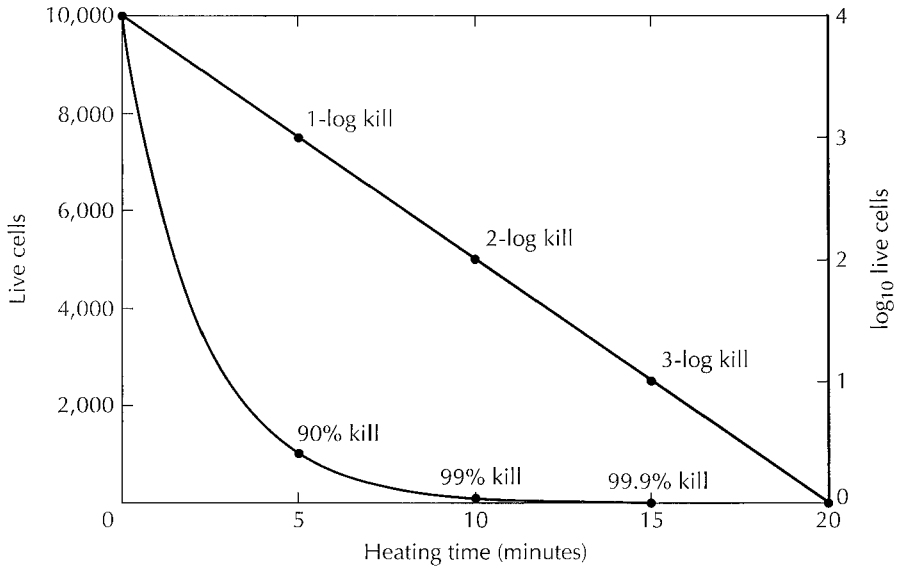
The death of a bacterial population by constant-temperature heating has logarithmic kinetics. That is, an equal time at a given temperature kills an equal percentage of the bacteria, regardless of the number of bacteria present. For example, if 5 min of boiling kills 90% of the bacteria, 10 min will kill 99% and 15 min will kill 99.9%. In theory, the kill can never reach 100%. Another way to look at this is that if the initial bacterial number were 100,000, after 5 min, the number would be reduced to 10,000. Similarly, if the initial number were 10,000,000, after 5 min, 1,000,000 would still be alive. It should be evident that, when many bacteria are present, a 90 or 99% reduction may not increase the safety of the product. (Figure 25.1 shows the arithmetic and log transform of lethality data.)

This negative exponential curve is expressed by the equation

$$N = N_0 e^{-k \cdot t}$$

In this equation, N is the number of viable cells after any time (t) of heating. N_0 is the initial number of viable cells, and k is the rate constant

Figure 25.1 Graph illustrating linear and logarithmic plots of the same data for viability of a bacterial population that is heated at a constant temperature. The black plot should be read using the linear (left) y axis. The purple plot corresponds to $\log_{10}$ values and should be read using the $\log_{10}$ (right) y axis.



(time^{-1}) for destruction at temperature T. If N_0, k, and t are known, N can be calculated. Similarly, if N_0 and k are known, the time t required to reduce a population to N bacteria can be determined.

The above may seem complicated. Fortunately, when the $\log_{10}$ number of viable bacteria is graphed against the time of heating, the curve is linear on the resultant survival curve (Fig. 25.2; Box 25.2). In this log-linear curve, the time it takes to reduce the number of viable bacteria 10-fold (i.e., 1 log unit or 90%) is called the D value, or *decimal reduction value*. In food microbiology, the D value replaces the concept of the rate constant k.

Survival curves show the rate of destruction of a specific organism in a specific medium or food at a specific temperature. Under given conditions, the death rate at any given temperature is constant and independent of the initial cell number. The logarithms of D values plotted against the heating temperatures give the thermal death time curve (Fig. 25.3). The thermal death time is the time necessary to kill a given number of organisms at a specified temperature.

Authors' note

The D *value is inversely related to* k *by the equation* $D_T = 2.3/k_T$.

The D value gives the microbe's heat resistance at a single temperature. However, in heat processing, microbes are exposed to many temperatures as the product heats up to the processing temperature, holds it, and then cools off. Higher temperatures have greater lethality (lower D values) than lower temperatures. The thermal death time curve shows how heat resistance changes at different temperatures. This enables one to calculate the cumulative lethality of each time-temperature combination encountered during the process.

The z value describes how lethality changes with temperature. The z value is the temperature increase (i.e., number of degrees) required to reduce the D value by a factor of 10. Graphically, it is equal to the slope of the thermal death time curve. The z value can also be calculated from the equation

$$t_2 = t_1 \times 10^{(T_1 - T_2)(z)-1}$$

where T_1 is the higher of the two temperatures at which the D value is t_1, T_2 is the lower temperature at which the D value is t_2, and z is the number of degrees required to change the D value by a factor of 10. For example, if the

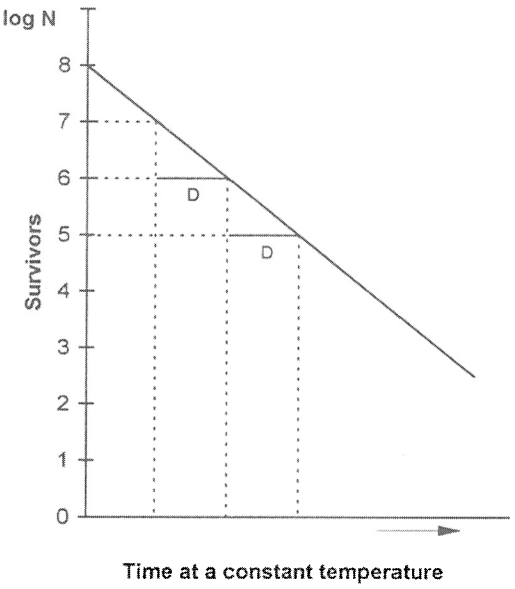


Figure 25.2 The bacterial survival curve shows the logarithmic order of bacterial death from which the D value is derived. Reprinted from J. Farkas, p. 567–592, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

BOX 25.2

How to calculate a *D* value

It is one thing to read in a book about *D* values, but another to actually calculate one. Here is a step-by-step illustrated example based on the heat resistance of *Bacillus* spores heated at 90°C.

1. Heat the spores at 90°C in an appropriate buffer or food. In this example, the spores were heated in milk.
2. At appropriate time intervals (0, 1, 2, 3, 5, and 10 min for this example) remove one 1-ml sample and place it in 9 ml of buffer to make a 10-fold dilution. From this, make 10-fold serial dilutions to a level that makes sense for the experiments.
3. Plate out the dilutions onto the appropriate agar, and incubate overnight at the appropriate temperature. Note that it is not necessary to plate every dilution. In the example below, the 10^1 and 10^2 dilutions were not plated until the 3-min time point because the initial spore number was known to be high. However, the "countable plate" can be missed if the right dilutions are not plated.
4. Record the data into a table such as the one shown in Fig. 1. Calculate the CFU per milliliter as the average of duplicate plates having between 30 and 300 colonies multiplied by the dilution factor. (There are different ways to record dilution data and calculate CFU per milliliter; if your instructor uses a different method, follow her instructions.) Note that in Fig. 1, the counts for 10 min are slightly below 30. These should be considered estimates.
5. Convert the data to $\log_{10}$.

Linear regression method

6. Do linear regression of the $\log_{10}$ data versus time using x as the time axis and $\log_{10}$ CFU/ml as the y axis. For the data in this example, the regression equation is $y = -0.2687x + 5.0089$. It is important to check the r^2 value for the linear regression line. The r^2 value is a "goodness of fit" or correlation constant, with values running from 0 (no correlation) to 1.0 (complete correlation). The r^2 value in this case is 0.98, which is very good. Low correlation coefficients (certainly below 0.85) suggest that the assumption of linearity should be reexamined, perhaps repeating the experiment.
7. The *D* value is the inverse of the negative value of the regression equation's slope:

$$D \text{ value} = 1/-(\text{slope}) = 1/-(-0.2687) = 3.72 \text{ min}$$

The linear regression method is done very easily with an Excel spreadsheet, as shown in Fig. 1.

Graphical method

D values can also be calculated graphically. Complete steps 1 through 5 above.

Time (min)	Dilution 10^1	10^2	10^3	10^4	10^5	cfu/mL	log cfu/mL
0			(87,107)	(7,12)	(3,0)	9.7E+04	4.987
1			(37,43)	(4,6)		8.0E+04	4.903
2			(23,36)	(5,2)		3.0E+04	4.477
3		(122,146)	(9,13)			1.3E+04	4.114
5		(32,35)	(1,8)			3.4E+03	3.531
10	(27,22)	(3,4)				2.5E+02	2.398
		Regression Equation: y=-0.2687*x+5.0089					
		D-value=1/-(slope)=1/-(-0.2687)=3.722 min.					

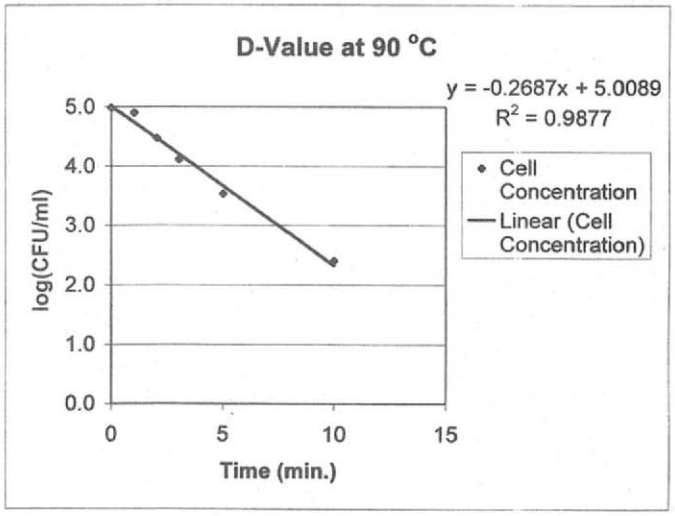
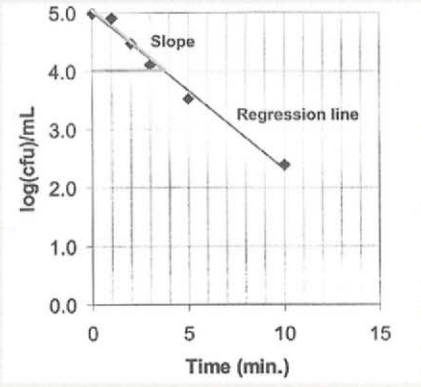


Figure 1

8. Use graph paper to plot the log CFU/ml versus time (Fig. 2).
9. Using the linear portion of the graph, find the amount of time it takes to make a 1-log reduction in CFU per milliliter. This time is the *D* value, about 3.7 min.

We thank Jon Cruz, an undergraduate in the Montville laboratory, for conducting the experiment and working out the example.

Figure 2



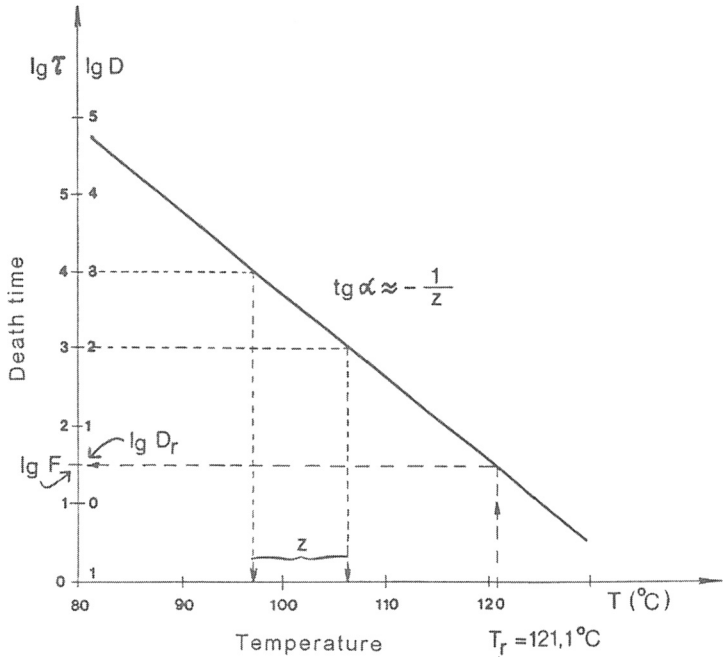


Figure 25.3 The thermal death time curve illustrates the z and F values. T_r, reference temperature; τ, death time. Reprinted from J. Farkas, p. 567–592, *in* M. P. Doyle, L. R. Beuchat, and T. J. Montville (ed.), *Food Microbiology: Fundamentals and Frontiers*, 2nd ed. (ASM Press, Washington, DC, 2001).

D value at 250°F was 2 min and the z value was 18°F (typical for many spores), then the D value at 232°F would be 20 min. It takes an 18° change in temperature to cause a 10-fold change in D value. What would be the D value at 268°F?

Know the difference:

The *D* value is the time (minutes) it takes at a given temperature to decrease viability by a factor of 10.

The *z* value is the number of degrees it takes to change the D value by a factor of 10.

The heat resistances (D values) and how they change with temperature (z values) are influenced by many factors (e.g., pH and a_w) and differ among microbial species. Bacterial spores are more heat resistant than vegetative cells of the same species. Some spores survive for minutes at 120°C or for hours at 100°C. Less heat-resistant spores have $D_{100°C}$ values of <1 min. Vegetative cells of bacteria, yeasts, and molds can have D values of <0.1 min at 70 to 80°C.

The age of the cells, their growth phase, the growth temperature, growth medium composition, exposure to prior stressors, and the heating environment also affect heat resistance. For example, one would expect the D value of a *Salmonella* sp. to be different in chocolate from what it is in milk. Thus, unlike chemical constants such as pK_a or molecular mass, D and z values depend on the situation of use. Since they depend on so many factors, great caution should be used in applying values obtained in one environment to some other food environment.

Generally speaking, the heat resistance of vegetative cells (Table 25.2) is much less than that of spores (Table 25.3). Vegetative cells of spore-forming bacteria are not especially heat resistant. The high heat resistance of spores is caused by their specific structures, which result in a relatively dehydrated

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Table 25.2 Relative heat resistances of some vegetative bacteria

Organism	Heating medium	*D* value (min) at:				*z* value (°C)
		70°C	65°C	60°C	55°C	
E. coli	Ringer solution, pH 7.0				4	
Lactobacillus plantarum	Tryptic soy broth + sucrose (a_w = 0.95)		4.7–8.1			
	Tomato juice, pH 4.5	11.0				12.5
Pseudomonas aeruginosa	Nutrient agar				1.9	
Pseudomonas fluorescens	Nutrient agar				1–2	
Salmonella enterica serovar Senftenberg (strain 775W)	Skim milk			10.8		6.0
	Phosphate buffer (0.1 M), pH 6.5		0.29			
	Buffer plus sucrose, 30:70 (%, wt/vol)		1.4–43			
	Same as above, plus glucose, 30:70 (%, wt/vol)		2.0–17.0			
	Milk chocolate	440				18.0
Salmonella enterica serovar Senftenberg	Heart infusion broth (a_w = 0.99), pH 7.4			6.1		6.8
	+ NaCl (a_w = 0.90, pH 7.4)			2.7		13
	+ sucrose (a_w = 0.90, pH 7.4)			75.2		8.9
Salmonella enterica serovar Typhimurium	Phosphate buffer (0.1 M), pH 6.5		0.056			
	Milk chocolate	816				19.0
Staphylococcus aureus	Custard or pea soup			7.8		4.5

spore core. (Remember that dry heat is much less lethal than wet heat.) The molecular basis of heat resistance in spores is covered in chapter 3.

Vegetative forms of yeasts are killed by 10- to 20-min heat treatment at 55 to 60°C. Increasing the temperature 5 to 10°C gives similar lethality for yeast ascospores. Vegetative propagules (a reproductive bud or offshoot) of

Table 25.3 Approximate heat resistance (*D* values) of some bacterial spores

Types of food and typical organisms	*D* value (min) at:		*z* value (°C)
	121°C	100°C	
Low-acid foods (pH > 4.6)			
Thermophilic aerobes			
Bacillus stearothermophilus	4.0–4.5	3,000	7
Thermophilic anaerobes			
Clostridium thermosaccharolyticum	3.0–4.0		12–18
Desulfotomaculum nigrificans	2.0 –3.0		
Mesophilic anaerobes			
Clostridium sporogenes	0.1–1.5		9–13
C. botulinum types A and B	0.1–0.2	50	10
Clostridium perfringens		0.3–20	10–30
Mesophilic aerobes			
Bacillus licheniformis		13	6
Bacillus subtilis		11	7
Bacillus cereus		5	10
Bacillus megaterium		1	9
Acid foods (pH ≤ 4.6)			
Thermotolerant aerobes			
Bacillus coagulans	0.01–0.1		
Mesophilic aerobes			
Bacillus polymyxa		0.1–0.5	
Bacillus macerans		0.1–0.5	
Clostridium butyricum (or *Clostridium pasteurianum*)		0.1–0.5	

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Industrial Strategies for Ensuring Safe Food

Introduction
GMPs
General Provisions (Subpart A)
Buildings and Facilities (Subpart B)
Equipment (Subpart C)
Production and Process Controls (Subpart E)
DALs (Subpart G)
Sanitation
SSOPs
HACCP
Conclusion
Summary
Suggested reading
Questions for critical thought

LEARNING OBJECTIVES

The information in this chapter will enable the student to:

- discuss the benefits of sanitation, good manufacturing practices (GMPs), and the Hazard Analysis Critical Control Point (HACCP) system with respect to human safety and food quality
- outline the process for development of a HACCP program
- recognize the limitations of HACCP
- outline the basic concepts of GMPs
- discuss the proper application of sanitizing agents
- integrate sanitation, GMPs, and HACCP
- identify practices that can result in the introduction of chemical, biological, or physical hazards into food
- identify segments of the food industry for which HACCP is mandatory

INTRODUCTION

Most people in the food industry are aware of Hazard Analysis Critical Control Point (HACCP) systems. Indeed, few students majoring in food science have not heard of HACCP by the end of their first year of college. This chapter focuses primarily on HACCP; however, before a HACCP program can be introduced or developed, basic building blocks must be in place. Specifically, in an industrial setting, implementation of good manufacturing practices (GMPs) and a sound sanitation program are key elements. The implementation of GMPs and of sanitation and HACCP systems is perhaps even more important now to address public concern that food supplies are safe and secure from intentional contamination by terrorists.

GMPs

In 1969, the Food and Drug Administration (FDA) published regulations on GMPs that provide basic rules for food plant sanitation. In 1986, the FDA published a revised version that expanded the original regulations. The GMPs can be found in the *Code of Federal Regulations* under Title 21, part 110, as "Current Good Manufacturing Practices in Manufacturing, Packing, or Holding Human Food" (referred to hereafter as CGMPs). To clarify what the *Code of Federal Regulations* contains, the following explanation may help. Proposed regulations are first published in a daily publication, the *Federal Register*, setting forth rulings by the executive branch and agencies of the federal government. These are then codified and published

as the *Code of Federal Regulations*. The *Code of Federal Regulations* is divided into 50 titles that represent areas that are subject to federal regulation.

The CGMPs are divided into seven subparts, each detailing requirements related to various operations or areas in food-processing facilities. The requirements are established to prevent contamination of a product from direct and indirect sources. Subparts D and F are reserved; therefore, only subparts A to C, E, and G are discussed here.

General Provisions (Subpart A)

Subpart A contains a list of definitions, the underlying reason for CGMPs linked to the Food, Drug and Cosmetic Act (referred to hereafter as the Act); responsibilities imposed on plant management regarding personnel; and exclusions. Definitions are provided to maintain consistency; however, they are often vague and open to interpretation. The criteria and definitions used to determine whether a food is *adulterated* (made unfit for human consumption by the addition of impurities) are covered under sections 402(a)(3) and 402(a)(4) of the Act. A food is considered adulterated if it contains any filth or putrid and/or decomposed material or is otherwise unfit for use as food. The Act goes on to state that food that has been prepared, packed, or held under unsanitary conditions; that may become contaminated with filth; or that has been rendered injurious to human health is adulterated.

Criteria for disease control, cleanliness, and the education and training of personnel are included. These requirements are designed to prevent the spread of disease-causing microorganisms from workers to food contact surfaces and the food itself. Workers in direct contact with food or areas that indirectly contact food must conform to hygienic practices. These include maintaining personal cleanliness, wearing clean clothing, removing jewelry, washing hands, and wearing appropriate protective garments (gloves, hairnets, and beard covers). Personnel responsible for plant sanitation should receive education and exhibit a level of competency so that they can pass the information on to workers in a facility.

The exclusions section holds exempt food establishments that are engaged solely in the harvesting, storage, or distribution of raw agricultural commodities [section 201(r) of the Act defines this as any food in its raw or natural state, including all fruits that are washed, colored, or otherwise treated in their unpeeled natural form prior to being marketed] that are ordinarily cleaned, prepared, treated, or otherwise processed before being marketed.

Buildings and Facilities (Subpart B)

Subpart B covers regulations for plant and grounds, which state that the grounds of a food plant should be maintained in a condition that will protect against the contamination of food. The plant design and construction must be such that operations within a plant can be separated and that materials used for walls, floors, etc. facilitate cleaning and sanitizing. The sanitary operations section covers general requirements related to general maintenance, cleaning and sanitizing compounds, toxic-material storage, pest control, sanitation of food contact surfaces, and storage and handling of cleaned equipment and utensils. The last section covers sanitary facilities and controls and provides minimum requirements for sanitary facilities and accommodations. Facilities and accommodations include, but are not limited to, the water supply, plumbing, sewage disposal, toilet facilities,

hand-washing facilities, and disposal of rubbish and *offal* (animal parts that are considered waste, generally applied to meat- and poultry-processing plants).

Equipment (Subpart C)

The equipment and utensils section specifies that the materials used and the design of plant equipment and utensils must allow adequate cleaning and proper maintenance. Emphasis is placed on preventing the adulteration of food by microbial contamination or by other contaminants, such as lubricants or metal fragments. Moreover, equipment used in the manufacturing, holding, or conveying of food must be designed and constructed for easy cleaning and maintenance. Cooling equipment (freezers and cold storage) and control instruments (e.g., pH meters and thermometers) must be accurate and adequately maintained. Finally, compressed gases that are introduced into food or used to clean food contact surfaces must be free of contaminants that are considered unlawful indirect food additives.

Production and Process Controls (Subpart E)

Subpart E is the most detailed subpart and is broken down into several sections and subsections that contain rules to ensure the fitness of raw materials and ingredients, to maintain manufacturing operations, and to protect finished products from deterioration. The processes and controls section stresses that all operations must be conducted in accordance with adequate sanitation principles. Quality control programs must be in place to ensure that food is suitable for human consumption and that packaging materials are safe and suitable. Quality control sanitation programs are required to ensure compliance with the Act.

The subsection on raw materials and ingredients contains descriptions for how those materials should be inspected, handled, washed, and segregated to minimize contamination and deterioration. Materials must not contain microorganisms or must be treated during manufacture to ensure that the levels of microorganisms that produce food poisoning or other human illness no longer present a problem and do not render the food adulterated. Compliance with regulations for natural toxins and other *deleterious* (harmful) substances is of the utmost importance and should be verifiable. Holding of raw materials must be adequate to protect against contamination and must be consistent with requirements for that material. Therefore, frozen materials must be kept frozen, dry materials must be protected against moisture and temperature extremes, and material scheduled for rework must be identified as such.

The handling of food in an appropriate way to prevent contamination is covered in the subsection on manufacturing operations. Required conditions are listed for the handling of foods that support the rapid growth of undesirable microbes, particularly those that cause foodborne illness. Requirements to protect several types of foods, for example, batters, sauces, and dehydrated foods, from contamination or adulteration are stated. Various operations, such as mechanical manufacturing (washing, cutting, and mashing), heat blanching, and filling, assembling, and packaging, must be performed to protect against contamination. To facilitate the tracking process, companies should code individual consumer packages with appropriate information (most companies routinely code each package). This is also important in the voluntary recall of products that are in violation of food laws and regulations.

The final section of subpart E, warehousing and distribution, is brief but nonetheless extremely important. The section states that the storage and transport of finished food must be conducted under conditions that prevent physical, chemical, and microbial contamination. The conditions must also prevent deterioration of the food and the container.

DALs (Subpart G)

The section of subpart G covering natural or unavoidable defects in food for human use that present no health hazards was established because it is not possible, and never has been possible, to grow (in open fields), harvest, and process crops that are totally free of natural defects. Certain defects that present no human health hazard, particularly in raw agricultural commodities, may be carried through to the finished product. The defect action level (DAL) represents the limits at or above which the FDA may take legal action to remove a product from the consumer market. The fact that the FDA has established DALs does not mean that a manufacturer need only stay below that level. Manufacturers cannot mix or blend a product containing any amount of defective food at or above the current DAL with another lot of the same or another product. This is not permitted by the FDA and renders the food unlawful regardless of the DAL of the finished food. Companies cannot use DALs to mask poor manufacturing practices. The FDA clearly states that the failure to operate under GMPs will result in legal action even though products might contain natural or unavoidable defects at levels lower than the currently established DALs.

An extensive list of DALs is available for products from allspice to wheat flour. The defect may be mold, rodent filth, insects, rot, or mammalian excreta. Below are a few examples of DALs (listed as product, defect, and action level):

- apple butter: rodent filth; average of four or more rodent hairs per 100 g of apple butter
- cocoa beans: mammalian excreta; mammalian excreta at 10 mg or more per pound
- fig paste: insects; >13 insect heads per 100 g of fig paste in each of two or more subsamples
- peaches, canned and frozen: moldy or wormy; average of 3% or more of fruit by count is wormy or moldy
- potato chips: rot; average of 6% or more of pieces by weight contain rot

Although not specified or required in the CGMPs, maintaining accurate records to document compliance with laws and regulations is implied. Records from all phases of the food operation should be maintained to verify that raw materials, food-packaging materials, and the finished product comply with the provisions of the Act. Records related to processes used to destroy or prevent the growth of microorganisms of public health concern are also important. To facilitate tracking of finished products and recalls, records related to distribution must be maintained. Record keeping also aids in the development of a HACCP program.

SANITATION

Many students, and for that matter food plant workers, incorrectly consider sanitation to be only the removal of trash and the cleaning of equipment. However, sanitation includes a host of activities that are ultimately

designed to prevent the adulteration of a product during processing. The true goal of any sanitation program is to protect the health of the consumer. In addition, a sound sanitation program increases the shelf life of a product, thereby minimizing economic loss, and prevents contamination of a product with materials that may offend the consumer (e.g., pieces of wood, rocks, or machinery). During the past 50 years, significant advances have been made in productivity and the ways that foods are processed. These advances have enhanced the safety and shelf life of products, but they require strict sanitation programs to ensure that contamination does not occur.

Before sanitation can be discussed, it must be defined with respect to the food industry. *Sanitation* is the creation of a comprehensive program designed to control microorganisms in a food-processing plant. A sanitation program must be based on sound science to ensure that healthy workers handle food in a clean environment. The goal of the program is to prevent contamination of food with pathogens that cause human illness and to limit the introduction and growth of microorganisms that result in spoilage. To achieve this goal, personal and environmental hygiene must be practiced. Sanitation is considered an applied science because of the human health issues that could arise should a breakdown in sanitation practices occur.

Within the sanitary-operations section of the CGMPs, general requirements for the maintenance of buildings, fixtures, and other physical facilities; pest control; sanitation of food contact surfaces; and storage and handling of cleaned equipment and utensils are covered. Depending on the product, the U.S. Department of Agriculture (USDA) may have jurisdiction over a food-processing facility, specifically those associated with meat, poultry, and eggs (according to the Federal Meat Inspection Act, Poultry Products Inspection Act, and Egg Products Inspection Act). Other agencies may also become involved. For instance, the Environmental Protection Agency may be involved if issues arise over water pollution (Federal Water Pollution Control Act) or air pollution (Clean Air Act) or through the use of pesticides (Federal Insecticide, Fungicide, and Rodenticide Act).

The benefits of sanitation can be numerous, but first and foremost they aid in protecting the consumer from illness. An effective sanitation program facilitates the development of an effective HACCP program. Inspections of food-processing plants, not only by federal inspectors but also by other companies (e.g., if company A buys a product from company B for use in a company A product), are becoming more stringent. An effective sanitation program can limit the number of negative consumer contacts related to food spoilage (a product has off odor or flavor) and foodborne illness. These types of problems weaken consumer confidence, causing decreased sales and increased claims. A stringent sanitation program can increase the storage life and shelf life of food, minimizing economic loss. Regular cleaning and maintenance of processing equipment and heating and cooling systems permit more efficient operation and reduce the microbial load in the facility and the product. In fact, the spread of microbes from cooling systems has been linked to the contamination of food and outbreaks of foodborne illness. Overall, a sound sanitation program can increase the quality of the product, improve customer relations, and increase the trust of inspectors associated with regulatory agencies.

Before sanitation standard operating procedures (SSOPs) can be developed, a sanitation audit must be conducted. A review of the facilities aids in identifying areas of noncompliance and establishing corrective actions. A

systematic approach should be used when conducting an audit. Starting at the receiving area, check food materials (ingredients) for temperature, odor, color, and appearance. Next, move to the general plant area and inspect walls, floors, drains, and footbaths. Processing equipment, tanks, pumps, pasteurization charts, and retort charts can also be inspected. While moving through the plant, observe employee practices and determine compliance. Check for employees who appear ill, have open lesions, are wearing jewelry, or are walking in restricted areas; check for the use of protective devices (hairnets, beard nets, and gloves) and footbaths. The building and the area around the outside of the building must also be checked for signs of pests, broken windows, a damaged roof, open doors and windows, standing water, and trash. Recall that sanitation is more than trash removal.

Microbes can be brought into processing facilities on raw materials or picked up by food during processing as it passes through or over handling equipment. Employees can spread microbes to equipment and to food. Microbes from process waters during washing, conveying, or preparation can contribute to the microbial load of the product. Microorganisms can also be present in ingredients that may be added to the product. This is particularly important if ingredients are added after a process designed to reduce or eliminate microbes has been completed.

Sanitization (using germicidal or sanitizing agents) of food contact surfaces is the key to maintaining a sanitary food-processing environment, thereby reducing the number of microorganisms associated with a food. Sanitization involves the treatment of a food contact surface so that the process effectively destroys vegetative cells of microorganisms capable of causing human illness and adequately reduces populations of other microbes. The process does not need to reduce populations of bacterial spores. The treatment should not have an adverse impact on the food (impart an off odor or color) or the safety of the consumer.

The main criterion for selecting a chemical sanitizing agent is the ability of the agent to kill microorganisms of concern within a specified time. While chlorine or other chemical sanitizing agents can be extremely effective in reducing the microbial load, they must be used appropriately. In food-processing facilities, chlorine compounds are the most widely used sanitizing agents. A number of factors can reduce or effectively inactivate sanitizers. Sanitizers can react with organic materials (soil, food, bacteria, oils, etc.) that have not been removed from equipment surfaces, reducing the effectiveness of the sanitizer. Impurities in water, including iron, manganese, nitrites, and sulfides, can react with a sanitizer and decrease its effectiveness. Although temperature is generally not a factor, temperatures below 10°C and above 90°C can dramatically influence the availability and action of some sanitizers. Generally, increasing the concentration of a sanitizer increases the rate of destruction of the microorganisms. However, in the case of hypochlorites (e.g., sodium hypochlorite), this is not the case. For example, the addition of hypochlorite to water to give 1,000 parts per million (ppm) of chlorine produces a solution that requires three times as long to kill the same number of bacteria as a solution containing 25 ppm. This is because hypochlorites are produced by combining chlorine with hydroxides. The addition of hypochlorite results in the addition of a proportional amount of alkali that raises the pH of the solution. A 1,000-ppm solution of chlorine has a pH of ~11.0, whereas a 25-ppm solution has a pH of 8.0. In general, the effectiveness of chlorine and iodine compounds decreases as the pH increases.

Chlorine sanitizers may be the most popular sanitizers, but a number of other chemical sanitizers are available that are more useful than chlorine compounds for certain applications. Iodophors are good as skin disinfectants, since they do not irritate the skin at recommended levels of use. They can be used to clean equipment but may stain epoxy and polyvinyl chloride and may cause discoloration of starchy food. Hydrogen peroxide is effective against biofilms and may be used on all types of surfaces (equipment, floors, and walls). Ozone, glutaraldehyde, and quaternary ammonium are also effective. The efficacies of these compounds against certain microbes differ from those of chlorine compounds. For example, quaternary ammonium sanitizers are effective against molds and gram-positive bacteria but are ineffective against most gram-negative bacteria.

Sanitizing can be accomplished without the use of chemical agents. Ultraviolet (UV) irradiation is used to destroy microorganisms in hospitals. In Europe, UV light units are commonly used to disinfect drinking water and food-processing waters. Steam can be used, but it is costly and not particularly effective. In fact, condensation from the application of steam may complicate the cleaning process and provide moisture for surviving microorganisms. Hot water may be useful for sanitizing small components (eating utensils and small containers), but like steam, the process is expensive because of high energy costs.

A sound sanitation program results in the manufacture of foods that are safe and wholesome and that meet legal obligations. Sanitation is the responsibility of every person in a food-processing plant. Sanitation must be practiced every day, a policy the food company must reinforce through proper training and encouragement of employees. If every person understands his or her responsibility in a sanitation program, individual efforts will ensure that a consistent product is produced and consumer confidence is maintained. If executed properly, sanitation should reduce (perhaps eliminate) concern about the spreading of human diseases and food poisoning. Moreover, if sanitation is maintained, a product with a reduced microbial population will be produced and waste and spoilage will be minimized.

SSOPs

SSOPs are guided by CGMPs detailing a specific sequence of events necessary to perform a task to ensure sanitary conditions. To prevent direct product contamination, SSOPs should contain a description of procedures that a food-processing facility will follow to address the elements of preoperational and postoperational sanitation. A set of written SSOPs must be developed and maintained by federally and state-inspected meat and poultry plants.

The USDA Food Safety and Inspection Service (FSIS) felt that SSOPs would permit each company to consistently follow effective sanitation procedures, thereby reducing the likelihood of direct product contamination or adulteration. The SSOPs must include information related to daily preoperational and operational procedures for the plant personnel responsible for monitoring daily plant sanitation activities. Companies must determine whether the SSOPs are effective and, if not, take corrective action. Corrective actions include procedures to ensure the appropriate disposition of contaminated products, restoring sanitary conditions, and preventing the recurrence of direct contamination or product adulteration. SSOPs written for meat and poultry plants must be signed and dated by the individual

with overall authority over the establishment to ensure that the company will implement the SSOPs.

Although written SSOPs are not required for all food production establishments, they are the cornerstones of HACCP. SSOPs go beyond specific processes. HACCP plans are developed as a proactive approach to ensure the safety of a product. Interfacing CGMPs and sanitation (SSOPs) permits the development of an effective HACCP program.

HACCP

The development of HACCP programs by companies has become routine since the mid-1990s. In some instances, the FDA and the USDA FSIS have published final rules mandating the development and implementation of HACCP programs to enhance the safety of domestic and imported seafoods, juices, and meat and poultry. In fact, HACCP can be used by all segments of the food industry to effectively and efficiently ensure farm-to-table food safety in the United States (or, for that matter, in any country). HACCP is a proactive, prevention-oriented program that addresses food safety through the analysis and control of biological (predominantly pathogenic microorganisms), chemical (including those naturally produced by microorganisms), and physical (metal, wood, and plastic) hazards. The hazards may come from raw material production or the handling, manufacturing, distribution, and consumption of the finished product.

Authors' note

The building blocks for development of a solid HACCP system require strict adherence to GMPs and SSOPs. Without implementation of GMPs and SSOPs, an effective HACCP system cannot be achieved.

A food processor must identify potential sources of food safety hazards for each product and production line (process). Preventative steps can then be identified to reduce or eliminate potential hazards. Since this is a proactive approach, the manufacturer determines in advance what to do with a product or process should a preventative step fail. This requires continuous monitoring and documentation to verify that the appropriate corrective action was taken. Records can be reviewed to aid in improvement of the plan. HACCP is a continuous system, and HACCP plans require updating as ingredients and processes change and as new technologies emerge.

Background

The HACCP concept was developed nearly 50 years ago by the National Aeronautics and Space Administration and Natick Laboratories as the Failure Mode Effect Analysis system for use in aerospace manufacturing. Through collaborative work by the Pillsbury Company, the National Aeronautics and Space Administration, and Natick Laboratories, the process was applied to foods produced for the U.S. space program. This was done to ensure that the food used would be 100% free of bacterial pathogens. Could you imagine being in space and developing a case of projectile vomiting and severe diarrhea? This initial work led to what is now HACCP.

Although HACCP was introduced to the public (food-manufacturing companies) in the early 1970s, the concept was met with limited interest. However, in 1985, a subcommittee of the Food Protection Committee of the National Academy of Sciences made a strong endorsement of HACCP. The National Advisory Committee on Microbiological Criteria for Foods was formed in 1988 and given the mission of developing material to promote the understanding of HACCP. HACCP is now practiced by food manufacturers and food service establishments, and even in the home.

HACCP Basics

The HACCP system is based on seven principles, or steps, which are discussed in more detail below. They are (i) conduct a hazard analysis, (ii) determine the critical control points (CCPs), (iii) establish critical limits, (iv)

establish monitoring procedures, (v) establish corrective actions, (vi) establish verification procedures, and (vii) keep records and documentation. There are a number of approaches that can be taken in the development of a HACCP program; however, they all stress a commonsense approach to food safety management. Template HACCP plans can be obtained through the FDA and USDA FSIS for specific products for which HACCP have been mandated.

Before a HACCP plan can be developed, there are a number of activities that must be completed. Obviously, management must support the development and implementation of HACCP. Without such support, efforts to develop and implement HACCP will likely prove futile. A HACCP team should be assembled and should include line workers and personnel involved in sanitation and in production and quality assurance. Including personnel involved in marketing and communication may also be appropriate. To balance the team and bring a fresh perspective, an outside expert may be beneficial. In small companies, it may be difficult to establish an effective HACCP team; this is when an outside expert(s) is especially useful.

The HACCP team must recognize that a generic HACCP plan cannot be developed for all food products produced in a manufacturing plant. A HACCP plan should be specific for each product. However, once a HACCP plan is developed, it may be appropriate to use the plan as a template for the development of subsequent plans. A description of the product should include the name, formulation, method of distribution, and storage requirements. The intended use and anticipated customers for the food should be established. The product may not be intended for use by the general public but rather by infants, the immunocompromised, or some other high-risk group.

The development and verification of a flow diagram describing the production process is an important step that is often overlooked. The flow diagram is essential for hazard analysis and establishing CCPs (Fig. 26.1). Moreover, the diagram becomes a guide and is essential in the verification process. Even though a flow diagram has been constructed, a "walk-through" of the production facility must be done to verify that the diagram is accurate. For example, new equipment, such as metal detectors, may have been added or the route for rework may not have been accounted for. The flow diagram should be modified if and when necessary.

The following narrative discusses the seven principles of HACCP, providing a brief description of each.

Principle 1: hazard analysis. Through hazard analysis, a list of hazards that are likely to cause injury or illness if not controlled is developed. This should be done systematically, starting with raw materials and following through each step of the food production flow diagram. Hazards that present little or no risk to the consumer should not be considered in a HACCP plan. Indeed, those hazards are generally addressed within GMP programs, a building block for the HACCP system. A hazard as set forth by the National Advisory Committee on Microbiological Criteria for Foods for HACCP is "a biological, chemical, or physical agent that is reasonably likely to cause illness or injury in the absence of its control." Even though a hazard may not be considered for inclusion in a HACCP plan, this does not mean that food processors can ignore the hazard, especially if the hazard fits the definition of adulteration.

Once the hazards have been identified through the review of ingredients, the entire production process, equipment, storage, and distribution,

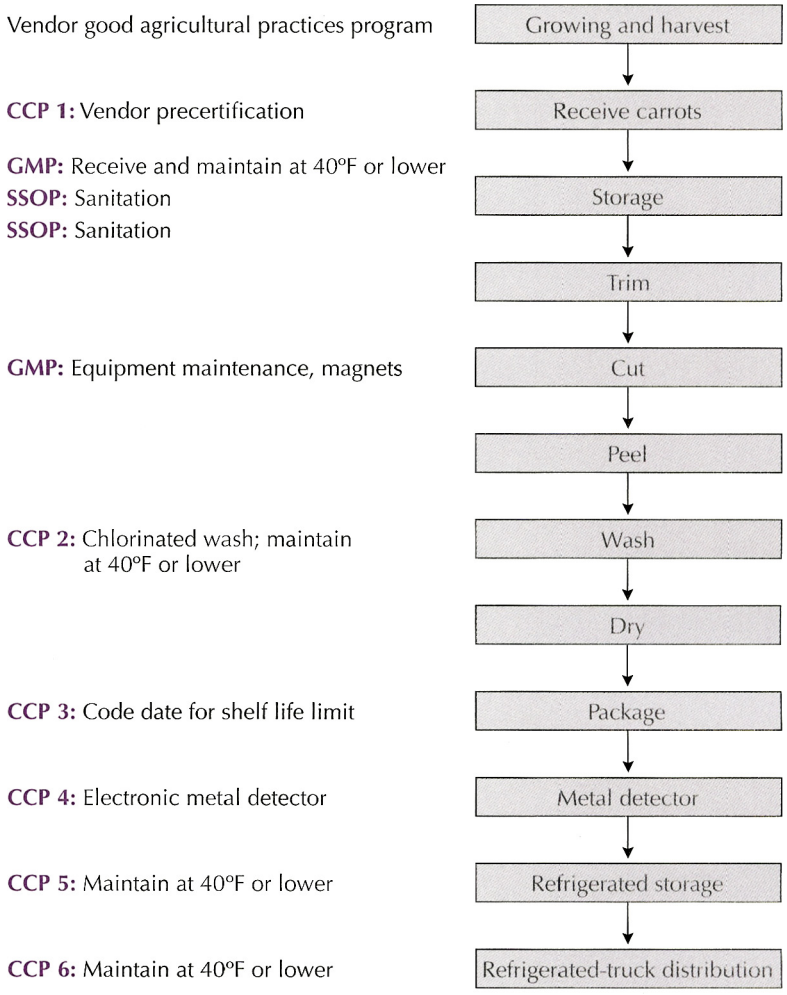

Figure 26.1 Representative flow diagram and CCPs for a nonthermally processed product. The flow diagram with CCPs is for peeled refrigerated minicarrots. Note that GAPs, GMPs, and a sanitation program are key elements in the production of a safe, wholesome product.

each hazard must be evaluated. During hazard evaluation, the HACCP team decides which hazards present a significant risk to consumers and must be addressed through the HACCP plan. Generally, this involves determining the severity of the hazard in terms of the illness or injury it could produce and the likelihood of occurrence. For example, the likelihood that an undercooked beef patty might be contaminated with *Escherichia coli* O157:H7 might be low, but the illness that could result is severe and the potential for subsequent illness (e.g., disease involving kidney failure) is high. Therefore, these hazards should be addressed through HACCP. Alternatively, the hazard may be small pieces of cardboard that enter when ingredient packages are opened during mixing. The likelihood of this occurring may be high, but the severity of the hazard is low (the cardboard may not be very tasty, but it presents little if any health hazard). This type of hazard can be addressed through GMPs. The evaluation of the severity of hazards can be based on epidemiological information, scientific research, company records, and information provided by government agencies.

During this process, the HACCP team must identify appropriate control measures. Since not all hazards can be prevented or eliminated, the term "control" is more appropriate. Control measures may vary significantly from one food process to another, even for the same hazard. To control a hazard, a change in processing, such as the addition of metal detectors, or a certificate of analysis from a supplier indicating the absence of a given hazard may be required. Additionally, existing plans, including GMPs and sanitation, can be reviewed to determine whether the hazard is controlled through those plans.

Principle 2: determining CCPs. Once all significant hazards and control measures have been identified, the CCPs to control each hazard must be identified. A CCP can be any point, step, or procedure at which control can be applied and a food safety hazard (chemical, biological, or physical) can be controlled (prevented, eliminated, or reduced to an acceptable level). A CCP may be a cooler, a metal detector, or a package label indicating storage and cooking instructions. Many processes may be redundant, for example, passing dry materials through multiple sieves to control entry of physical hazards into the final product or the use of multiple metal detectors on a production line. In either case, the last sieve or metal detector is considered the CCP.

The temptation in establishing CCPs in a process is to assign each step as a CCP. This is particularly true for raw products. The number of CCPs for any process should be kept to a minimum to facilitate monitoring and documentation (Fig. 26.1). Otherwise, records are generated that have little meaning with respect to the effectiveness of the HACCP plan. Keep in mind that a CCP may control a hazard that has occurred upstream in the process. For example, the hazard might be enteric pathogens in ground beef being used for cooked beef patties. The cooking process is the CCP, since this step will kill pathogens of concern. This does not mean that the ground beef can be left at room temperature or otherwise abused prior to or after being cooked. Indeed, keeping the ground beef at $\leq 4°C$ may also be a CCP.

Principle 3: critical limits. Critical limits must be set for each CCP. A critical limit is based on predetermined tolerances that must be met at a CCP to prevent, eliminate, or reduce a biological, chemical, or physical hazard to an acceptable level. Critical limits for preventative measures may be set for such factors as time, temperature, pH, water activity, salt concentration, moisture level, and even the presence or concentration of preservatives. Each food company is responsible for ensuring that the critical limits indicated in its HACCP plan will control the identified hazard. Critical limits may be based on in-house experimental studies or derived from regulatory standards and guidelines or review of the scientific literature. Validation (see principle 6 below) that the critical limits are effective and being met is the key to HACCP.

Principle 4: monitoring procedures. Monitoring is conducted to ensure that a CCP and its limits are effective. The results of monitoring are documented and provide a record for future use in verification. Monitoring may be done automatically (e.g., by computer) or manually. Monitoring activities include measurement of the pH, temperature, and moisture level or water activity. Monitoring must be a planned sequence of observations or measurements to assess whether a CCP is under control. If it is not conducted as

a routine, the HACCP plan becomes compromised and contaminated products may reach the consumer. Ideally, monitoring should be continuous and allow advance warning of a problem before violation of a critical limit occurs. However, when continuous monitoring is impractical, a monitoring interval must be established, procedures for the collection of data and equipment to be used must be defined, and the person responsible for monitoring must be identified. The frequency of monitoring will depend in large part on the product and the type of measurement made.

The monitoring procedure(s) for a given CCP should be rapid, which rules out the use of time-consuming analytical tests. Therefore, microbial testing is rarely used for monitoring CCPs. Personnel involved in monitoring must be properly trained, understand the importance of monitoring, and know the proper action to take should deviation from critical limits occur. For example, if ground beef patties were to be cooked for 60 s but the belt increased in speed and the patties were cooked for only 30 s, what should be done? To address this deviation, corrective actions must be established.

Principle 5: corrective actions. HACCP is a proactive system to control hazards that may be associated with a food product. Therefore, corrective actions designed to address deviations from established critical limits should be written into the HACCP plan. Corrective actions should address how the company will fix or correct the cause of deviation to ensure that the process is brought under control and that critical limits are achieved, indicate what is to be done with a product in which deviation from the critical limit occurred, and facilitate the process for determining whether adjustments to the HACCP plan are required.

Corrective action for the beef patties that were not cooked adequately would include correcting the equipment problem, i.e., adjusting the conveyor belt speed. Moreover, the product involved in the deviation must not be released. The product could be sent through the oven again to receive the proper time-temperature treatment, disposed of as waste, or perhaps used in some other product.

Principle 6: verification procedures. Verification within and of the HACCP plan is conducted at several levels. The verification process determines whether the HACCP plan is being followed and whether records of monitoring activities are accurate. The entire HACCP plan should be revalidated through the use of outside auditors. Verification should be done to ensure that the HACCP plan is functioning satisfactorily and in compliance, if required, with government regulations. In the case of cooking ground beef patties, the time-temperature parameters must be verified to kill, for example, *E. coli* O157:H7.

Verification at some level, a given critical limit or the entire HACCP plan, should take place on a regular and predetermined schedule. This may be daily, weekly, monthly, or yearly. The frequency is determined by the operation being verified.

Principle 7: record keeping. Record keeping is the heart of a HACCP plan. All records generated in association with a HACCP plan must be on file at the food establishment, particularly those relating to CCPs and any action on critical deviations and product disposition. Records provide a means for implementation of changes by the HACCP team to assess and

document the safety of products. HACCP plan records should include the names of team members, a product description and intended use, flow diagram with CCPs indicated, types of hazards, critical limits, and a description of monitoring, corrective actions, and types of records.

HACCP: beyond the Food Processor

Whether dining at a fine restaurant or enjoying a meal at the local burger establishment, customers expect that the food served will be wholesome and safe. Operators of food service and retail establishments are not required to utilize HACCP. However, many such establishments have voluntarily implemented a food safety management system based on HACCP principles. This proactive approach is encouraged by the FDA to ensure that the food served or sold at a given establishment is safe. In fact, utilizing HACCP-based systems can help food service and retail establishments meet regulations associated with the Food Code.

The flow of food through a food service operation is critical to limiting the contamination of the finished product. Of particular concern is cross-contamination, whereby a cooked or ready-to-eat food comes into contact with an uncooked or raw product. The holding temperature of cold or hot foods is also critical to preventing or limiting the growth of microbes that may be present on a food. In order to ensure that the food is handled appropriately, a flow diagram must be developed (Fig. 26.2). The process for developing a HACCP plan for a food service facility is clearly very similar to that used by the food processor. In fact, HACCP plans can be developed for the home by using the same approach.

HACCP Established for Specific Food Industries

In 1996, the USDA FSIS established the Pathogen Reduction and HACCP system program for meat- and poultry-processing plants. The Pathogen Reduction and HACCP rule encompasses more than HACCP. The plan requires all meat and poultry plants to develop and implement HACCP to improve the safety of their products, sets pathogen reduction performance standards for *Salmonella,* requires all meat and poultry plants to develop and implement written SSOPs, and requires animal- and poultry-slaughtering plants to conduct microbial testing for generic *E. coli* to verify the adequacy of their process controls for the prevention of fecal contamination. The plan was precipitated by a large outbreak of foodborne illness caused by the consumption of undercooked hamburgers that were contaminated with *E. coli* O157:H7.

In 1996, the FDA published the *Fish and Fishery Products Hazards and Control Guide* to assist processors in the development of HACCP plans. The plans must address the various hazards to which seafood can be exposed from the fishery to the table, including viruses, bacteria, parasites, natural toxins, and chemical contaminants. Under the regulation, seafood companies must also write SSOPs. The FDA will periodically inspect seafood processors and review HACCP records to determine how well a company is complying over time. The action by the FDA to implement a mandatory HACCP system for the seafood industry is in part linked to the knowledge that more than one-half of the seafood eaten in the United States is imported from almost 135 countries.

The FDA Center for Food Safety and Applied Nutrition proposed the HACCP fruit and vegetable juice regulation in 1998, which applies to juice products in both interstate and intrastate commerce. As in other HACCP

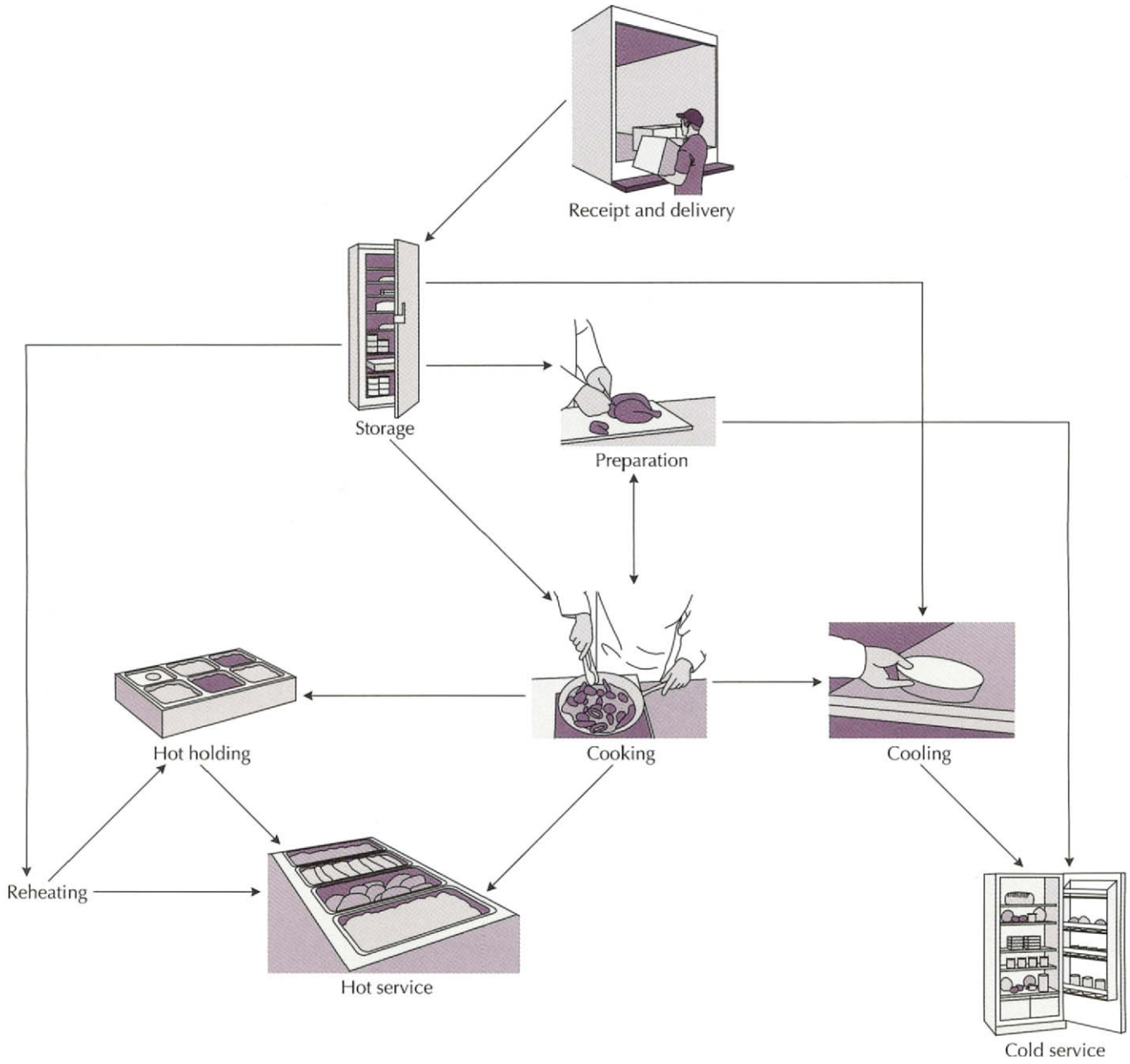

Figure 26.2 The flow and handling of product in a food service facility are of the utmost importance to ensuring that the product served to the consumer is wholesome and safe. The flow diagram outlines the flow of foods intended to be served hot or cold.

regulations, juice processors are required to evaluate their manufacturing processes to determine whether any chemical, biological, or physical hazards could contaminate their products. The real concern is microbiological hazards. The most notable aspect of this regulation is that processors must implement control measures that achieve a 5-log-unit reduction in the numbers of the most resistant pathogens in their finished products compared with levels that may be present in untreated juice. A combination of

methods can be used to achieve this goal. Citrus processors can use surface treatments that achieve the 5-log-unit pathogen reduction. Processors are exempt from the microbiological-hazard requirements of the HACCP regulation if they are making shelf-stable juices or concentrates by using a single thermal-processing step. Finally, retail establishments that package and sell juice directly to consumers are not required to comply with the regulation. A driving force behind this HACCP regulation was a large outbreak of foodborne illness linked to the consumption of unpasteurized apple juice that was contaminated with *E. coli* O157:H7.

CONCLUSION

Implementation of GMPs and sanitation and HACCP programs prevents the sale of adulterated food products to the consumer. These measures also have a positive impact by decreasing the number of cases of foodborne illness. In today's global marketplace, measures need to be in place to prevent intentional contamination of food or to prevent food that is potentially contaminated from reaching the consumer. The programs discussed above have mainly concentrated on foods postharvest. On the farm, measures such as good agricultural practices (GAPs) are being used to decrease biological, chemical, and physical hazards that may be associated with raw agricultural products. Indeed, many fruit and vegetable processors are now requiring producers (farmers) to implement GAPs. Reducing the microbiological load of a food at the farm level should aid in keeping the microbiological load of the finished product in an acceptable range.

Summary

- GMPs and sanitation are the building blocks of a HACCP program.
- GMPs are designed to prevent the manufacture of food under unsanitary conditions and the sale of adulterated food.
- Sanitation includes personal and environmental hygiene.
- HACCP is a proactive approach to effectively and efficiently ensure farm-to-table food safety.
- HACCP can be effectively applied to food service facilities to ensure safety of prepared food.
- HACCP is mandatory for meat- and poultry-processing plants, fruit and vegetable juice producers, and seafood processors.

Suggested reading

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Marriott, N. G. (ed.). 2006. *Principles of Food Sanitation,* 5th ed. Springer Publishing, New York, NY.

National Advisory Committee on Microbiological Criteria for Foods. 1998. Hazard analysis and critical control point principles and application guidelines. *J. Food Prot.* **61**:762–775.

U.S. Department of Health and Human Services, Food and Drug Administration, Center for Food Safety and Applied Nutrition. 2006. *Managing Food Safety: a Manual for the Voluntary Use of HACCP Principles for Operators of Food Service and Retail Establishments.* http://www.cfsan.fda.gov/~acrobat/hret2.pdf.

Questions for critical thought

1. Why is it necessary to have DALs for raw agricultural products?
2. What are the seven HACCP principles? Before a HACCP plan can be developed, what needs to be done?
3. Explain why GMPs and sanitation are considered building blocks (prerequisites) for the development of HACCP.
4. Select a product, such as bagged chopped lettuce or precooked hamburger patties, and develop a HACCP plan for it.
5. When sanitizing agents are used, what factors can influence their efficacy? Explain why.
6. The FDA and USDA FSIS made HACCP mandatory for meat and poultry processors, fruit and vegetable juice producers, and the seafood industry. Write one page for each, detailing events that necessitated mandatory HACCP for those industries.
7. Define adulteration of a food product. Provide a scenario that could lead to a product being considered adulterated.
8. You have been hired by a small company to develop and direct the company's HACCP program. The owner of the company was against hiring you, since he believes that GMPs and sanitation already cover all aspects of HACCP. You must explain to him what a HACCP program does that GMPs and sanitation do not. He requests that you provide him with a two-page report.
9. Establishing a HACCP program for a restaurant can be difficult since food may be served cooked (grilled steak) or raw (house salad). If possible, visit the kitchen of one of your school's dining halls and record the flow of food through the kitchen. Now determine CCPs for hot and cold foods. Consider whether the flow of food needs to be changed to ensure that raw foods do not cross-contaminate cooked foods. This exercise could be conducted in teams, and then the teams could compare the HACCP programs that they have developed.

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Answers to Puzzles

CROSSWORD PUZZLES

Chapter 2

Across

3. homeostasis
4. homolactic
7. five
9. energy
11. Baird-Parker
12. selective

Down

1. no
2. psychrophile
5. oxidative
6. one
8. intrinsic
10. hurdle

Chapter 6

Across

2. stringency
3. quantitative
5. manufacturing
8. Howard
9. direct
10. Codex
11. standard

Down

1. specification
4. indicator
5. mandatory
6. cadaverine
7. guideline

Chapter 15

Across

3. inactivated
5. BOTOX
8. anaerobic
9. psychrophilic
10. soil
11. infant

Down

1. neurotoxin
2. wound
4. neurological
6. nitrite
7. diaphragm

Chapter 19

Across

1. bacteriophage
5. ATP
6. fermentation
9. nitrite
12. homofermentative
14. raid
15. eyes
18. oxygen
19. seed
20. yeast

Down

2. acidification
3. carbohydrates
4. vinegar
7. cit
8. pentoses
10. proteins
11. *Lactococcus*
13. oxidized
16. sweats
17. lytic

Chapter 25

Across

1. Xerox
3. fee
5. *D* value
7. freeze
8. hydro
9. spore
10. rain
12. ORP
14. pasteurization
17. radiation
18. HUS
19. vent
21. AM
22. torr
23. microwave

Down

1. xero
2. xerotolerant
3. freeze drying
4. energy
5. dehydration
6. add
9. SOS
11. nonthermal
13. pressure
15. activity
16. ziti
20. pico
21. a_w
23. MW

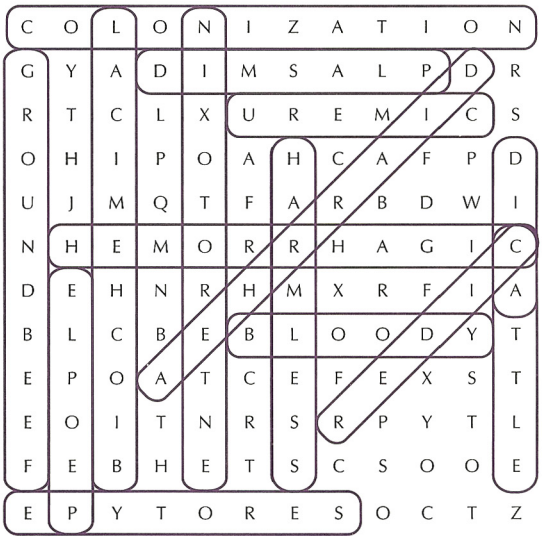
WORD SCRAMBLE PUZZLE

Chapter 3

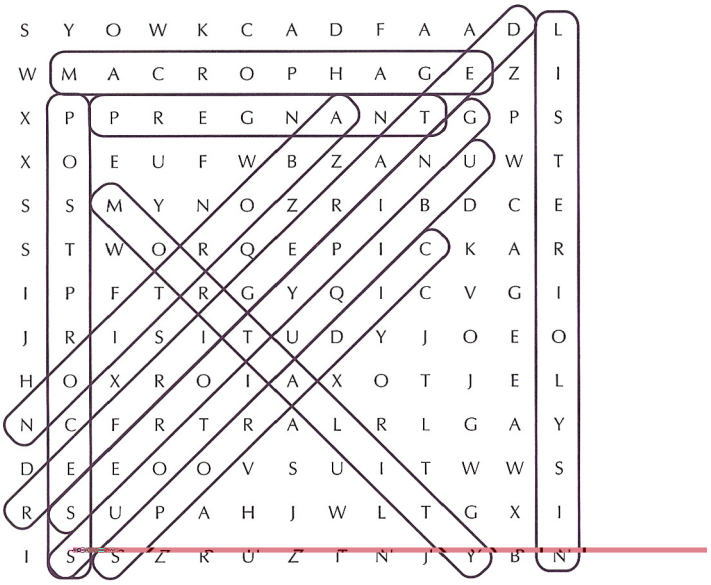
1. Appert
2. aseptic
3. *Bacillus*
4. *botulinum*
5. *Clostridium*
6. commercial
7. cortex
8. exosporangium
9. germination
10. resistance
11. spoilage
12. sporulation

WORD SEARCH PUZZLES

Chapter 9



Chapter 13



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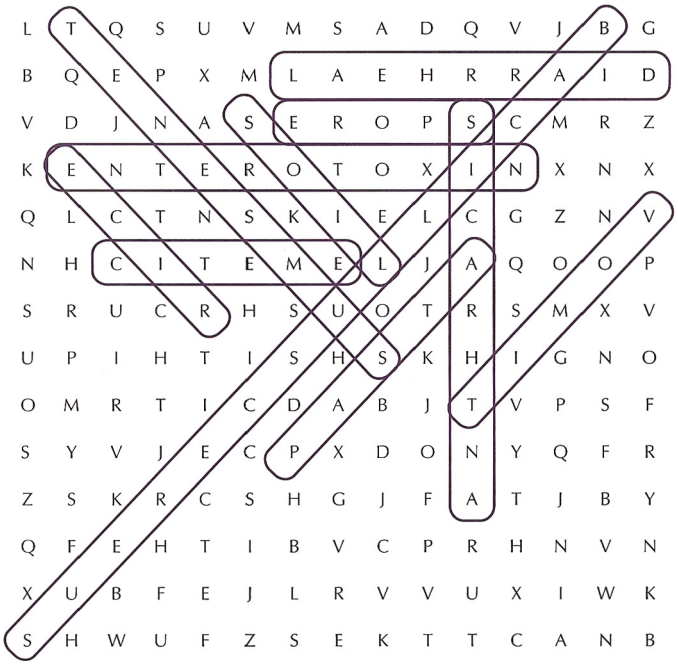

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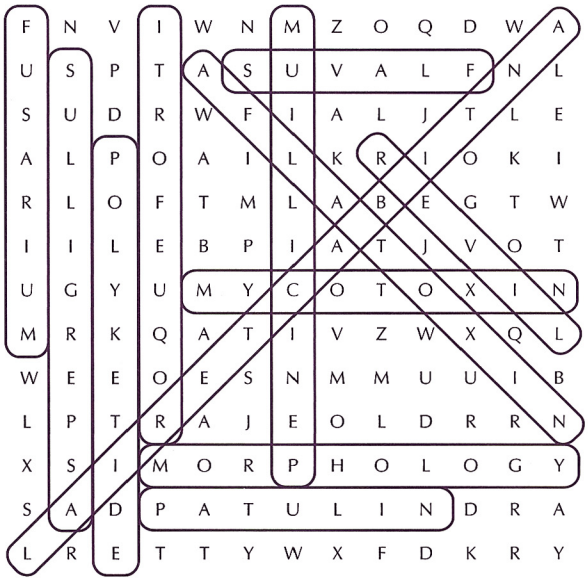
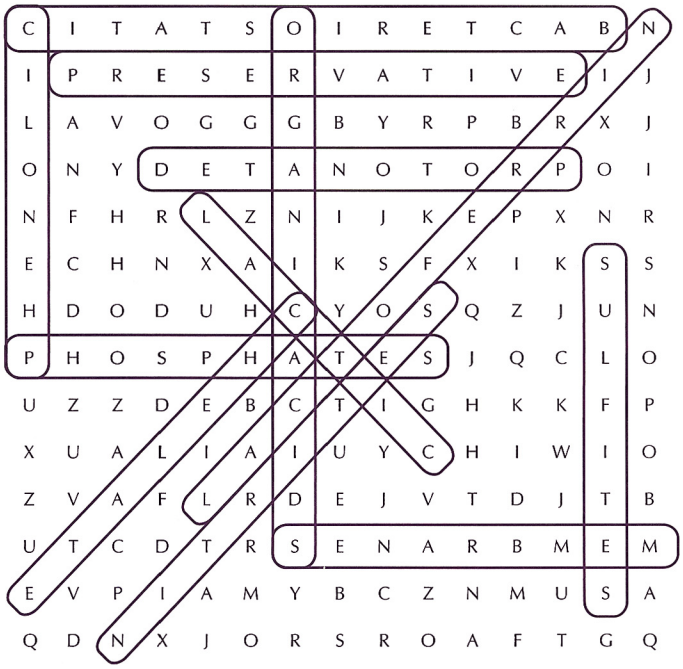


Chapter 23



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Glossary

Acid tolerance response The response of cells to an initial acid treatment which allows them to survive more severe acid treatments. Abbreviated ATR.

Adenosine triphosphate A compound that serves as cellular energy currency. Abbreviated ATP.

Aerobe An organism that requires oxygen to grow.

Aerobic plate count A count that provides an estimation of the number of microorganisms in a food. Abbreviated APC.

Anaerobe An organism that cannot grow in the presence of air.

Antigenic Capable of eliciting an immune (antibody) response. Small molecules may be nonantigenic. Large molecules may have several antigenic sites.

AOAC International An organization involved in validation of testing methods. Formerly the Association of Official Analytical Chemists.

APC *See* Aerobic plate count.

Ascospore A heat-resistant reproductive fungal spore.

ATP *See* Adenosine triphosphate.

ATR *See* Acid tolerance response.

Attaching and effacing lesions Lesions that occur when a bacterium (e.g., *Escherichia coli* O157:H7) adheres to the surface of an intestinal epithelial cell, resulting in loss of microvilli (effacement).

Autoxidation The spontaneous oxidation of a substance.

a_w *See* Water activity.

Bacteremia The presence of bacteria in the blood.

Bureau of Alcohol, Tobacco, Firearms and Explosives An agency of the U.S. Department of Justice that has regulatory authority over beer, wine, and liquor (as well as firearms, explosives, and tobacco).

CDC *See* Centers for Disease Control and Prevention.

Centers for Disease Control and Prevention The agency within the Department of Health and Human Services that tracks foodborne illness and helps solve outbreaks. It has no regulatory authority. Abbreviated CDC.

CFU *See* Colony-forming unit.

Chelate To bind ions.

Chlamydospores Asexually produced resting spores of certain fungi.

-cidal A suffix indicating the ability to kill.

Colony-forming unit A colony on an agar plate that in theory arises from a single bacterial cell. Abbreviated CFU.

Commercial sterility A level of sterility that indicates that an item is free of organisms that can cause illness.

Compatible solutes "Harmless" compounds which cells accumulate to equilibrate their internal water activity with the water activity of the environment.

Cortex The spore structure responsible for resistance properties, presumably through dehydration of the core.

Cucurbits A group of plants that includes squashes, cucumbers, and pumpkins.

Cytokines Substances or compounds that are produced and secreted by cells of the immune system. These substances play a significant role in the immune response.

Cytotoxic Lethal to cells.

Differential media Media that allow specific bacteria to be visualized (e.g., through a color reaction) among a population of other bacteria.

Disulfide linkage A covalent bond formed between two cysteine molecules in different parts of a protein; serves to stabilize the protein shape.

Emetic toxin A toxin that causes vomiting.

Endospore A bacterial spore formed in the body of the mother cell.

Endotoxin A toxin structurally associated with the cell.

Enterocolitis Inflammation of the lining of the intestine.

Enterotoxin A toxin that acts in the gastrointestinal tract.

Environmental Protection Agency A federal government agency that regulates food-related issues, such as sanitizer efficacy and pesticides. Abbreviated EPA.

EPA *See* Environmental Protection Agency.

Epidemiology The study of epidemics; it is used to determine the factors that lead to an outbreak of foodborne illness.

Exotoxin An excreted toxin.

Extrinsic factor An external factor, such as temperature or atmosphere, that influences the ability of microbes to grow in a food.

Facultative Having the ability to do something that is not the preferred mode. For example, a facultative anaerobe can grow in the absence of oxygen but grows better in its presence.

FDA *See* Food and Drug Administration.

Fecal-oral route A route of disease transmission from fecal matter to the body via the oral cavity.

Fecal-oral transmission The transmission of pathogens from one person's feces to another person's mouth, usually due to poor hand washing.

50% lethal dose The concentration of a substance that will kill 50% of a population. Abbreviated LD_{50}.

Food and Drug Administration The U.S. government agency that has legal authority over all foods except meat, poultry, eggs, and alcohol. It is part of the Department of Health and Human Services. Abbreviated FDA.

Food Safety and Inspection Service The food safety regulatory arm of the U.S. Department of Agriculture. It inspects all meat and poultry processing plants. Abbreviated FSIS.

Food safety objective A quantitative goal for the frequency of a particular foodborne illness. Abbreviated FSO.

FSIS *See* Food Safety and Inspection Service.

FSO *See* Food safety objective.

GAPs *See* Good agricultural practices.

Gastroenteritis Broadly speaking, a disease or illness that originates in the gut.

Generally recognized as safe A legal classification of food additives in use before 1958; it includes additives affirmed as safe since that time. Abbreviated GRAS.

Genetic fingerprinting A nucleic acid-based technique that provides specific identification (a "fingerprint") of a microorganism.

Germinant A compound that induces spore germination.

Germination The first irreversible step in the process by which a spore becomes a vegetative cell.

Glyco- A prefix meaning "containing a sugar."

GMPs *See* Good manufacturing practices.

Good agricultural practices Prescribed practices, such as the use of potable water for rinses, prohibition against fertilizing with human manure, and good worker hygiene, that help ensure the microbial safety of food at the farm level. Abbreviated GAPs.

Good manufacturing practices Prescribed practices, such as rodent control programs, use of hair nets and gloves, and prohibition of jewelry, that help ensure the safety of food in food-processing plants. Abbreviated GMPs.

GRAS *See* Generally recognized as safe.

HACCP *See* Hazard Analysis Critical Control Point.

Halotolerant Able to tolerate high salt concentrations.

Hazard Analysis Critical Control Point A proactive, prevention-oriented program that addresses food safety through the analysis and control of biological, chemical, and physical hazards. Abbreviated HACCP.

Hemolysin A compound that causes lysis of red blood cells.

Hemolytic Able to break open red blood cells.

Hemorrhagic colitis A disease characterized by bloody diarrhea.

Hermetically sealed Sealed under a vacuum.

Heterofermentative Forming lactic acid, acetic acid, and ethanol as fermentation products. Also called heterolactic.

Homeostasis An attempt by a microorganism to maintain a constant intracellular state, e.g., maintenance of pH.

Homolactic Forming only lactic acid as a fermentation product. Also called homofermentative.

Host An organism that provides an environment for a second organism to live in. In the case of foodborne pathogens, the host is the victim.

Immunocompromised Having an immune system that is unable to combat normal disease processes.

Inoculum (pl., inocula) The bacteria initially present that initiate growth.

Intrinsic factor A property, such as pH or water activity, inherent in a food.

Isoelectric point The pH at which a protein has no net charge.

Isolate A strain of bacteria obtained ("isolated") from a specific site. Strains of the same species that are environmental or clinical isolates can be quite different.

kGy *See* Kilogray.

Kilogray A unit of absorbed radiation equal to 1 joule of energy. Abbreviated kGy.

LD_{50} *See* 50% lethal dose.

Low-acid food A food with a pH of >4.6 and a water activity of >0.85.

Lyse To break open.

Lysis The breakage of cells.

Lysozyme An enzyme that degrades cell walls.

MAP *See* Modified-atmosphere packaging.

Meningitis Inflammation of the tissue (meninges) surrounding the brain and spinal cord.

Mesophile An organism with an optimal growth range of 20 to 45°C.

4-Methylumbelliferyl-β-D-glucuronide A substrate used to determine production of β-D-glucuronidase by *Escherichia coli* O157:H7. Abbreviated MUG.

Modified-atmosphere packaging Storage or packing of foods in increased amounts of CO_2. Abbreviated MAP.

Monoclonal antibody An antibody that detects a specific cell target.

Most probable number A statistical method for estimating small populations of bacteria. Abbreviated MPN.

MPN *See* Most probable number.

MUG *See* 4-Methylumbelliferyl-β-D-glucuronide.

Mycotoxins A generic term for the chemical toxins formed by fungi.

Necrotic Dead.

Neurotoxin A toxin that acts on nerves.

New-variant Creutzfeldt-Jakob disease A human neurological disease caused by prions. Abbreviated nvCJD.

nvCJD *See* New-variant Creutzfeldt-Jakob disease.

Opportunistic pathogen A bacterium that causes illness only in people with some preexisting medical condition.

Osmotolerant Tolerant of conditions (i.e., high salt or sugar concentrations) where there is little available water (low water activity).

Outgrowth The process by which a germinated spore becomes a vegetative cell.

Pandemic Epidemic over an especially wide geographic area.

Pasteurization A heat process designed to kill pathogens but not necessarily spoilage organisms.

Plasmid A piece of circular DNA, separate from the chromosome, that contains genes for traits such as virulence, toxins, bacteriocin production, and various biochemical traits. Plasmids can be transmitted among different species of bacteria.

Plasmolysis Shrinkage of a cell due to water loss.

Poliomyelitis A degenerative disease of the muscles that causes paralysis. Also called polio.

Polyclonal antibody An antibody that detects many cellular targets.

Pomaceous fruits A group of fruits that includes apples and pears.

Prion An unusual type of protein which, by changing shape, causes transmissible encephalopathies.

Prostration An extreme state of exhaustion.

Proteolytic Producing enzymes that degrade proteins.

Pseudoappendicular syndrome A syndrome in which, in broad terms, an individual experiences symptoms commonly associated with appendicitis.

Psychrophile An organism that "loves" to grow in the cold, has an optimum growth temperature of 15°C, and cannot grow at 30°C.

Psychrotroph An organism that can grow in the cold but has an optimum growth temperature of >20°C and can grow at 30°C.

Ready-to-eat food A food that can be eaten without further cooking, such as deli meats.

Reiter's syndrome An autoimmune disease characterized by arthritis and inflammation around the eye.

***rpoS* genes** Genes that encode DNA-dependent RNA polymerase.

Saprophyte An organism that survives by living off dead or decaying plant material.

SASP *See* Small acid-soluble proteins.

SEA *See* Staphylococcal enterotoxin A.

Selective media Media that select for the growth of specific bacteria by inhibiting the growth of other bacteria that may be present.

Septicemia A gross, whole-body infection.

Serological Able to cause an antibody response.

Serotypes Varieties of a bacterial species that respond to different antibodies.

Sigma factor A protein that binds to a DNA-dependent RNA polymerase.

Small acid-soluble proteins Spore proteins that confer resistance properties. Abbreviated SASP.

Sporadic Occurring randomly.

Sporulation The process by which a vegetative cell forms and releases a spore.

Staphylococcal enterotoxin A One serological type of staphylococcal enterotoxin. Abbreviated SEA.

-static A suffix indicating the ability to inhibit or stop something.

Temperature abuse The holding of food at temperatures that permit microbial growth, i.e., 40 to 140°F.

Tenesmus The sensation of an urgent need to defecate while being unable to do so.

Thermophile An organism that grows at high temperatures.

Transmissible spongiform encephalopathy A disease, such as scrapie, kuru, or "mad cow disease," that is caused by prions. Abbreviated TSE.

TSE *See* Transmissible spongiform encephalopathy.

Turkey "X" disease A disease of turkeys for which the causal agent was not initially known.

USDA *See* U.S. Department of Agriculture.

U.S. Department of Agriculture A cabinet-level government department that has legal authority over meat, poultry, and eggs. Abbreviated USDA.

Vehicle A source or carrier.

Viable but nonculturable A term applied to cells that cannot be cultured by conventional methods but that still cause illness if ingested. Abbreviated VNC.

Virulent Causing illness.

VNC *See* Viable but nonculturable.

Water activity The measure of water available for microbial growth and chemical reactions, defined as the equilibrium relative humidity of a product. Abbreviated a_w.

Xerotolerant Capable of tolerating dry conditions.

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